Cloud Data Integration for Developers

onDemand Lab Guide

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Cloud Data Integration for Developers (onDemand)

November 2024

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Document Conventions

This guide uses the following formatting conventions:

If you see…	It means…	Example
>	Indicates a sub menu to navigate to.	Click Repository > Connect. In this example, you should click the Repository menu or button and choose Connect.
boldfaced text	Indicates text you need to type or enter.	Click the Rename button and name the new source definition **S_EMPLOYEE**.
UPPERCASE	Database tables and column names are shown in all UPPERCASE.	T_ITEM_SUMMARY
italicized text	Indicates a variable you must replace with specific information.	Connect to the Repository using the assigned *login_id*.
Note:	The following paragraph provides additional facts.	**Note:** You can select multiple objects to import by using the Ctrl key.
Tip:	The following paragraph provides suggested uses or a Velocity best practice.	**Tip:** The m_ prefix for a mapping name is…

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In addition to the student and lab guides, Informatica provides these other resources:

- Documentation and Knowledge Base
- Global Customer Support
- Professional Certification

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https://www.informatica.com/services-and-training/support-services/contact-us.html

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You can take and pass exams provided by Informatica to obtain Informatica Professional Certification. For more information, go to

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Table of Contents

Pre-requisite Labs

Lab 0-1: Creating Informatica Intelligent Cloud Services (IICS) Trial Account 1

Lab 0-2: Setting up Lab Environment 8

Lab 0-3: Configure SQL Server Database 12

Module 1: Informatica Cloud Overview

Lab 1-1: Navigating the IICS Interface 18

Module 2: Introduction to Data Integration Service

Lab 2-1: Installing Secure Agent 30

Lab 2-2: Creating Flat File Connections 46

Module 3: Synchronization Task

Lab 3-1: Creating a Salesforce Connection 49

Lab 3-2: Creating SQL Server Connections 56

Lab 3-3: Creating a Synchronization Task 60

Lab 3-4: Using Filter, Expression, and Lookup in a Synchronization Task 70

Lab 3-5: Using Saved Query in a Synchronization Task 82

Lab 3-6: Using Upsert Operation in Synchronization Task 88

Module 4: Data Transfer Task

Lab 4-1: Creating a Data Transfer Task 98

Module 6: IICS Transformations

Lab 6-1: Classifying the Product Shipping Details Using Various Transformations 105

Lab 6-2: Using Union, Sorter, and Rank Transformations in a Mapping 124

Lab 6-3: Using Expression Macro in a Mapping 132

Lab 6-4: Using Vertical Macro in a Mapping 148

Lab 6-5: Using Horizontal Macro in a Mapping 156

Module 7: Joiner and Lookup Transformations

Lab 7-1: Using Lookup Transformation in a Mapping 166

Lab 7-2: Merging Current and Previous Data Loads Using Full Outer Join 190

Module 8: Mapplets

Lab 8-1: Using Mapplet Transformation in a Mapping 210

Lab 8-2: Create Mapplet and Use It in the Existing Mapping 225

Lab 8-3: Creating a Mapping Using Existing Mapplet and Aggregator Transformations. 239

Module 9: Mapping Parameterization

Lab 9-1: Performing Complete Parameterization 252

Lab 9-2: Using In-Out Parameters for Incremental Data Loading 270

Module 10: Mapping Task

Lab 10-1: Creating a Mapping Task to Perform Basic History Tracking And Configure Advanced Settings. 283

Module 11: Replication Task

Lab 11-1: Replicating Data to a Flat File. 306

Module 12: Masking Task

Lab 12-1: Creating a Masking Task..... 312

Lab 12-2: Using Data Masking Transformation in a Mapping. 319

Module 13: Taskflows

Lab 13-1: Creating a Linear Taskflow. 327

Lab 13-2: Creating a Taskflow with Conditional Logic..... 334

Module 14: IICS REST APIs

Lab 14-1: Creating a Mapping Using a REST V2 Connector. 350

Lab 14-2: Using Web Services Transformation in a Mapping. 362

Lab 14-3: Running a Mapping Task Using REST API..... 375

Lab 14-4: Monitoring Tasks Using REST API 384

Pre-requisite Labs

Lab 0-1: Creating Informatica Intelligent Cloud Services (IICS) Trial Account

Overview:

To use the different services that are available in the IICS platform, users must have a valid IICS account. The trial account is valid for 90 days.

This document lists the steps to create an IICS trial account to use in the onDemand training course. The course content and Labs are activated at the time of registration and are available for 90 days or until all the lab hours are consumed.

Objective:

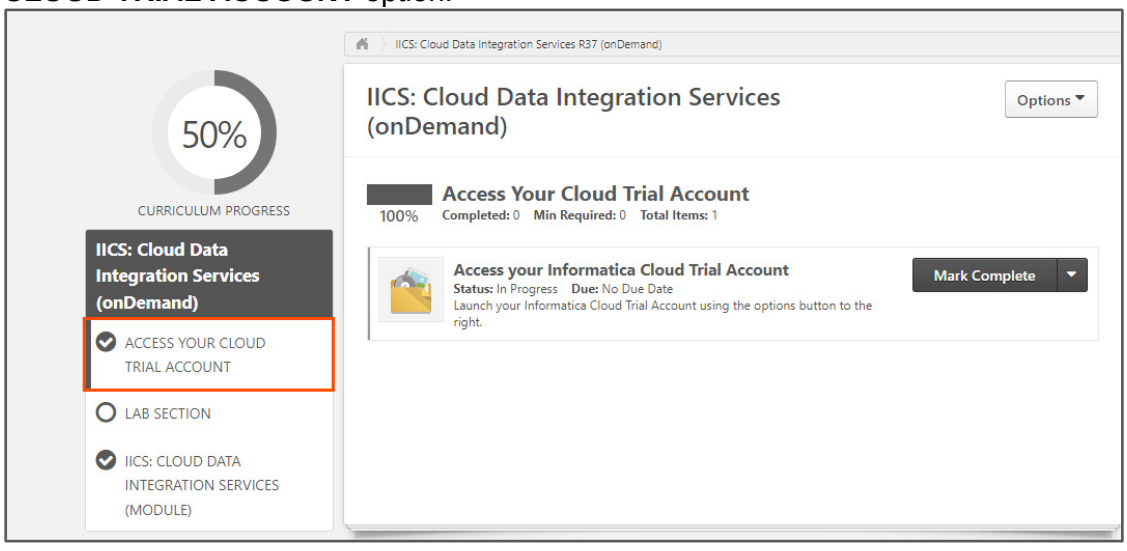
- Create an IICS trial account

Important:

The screenshots used in this lab guide is for Cloud Data Integration course. The name of course may differ depending upon the enrolled course. However, the steps for creating the IICS account will remain the same.

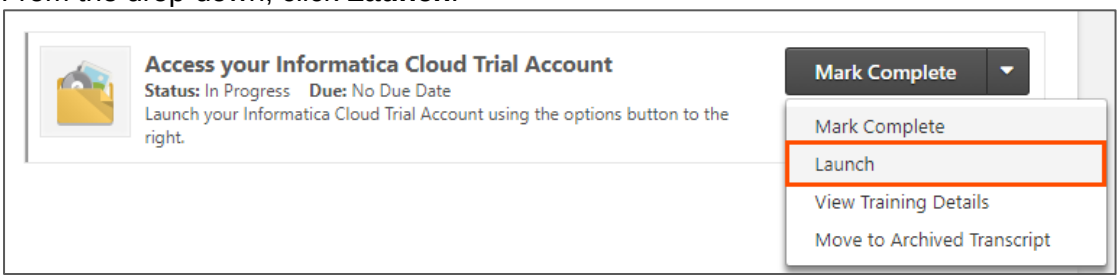
Creating an IICS account

1. From CSOD (the page where you accessed the course), click on the **ACCESS YOUR CLOUD TRIAL ACCOUNT** option.



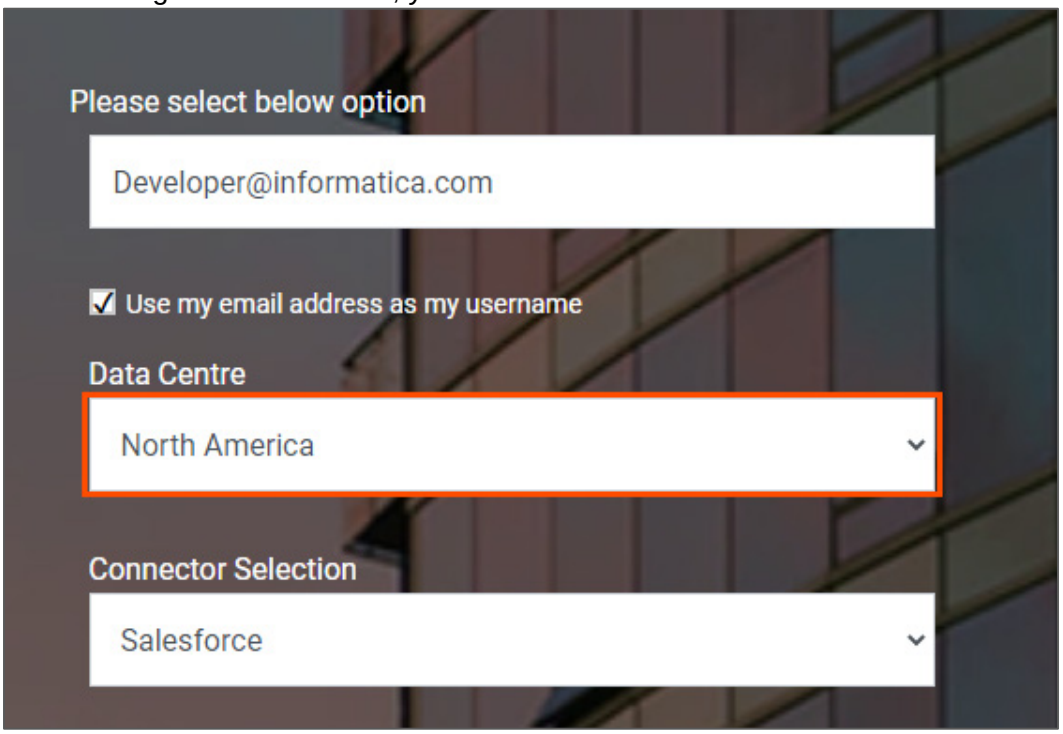
Note: The Course Name shown on the screen will differ according to the course you have enrolled for.

2. From the drop-down, click **Launch**.

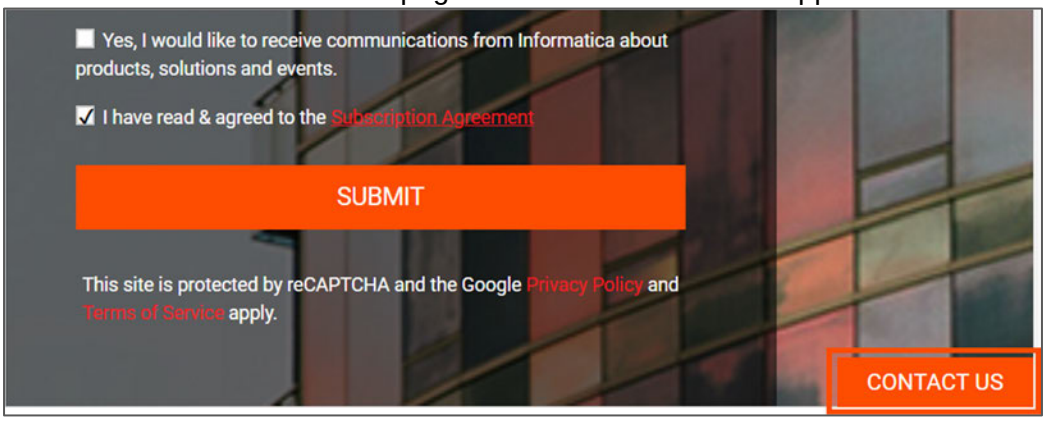


Note: If the option menu is labeled anything other than **Launch**, click the down arrow to the right and click **Launch**.

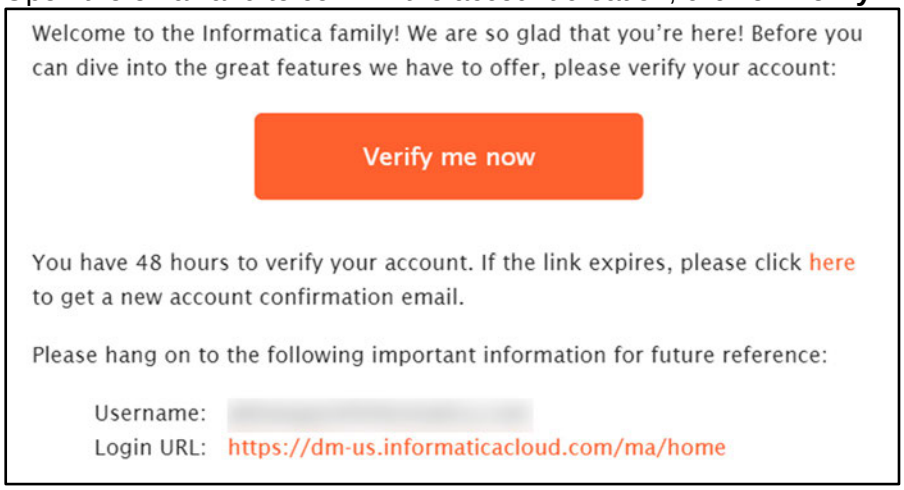
3. A new University Trial window opens in a new tab. Use the new window to sign up for an IICS trial account.
4. On the signup page, enter a valid email id and other required information.
Note: During account creation, you must select the Data Centre as **North America**.



5. If you are unable to register your email-id while creating the account, use the **CONTACT US** button at the bottom of the page to contact Informatica Support.



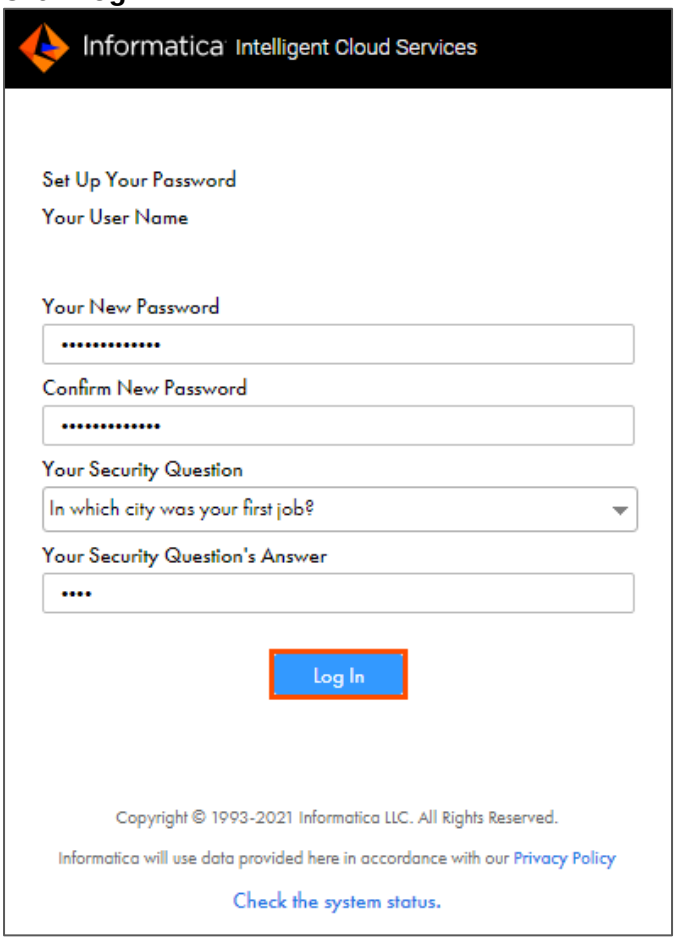
6. Once the required information is entered, check the "**I have read & agreed to the Subscription Agreement**" checkbox and click **SUBMIT**.
7. After you submit the request, in a few seconds you will get an email from admin@informaticacloud.com in your registered mailbox.
8. Open the email and to confirm the account creation, click on **Verify me now** link.



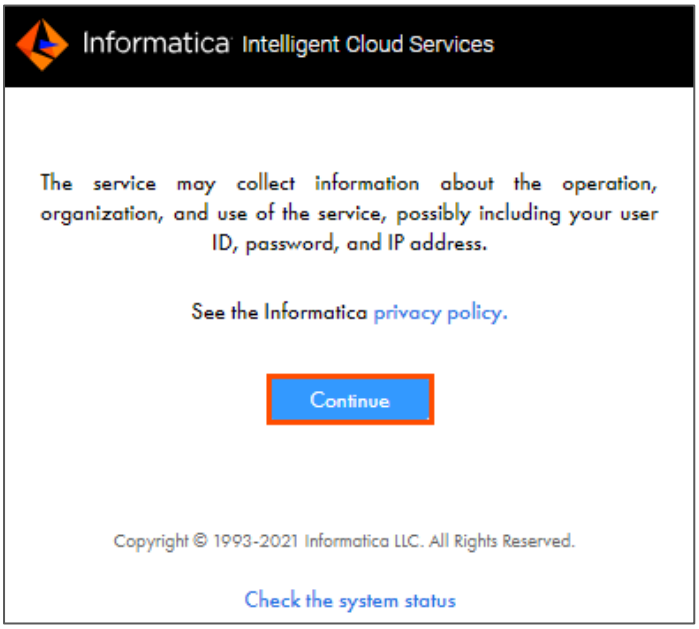
Note: You can bookmark the URL link to access the IICS login page for future access.

9. In the password setup page, enter a suitable password and a security question for the account.

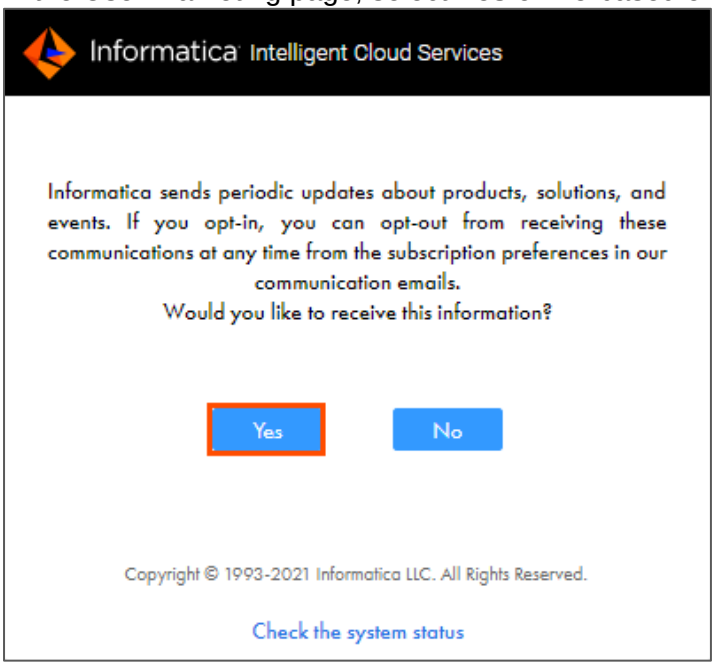
10. Click **Log In**.



11. Click **Continue**.

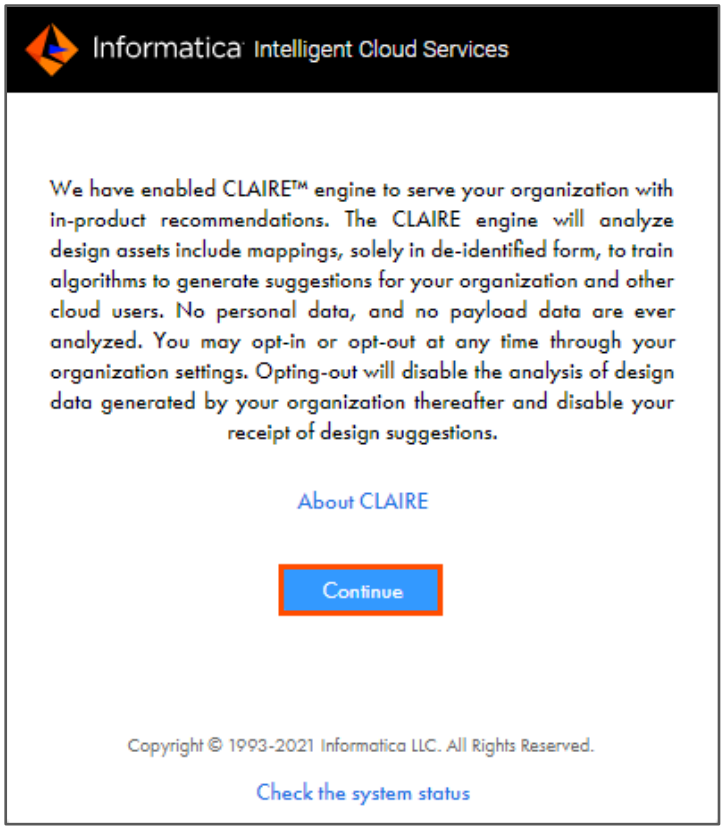


12. In the User Marketing page, select **Yes** or **No** based on your preference.

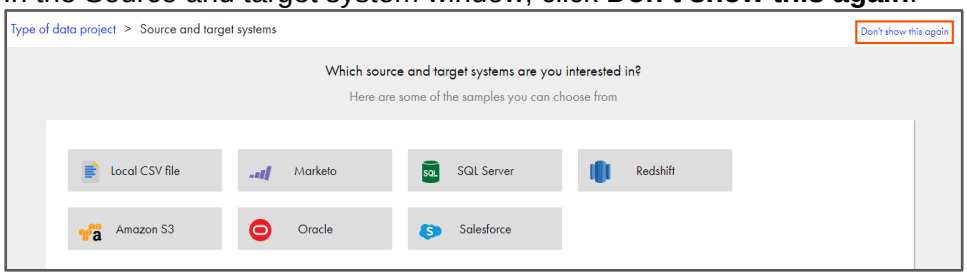


Note: It is recommended to select Yes to receive regular updates about product and other related information.

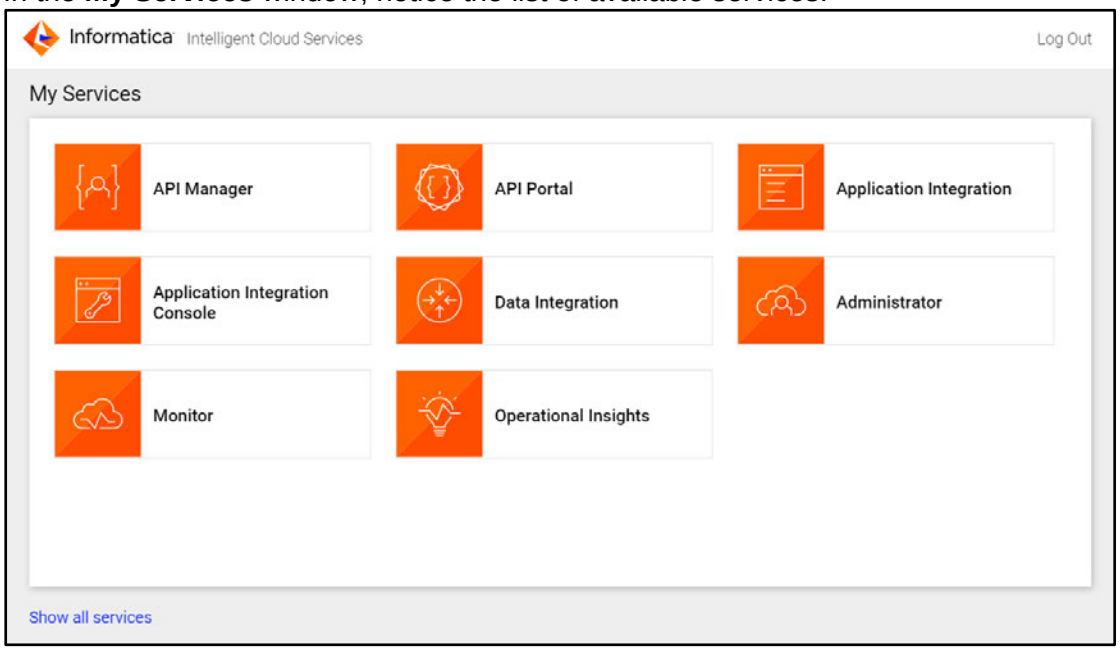
13. Click **Continue**.



14. In the Source and target system window, click **Don't show this again**.

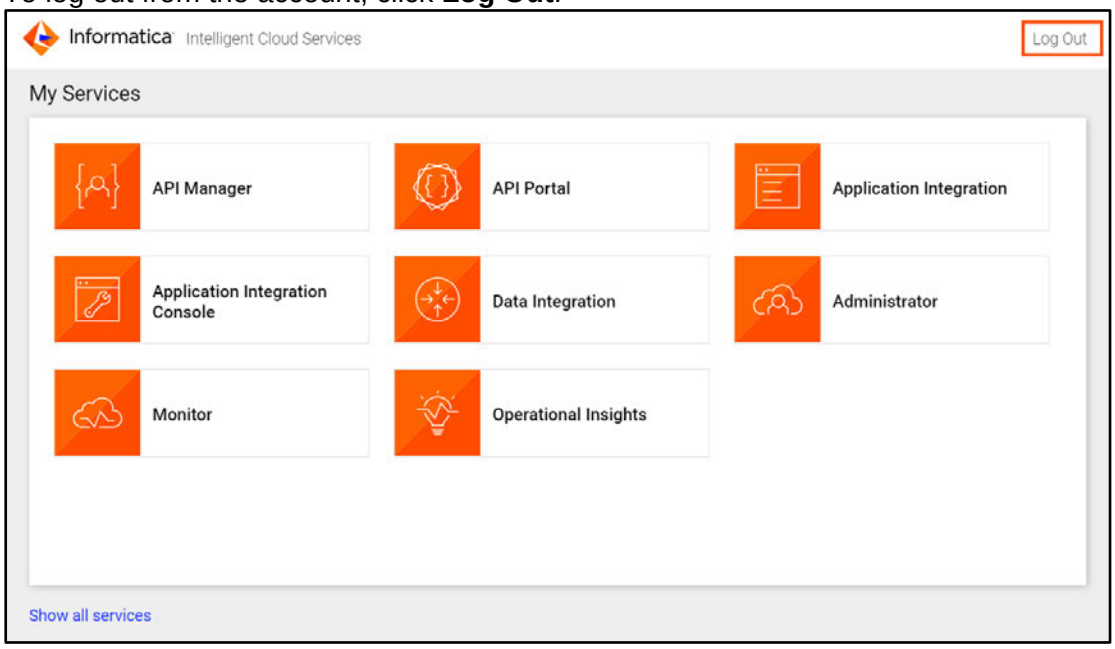


15. In the **My Services** window, notice the list of available services.



Note: The number of services that you can see, vary from Org to Org depending upon the licenses and configuration.

16. To log out from the account, click **Log Out**.



This concludes the lab.

Pre-requisite Labs

Lab 0-2: Setting up Lab Environment

Overview:
To execute the labs for this course, there are certain tools and files that must be present in a defined location on your local machine. This document provides steps to setup the lab environment for performing the lab exercises of this course.

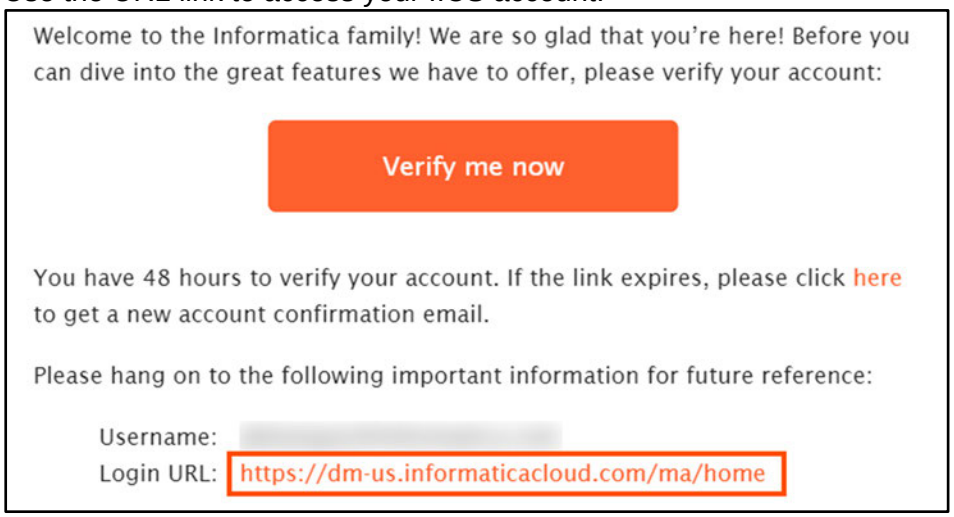
Objective:

- Enable Connector
- Copy Lab Prep Files

Duration:
20 minutes

Enable Connector

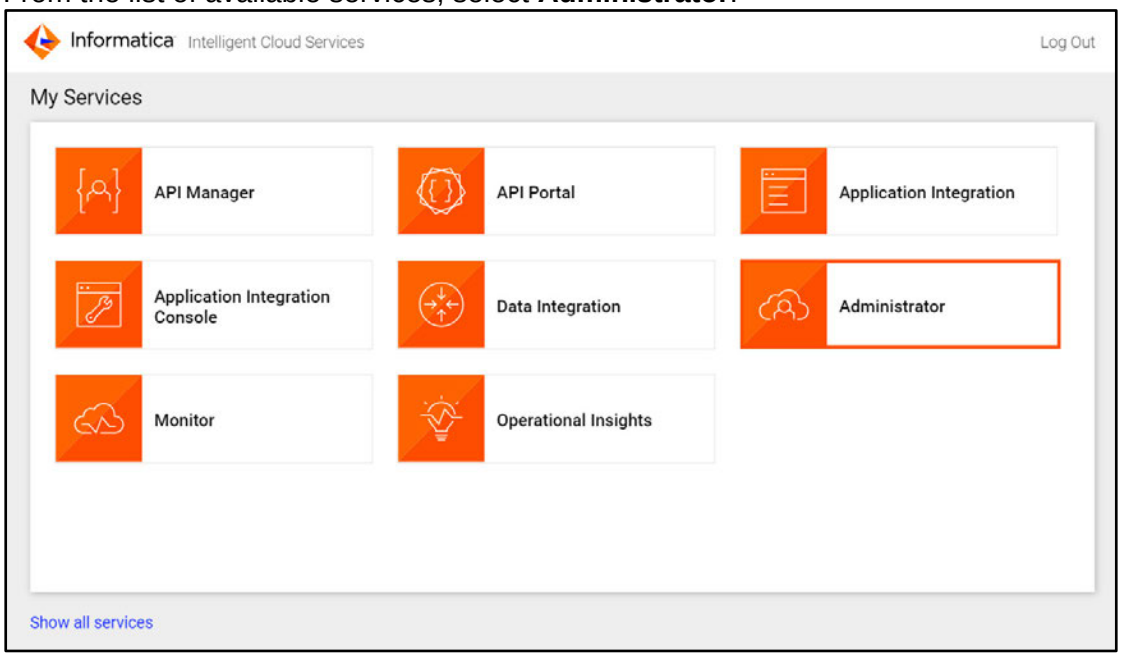
1. For accessing the IICS login page, open the email from **admin@informaticacloud.com** in your mailbox.
2. Use the URL link to access your IICS account.



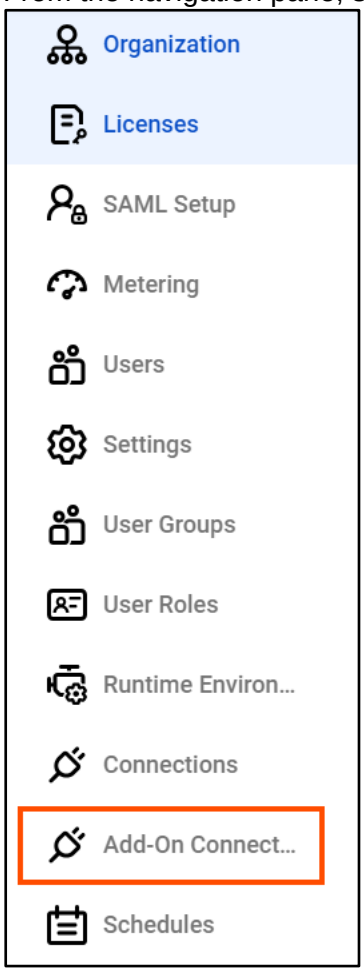
Note: You must bookmark the above URL for future use.

3. Enter your login credentials and click **Log In**.

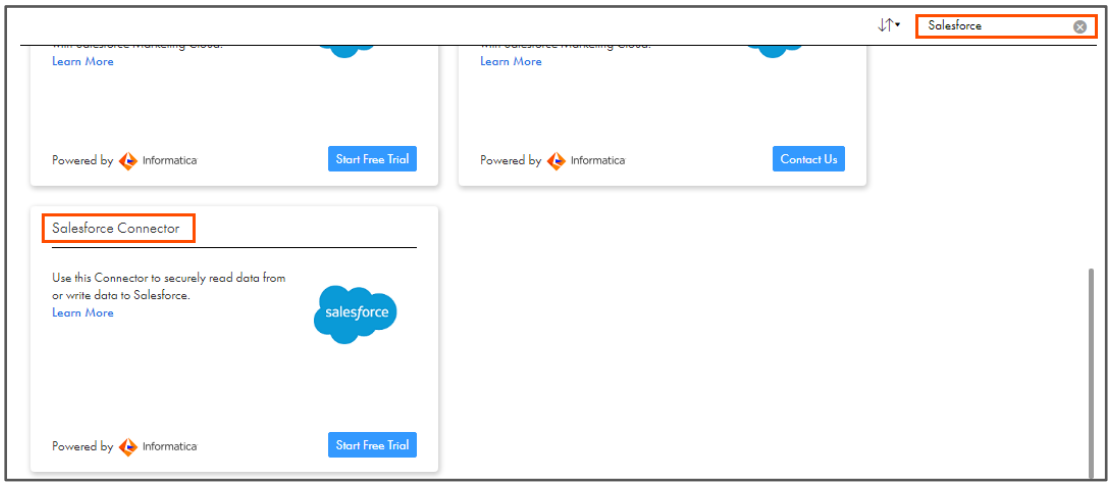
4. From the list of available services, select **Administrator**.



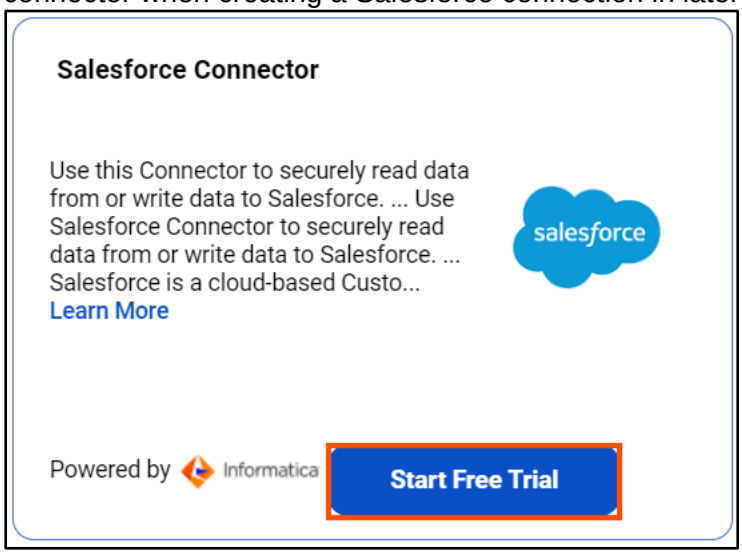
5. From the navigation pane, select **Add-on Connectors** tab.



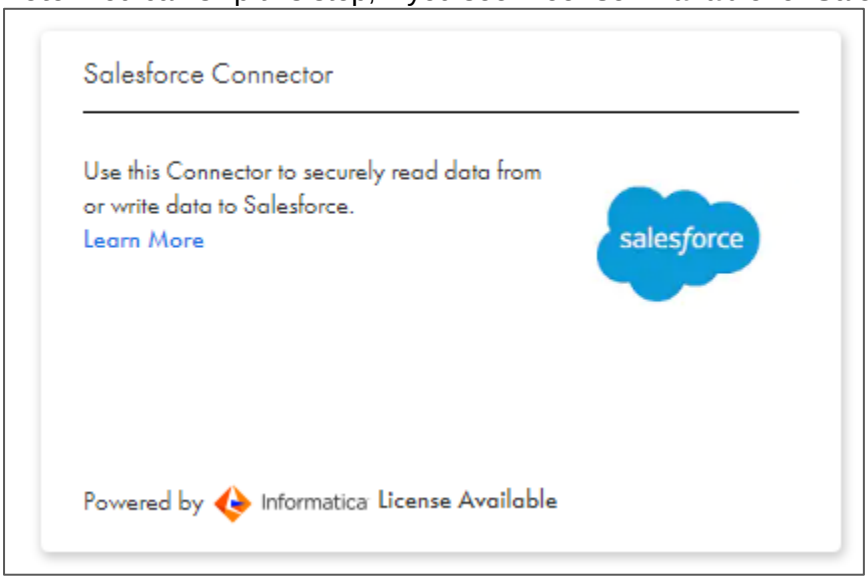
6. Use the find section to locate the **Salesforce** connector.



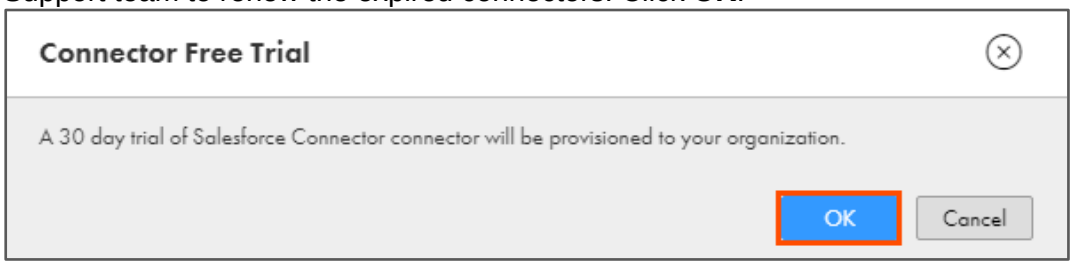
7. Click **Start Free Trial** for the Salesforce Connector. This enables you to view the connector when creating a Salesforce connection in later lab.



Note: You can skip this step, if you see **License Available** for Salesforce connector.



8. Note that the trial period for add on connectors is 30 days. Reach out to the Informatica Support team to renew the expired connectors. Click **OK**.



9. Logout of the IICS Org.

Copy Lab Prep Files

10. From the Informatica training portal, download the **Lab Prep Files**. Unzip the Lab Prep Files folder and observe that there are 3 sub-folders named **SrcFiles**, **TgtFiles**, and **Tables**.
    Note: The prep files are the source files required to perform the lab exercise. You must save these files in a defined location for smooth execution of the labs.
11. In the **C:** drive of your secure agent machine, create a new folder and name it as **IICSLabFiles**.
12. Copy the **SrcFiles** and **TgtFiles** folders downloaded from the training portal and paste them in the newly created **IICSLabFiles** folder.

This concludes the lab.

Pre-requisite Labs

Lab 0-3: Configure SQL Server Database

Overview:

MS SQL Server is a relational database management system. It is commonly used for running online transaction processing (OLTP), data warehousing (DW), and mixed (OLTP & DW) database workloads.

This document lists the steps to configure the SQL Server database for performing lab exercises for this course.

Objective:

- Create User
- Create Tables

Duration:

30 minutes

Important

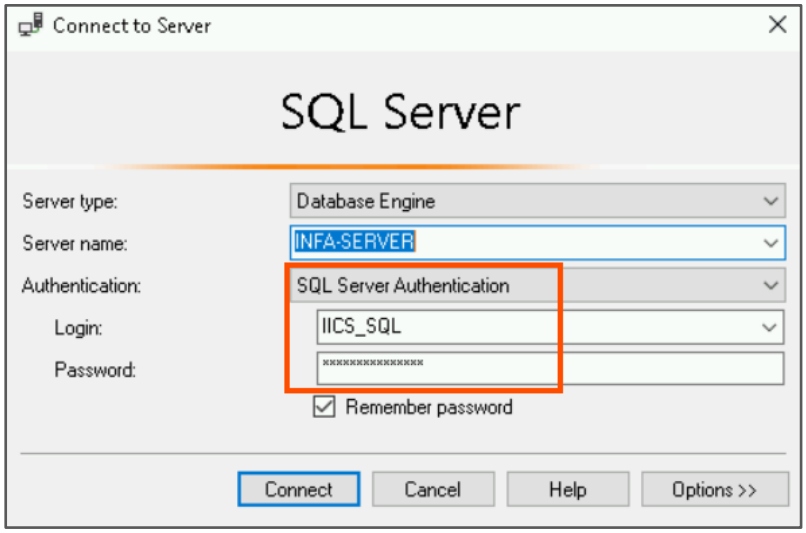
1. It is recommended that your system has a minimum of 8 GB RAM.

2. In the lab exercises for this course, the following lab guides are designed and documented using SQL Server database. We therefore recommend that you have SQL Server database installed on your system (Secure Agent machine) to execute the labs.

List of SQL Server Database Dependent Labs
Lab 3-4: Using Saved Query in a Synchronization Task
Lab 4-1: Creating a Data Transfer Task
Lab 6-1: Classifying the Product Shipping Details Using Various Transformations
Lab 7-1: Using Lookup Transformation in a Mapping
Lab 8-2: Create Mapplet and Use it in the Existing Mapping
Lab 8-3: Creating a Mapping Using Existing Mapplet and Aggregator Transformations
Lab 9-2: Using In-Out Parameters for Incremental Data Loading
Lab 10-1: Create a Mapping Task and Configure Advanced Settings
Lab 13-2: Creating a Taskflow with Conditional Logics
Lab 14-3: Running a Mapping Task Using REST API

Note: If you do not have SQL Server installed on your system, you can skip the labs that are SQL Server database dependent and continue with the rest of the course and lab exercises.

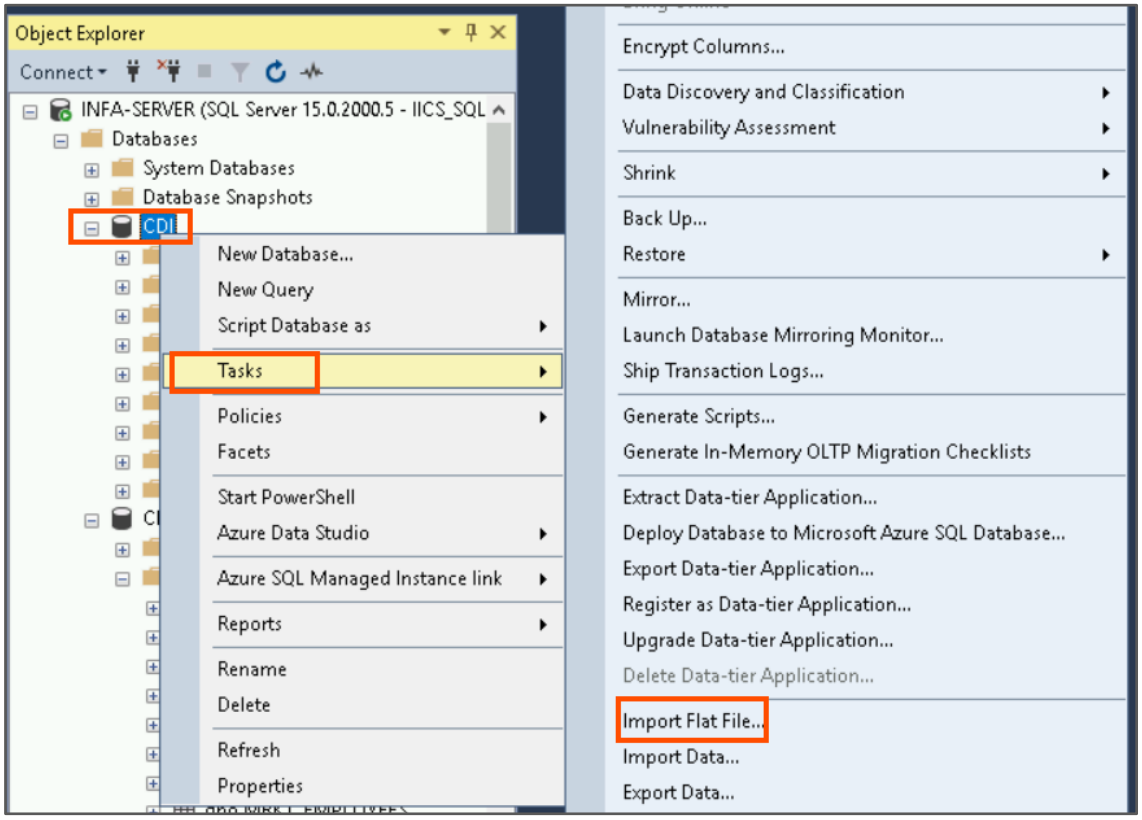
3. The labs for this course are created on **SQL Server 2019** and using the client **Microsoft SQL Server Management Studio** version 19.0.
4. Get in touch with your Organization's IT team and in the SQL Server Management Studio tool, create an **SQL Server Authentication** user with login username as **IICS_SQL** and password as **IICS_SQL**.



5. In MS SQL Server Management Studio, connect to your using the login credentials **IICS_SQL/IICS_SQL**.
6. Expand Databases and create 2 databases named **CDI**, and **CDI_STUDENT**. **Note**: CDI is the source database and CDI_STUDENT will be your target database where the target tables will get saved. In this case, you will need to create 1 source SQL Server connection and 1 target SQL Server connection.

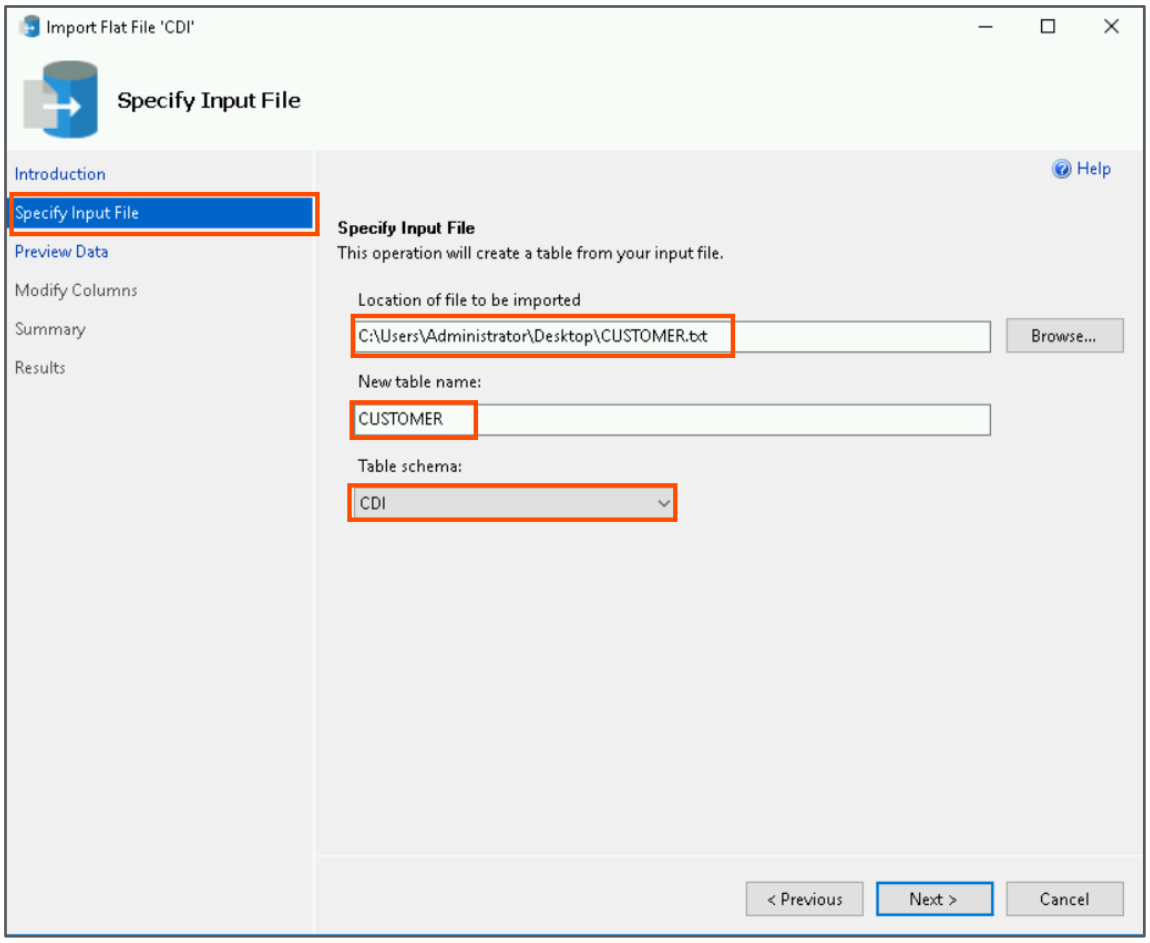
Create Tables

7. From the downloaded Lab Prep Files folder, go to the **Tables** folder. You will see a folder named CDI. These folders contain source tables in a flat file format. You will now import the data tables into the MS SQL Server Management Studio tool.
8. Navigate to the SQL Server client tool and right-click the **CDI** database.
9. Select **Tasks** > **Import Flat File**.



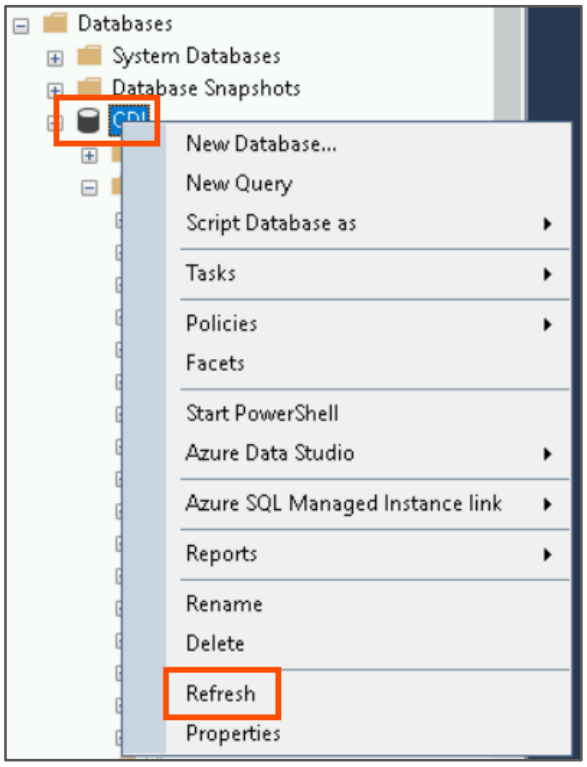
10. The Import Flat File 'CDI' wizard opens. Click **Next**.
11. In the Specify Input File tab, click **Browse** to navigate to the CDI database folder, and select the flat file **CUSTOMER**.

12. Retain the table name as **CUSTOMER** and change the Table Schema to **CDI**.

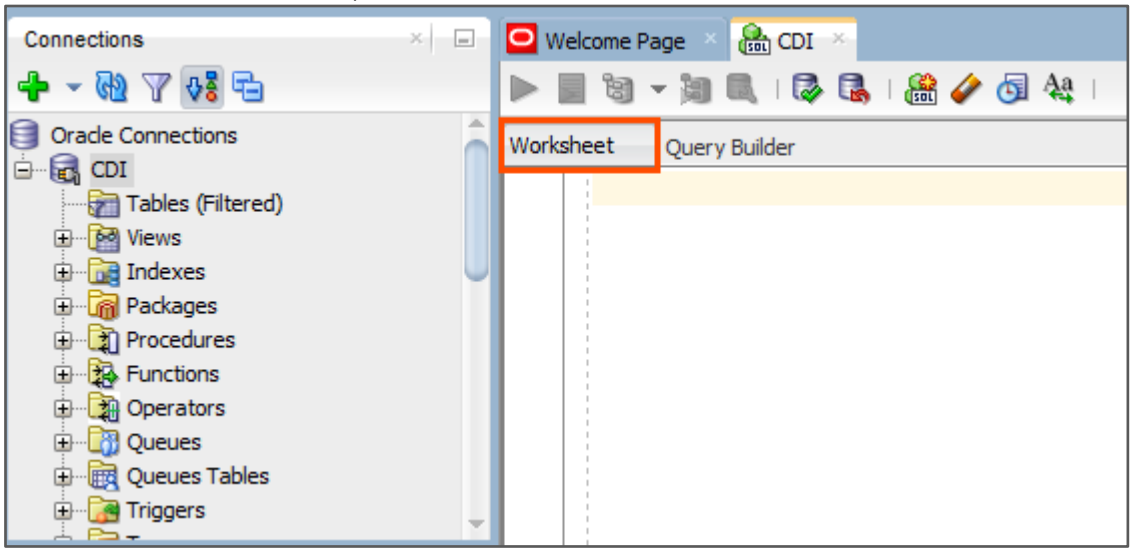


13. Click **Next**.
14. In the Preview Data tab, view the data, and click **Next**.
15. In the Modify Columns tab, click **Next**.
16. In the Summary tab, click **Finish**. This will complete the data import operation.
17. Click **Close** to close the data import wizard.

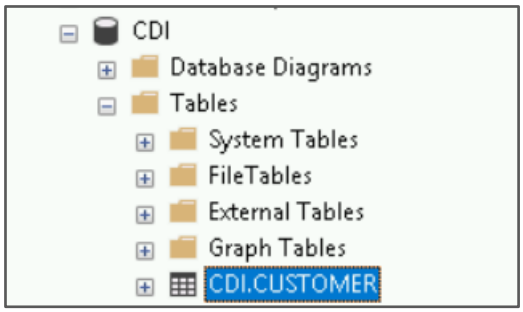
18. To populate the newly added CUSTOMER database table, right-click the database **CDI** and click **Refresh**.



19. To create the tables in CDI, select the **Worksheet** tab.



20. Observe that the CUSTOMER table is successfully created in the CDI database.



21. Similarly, create the following tables in the CDI database (using **CDI** as schema):

    Note: You will find all the required Flat Files of the tables in the Tables > CDI folder of Lab Prep Files.

    a. ORDERS
    b. TRANSACTION
    c. DEALERSHIP
    d. EMPLOYEE
    e. SF_ACCOUNT

22. Now, create another database and name it as **CDI_STUDENT**. This will be your target database.
23. Into this target database CDI_STUDENT, create a target table named **CUSTOMER_MASTER** using the flat file **CUSTOMER_MASTER.txt**. After creating the table, make the port **CUST_ID** as the primary key.

    Note: You will find all the required Flat Files of the tables in the Tables > CDI_STUDENT folder of Lab Prep Files
24. Similarly, create the table named **DIM_CUSTOMER** into the target database CDI_STUDENT using the flat file **DIM_CUSTOMER.txt**.
25. Close the SQL Server client tool.

Thus, you have successfully created databases and database tables in SQL Server.

This concludes the lab.

Module 1: Informatica Cloud Overview

Lab 1-1: Navigating the IICS Interface

Overview:

Informatica Intelligent Cloud Services (IICS) is a platform that helps in Enterprise Data Management through a suite of Intelligent Cloud services. To effectively manage data, the IICS platform provides Data Integration, Administrator, and Monitor services.

The Data Integration service synchronizes data between a heterogeneous source and a target.

The Administrator service provides organization management capabilities such as managing security, licenses, users, user groups, user roles, connections, schedules, add-on bundles, and swagger files.

Objective:

- Log in to the Informatica Cloud Org
- Navigate between services
- Access the Informatica Cloud online help
- Search the online help

Scenario:

After hearing about IICS, Ruby wants to use Informatica Cloud Data Integration Service to improve the performance of her store. So, to introduce Ruby to various features of IICS, John tells Ruby how to access the IICS interface and navigate between the services. He also explains the procedure to access the online help option to Ruby.

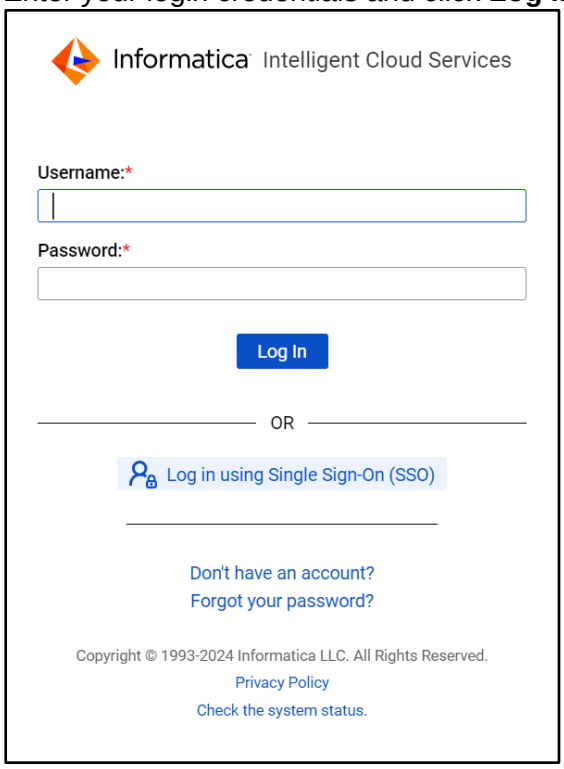
In this lab, Ruby will access the IICS interface and access the online help option. Ruby will also explore various IICS services.

Duration:
10 minutes

Tasks

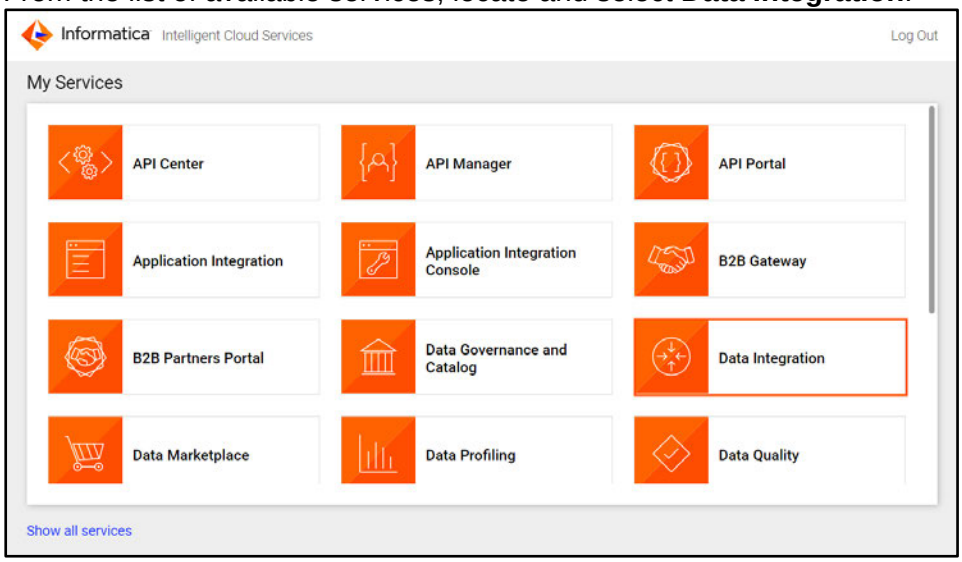
Access IICS Interface

1. Open a web browser and copy the following **Informatica Cloud** URL into the browser: https://dm-us.informaticacloud.com/identity-service/home
2. Enter your login credentials and click **Log In**.



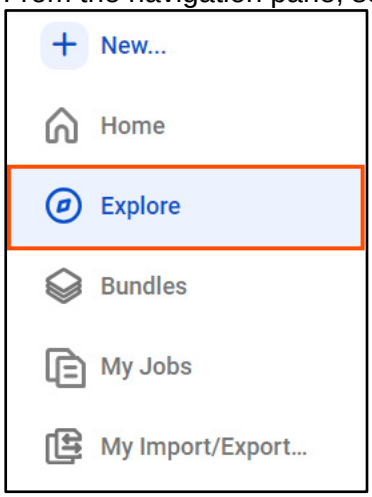
The **My Services** window appears.

3. From the list of available services, locate and select **Data Integration**.



Note: The number of services that you can see in the list varies from Org to Org depending upon the licenses and configuration.

4. From the navigation pane, select **Explore**.

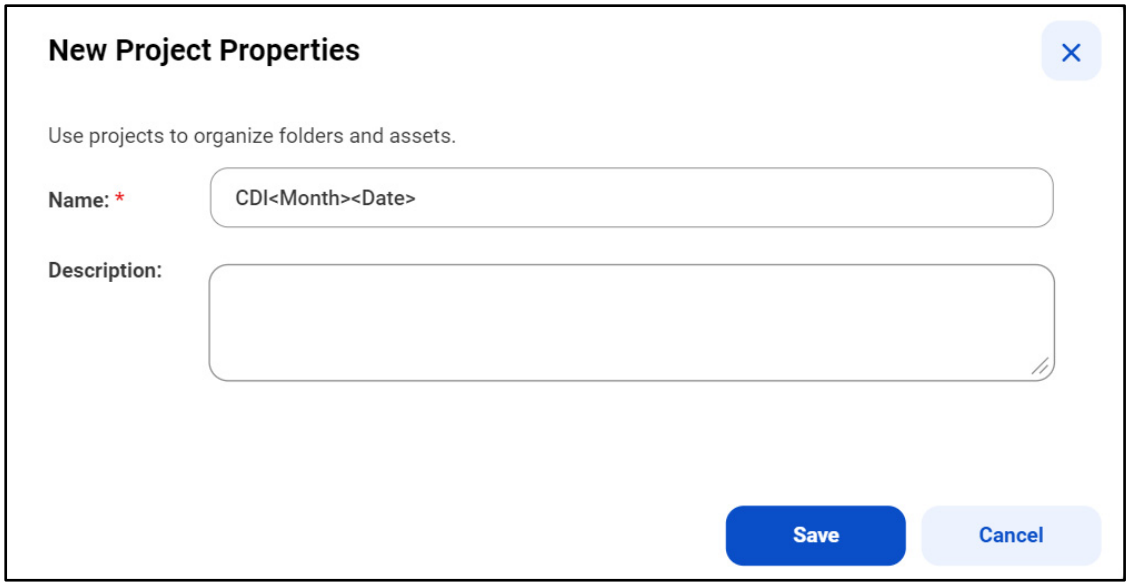


The Explore page shows all the projects and assets built using IICS.

Create a Working Folder

5. Click **New Project**.

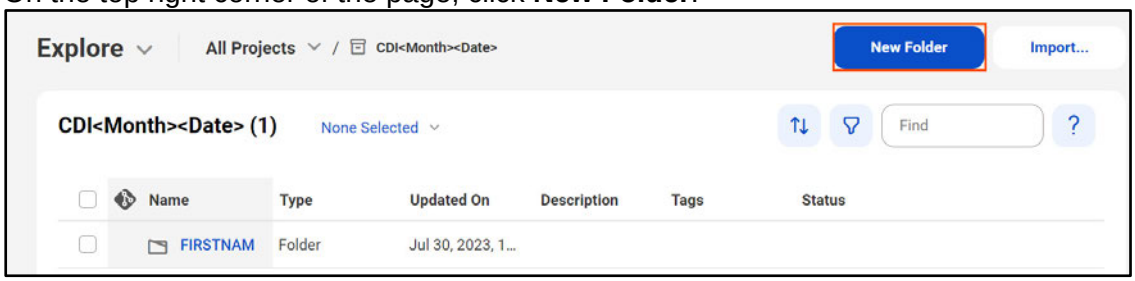
6. Name the project as **CDI<Month><Date>**. Here replace <Month> and <Date> with the month and date on you started the training. For example, if you started the CDI training on **19th January**, the course name would be CDI Jan 19. Click **Save**.



7. Go to your CDI training project.

Note: You will create your working folder inside this training project.

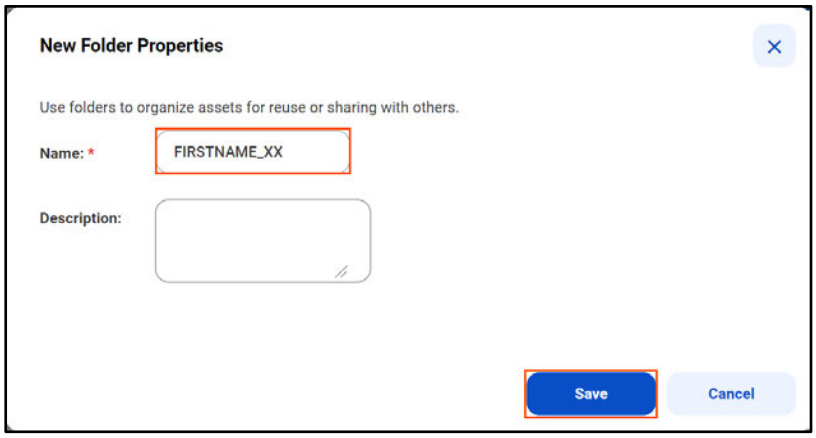
8. On the top right corner of the page, click **New Folder**.

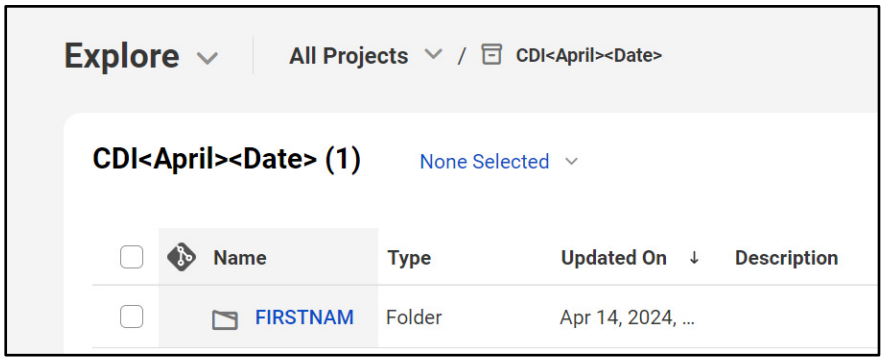


9. Name the folder as **FIRSTNAME_XX**.

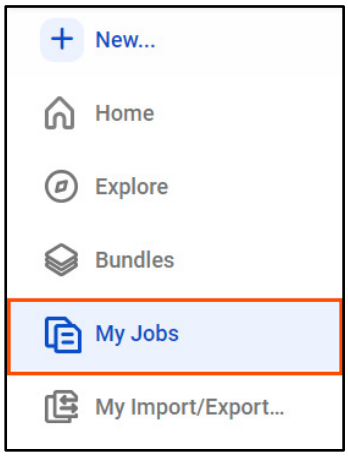
   Note: Here, replace **FIRSTNAME** with your original first name and **XX** with your DOB to avoid confusion between various working folders.

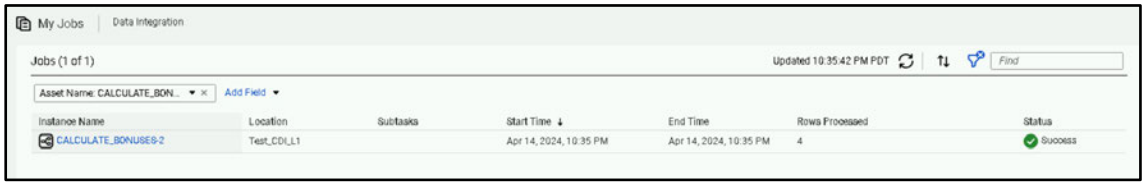
10. Click **Save**. Verify that you can see your working folder.





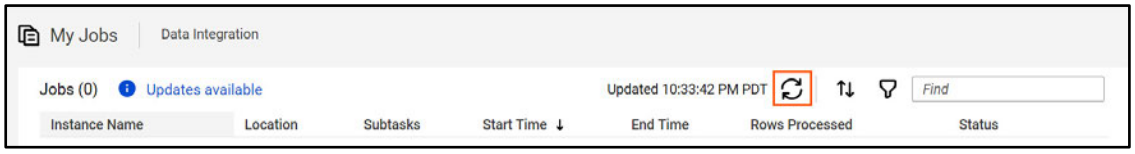
11. To check the status of a job, from the navigation pane, select **My Jobs**.



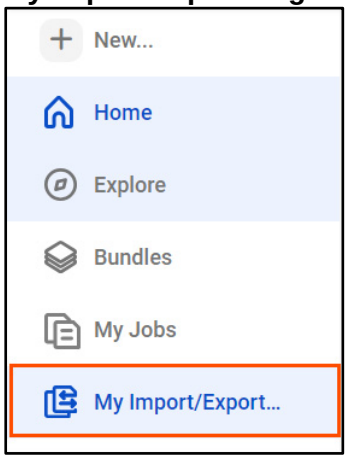


Note: If you do not find any jobs on the My Jobs page, it means that you have no previously executed jobs in your IICS account. As you progress with the labs, you will find the executed jobs status here.

12. In the subsequent lab exercises, to check the updated status of a running task, use the Refresh icon .

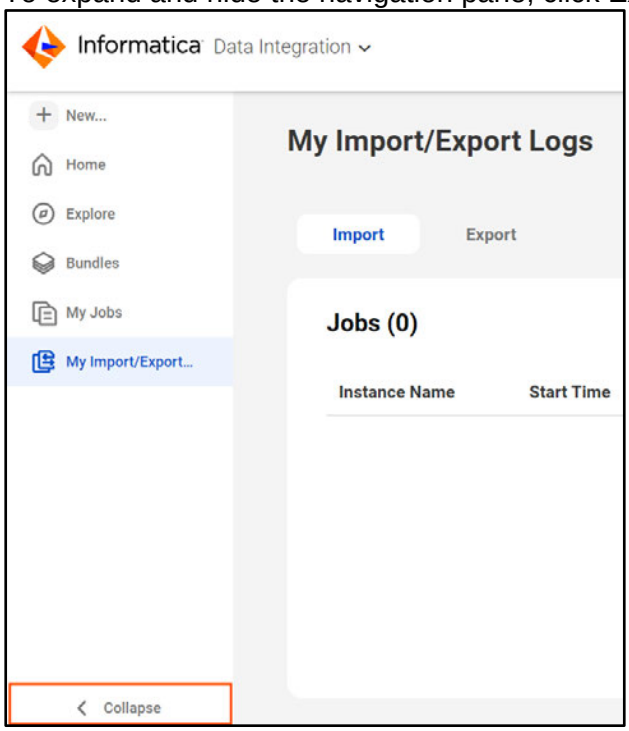


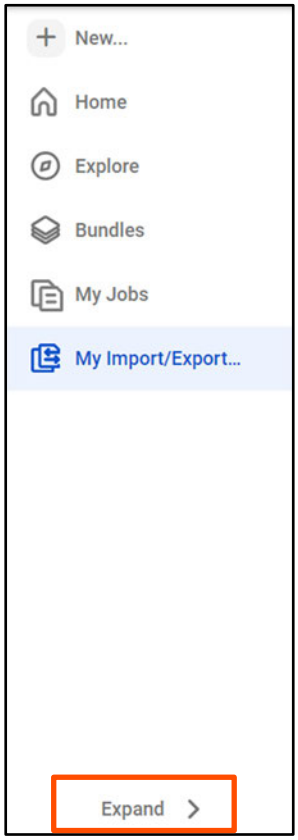
13. To list import/export tasks started by the current user, from the navigation pane, select **My Import/Export Logs**.



The **Import** tab lists all the import jobs, while the **Export** tab shows the export job started by the user.

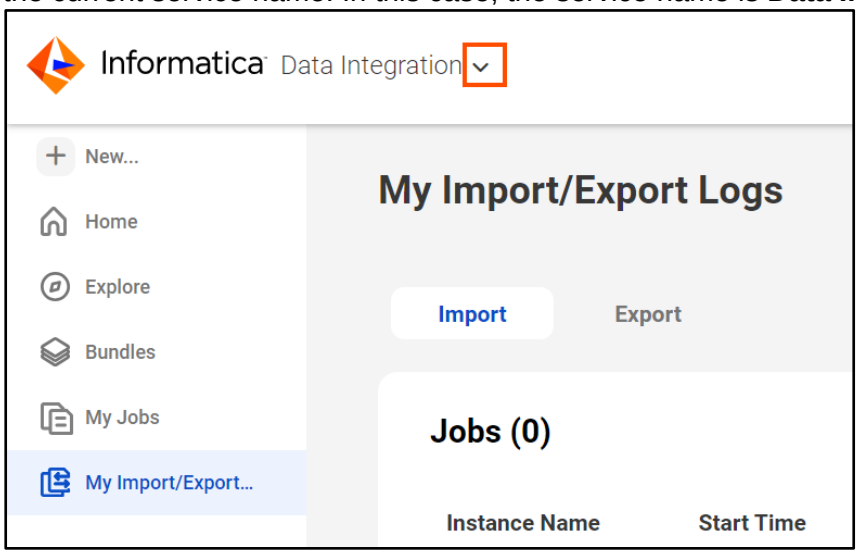
14. To expand and hide the navigation pane, click Expand and Collapse.





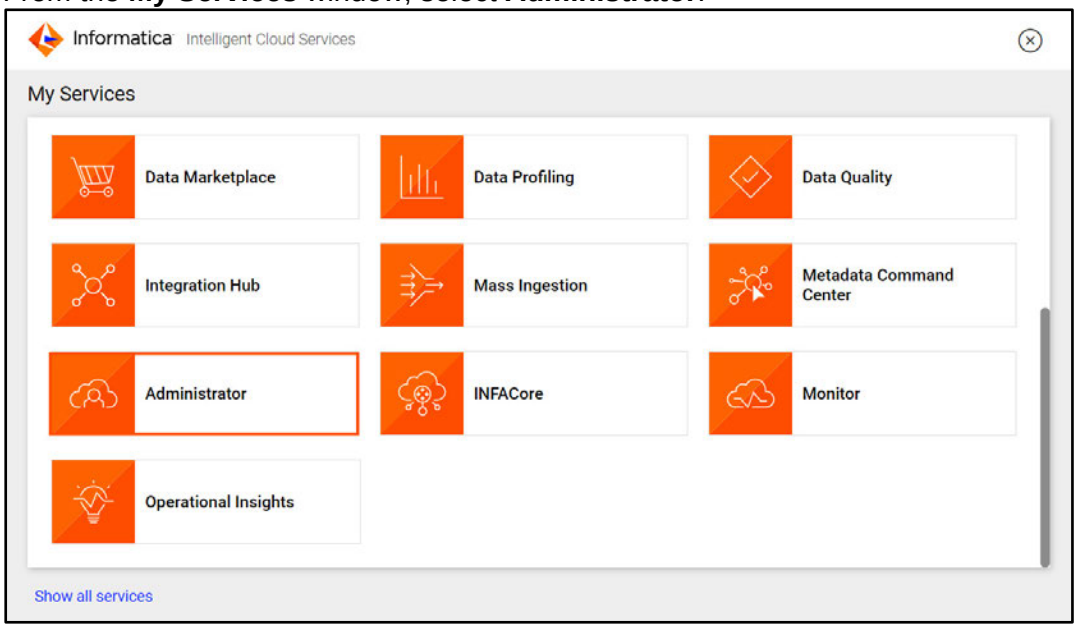
Switching Between Services

15. To switch between the available services, from the toolbar, select the drop-down next to the current service name. In this case, the service name is **Data Integration**.

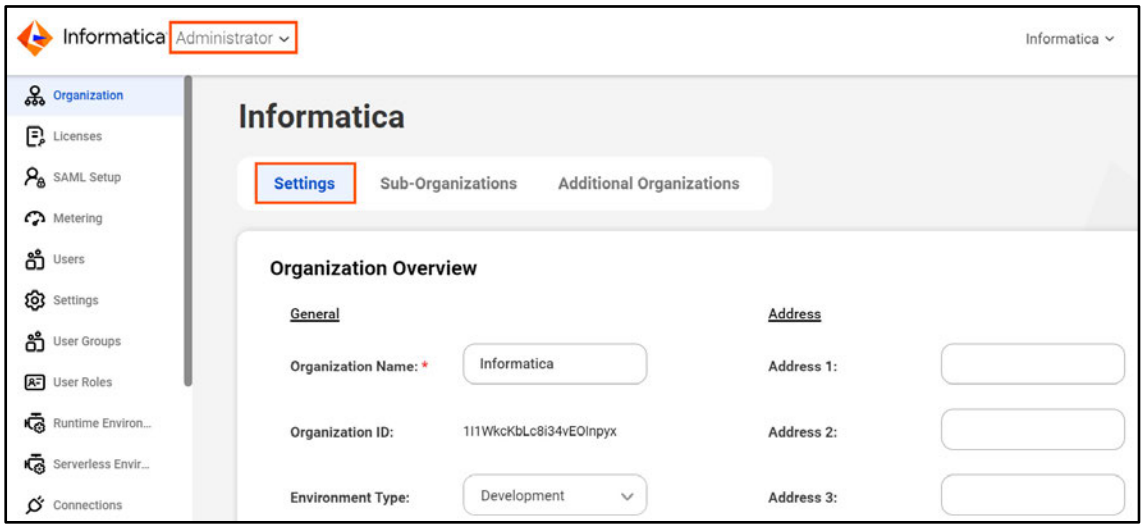


Note: You can select any available service at the time of login as well.

16. From the **My Services** window, select **Administrator**.

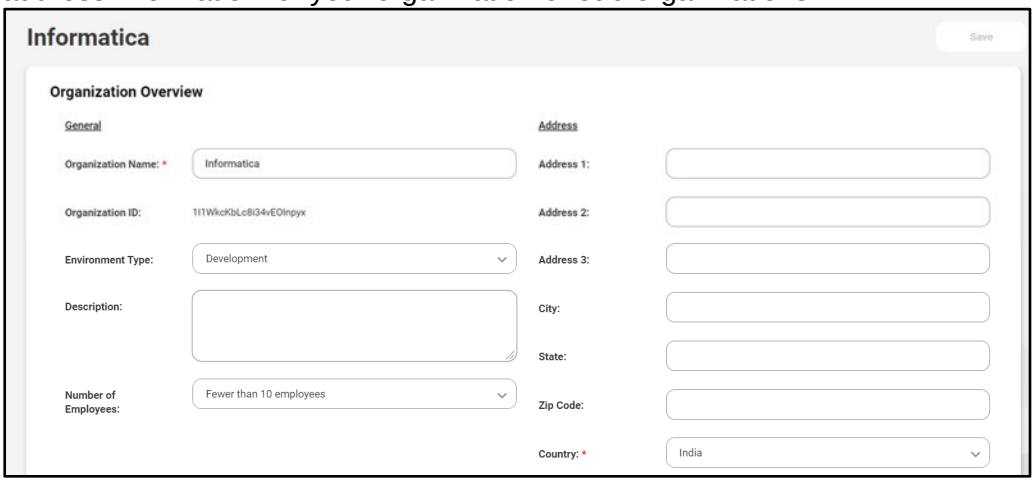


17. If you are not the admin of the IICS Org, then, by default, the Settings page appears.



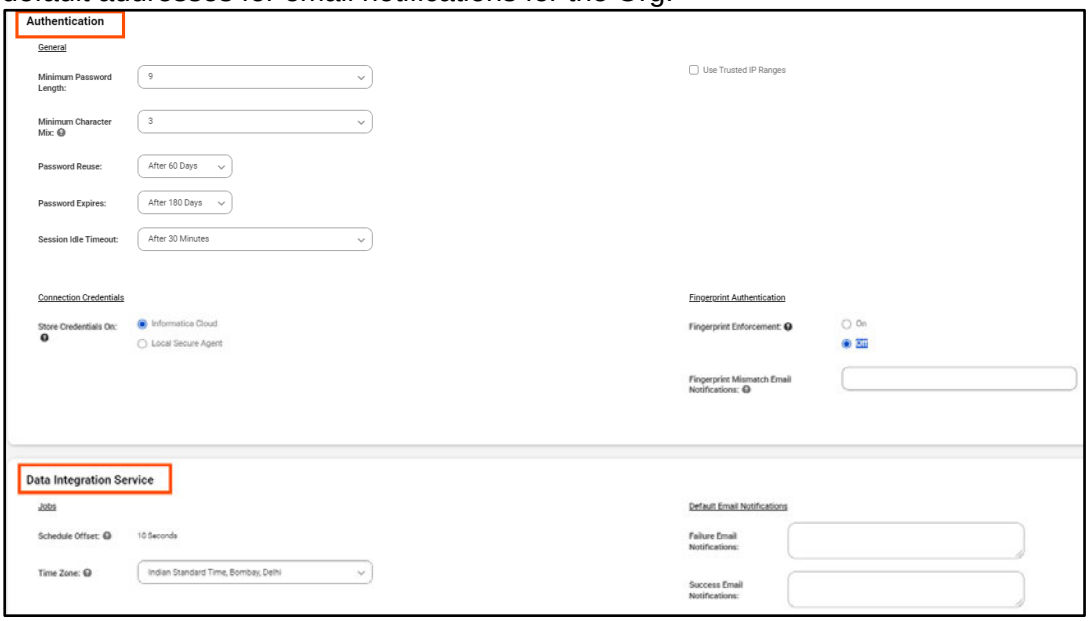
Note:

a. The tabs you view depend on the privileges and roles you have. If you are an Admin, then, by default, the **Organization** page appears. You can configure Organization properties such as organization name, description, number of employees, and address information for your organization or sub-organizations.

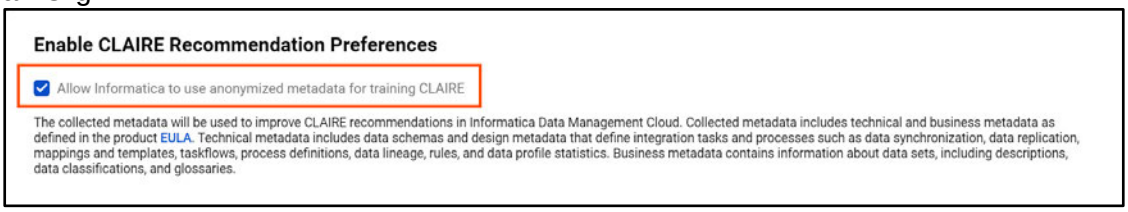


The **Authentication** section is used to update password policies for your Org and connection properties storage.

b. The **Data Integration service** properties section is used to set the time zone and default addresses for email notifications for the Org.



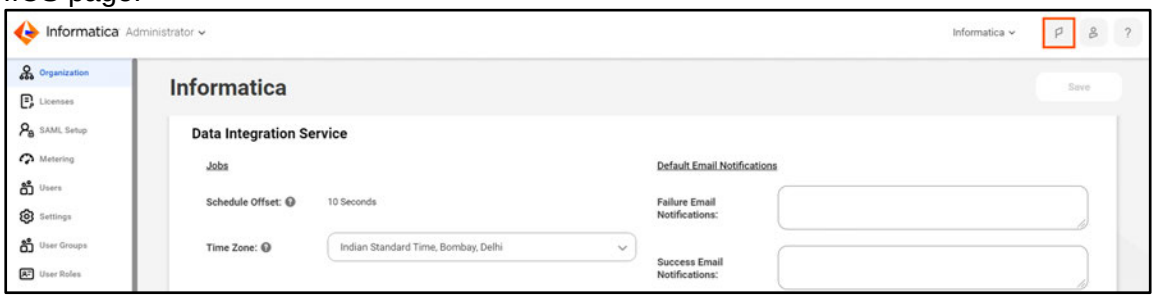
c. Towards the bottom of the page, you will have the **CLAIRE Recommendation Preferences** section and by default, the CLAIRE recommendations are enabled in an Org.



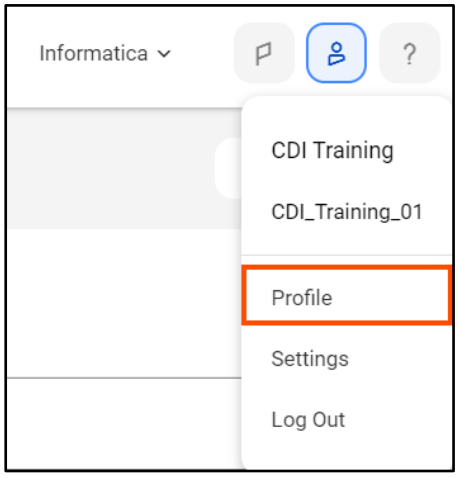
Important: The course is built using the **Admin** role user. Thus, you may see additional tabs/pages in the screenshots.

Accessing Notifications, User details, and Online Help

18. To view the notifications, select the flag icon ( ) on the upper right corner of the IICS page.



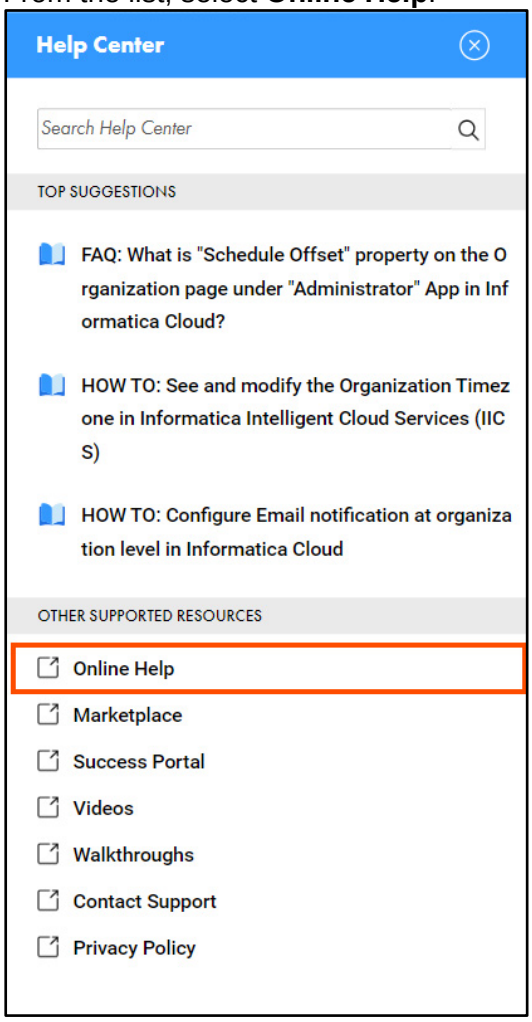
19. To view user-related information, click the user icon ( ).



Note: You can use the Profile option to update user information like first and last name, job title, password, and security question for the user account. In the above screenshot, the name of the organization and email address have been masked.

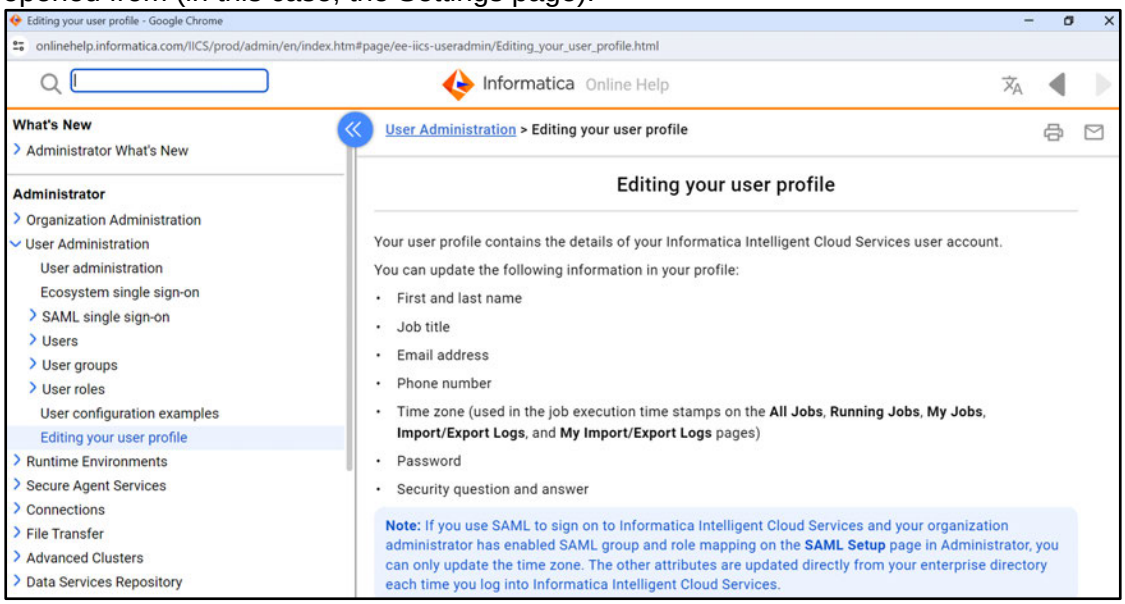
20. To access online help, click the question mark icon ( ).

21. From the list, select **Online Help**.



Note: The Online Help opens in a new browser window or tab.

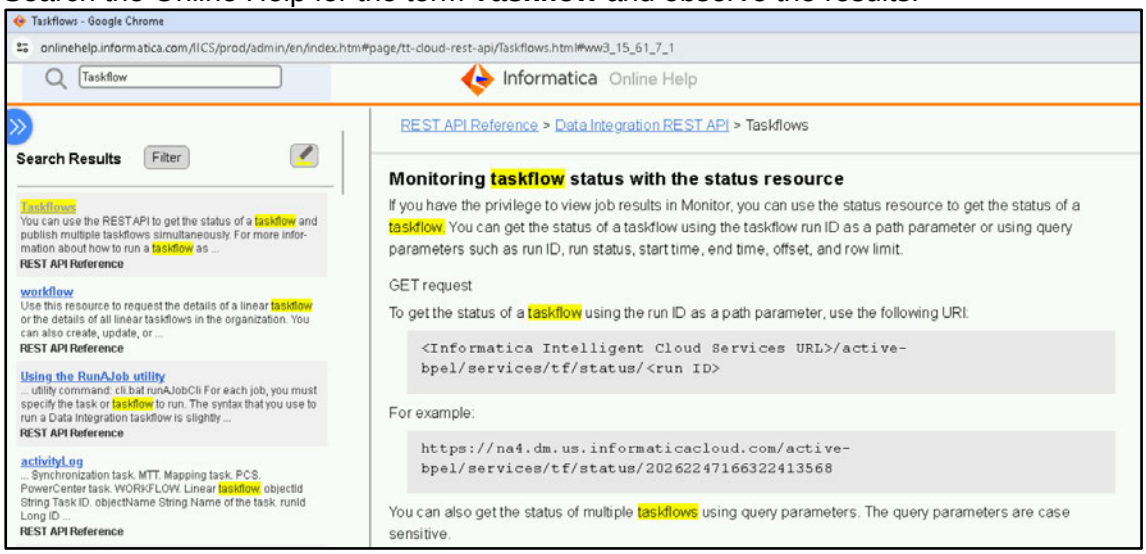
22. By default, Online Help shows the information related to the page it was originally opened from (in this case, the Settings page).



Note: You can browse the Online Help as per the Content, Index, Search, and navigate to the Next and Previous help articles.

If you have performed the **Note** point of step 17, then you will land on the User Profile page in the Online Help section.

23. Search the Online Help for the term **Taskflow** and observe the results.



24. Close the online help and the Help Center window.

This concludes the lab.

Module 2: Introduction to Data Integration Service

Lab 2-1: Installing Secure Agent

Overview:

The Secure Agent is a lightweight, self-upgrading program that runs inside your network. It is responsible for moving data from the source to a target. IICS Secure Agent runs all tasks and enables a secure communication between your organization and Informatica Cloud. You can install and run one Secure Agent on a physical or virtual machine. After the Secure Agent is installed, all the users in the organization share the Secure Agent.

In this lab, you will set up the Secure Agent.

Objective:

- Download the Secure Agent
- Install the Secure Agent
- Assign Administrative rights to Secure Agent

Scenario:

John informs Ruby that to integrate various data sources with IICS, she needs to install the Secure Agent in her IICS Org.

In this lab, Ruby will download and install a Secure Agent in IICS. She will also assign Administrative rights to the Secure Agent to access the files present on her machine.

Important:

You can install Secure Agent on machines running on **Windows** or **Linux** operating systems only. This lab guide contains steps to install Secure Agent on Windows machine. For installing Secure Agent on Linux OS, refer the link: https://knowledge.informatica.com/s/article/513826

Duration:

30 minutes

Secure Agent Minimum Requirement:

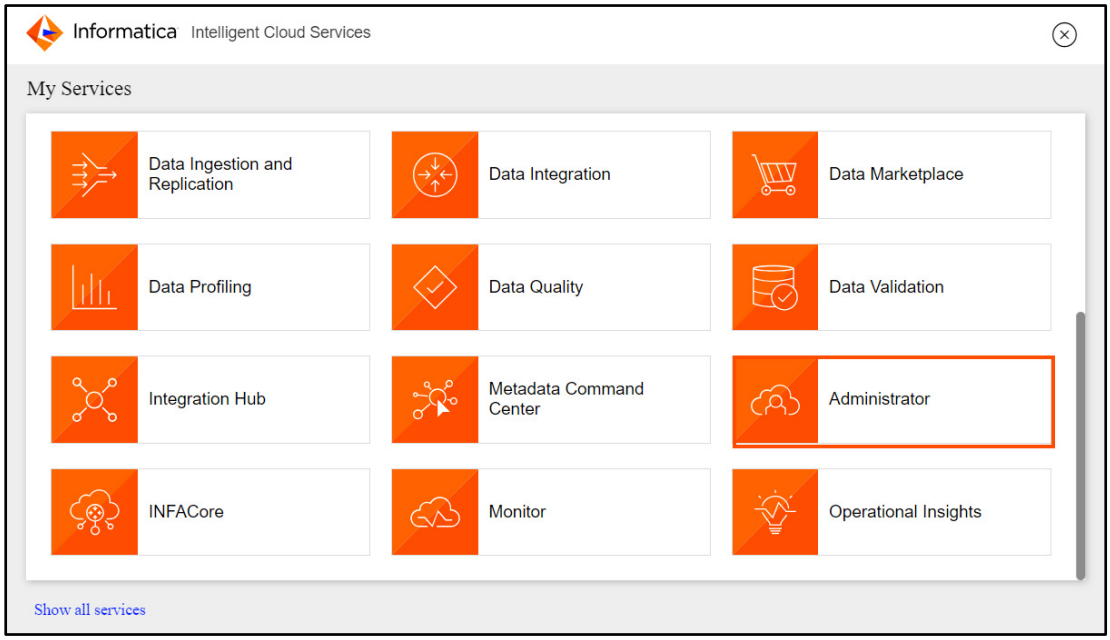
You can install the Secure Agent on any machine that has internet connectivity and can access Informatica Intelligent Cloud Services.

- Verify that the machine where you install the Secure Agent has at least 5 GB of free disk space.
- Verify that the account you use to install the Secure Agent has access to all remote directories that contain flat source or target files.
- Verify that no other Secure Agent is installed on the machine. If another Secure Agent is installed on the machine, you must uninstall it first.

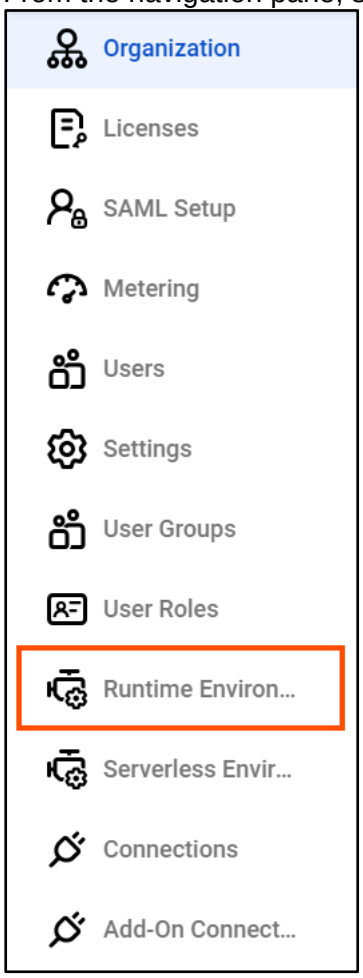
Tasks

Download the IICS Secure Agent

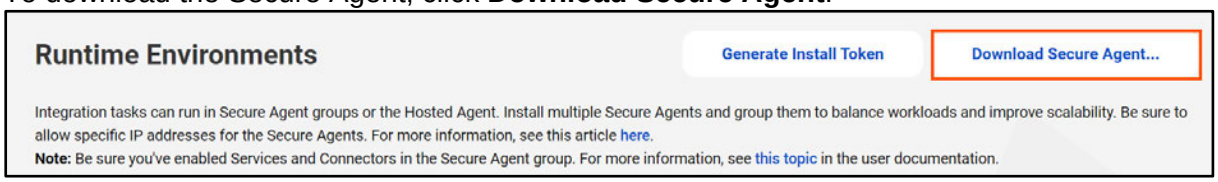
1. Enter your IICS login credentials and click **Log In**.
2. From the **My Services** window, select **Administrator**.



3. From the navigation pane, select **Runtime Environments**.

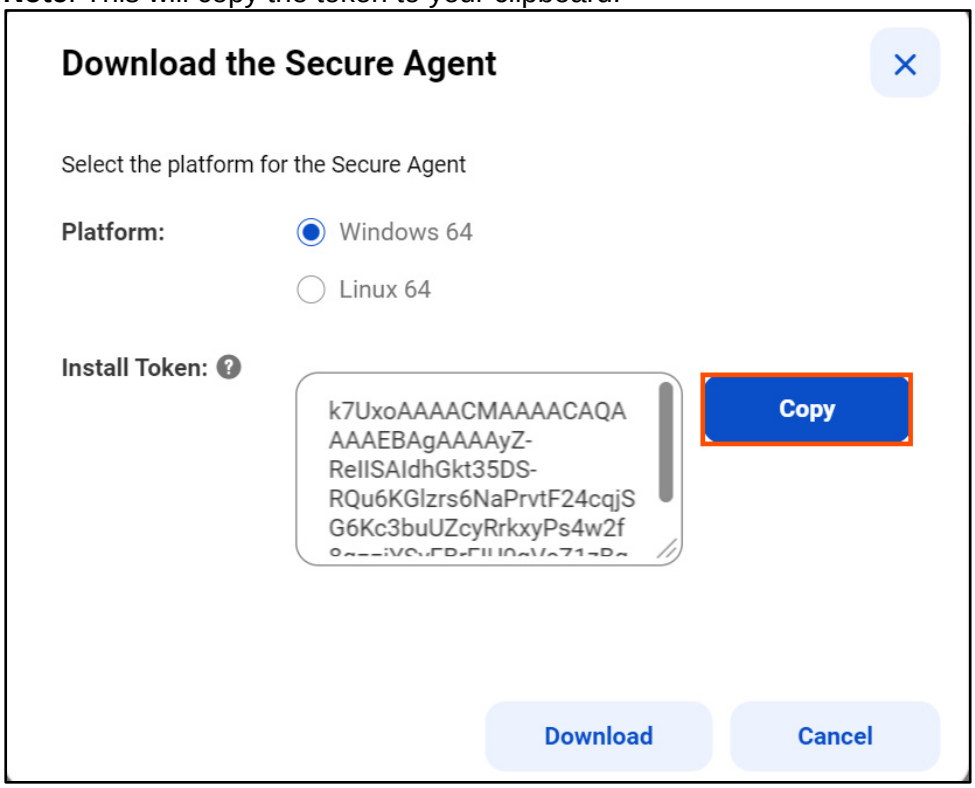


4. To download the Secure Agent, click **Download Secure Agent**.



5. From the Download the Secure Agent window, select **Windows 64**, and from the Install Token field, click **Copy**.

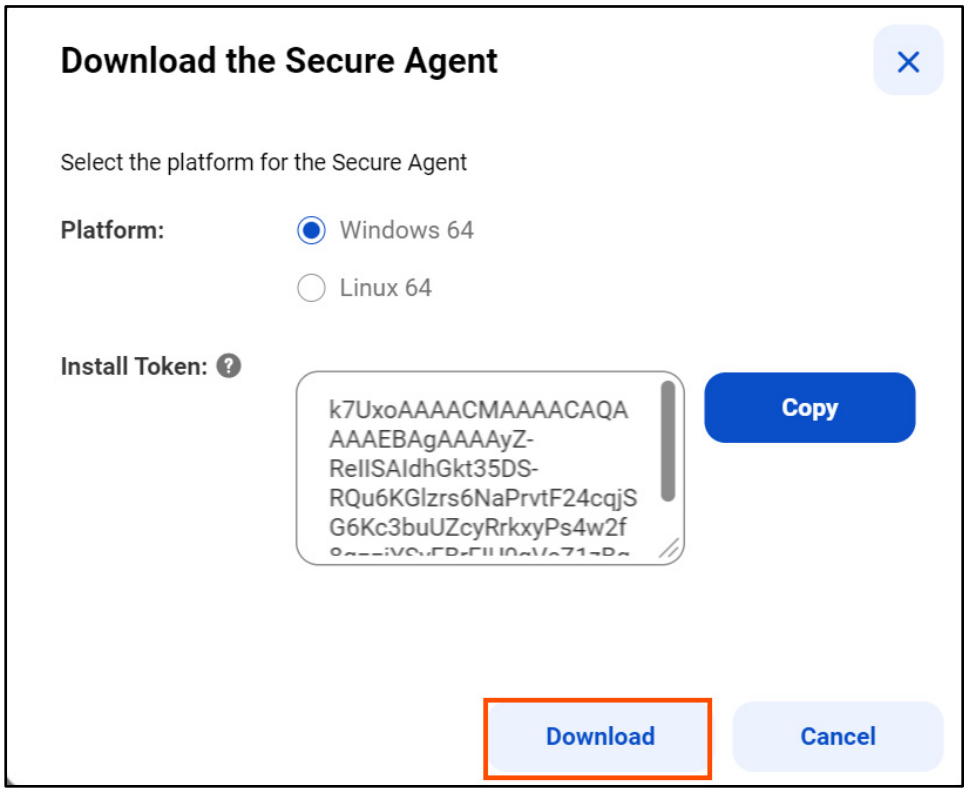
Note: This will copy the token to your clipboard.



Note: If you are installing Secure Agent on Linux platform, select **Linux 64** in the Download the Secure Agent window.

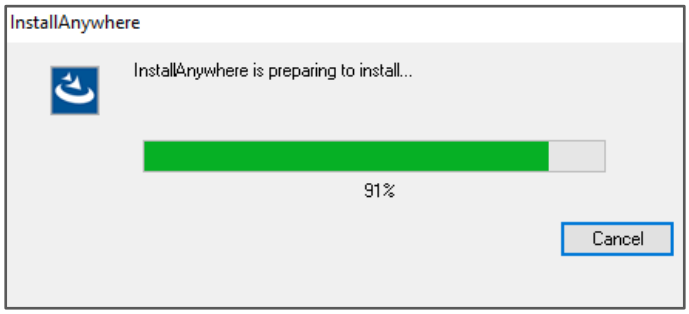
6. Paste the copied token in a text file.

7. Click **Download**.



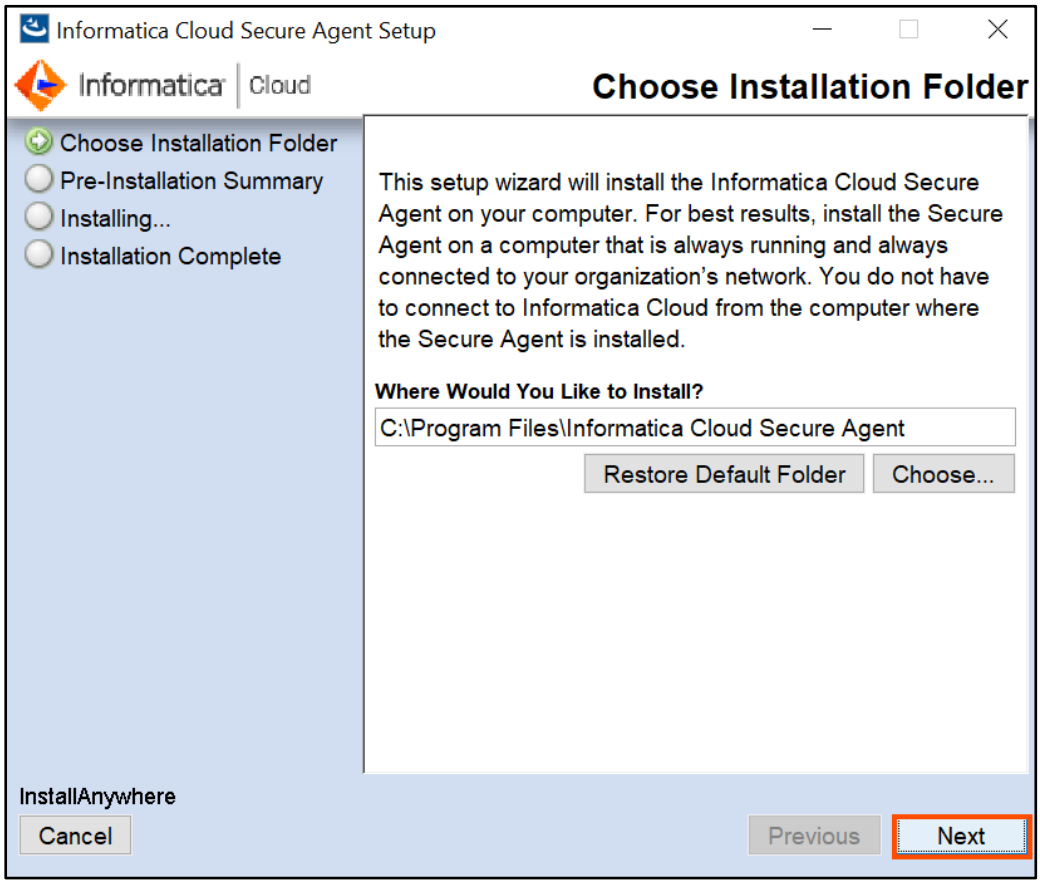
Install the Secure Agent

8. Go to the download directory on your machine and locate the agent installation file.
9. To install the Secure Agent, run the executable file **agent64_install_ng_ext.exe**.

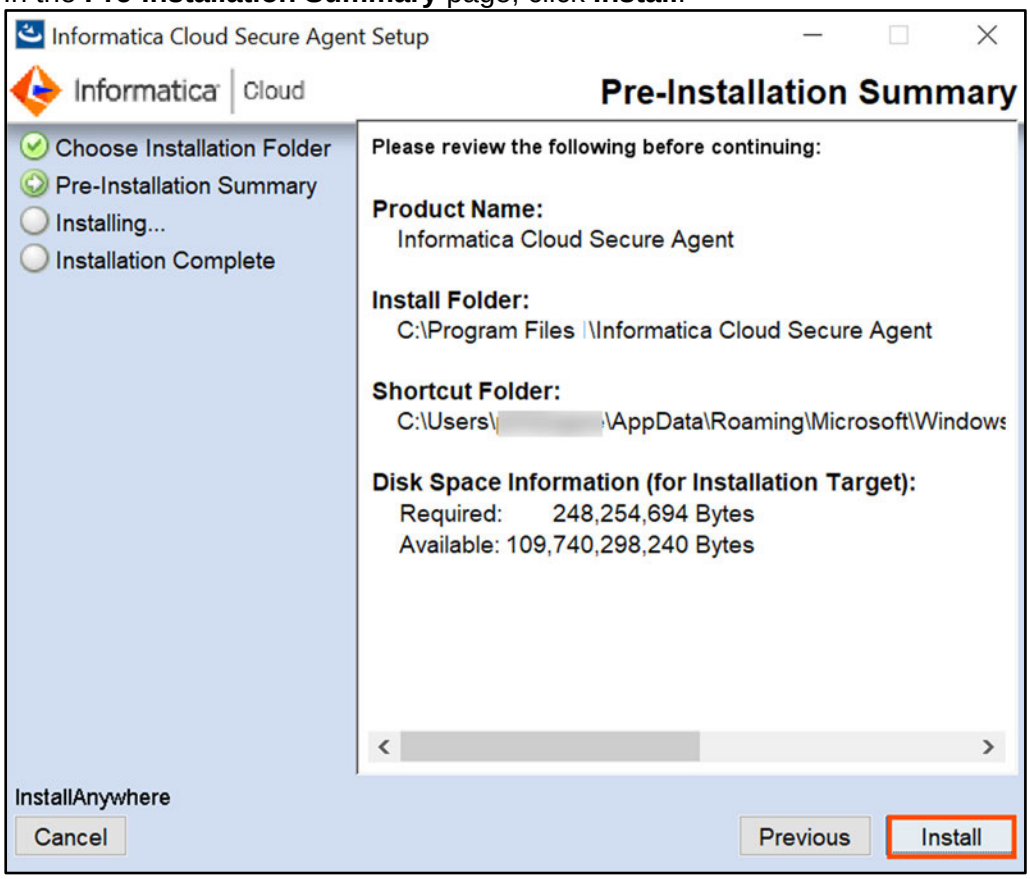


Note: When the Secure Agent initiates the installation, it displays the **InstallAnywhere** window.

10. In the **Choose Installation Folder** window, retain the default installation location and click **Next**.

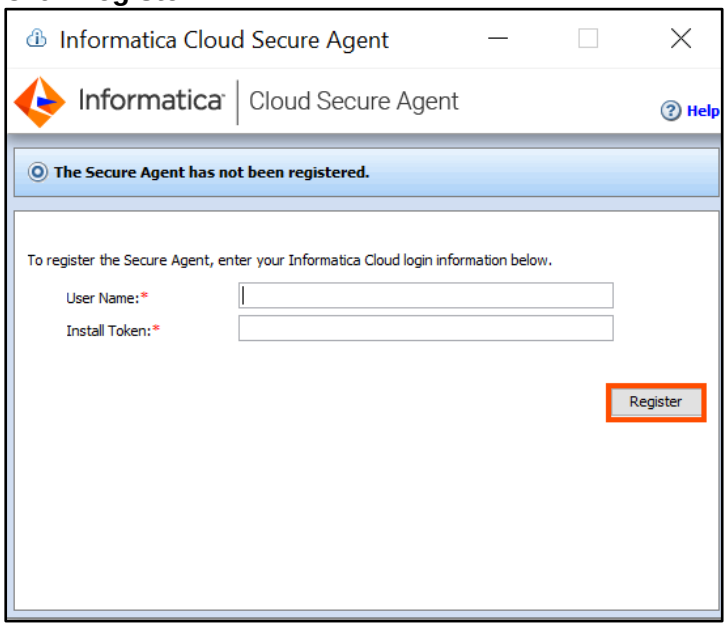


11. In the **Pre-Installation Summary** page, click **Install**.



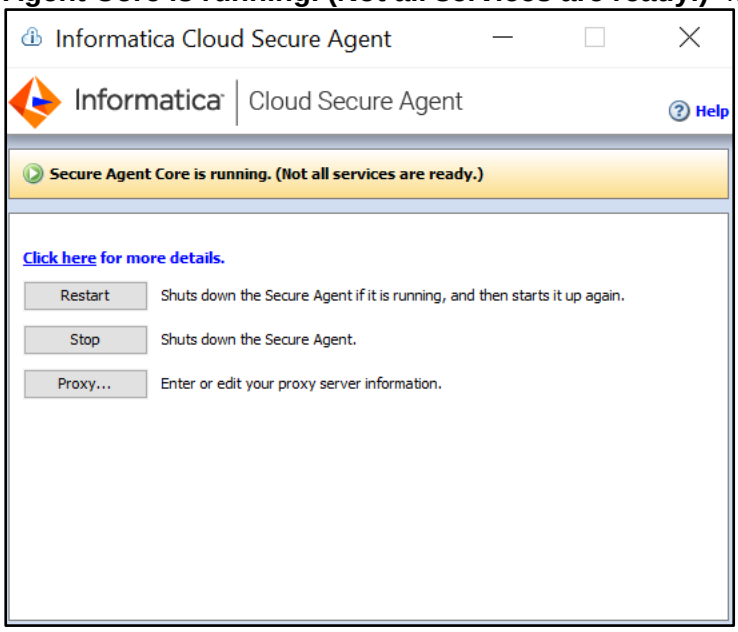
Note: After the installation process is complete, the Secure Agent registration page appears with the message **The Secure Agent has not been registered**.

12. To register your Secure Agent, enter your IICS Org username and paste the Install Token copied earlier in the Install Token field.
13. Click **Register**.



Note: In the above image, the User Name and Install Token is masked. After successful registration, the Secure Agent will download the necessary files for the connectors.

14. When the configuration of the Secure Agent services is in progress, it displays "**Secure Agent Core is running. (Not all services are ready.)**" message.

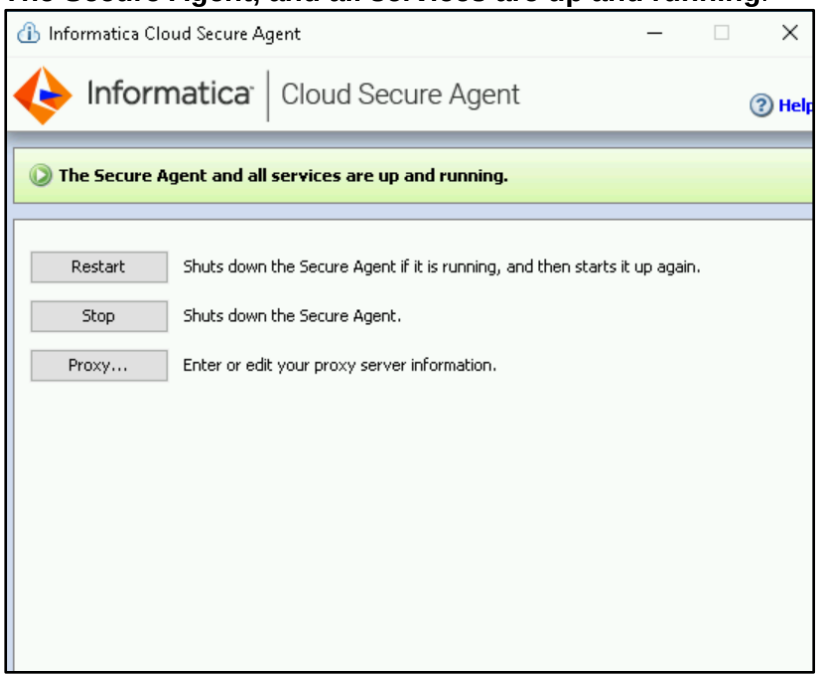


Note:

a. It takes about 10 to 15 minutes for the secure agent to come up and for all the services to be in a running state.

b. If your organization uses an outgoing proxy server to connect to the internet, you must configure the proxy server settings in the **Proxy** tab of the Secure Agent installer.

15. After the Secure Agent configures all the services, the agent status message changes to **The Secure Agent, and all services are up and running**.



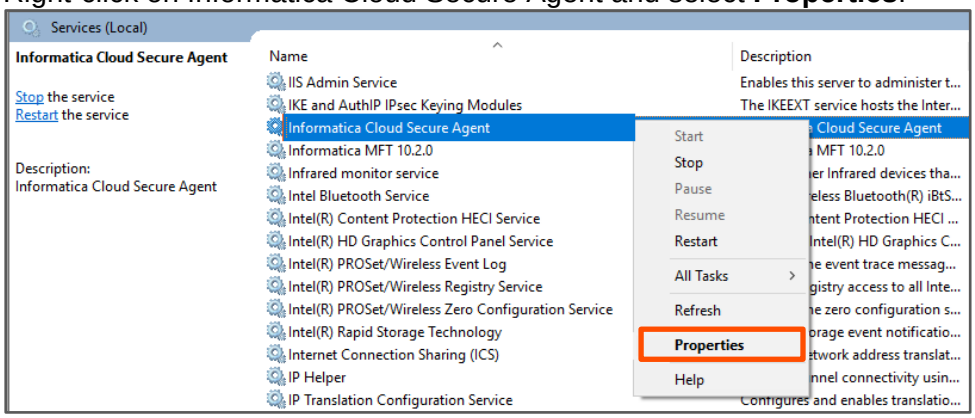
16. Minimize the Secure Agent window.

Setting Administrative Rights for Secure Agent

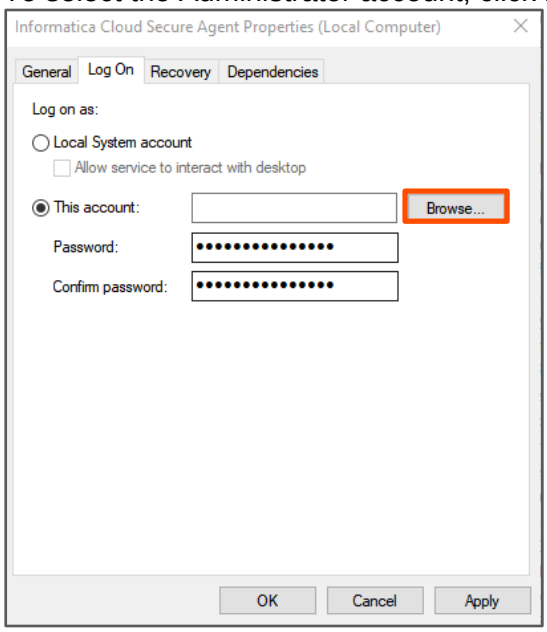
When you install Secure Agent on Windows platform, you need to assign Administrative rights to the Secure Agent service. This allows the Secure Agent to access files and directories present on the machine on which the Secure Agent is installed. This feature is useful when you configure connections or run tasks that use Flat File or FTP/SFTP connection types, that require read and write permissions on the related directories.

17. Select the windows **Start** menu.
18. In the search bar, type **services**, and press **enter** on your keyboard.
19. In the services page, from the list of services, select **Informatica Cloud Secure Agent**.

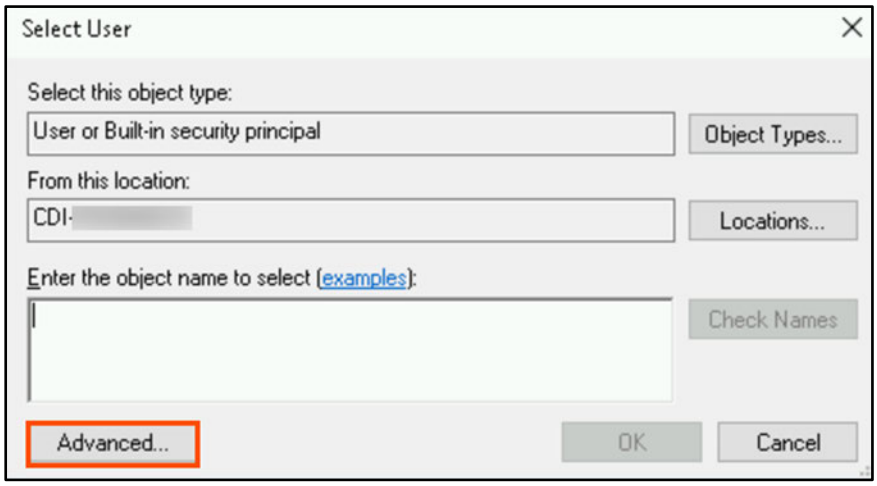
20. Right-click on Informatica Cloud Secure Agent and select **Properties**.



21. Go to **Log On** tab and select **This account**.
22. To select the Administrator account, click **Browse**.



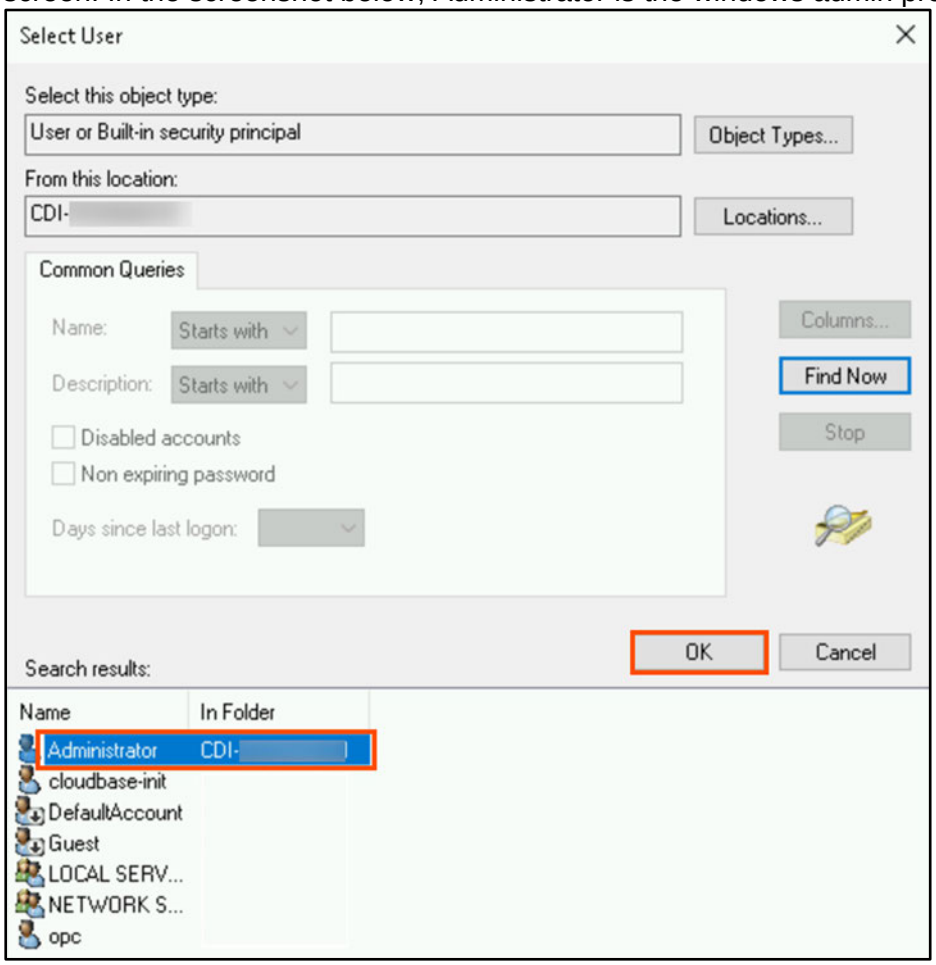
23. Click **Advanced**.



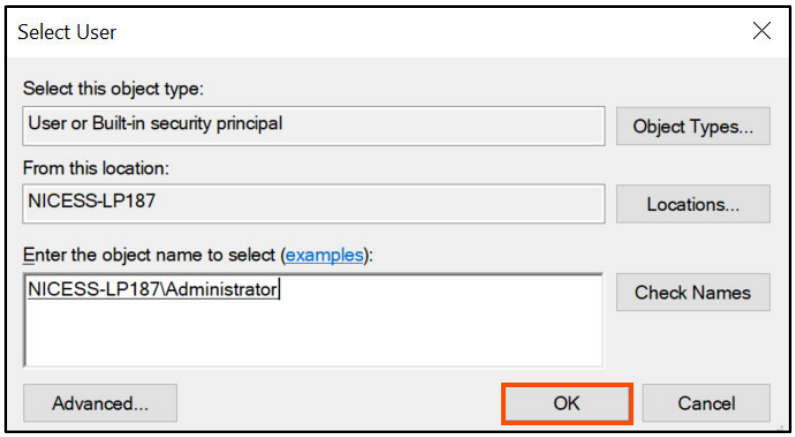
24. In the Select User window, select **Find Now**.

25. From the list, select **Administrator**, and click **OK**.

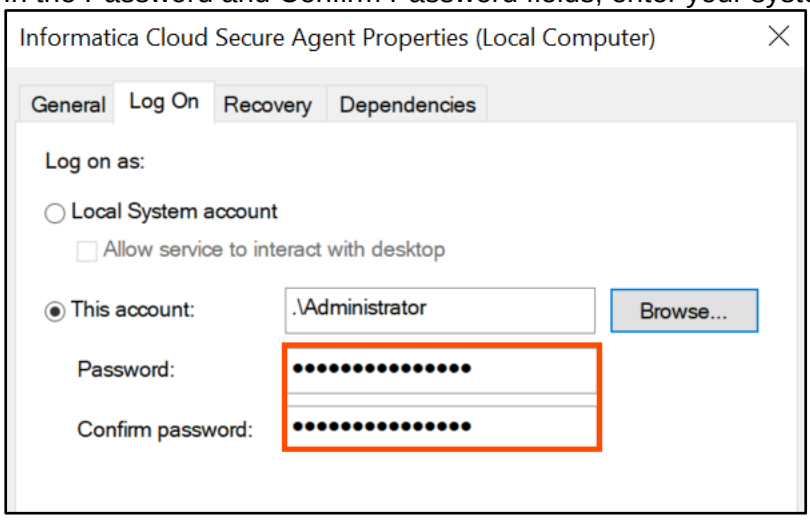
    Important Note: You must select your system admin user from the list shown on your screen. In the screenshot below, Administrator is the windows admin profile user.



26. Click **OK**.

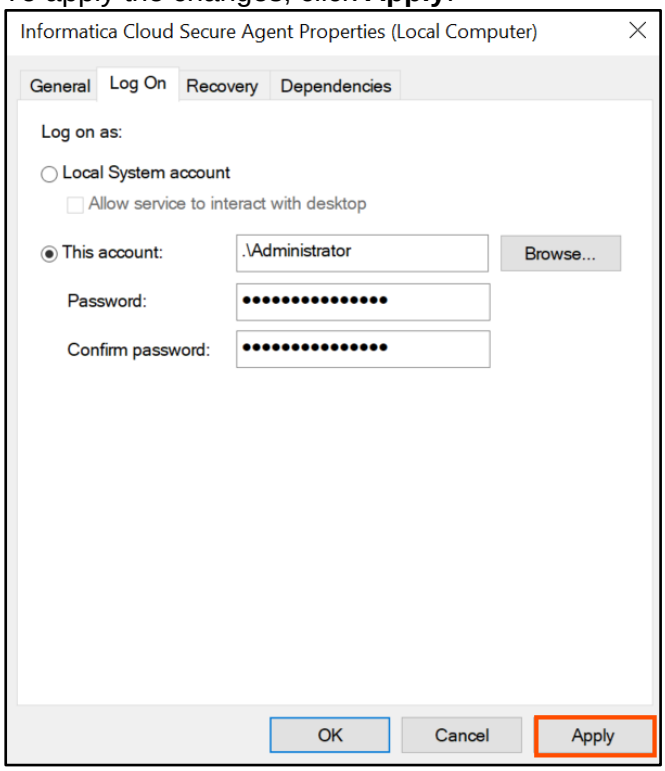


27. In the Password and Confirm Password fields, enter your system user's password.

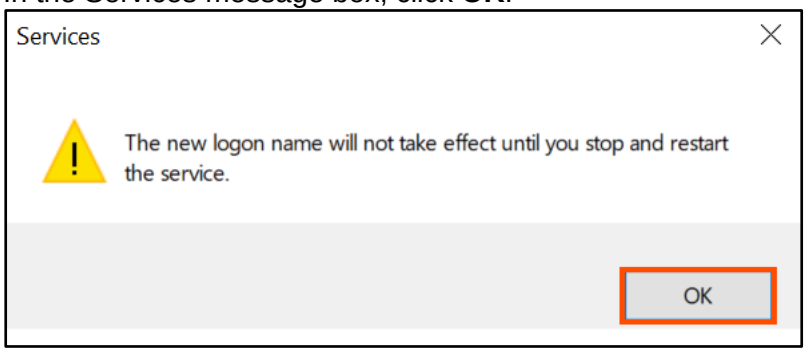


Important Note: This is the password of Windows user used to log in to the machine on which the Secure Agent is installed. If you are installing the agent on your local machine, use your Windows user password in the Password and Confirm Password fields.

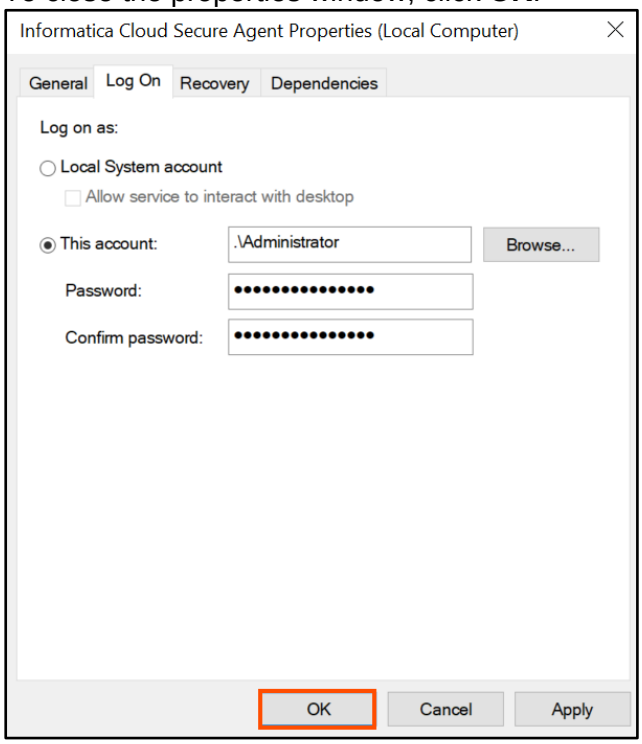
28. To apply the changes, click **Apply**.



29. In the Services message box, click **OK**.



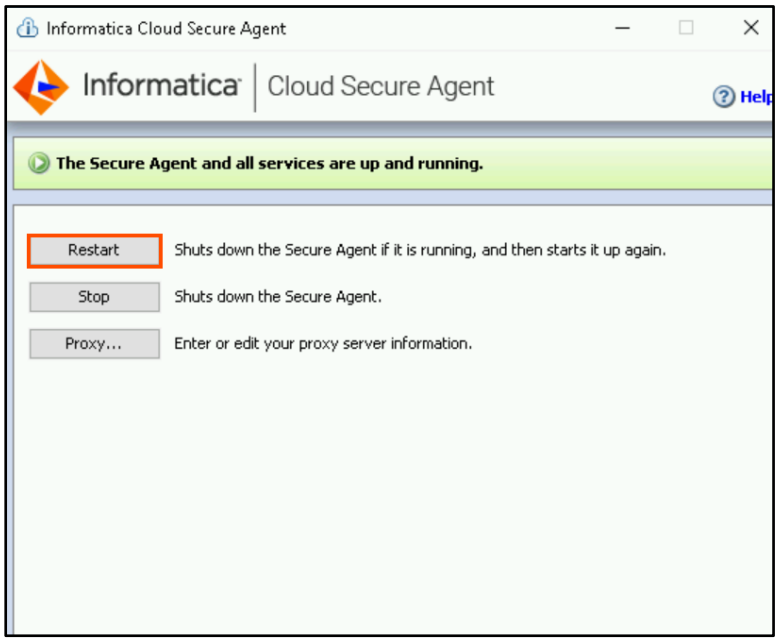
30. To close the properties window, click **OK**.



Note: After this, you must restart the Secure Agent for the changes to take effect.

31. Close the Services window.
32. In the desktop, right-click on the Secure Agent, and select **Run as administrator**.
Note: Skip this step if the Secure Agent window is already open.

33. Click **Restart**.



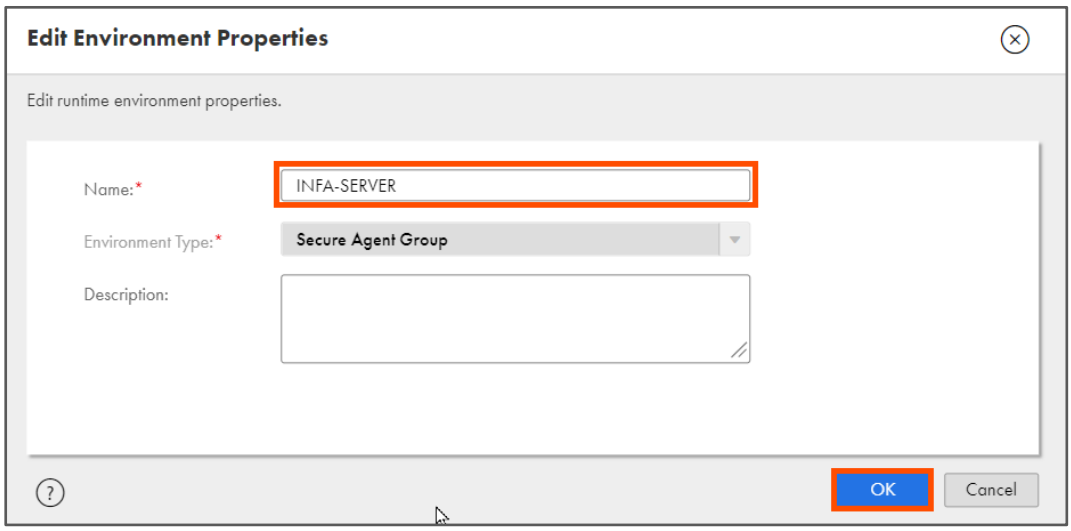
Note: It takes 10-15 minutes for the secure agent to come up. You must wait for the Secure Agent to get back to the running state.

View the Secure Agent and Rename it in IICS

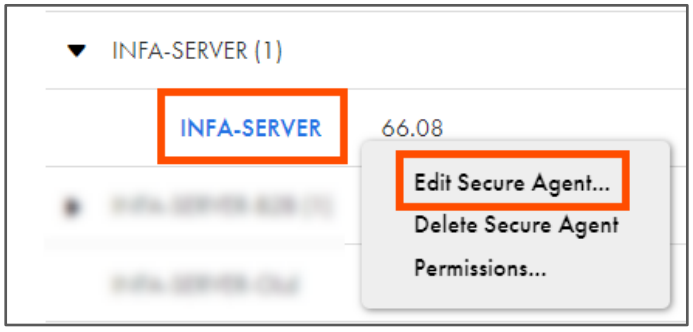
34. Navigate back to the browser, and from the Administrator service navigation pane, select **Runtime Environments**.
35. Refresh the page if its already opened and observe that the Secure Agent status is **Running**.

    Note: By default, the Secure Agent takes the name of the computer it is installed on. If the Secure Agent does not appear in **Runtime Environments** page, you can refresh the webpage to view the updated status of the page.
36. Right-click on the secure agent group and select **Edit Environment Properties**.

37. Rename the agent group to **INFA-SERVER** and click **OK**.

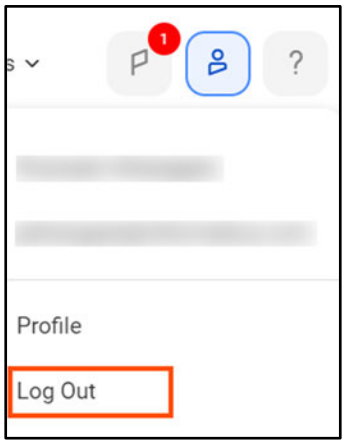


38. Similarly, expand the agent group, right-click on the secure agent, and click **Edit Secure Agent**. Enter **INFA-SERVER** as Agent Name and click **Save**.



Note: It is NOT mandatory to rename the secure agent while working with it. We have asked you to rename secure agent to maintain uniformity of the screenshots you see in the lab guides and the screen you see in your machine. All the lab exercises are built with the secure agent named as **INFA-SERVER**.

39. To log out from the Org, click the **User** icon, and select **Log Out**.



This concludes the lab.

Module 2: Introduction to Data Integration Service

Lab 2-2: Creating Flat File Connections

Overview:

Flat file connections store the information to create, access, and store flat files. There may be multiple flat files that you need to access from a local system.

Objective:

- Create flat file connections

Scenario:

As mentioned earlier, different outlets of NH Suppliers manage data on different data sources. The outlet in California uses flat files to organize the everyday sales data. So, in this lab, Ruby will create a Flat File connection to access flat files on her local machine.

Duration:

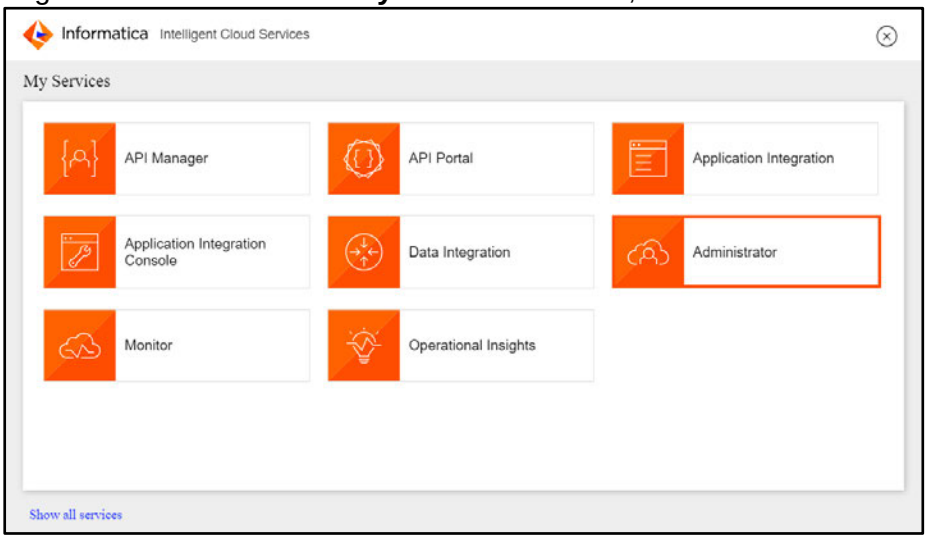
10 minutes

Tasks

Create Source and Target Flat File Connections

You have already created **IICSLabFiles** folder and pasted a source folder named **SrcFiles** – which has all the source files required for the training. Also, you have pasted a target folder named **TgtFiles** – this is the target folder to which you will write the target files/data. Thus, in this section, you will create 2 flat file connections – 1 source connection and 1 target connection pointing to their respective directories.

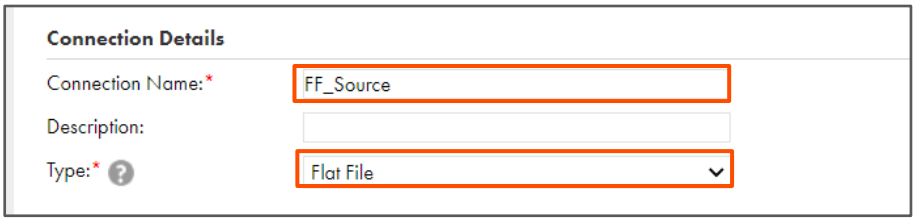
1. Login to IICS and from the **My Services** window, select **Administrator**.



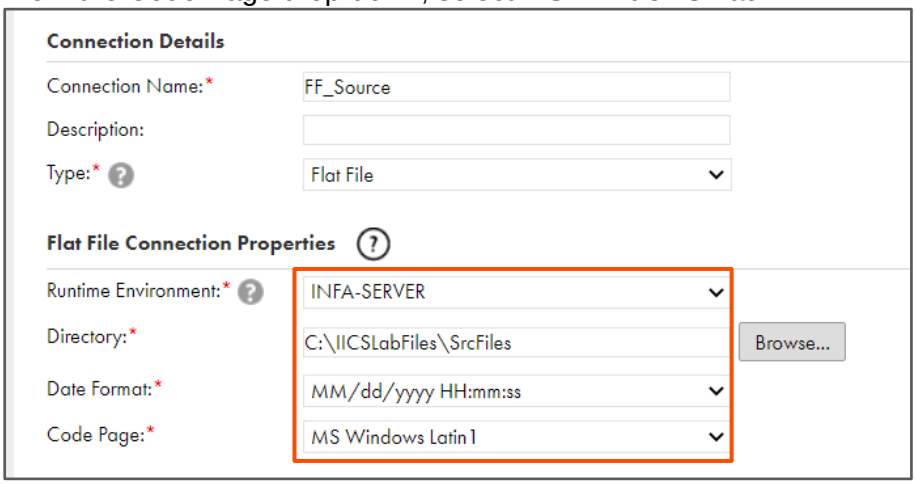
2. From the navigation pane, select **Connections**.

3. To create a new connection, select **New Connection**.

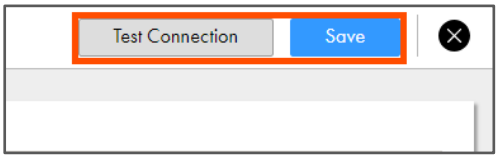
4. Enter the Connection Name as **FF_Source**. This is the source flat file connection.
5. From the Type drop-down, select **Flat File**.

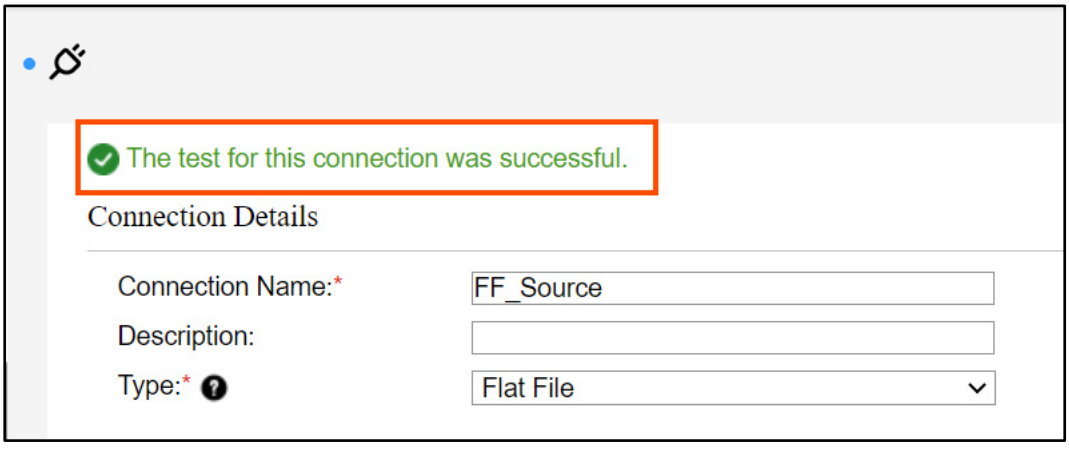


6. From the **Runtime Environment** drop-down, select your secure agent group.
7. In the Directory field, enter **C:\IICSLabFiles\SrcFiles**.
8. From the Code Page drop-down, select **MS Windows Latin1**.

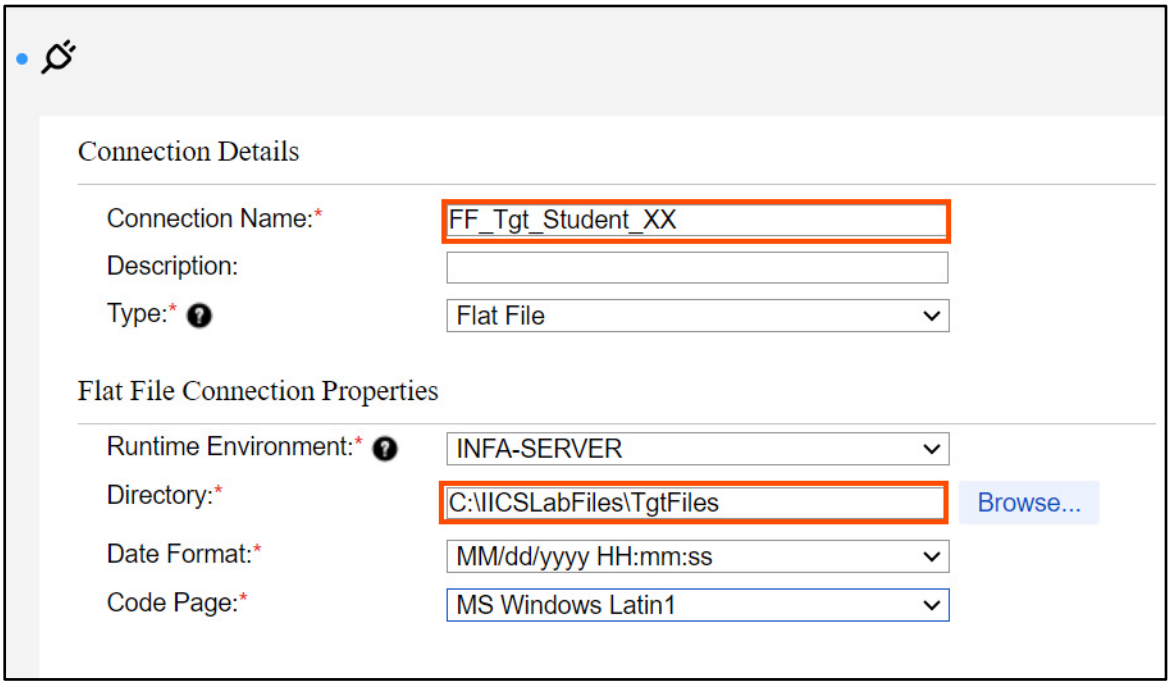


9. Test and save the connection.

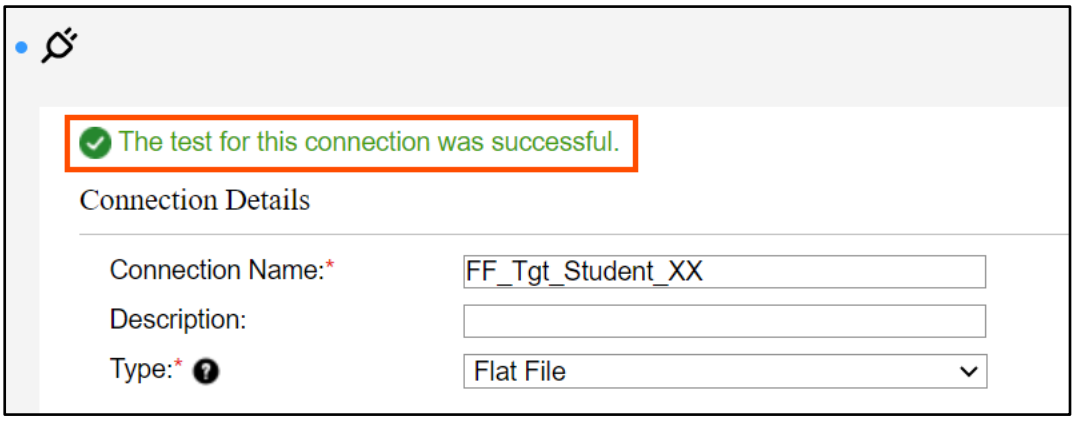




10. Similarly, create target flat file connection and name it as **FF_Tgt_Student_XX**. Here, in the asset name, replace XX with your Data of Birth (DOB).
11. Enter the target flat file directory as **C:\IICSLabFiles\TgtFiles**.



12. Test and save the connection.



Thus, you have created your source and target flat file connections. Using these flat file connections, you will access the source and target files.

GENERAL INSTRUCTIONS: In the course, all the assets are named in a certain format by following the Informatica naming conventions. They all "XX" in their asset name and a few of them have FirstName in the asset names. Replace **XX** with your DOB or any random number and replace **FirstName** with your **first name**. For example, if your DOB is January 19, then replace **XX** with **19** in the asset names.

This concludes the lab.

Module 3: Synchronization Task

Lab 3-1: Creating a Salesforce Connection

Overview:

In IICS, a connection allows you to gain access to data that is available on Cloud and on-premise applications such as platforms, databases, and flat files. After you create a connection in IICS, it is available to all users in the organization.

A Salesforce connection allows you to securely read data from or write data to Salesforce sources or targets.

Objective:

- Create a Salesforce connection

Scenario:

Now that Ruby has installed the secure agent, John informs her that she needs to create a connection on IICS to connect to a data source. One of the outlets of NH Retails manages data on Salesforce. So, Ruby needs to create a Salesforce connection to read data from Salesforce.

In this lab, Ruby will create a Salesforce connection.

Duration:

20 minutes

IMPORTANT: Before starting this lab, you must have a functional Salesforce **Developer** account. If you do not have a Salesforce Developer account, create one by using the following URL: https://developer.salesforce.com/signup

You must only create a Developer account and any existing Salesforce accounts or “free trial” accounts will not work for this training.

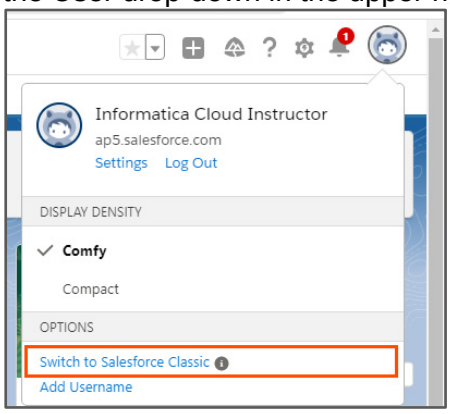
IICS accesses Salesforce.com through APIs. You must reset your Salesforce.com security token (unless you have previously accessed the API from your current machine). You must execute this process only once.

Tasks

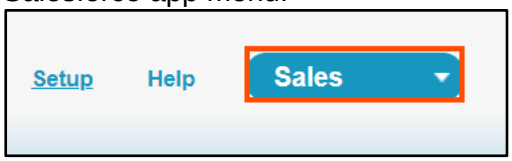
Reset your Security Token in Salesforce.com

1. Open a new tab in the web browser.
2. Log in to Salesforce.com using your Salesforce Developer credentials: https://login.salesforce.com/?locale=in
   Note: To log in to Salesforce you must register your phone number on Salesforce.

3. This lab is based on the Classic version of Salesforce. If you are using the Lightning version, switch to the Classic version. The option to switch versions is available under the User drop-down in the upper right corner of the user interface.



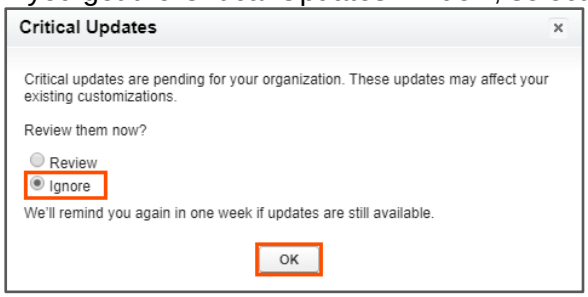
Note: For better visibility of tabs in Salesforce, you must select **Sales** from the Salesforce app menu.



4. From the Username drop-down, select **My Settings**.

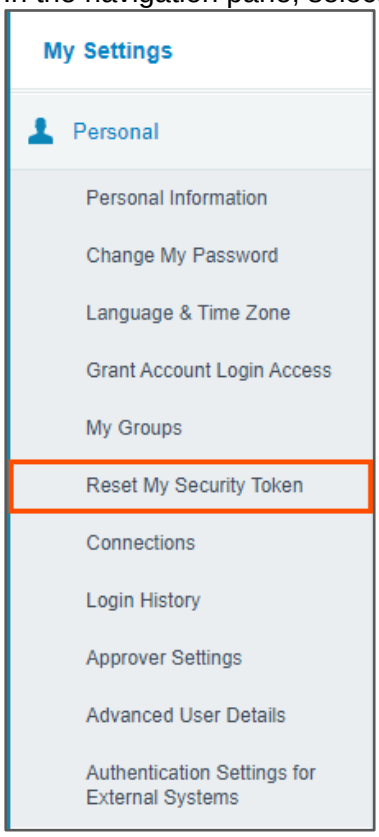
Note: The Menue option is under your Username.

5. If you get the Critical Updates window, select **Ignore** and click **OK**.

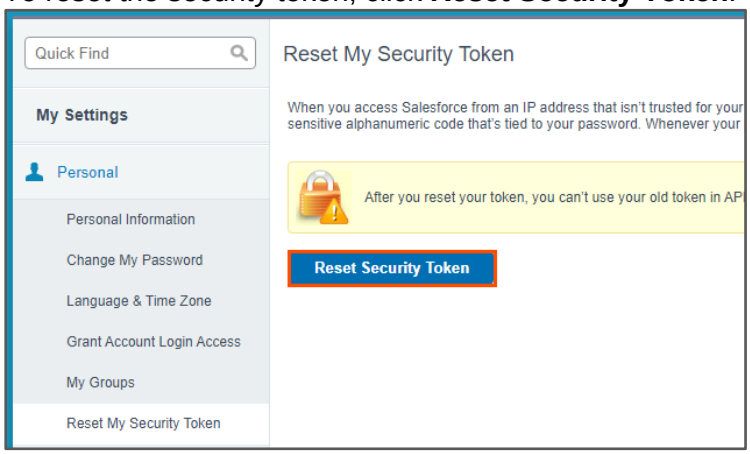


6. In the **My Settings** page, drill down to **Personal**.

7. In the navigation pane, select **Reset My Security Token**.



8. To reset the security token, click **Reset Security Token**.

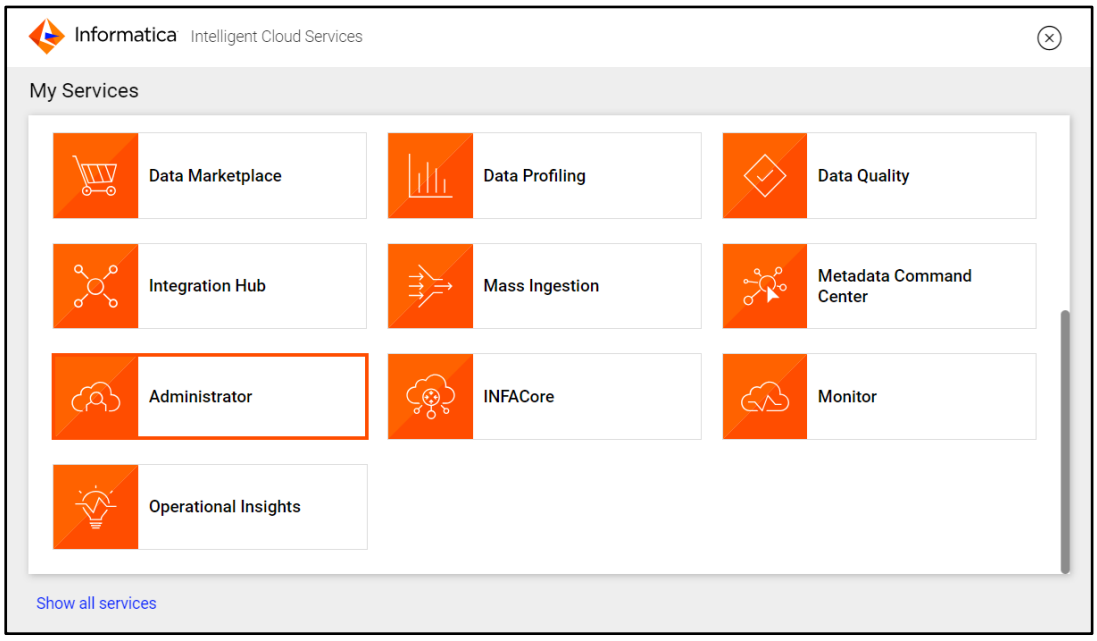


Copy Security Token

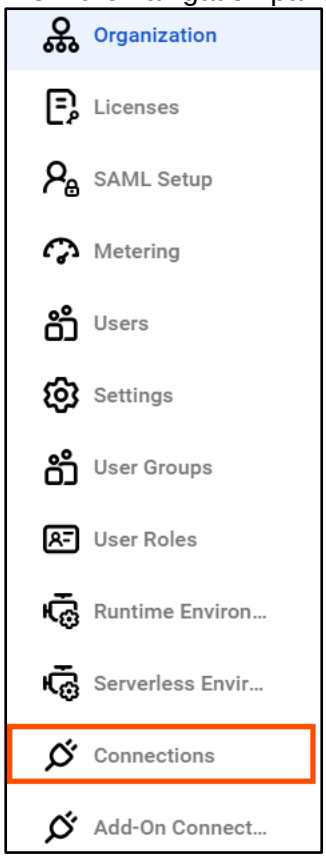
9. After you reset the security token, you will receive an email in your email id registered with Salesforce.
10. Open the email from **support@salesforce.com** from your mailbox.
11. Copy your security token and paste it into a text file.

Create a Salesforce Connection

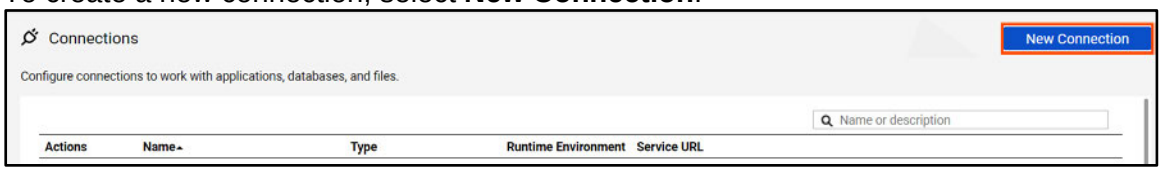
12. If you have logged out of IICS, login back to IICS, and from the **My Services** window, select **Administrator**.



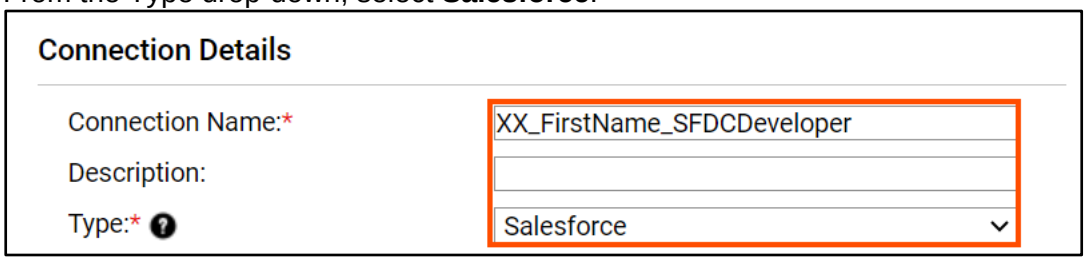
13. From the navigation pane, select **Connections**.



14. To create a new connection, select **New Connection**.



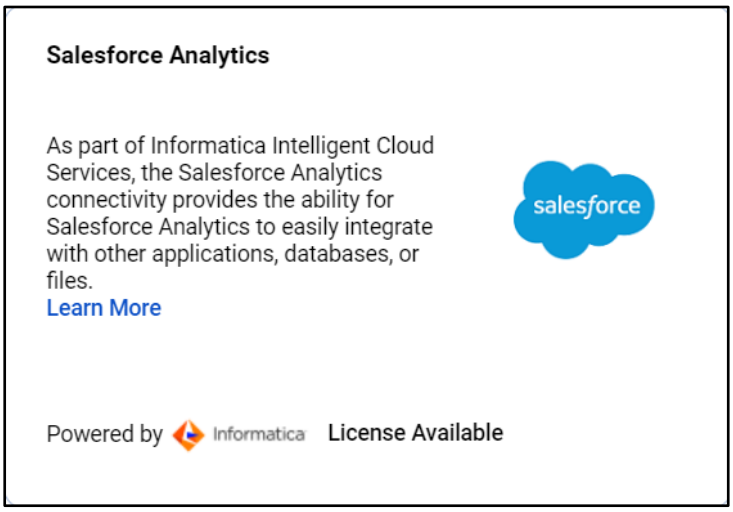
15. Enter the Connection Name as **XX_FirstName_SFDCDeveloper**
16. From the Type drop-down, select **Salesforce**.



Note:

a. If you do not see the desired connector type in the Type drop-down. You can enable the free trial for a connector from the **Add-on Connectors** tab of the Administrator service.

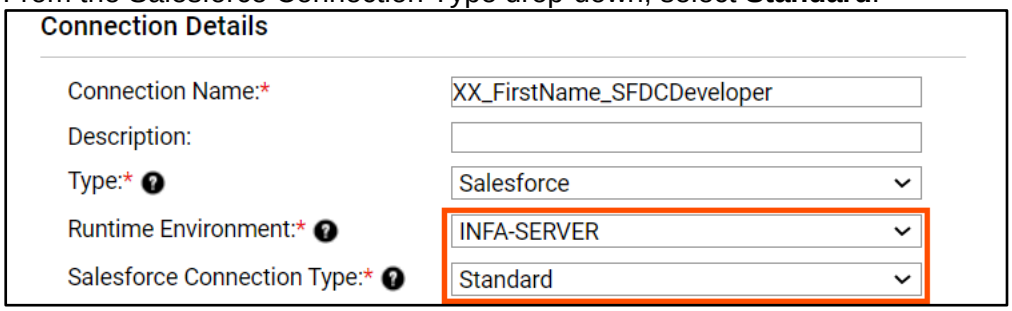
b. In the Add-on Connectors page, search for **Salesforce Connector** and click **Start Free Trial** to enable the connector. If the connector is available, its status shows as '**License Available'**.



Note: If the connector is already available for the org, you can skip the steps to enable the connector.

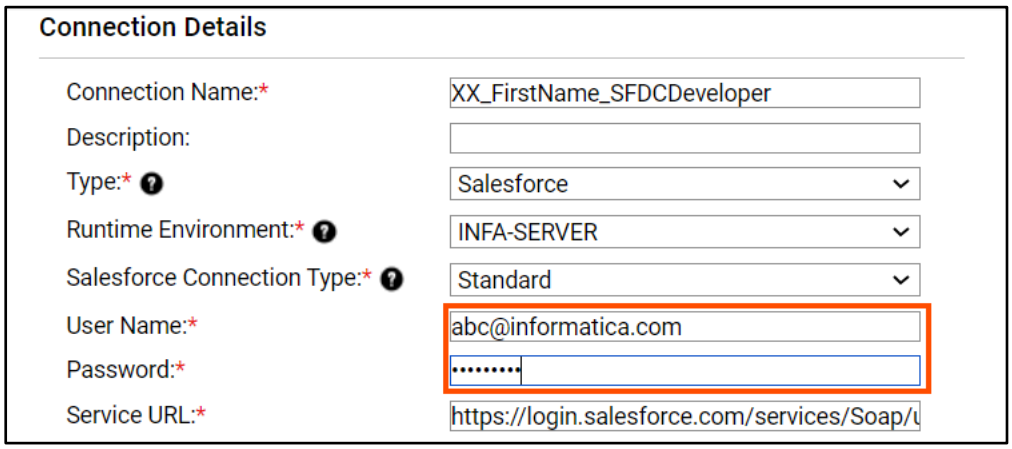
17. From the **Runtime Environment** drop-down, select **INFA-SERVER** as the secure agent group.

18. From the Salesforce Connection Type drop-down, select **Standard**.

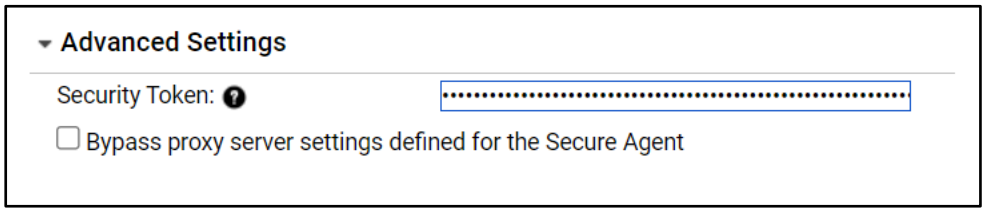


Note: Throughout the course, the runtime agent shown in the screenshots may be from what you are viewing in your environment. Kindly select the runtime environment that is created for you. Get in touch with your Instructor to know about your agent name.

19. In the **User Name** and **Password** fields, enter your Salesforce credentials.
20. Retain the value in **Service URL**.

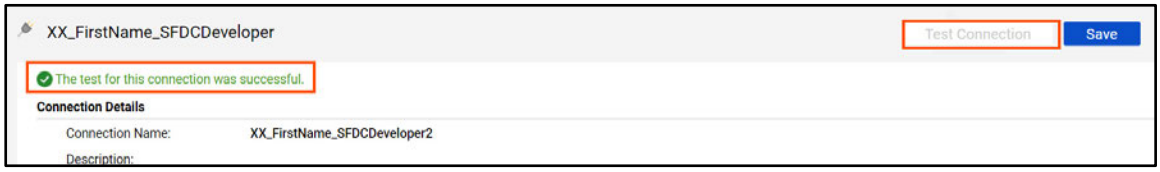


21. In the **Security Token** field, paste your Salesforce security token copied earlier from your email.

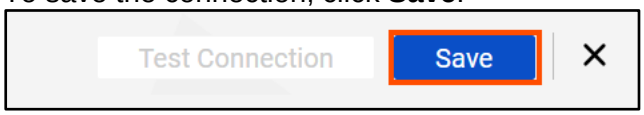


Note: By default, the Service URL contains the latest Salesforce API version. It is recommended not to change the Salesforce API version in Service URL, as some of the features are not available for older versions of Salesforce API.

22. To test the connection, click **Test Connection**.



23. To save the connection, click **Save**.



24. Close the connection.

This concludes the lab.

Module 3: Synchronization Task

Lab 3-2: Creating SQL Server Connections

Overview:

In IICS, you can connect to SQL server Database through a SQL server connection.

In this lab, you will create a SQL server Connection in IICS.

Objective:

- Create a SQL server connection

Scenario:

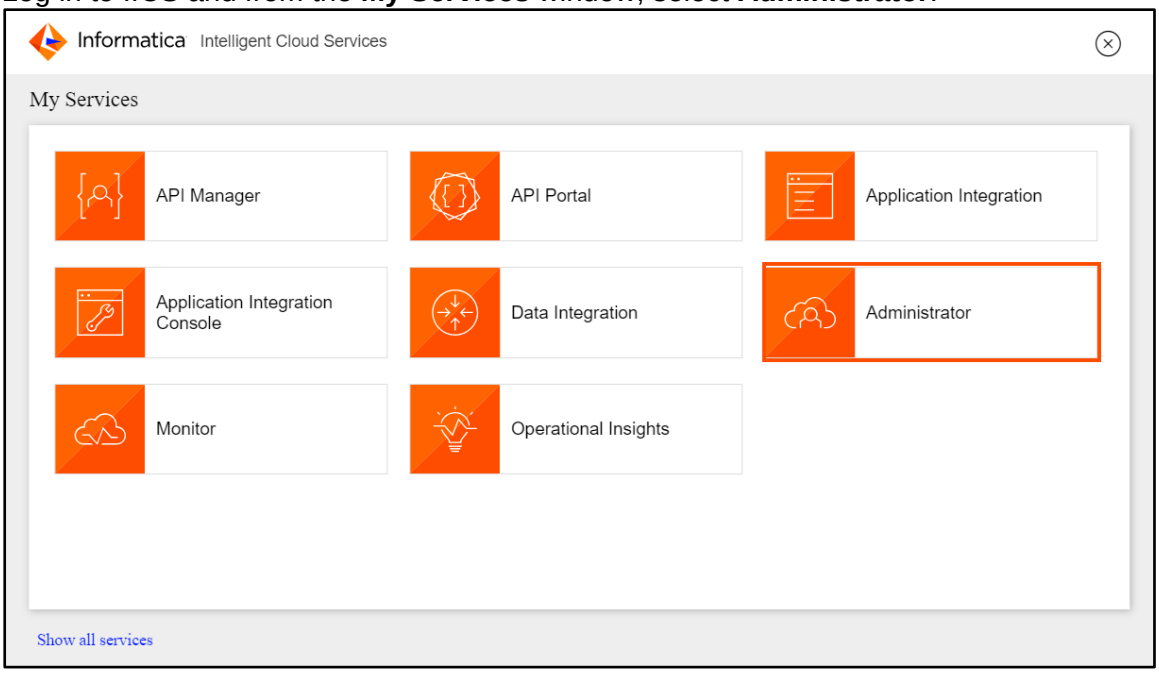
The Alaska outlet of NH suppliers uses SQL Server Management Studio database to manage the sales data. So, to integrate data from SQL server database, Ruby must create a SQL Server Connection.

Duration:
10 minutes

Tasks:

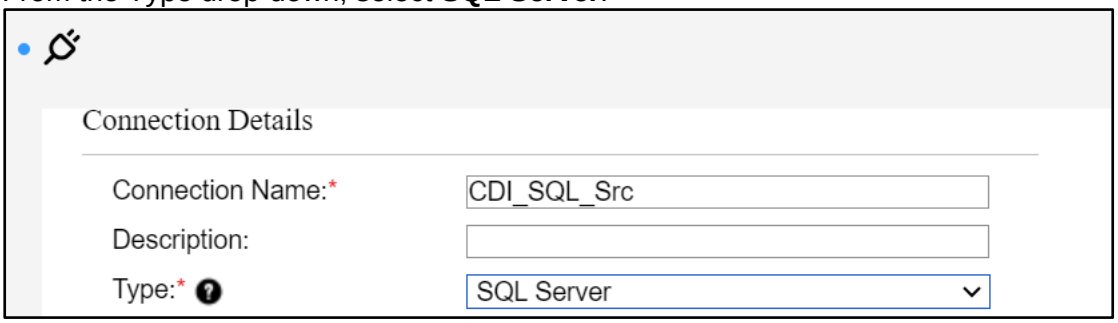
Create Source and Target SQL Server Database Connections

1. Log in to IICS and from the **My Services** window, select **Administrator**.



2. From the **Connections** tab create a new connection.

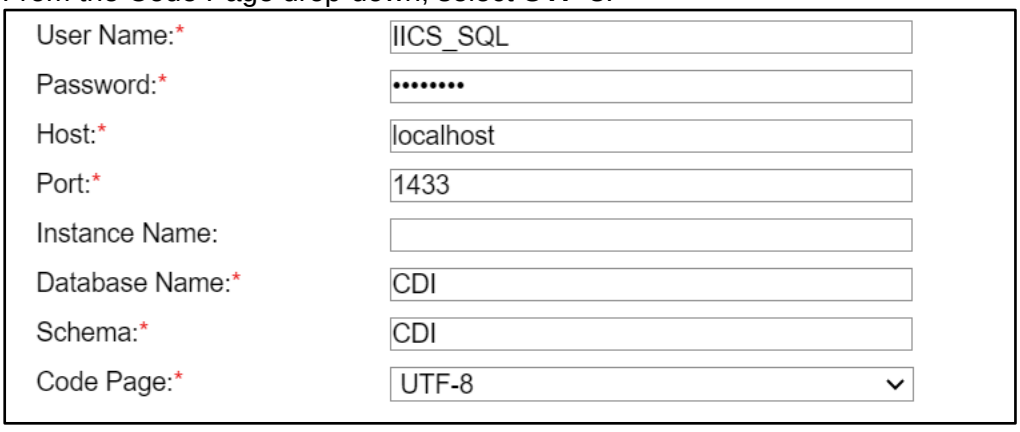
3. In the Name field, enter **CDI_SQL_Src**. This is the source SQL Server connection to the **CDI** database.
4. From the Type drop-down, select **SQL Server**.



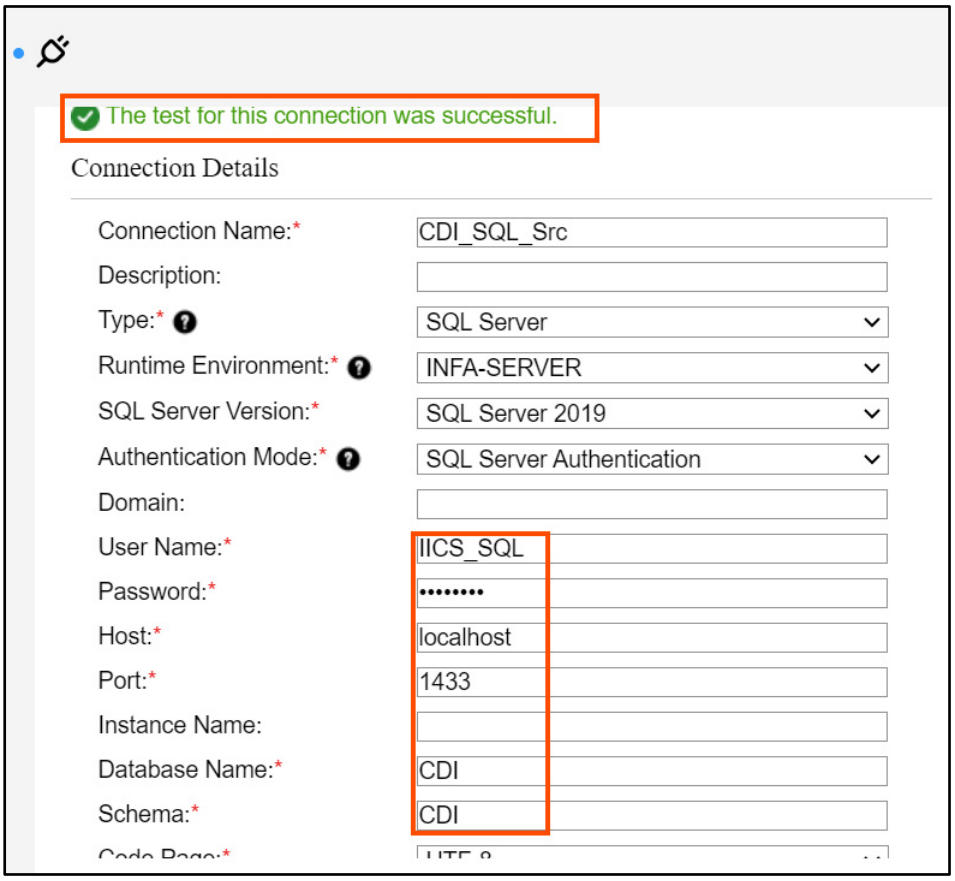
5. From the Runtime Environment drop-down, select your secure agent **INFA-SERVER**.
6. From the SQL Server Version drop-down, **select SQL Server 2019**. Select the version applicable for your software.
7. From the Authentication mode drop-down, select **SQL Server Authentication**.

8. Enter the Username and Password as **IICS_SQL**.

   Note: The Password is case-sensitive.
9. In the Host field, enter **localhost**, and retain the Port as **1433**.
10. In the Database Name field, enter **CDI**.
11. Enter the Schema name as **CDI**.
12. From the Code Page drop-down, select **UTF-8**.

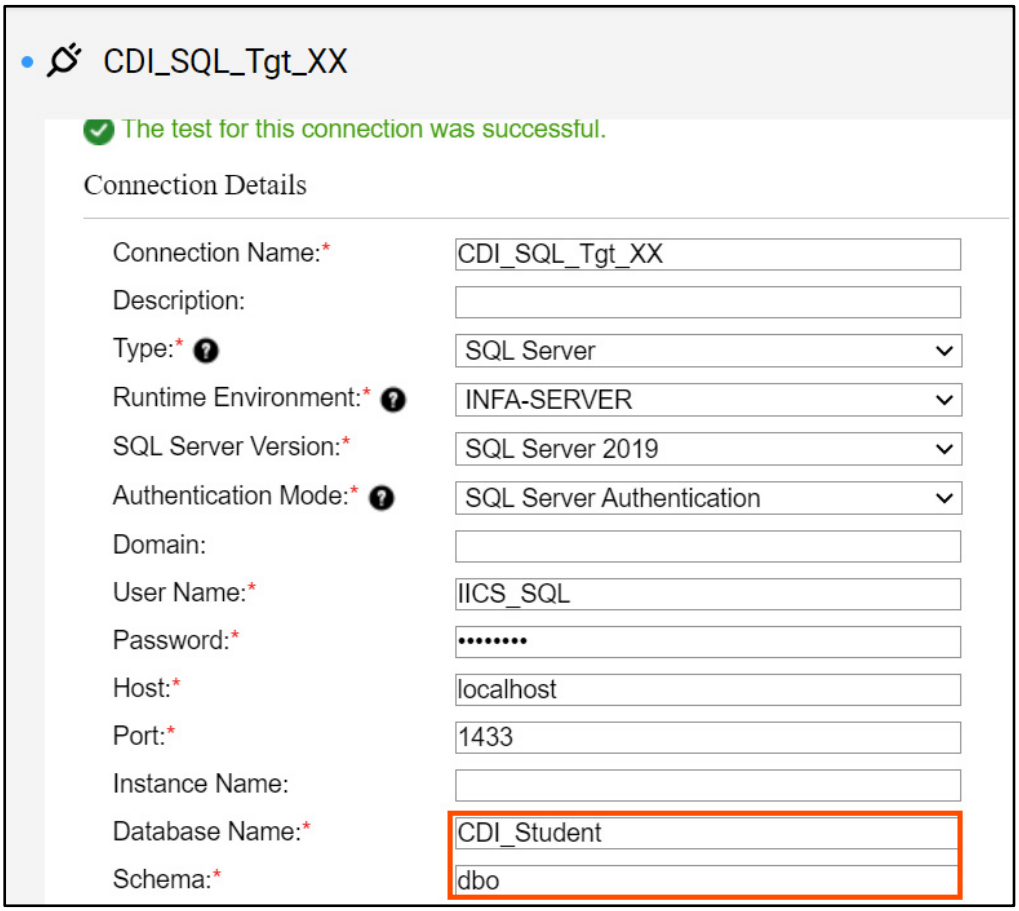


13. Test and save the connection.



14. Similarly, create another SQL Server database connection to access the **CDI_STUDENT** database and name it as **CDI_SQL_Tgt_XX**.

15. Provide the same connection details as above except for Schema and Database Name. For Database Name, enter the value as **CDI_STUDENT** and for Schema, enter **dbo**.



16. Test and save the connection.
17. You have successfully created 1 source connections to the CDI database and 1 target database connection to the CDI_STUDENT database.

This concludes the lab.

Module 3: Synchronization Task

Lab 3-3: Creating a Synchronization Task

Overview:

The Synchronization task synchronizes data between a source and a target. The Data Synchronization application supports Insert, Update, Upsert, and Delete operations.

Objective:

- Create and configure a synchronization task
- Run the task and validate the results in Salesforce

Scenario:

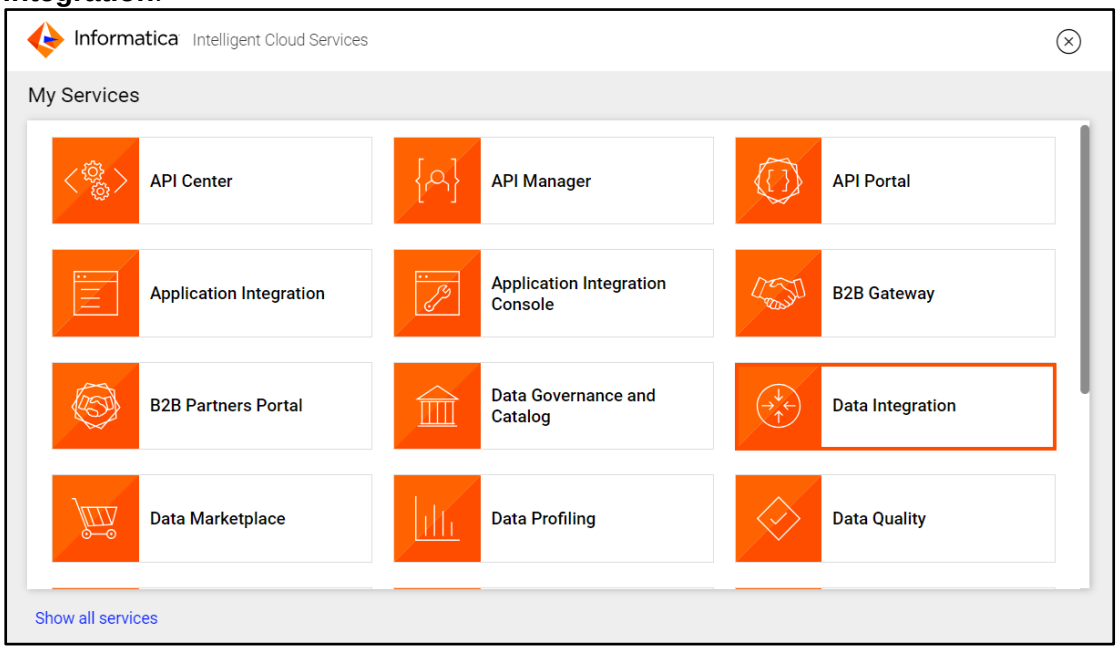
After Ruby creates the connections in IICS, she asks Joseph about the process to integrate data between various data sources. To this, John says that she must create a Synchronization task in IICS. In this lab, Ruby will create a synchronization task to load outlet data from a CSV file to the Account object in Salesforce.

Duration:
15 minutes

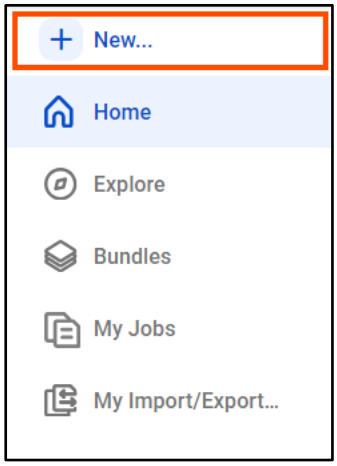
Tasks

Create a Synchronization Task in IICS

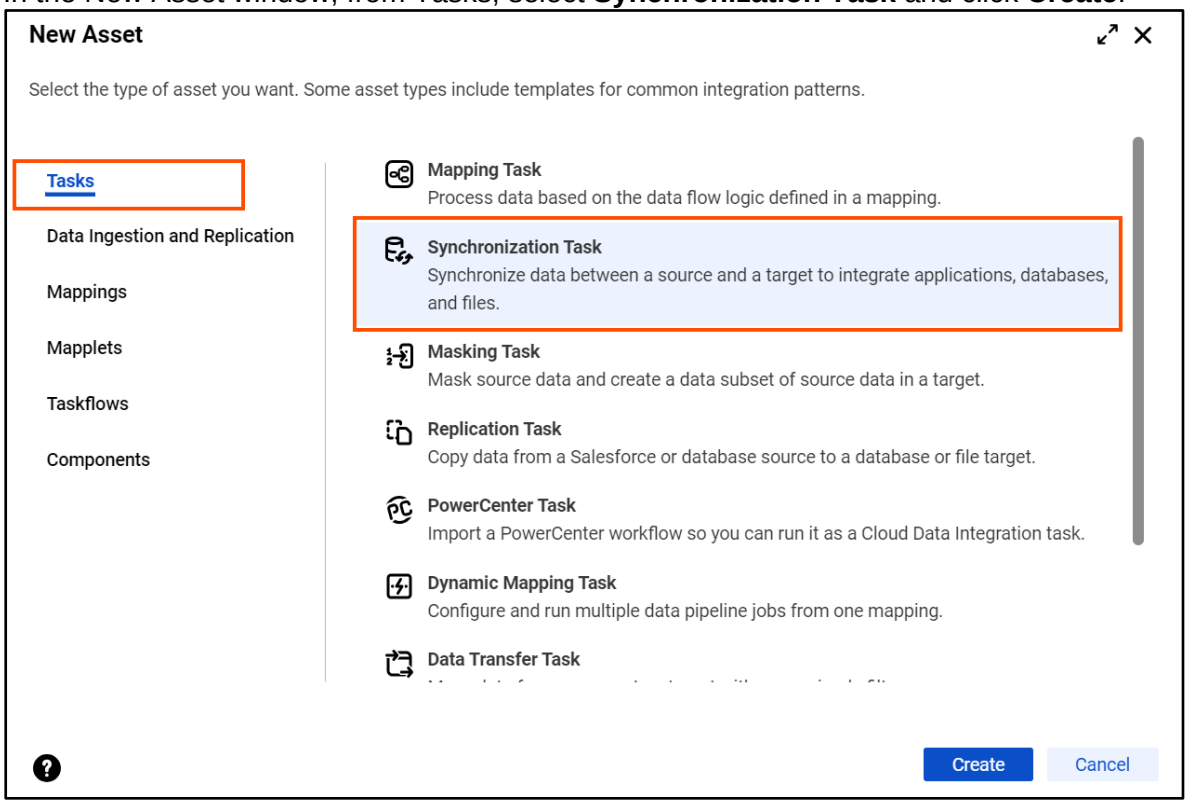
1. If navigated away, login to IICS and from the **My Services** window, select **Data Integration**.



2. To create a new asset, from the navigation pane, select **New**.

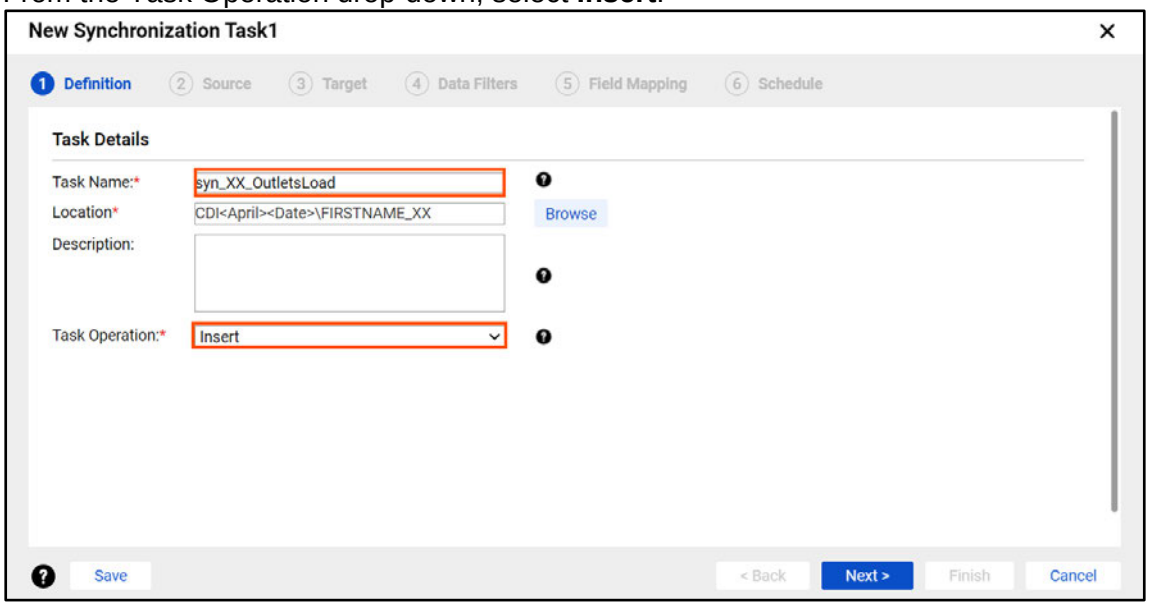


3. In the New Asset window, from Tasks, select **Synchronization Task** and click **Create**.



Specify Definition Information

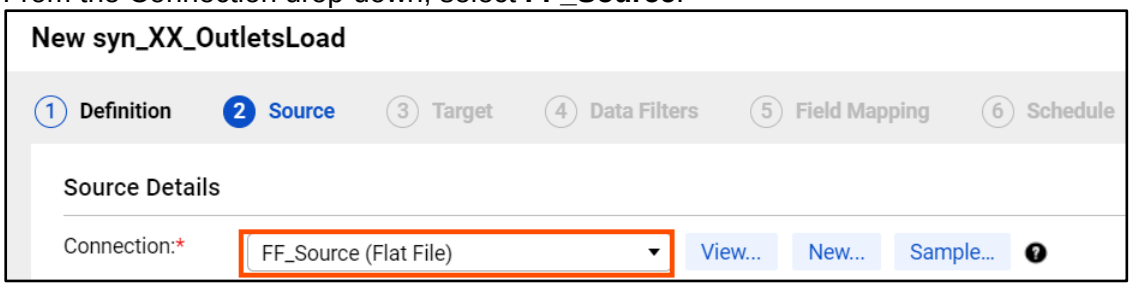
4. In the Task Name field, enter **syn_XX_OutletsLoad**.
   Note: Replace **XX** with your DOB.
5. Use the **Browse** option to save the asset in your working directory.
6. From the Task Operation drop-down, select **Insert**.



7. Click **Next**.

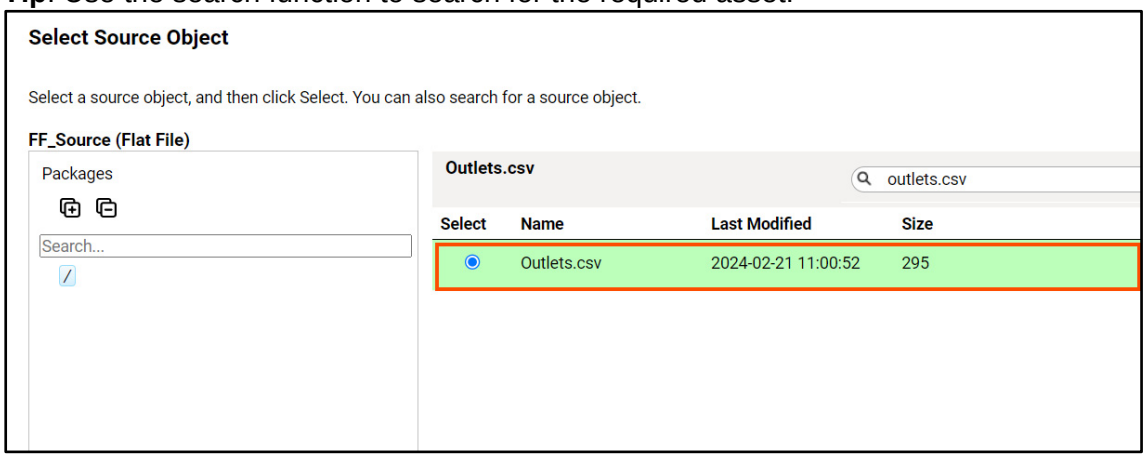
Specify Source Information

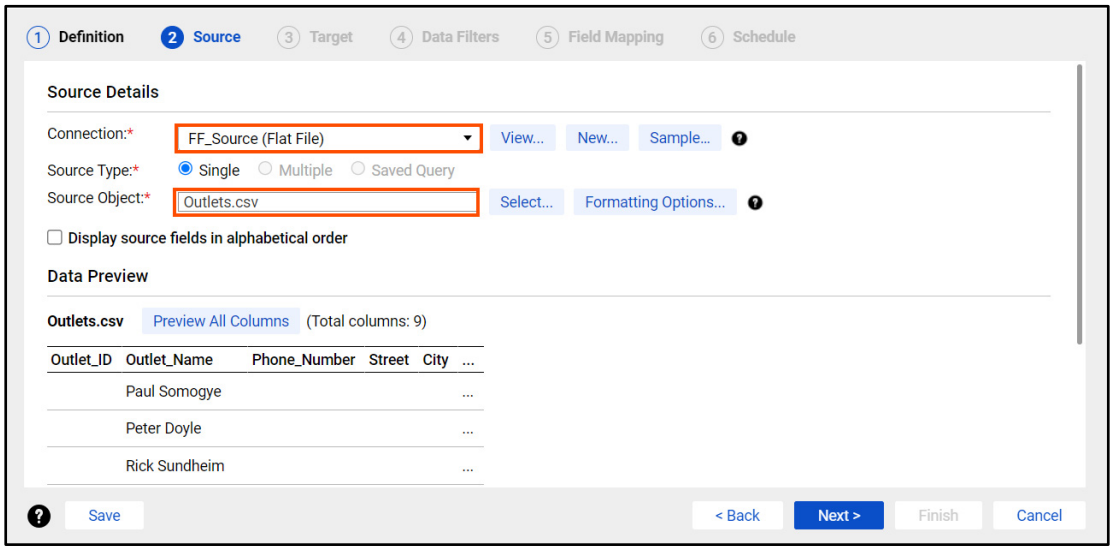
8. From the Connection drop-down, select **FF_Source**.



9. From the Source Object drop-down, click **Select** and choose **Outlets.csv**.

Tip: Use the search function to search for the required asset.



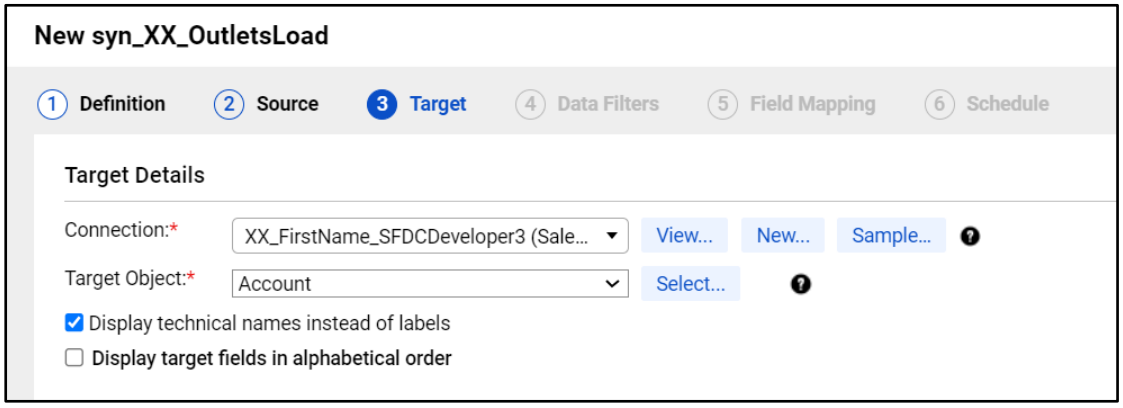


Note: The Data Preview section appears. It shows the first ten rows of the first five columns in the object and displays the total number of columns in the object.

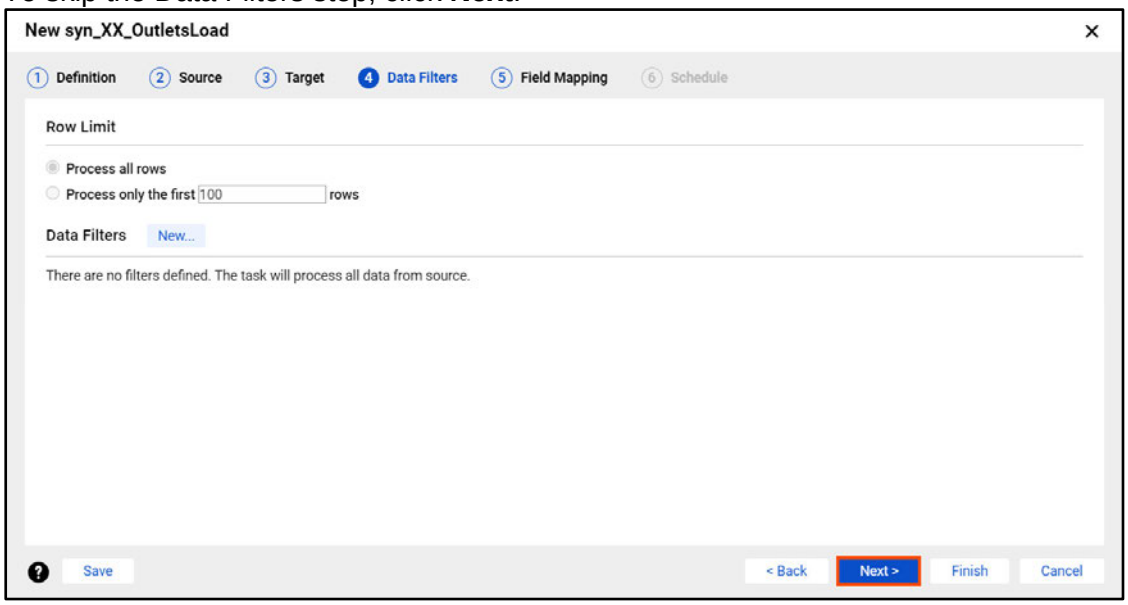
10. Click **Next**.

Specify Target Information

11. From the Connection drop-down, select your Salesforce connection (present in the format **XX_FirstName_SFDCDeveloper**).
12. From the Target Object drop-down, select **Account**.



13. Click **Next**.
14. To skip the Data Filters step, click **Next**.

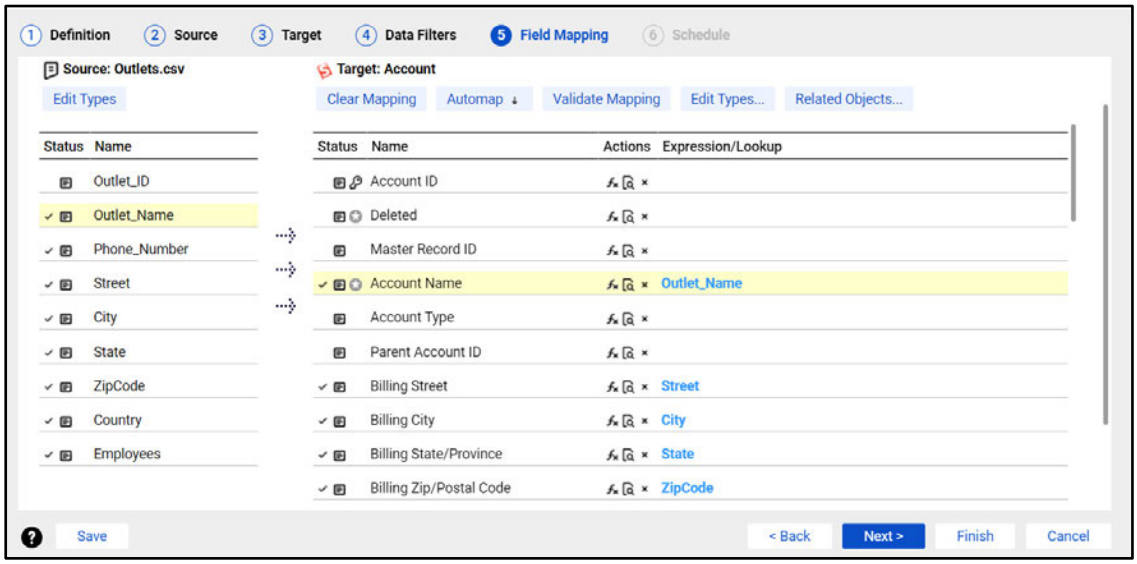


Define Field Mappings

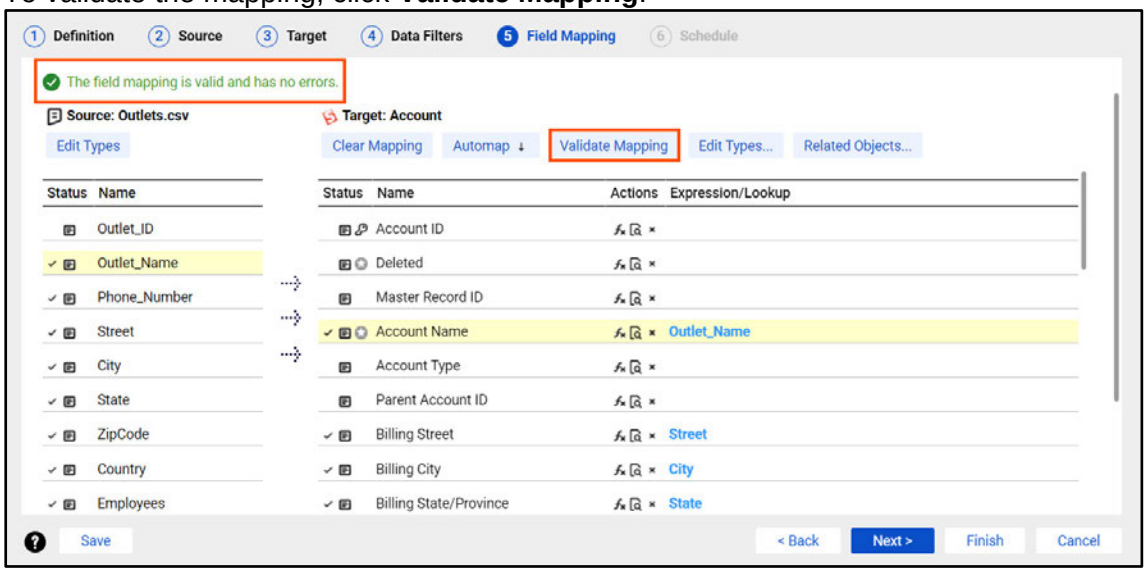
15. Map the Source field with the Target field, as shown in the table below:

Source Field Name	Target Field Name
Outlet_Name	Account Name (Instead of Account Name, you may see “Name”. Map to which ever field you see.)
Phone_Number	Account Phone
Street	BillingStreet
City	BillingCity
State	BillingState
ZipCode	BillingPostalCode
Country	BillingCountry
Employees	Employees

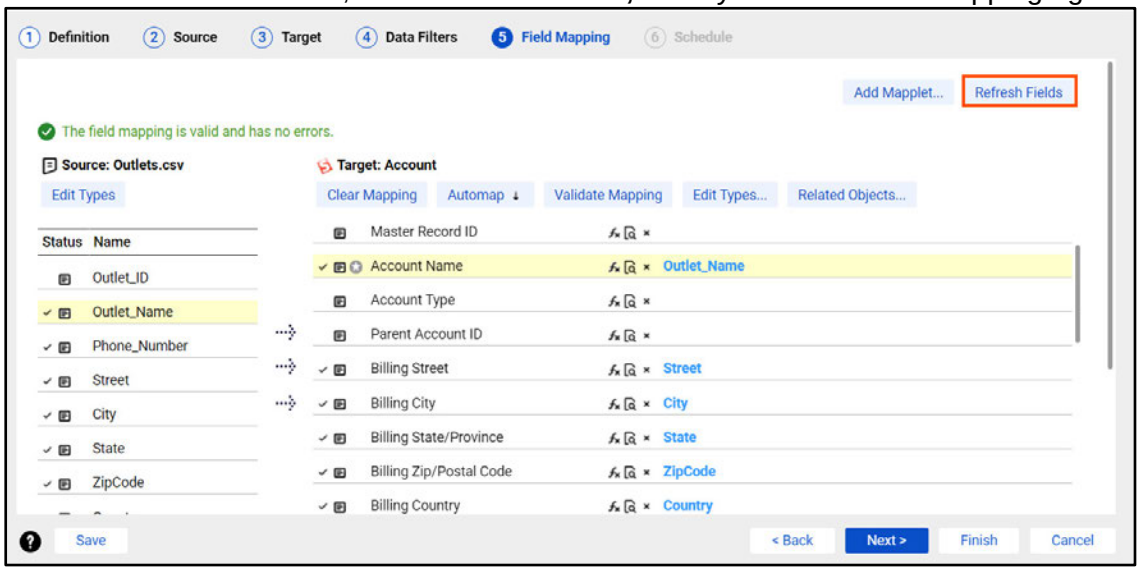
Note: To map the fields, drag a Source field and drop it onto a Target field. If some of the fields are mapped automatically, do not map the fields again. Sometimes, you may see a Target field name that is slightly different from the name mentioned in the table above. However, select the appropriate field. For example, you may see **NumberOfEmployees** instead of **Employees** in the Target object Account.



16. To validate the mapping, click **Validate Mapping**.

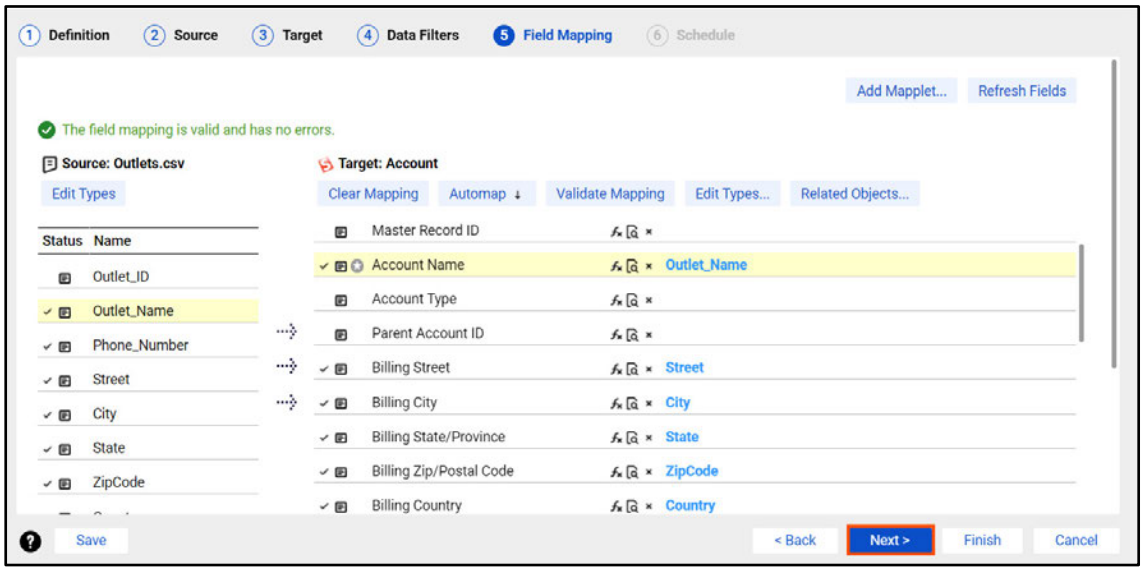


Note: If the validation fails, click **Refresh Fields,** and try to validate the mapping again.



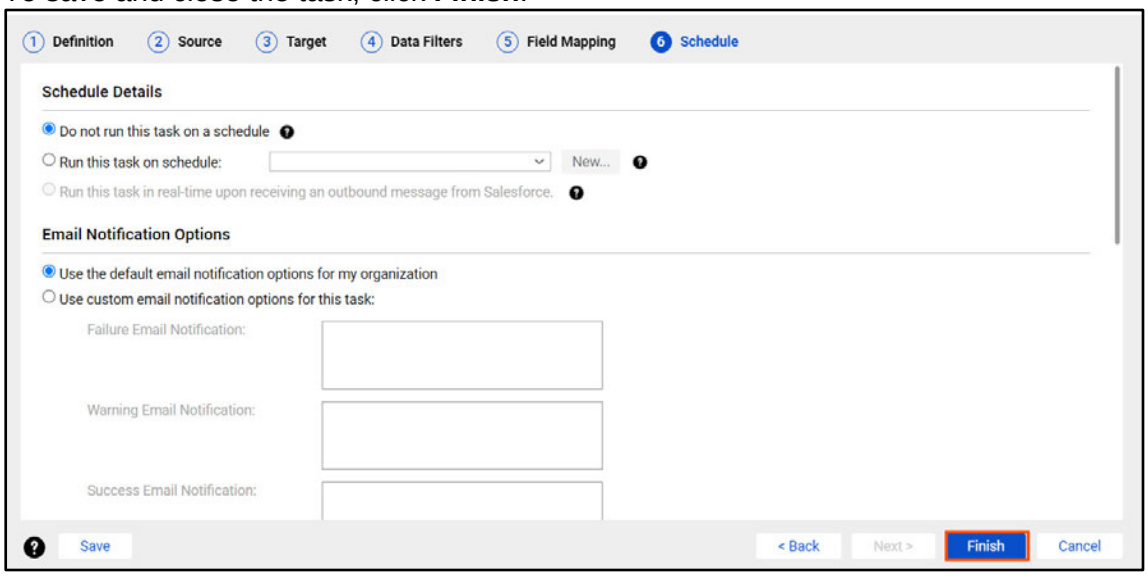
The **Refresh Fields** option updates the data integration cache and shows the latest field attributes.

17. Click **Next**.



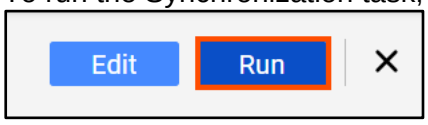
In this lab, you will not define a Schedule.

18. To save and close the task, click **Finish**.



When you click Finish, the Synchronization Task Asset appears in the navigation pane and displays the task details on the page.

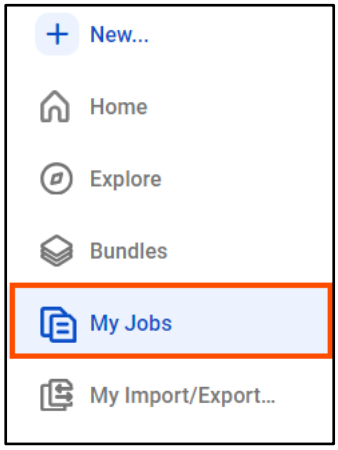
19. To run the Synchronization task, click **Run**.



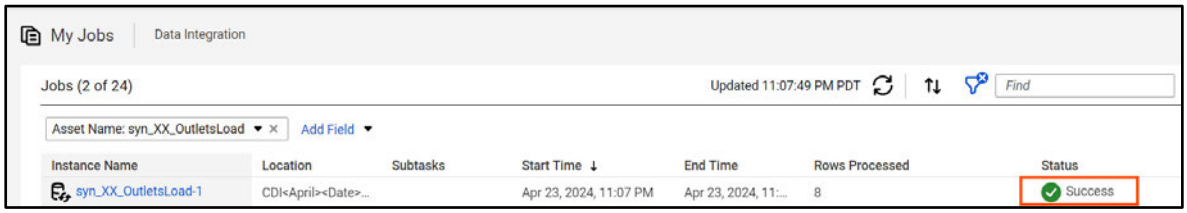
Important Note: If you want to re-run a task, delete the records from Salesforce inserted by the task in the first run to avoid getting duplication errors.

Monitor the Synchronization Task

20. To monitor the task, from the navigation pane, click **My Jobs**.

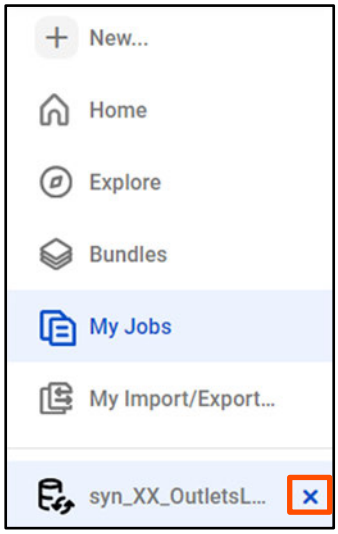


21. When the task completes, the status changes to **Success**.



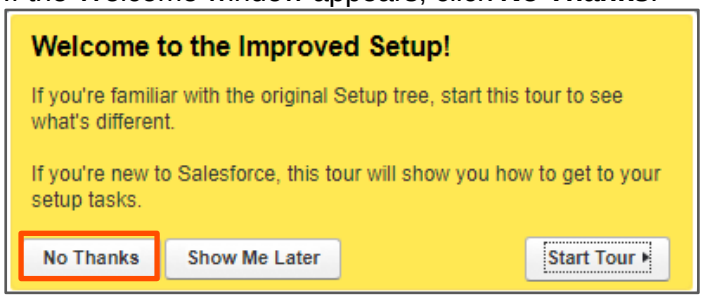
Note: You can use the button to refresh the page if the status does not change automatically. If you see the Job Status as Warning instead of Success, it means that you have duplicate records in your Salesforce account. In this case, do a Mass Deletion of all records in the Accounts object (in Salesforce) and re-run the job in DI.

22. Close the asset from the navigation pane.

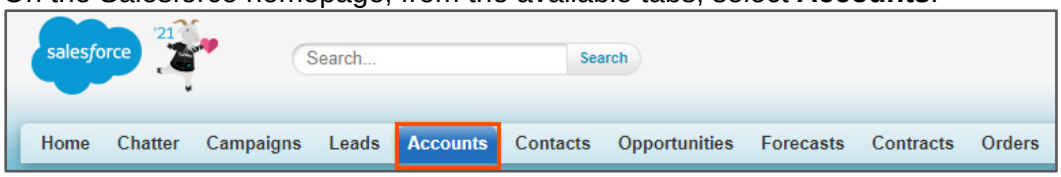


Verify the Results

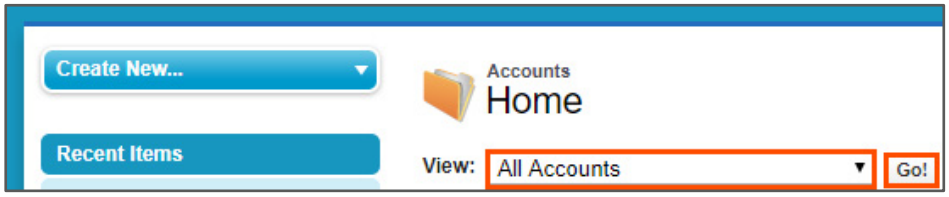
23. Log in to the Salesforce Developer account using your credentials.
24. If the Welcome window appears, click **No Thanks**.



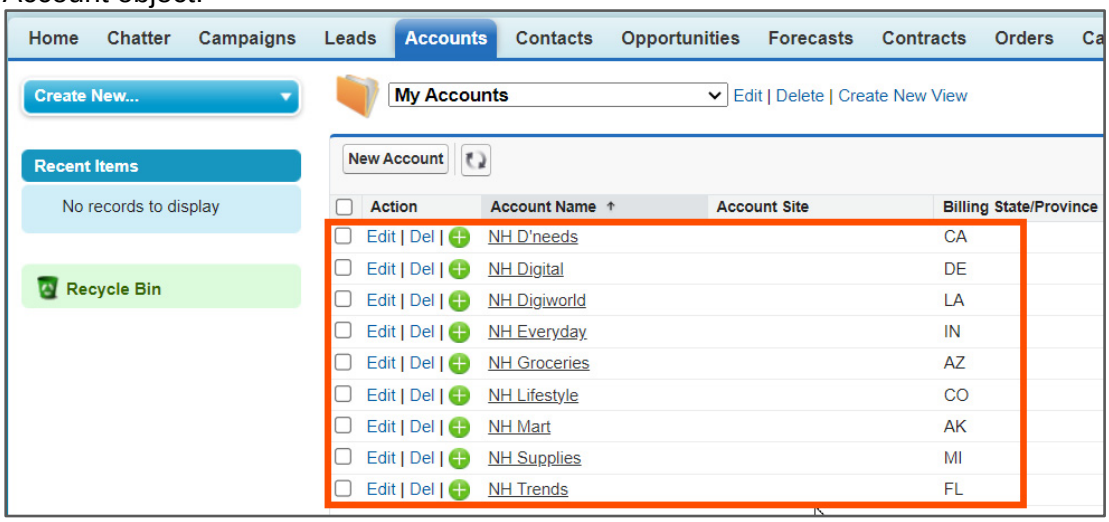
25. On the Salesforce homepage, from the available tabs, select **Accounts**.



26. From the drop-down, select **All Accounts**.
27. Click **Go!**



28. Verify that all the Accounts in the source file (Outlets.csv) are now in the Salesforce Account object.



This concludes the lab.

Module 3: Synchronization Task

Lab 3-4: Using Filter, Expression, and Lookup in a Synchronization Task

Overview:

Data filters help you to fetch the required data from an object. The synchronization task uses the data filters to process the data as per the data filter assigned to that object. A lookup returns values based on a lookup condition. You can create a lookup condition based on the information in the source.

Objective:

- Create data filter
- Create field expressions
- Use a lookup to relate outlet name and account name

Scenario:

Now that Ruby has inserted the data in the Salesforce Accounts object, she wants to load employee data on Salesforce as well.

However, the format of employee data is not compatible with Salesforce. So, John informs Ruby that he can use various features of the IICS synchronization task to transform the data and load it into Salesforce.

In this lab, John will use the data filters to skip loading the sales department's employee data. He will also use the field expression to separate first and last names and to perform a lookup in Salesforce using the account name.

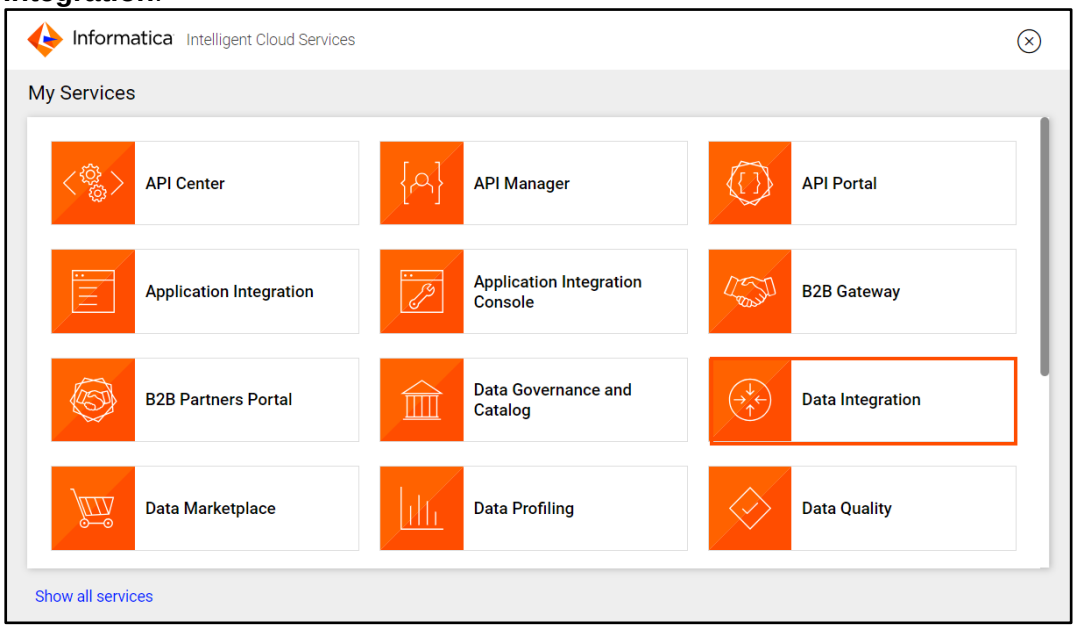
Duration:

20 minutes

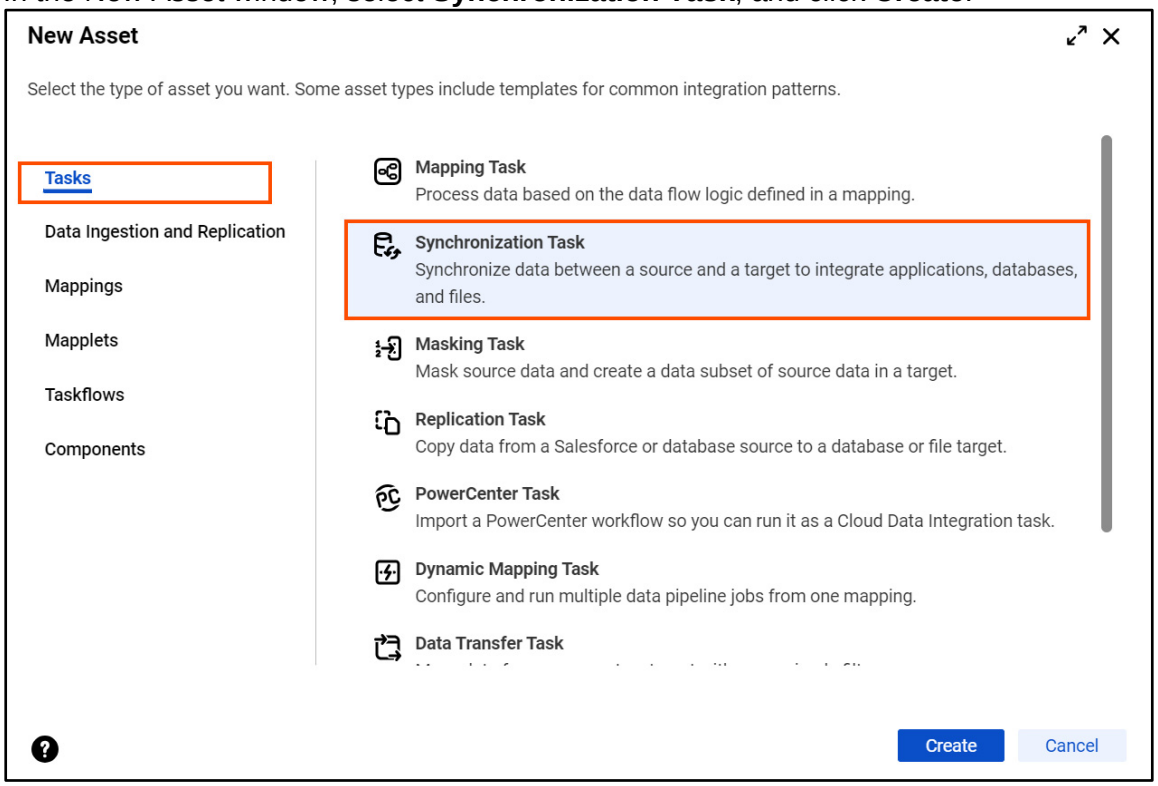
Tasks:

Create a Synchronization Task in IICS

1. If navigated away, login into IICS and from the **My Services** window, select **Data Integration**.

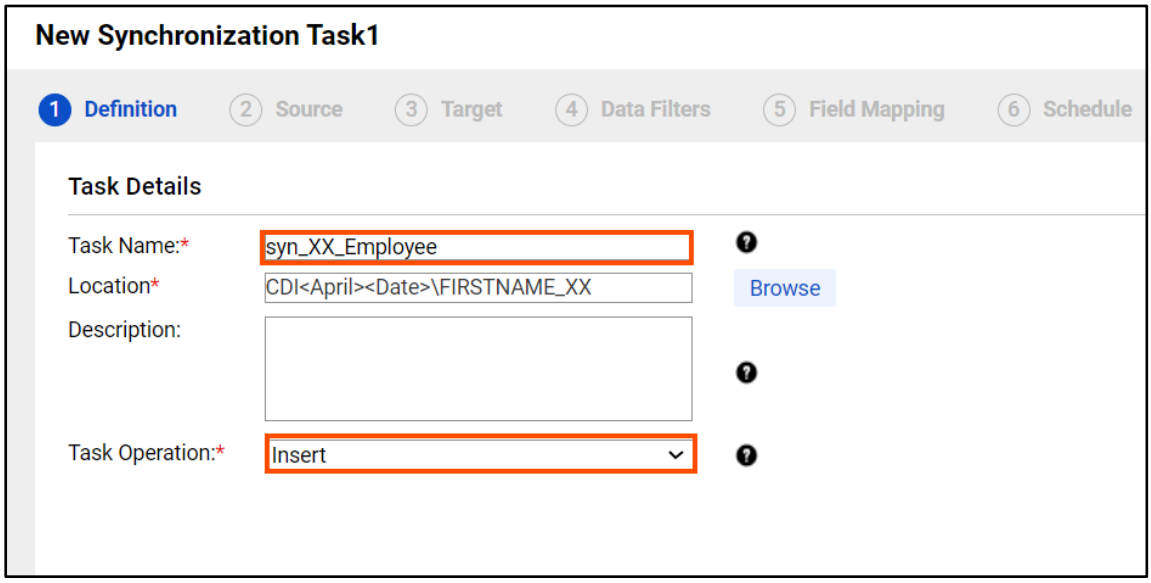


2. To create a new asset, from the navigation pane, select **New**.
3. In the New Asset window, select **Synchronization Task**, and click **Create**.



Specify Definition Information

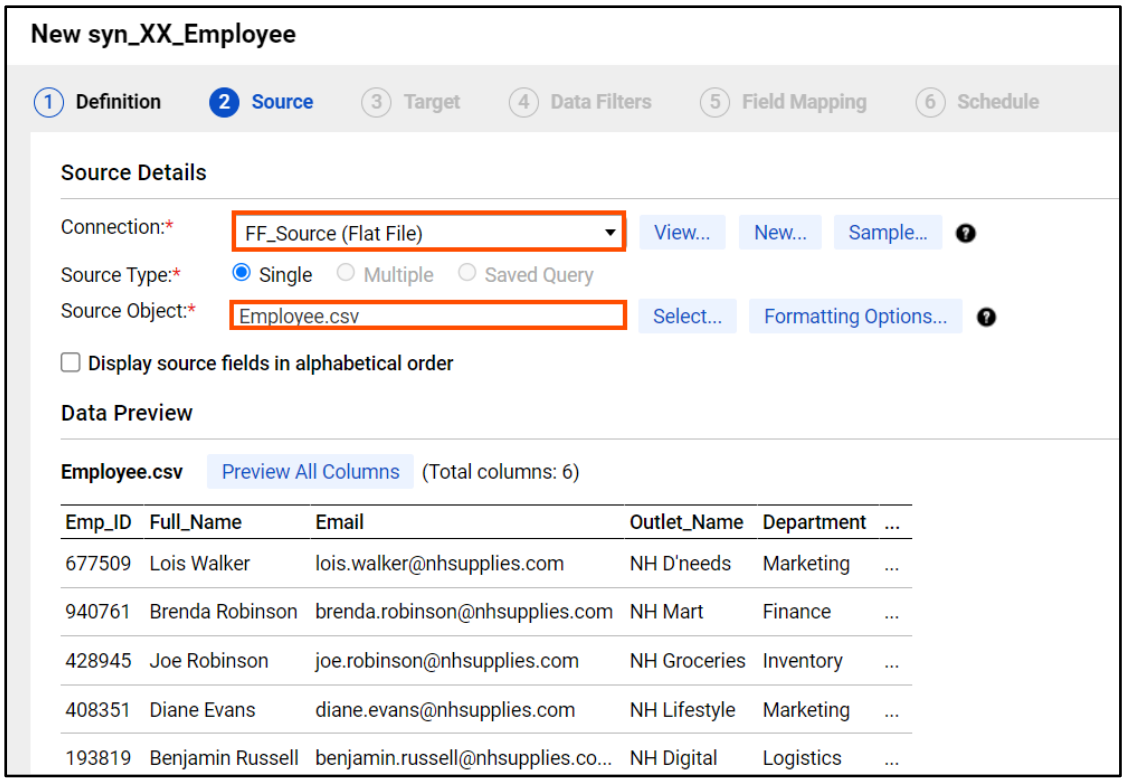
4. In the Task Name field, enter **syn_XX_Employee**. Replace **XX** with your DOB.
5. From the Task Operation drop-down, select **Insert**.



6. Verify that the asset Location is your working directory.
7. Click **Next**.

Specify Source Information

8. From the Connection drop-down, select **FF_Source**.
9. From the Source Object field, click **Select** and choose **Employee.csv**.

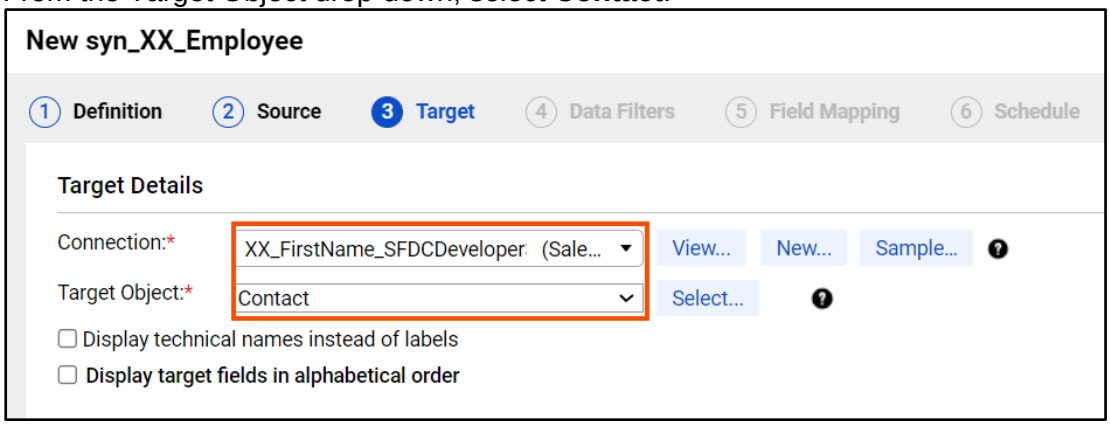


Note: The Data Preview section appears. It shows the first ten rows of the first five columns in the object and displays the total number of columns in the object.

10. Click **Next**.

Specify Target Information

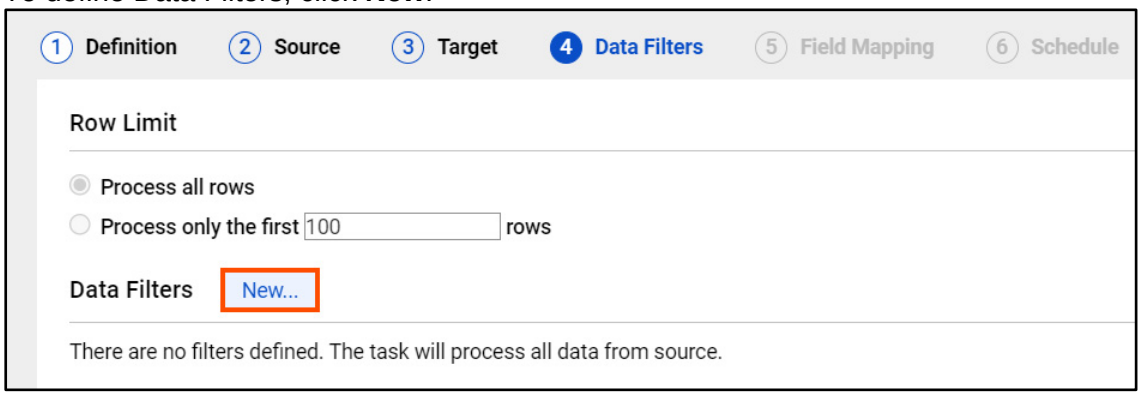
11. From the Connection drop-down, select your Salesforce connection (present in the format **XX_FirstName_SFDCDeveloper**).
12. From the Target Object drop-down, select **Contact**.



13. Click **Next**.

Define Data Filters to Skip Contacts with Department as Sales

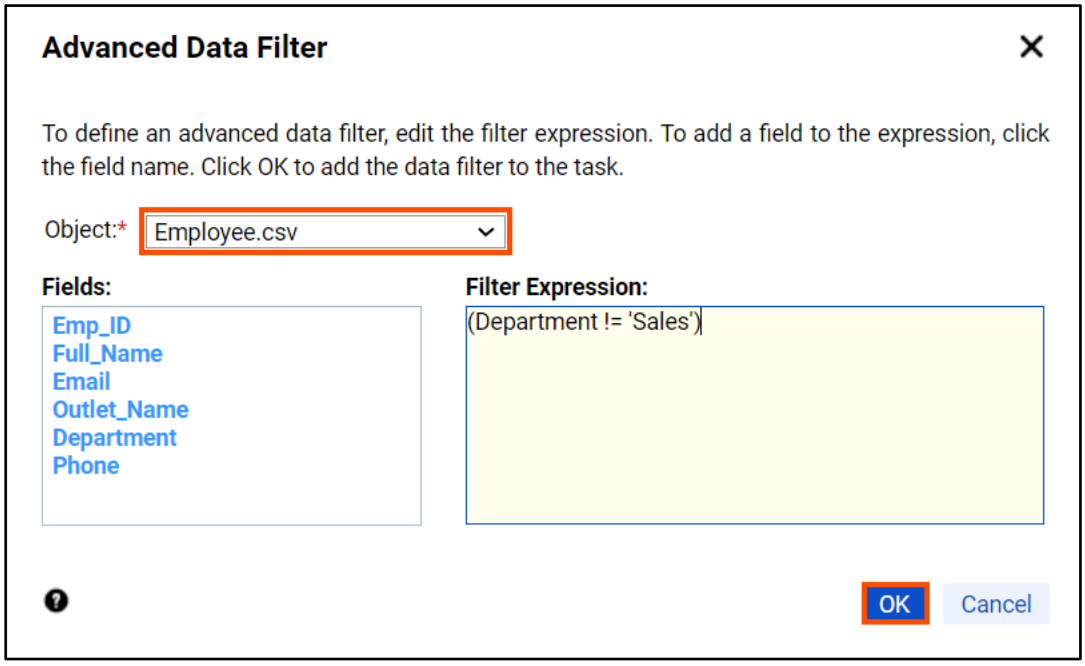
14. To define Data Filters, click **New**.



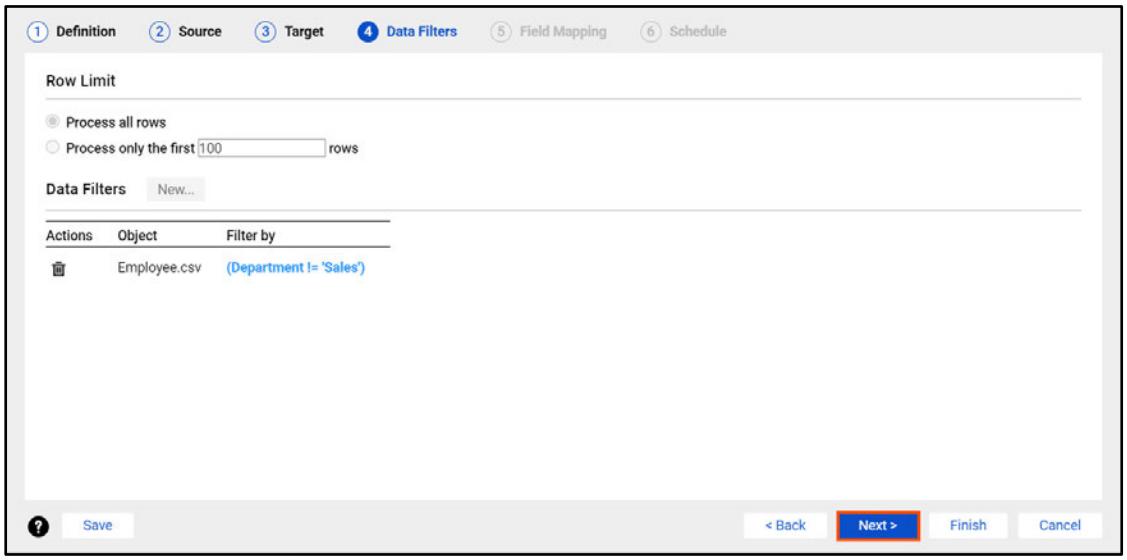
Note: The Advanced Data Filter window appears. You cannot apply a simple filter if the source connection is a flat-file connection.

15. From the Object drop-down, select **Employee.csv**.
16. In the Filter Expression field, enter the following expression. This expression filters out all the entries that have Department as Sales in the Employee.csv file.
(Department != 'Sales')

17. Click **OK**.



18. Click **Next**.

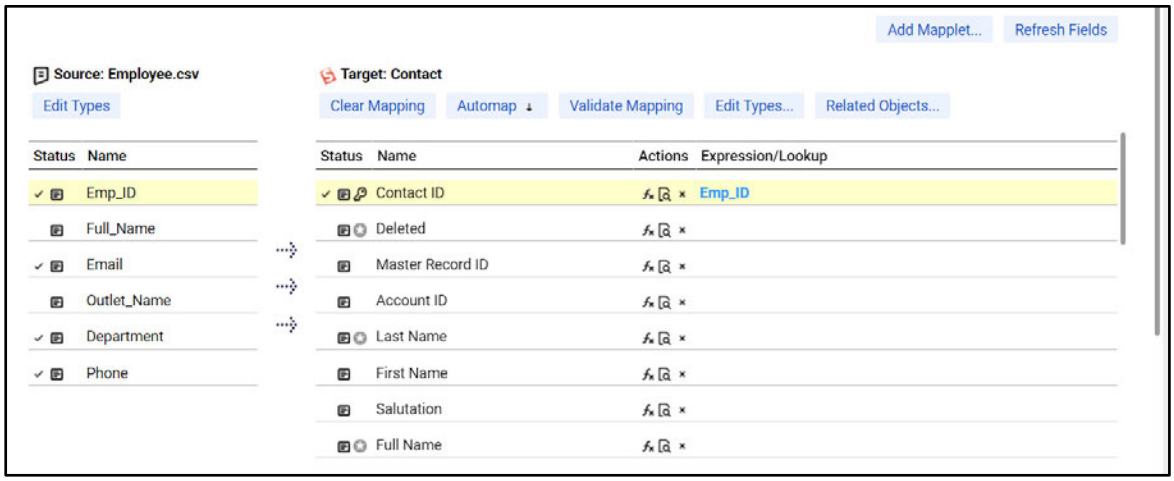


Define Field Mappings

19. Map the Source field with the Target field, as shown in the table below:

Source Field Name	Target Field Name
Emp_ID	Contact ID (or ID, whichever you see)
Email	Email
Department	Department
Phone	Business Phone (or Phone, whichever you see)

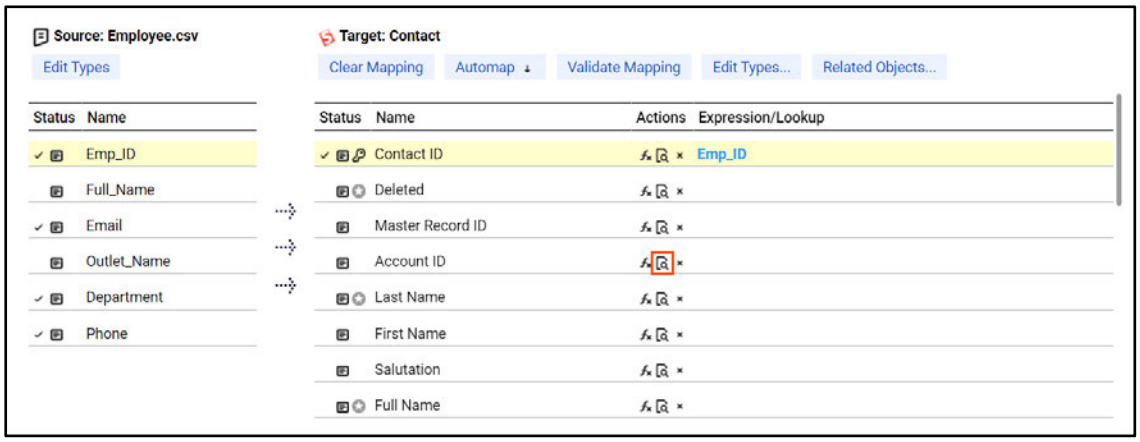
Note: For this lab, do not map **Full_Name** and **Outlet_Name** fields. Some of the fields might be mapped automatically. For fields that are already mapped, do not map them again.



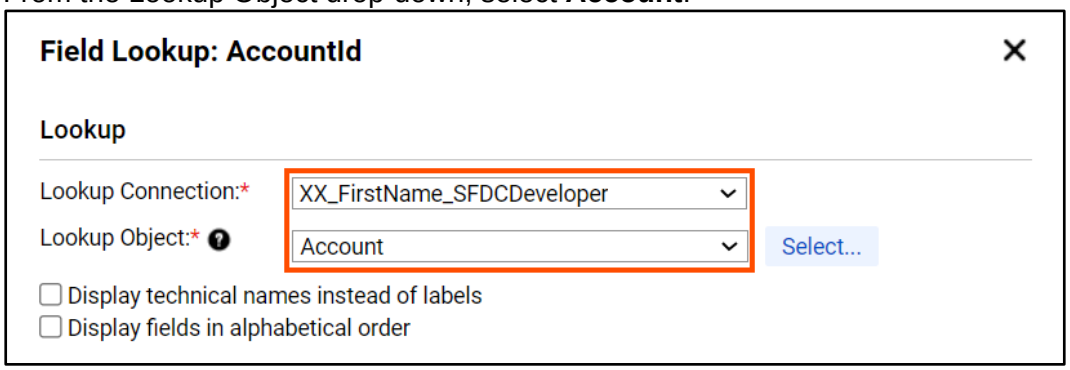
Add a Lookup Condition

To get the Account ID from the Salesforce Account object, create a lookup in the Field Mapping page of the synchronization task wizard. The lookup returns values based on a lookup condition. Here, the lookup condition returns the Account ID for the fields where Outlet_Name is equal to Account Name.

20. To define a lookup condition, from the Target: Contact section, select for **Account ID**.

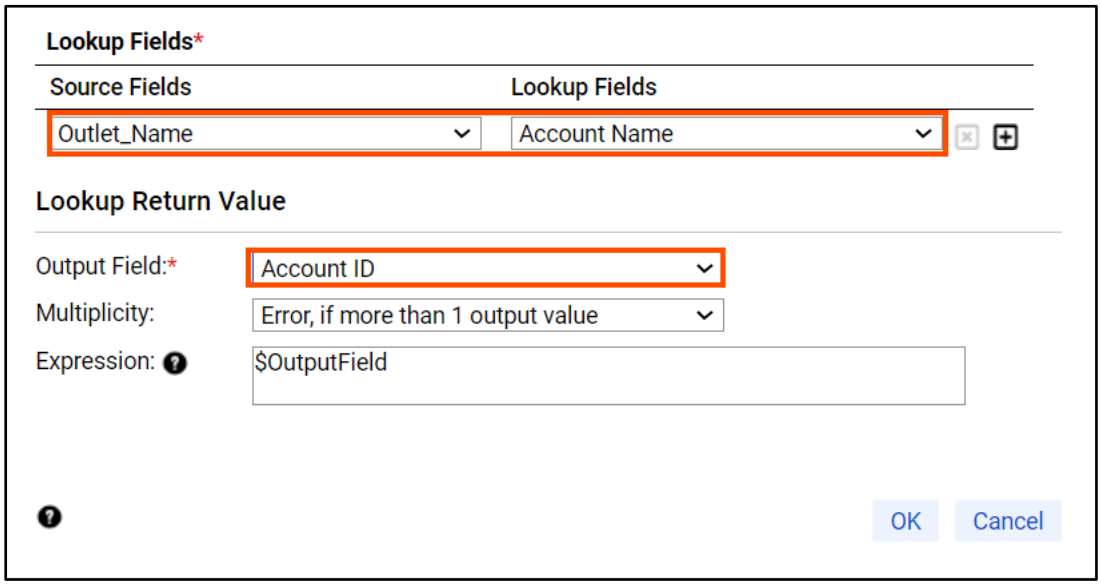


21. From the Lookup Connection drop-down, select **XX_FirstName_SFDCDeveloper** (your Salesforce connection).
22. From the Lookup Object drop-down, select **Account**.



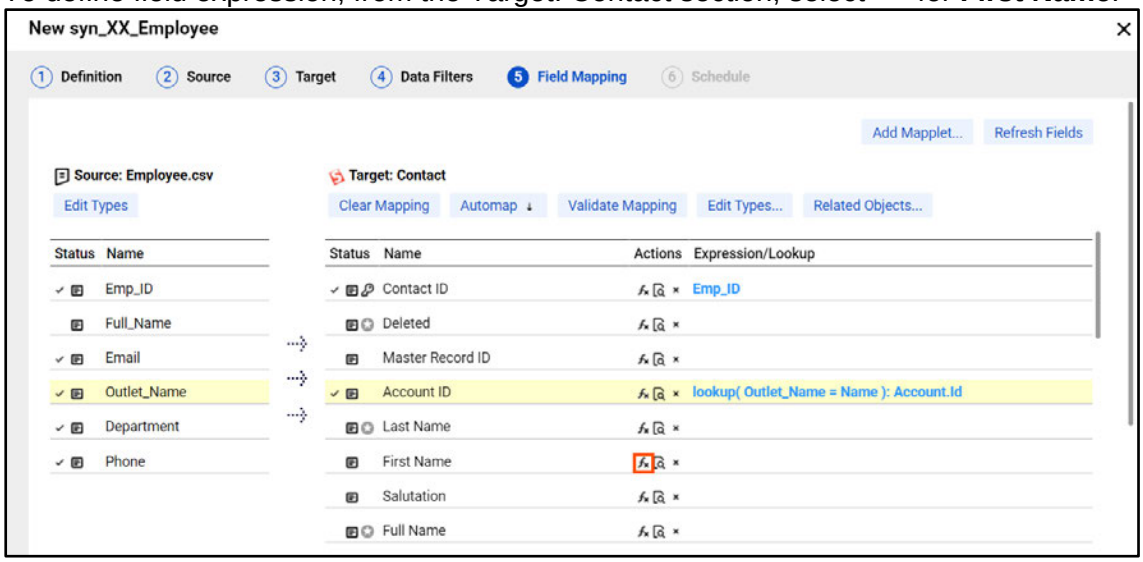
23. In the Source Fields drop-down, select **Outlet_Name**.
24. In the Lookup Fields drop-down, select **Account Name**.

25. In the Lookup Return Value section, from the Output Field drop-down, select **Account ID** and click **OK**.



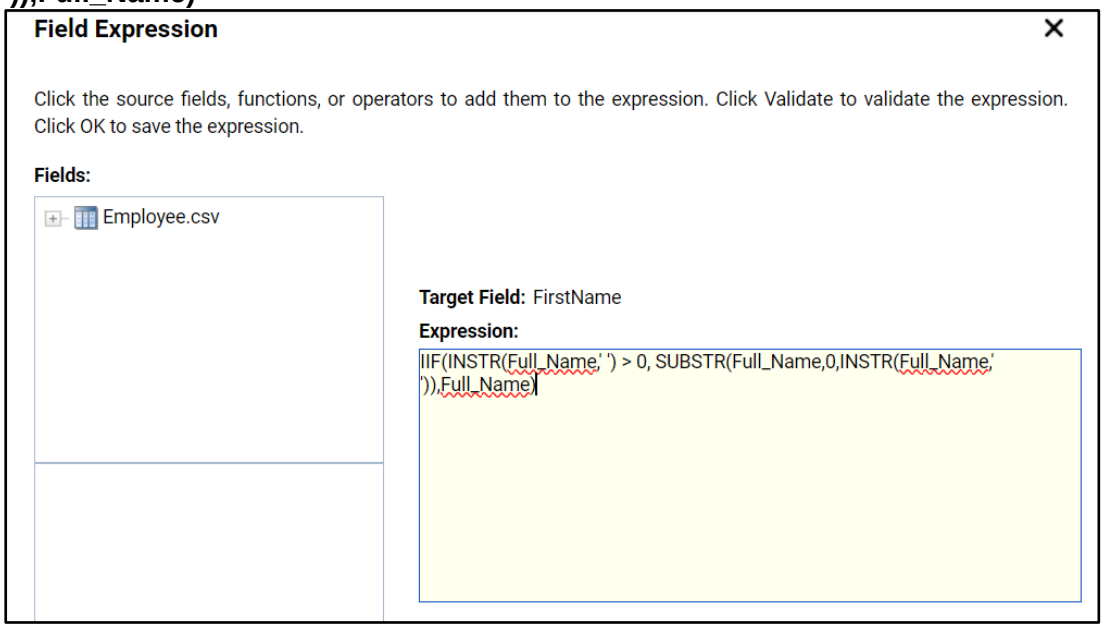
Define Field Expression to Split Full Name into First Name and Last Name

26. To define field expression, from the Target: Contact section, select for **First Name**.

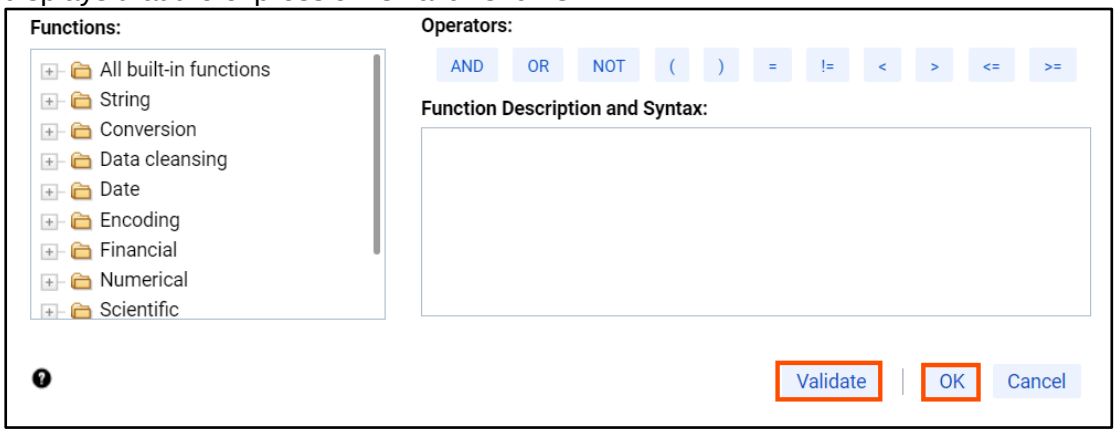


27. In the **Expression** field, enter the following expression:
IIF(INSTR(Full_Name,' ') > 0, SUBSTR(Full_Name,0,INSTR(Full_Name,'

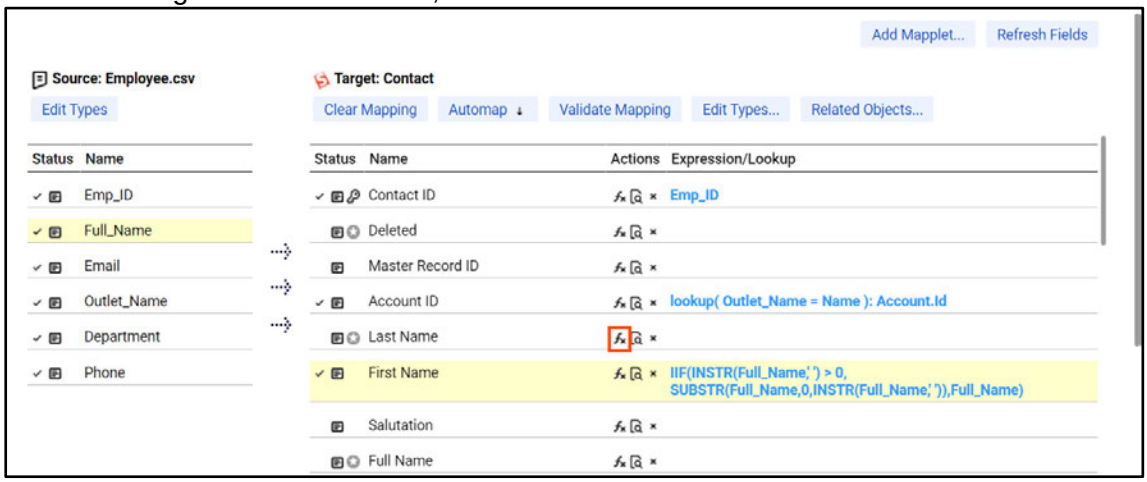
')),Full_Name)



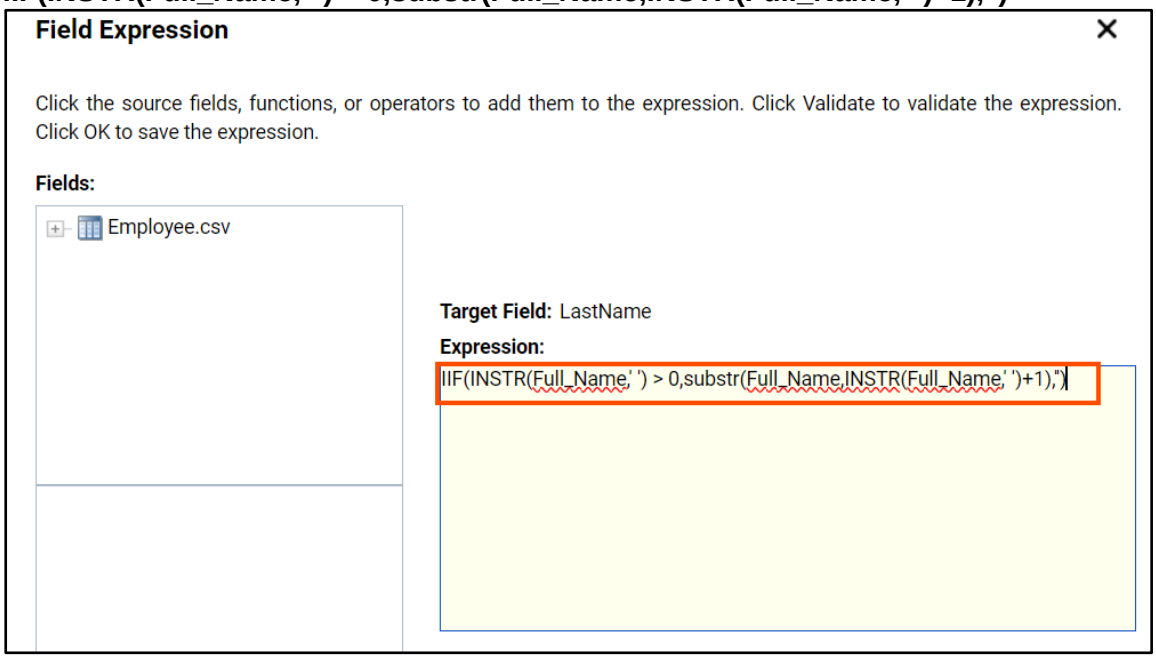
28. To validate the expression, scroll down, and click **Validate**. At the top of the window, it displays that the expression is valid. Click **OK**.



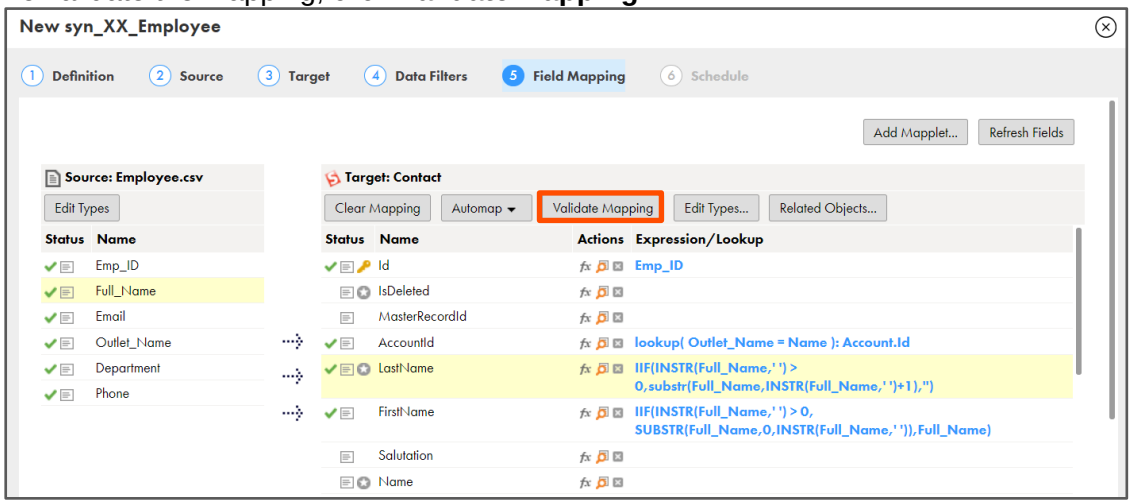
29. From the Target: Contact section, select for **Last Name**.



30. In the **Expression** field, enter the following expression:
IIF(INSTR(Full_Name,' ') > 0,substr(Full_Name,INSTR(Full_Name,' ')+1),'')



31. **Validate** the expression and click **OK**.
32. To validate the mapping, click **Validate Mapping**.

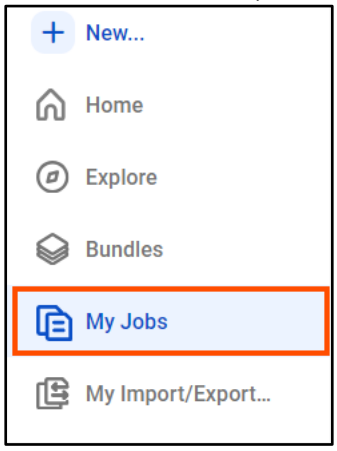


33. To save and close the task, click **Finish**.
Note: When you click Finish, the Synchronization Task Asset appears in the navigation pane and displays the task details on the page.
34. Run the synchronization task.

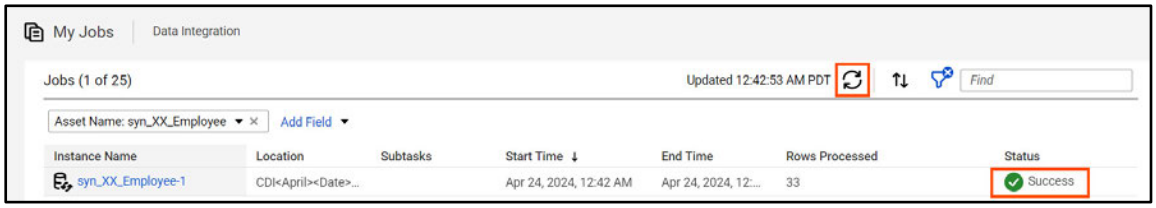
Important Note: If you want to re-run a task, delete the records inserted by the task in the first run to avoid getting duplication errors.

Monitor the Synchronization Task

35. To monitor the task, from the navigation pane, click **My Jobs**.



36. When the task completes, the status changes to **Success**.

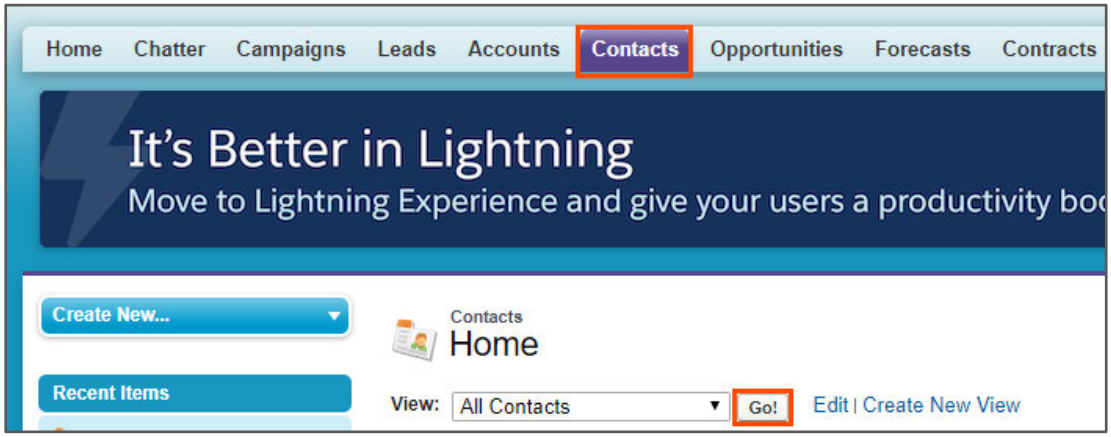


Note: Verify that 33 rows are processed by the task. You can use the refresh option if the status of the task does not change automatically.

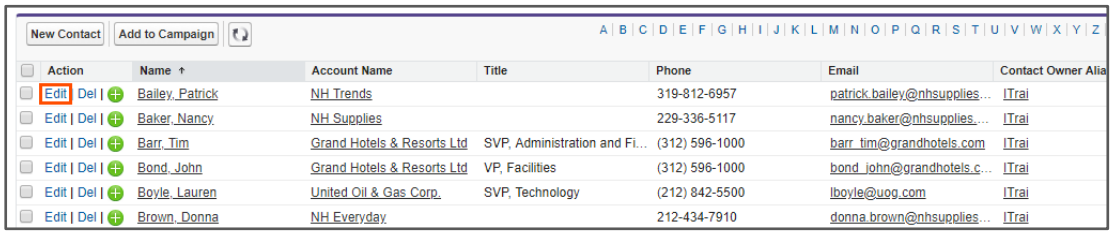
37. Close the asset from the navigation pane.

Verify the Results

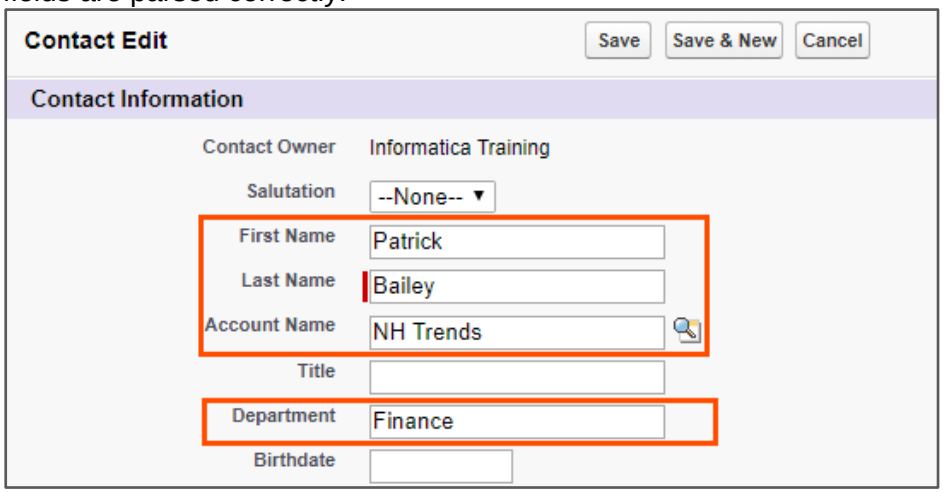
38. Log in to your Salesforce Developer account using your credentials.
39. To view the new account records that were inserted, click **Contacts**.
    Note: Verify that the View field is set to **New This Week**.
40. Click **Go!**.



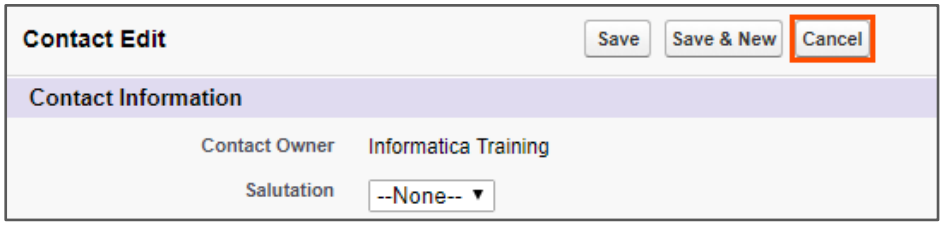
41. To verify that the First Name and Last Name are parsed correctly, select a contact and click **Edit**.



42. Observe that the entries in the First Name, Last Name, Account Name, and Department fields are parsed correctly.



43. Click **Cancel**.



Note: If you do not find the Cancel button at the top of the page, you can find it at the bottom.

Also, you can perform a random check among the records loaded into Salesforce to verify that there is no record with the **Department** category as **Sales** (as you have added the Data filter to not load the records that belong to the Sales department).

This concludes the lab.

Module 3: Synchronization Task

Lab 3-5: Using Saved Query in a Synchronization Task

Overview:

The Synchronization task synchronizes data between a source and a target. The Data Synchronization application supports Insert, Update, Upsert, and Delete operations.

Objective:

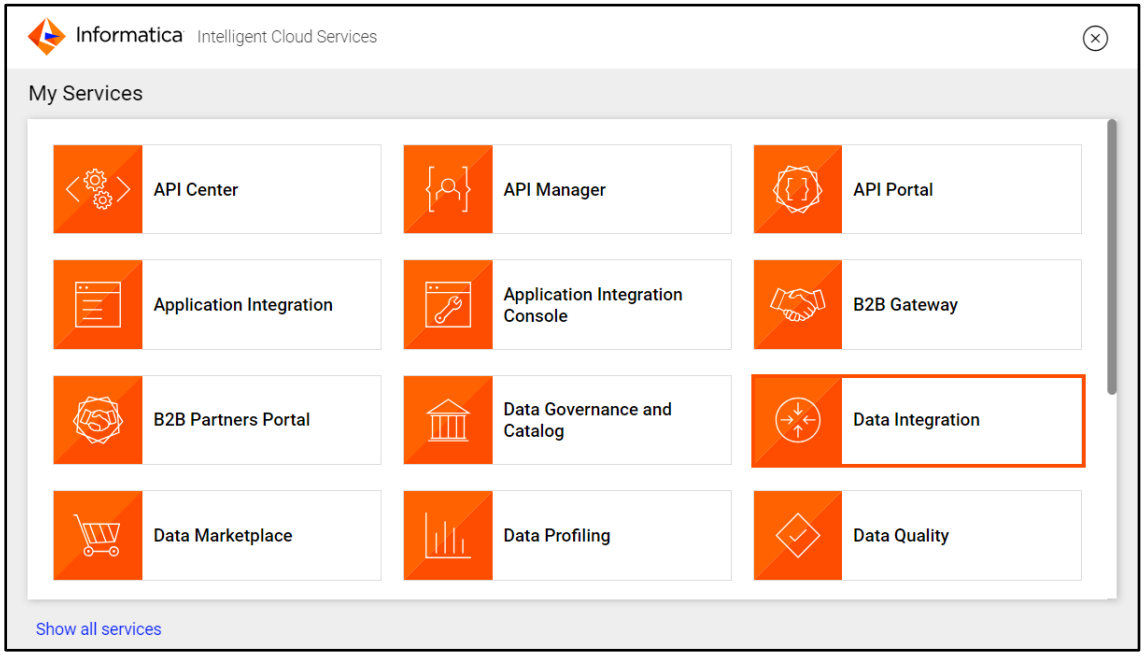
- Create a saved query
- Use the saved query in the Synchronization task

Duration:
10 minutes

Tasks

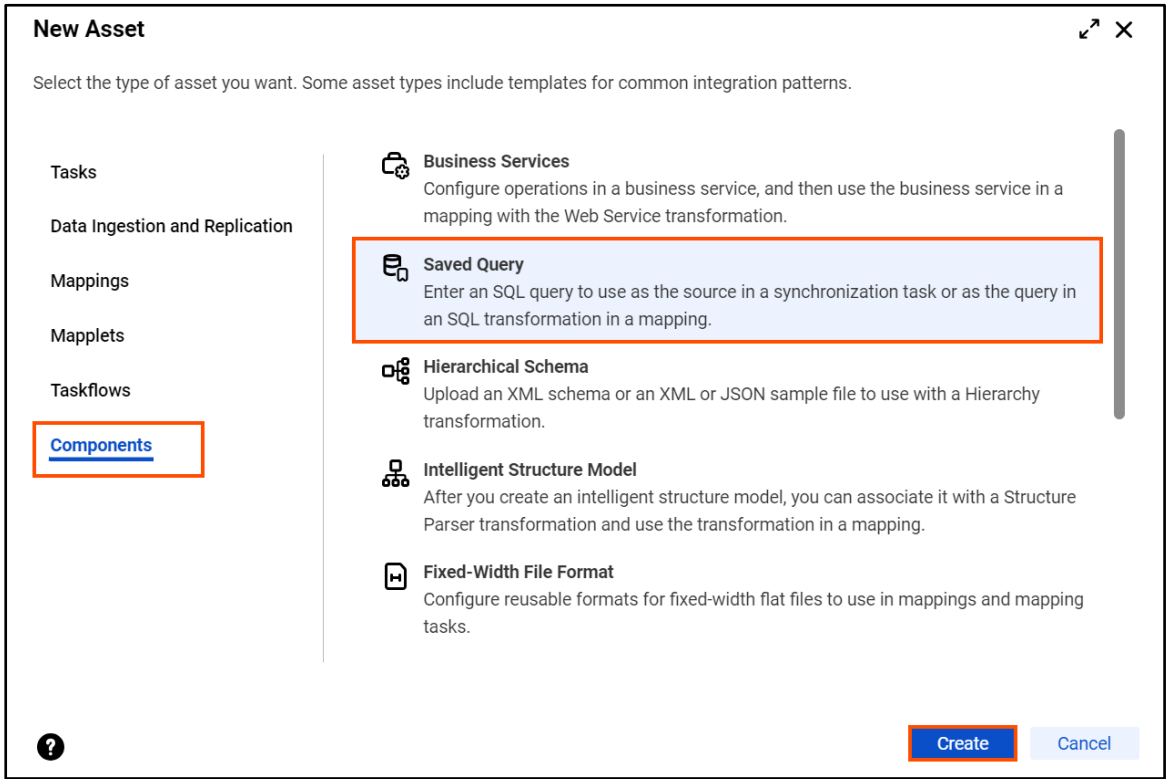
Create a saved query

1. Login into IICS and from the **My Services** window, select **Data Integration**.



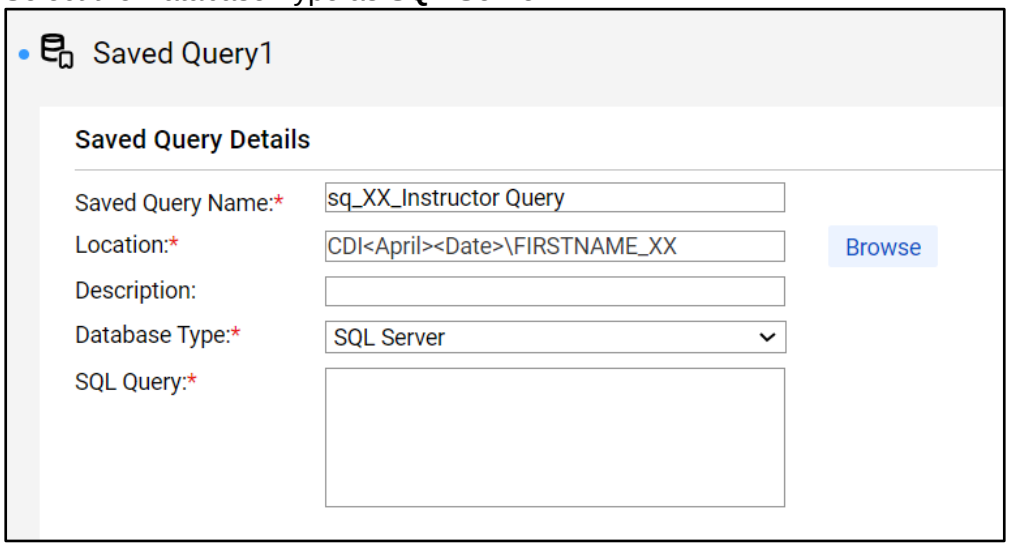
2. To create a new saved query, from the navigation pane, select **New**.

3. From the **Components** tab, select **Saved Query**.
4. Click **Create**.



5. In the saved Query Name enter the name as **sq_XX_Instructor Query**.

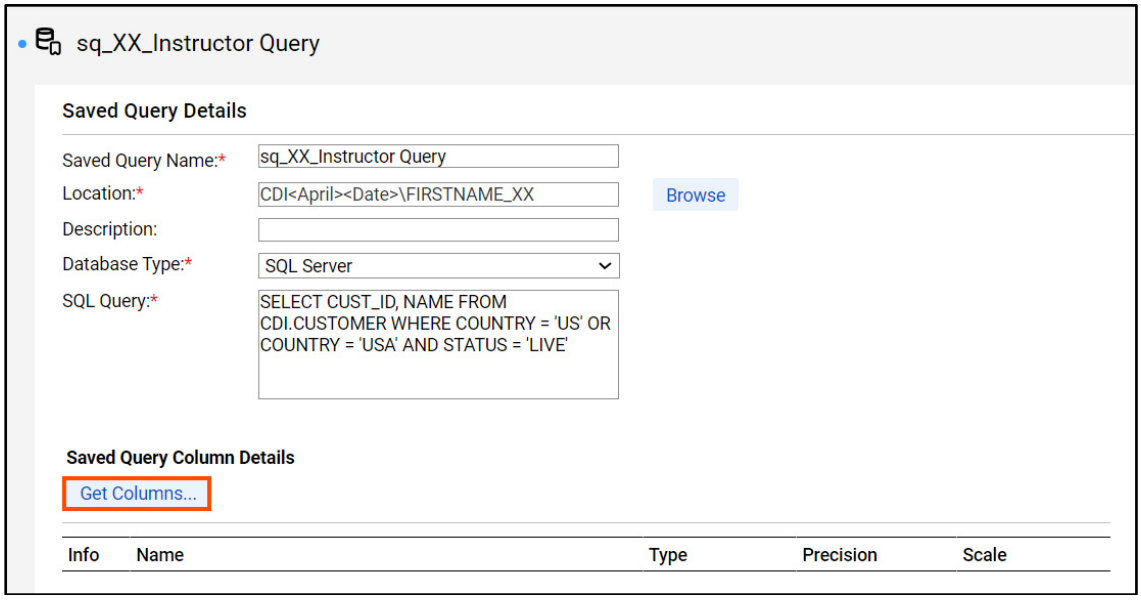
   Note: Replace **XX** with your DOB.
6. Use the Browse option to save the asset in your working directory.
7. Select the Database Type as **SQL Server**.



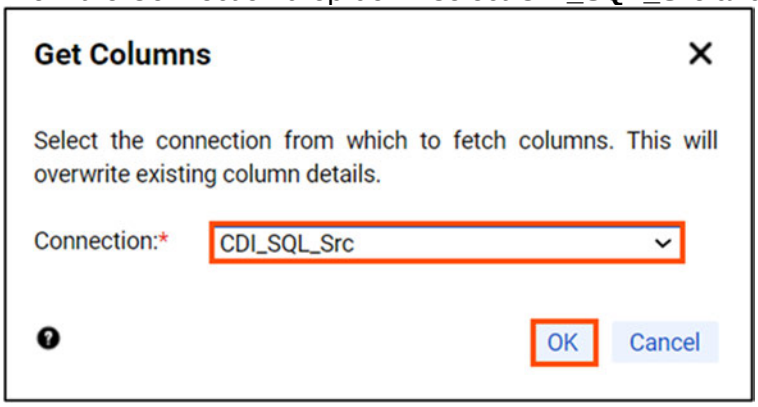
8. In the SQL Query enter the Query as mentioned below:

   SELECT CUST_ID, NAME FROM CDI.CUSTOMER WHERE COUNTRY = 'US' OR COUNTRY = 'USA' AND STATUS = 'LIVE'

9. Click on the **Get Columns** tab.



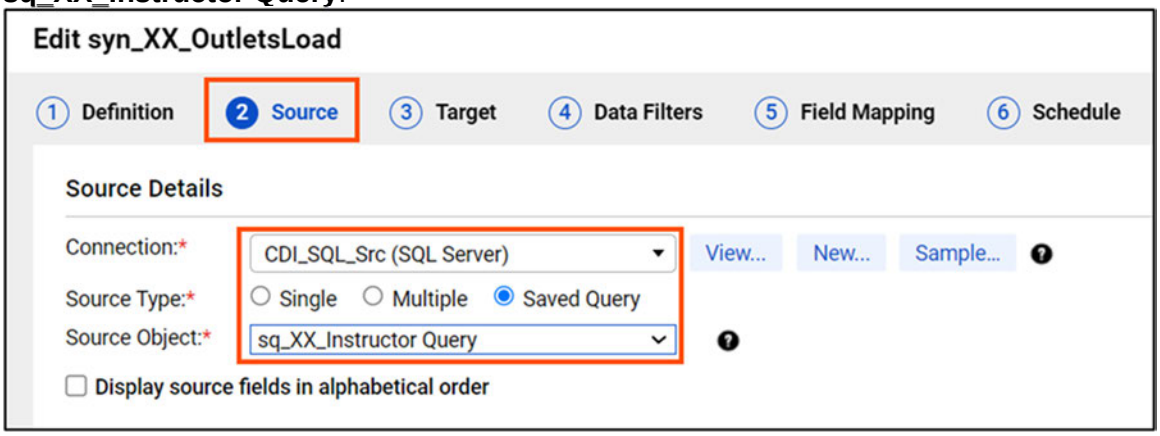
10. From the Connection drop-down select **CDI_SQL_Src** and click **OK**.



11. **Save** the SQL Query.

12. Now, in the Explore folder, in your directory, open the existing Synchronization task **syn_XX_OutletsLoad**.
13. To modify the task, click **Edit**.

14. Go to the Source tab and in the source details field, select the Connection as **CDI_SQL_Src**.
15. Select the Source Type as **Saved Query**.
16. From the source object property, select your saved query created earlier, that is, **sq_XX_Instructor Query**.

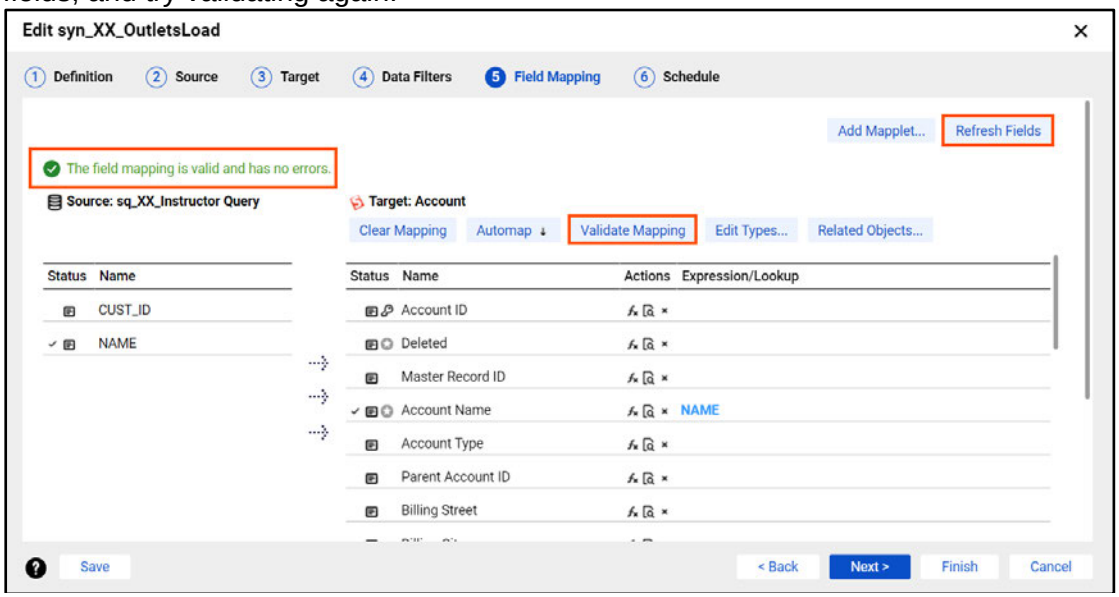


17. Navigate to the Field Mapping tab.
18. Clear the existing fields mapping using the **Clear Mapping** option and map the fields as shown below:

Source Field Name	Target Field Name
Name	Account Name (or Name, whichever you see)

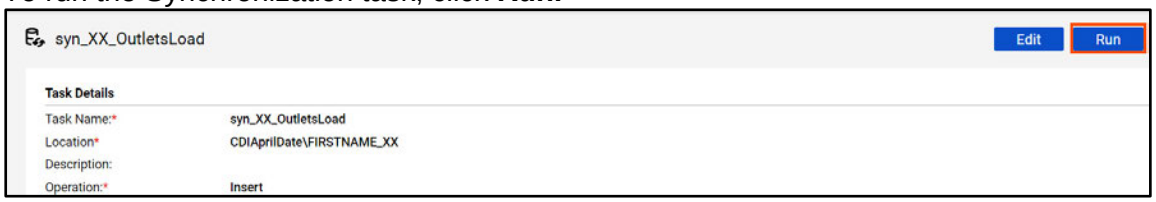
19. Validate the mapping.

    Note: If you see an Internal Error after validating the mapping, Refresh the mappings fields, and try validating again.



20. Click **Finish**.

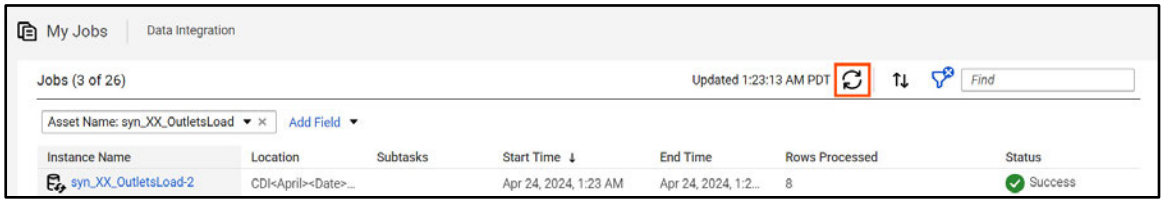
21. To run the Synchronization task, click **Run.**



Monitor the Synchronization Task

22. To monitor the task, from the navigation pane, click **My Jobs**.

23. When the task completes, the status changes to **Success**.

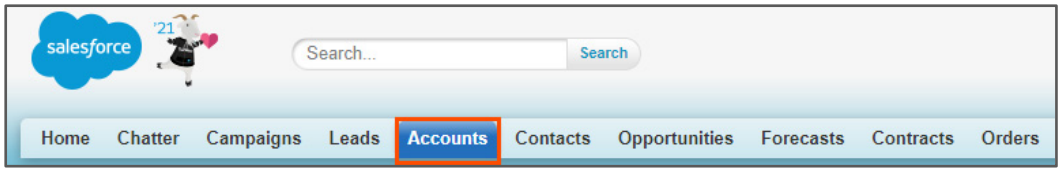


Note: You can use the icon to refresh the page if the status does not change automatically.

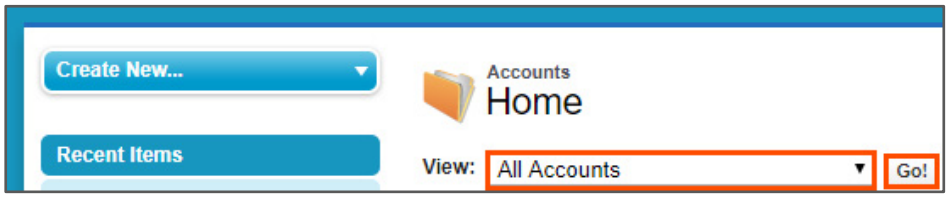
24. Close all the assets from the navigation pane.

Verify the Results

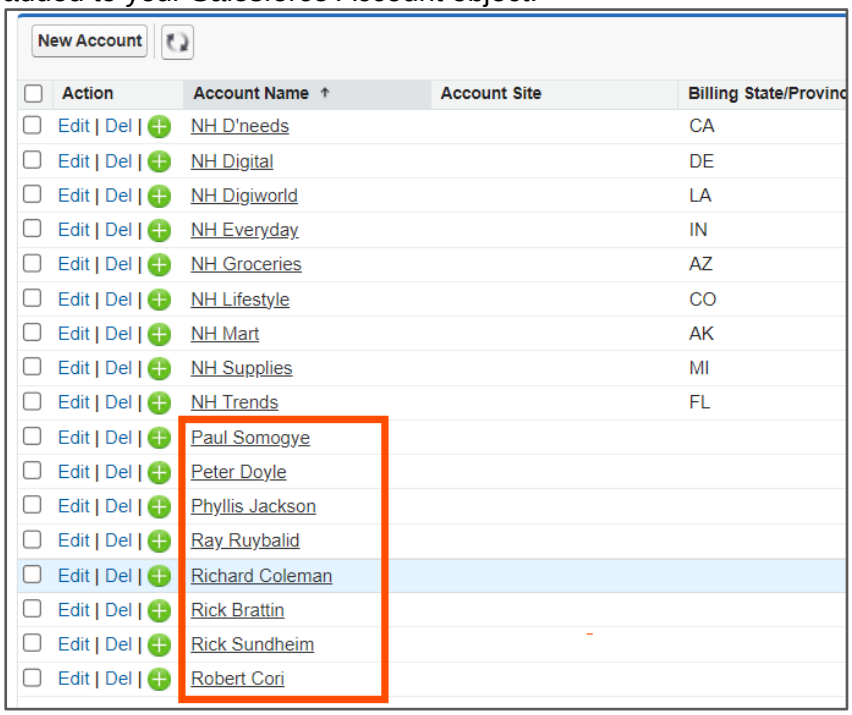
25. Log in to the Salesforce Developer account using your credentials.
26. On the Salesforce homepage, from the available tabs, select **Accounts**.



27. From the drop-down, select **All Accounts**.
28. Click **Go!**.



29. Verify that all the Accounts in the source file (SQL server database tables) are now added to your Salesforce Account object.



Note: The number of records shown in the screenshot above may vary depending on the records present in your Salesforce account.

This concludes the lab.

Module 3: Synchronization Task

Lab 3-6: Using Upsert Operation in Synchronization Task

Overview:

A Synchronization Task can be configured with the Task Operation property set to Upsert (Update or Insert) for a relational target table that has a primary key. This enables new records to be inserted while records that already exist in the target (same primary key value) will generate an update operation that can change the data in the target to match the source data for that record. This feature is available in Salesforce targets configured with an External ID.

Objectives:

- Create a Synchronization task with the Task Operation property set to Upsert
- Modify the Field Mapping after the source file is changed
- Observe how the upsert logic impacts the target data

Scenario:

A database table is set up to hold customer records with individual customer attributes such as address fields, phone numbers, and email. A CUST_ID field uniquely identifies each customer, and since it is a primary key on the database, duplicate records (same CUST_ID value) cannot be inserted. We need to keep the data in this table updated according to changes in the source data, and we also want to avoid writing duplicate records that would be rejected by the target database. Finally, we need to capture new customer records when they appear in the data.

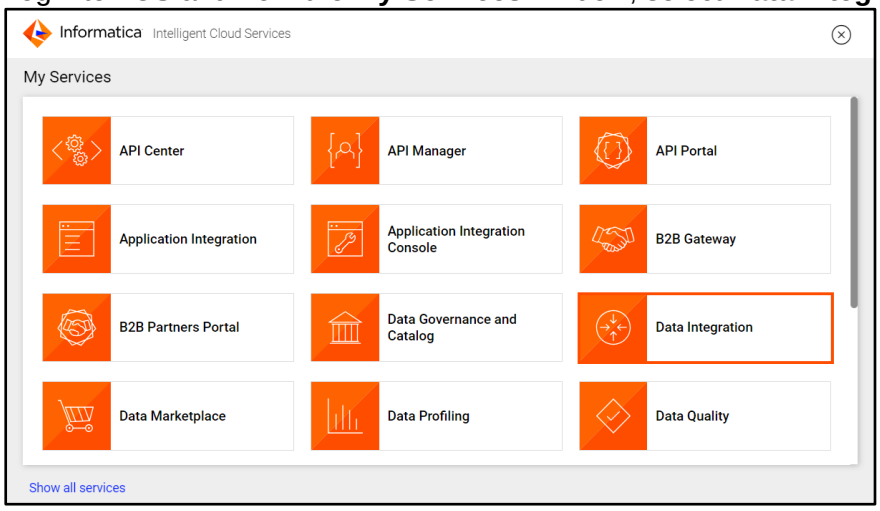
Duration:

20 minutes

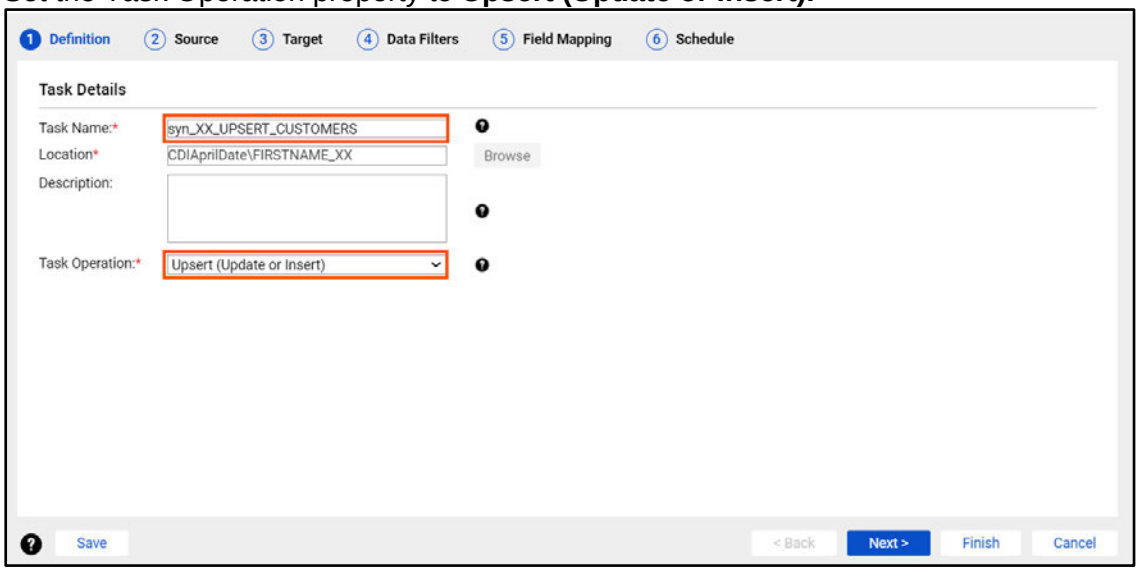
Tasks

Create Synchronization Task

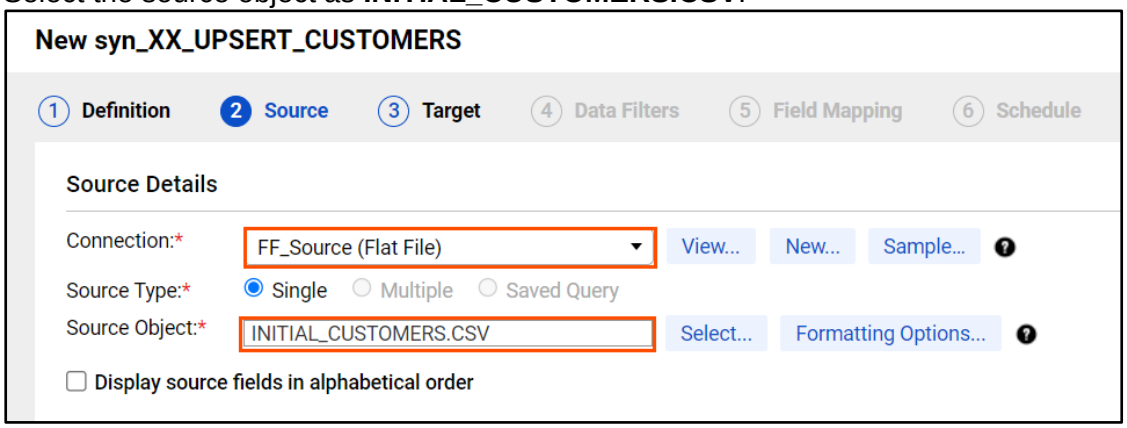
1. Login to IICS and from the **My Services** window, select **Data Integration**.



2. Create a new Synchronization Task and name it **syn_XX_UPSERT_CUSTOMERS**. Replace XX with your DOB.
3. Verify that you are saving the asset to your working folder.
4. Set the Task Operation property to **Upsert (Update or Insert).**

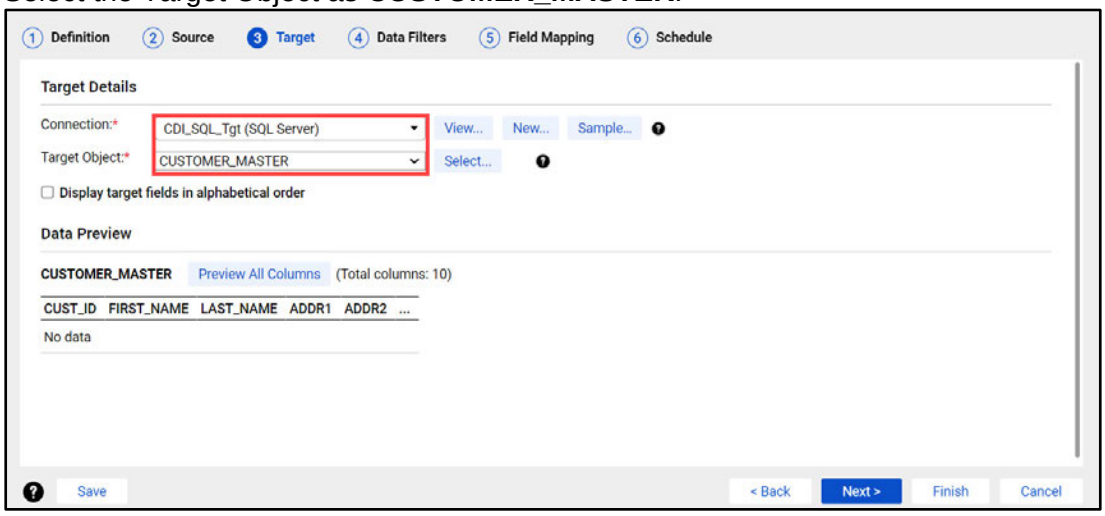


5. Click **Next**.
6. In the Source tab, set the Connection to **FF_Source**.
7. Select the source object as **INITIAL_CUSTOMERS.CSV**.



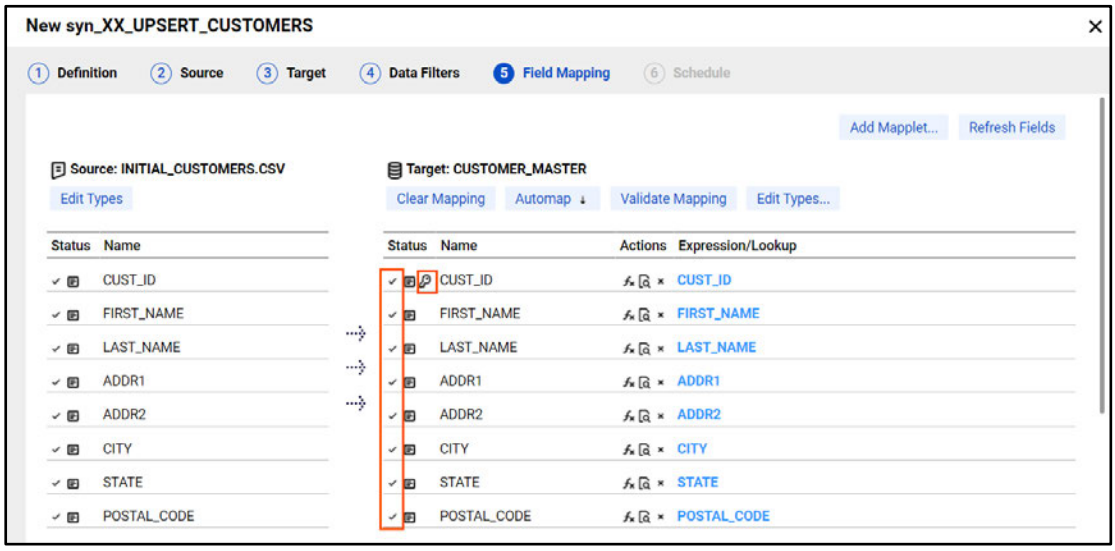
Note: The source file has ten records, each with a unique **CUST_ID**.

8. Click **Next**.
9. In the Target tab, set the Connection to your target SQL connection.
10. Select the Target Object as **CUSTOMER_MASTER**.

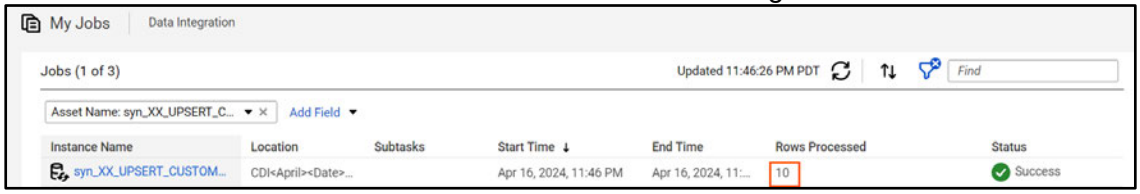


Note: The table currently has no data.

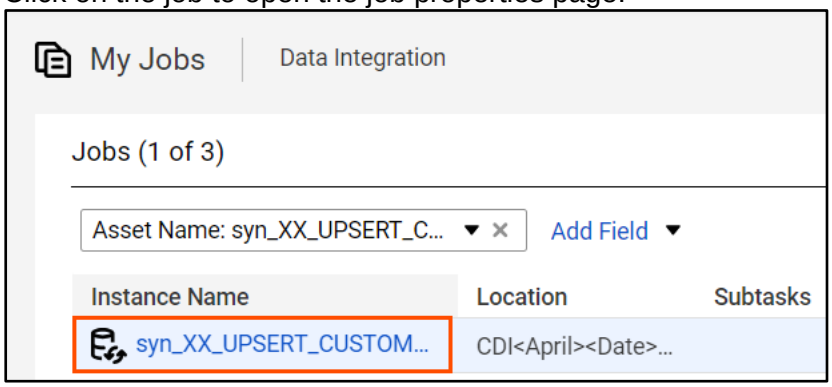
11. Click **Next**.
12. Skip the Data Filters tab. Click **Next**.
13. In the Field Mapping tab, note the CUST_ID is a primary key on the target table. Also, note each of the target fields has been mapped automatically because the target field names match the source fields.



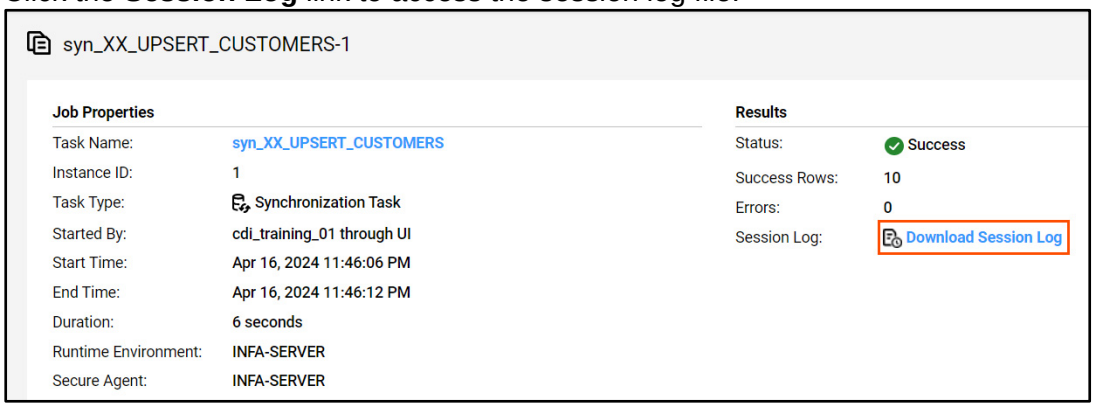
14. Click **Finish** to save the Synchronization Task.
15. **Run** the Synchronization Task and navigate to the **My Jobs** page.
16. Observe that all ten records have been inserted into the target table.



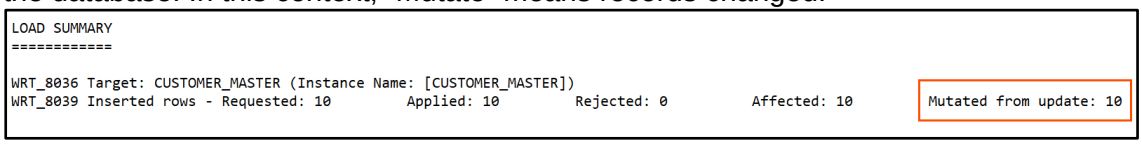
17. Click on the job to open the job properties page.



18. Click the **Session Log** link to access the session log file.

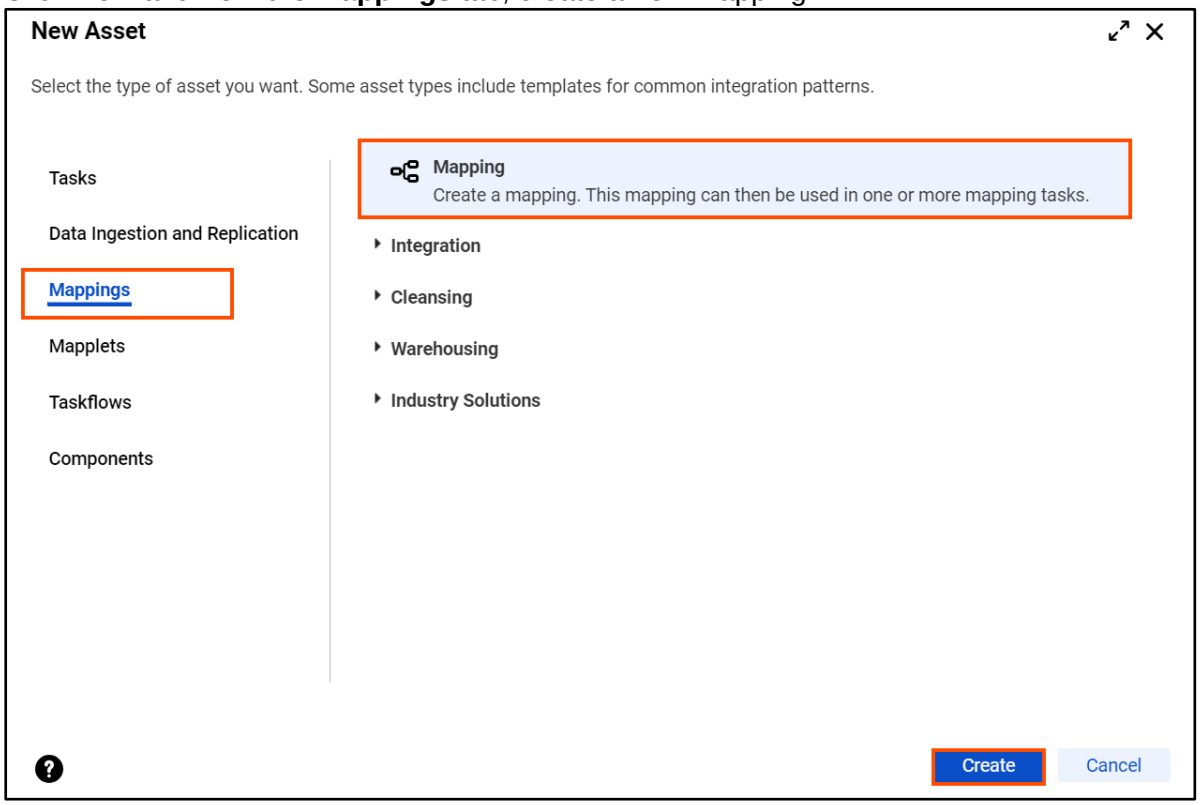


19. Open the downloaded log file and scroll down to the bottom to observe the **Load Summary**. Note that ten records have been inserted. Also, note that there is a value under "**Mutated from update**" on the right, even though no update records were sent to the database. In this context, "Mutate" means records changed.



Create a Dummy Mapping To View Data

20. Click **New** and from the **Mappings** tab, create a new mapping.

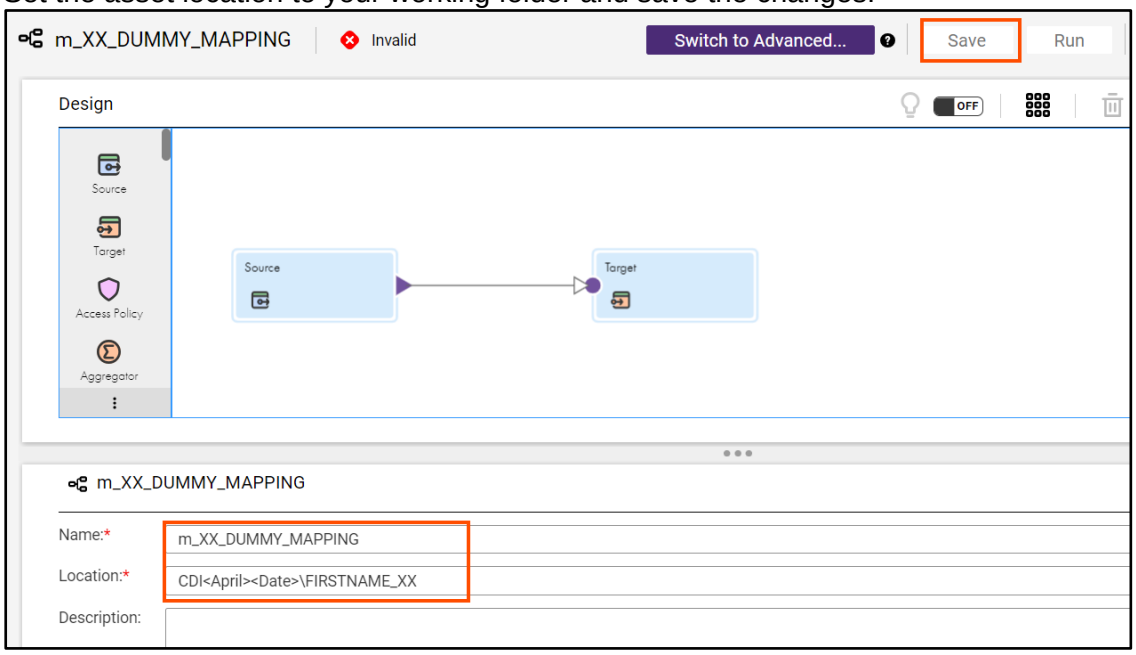


Note: You will about mappings in the succeeding modules. You will just create a dummy mapping to view the data that is written to the target table.

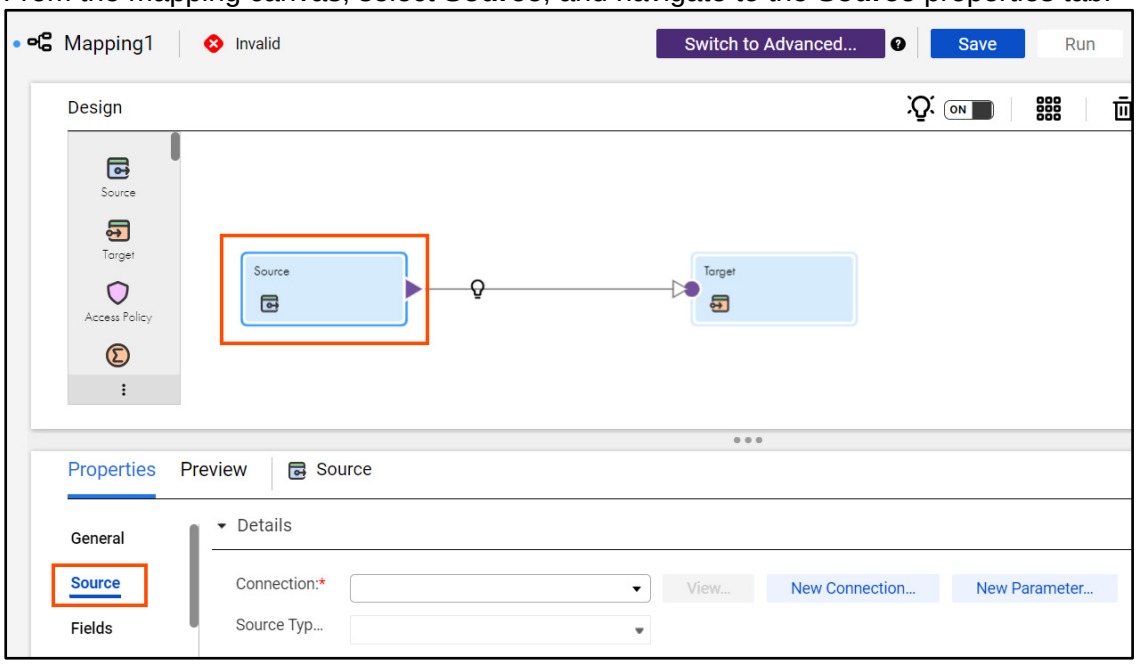
By default, you will have a source and target linked to each other on the mapping canvas.

21. Name the mapping as **m_XX_DUMMY_MAPPING**. Replace XX with your DOB.

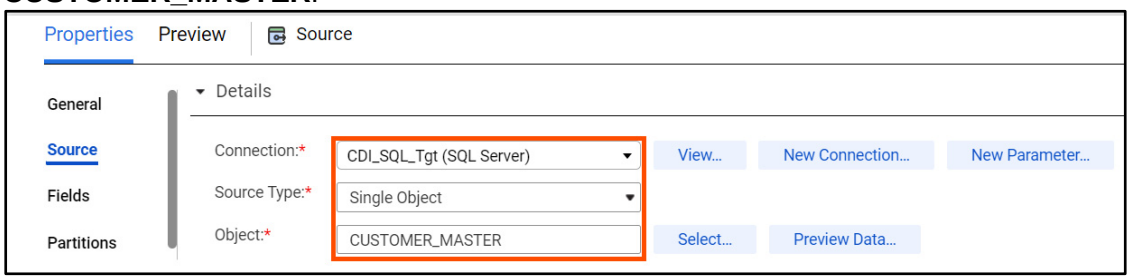
22. Set the asset location to your working folder and save the changes.



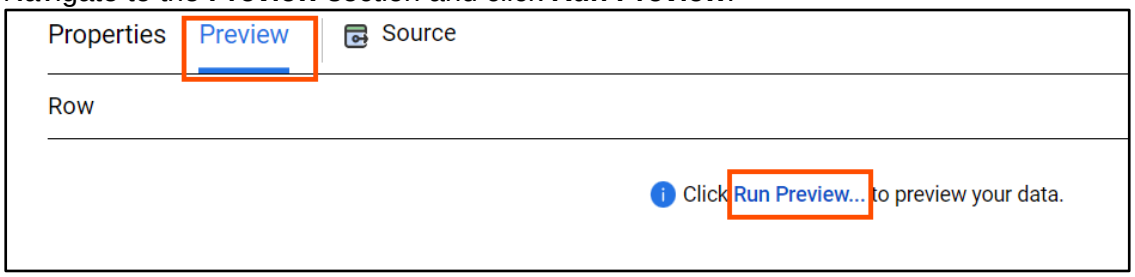
23. From the mapping canvas, select **Source**, and navigate to the **Source** properties tab.



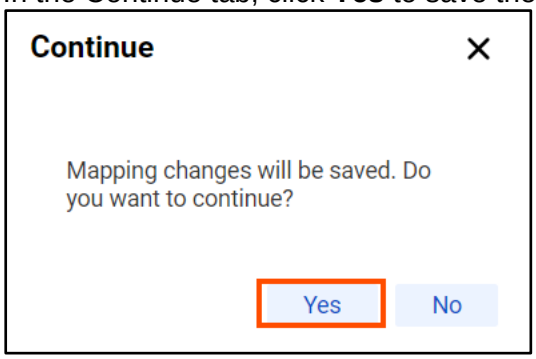
24. Select the Connection to your target SQL connection and the Obejct as **CUSTOMER_MASTER**.



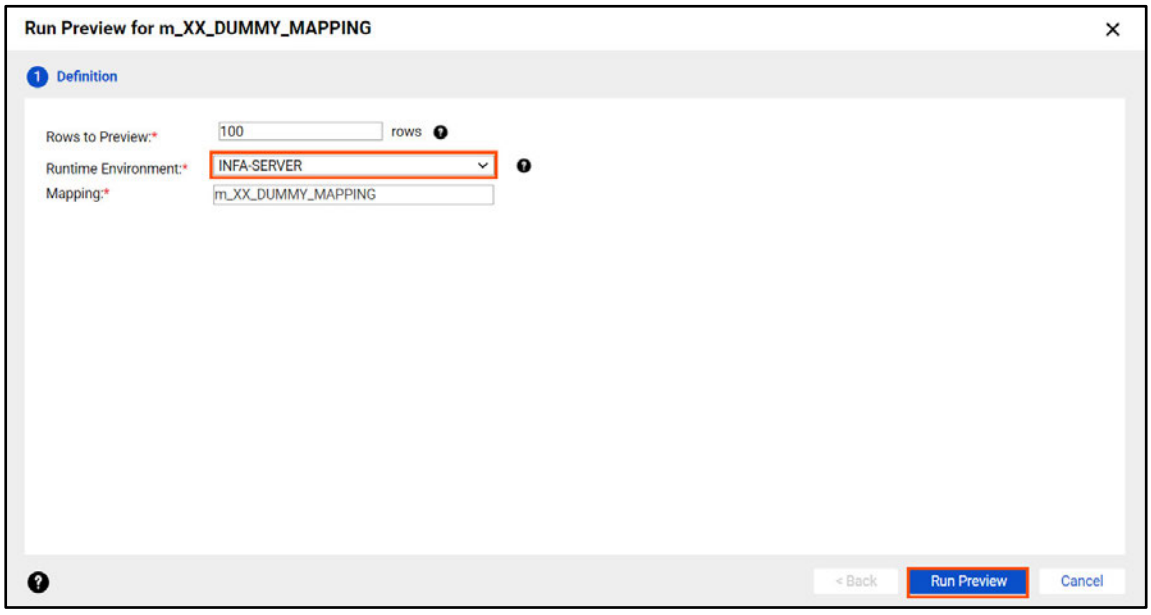
25. Navigate to the **Preview** section and click **Run Preview**.



26. In the Continue tab, click **Yes** to save the changes.



27. If you are prompted with the Run Preview window, select the Runtime Environment as **INFA-SERVER** and click the **Run Preview** button.



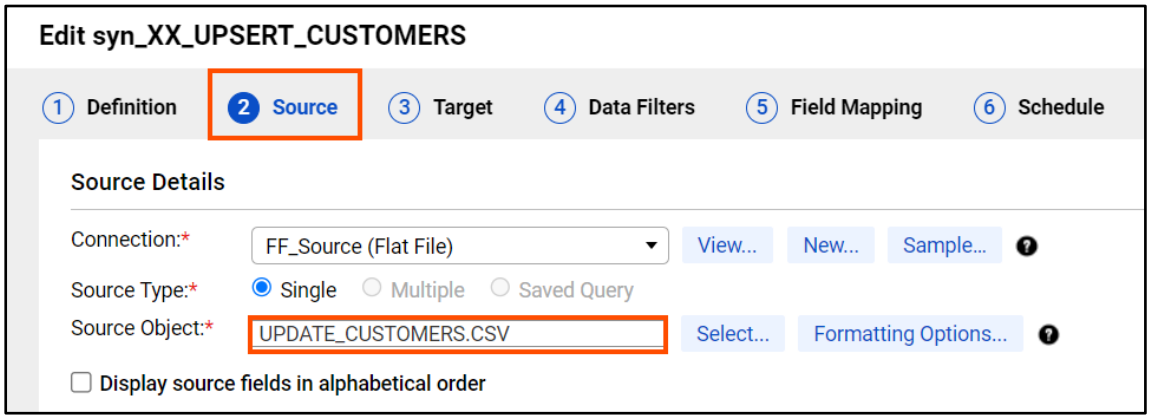
Note: If you have only 1 secure agent configured in your Org, then you will not be prompted with the Run Preview window to select the runtime environment.

28. Wait for a couple of seconds for the results to load in the Preview tab. Maximize the Preview tab to view all the records (use the Minimize and Maximize icons present at the top right corner of the section for a better view). Note that customer number three has failed to provide phone and email contact data.

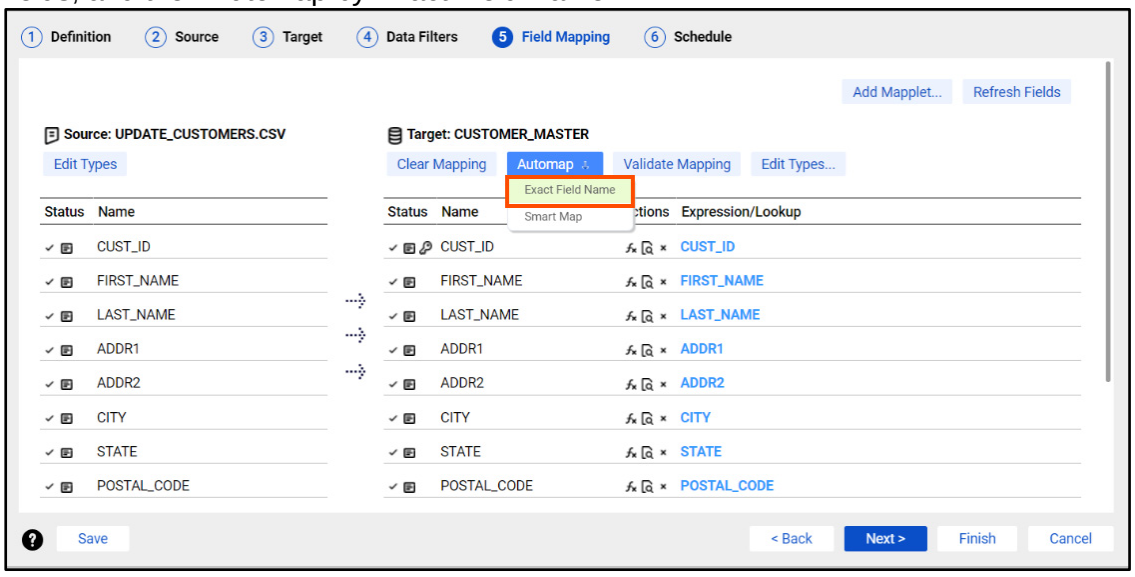
CUST_ID	FIRST_NAME	LAST_NAME	ADDR1	ADDR2	CITY	STATE	POSTAL_CODE	PHONE	EMAIL
1	Neil	Armstrong	100 Maple St		Cincinatti	OH	45634-3478	4582347809	apollo11comma...
2	Michael	Collins	3612 Broadway	Box 278	New York	NY	05354-4563	2124635563	columbiapilot@...
3	Buzz	Aldrin	453 Galaxy Way		Reno	NV	45323-0000	NOYB	NOYB
4	Edward	Mitchell	1290 Paranorma...		San Pedro	CA	90345-3489	4574329087	edmitchell@aol....
5	John	Glenn	456 Valley Way		Houston	TX	45342-5622	6783346723	bigjohn@outloo...
6	John	Young	74 Harrison Ave		West Orange	NJ	07052-0000	3491238976	shuttlepilot@na...
7	Alan	Shepard	3000 Park Ave	Box 4588	New York	NY	08934-3482	3462219754	golfball@gmail.c...
8	Harry	Schmitt	8944 William St.		Naples	FL	43890-0000	4904357623	harrysmoonwalk...
9	Pete	Conrad	23 Cherry St.		Dallas	TX	53555-3422	5692345612	gaptooth@nasa...

29. Restore the size of the Preview tab, if you have maximized it.
30. Open the Synchronization Task **syn_XX_UPSERT_CUSTOMERS** in the edit mode.

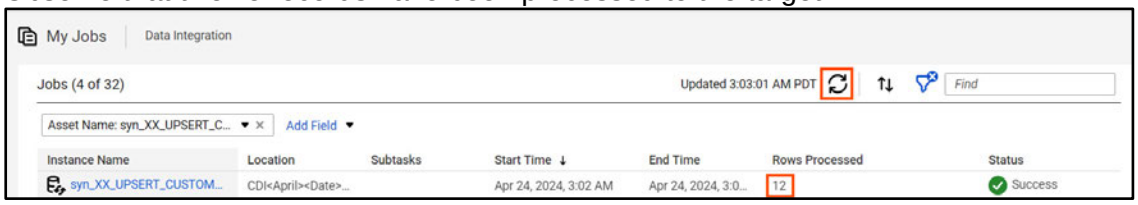
31. Go to the Source tab and change the source file to **UPDATE_CUSTOMERS.CSV**. This file contains a total of ten records, eight of which include data from the ten customers already loaded in the target, plus two new customer records.



32. In the Field Mapping tab, you must click the **Clear Mapping** button above the target fields, and then Automap by Exact Field Name.



33. Validate, save, and run the Synchronization Task.
34. Observe that twelve records have been processed to the target.



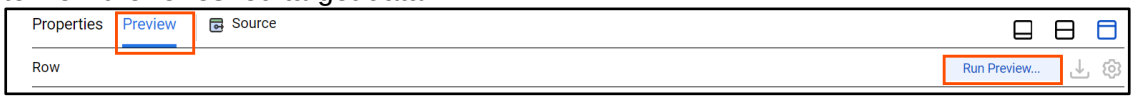
35. Download the Session log file and open it.

36. Scroll down to the bottom to observe the Load Summary. Note that two records have been inserted into the target table, while eight records generated updates because the corresponding CUST_ID values already existed in the target (existing customers). Any changes to the data should have been captured by the target database. According to the “Mutated from update” statistic, two records changed.

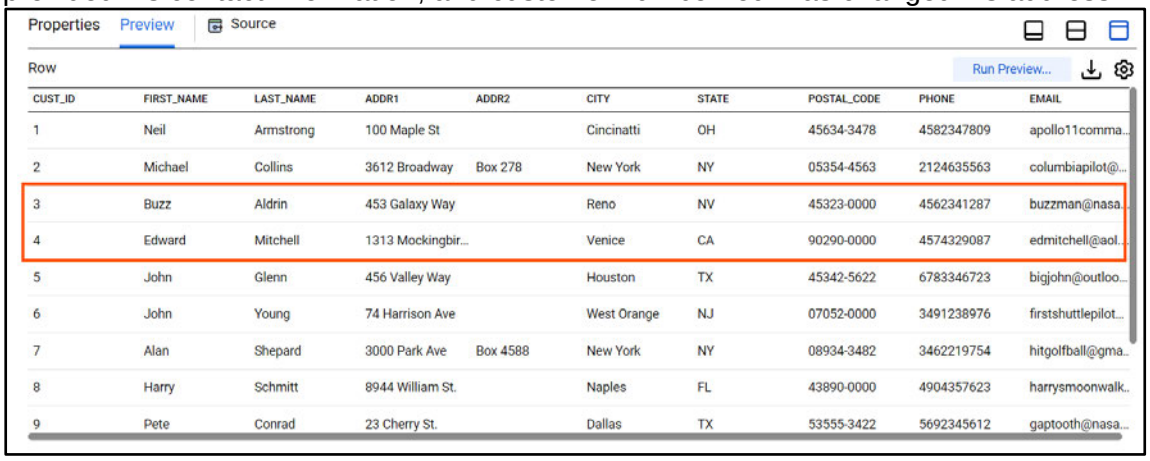
```
LOAD SUMMARY
============

WRT_8036 Target: CUSTOMER_MASTER (Instance Name: [CUSTOMER_MASTER])
WRT_8039 Inserted rows - Requested: 2          Applied: 2          Rejected: 0          Affected: 2          Mutated from update: 2
WRT_8041 Updated rows  - Requested: 10         Applied: 10         Rejected: 0          Affected: 10
```

37. Navigate to your dummy mapping and run the Preview in the Preview tab of the source to view the refreshed target data.



38. If prompted, select the runtime environment as INFA-SERVER and run the preview.s
39. Note that two new records were loaded. Also, note that customer number three has now provided his contact information, and customer number four has changed his address.



CUST_ID	FIRST_NAME	LAST_NAME	ADDR1	ADDR2	CITY	STATE	POSTAL_CODE	PHONE	EMAIL
1	Neil	Armstrong	100 Maple St		Cincinatti	OH	45634-3478	4582347809	apollo11comma...
2	Michael	Collins	3612 Broadway	Box 278	New York	NY	05354-4563	2124635563	columbiapilot@...
3	Buzz	Aldrin	453 Galaxy Way		Reno	NV	45323-0000	4562341287	buzzman@nasa...
4	Edward	Mitchell	1313 Mockingbir...		Venice	CA	90290-0000	4574329087	edmitchell@aol...
5	John	Glenn	456 Valley Way		Houston	TX	45342-5622	6783346723	bigjohn@outloo...
6	John	Young	74 Harrison Ave		West Orange	NJ	07052-0000	3491238976	firstshuttlepilot...
7	Alan	Shepard	3000 Park Ave	Box 4588	New York	NY	08934-3482	3462219754	hitgolfball@gma...
8	Harry	Schmitt	8944 William St.		Naples	FL	43890-0000	4904357623	harrysmoonwalk...
9	Pete	Conrad	23 Cherry St.		Dallas	TX	53555-3422	5692345612	gaptooth@nasa...

The Upsert setting has added new records and updated (changed) existing records with the source data while avoiding duplicate records and database uniqueness constraint rejections.

40. Save and close all the assets from the navigation pane.

This concludes the lab

Module 4: Data Transfer Task

Lab 4-1: Creating a Data Transfer Task

Overview:

The data transfer task allows you to transfer data from a source to a target. For example, you might use a data transfer task to transfer data from an on-premises database to a cloud data warehouse.

Objective:

- Create a data transfer task to copy fields from SQL to flat-file

Scenario:

Ruby wants to create a backup of the CUSTOMER SQL table for the fields with TIER as not null. So, John suggests creating a data transfer task to copy data to a flat file.

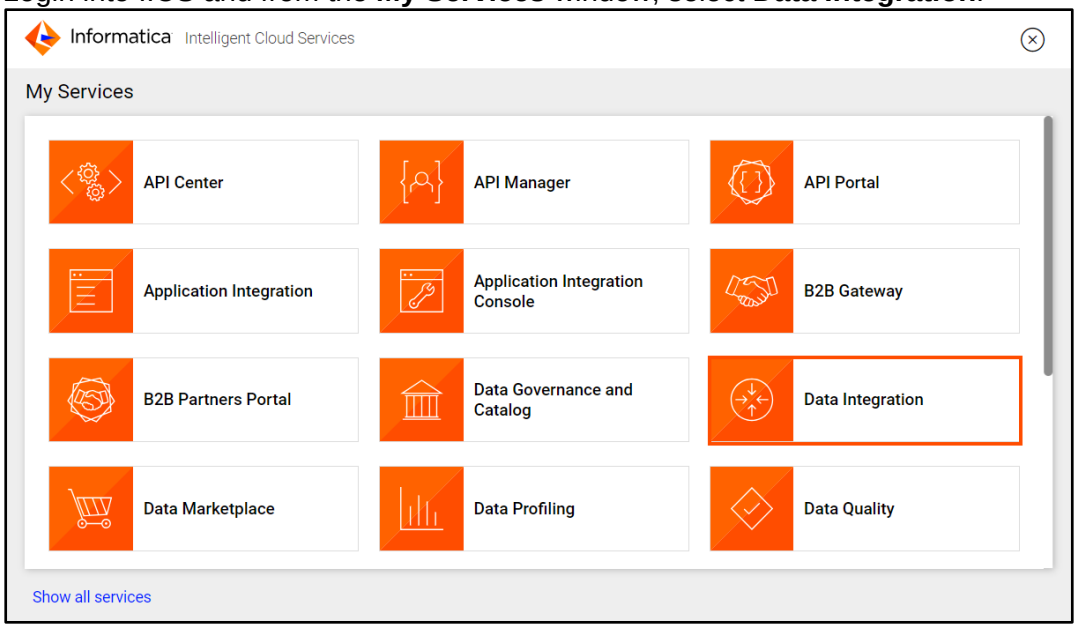
Duration:

10 minutes

Tasks

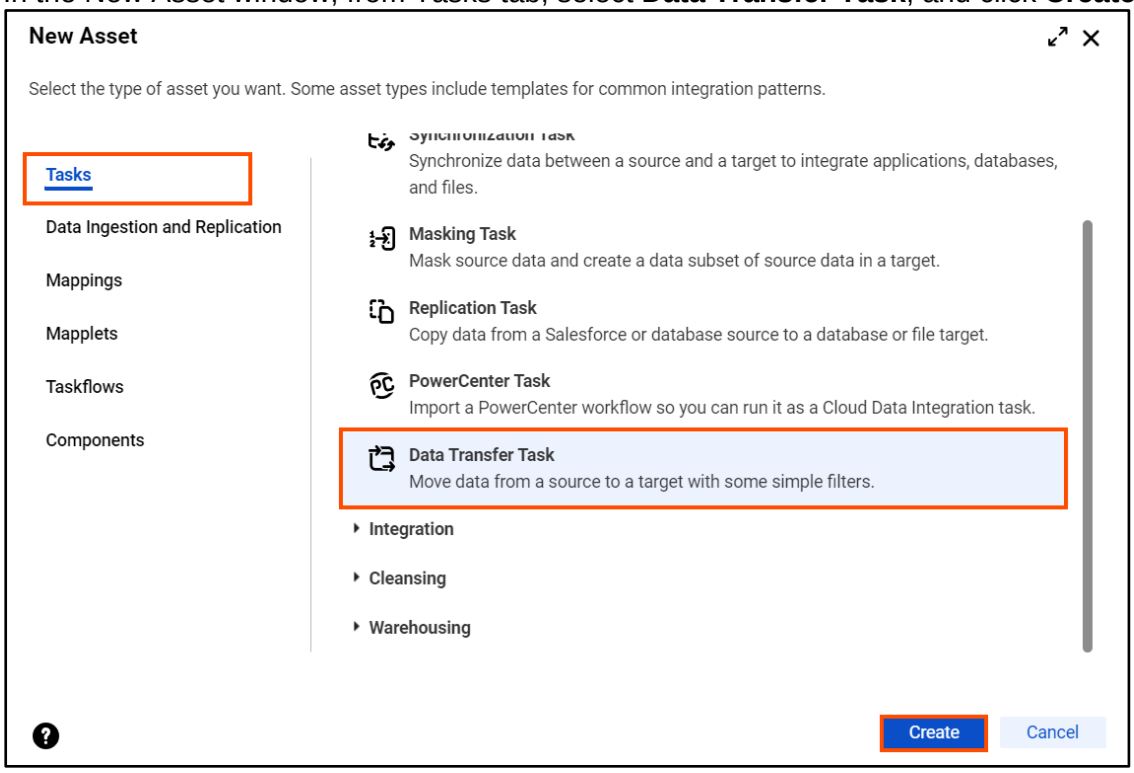
Create a Data Transfer in IICS

1. Login into IICS and from the **My Services** window, select **Data Integration**.

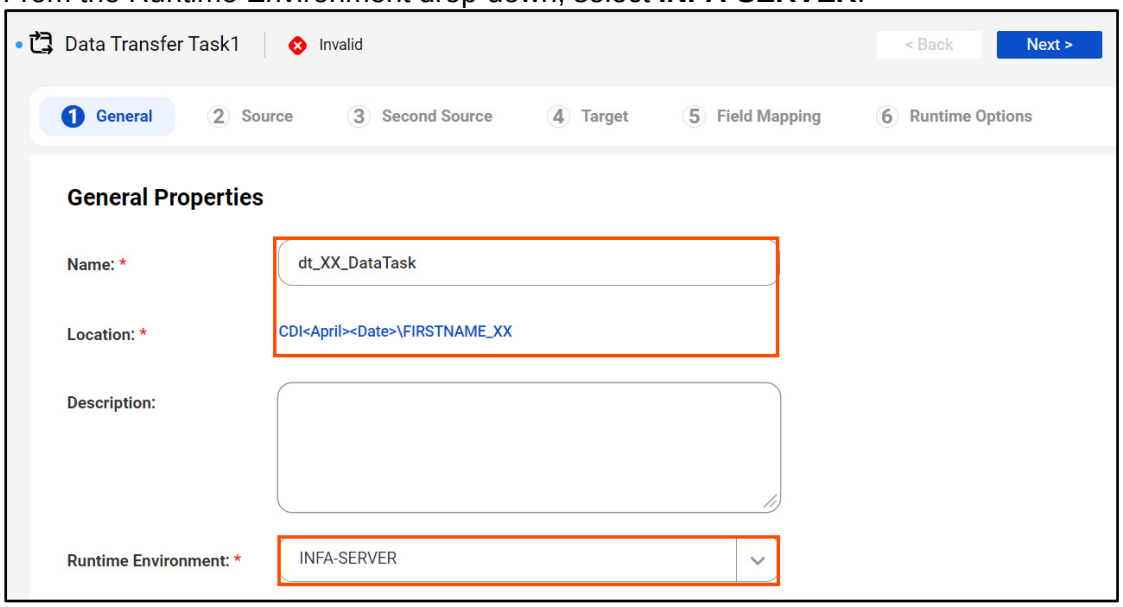


2. To create a new asset, from the navigation pane, select **New**.

3. In the New Asset window, from Tasks tab, select **Data Transfer Task**, and click **Create**.

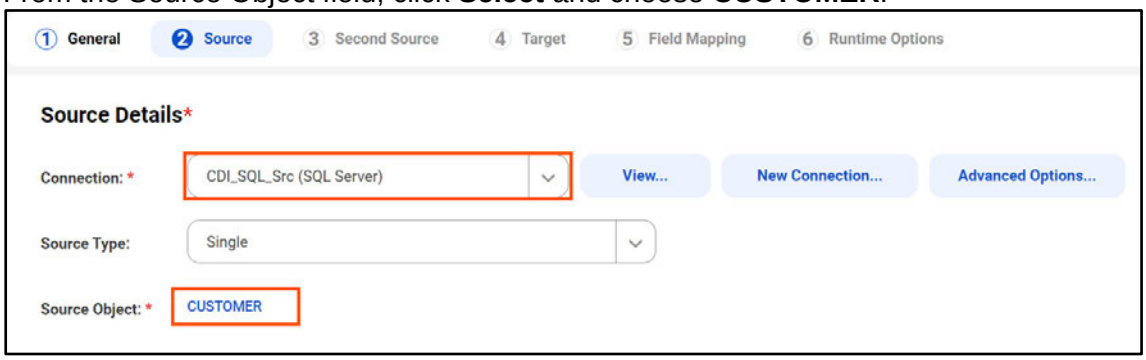


4. In the Task Name field, enter **dt_XX_DataTask**. Replace XX with your DOB.
5. Verify that the asset Location is your working directory.
6. From the Runtime Environment drop-down, select **INFA-SERVER**.



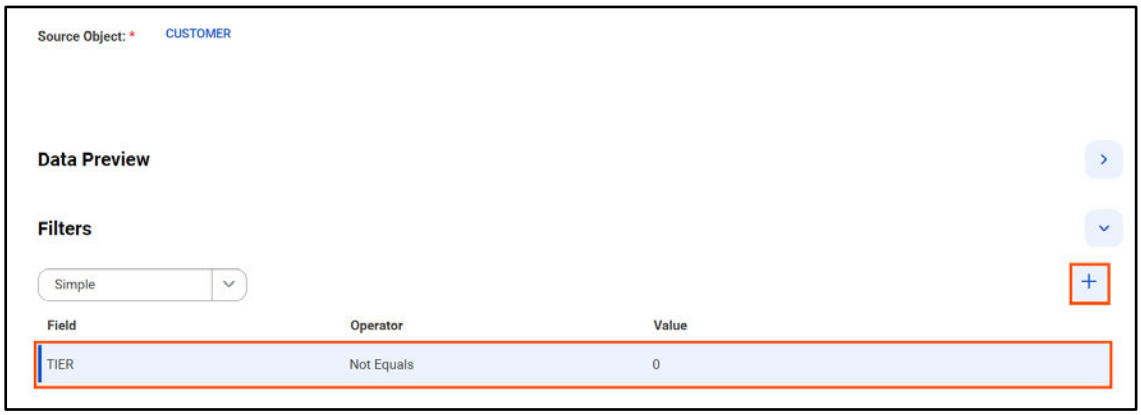
7. Click **Next**.
8. From the Connection drop-down, select **CDI_SQL_Src**.

9. From the Source Object field, click **Select** and choose **CUSTOMER**.



10. To apply a filter, expand the **Filters** section and click to add a filter condition.
11. Enter the filter condition, as shown in the table below:

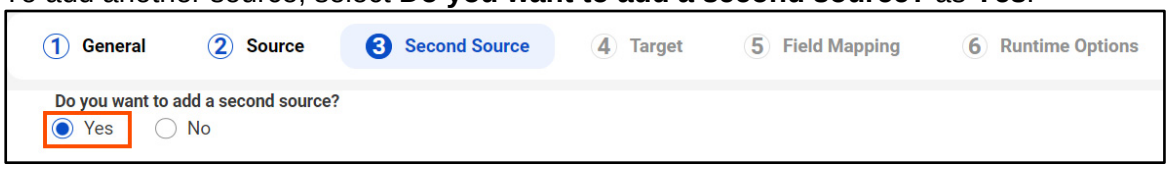
Field	Operator	Value
TIER	Not Equals	0



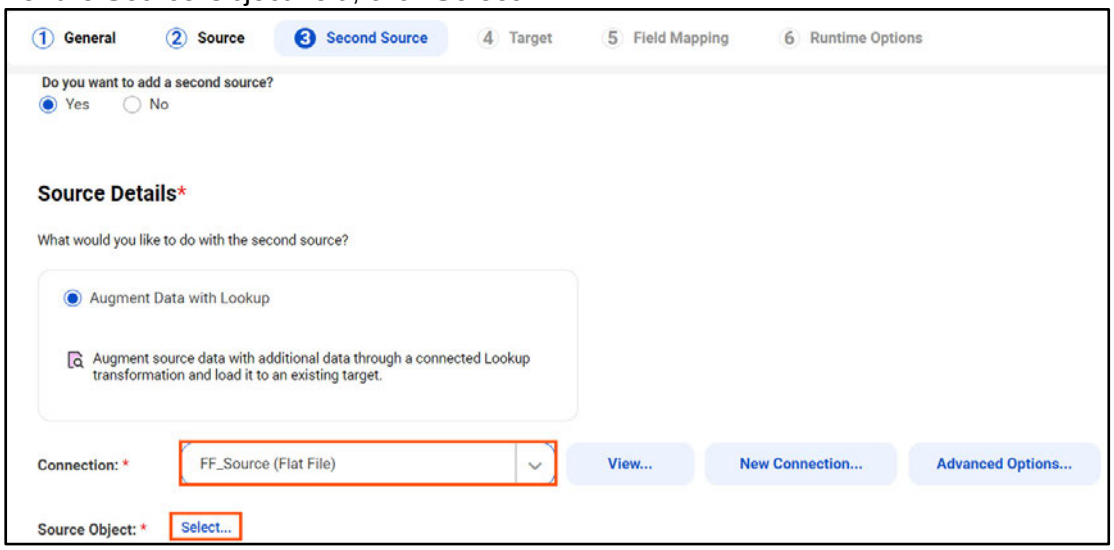
12. To sort the fields, expand the **Sorting** section and click the icon to add the sort condition.

13. Click in the field and from the field drop down, select **NAME**.

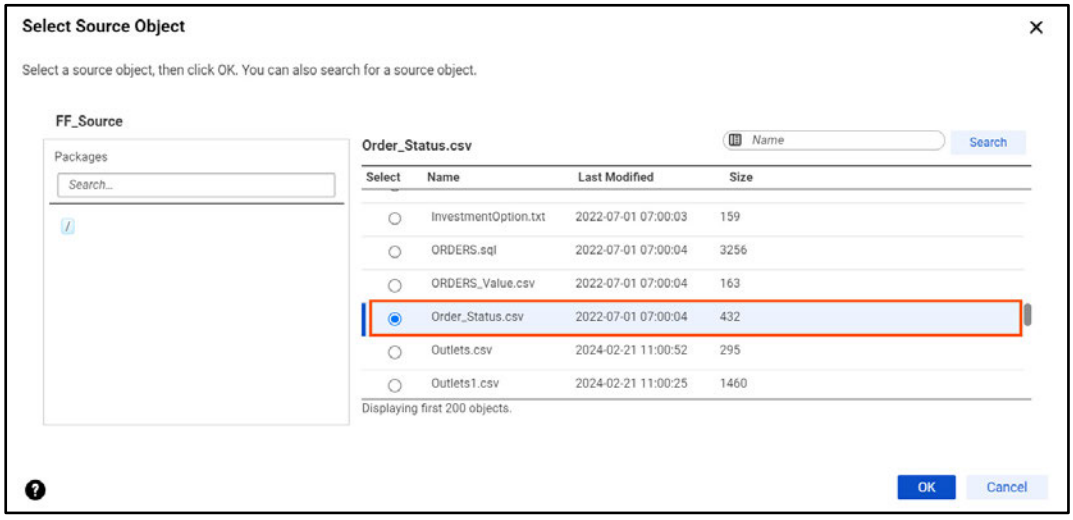
14. Click **Next**.
15. To add another source, select **Do you want to add a second source?** as **Yes**.



16. In the Source Details section, from the Connection drop down, select **FF_Source**.
17. For the Source Object field, click **Select**.



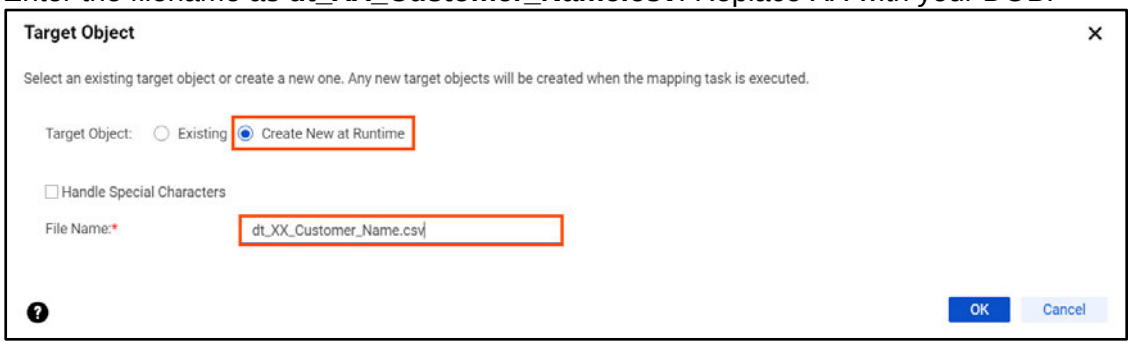
18. From the Select Source Object window, click **Order_Status.csv** and **OK**.



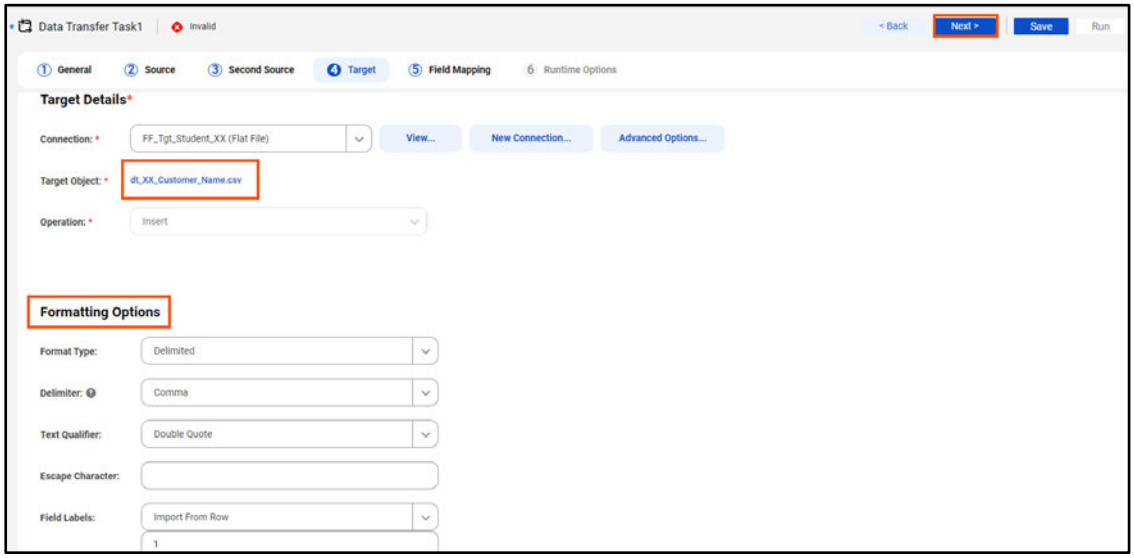
19. Scroll down to the Lookup Conditions section, and click + to add a lookup condition as shown in the table below:

Lookup Field	Operator	Incoming Field
CUST_ID	Equals	SRC_CUST_ID

20. Click **Next**.
21. In the Target Details section, select your **target Flat File connection**.
22. From the Target Object, click **Select**.
23. In the Target Object window, click **Create New at Runtime**.
24. Enter the filename as **dt_XX_Customer_Name.csv**. Replace XX with your DOB.



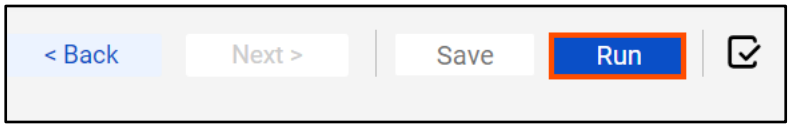
25. Click **OK**.
26. Retain the default values for Formatting Options.
27. Click **Next**.



28. Verify that the source and target fields are mapped. Click **Next**.

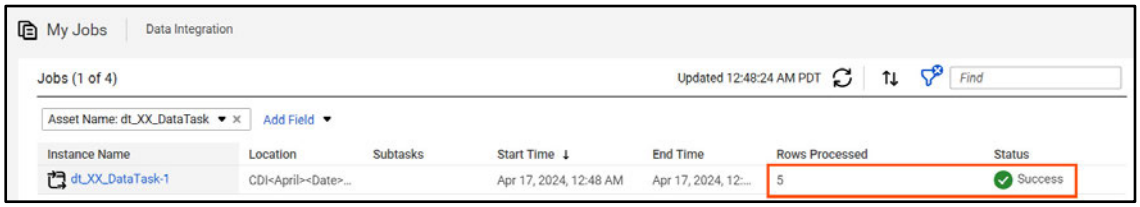
29. You can configure the task to run on schedule in the Runtime Options tab. For this lab, let us retain the default option and click **Save**.

30. Click **Run** to run the task.



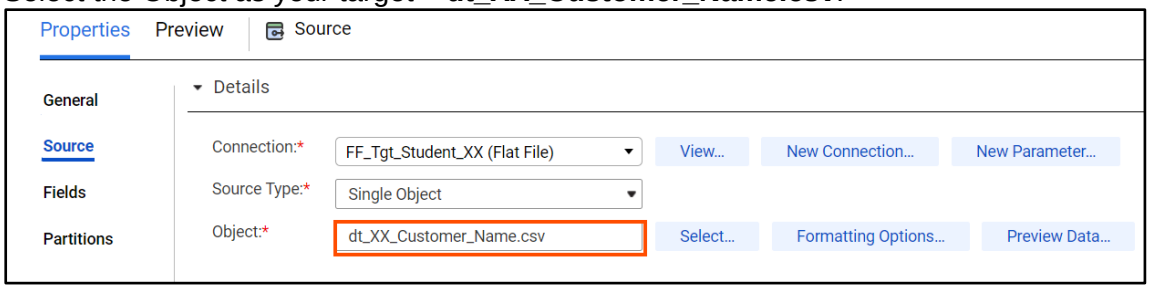
Monitor Task

31. To monitor the task, from the navigation pane, click **My Jobs**.
32. When the task completes, the status changes to **Success**.



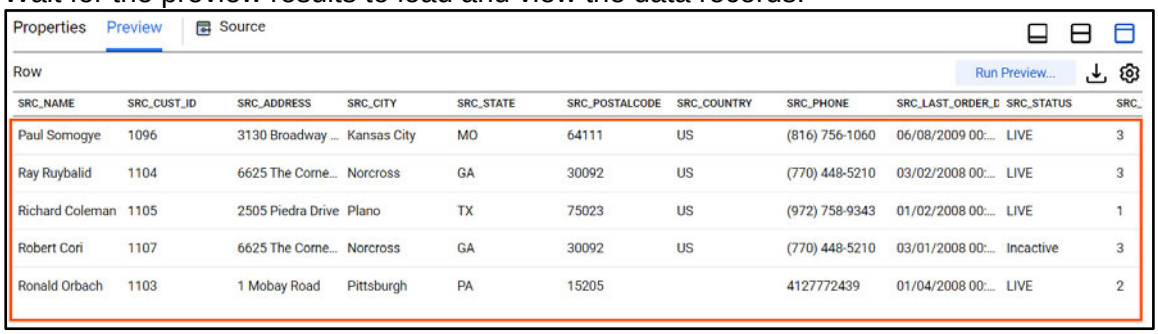
Optional - Open the Dummy Mapping and View the Data

33. Navigate to your working folder and open the dummy mapping **m_XX_DUMMY_MAPPING**.
34. From the mapping canvas, select **Source** and go to the **Source** properties tab.
35. Change the Connection to your target Flat File connection. Click **Yes** in the Change Connection window.
36. Select the Object as your target – **dt_XX_Customer_Name.csv**.



37. Go to the Preview section and click **Run Preview**. In the Continue window, click **Yes**.
38. If prompted, select runtime environment as **INFA-SERVER** and click **Run Preview**.

39. Wait for the preview results to load and view the data records.



SRC_NAME	SRC_CUST_ID	SRC_ADDRESS	SRC_CITY	SRC_STATE	SRC_POSTALCODE	SRC_COUNTRY	SRC_PHONE	SRC_LAST_ORDER_D	SRC_STATUS	SRC_
Paul Somogye	1096	3130 Broadway ...	Kansas City	MO	64111	US	(816) 756-1060	06/08/2009 00:...	LIVE	3
Ray Ruybalid	1104	6625 The Corne...	Norcross	GA	30092	US	(770) 448-5210	03/02/2008 00:...	LIVE	3
Richard Coleman	1105	2505 Piedra Drive	Plano	TX	75023	US	(972) 758-9343	01/02/2008 00:...	LIVE	1
Robert Cori	1107	6625 The Corne...	Norcross	GA	30092	US	(770) 448-5210	03/01/2008 00:...	Incactive	3
Ronald Orbach	1103	1 Mobay Road	Pittsburgh	PA	15205		4127772439	01/04/2008 00:...	LIVE	2

40. Close all the assets from the navigation pane.

This concludes the lab.

Module 6: IICS Transformations

Lab 6-1: Classifying the Product Shipping Details Using Various Transformations

Overview:
The Mapping Designer feature of IICS allows you to create a mapping and use it in a Mapping Task. You can also create a custom query to perform complicated joins of multiple tables or to reduce the number of fields that enter the data flow.

Objective:

- Create a Mapping
- Add Joiner, Expression, and Filter transformations and complete the logic

Scenario:
NH Suppliers recently started offering express delivery to their customers with an order value of more than 1000 USD. However, Ruby, the owner of NH suppliers, wants to automate the delivery selection process. John, who is the Lead Developer in NH Suppliers, suggests using the IICS Mapping Designer to fulfill this requirement.

In this lab, John will combine the Products and Order details to calculate the total order amount. This calculated amount will be utilized to process the orders as normal shipping or expedited shipping.

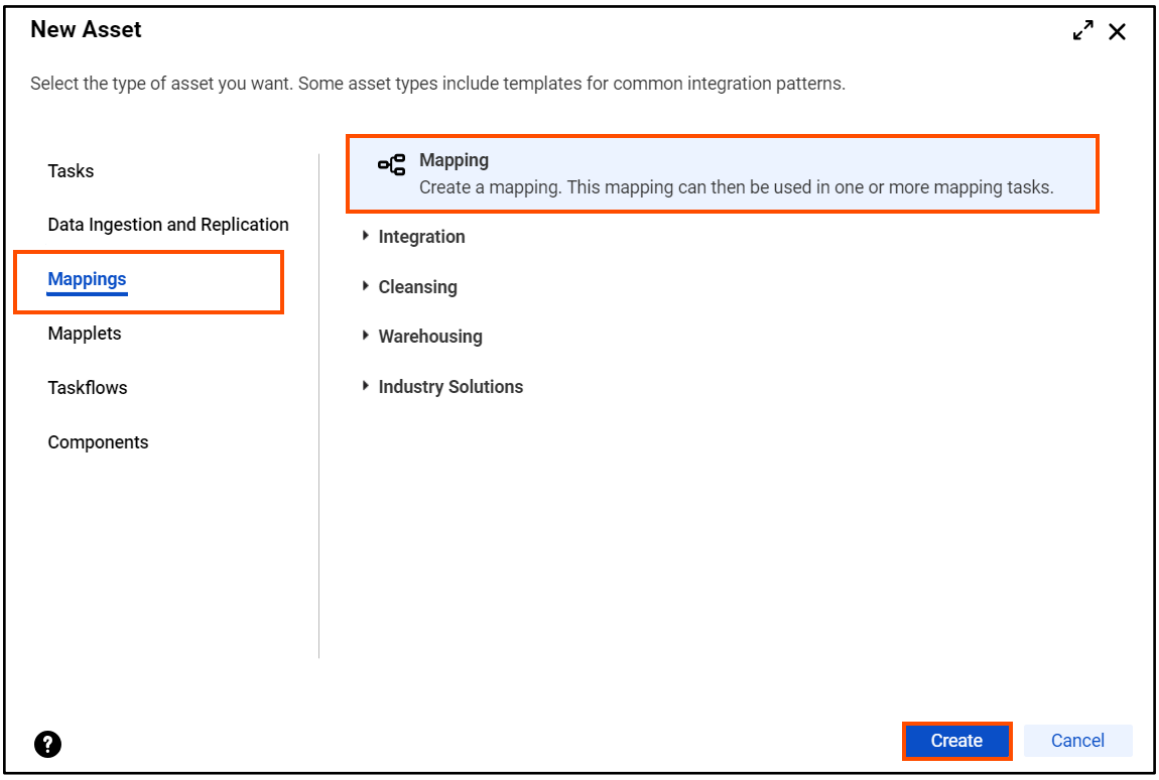
Duration:
35 minutes

Tasks

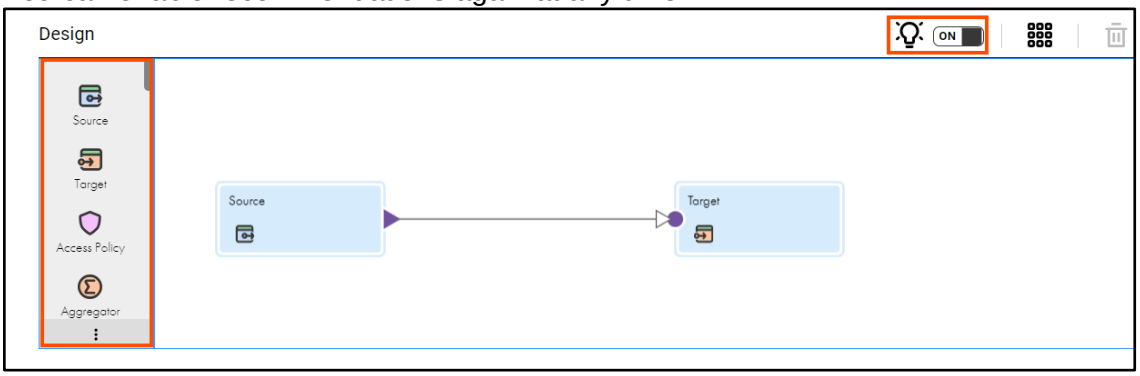
Create Mapping

1. Login into IICS and from the My Services window, select **Data Integration**.
2. From the navigation pane, select **New**.
3. From the New Asset window, click the **Mappings** tab, and select **Mapping**.

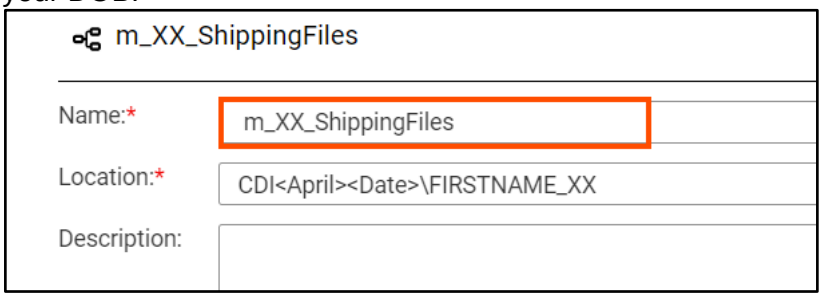
4. Click **Create**.



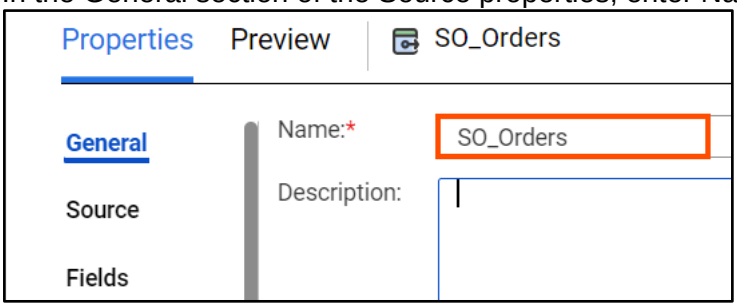
Note: The following Mapping page appears. By default, the mapping canvas has a Source and Target transformation. You can add transformations on the canvas from the Transformation palette on the left side of the canvas. If your IICS Org has CLAIRE recommendations enabled, you can receive recommendations during mapping design. To disable recommendations for the current mapping, use the recommendation toggle. You can enable recommendations again at any time.



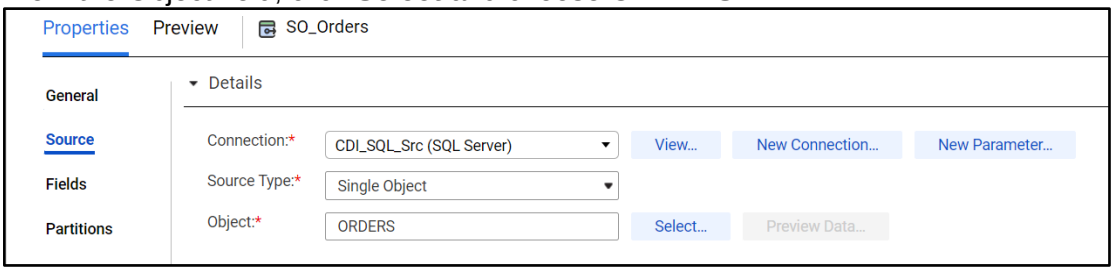
5. In the Name field, enter the mapping name as **m_XX_ShippingFiles**. Replace XX with your DOB.



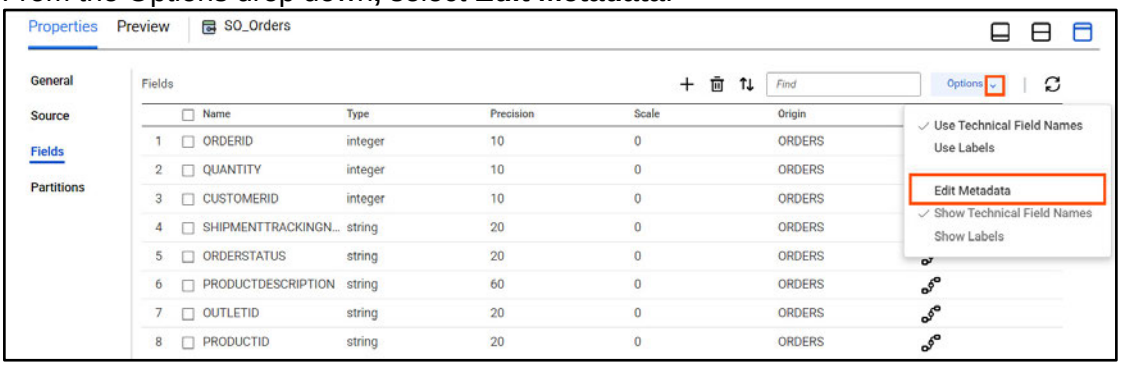
6. To configure the source, from the mapping canvas, click the **Source** transformation.
7. In the General section of the Source properties, enter Name as **SO_Orders**.



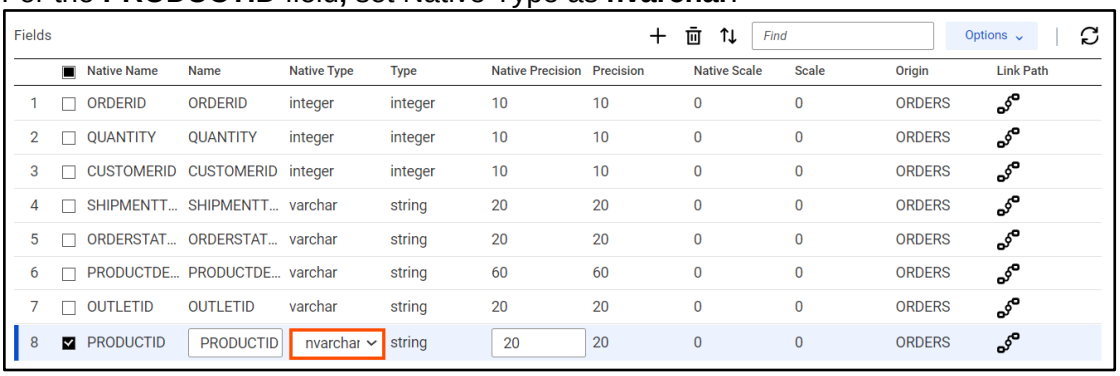
8. From the properties pane, click **Source**.
9. From the Connection drop-down, select **CDI_SQL_Src**.
10. From the Source Type field, retain Single Object.
11. From the Object field, click **Select** and choose **ORDERS**.



12. From the properties pane click **Fields**.
13. From the Options drop down**,** select **Edit Metadata**.



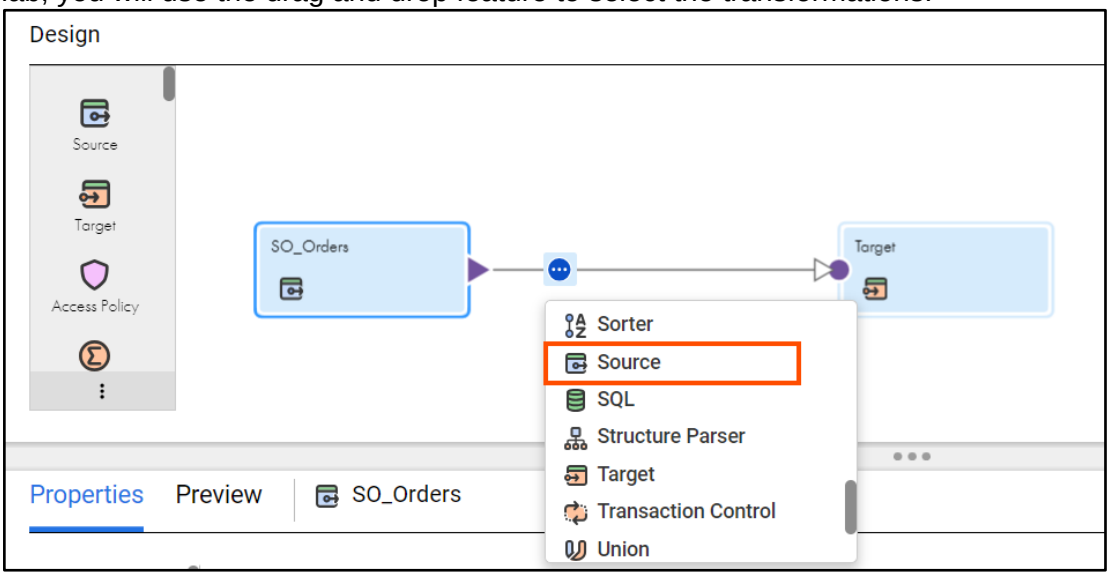
14. For the **PRODUCTID** field**,** set Native Type as **nvarchar**.



15. From the Options drop-down, uncheck **Edit Metadata**.

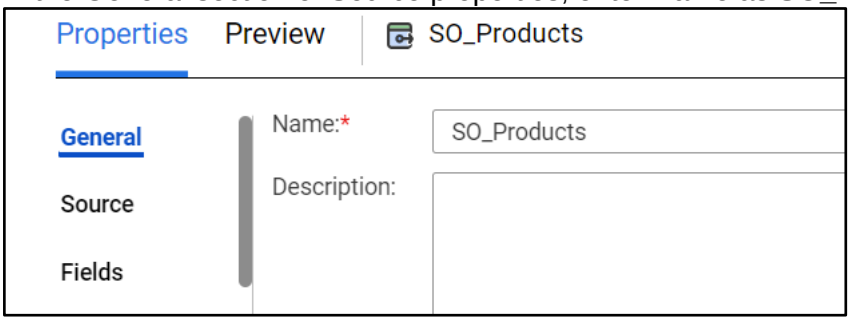
Add Source Transformation

16. From the list of available transformations, drag and drop a **Source** transformation onto the mapping canvas.
    Note: You can also use the Add Transformation icon (the orange bulb icon on the link) to add transformations to a mapping directly on the mapping canvas. The Add Transformation icon appears when you hover over the link between the transformations only when the CLAIRE recommendations option is enabled for the mapping. For this lab, you will use the drag and drop feature to select the transformations.

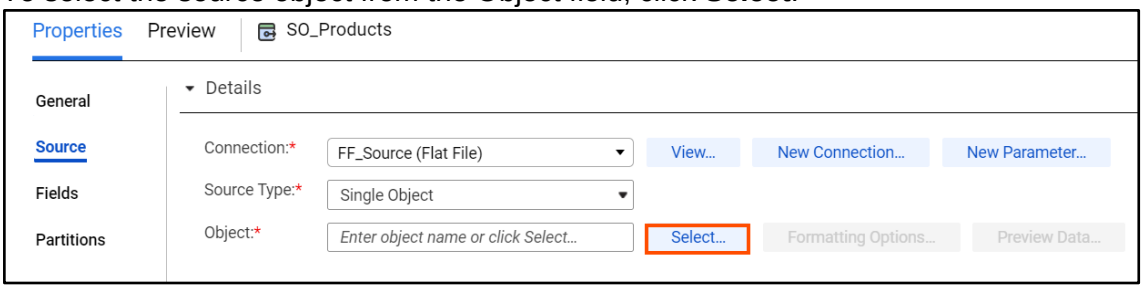


17. Select **Source** from the mapping canvas.

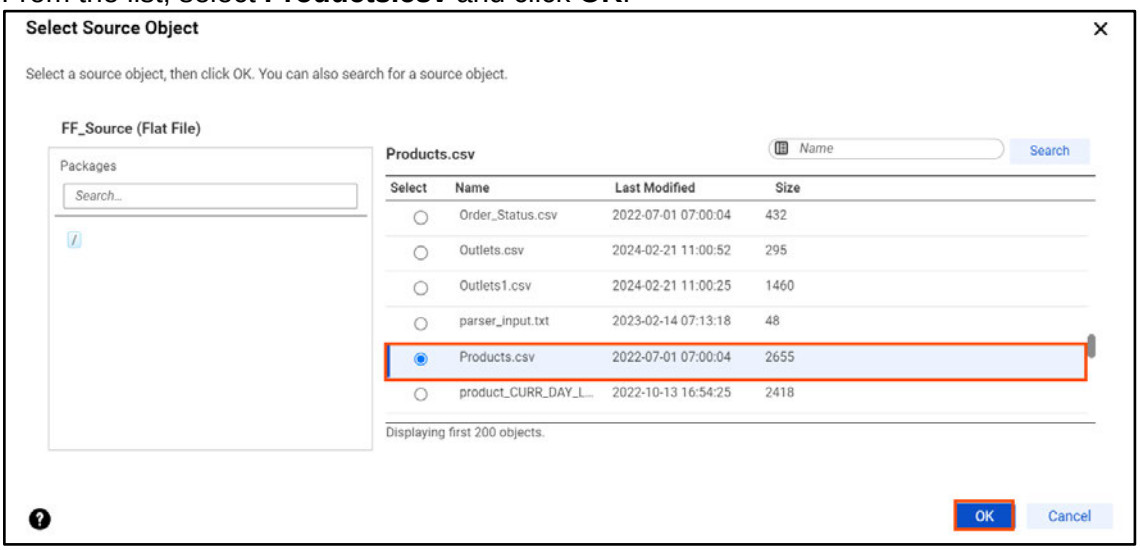
18. In the General section of Source properties, enter Name as **SO_Products**.



19. From the properties pane, click **Source**.
20. From the Connection drop-down, select **FF_Source**.
21. Retain Source Type as **Single Object**.
22. To select the source object from the Object field, click **Select**.



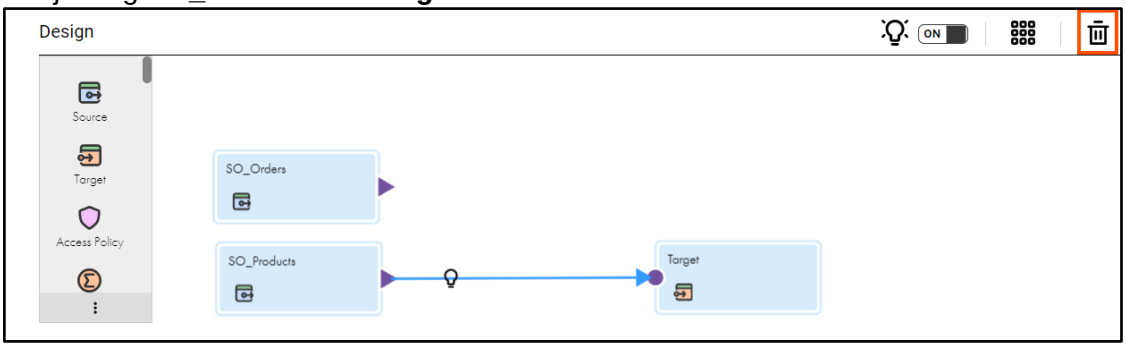
23. From the list, select **Products.csv** and click **OK**.



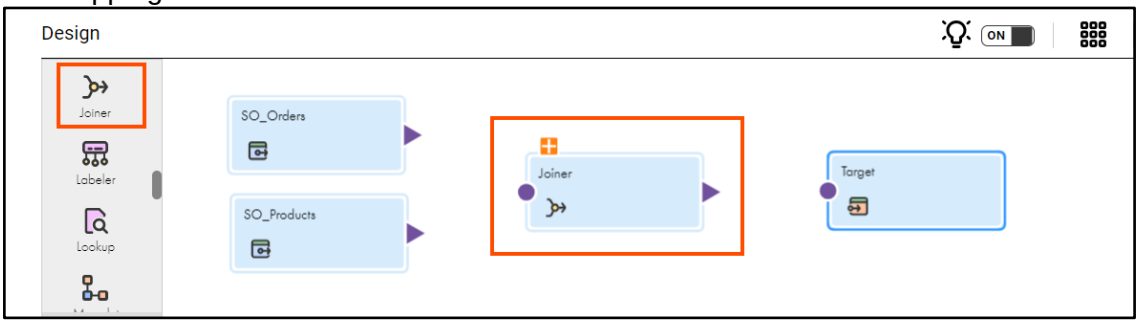
Add Joiner Transformation

To join data from the Orders table and Products CSV file, use the Joiner transformation. The Joiner transformation joins data based on the join conditions and the join type. A join condition matches fields between the two sources. Here, you will join the data using the Product ID value for both sources.

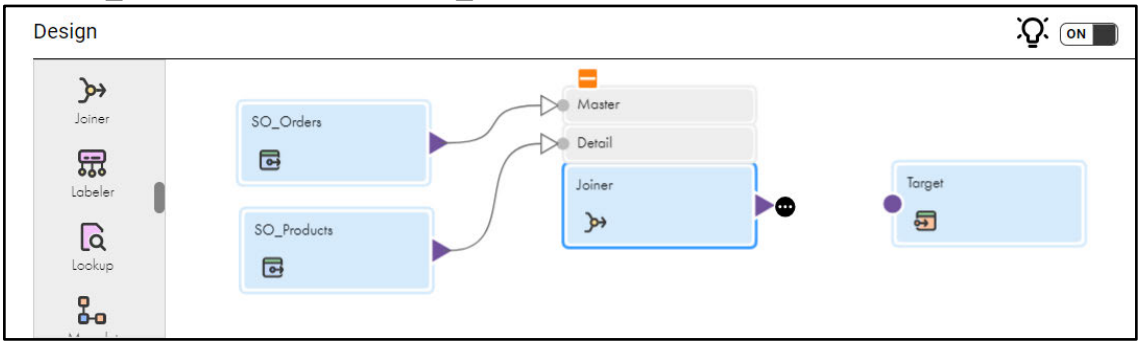
24. Click the link joining the **SO_Products** with **Target** and click .
    Note: If you used the drag-and-drop feature to add the Source transformation, click the link joining **SO_Orders** with **Target** and click .



25. From the list of available transformations, drag and drop a **Joiner** transformation onto the mapping canvas.

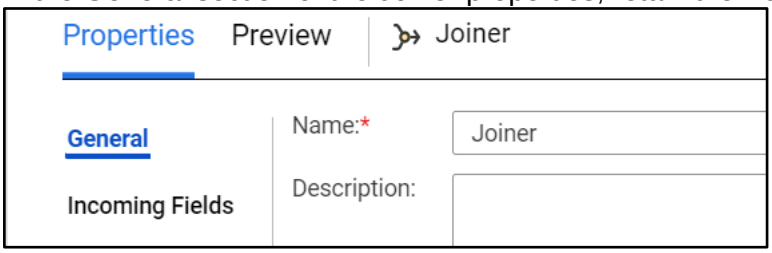


26. To expand the join options, click .
27. Link **SO_Orders** to Master and **SO_Products** to Detail.



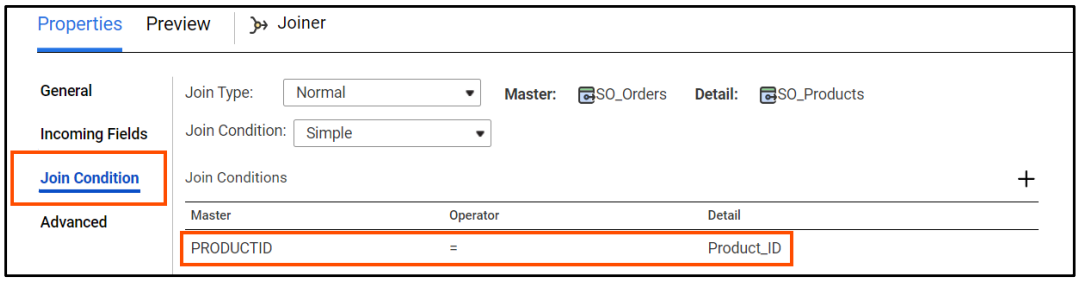
28. To configure the Joiner transformation, from the mapping canvas, click the **Joiner** transformation.

29. In the General section of the Joiner properties, retain the Name as **Joiner**.



30. From the properties pane, click **Join Condition**.
31. To add a new join condition, click ⊞.
32. Enter the join condition, as shown in the table below:

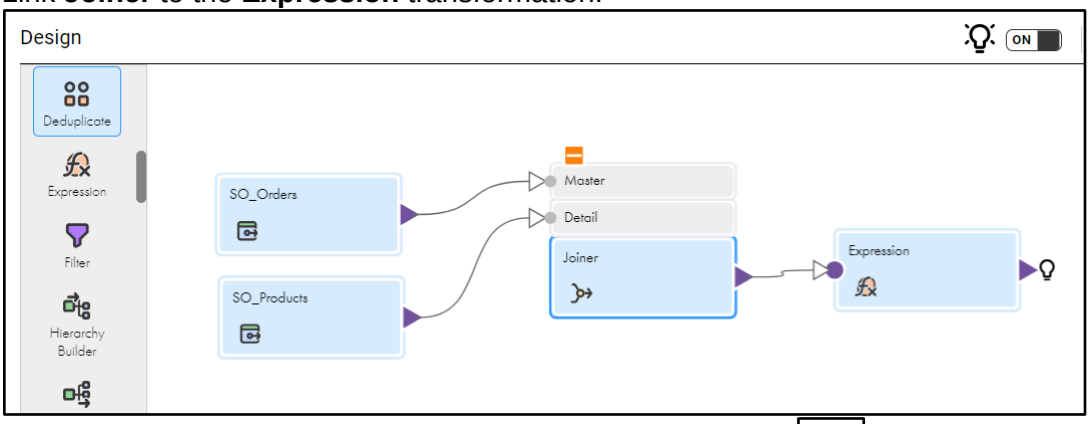
Master	Operator	Detail
PRODUCTID	=	Product_ID



Add Expression Transformation to calculate the order total

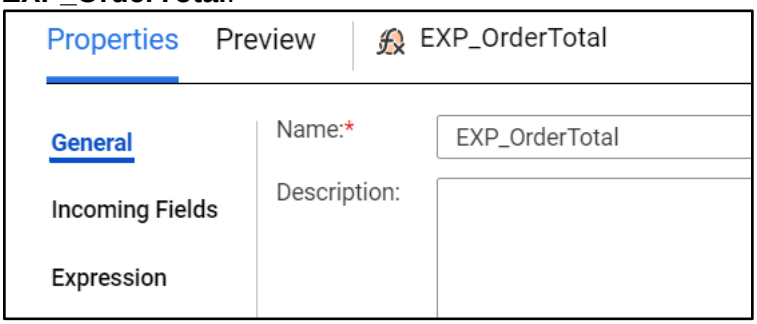
33. From the list of available transformations, drag and drop an **Expression** transformation onto the mapping canvas.

34. Link **Joiner** to the **Expression** transformation.



Note: To arrange all the transformations on the canvas, click .

35. Select the Expression transformation from the mapping canvas.
36. In the General section of the Expression properties, enter the Name as **EXP_OrderTotal**.

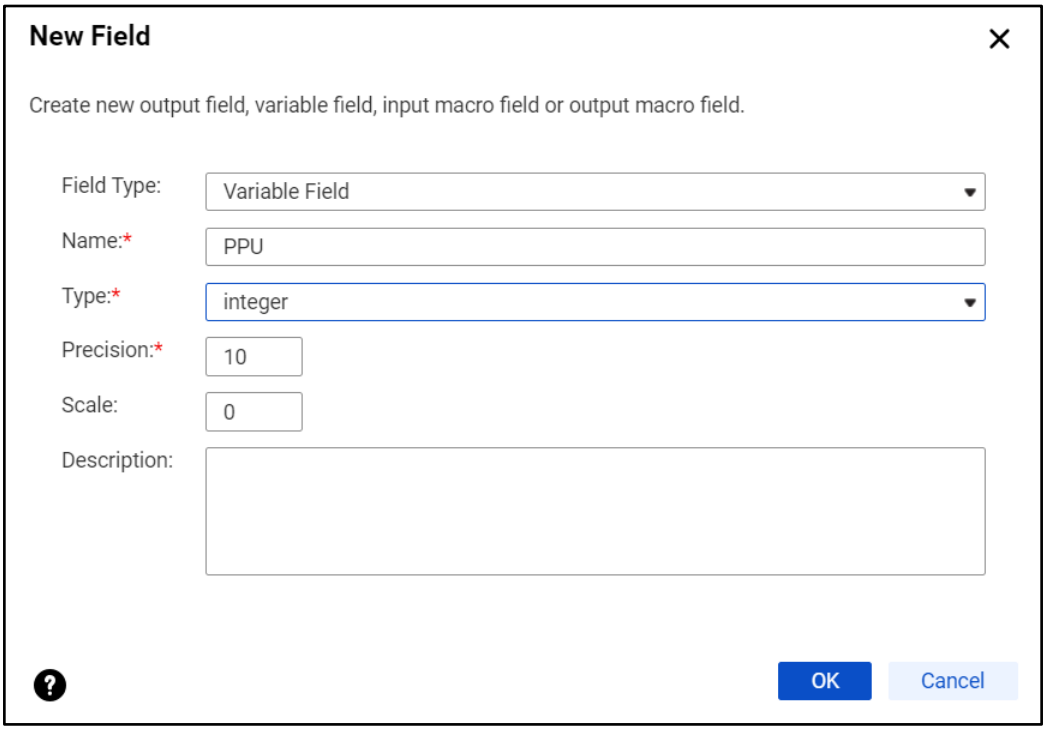


Note: You must create two variable expressions to convert the incoming fields (UnitPrice and Quantity) to integers before you can use them in the output field to calculate the Order Total.

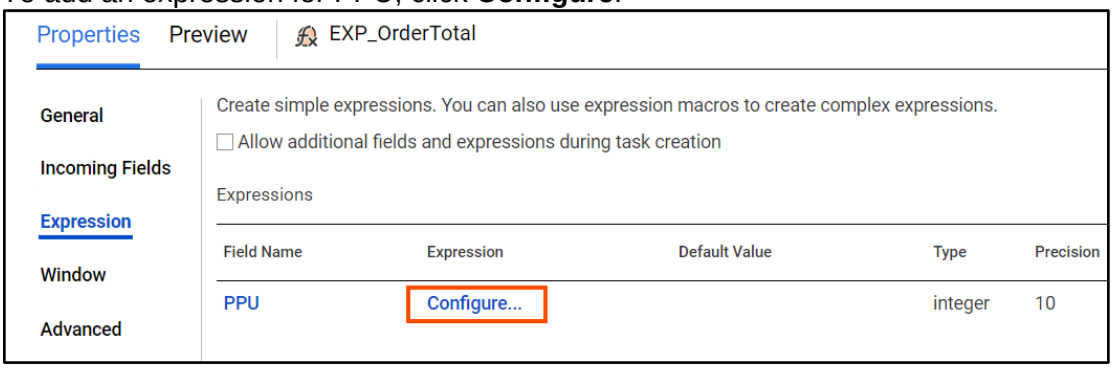
37. From the properties pane, click **Expression**.
38. To add a new expression, click .

39. Enter the details as shown in the table below and click **OK**.

Field Type	Name	Type	Precision	Scale
Variable Field	PPU	integer	10	0



40. To add an expression for PPU, click **Configure**.

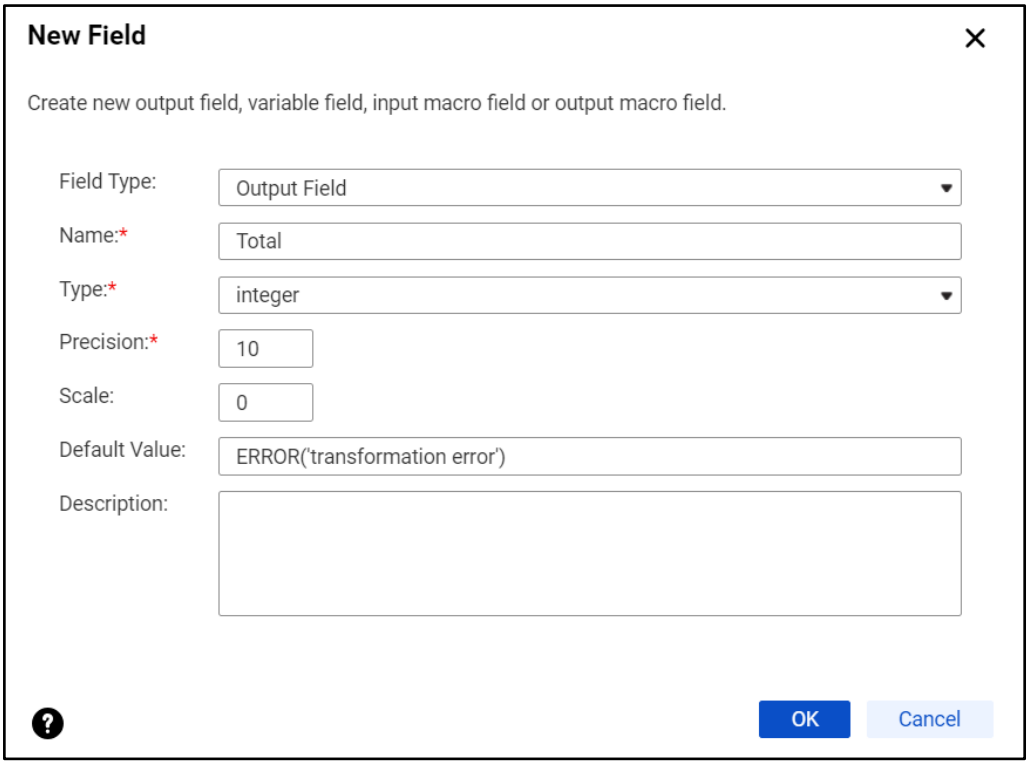


41. In the expression window, enter the following expression:
TO_INTEGER(UnitPrice)

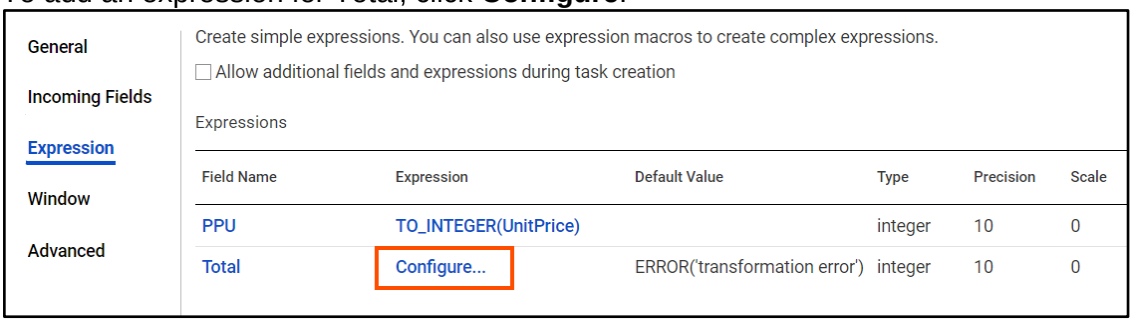
42. Click **OK**.

43. Add another expression as shown in the table below and click **OK**.

Field Type	Name	Type	Precision	Scale
Output Field	Total	integer	10	0



44. To add an expression for Total, click **Configure**.

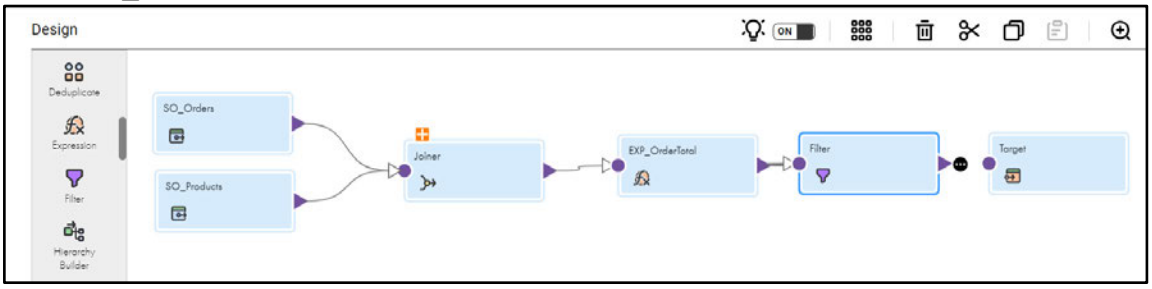


45. In the expression window, enter the following expression:
PPU*QUANTITY
46. Click **OK**.

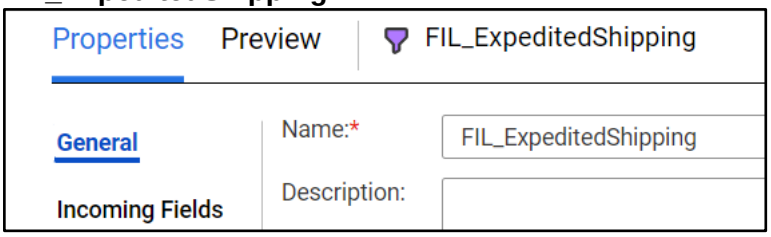
Add Filter Transformation

To process the orders as normal shipping or expedited shipping, use the Filter transformation to filter data based on the Total value of the order.

47. From the list of available transformations, drag and drop a **Filter** transformation onto the mapping canvas.
48. Link **EXP_OrderTotal** to the **Filter** transformation.

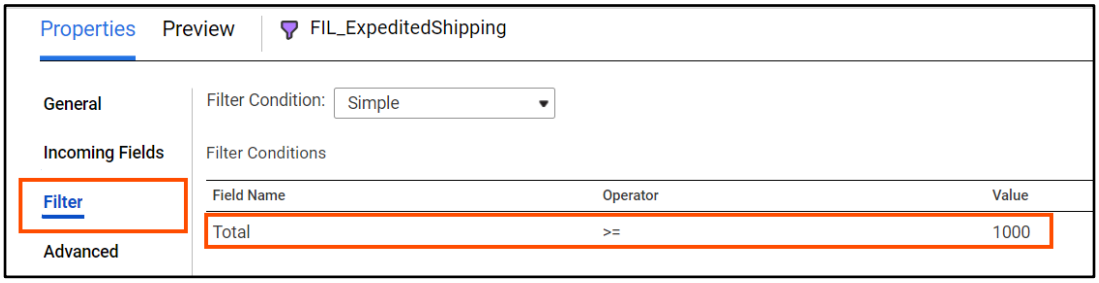


49. In the General section of the Filter properties, enter the Name as **FIL_ExpeditedShipping**.



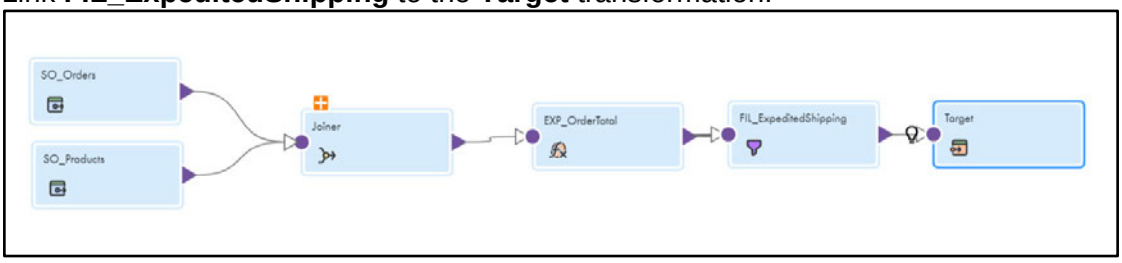
50. From the properties pane, click **Filter**.
51. To add a new filter, click ⊞.
52. Enter filter condition as shown in the table below:

Field Name	Operator	Value
Total	>= (Greater than or equal to)	1000

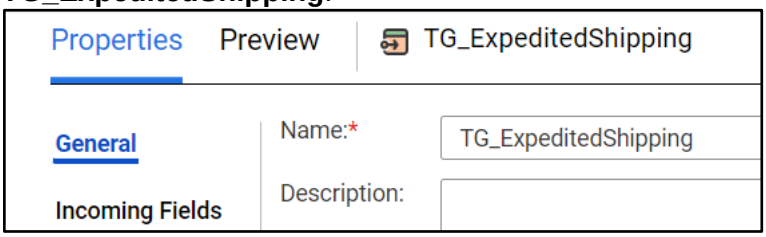


Configure Target Transformation

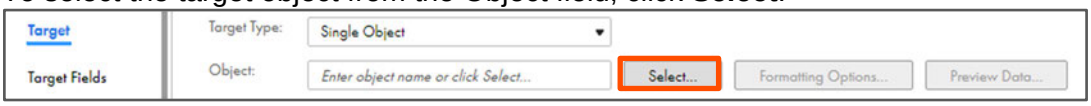
53. Link **FIL_ExpeditedShipping** to the **Target** transformation.



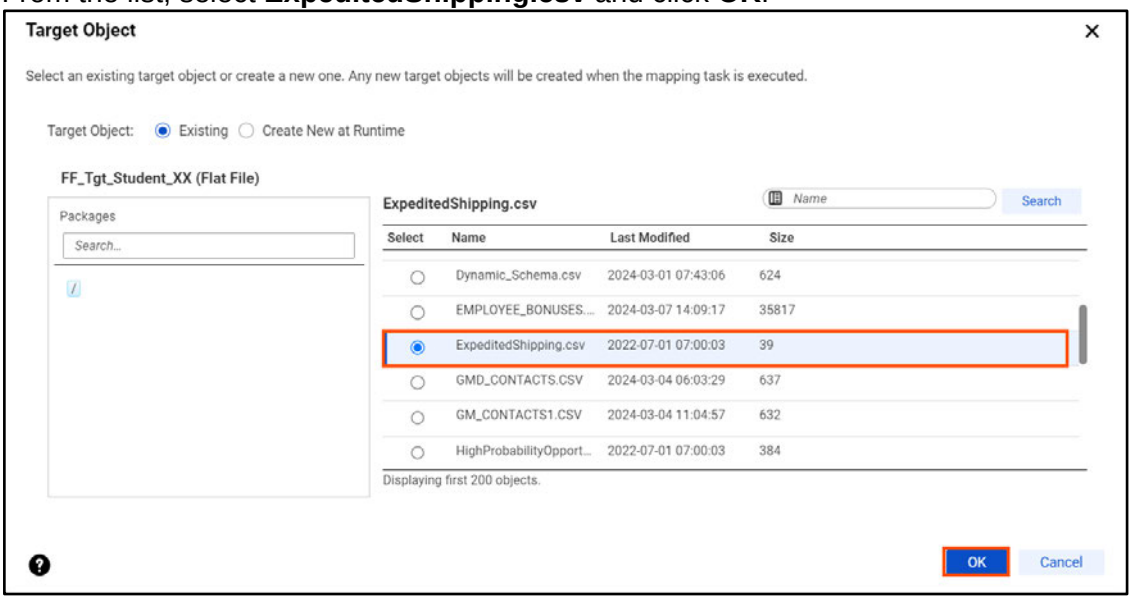
54. Select the **Target** transformation from the mapping canvas.
55. In the General section of the Target properties, enter the Name as **TG_ExpeditedShipping**.



56. From the properties pane, click **Target**.
57. From the Connection drop-down, select **your target flat file connection**.
58. Retain Target Type as **Single Object**.
59. To select the target object from the Object field, click **Select**.



60. From the list, select **ExpeditedShipping.csv** and click **OK**.



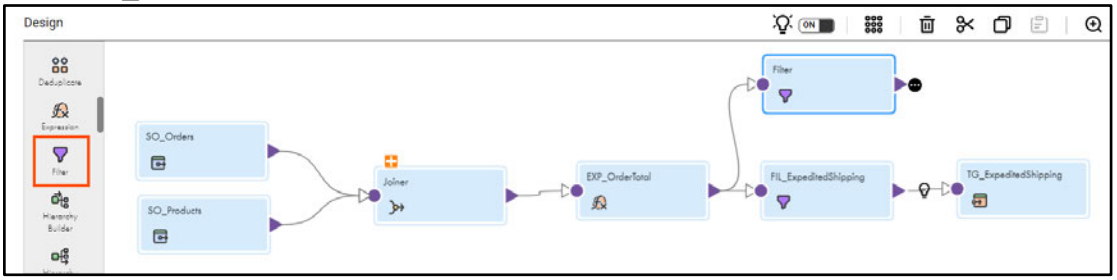
61. From the properties pane, click **Field Mapping**.
Note: The Field Mapping tab is used to map source fields with the target fields for writing the mapping output in the target object. In this case, ExpeditedShipping.csv.

62. Match the fields as shown in the following table:
    Note: Click and drag the fields to map them. Some of the fields might be mapped automatically. For the already mapped fields, do not map them again.

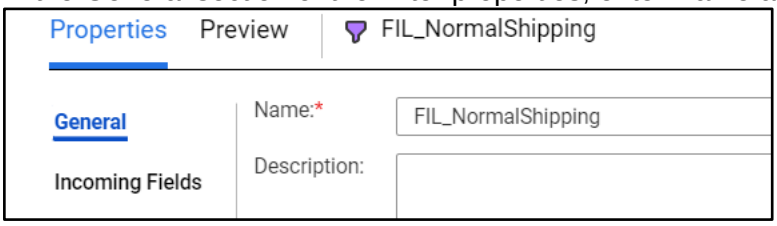
Incoming Field	Target Field
ORDERID	OrderID
PRODUCTDESCRIPTION	ProductDescription
Total	OrderTotal

Add Filter Transformation

63. From the list of available transformations, drag and drop another **Filter** transformation onto the mapping canvas.
64. Link **EXP_OrderTotal** to the **Filter** transformation.



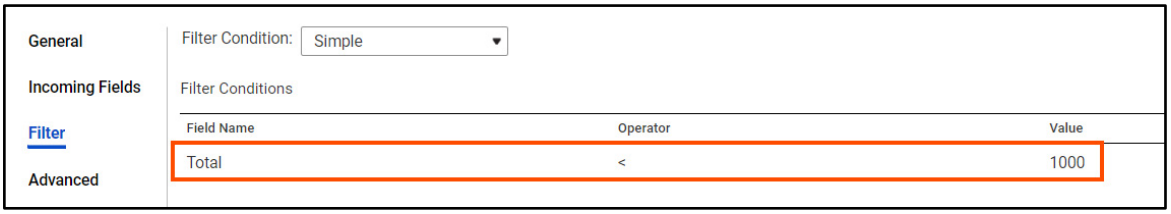
65. Select the **Filter** transformation from the mapping canvas.
66. In the General section of the Filter properties, enter Name as **FIL_NormalShipping**.



67. From the properties pane, click **Filter**.
68. To add a new filter, click +.

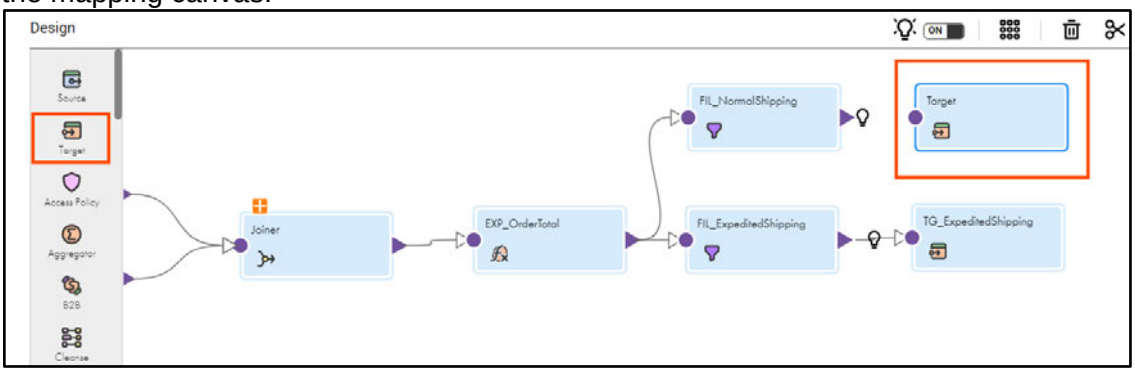
69. Enter filter condition as shown in the table below:

Field Name	Operator	Value
Total	< (Less than)	1000

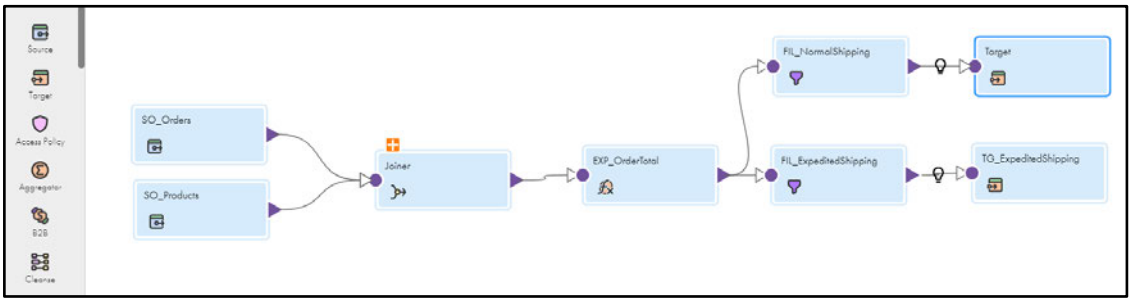


Configure Target Transformation

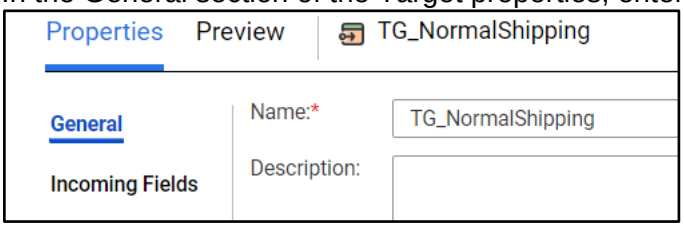
70. From the list of available transformations, drag and drop a **Target** transformation onto the mapping canvas.



71. Link **FIL_NormalShipping** to **Target** transformation.



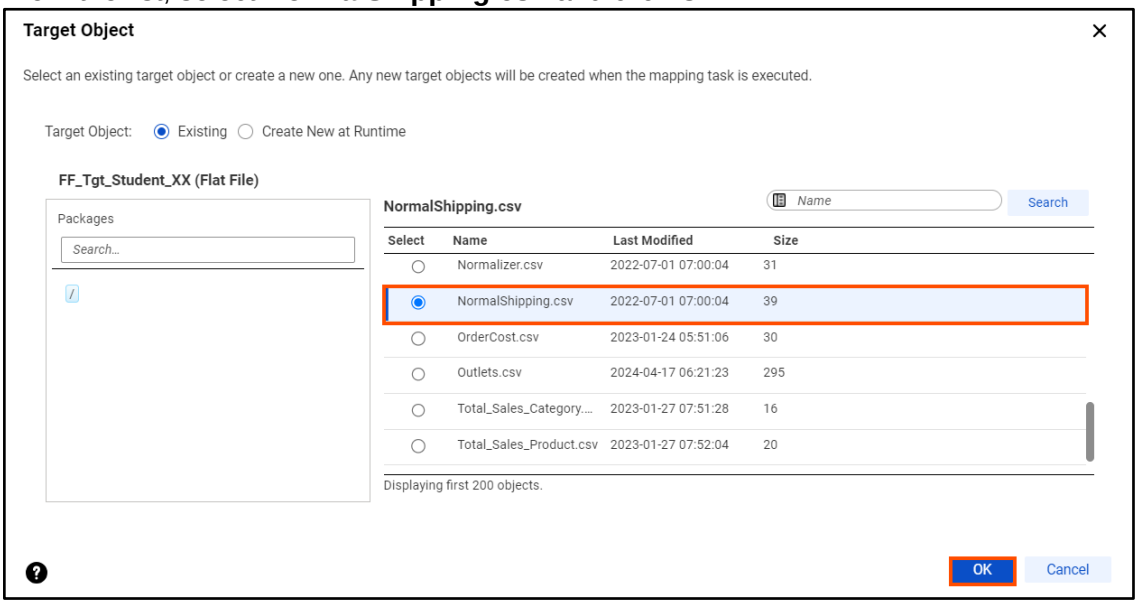
72. In the General section of the Target properties, enter Name as **TG_NormalShipping**.



73. From the properties pane, click **Target**.
74. From the Connection drop-down, select **your target flat file connection**.

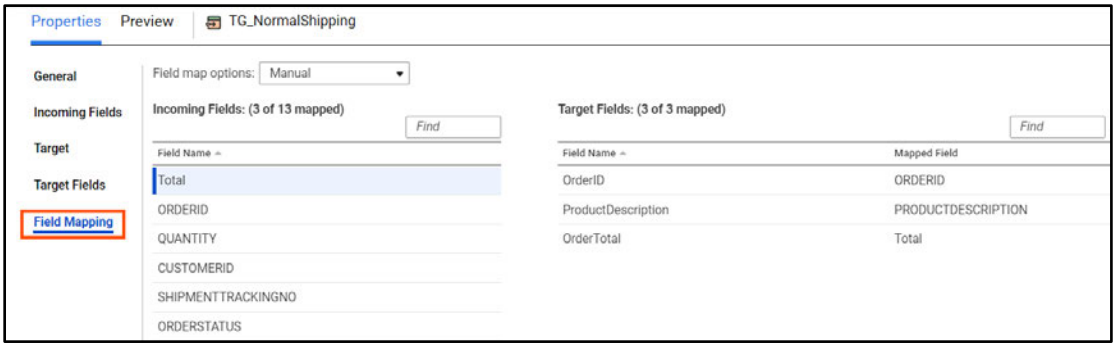
75. To select the target object from the Object field, click **Select**.

76. From the list, select **NormalShipping.csv** and click **OK**.

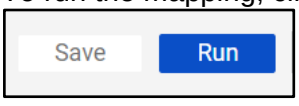


77. Match the fields as shown in the table below:

Incoming Field	Target Field
ORDERID	OrderID
PRODUCTDESCRIPTION	ProductDescription
Total	OrderTotal

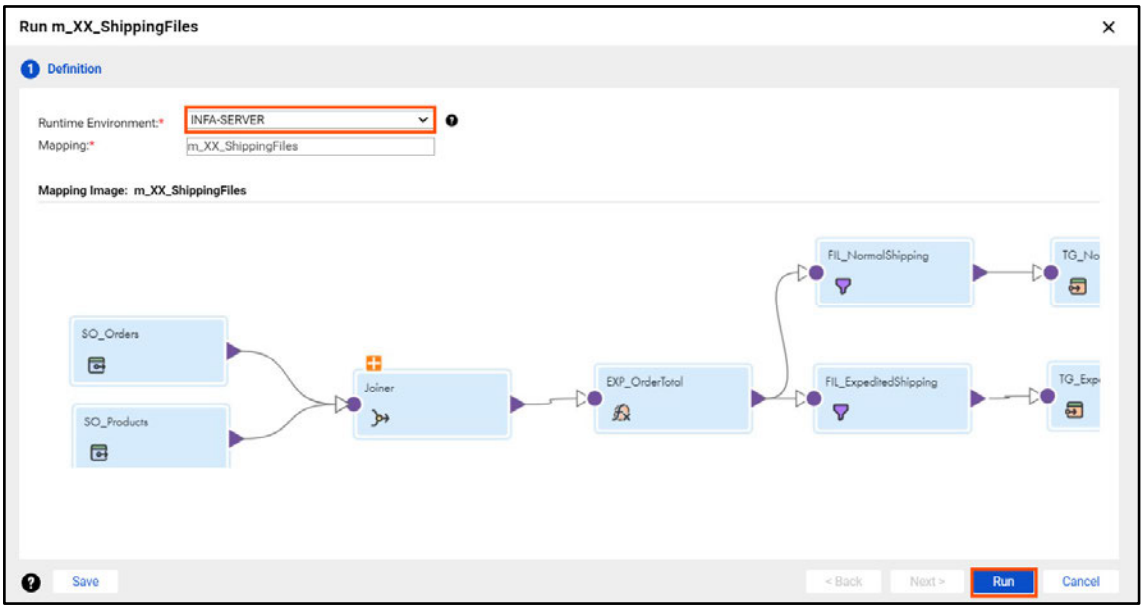


78. To save the mapping, click **Save**.
79. To run the mapping, click **Run**.



80. From the Runtime Environment drop-down, select **INFA-SERVER**.

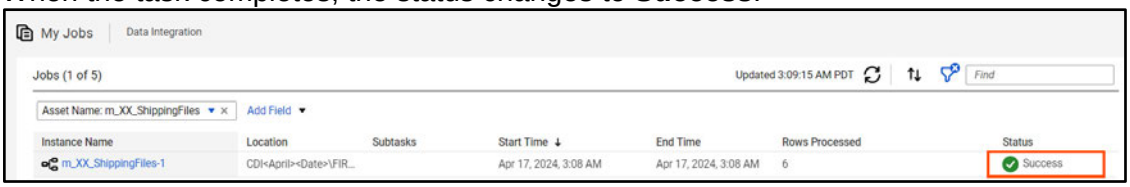
81. Click **Run**.



Note: At this point, the IICS generates a temporary Mapping Task for the mapping and runs it.

Monitor Status

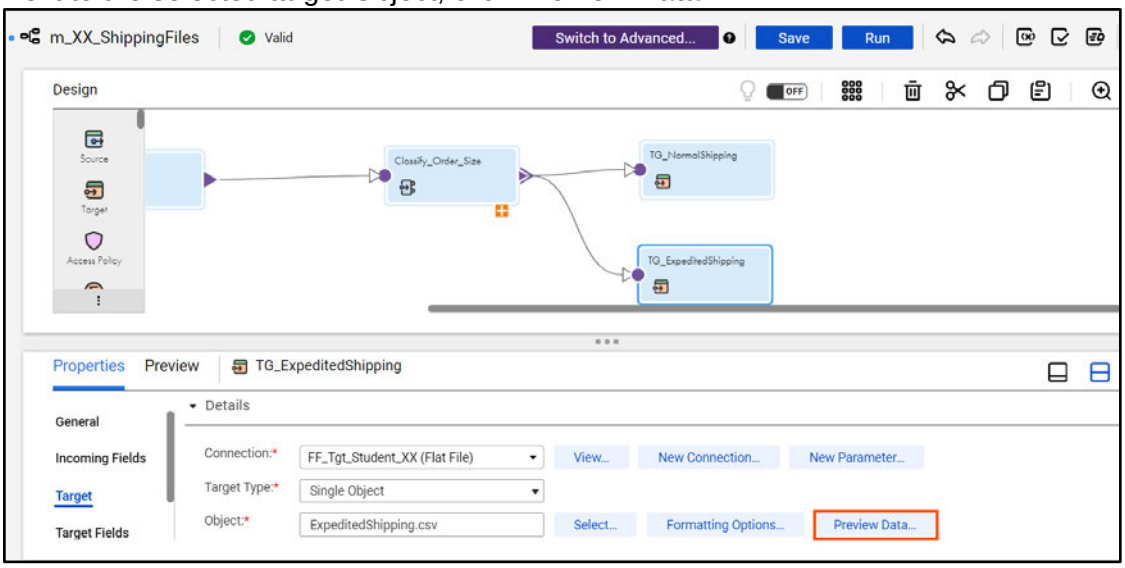
82. Monitor the task status from the **My Jobs** page.
83. When the task completes, the status changes to **Success**.



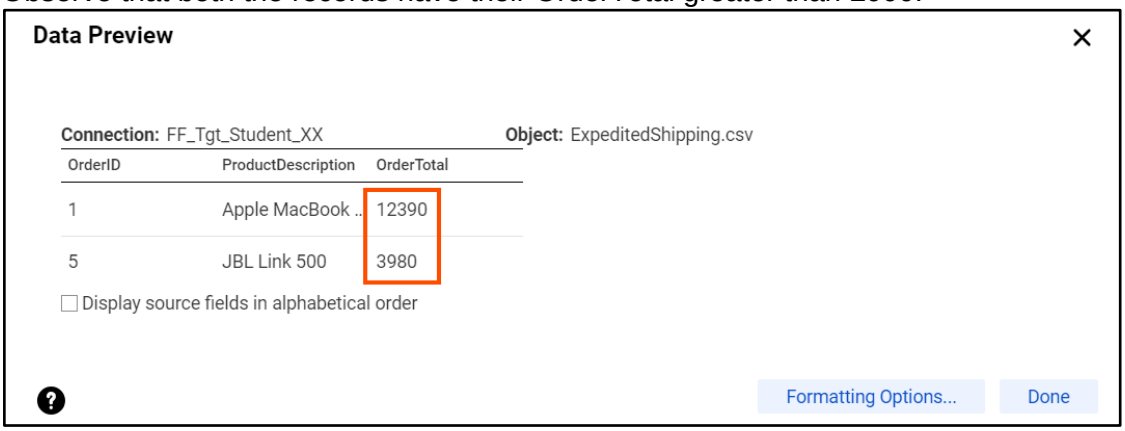
Note: You can refresh the job status if it does not change automatically.

Optional - View Target Data

84. Navigate back to your mapping **m_XX_ShippingFiles**.
85. From the mapping canvas, select the target object **TG_ExpeditedShipping** and navigate to the **Target** properties tab.
86. Next to the selected target Object, click **Preview Data**.

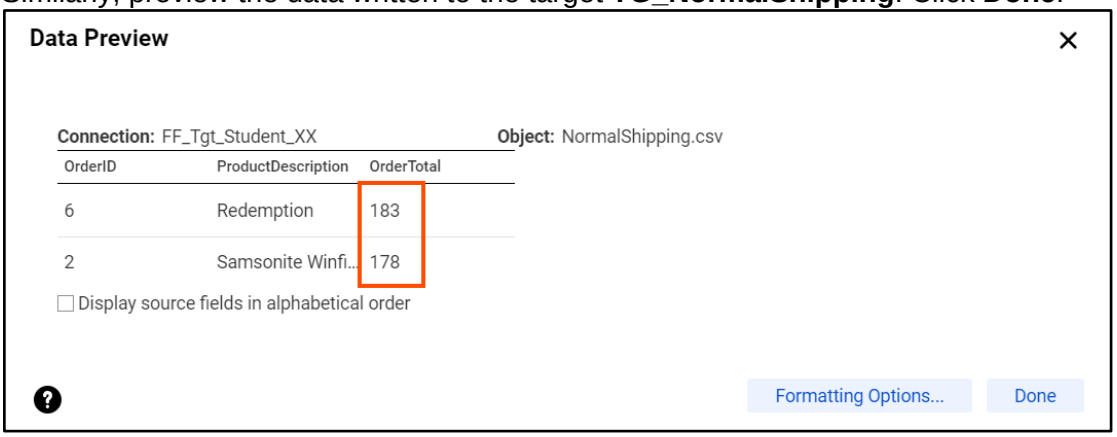


87. Observe that both the records have their OrderTotal greater than **1000**.



88. Click **Done**.

89. Similarly, preview the data written to the target **TG_NormalShipping**. Click **Done**.



90. Save and close the asset from the navigation pane.

This concludes the lab.

Module 6: IICS Transformations

Lab 6-2: Using Union, Sorter, and Rank Transformations in a Mapping

Overview:
The Union transformation is used to read data from multiple pipelines and merge data into a single pipeline. You can use the Rank transformation to return the largest or smallest numeric values in a group. Sorter transformation sorts the data based on the incoming fields either in ascending or descending order.

Objective:

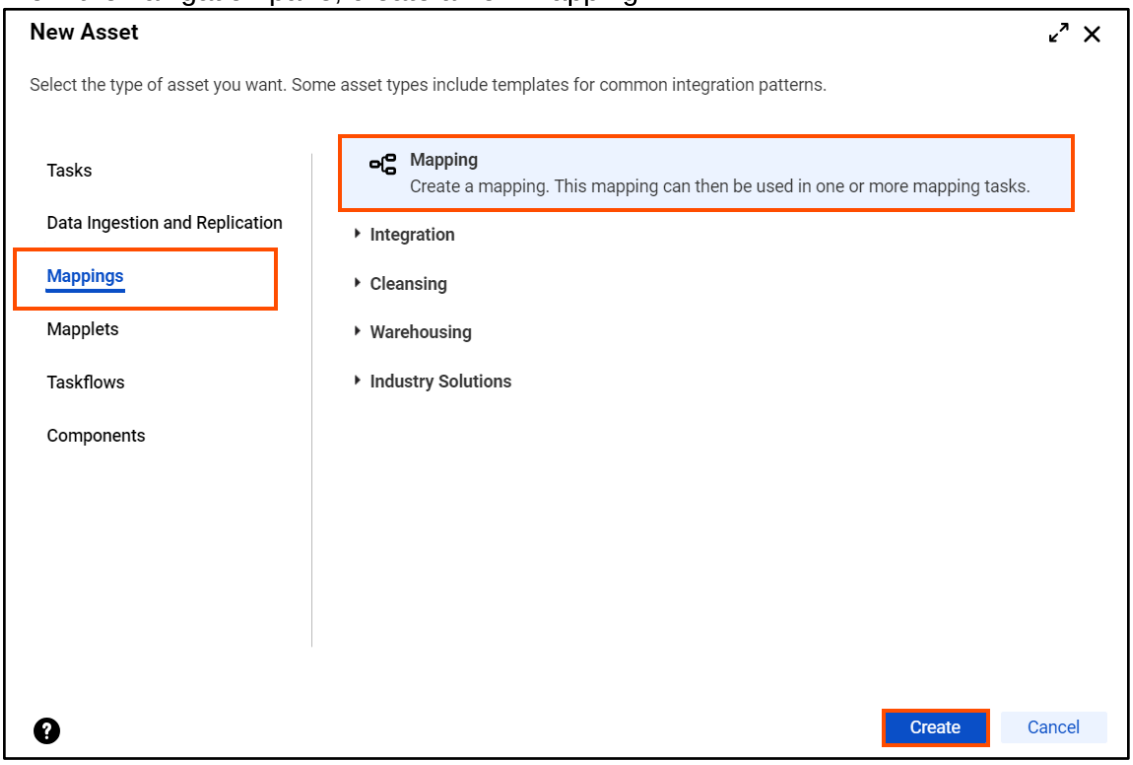
- Create a Mapping using the Union, Rank, and Sorter Transformations

Duration:
15 minutes

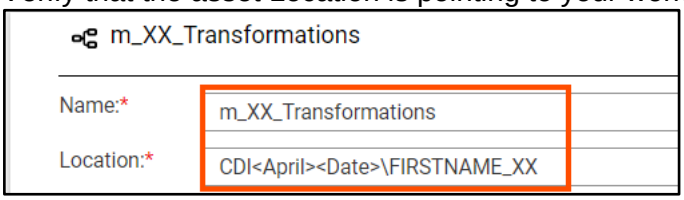
Tasks

Create Mapping

1. Login into IICS and from the My Services window, select **Data Integration**.
2. From the navigation pane, create a new mapping.

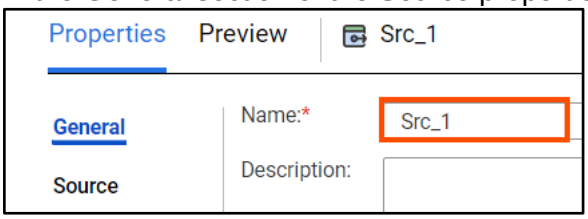


3. In the Name field, enter **m_XX_Transformations**. Replace XX with your DOB.
4. Verify that the asset Location is pointing to your working folder.

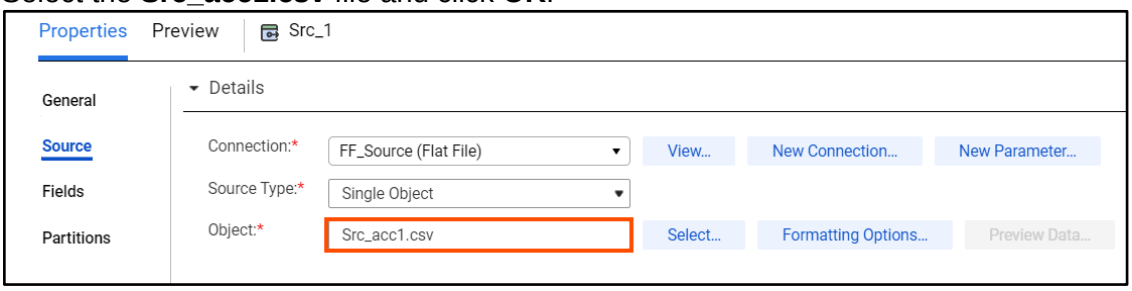


Configure Source Transformation

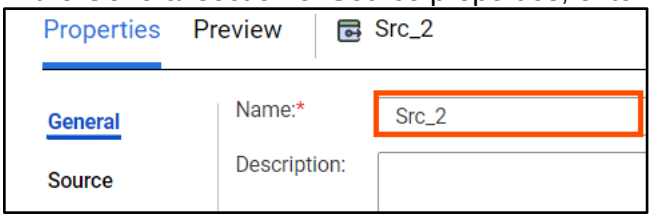
5. Delete the existing link between the source and the target objects in the mapping canvas.
6. To configure the source, from the mapping canvas, click the **Source** transformation.
7. In the General section of the Source properties, enter Name as **Src_1**.



8. From the properties pane, click **Source**.
9. From the Connection drop-down, select **FF_Source**.
10. From the Object Type field, click **Select**.
11. Select the **Src_acc1.csv** file and click **OK**.



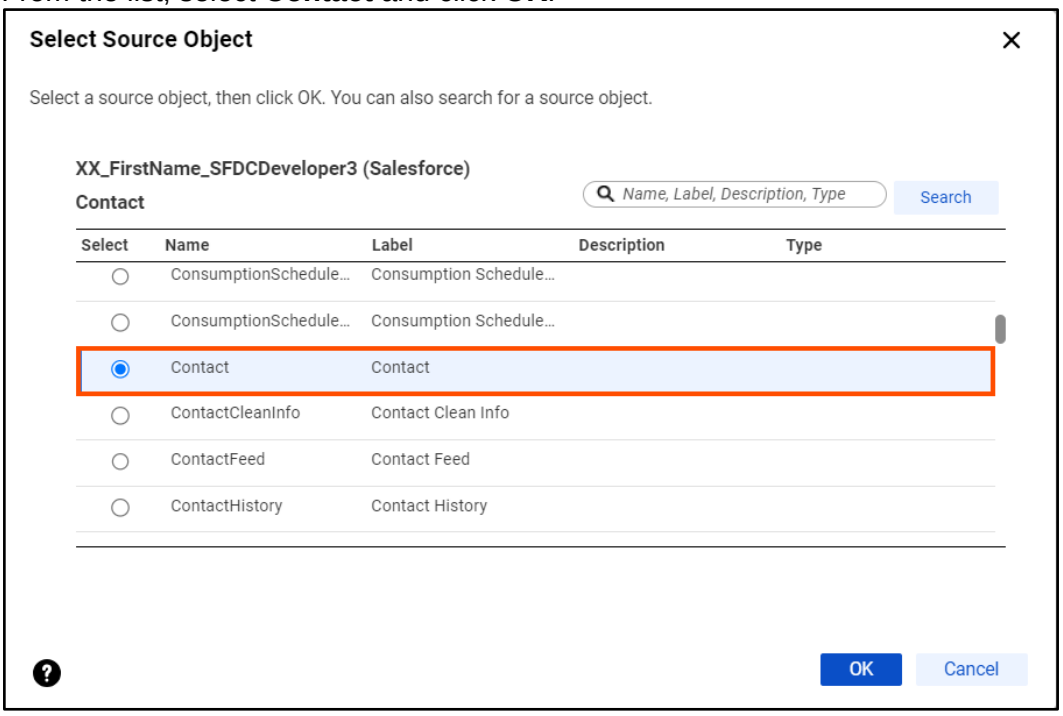
12. Drag and drop another source transformation onto the mapping canvas.
13. In the General section of Source properties, enter Name as **Src_2**.



14. From the properties pane, click **Source**.

15. From the Connection drop-down select your **Salesforce** Connection.

16. Retain Source Type as **Single Object**.
17. To select the source object from the Object field, click **Select**.
18. From the list, select **Contact** and click **OK**.

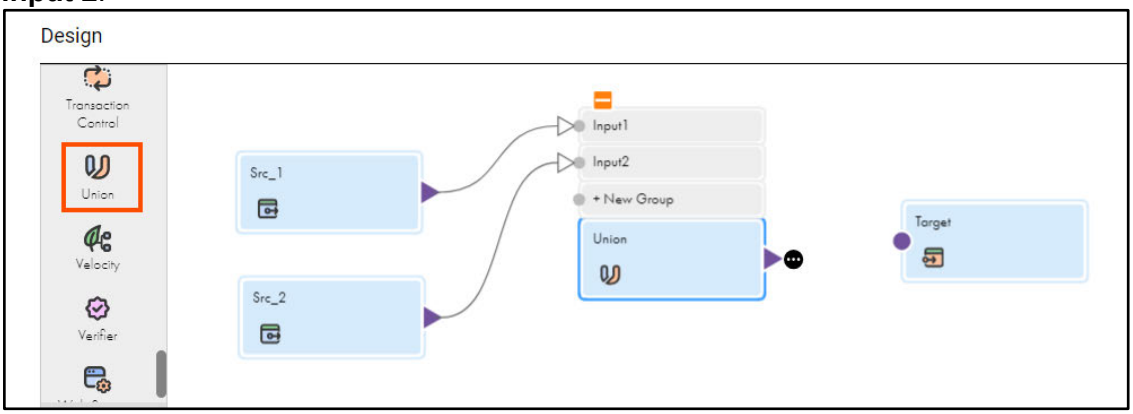


Add Union Transformation

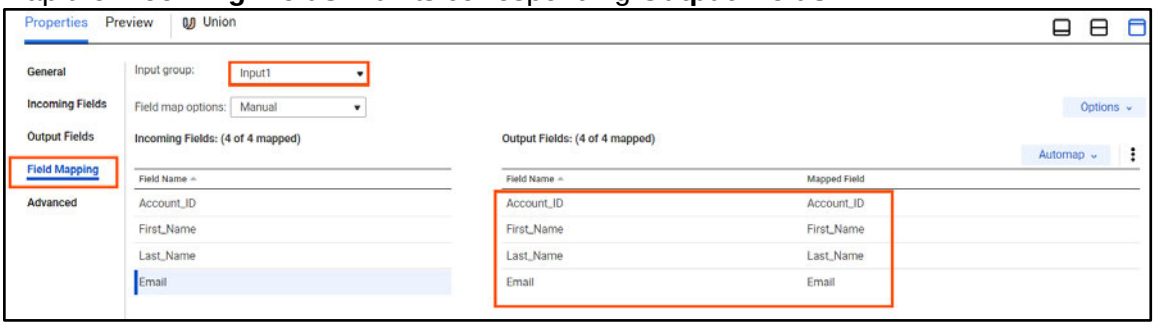
The union is used to combine two or more data sources into a single stream that includes the union of all rows.

19. Drag and drop Union transformation onto the mapping canvas.

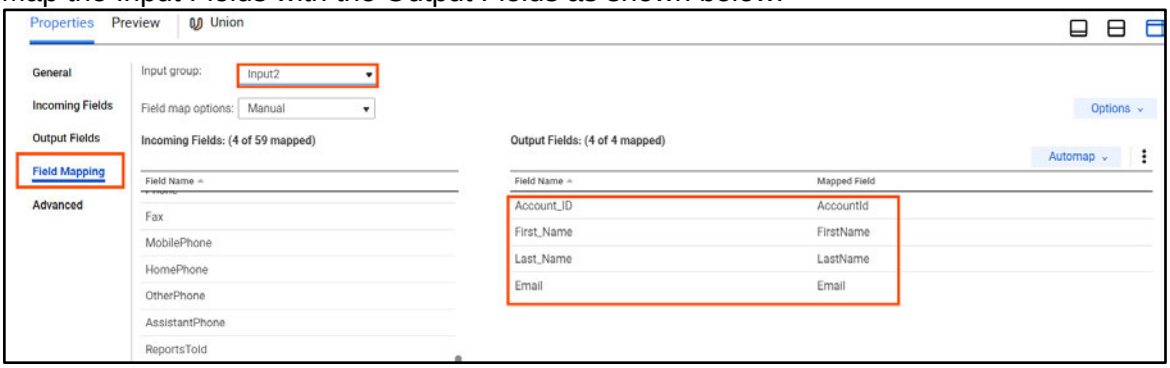
20. Click on the **+** sign of the Union transformation and join **Src_1** to **Input1** and **Src_2** to **input 2**.



21. Select the Union transformation and from the properties pane select **Field Mapping**.
22. In the Input group, select **Input1**.
23. Map the **Incoming Fields** with its corresponding **Output Fields**.

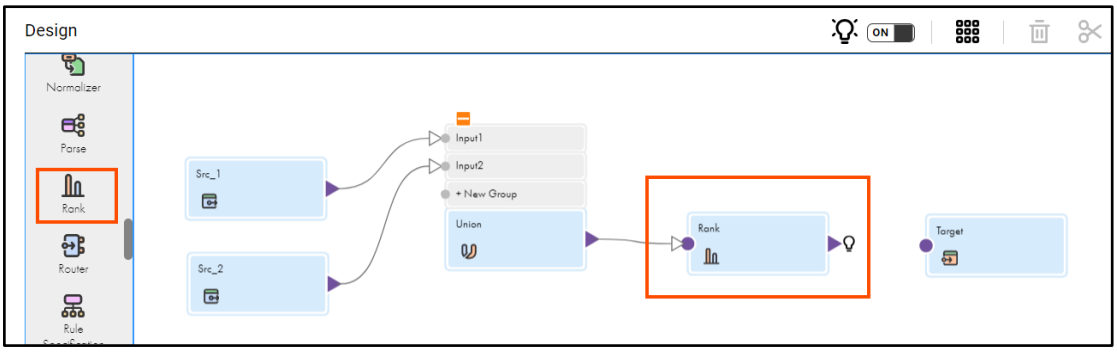


24. Similarly, from the Input group drop-down, select **Input2**.
25. Map the Input Fields with the Output Fields as shown below:

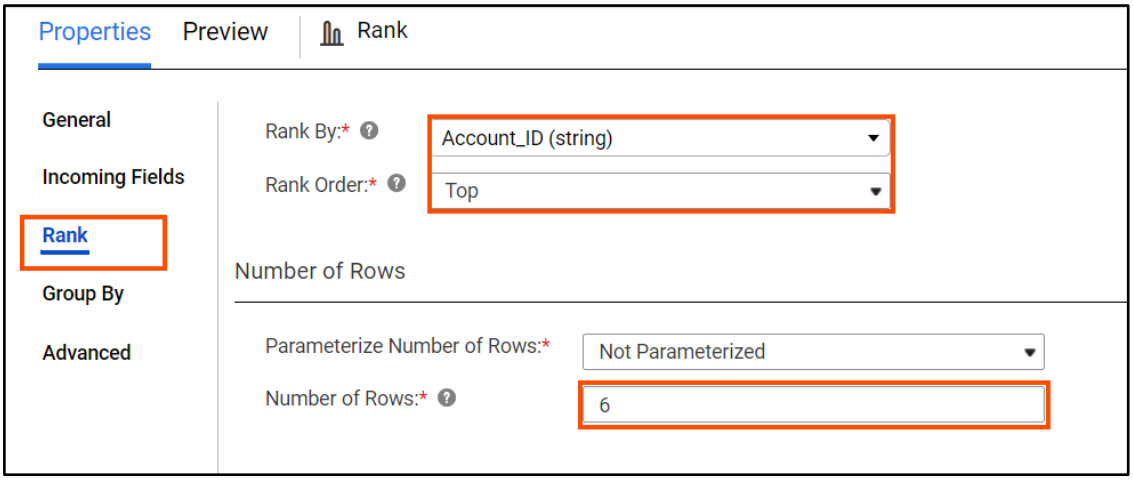


Add Rank Transformation

26. From the list of available transformations, drag and drop a **Rank** transformation onto the mapping canvas.
27. Join the **Union** and the **Rank** Transformation.

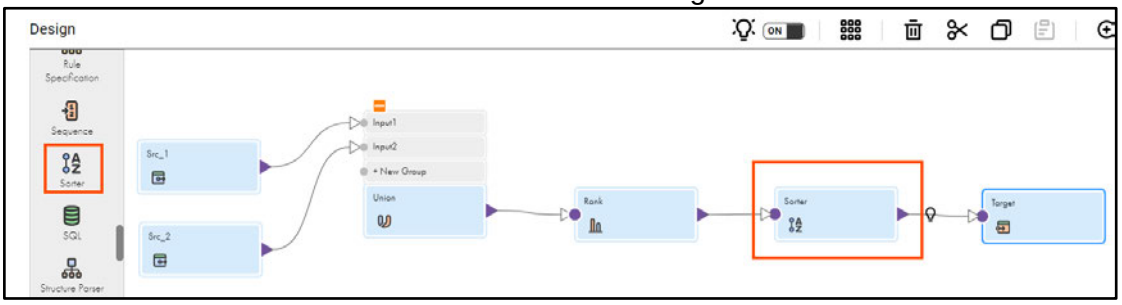


28. Select the **Rank** transformation on the mapping canvas.
29. From the properties pane, click **Rank**.
30. From the Rank By drop-down, select **Account_ID**
31. Retain the Rank Order as **Top** and set the Number of Rows as **6**.



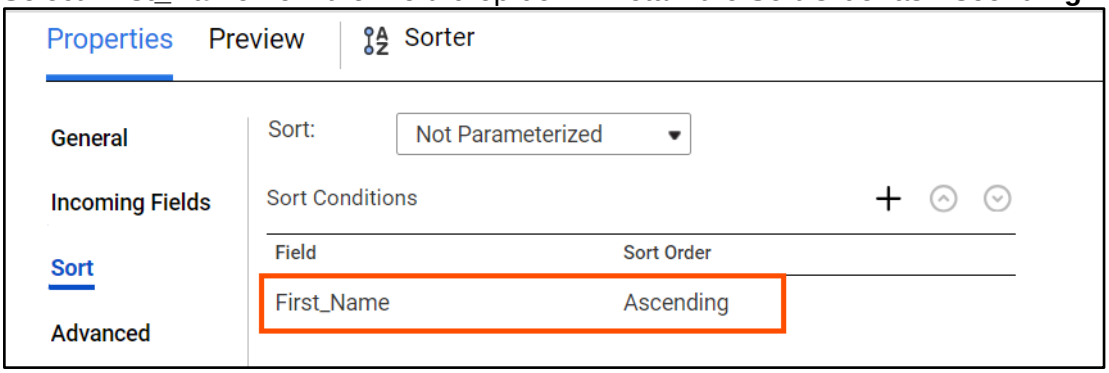
Add Sorter Transformation

32. From the list of available transformations, drag and drop a **Sorter** transformation onto the mapping canvas.
33. Join the **Sorter** transformation with the Rank and Target transformation.



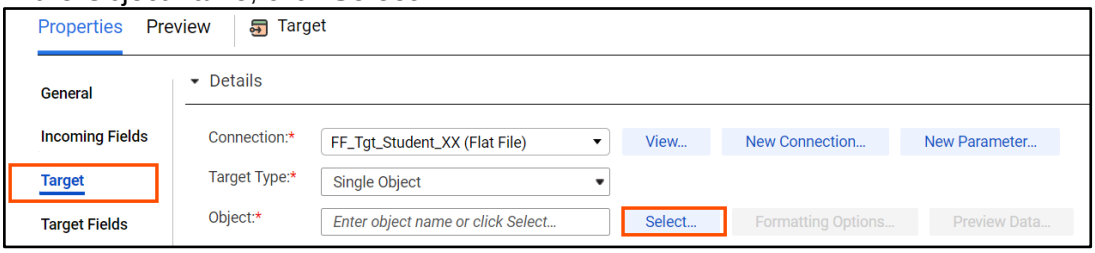
34. Select the **Sorter Transformation** from the mapping canvas.
35. From the properties pane, select **Sort**.
36. From the Sort Conditions, select the Add icon.

37. Select **First_Name** from the **Field** drop-down. Retain the Sort Order as **Ascending**.



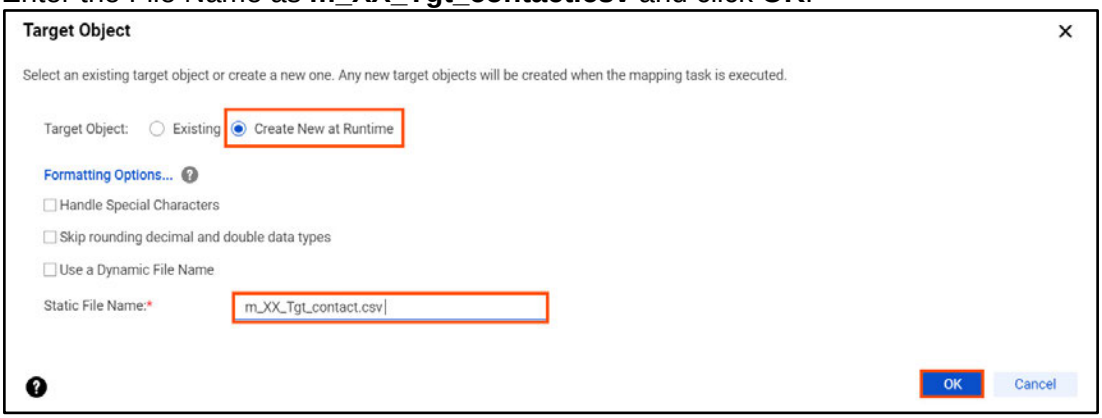
Configure the Target transformation

38. Select the Target transformation on the mapping canvas.
39. From the properties pane, select **Target**.
40. From the connection drop-down, select the connection as your **target flat file** connection.
41. Retain the Target Type as Single Object.
42. In the Object name, click **Select**.



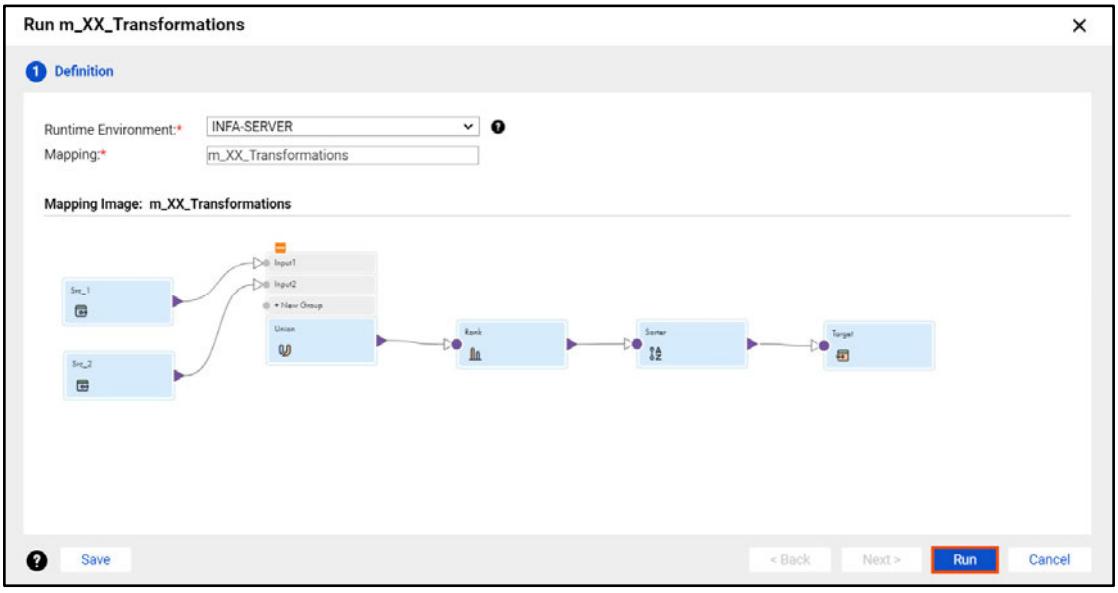
43. In the Target Object window, Select the **Create New at Runtime** option.

44. Enter the File Name as **m_XX_Tgt_contact.csv** and click **OK**.



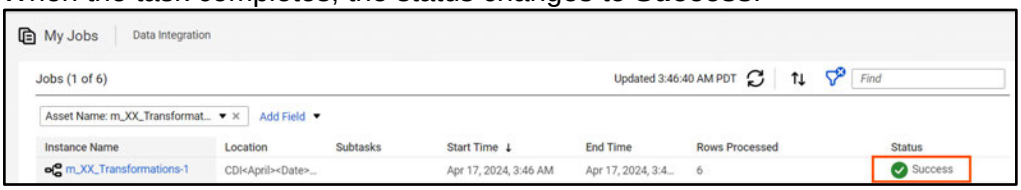
Extra Information: On the mapping canvas, towards the bottom of the space, you will see a floating icon . This is called the Birds Eye View. Click on the icon, and scroll up and down on the canvas page. This will help you to provide a better and clear visibility of all your mapping assets. Using this option, you can position your mapping assets so that you can locate and view your mapping. Click the cross mark to close the Birds Eye View window.

45. **Save** and **Run** the Mapping.
46. Select the Runtime Environment as **INFA-SERVER** and click **Run**.



Monitor Status

47. Monitor the task status from the **My Jobs** page.
48. When the task completes, the status changes to **Success**.

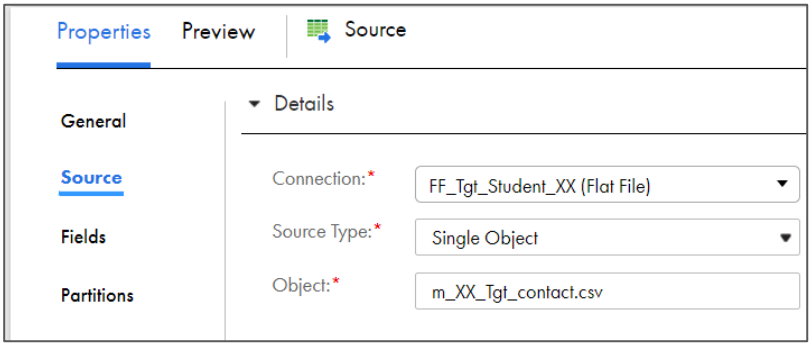


Note: You can refresh the job status if it does not change automatically.

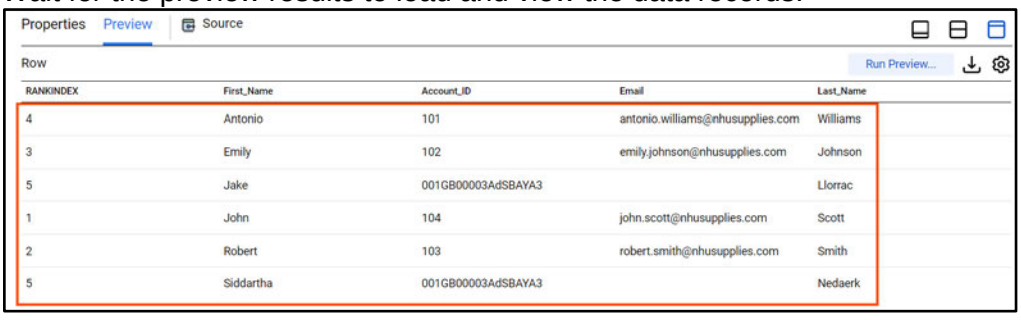
49. Close the asset from the navigation pane.

Optional - Open the Dummy Mapping and View the Data

50. Navigate to your working folder and open the dummy mapping **m_XX_DUMMY_MAPPING**.
51. From the mapping canvas, select **Source** and go to the **Source** properties tab.
52. Change the Connection to your target Flat File connection. Click **Yes** in the Change Connection window.
53. Select the Object as your target – **m_XX_Tgt_Contact.csv**.



54. Go to the Preview section and click **Run Preview**. In the Continue window, click **Yes**.
55. If prompted, select runtime environment as **INFA-SERVER** and click **Run Preview**.
56. Wait for the preview results to load and view the data records.



57. Save and close all the assets from the navigation pane.

This concludes the lab.

Module 6: IICS Transformations

Lab 6-3: Using Expression Macro in a Mapping

Overview:

An expression macro is a macro that you use to create repetitive or complex expressions in mappings. You can use an expression macro to perform calculations across a set of fields or constants. For example, you might use an expression macro to replace null values in a set of fields or to label items based on a set of sales ranges. In an expression macro, one or more input fields represent source data for the macro. An expression represents the calculations that you want to perform. And an output field represents the results of the calculations. At run time, the task expands the expression to include all the input fields and constants, and then writes the results to the output fields.

Objective:

- Use an Expression Macro in a mapping

Scenario:

In one of her review meetings, Ruby observed that the date format for NH Digital sales data was different from other outlets. Due to this reason, the quarterly sales report was showing incorrect results. So, Ruby asks John to use the NH Digital data and convert the data to the MM/DD/YYYY format.

In this lab, John uses the IICS expression macros to define an expression to convert the dates into the correct format.

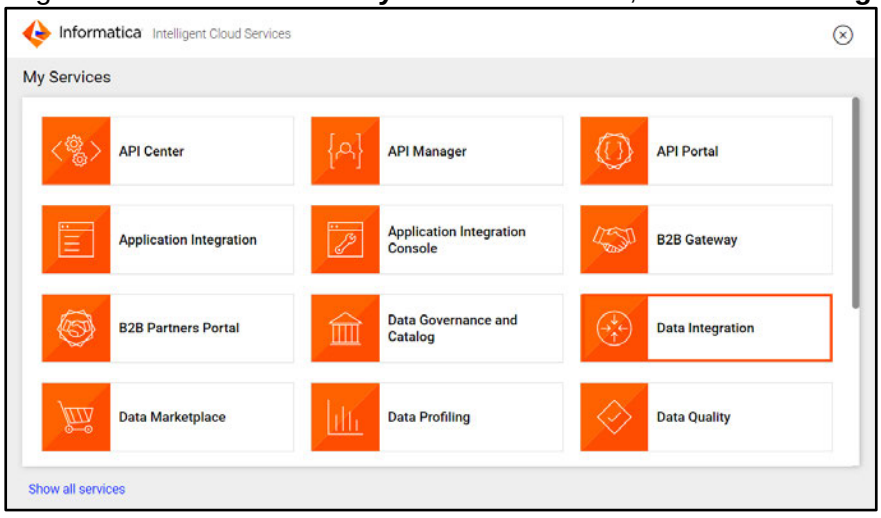
Duration:

20 minutes

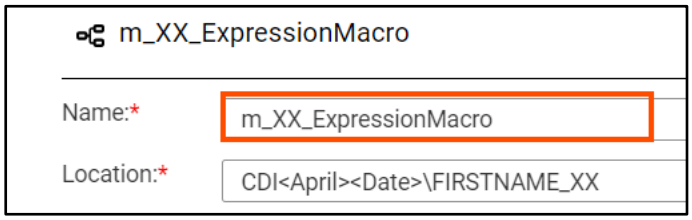
Tasks

Create Mapping

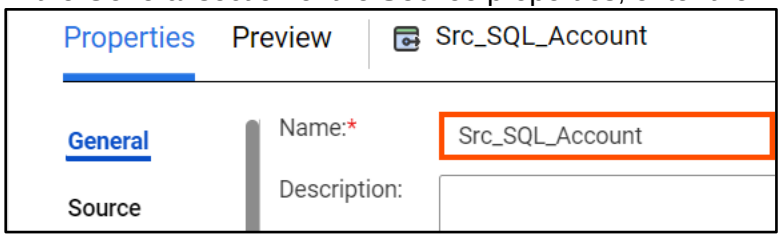
1. Login to IICS and from the **My Services** window, select **Data Integration**.



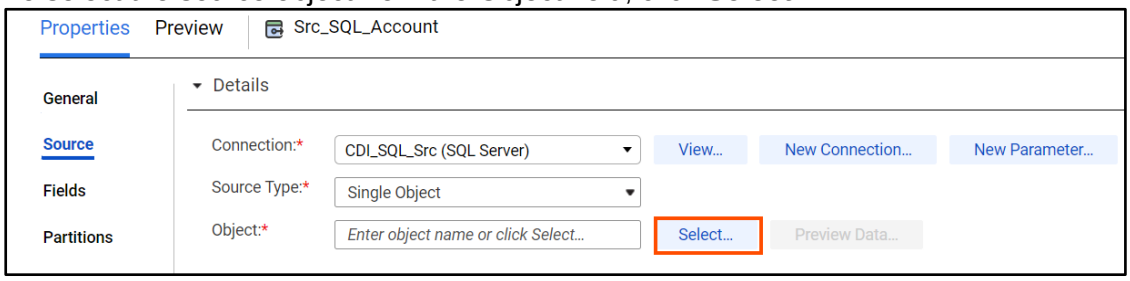
2. Create a new Mapping and name it as **m_XX_ExpressionMacro**. Replace XX with your DOB.



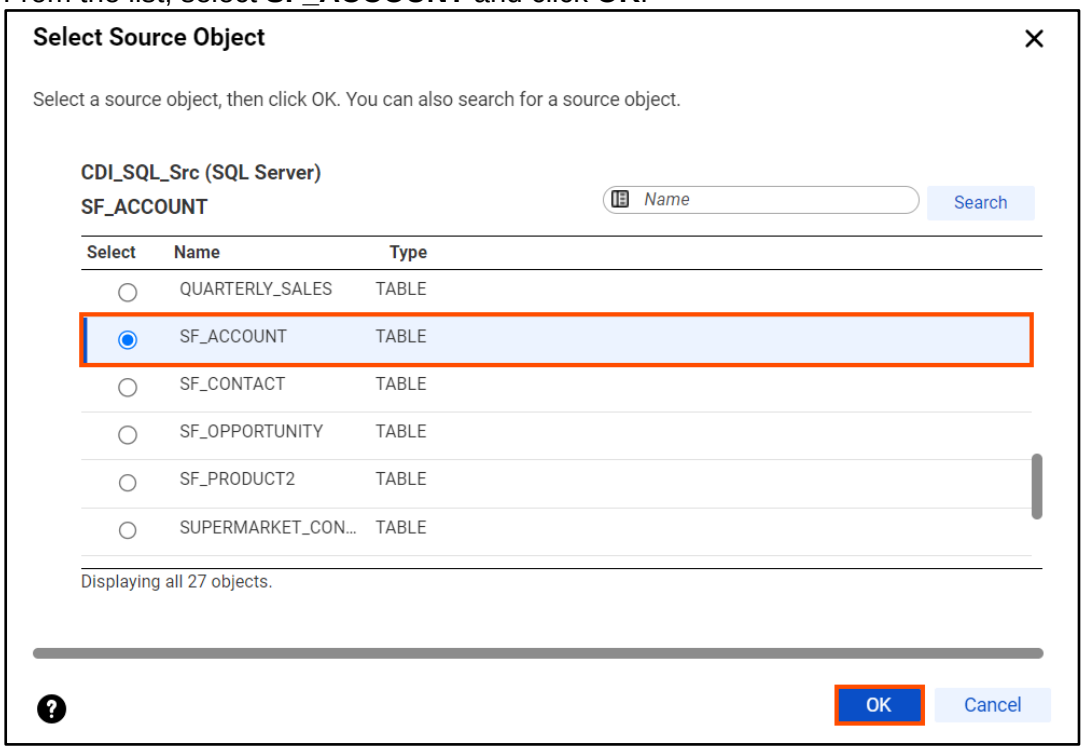
3. Verify that the asset is being saved to your working folder.
4. To configure the source, from the mapping canvas, click the **Source** transformation.
5. In the General section of the **Source** properties, enter the Name as **Src_SQL_Account**.



6. From the properties pane, click **Source**.
7. From the Connection drop-down, select the Source connection as **CDI_SQL_Src**.
8. Retain the Source Type as Single Object.
9. To select the source object from the Object field, click **Select**.

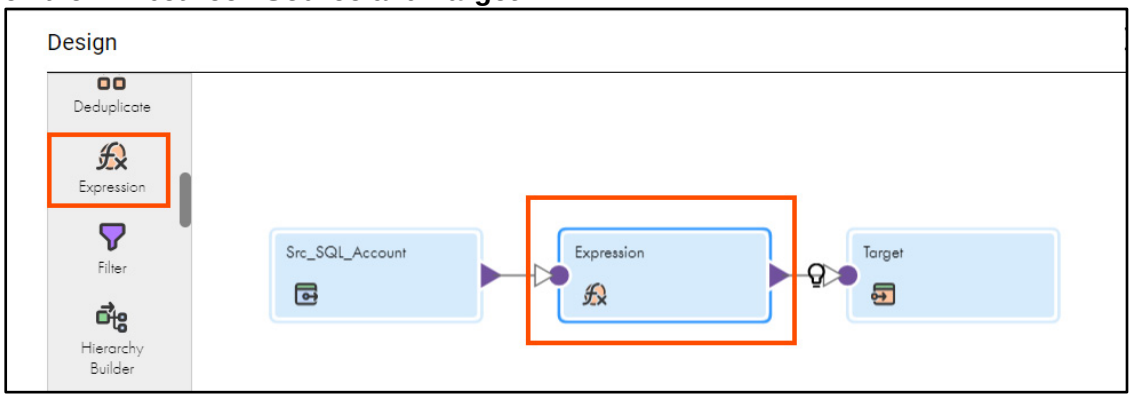


10. From the list, select **SF_ACCOUNT** and click **OK**.



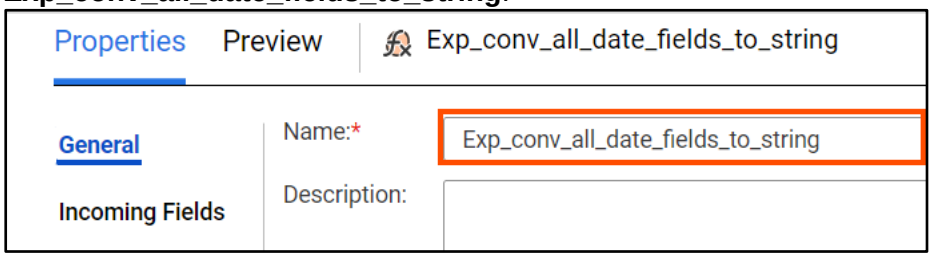
Add Expression Transformation

11. From the list of available transformations, drag and drop an **Expression** transformation on the link between **Source** and **Target**.

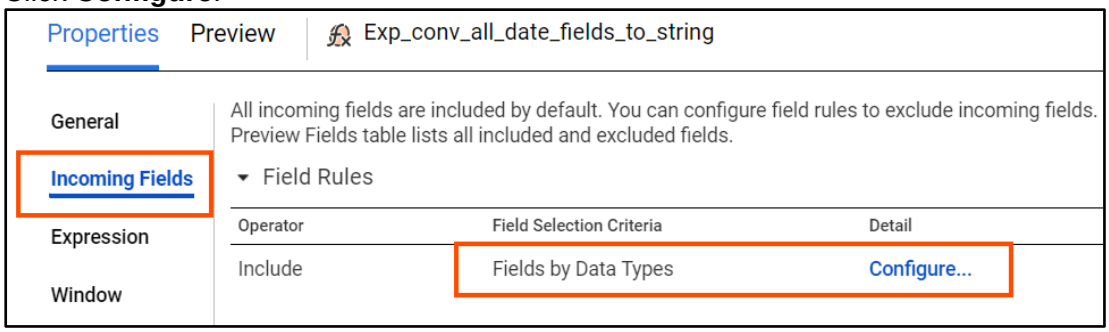


12. Select the **Expression** transformation from the mapping canvas.

13. In the **General** section of the Expression properties, enter the Name as **Exp_conv_all_date_fields_to_string**.



14. From the properties pane, click **Incoming Fields**.
15. In the Field Rules section, from Field Selection Criteria drop-down, select **Fields by Data Types**.
Note: You might need to click outside of Fields by Data Types drop-down, in a black space to enable the Configure option.
16. Click **Configure**.

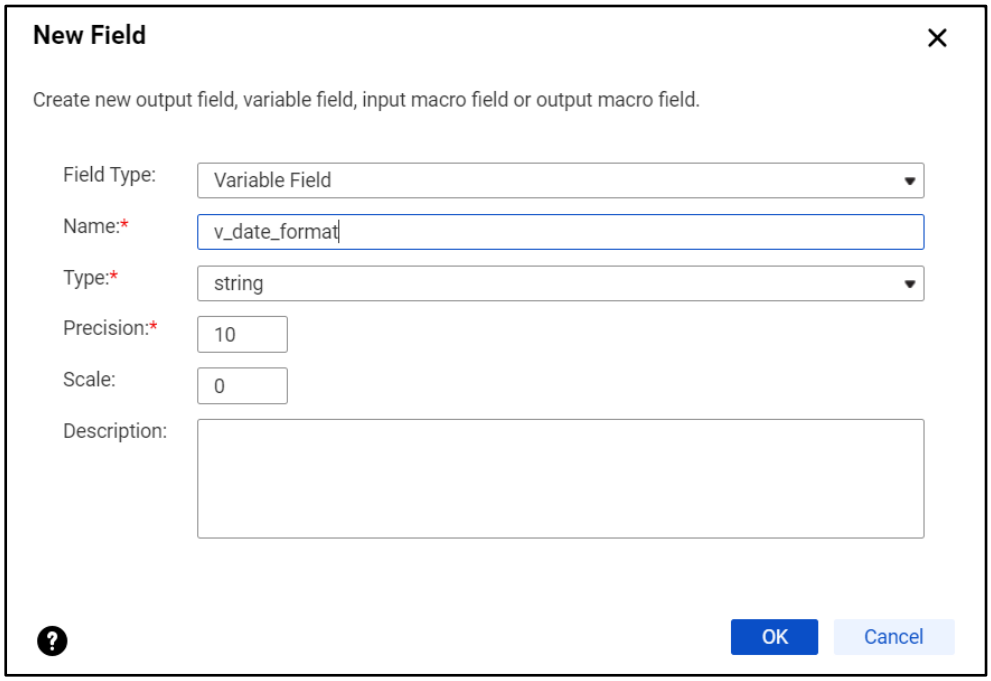


17. Select **date/time** and click **OK**.

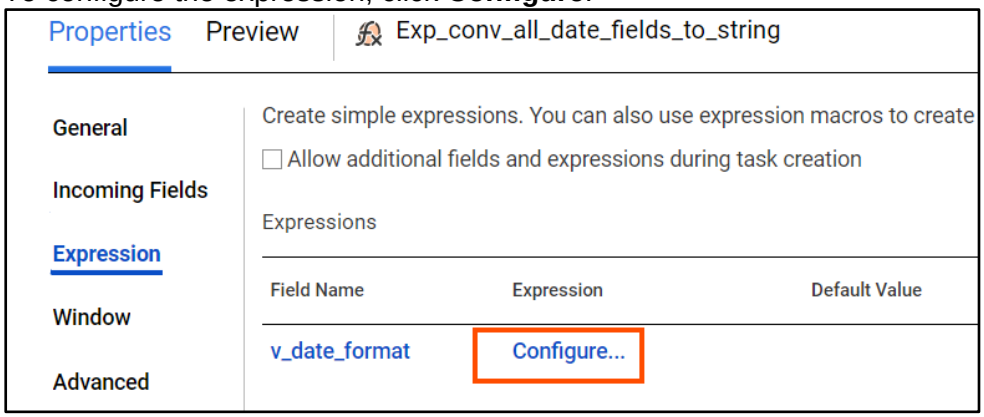
18. From the properties pane, click **Expression**.
19. To add a new expression, click +.

20. Enter the details as shown in table below and click **OK**.

Field Type	Name	Type	Precision	Scale
Variable Field	v_date_format	string	10	0



21. To configure the expression, click **Configure**.



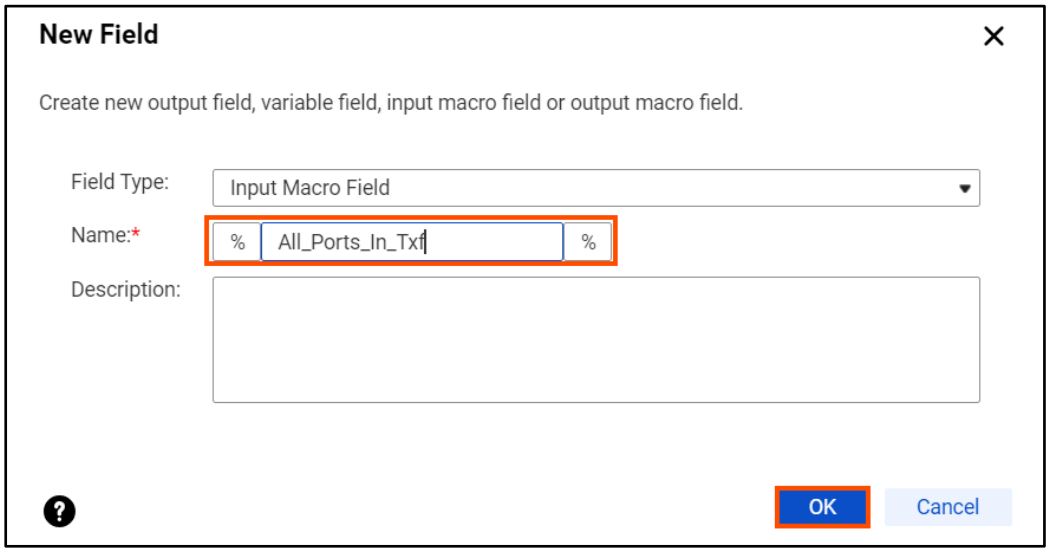
22. In the Expression field, enter the following expression:
'mm/dd/yyyy'

23. Validate the expression and click **OK**.

Note: If you copy and paste the above expression, you need to verify the expression to avoid getting an apostrophe error.

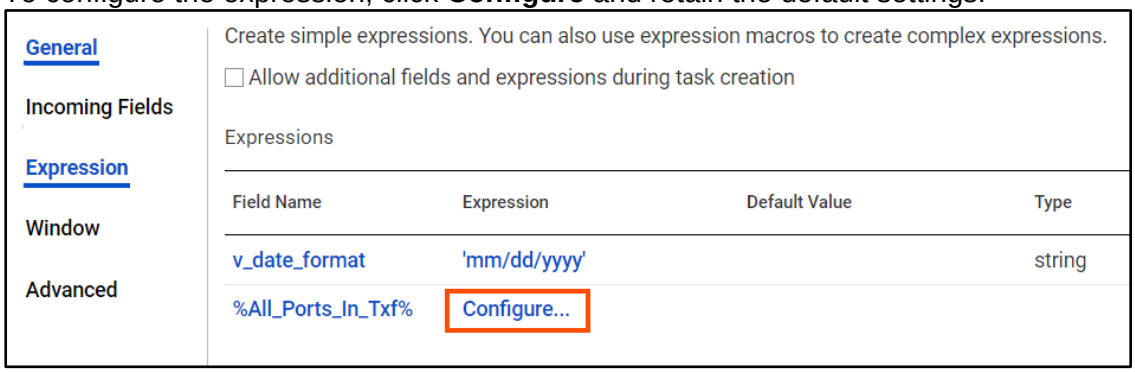
24. Add another expression field as shown in the table below and click **OK**.

Field Type	Name
Input Macro Field	All_Ports_In_Txf

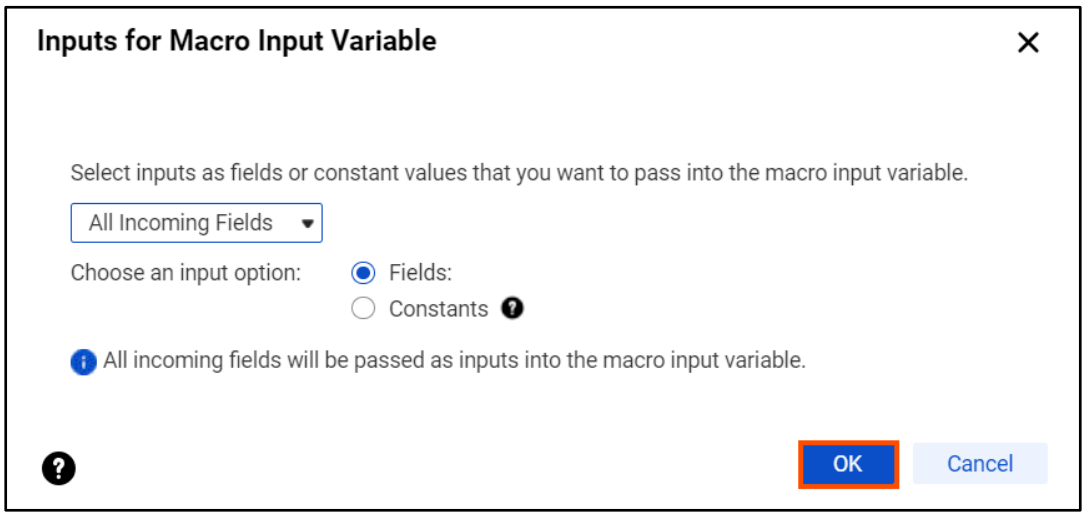


Note: A macro input field is a field that represents input that you want to use in the expression macro.

25. To configure the expression, click **Configure** and retain the default settings.

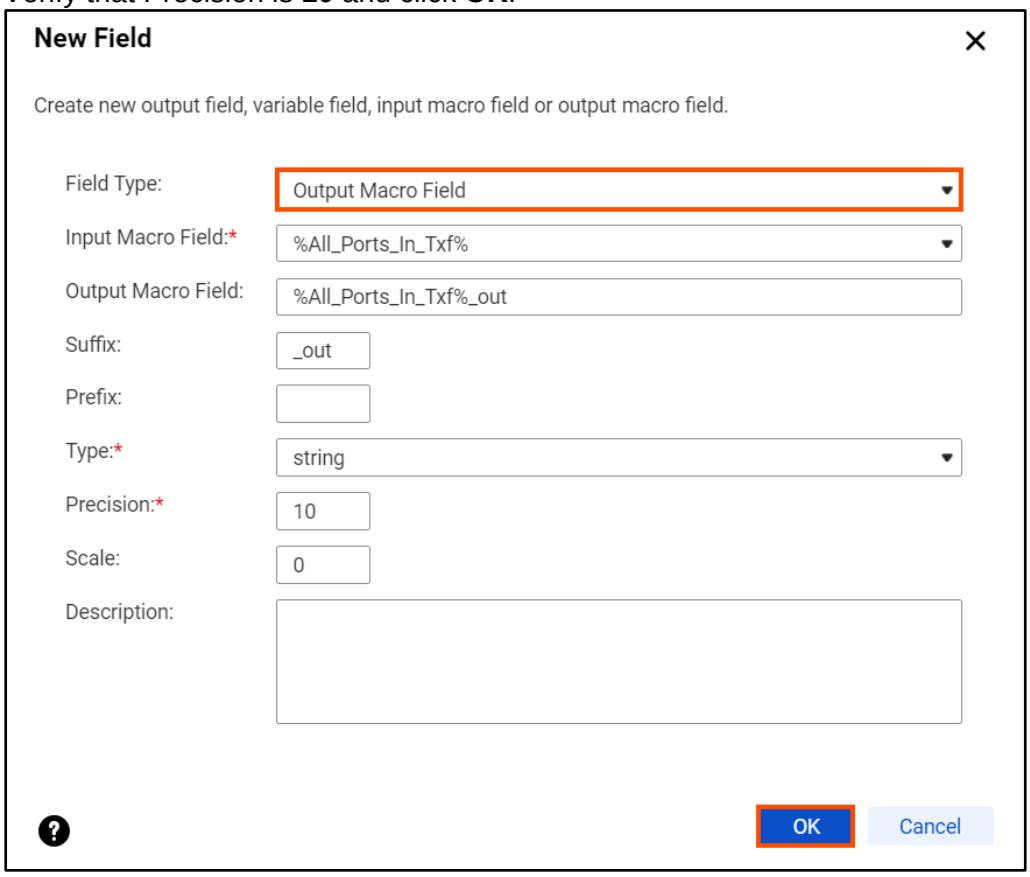


26. Click **OK**.

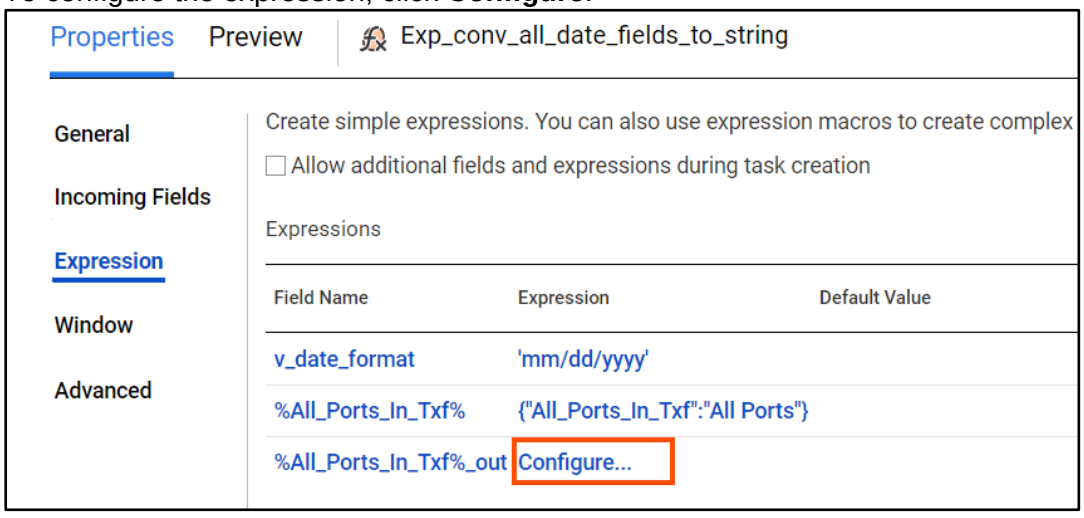


27. To create a new macro-output field, click +.
28. From the Field Type drop-down, select **Output Macro Field**.

29. Verify that Precision is **10** and click **OK**.

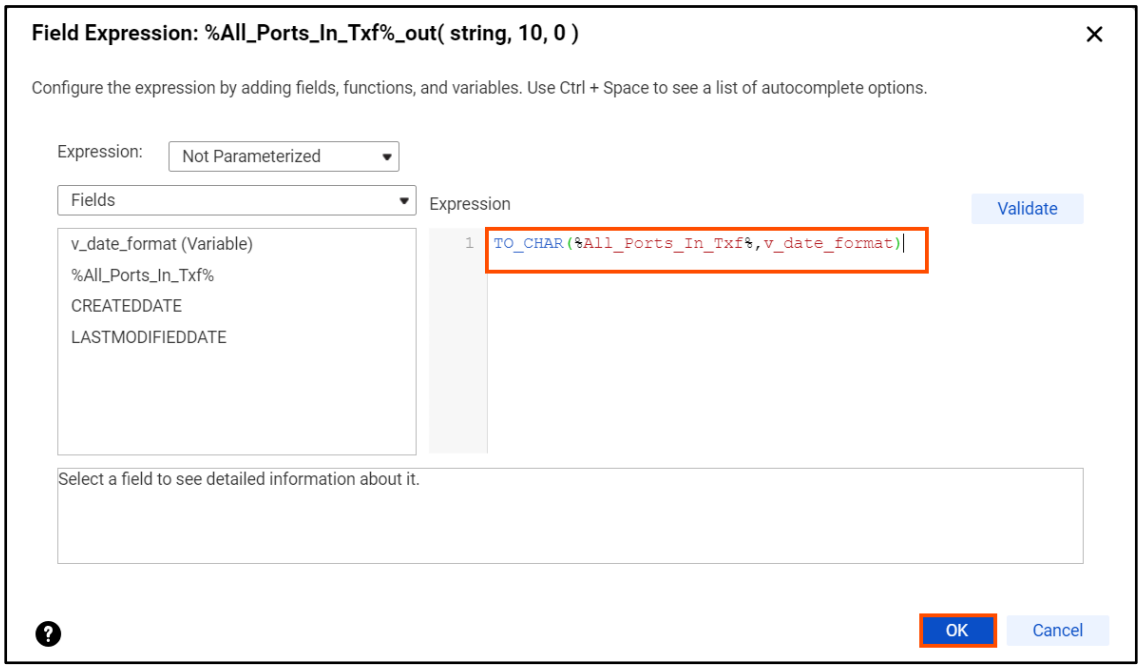


30. To configure the expression, click **Configure**.



31. In the Expression field, enter the following expression:
TO_CHAR(%All_Ports_In_Txf%,v_date_format)

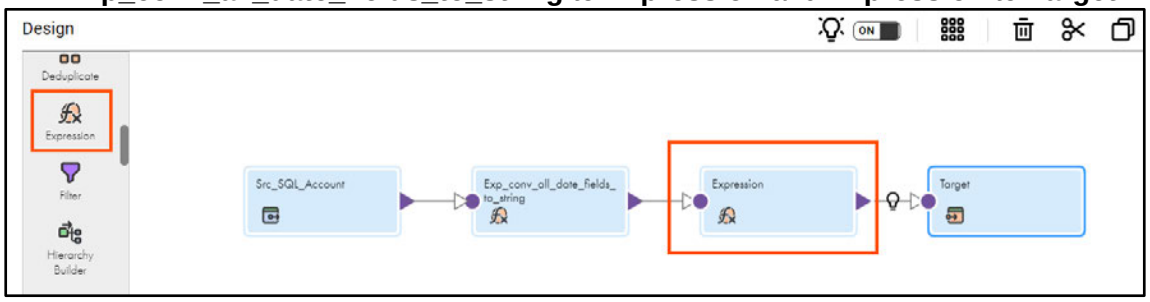
32. Click **OK**.



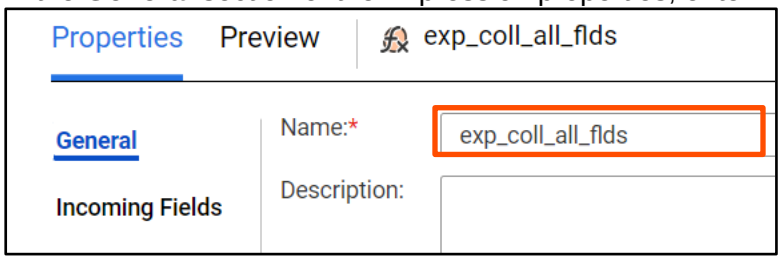
Note: You cannot validate a macro field's expression.

Add Expression Transformation

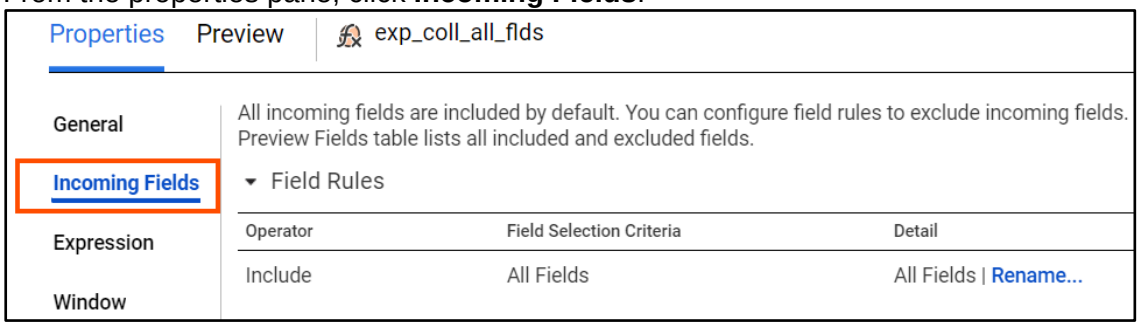
33. From the list of available transformations, drag and drop an **Expression** transformation on to the mapping canvas.
34. Delete the existing link between **Exp_conv_all_date_fields_to_string** and **Target**.
35. Link **Exp_conv_all_date_fields_to_string** to **Expression** and **Expression** to **Target**.



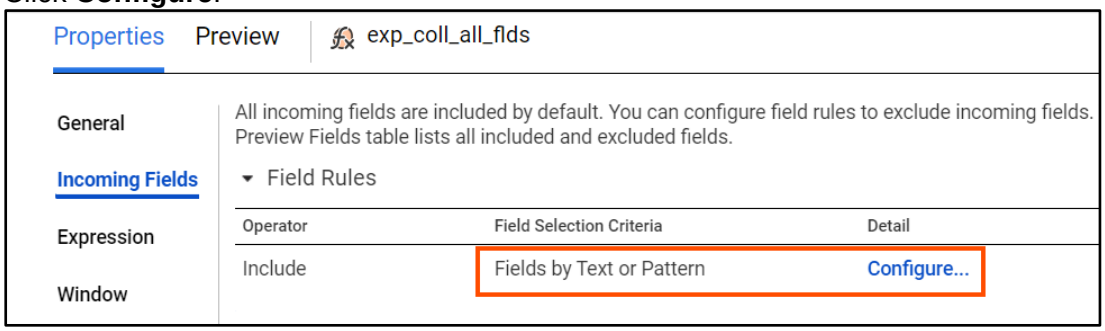
36. Select the **Expression** transformation from the mapping canvas.
37. In the **General** section of the Expression properties, enter Name as **exp_coll_all_flds**.



38. From the properties pane, click **Incoming Fields**.



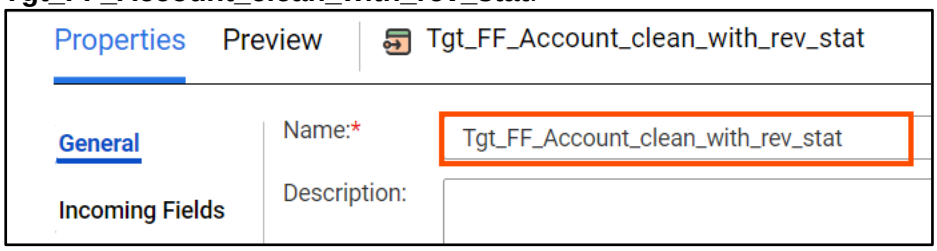
39. From the Operator drop-down, retain Include.
40. From the Field Selection Criteria drop-down, select **Fields by Text or Pattern**.
41. Click **Configure**.



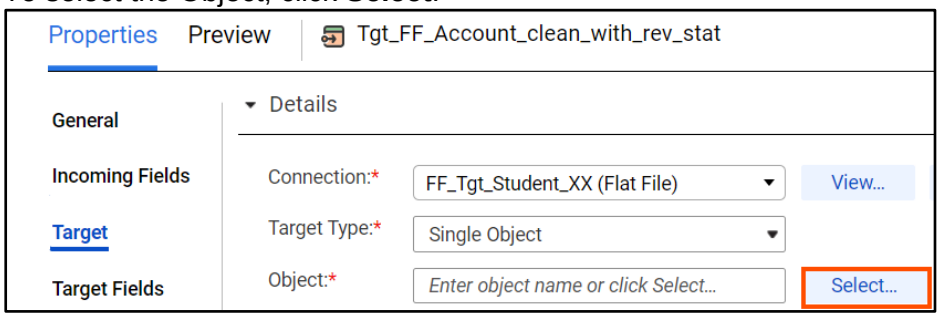
42. From the Select fields by drop down, select **Suffix**.
43. In the Specify field, enter _**out** and click **OK**.

44. To configure the target, from the mapping canvas, click the **Target** transformation.

45. In the **General** section of Target properties, enter the Name as **Tgt_FF_Account_clean_with_rev_stat**.

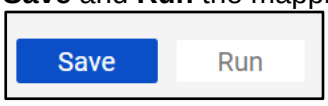


46. From the properties pane, click **Target**.
47. From the Connection drop down, select your **target flat file** connection.
48. Retain the Target Type as Single Object.
49. To select the Object, click **Select**.



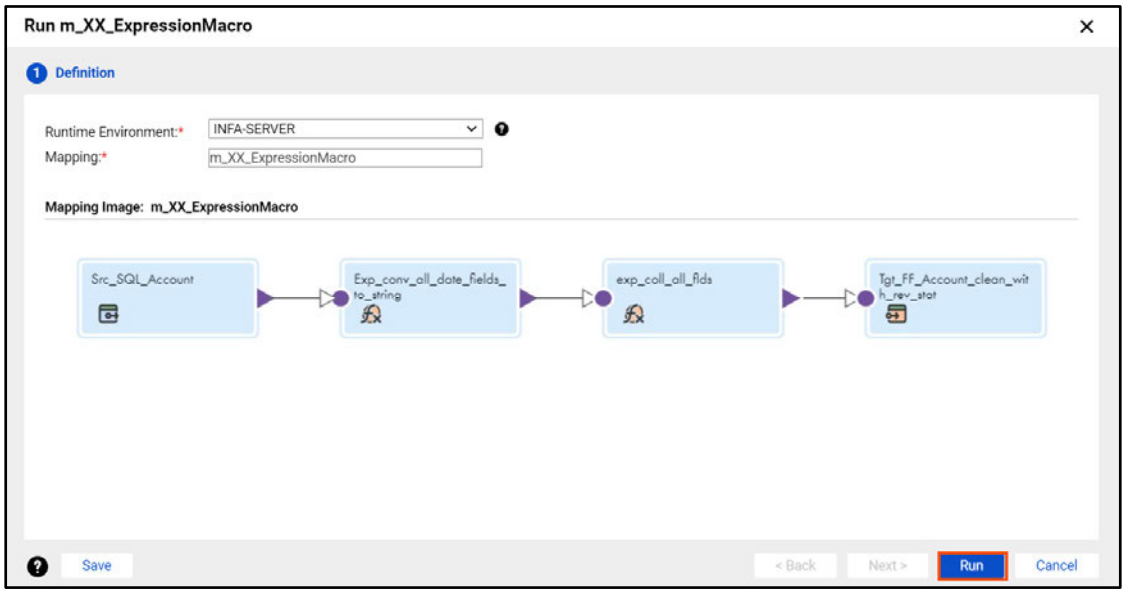
50. From the Target Object, select **Create New at Runtime**.
51. Enter the Static File Name as **S_XX_SQL_Account_test_data.csv** and click **OK**. Replace XX with your DOB.

52. **Save** and **Run** the mapping.



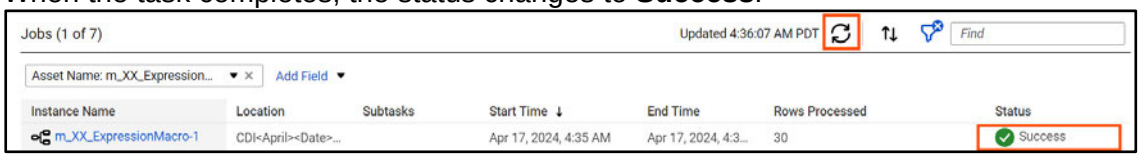
53. From the Runtime Environment drop-down, select **INFA-SERVER**.

54. Click **Run**.


Run m_XX_ExpressionMacro
Definition
Runtime Environment:*
INFA-SERVER
Mapping:*
m_XX_ExpressionMacro
Mapping Image: m_XX_ExpressionMacro
Src_SQL_Account
Exp_conv_all_date_fields_to_string
exp_call_all_flds
Tgt_FF_Account_clean_with_rev_stat
Save
< Back
Next >
Run
Cancel

Monitor Status

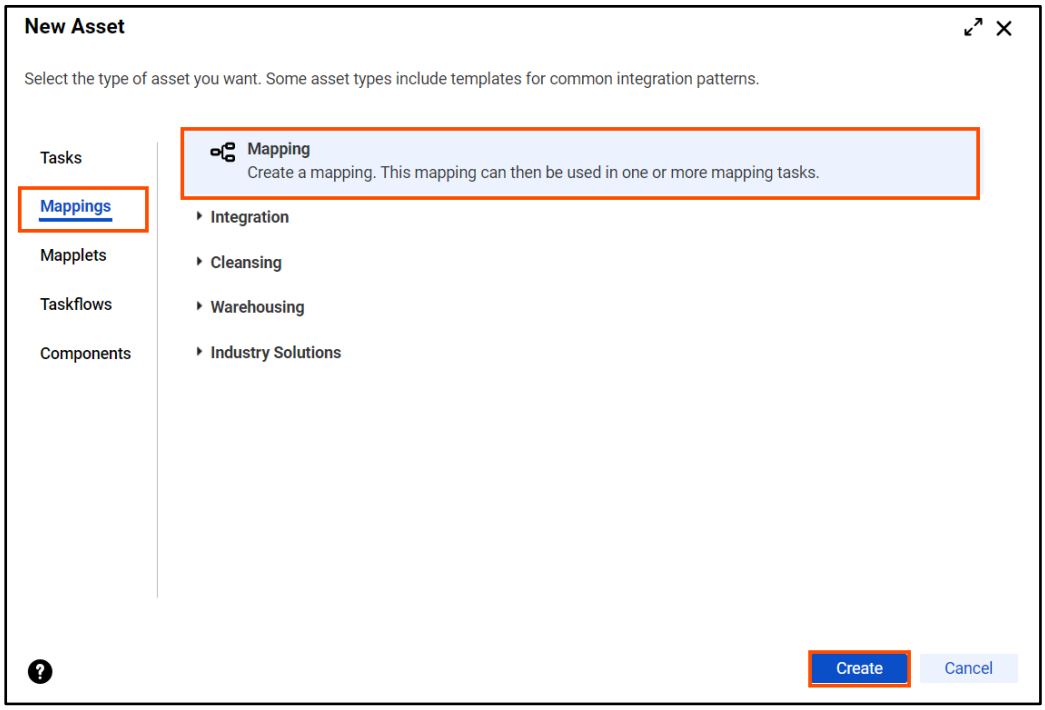
55. To monitor the mapping status, from the navigation pane, click **My Jobs**.
56. When the task completes, the status changes to **Success**.



57. Close the asset from the navigation pane.
58. Go to your Target location and open the **SQL_Account_test_data.csv** file.

Create a Dummy Mapping to View Data

59. Click **New** and from the **Mappings** tab, create a new mapping.

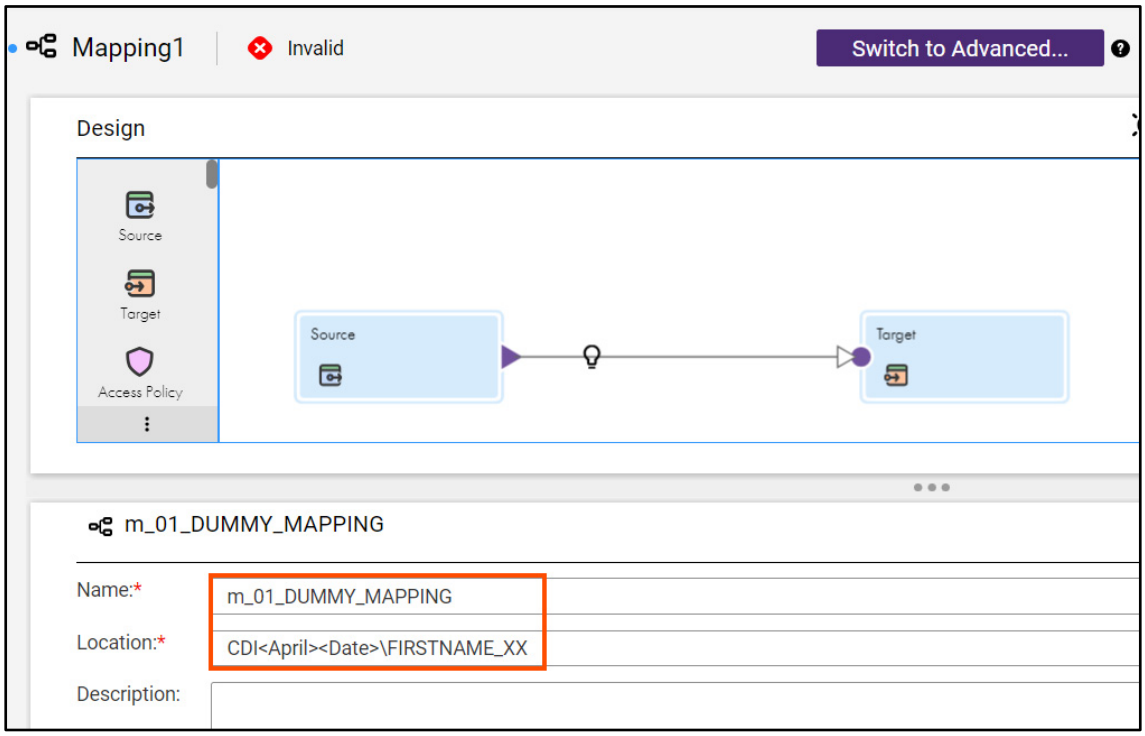


Note: You will know about mappings in the succeeding modules. You will just create a dummy mapping to view the data that is written to the target table.

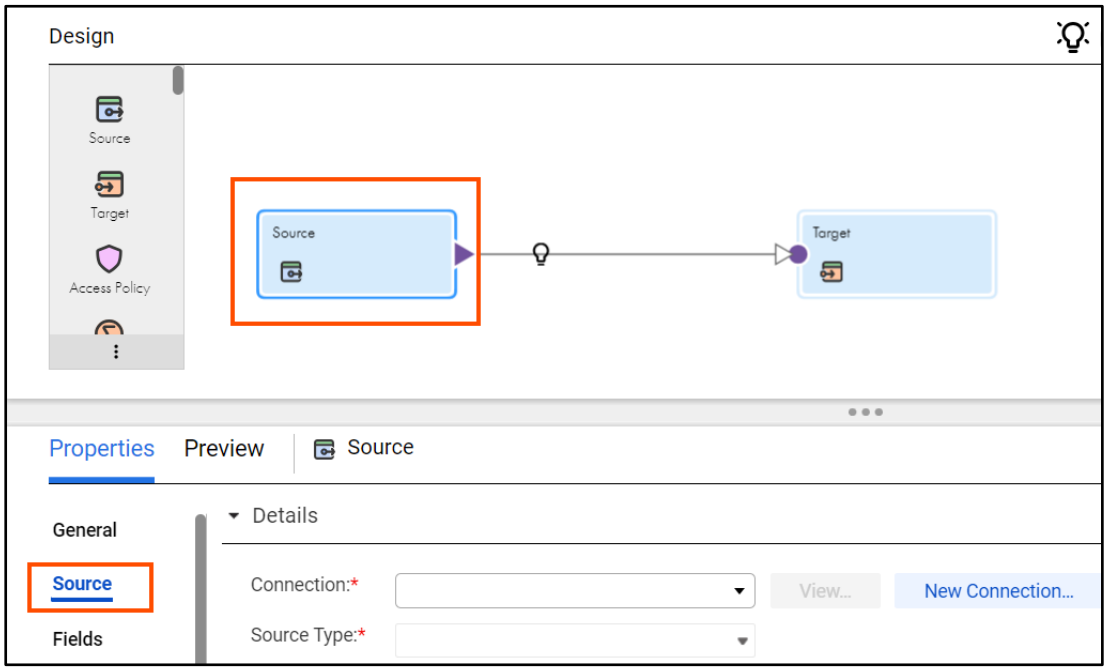
By default, you will have a source and target linked to each other on the mapping canvas.

60. Name the mapping as **m_XX_DUMMY_MAPPING**. Replace XX with your DOB.

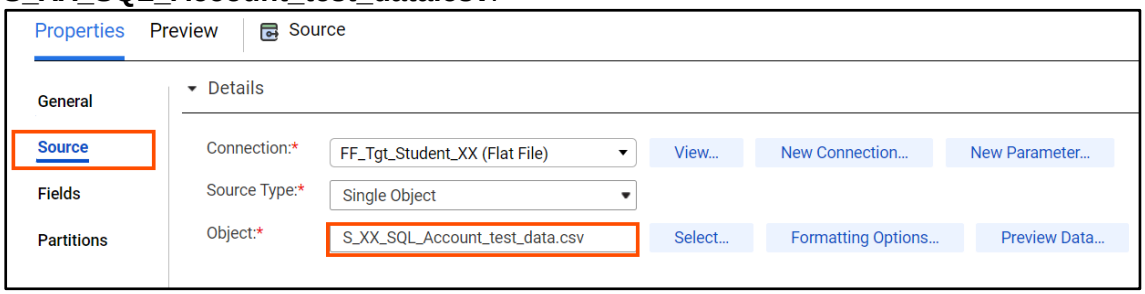
61. Set the asset location to your working folder and save the changes.



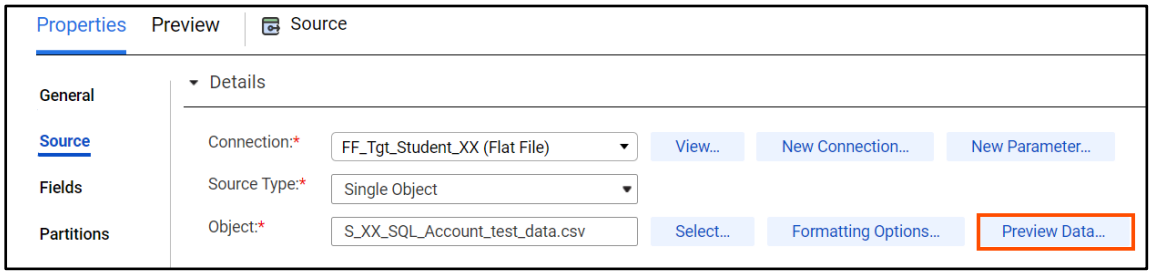
62. From the mapping canvas, select **Source**, and navigate to the **Source** properties tab.



63. Select the Connection as **your target Flat File** connection and the Object as **S_XX_SQL_Account_test_data.csv**.



64. Click **Preview Data**.



65. Verify that the dates are written in the “MM/DD/YYYY” format. View the data and close the window.

66. Save and close all the assets.

This concludes the lab.

Module 6: IICS Transformations

Lab 6-4: Using Vertical Macro in a Mapping

Overview:

A vertical macro expands an expression vertically. The vertical macro generates a set of similar expressions to perform the same calculation on multiple incoming fields.

Use a vertical macro to apply a macro expression to a set of incoming fields.

Objective:

- Use a Vertical Macro in a mapping.

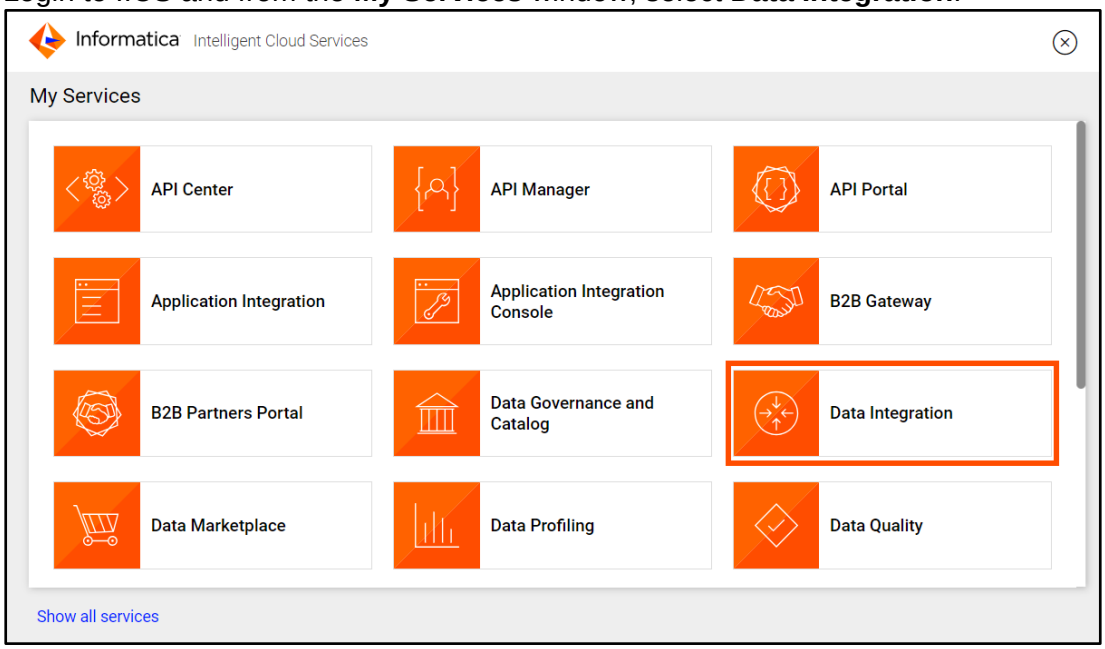
Duration:

10 minutes

Tasks

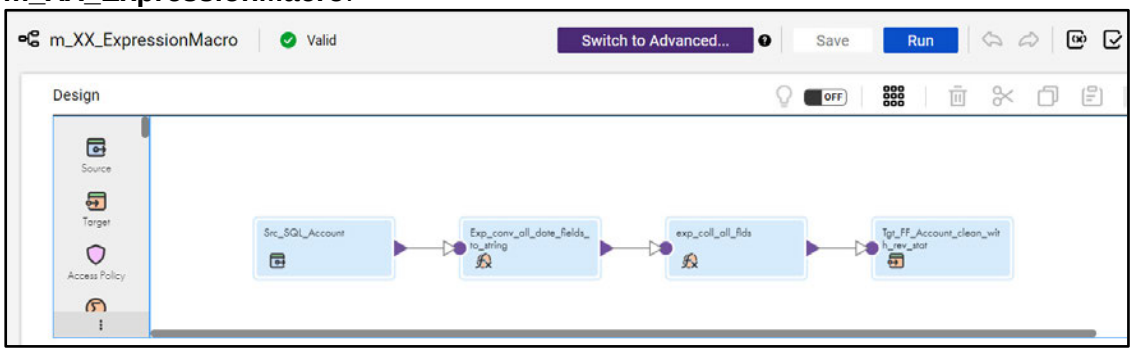
Create Vertical Mapping

1. Login to IICS and from the **My Services** window, select **Data Integration**.



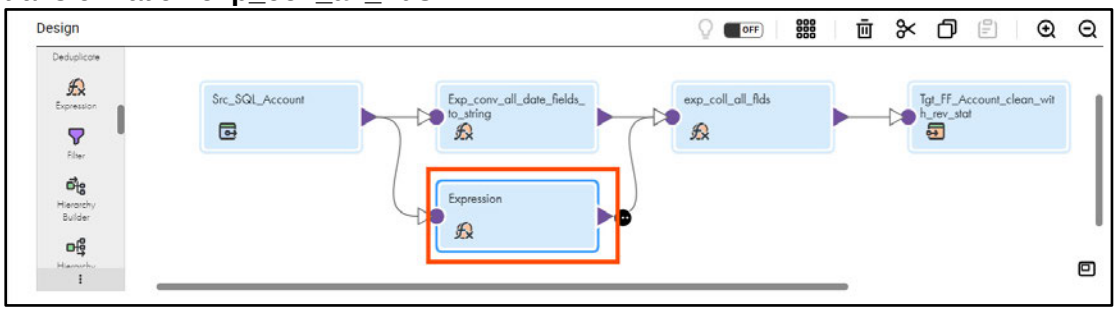
2. From the navigation pane, select **Explore**.

3. From your working directory, open the Expression macro mapping **m_XX_ExpressionMacro**.



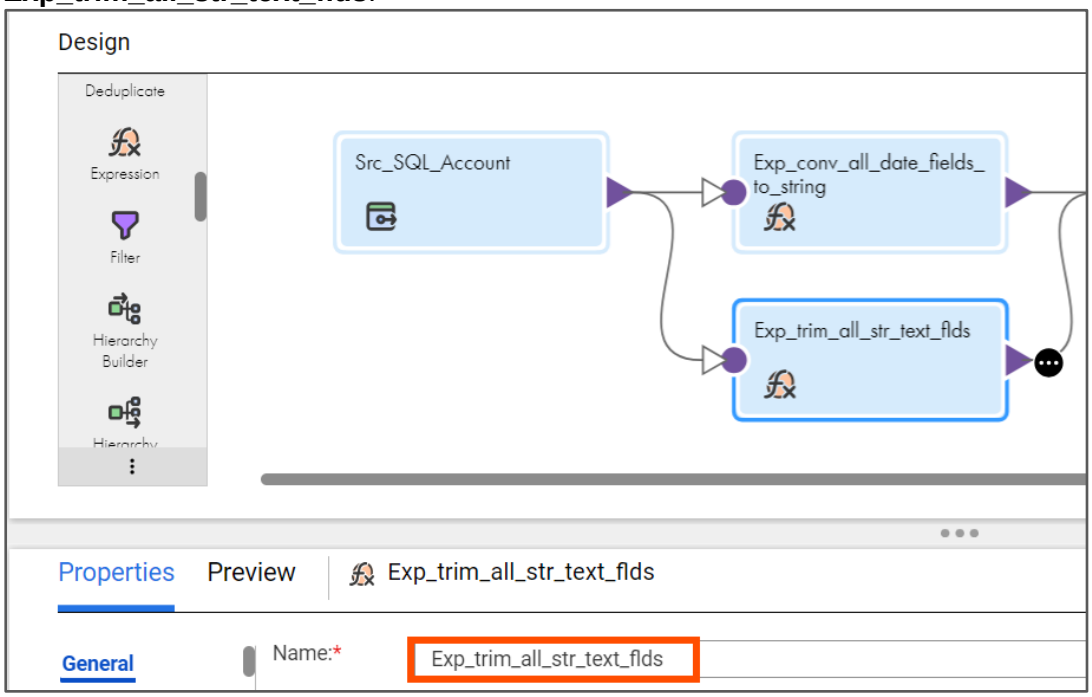
Add Expression Transformation

4. From the list of available transformations, drag and drop an **Expression** transformation on the mapping canvas.
5. Link the Expression transformation with the source **Src_SQL_Account** and expression transformation **exp_coll_all_flds**.

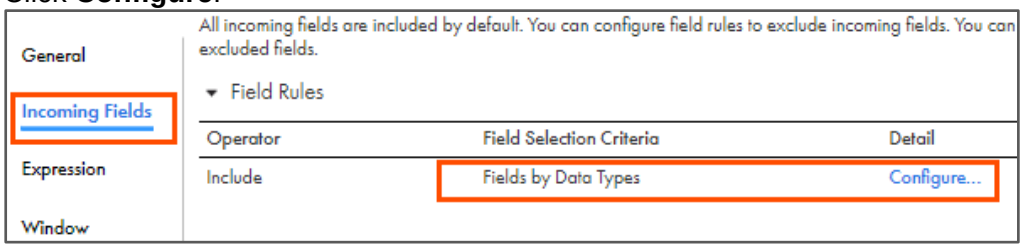


6. Select the **Expression** transformation from the mapping canvas.

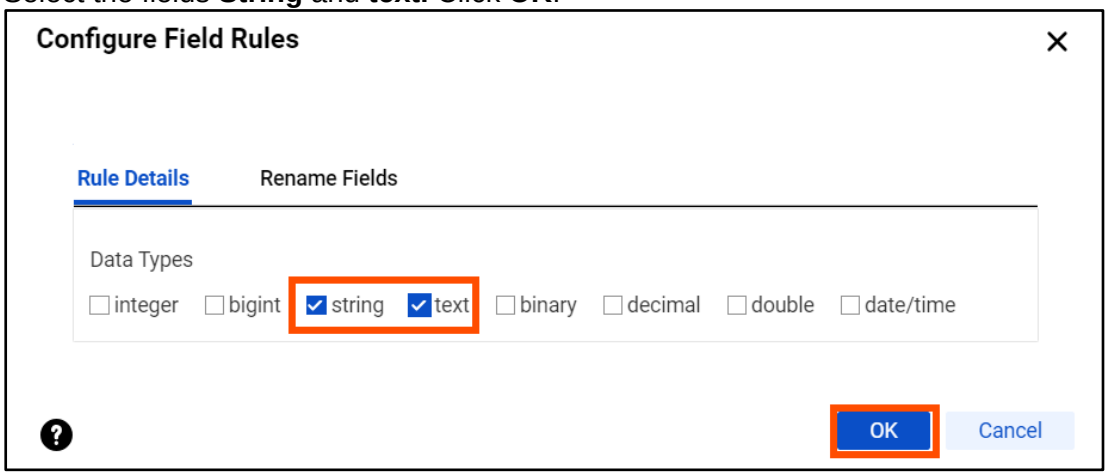
7. In the **General** section of the Expression properties, enter the Name as **Exp_trim_all_str_text_flds**.



8. From the properties pane, click **Incoming Fields**.
9. In the Field Rules section, from Field Selection Criteria drop-down, select **Fields by Data Types**.
   Note: You might need to click outside of Fields by Data Types drop-down, in a black space to enable the Configure option.
10. Click **Configure**.

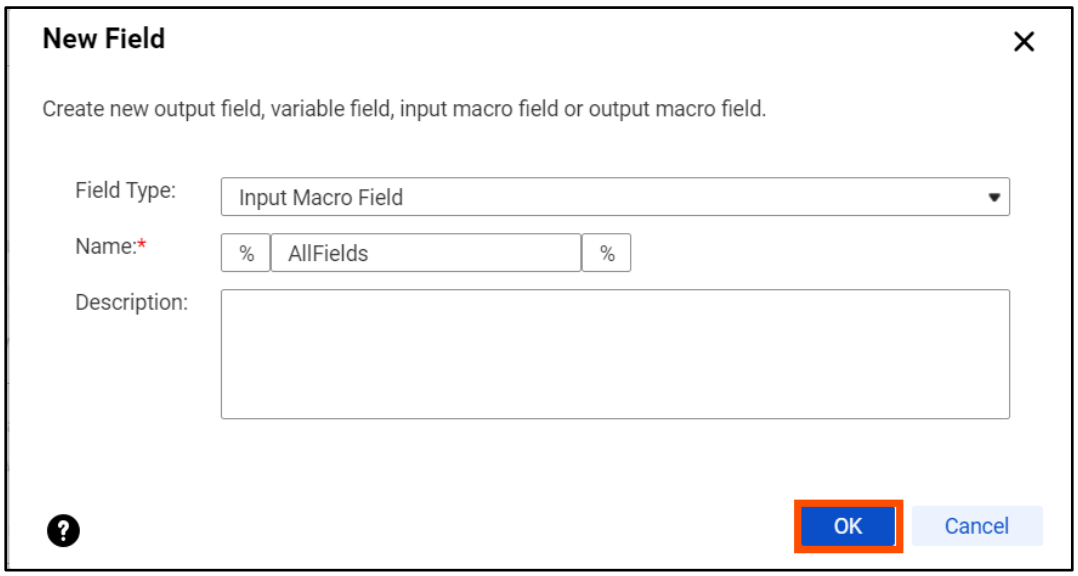


11. Select the fields **String** and **text.** Click **OK**.

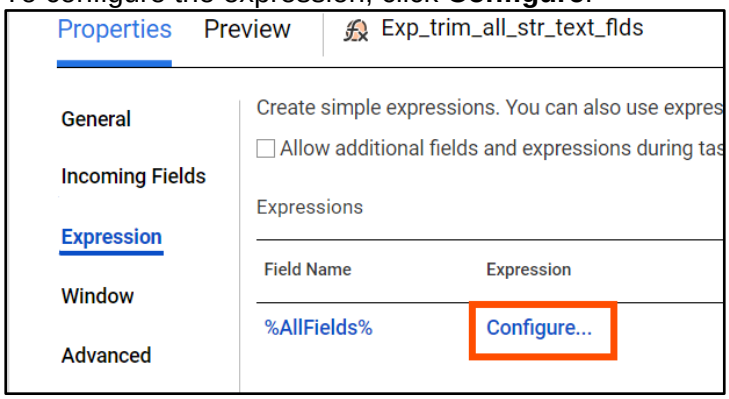


12. From the properties pane, click **Expression**.
13. To add a new expression, click +.
14. Enter the details as shown in the table below and click **OK**.

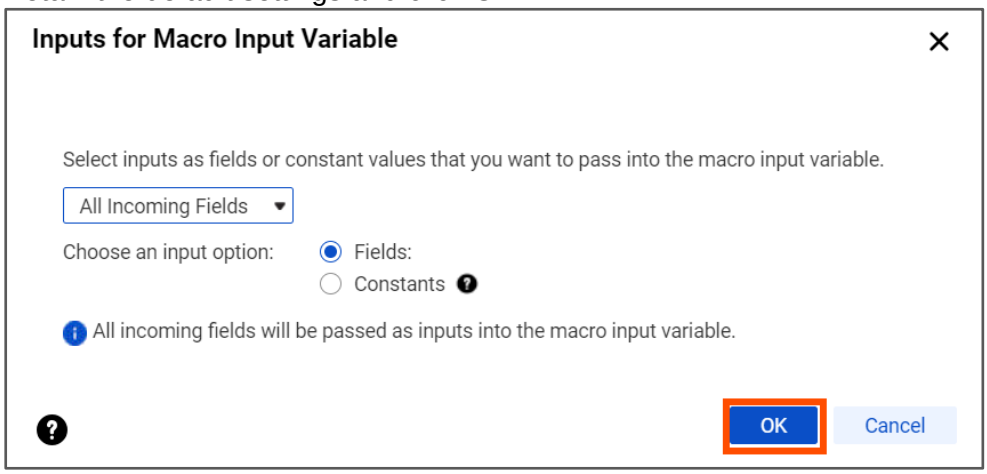
Field Type	Name
Input Macro Field	AllFields



15. To configure the expression, click **Configure**.

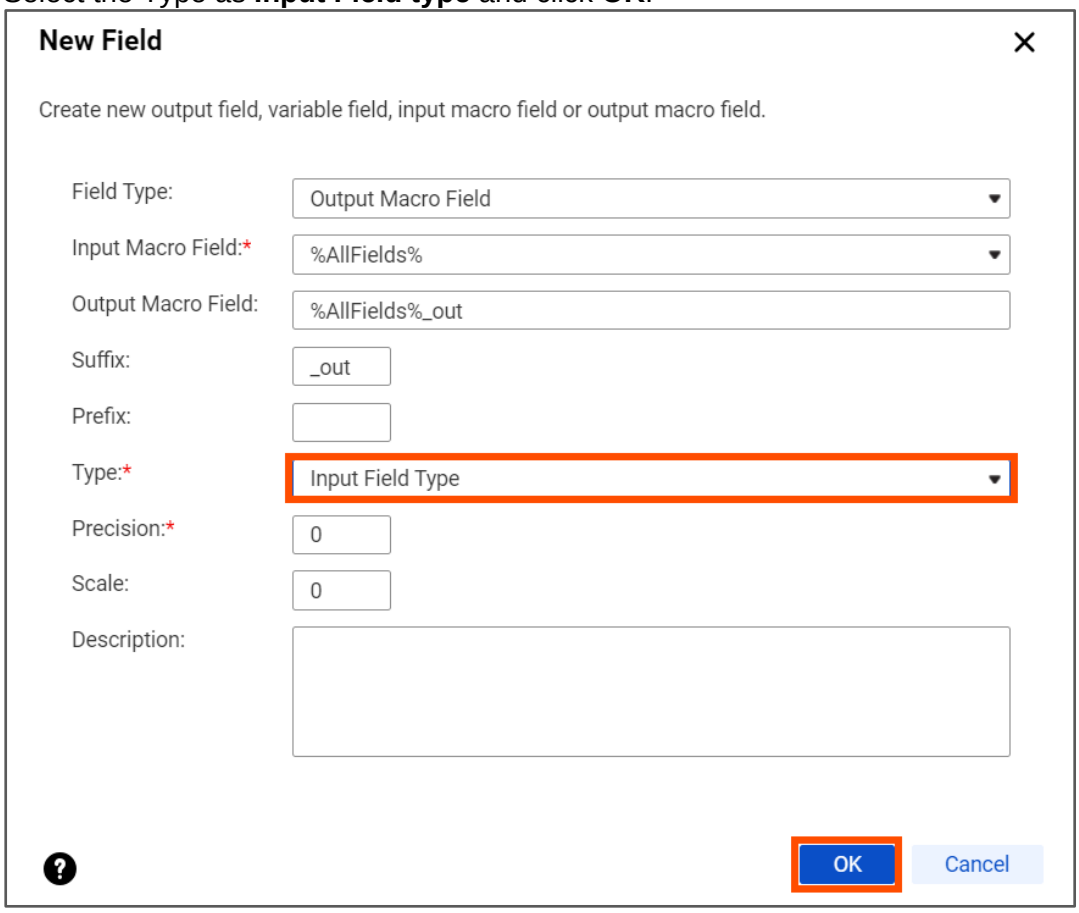


16. Retain the default settings and click **OK**.

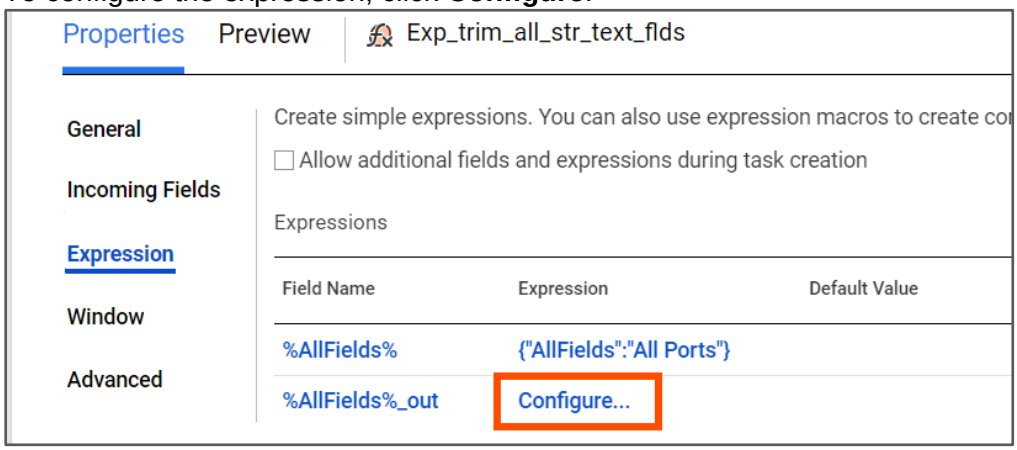


17. To create a new macro output field, click +.
18. From the Field Type drop-down, select **Output Macro Field**.

19. Select the Type as **Input Field type** and click **OK**.



20. To configure the expression, click **Configure**.

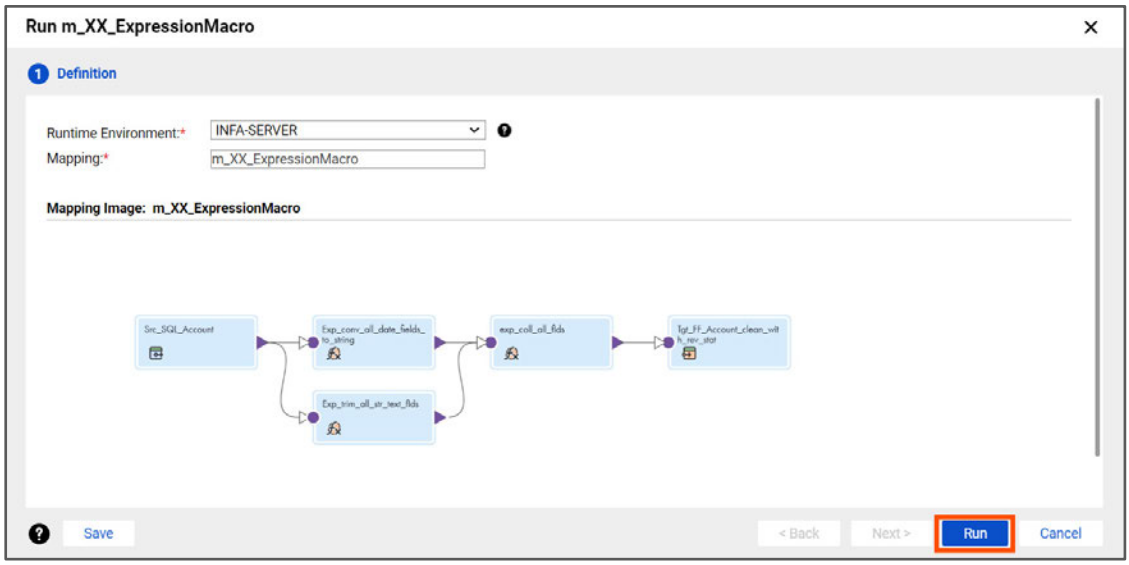


21. In the Expression field, enter the following expression:
LTRIM(RTRIM(%AllFields%))

22. Click **OK**.

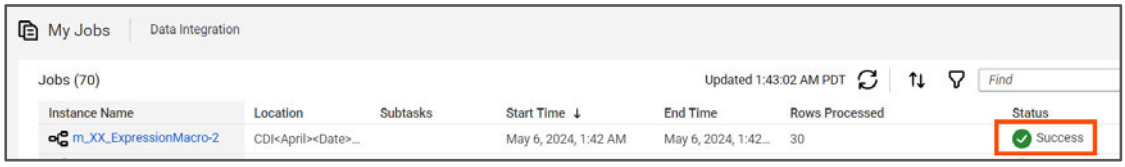
Note: As the field is macro field, the Validate option is disabled.

23. **Save** and **Run** the mapping.
24. From the Runtime Environment drop-down, select **INFA-SERVER**.
25. Click **Run**.



Monitor Status

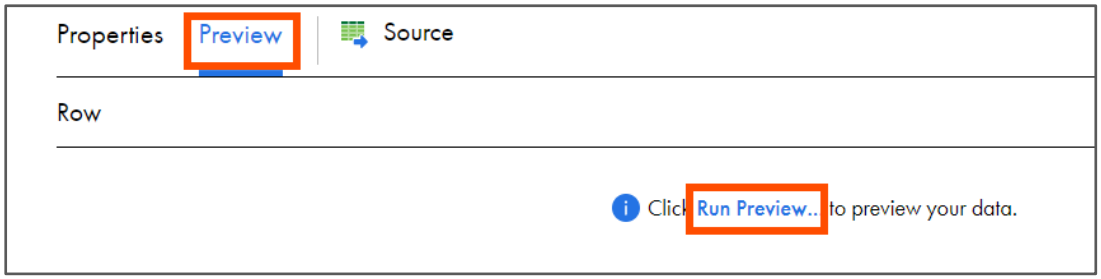
26. To monitor the mapping status, from the navigation pane, click **My Jobs**.
27. When the task is completed, the status changes to **Success**.
    Note: The number of rows processed may change depending on the data in the Salesforce Account object.



28. Close the asset from the navigation pane.

Self-Try Exercise: View the Target Data

29. From your working folder, open the dummy mapping.
30. Make sure that your target flat file connection and the target object **S_XX_SQL_Account_test_data.csv** is selected.
31. To view the latest mapping run result, go to the **Preview** tab and click **Run Preview**.



32. In the Continue tab, click **Yes** to save the changes.
33. If you are prompted with the Run Preview window, select the Runtime Environment as **INFA-SERVER,** and click the **Run Preview** button.
34. View the target data and close all the assets.

This concludes the lab.

Module 6: IICS Transformations

Lab 6-5: Using Horizontal Macro in a Mapping

Overview:

A horizontal macro expands an expression horizontally. The horizontal macro generates one extended expression that includes a set of fields or constants.

Objective:

- Use a horizontal Macro in a mapping

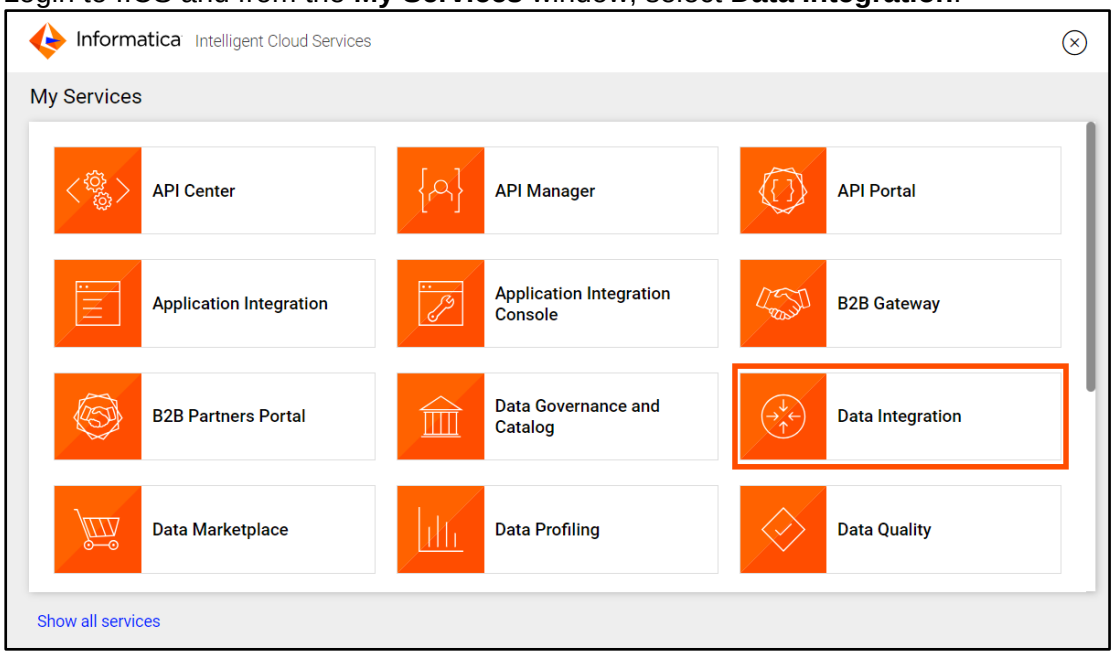
Duration:

20 minutes

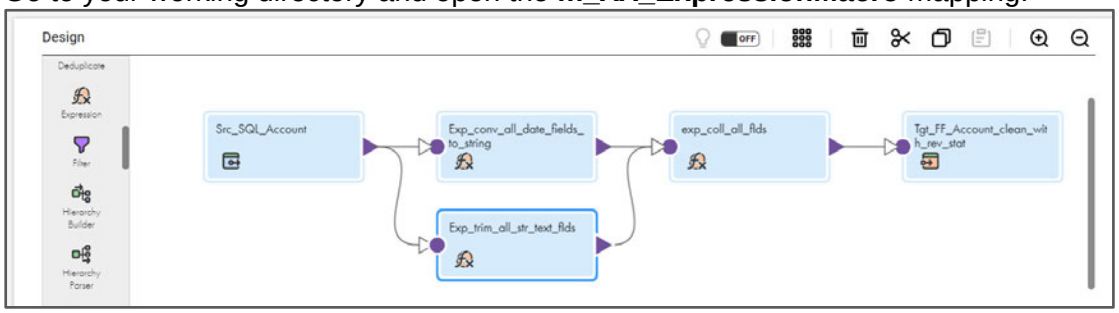
Tasks

Create Mapping

1. Login to IICS and from the **My Services** window, select **Data Integration**.

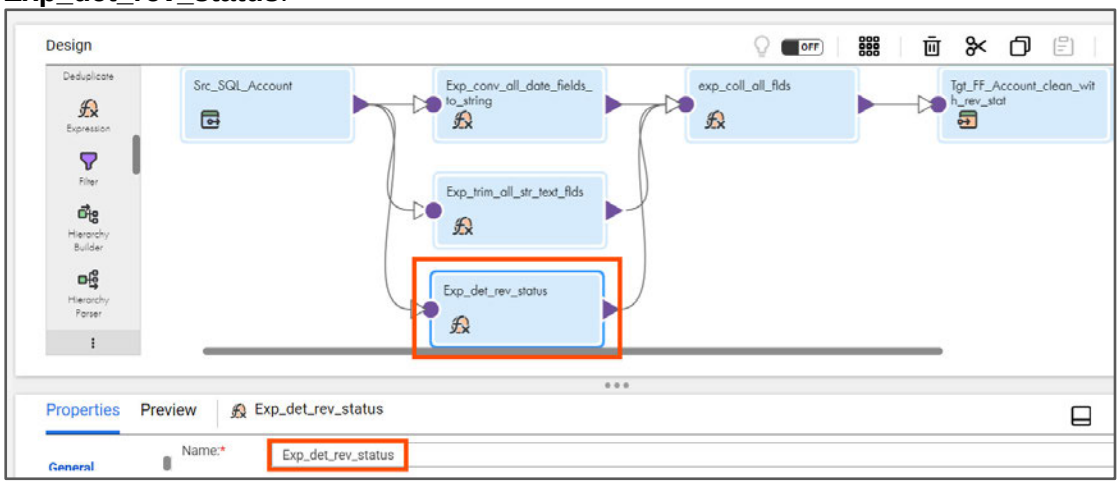


2. From the navigation pane, navigate to the **Explore** page.
3. Go to your working directory and open the **m_XX_ExpressionMacro** mapping.



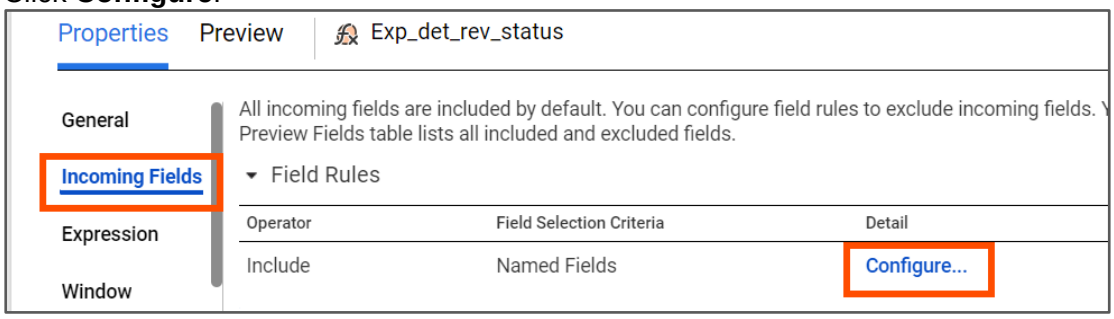
Add Expression Transformation

4. From the list of available transformations, drag and drop an **Expression** transformation on the mapping canvas.
5. Link it to Source **Src_SQL_Account** and Expression **exp_coll_all_flds**.
6. Select the **Expression** transformation from the mapping canvas.
7. In the **General** section of the Expression properties, enter the Name as **Exp_det_rev_status**.

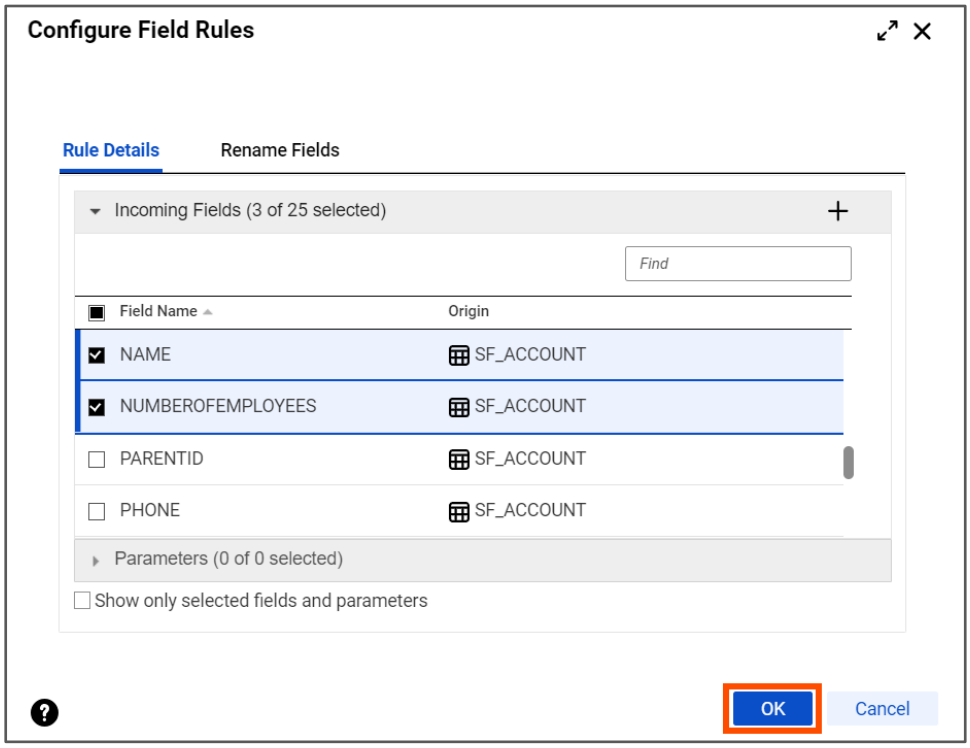


8. From the properties pane, click **Incoming Fields**.
9. In the Field Rules section, from Field Selection Criteria drop-down, select **Named Fields**.
   Note: You might need to click outside of Fields by Data Types drop-down, in a black space to enable the Configure option.

10. Click **Configure**.

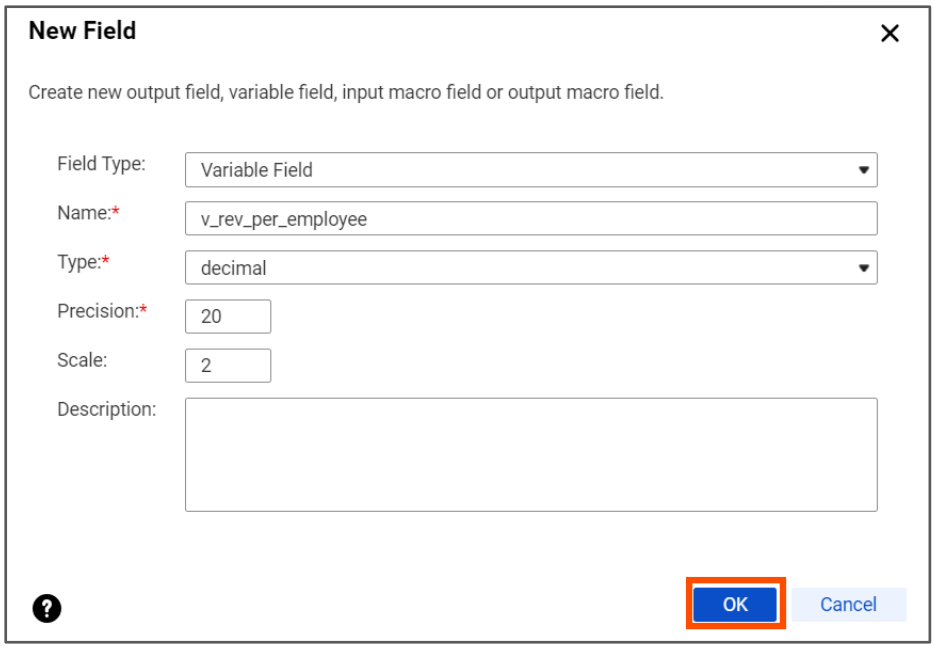


11. In the Configure Field Rules window, select **ANNUALREVENUE**, **NAME** and **NUMBEROFEMPLOYEES** and click **OK**.

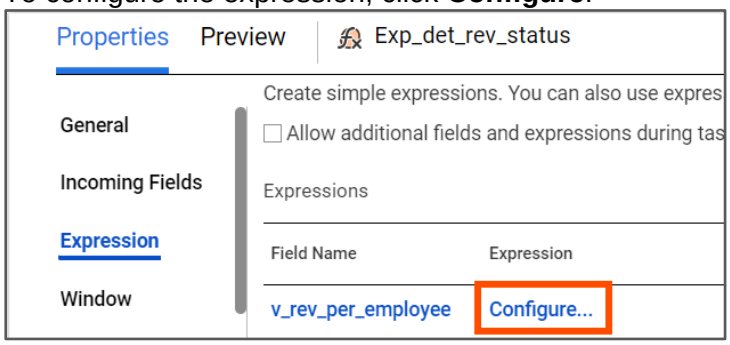


12. From the properties pane, click **Expression**.
13. To add a new expression, click +.
14. Enter the details as shown in table below and click **OK**.

Field Type	Name	Type	Precision	Scale
Variable Field	v_rev_per_employee	decimal	20	2

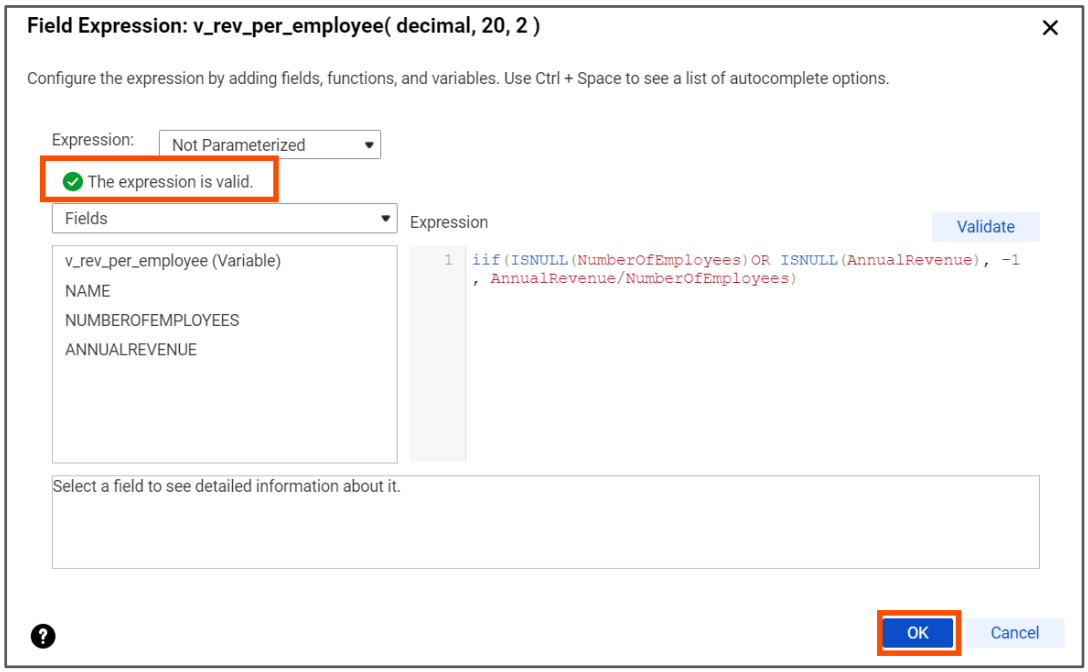


15. To configure the expression, click **Configure**.



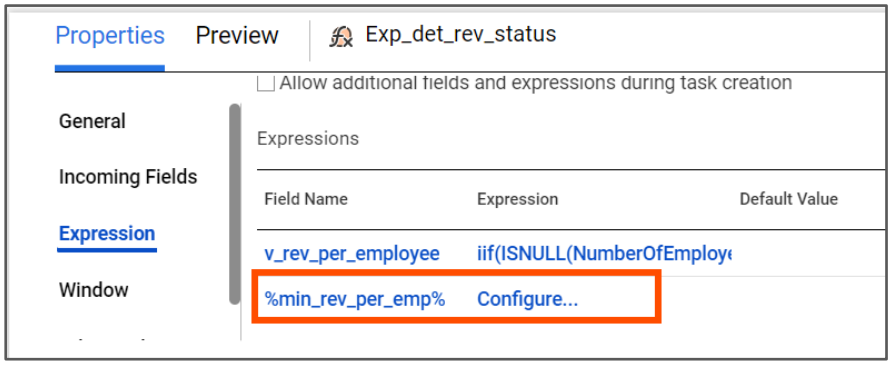
16. In the Expression field, enter the following expression:
iif(ISNULL(NumberOfEmployees)OR ISNULL(AnnualRevenue), -1 , AnnualRevenue/NumberOfEmployees)

17. Click **Validate** and then click **OK**.



18. Add another expression as shown in the table below and click **OK**.

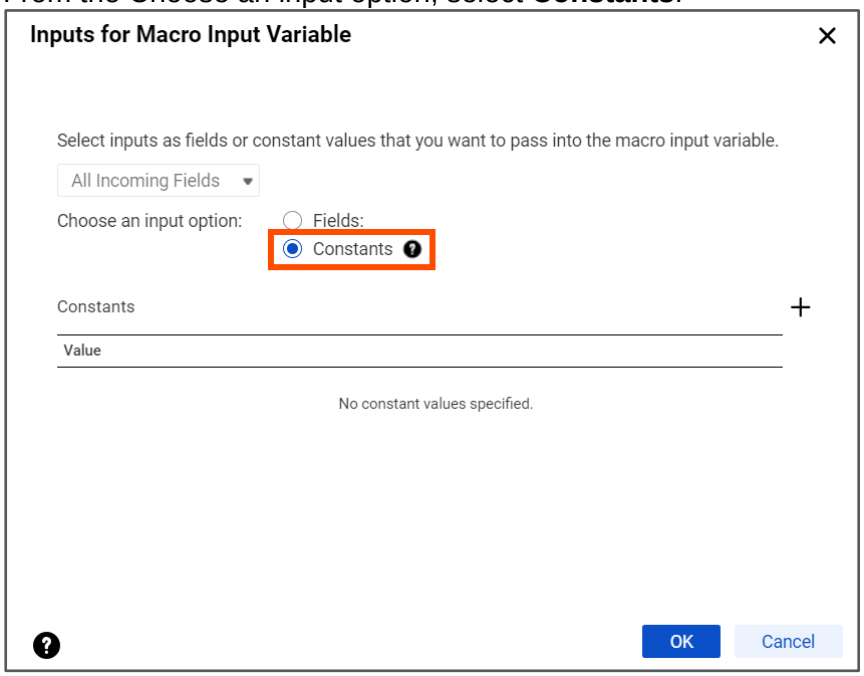
Field Type	Name
Input Macro Field	min_rev_per_emp



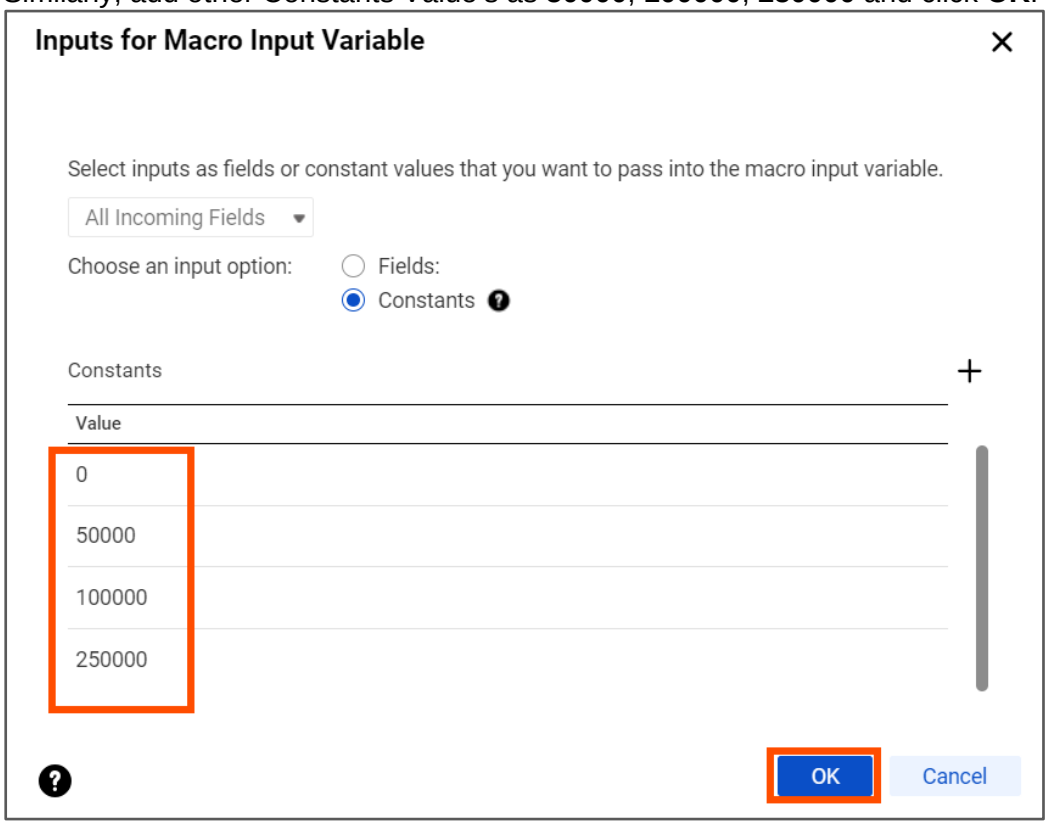
Note: A macro input field is a field that represents input that you want to use in the expression macro.

19. To configure the expression, click **Configure**.

20. From the Choose an input option, select **Constants**.



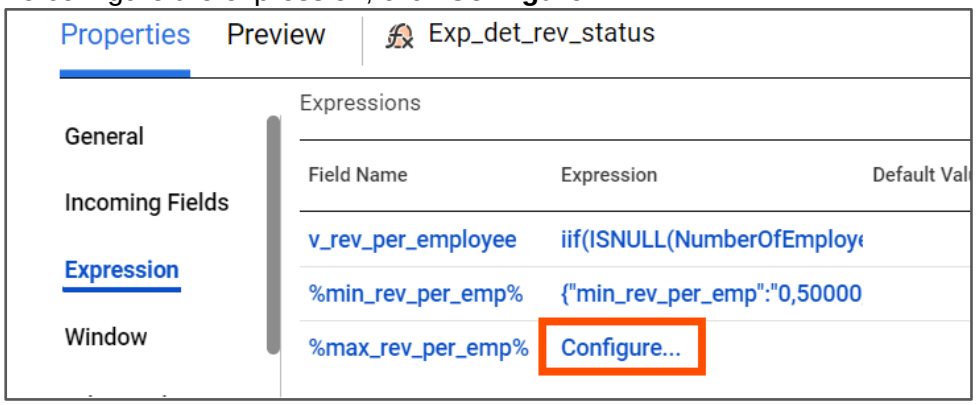
21. To add a new Constant, click [+].
22. In the Value field, enter the newConstant as **0**.
23. Similarly, add other Constants Value's as **50000**, **100000**, **250000** and click **OK**.



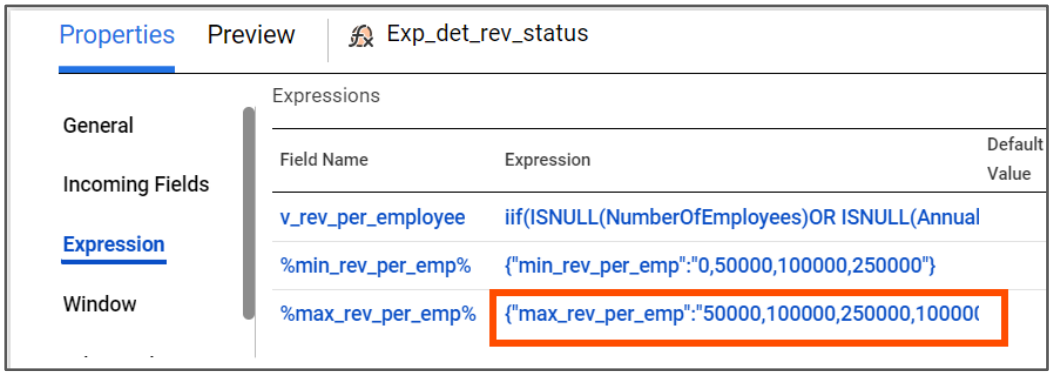
24. Add another expression as shown in the table below and click **OK**.

Field Type	Name
Input Macro Field	max_rev_per_emp

25. To configure the expression, click **Configure**.



26. From the Choose an input option, select **Constants**.
27. To add a new Constant, click +.
28. In the Value field, enter the newConstant as **50000**.
29. Similarly, add other Constant Values as **100000**, **250000**, **1000000** and click **OK**.

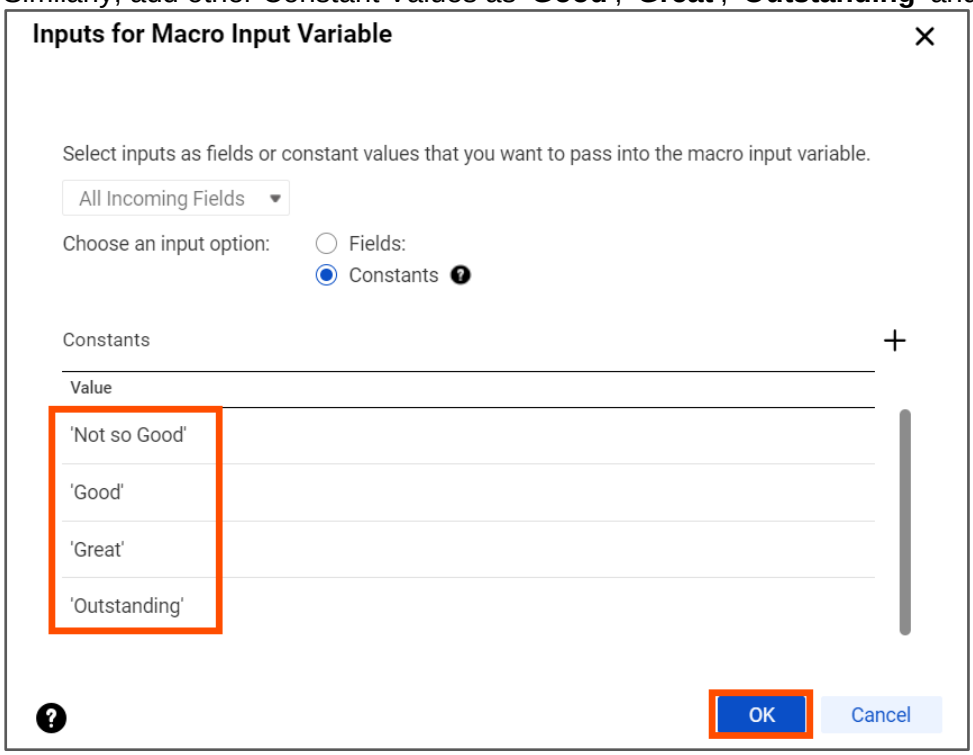


30. Add another expression as shown in the table below and click **OK**.

Field Type	Name
Input Macro Field	rev_per_emp_status

31. To configure the expression, click **Configure**.
32. In the macro input variable field, select **Constants**.
33. To add a new Constant, click + .
34. In the Value field, enter the newConstant as '**Not so Good**'.

35. Similarly, add other Constant Values as '**Good**', '**Great**', '**Outstanding**' and click **OK**.



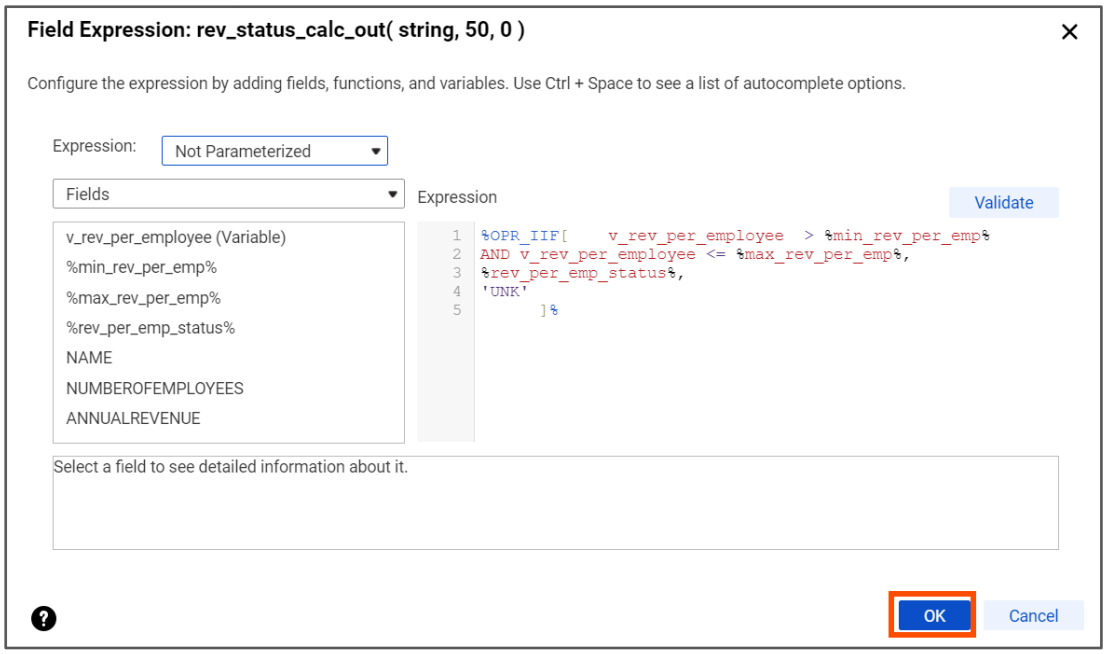
36. To add another expression, enter the details as shown in table below and click **OK**.

Field Type	Name	Type	Precision	Scale
Output Field	rev_status_calc_out	string	50	0

Note: As the field is macro field, the Validate option will not work.

37. To configure the expression, click **Configure**.
38. In the Expression field, enter the following expression and click **OK**.

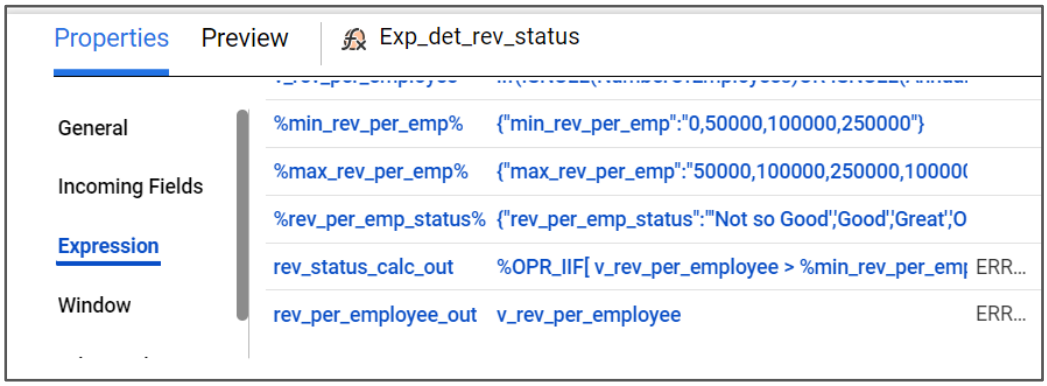
%OPR_IIF[v_rev_per_employee > %min_rev_per_emp%
AND v_rev_per_employee <= %max_rev_per_emp%,
%rev_per_emp_status%,
'UNK'
]%



39. To add another expression, enter the details as shown in table below and click **OK**.

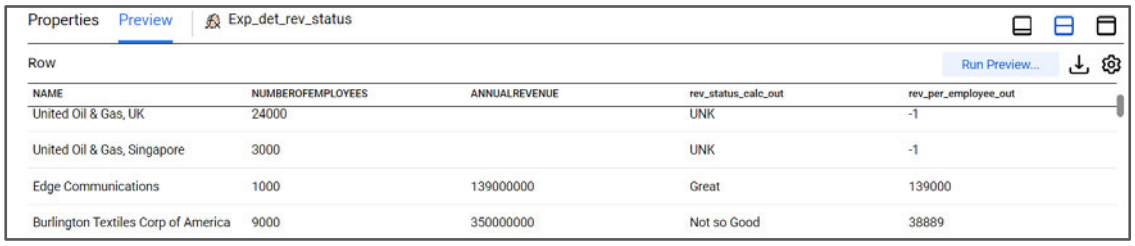
Field Type	Name	Type	Precision	Scale
Output Field	rev_per_employee_out	decimal	20	0

40. To configure the expression, click **Configure**.
41. In the Expression field, enter the following expression and click **OK**.
v_rev_per_employee



42. Go to the Preview tab and click on **Run Preview**.

43. Click **Yes** and **Run Preview** to check the data.

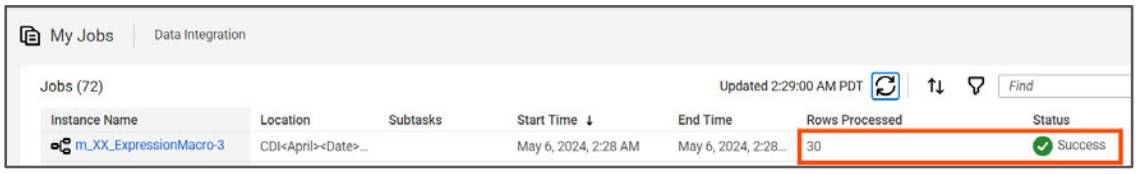


Note: Your data may differ from the one shown in the above screen shot.

44. Save and run the mapping.
45. From the Runtime Environment drop-down, select **INFA-SERVER**.
46. Click **Run**.

Monitor Status

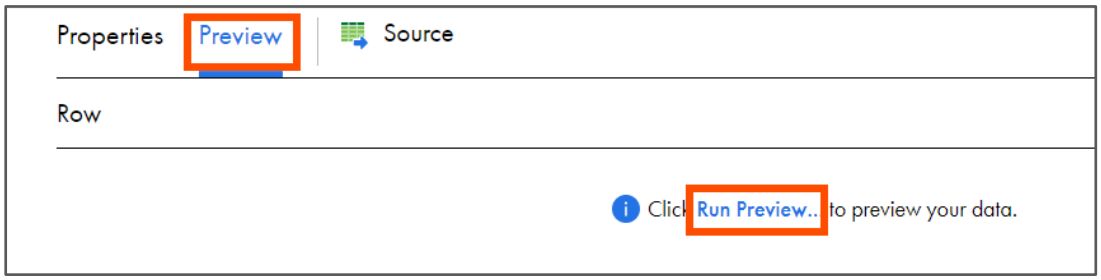
47. To monitor the mapping status, from the navigation pane, click **My Jobs**.
48. When the task completes, the status changes to **Success**.



49. Close the asset from the navigation pane.

Self-Try Exercise: View the Target Data

50. From your working folder, open the dummy mapping.
51. Make sure that your target flat file connection and the target object **S_XX_SQL_Account_test_data.csv** is selected.
52. To view the latest mapping run result, go to the **Preview** tab and click **Run Preview**.



53. In the Continue tab, click **Yes** to save the changes.
54. If you are prompted with the Run Preview window, select the Runtime Environment as **INFA-SERVER,** and click the **Run Preview** button.
55. View the target data and close all the assets.

This concludes the lab.

Module 7: Joiner and Lookup Transformations

Lab 7-1: Using Lookup Transformation in a Mapping

Overview:

Lookup transformation in Informatica Cloud (IICS) is used to look up a database table or a flat file based on a condition and retrieve value from the object used as a lookup.

Objective:

- Use Lookup transformation in the mapping

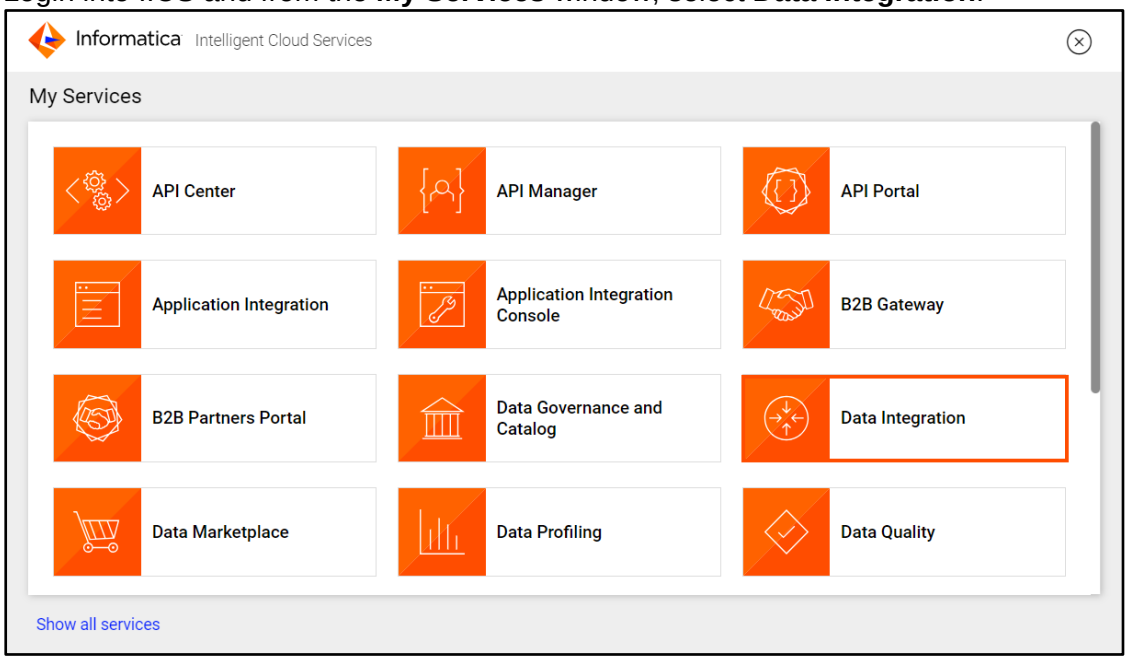
Duration:

50 minutes

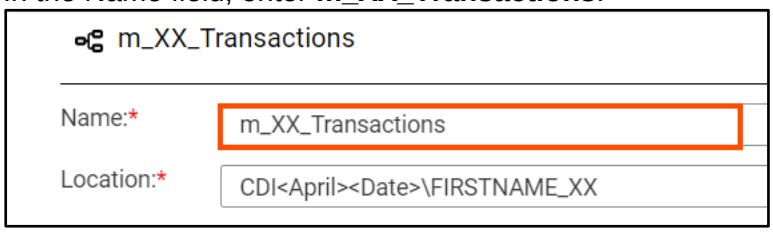
Tasks

Create Mapping

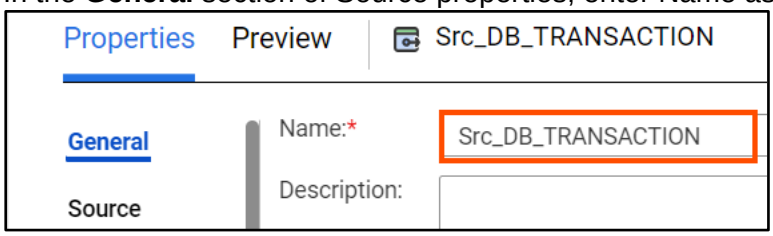
1. Login into IICS and from the **My Services** window, select **Data Integration**.



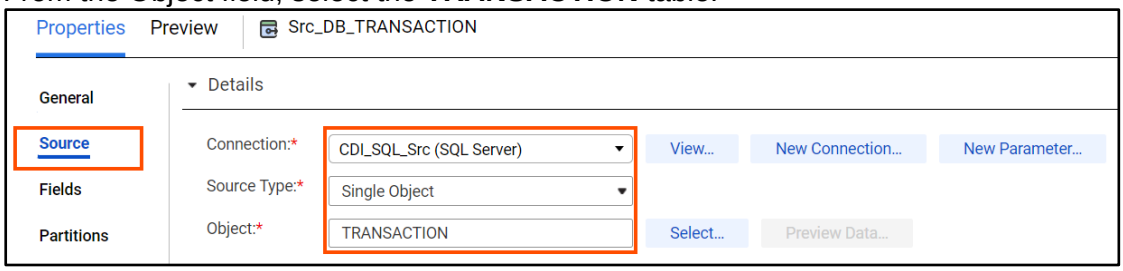
2. Create a new Mapping.
3. In the Name field, enter **m_XX_Transactions**.



4. Verify that the asset Location is pointing to your working folder.
5. To configure the source, from the mapping canvas, click the **Source** transformation.
6. In the **General** section of Source properties, enter Name as **Src_DB_TRANSACTION**.



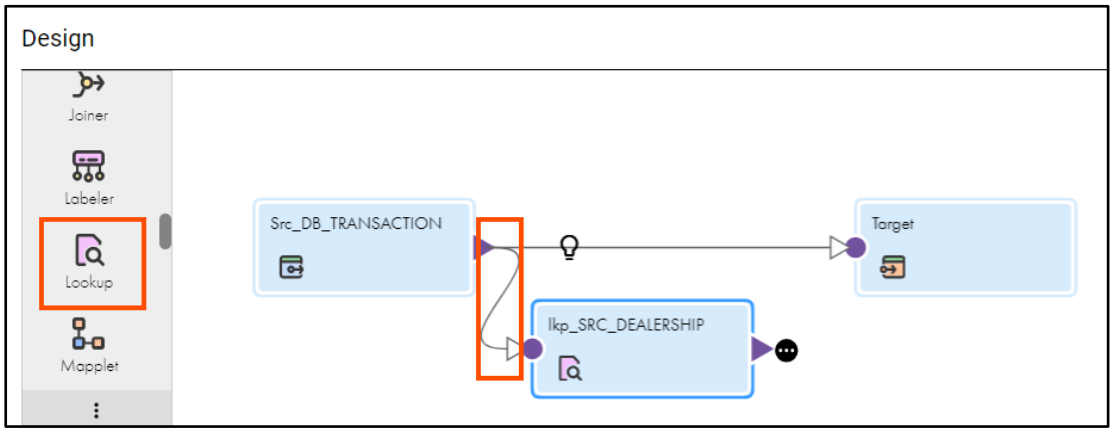
7. From the properties pane, click **Source**.
8. From the Connection drop-down, select **CDI_SQL_Src**.
9. From the Object field, select the **TRANSACTION** table.



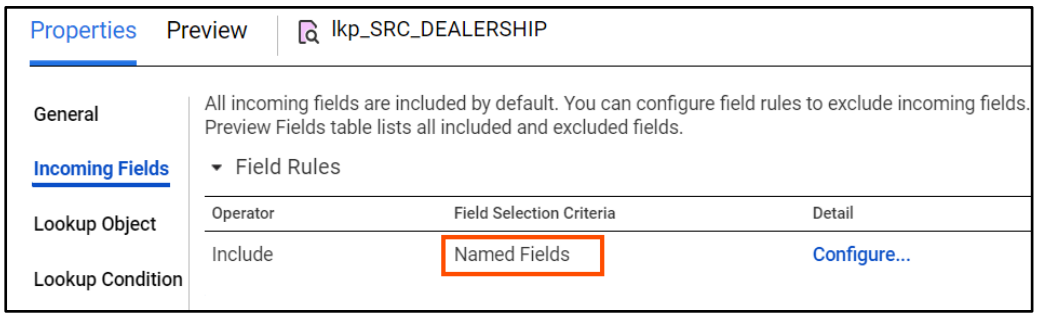
Add Lookup Transformation

10. From the list of available transformations, drag and drop a Lookup transformation on the mapping canvas.
11. Select the Lookup transformation on the mapping canvas.
12. In the **General** section of Lookup properties, enter Name as **lkp_SRC_DEALERSHIP**.

13. Link the **Src_DB_TRANSACTION** with the **lkp_SRC_DEALERSHIP** on the mapping canvas.

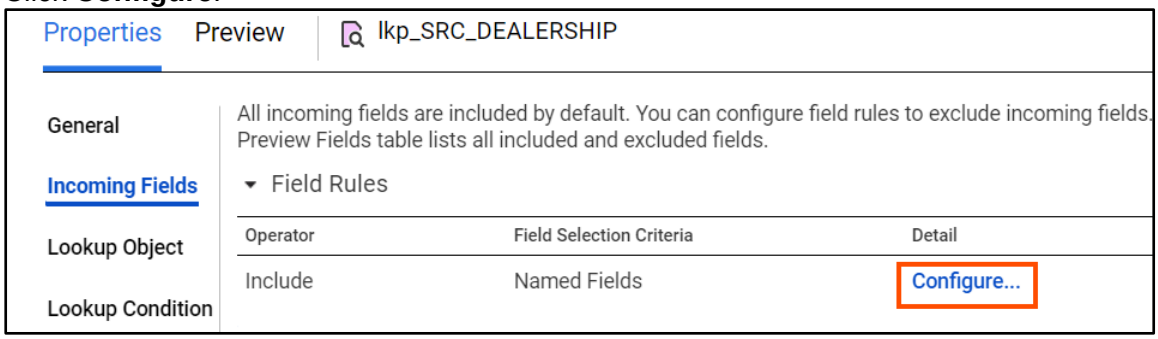


14. From the properties pane, go to **Incoming Fields**.
15. In the Field Rules section, from the **Field Selection Criteria** drop-down, select **Named Fields**.



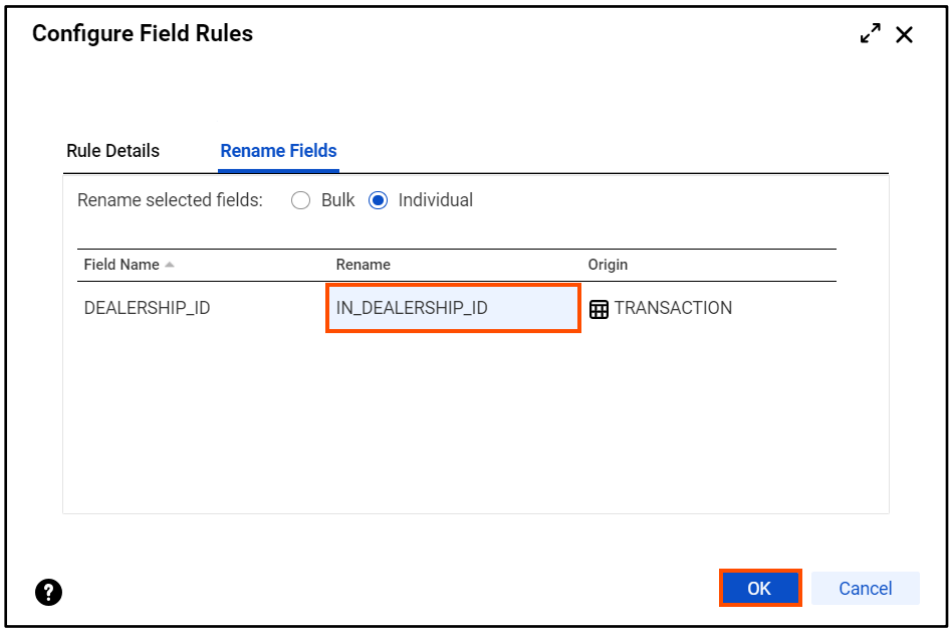
Note: After selecting Named Fields, click anywhere on the blank space to enable the Configure link.

16. Click **Configure**.

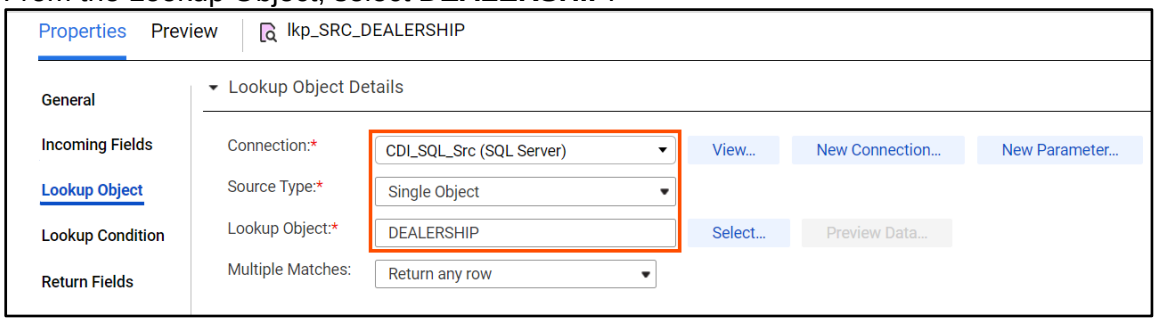


17. Select **DEALERSHIP_ID** from the **Rule Details** tab.
18. Select the **Rename Fields** tab and in the **Rename** field, enter **IN_DEALERSHIP_ID**.

19. Click **OK**.

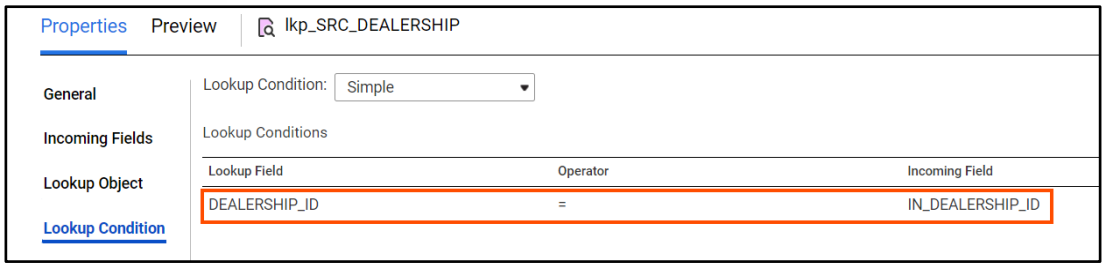


20. From the properties pane, select **Lookup Object**.
21. From the connection dropdown, select **CDI_SQL_Src**.
22. From the Lookup Object, select **DEALERSHIP**.

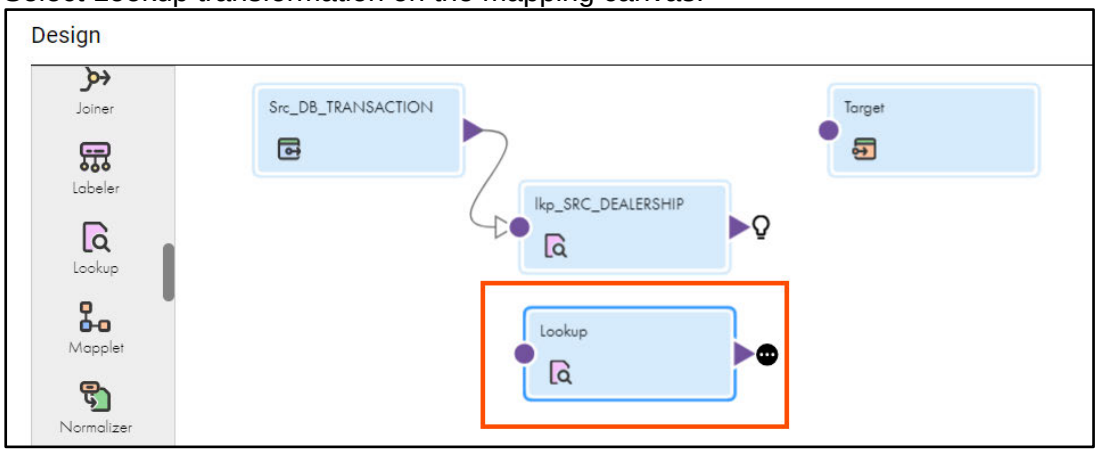


23. Select **Lookup Condition**, from the properties pane.
24. To add a new Lookup condition, click ⊞.
25. Enter the details as shown in the table below and click **OK**.

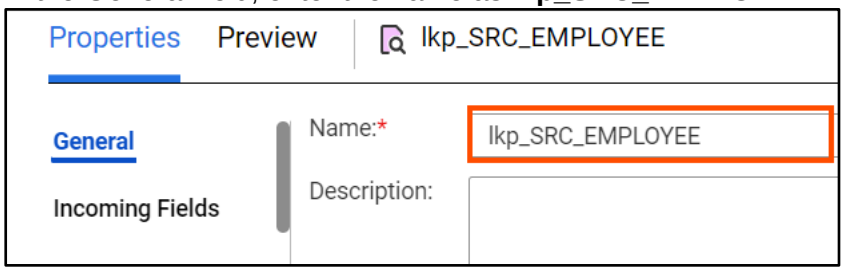
Lookup Field	Operator	Incoming Field
DEALERSHIP_ID	=	IN_DEALERSHIP_ID



26. From the list of available transformations, drag and drop another **lookup** transformation on the mapping canvas.
27. Delete the link between Source and Target objects.
28. Select Lookup transformation on the mapping canvas.



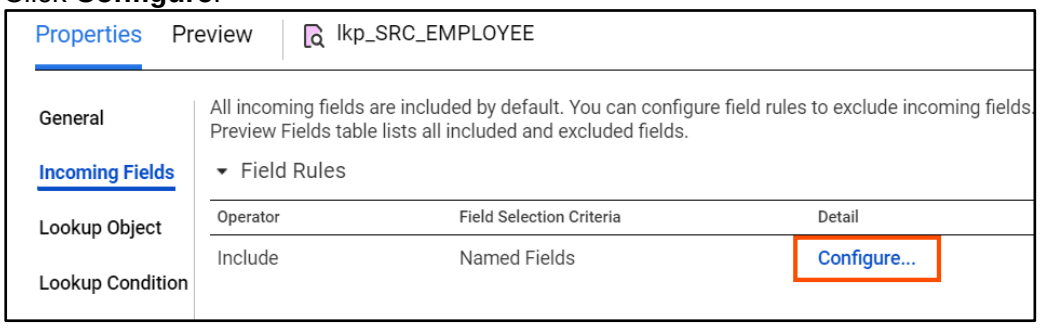
29. In the General field, enter the name as **lkp_SRC_EMPLOYEE**.



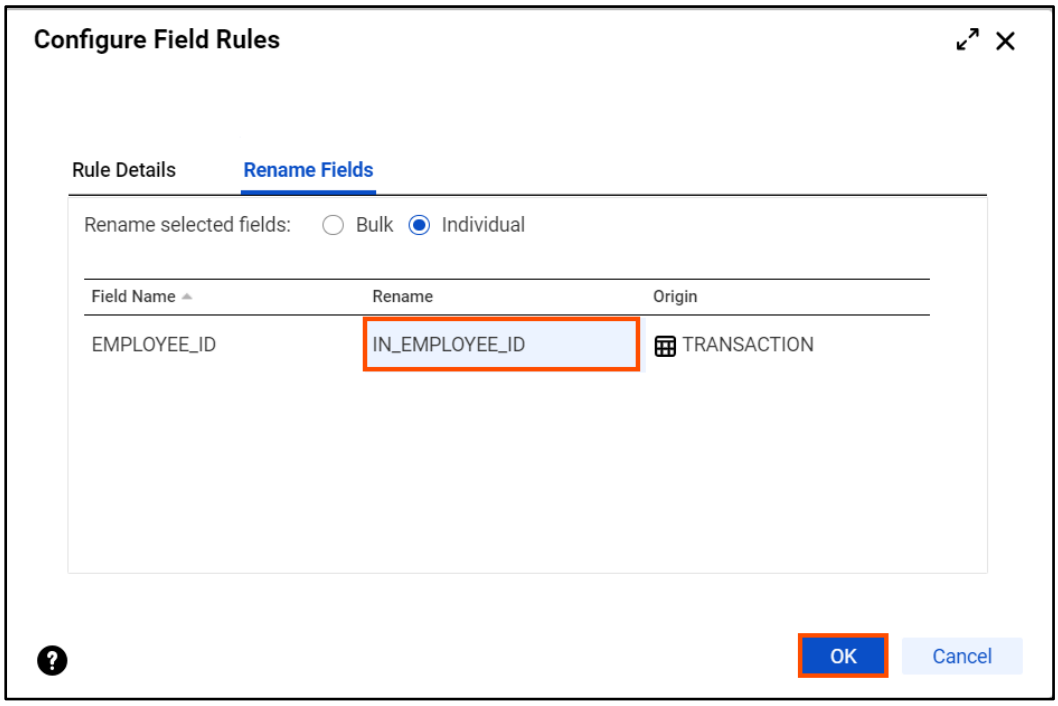
30. Link the **Src_DB_TRANSACTION** with the **lkp_SRC_EMPLOYEE**
31. From the properties pane, click on **Incoming Fields**.
32. In the **Field Rules** section, from the Field Selection Criteria drop-down, select **Named Fields**.

    Note: After selecting Named Fields, click anywhere on the blank space to enable the Configure link.

33. Click **Configure**.

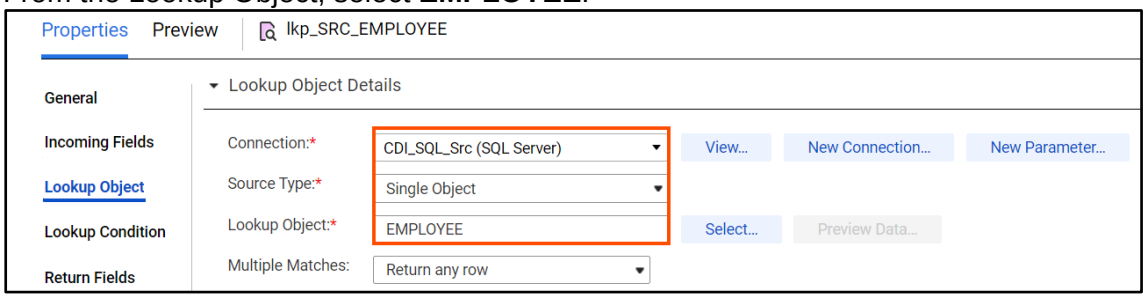


34. Select **EMPLOYEE_ID** from the Rule Details tab.
35. Select the Rename Fields tab, and in the Rename field enter **IN_EMPLOYEE_ID**.
36. Click **OK**.



37. From the properties pane, select **Lookup Object**.
38. From the connection dropdown, select **CDI_SQL_Src**.

39. From the Lookup Object, select **EMPLOYEE**.



40. Select **Lookup Condition**, from the properties pane.
41. To add a new Lookup condition, click ⊞.
42. Enter the details as shown in the table below and click **OK**.

Lookup Field	Operator	Incoming Field
EMPLOYEE_ID	=	IN_EMPLOYEE_ID

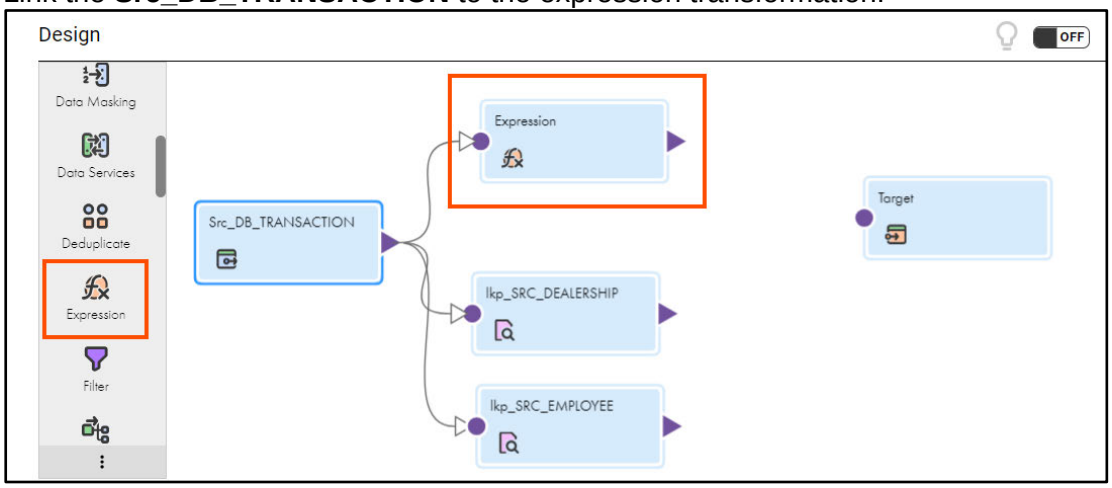
Add Expression Transformation

Expression transformation calculates value within a single row. Expression transformation can also be used to test conditional statements before passing the data to other transformations.

When you configure an Expression transformation, create an expression field for the output of each calculation that you want to use in the data flow.

43. From the list of available transformations, drag and drop the Expression transformation onto the mapping canvas.

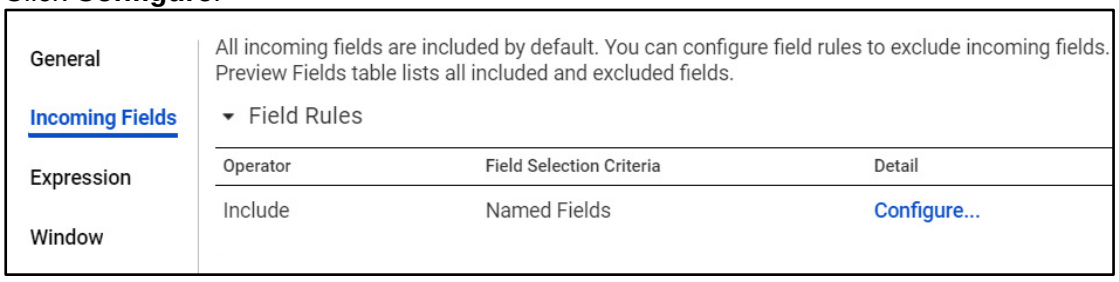
44. Link the **Src_DB_TRANSACTION** to the expression transformation.



45. Select the Expression transformation from the mapping canvas.
46. From the properties, in the General section, enter the name as **exp_CALC_REVENUE**.

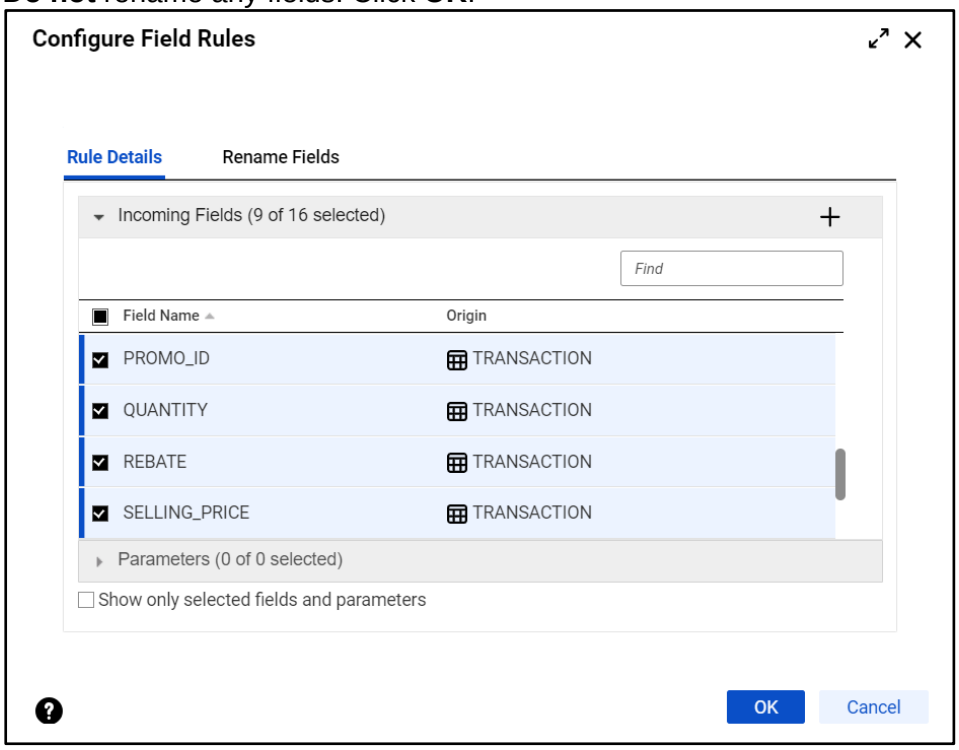
47. Select the Expression transformation on the mapping canvas. From the properties pane, click on **Incoming Fields**.
48. In the Field Rules section, from the Field Selection Criteria drop-down, select **Named Fields**.

    Note: After selecting Named Fields, click anywhere on the blank space to enable the Configure link.
49. Click **Configure**.

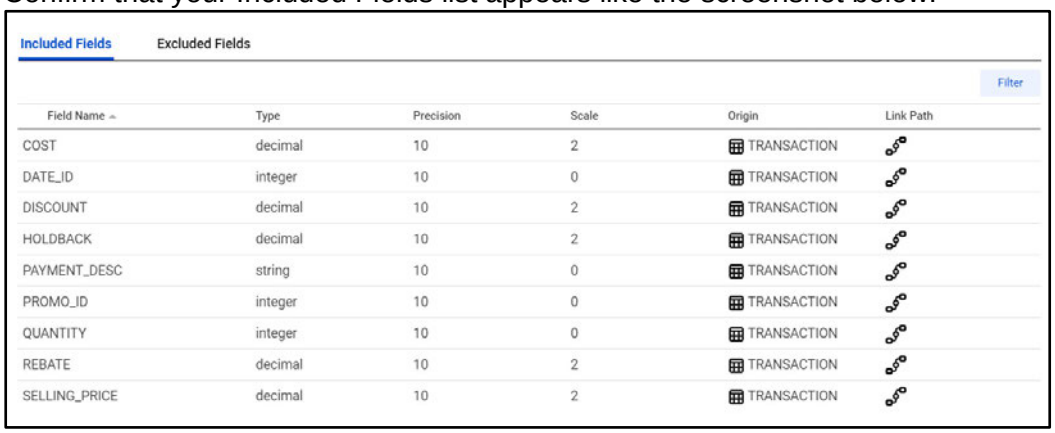


50. Select **COST**, **DATE_ID**, **DISCOUNT**, **HOLDBACK**, **PAYMENT_DESC**, **PROMO_ID**, **QUANTITY**, **REBATE**, and **SELLING PRICE** from the Rule Details tab.

51. **Do not** rename any fields. Click **OK**.

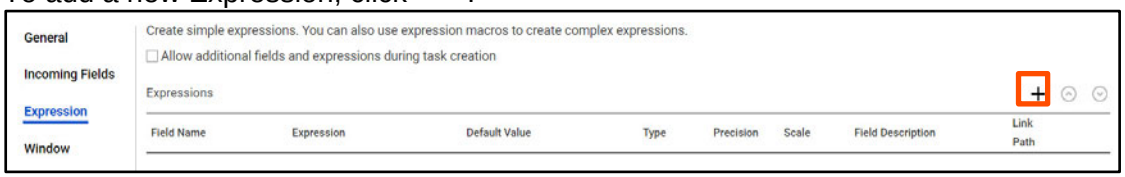


52. Confirm that your Included Fields list appears like the screenshot below:



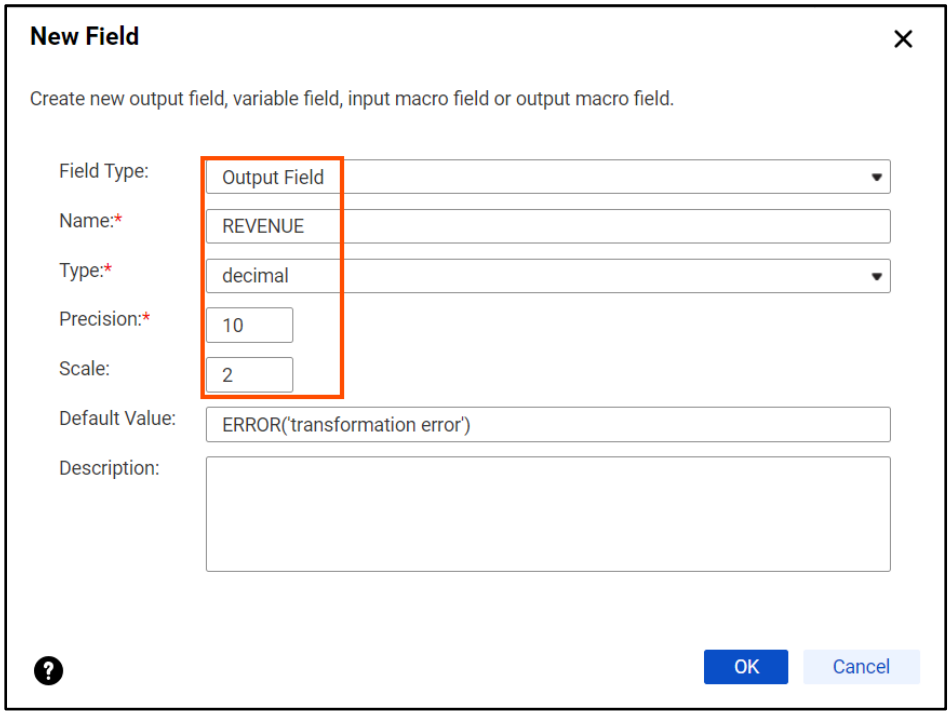
53. From the properties pane, select **Expression**.

54. To add a new Expression, click .

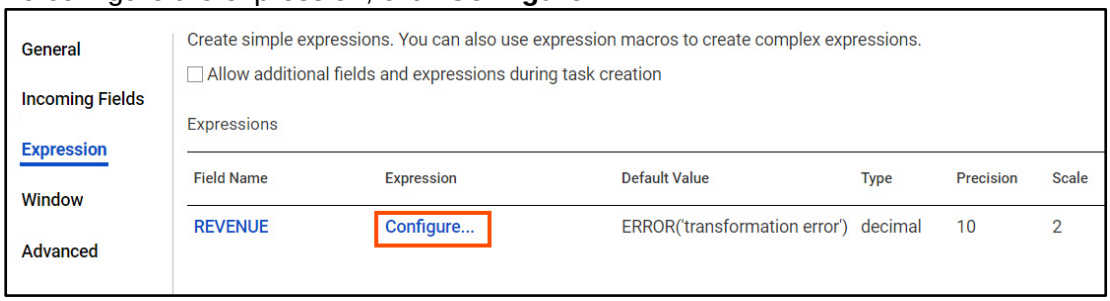


55. Enter the details as shown in the table below and click **OK**.

Field Type	Name	Type	Precision	Scale
Output Field	REVENUE	decimal	10	2

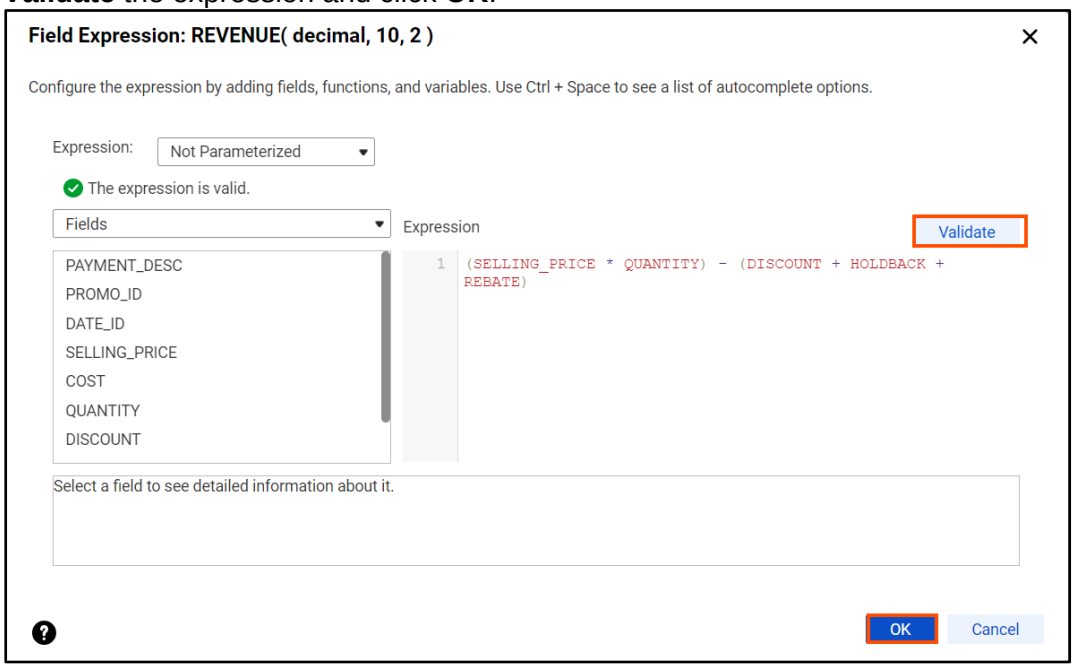


56. To configure the expression, click **Configure**.

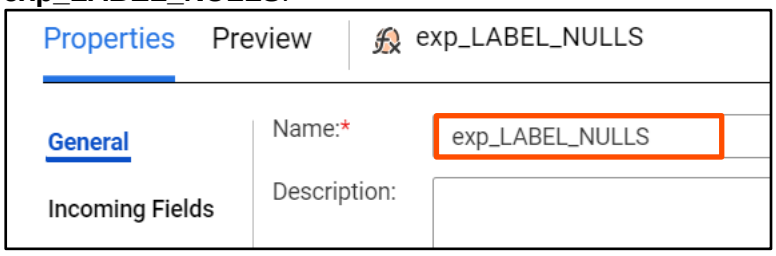


57. In the Expression field, copy and paste the following expression:
(SELLING_PRICE * QUANTITY) - (DISCOUNT + HOLDBACK + REBATE)

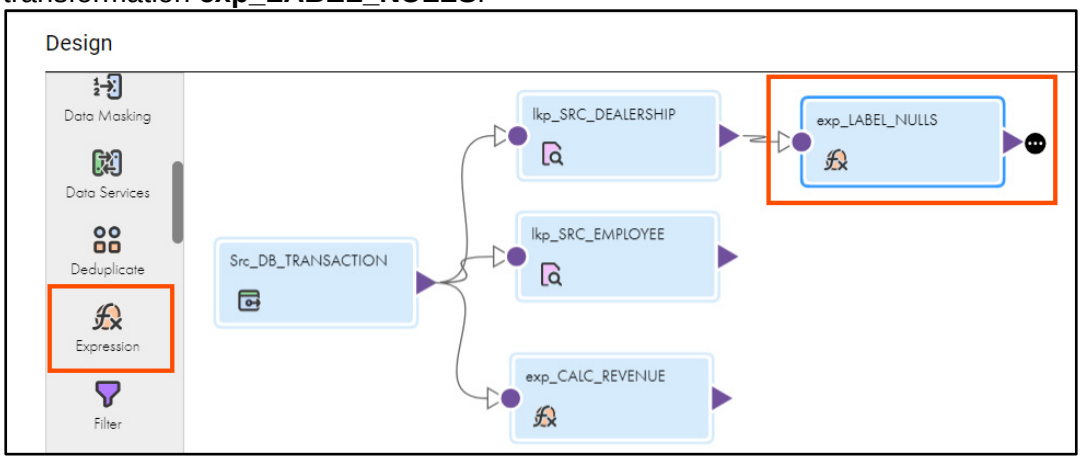
58. **Validate** the expression and click **OK**.



59. From the list of available transformations, drag and drop another **Expression** transformation on the mapping canvas.
60. Select the new Expression transformation from the mapping canvas.
61. From the properties pane, in the General section, enter the name as **exp_LABEL_NULLS**.

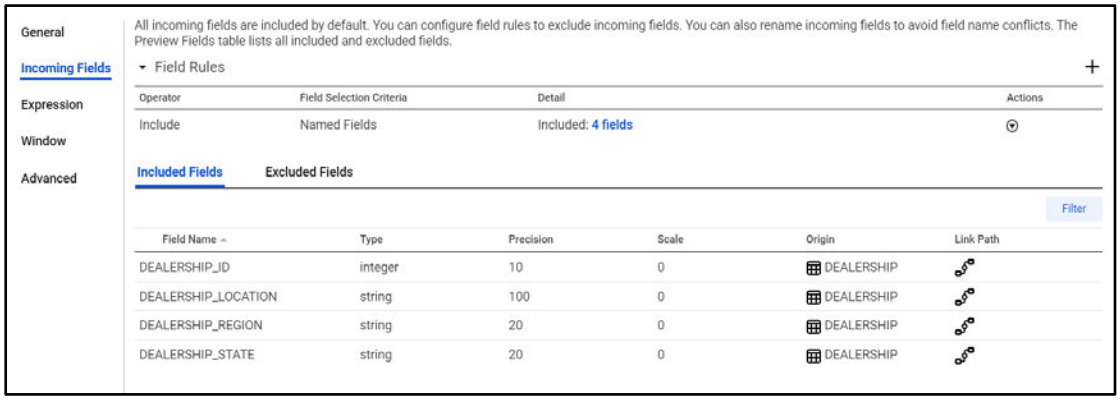


62. Link the output of the **lkp_SRC_DEALERSHIP** lookup to the new Expression transformation **exp_LABEL_NULLS**.

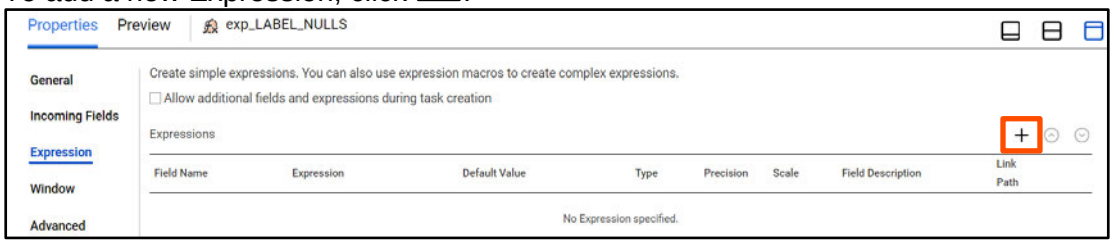


63. From the properties pane, click on **Incoming Fields**.

64. From the **Field Selection Criteria** drop down, select the **Named Fields**.
65. **Include** the fields DEALERSHIP_ID, DEALERSHIP_LOCATION, DEALERSHIP_REGION, and DEALERSHIP_STATE. Ensure the other fields are unchecked. Your Incoming Fields properties should match the screenshot below.

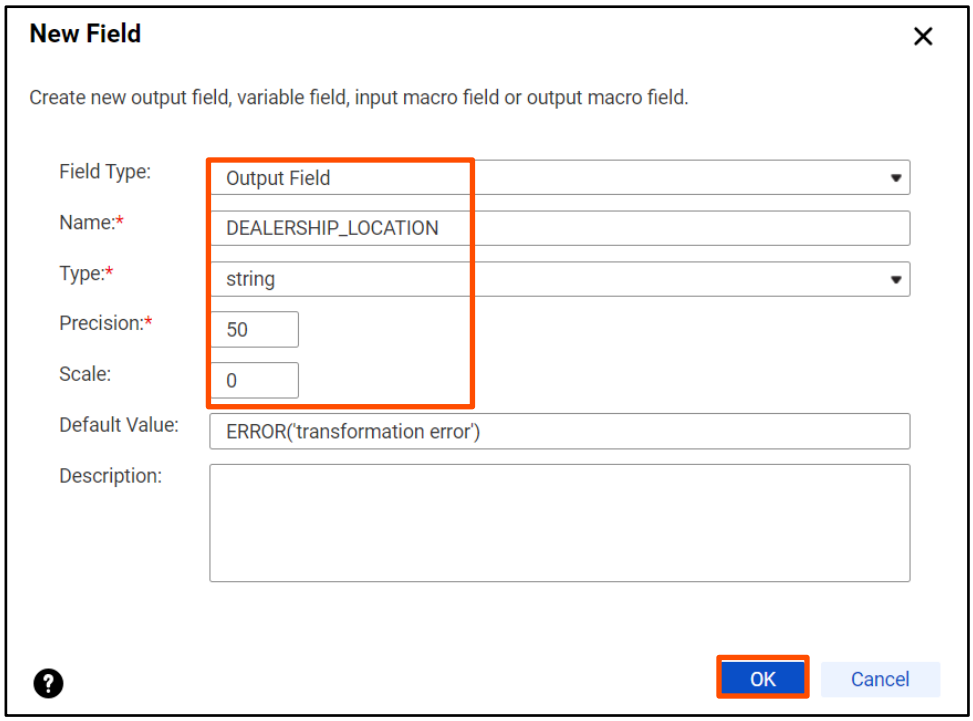


66. Navigate to the Expression tab.
67. To add a new Expression, click +.

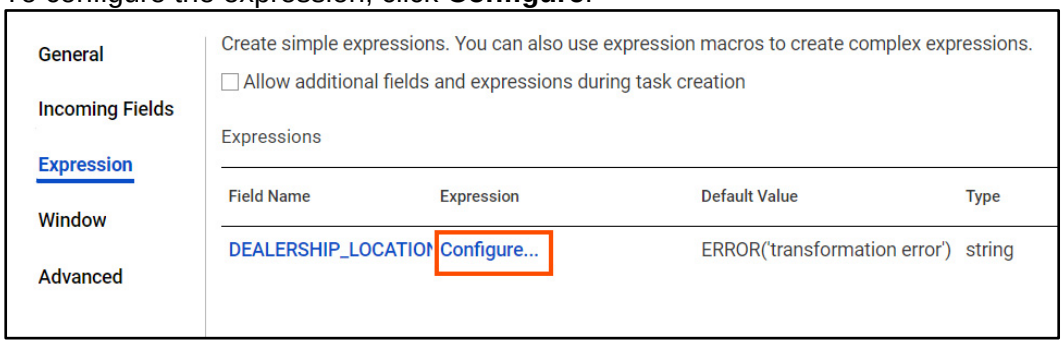


68. Set the new expression properties as shown in the table below and click **OK**.

Field Type	Name	Type	Precision	Scale
Output Field	DEALERSHIP_LOCATION	string	50	0

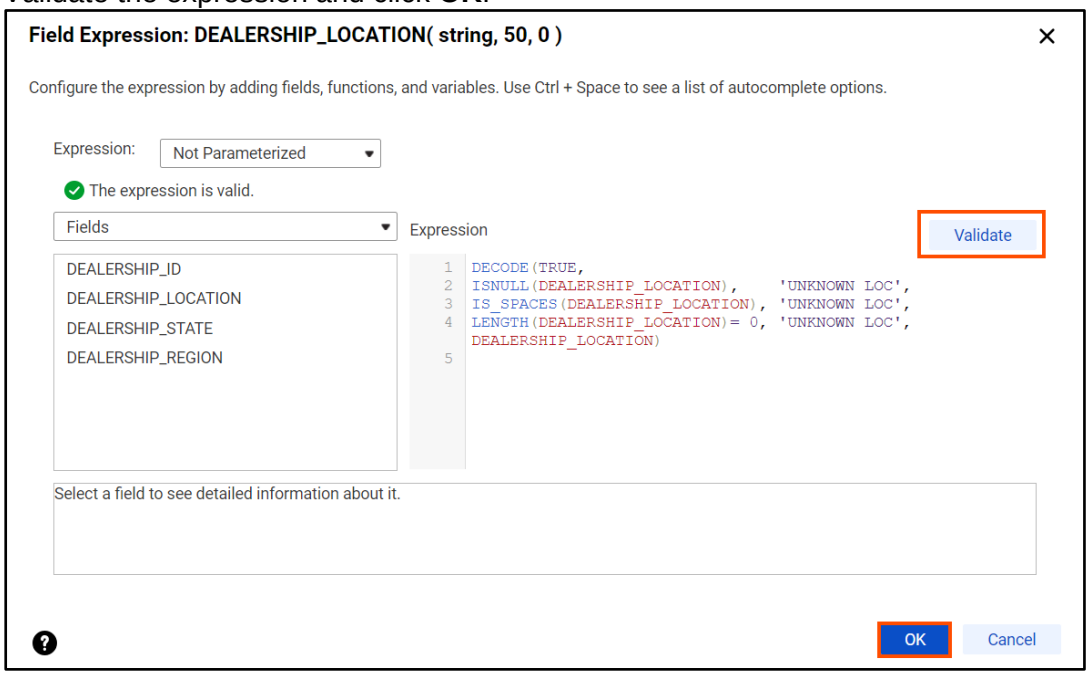


69. To configure the expression, click **Configure**.



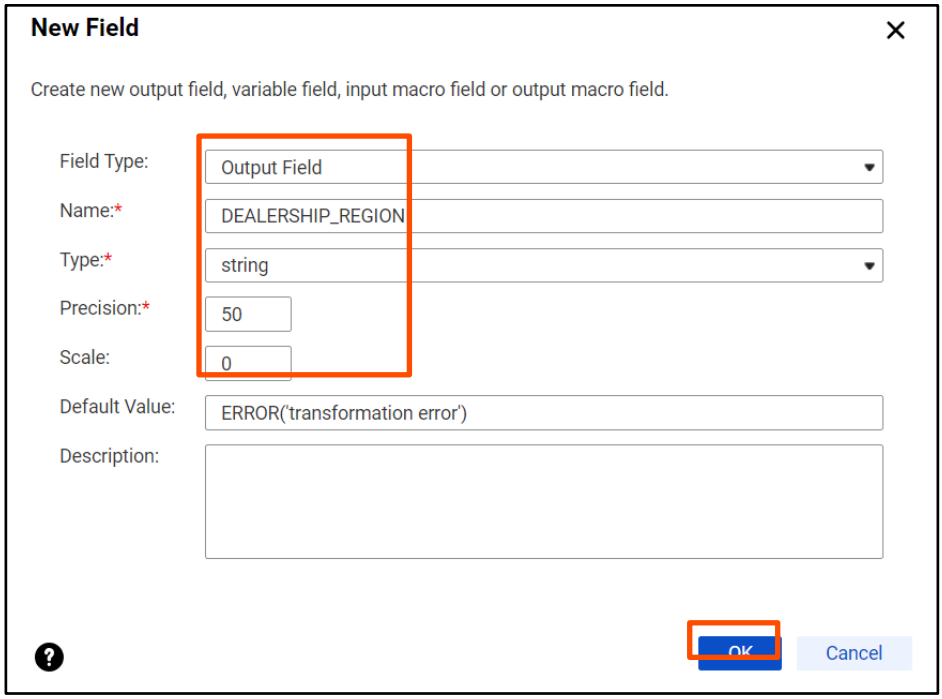
70. In order to identify and label NULL or blank values in the DEALERSHIP_LOCATION value, in the Expression field, enter the following expression:
DECODE(TRUE,
ISNULL(DEALERSHIP_LOCATION), 'UNKNOWN LOC',
IS_SPACES(DEALERSHIP_LOCATION), 'UNKNOWN LOC',
LENGTH(DEALERSHIP_LOCATION)= 0, 'UNKNOWN LOC',
DEALERSHIP_LOCATION)

71. Validate the expression and click **OK**.



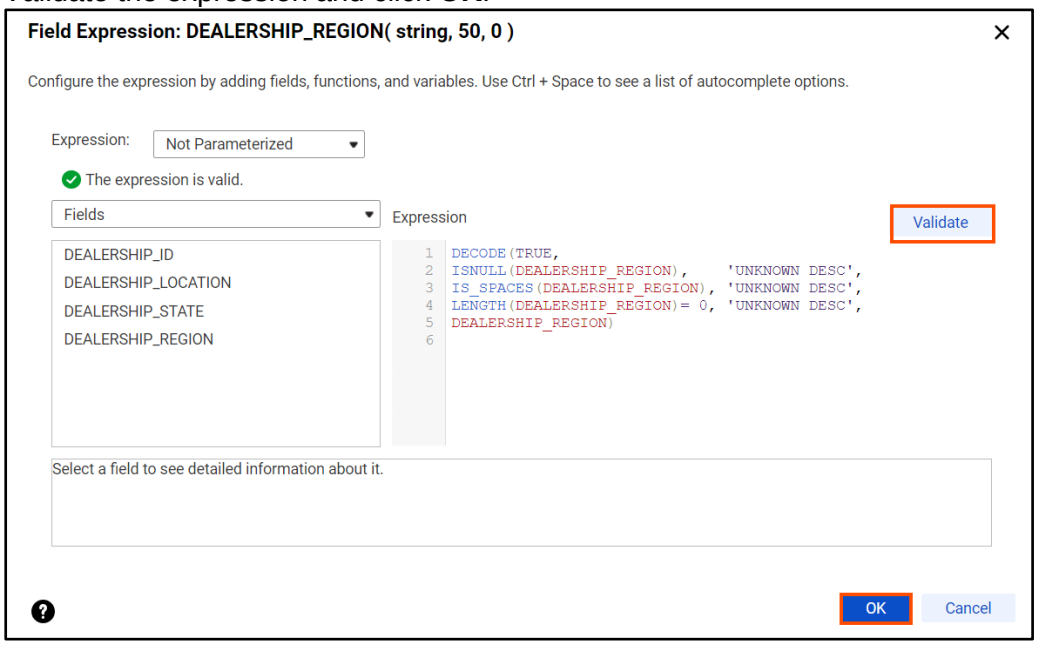
72. In the Expression tab, add another expression and set its properties as shown in the table below. Click **OK**.

Field Type	Name	Type	Precision	Scale
Output Field	DEALERSHIP_REGION	string	50	0

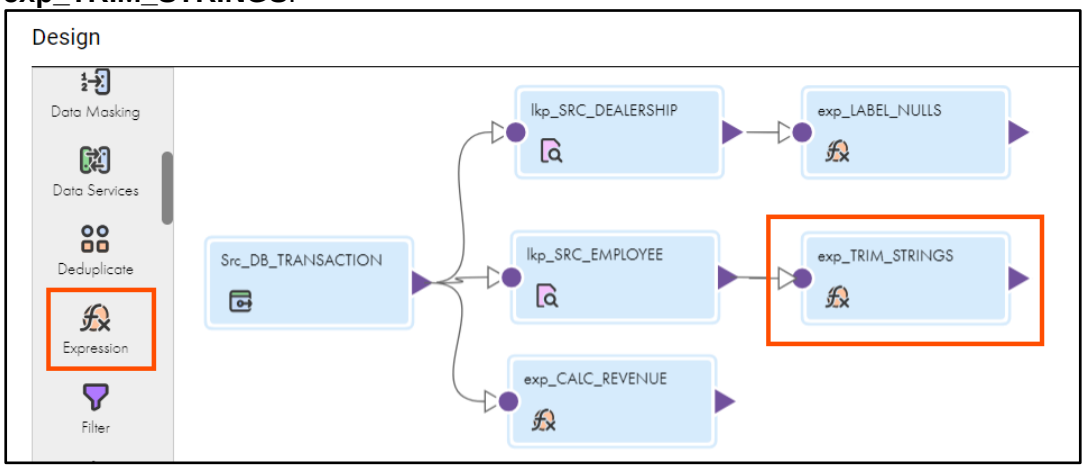


73. To configure the expression, click **Configure**.

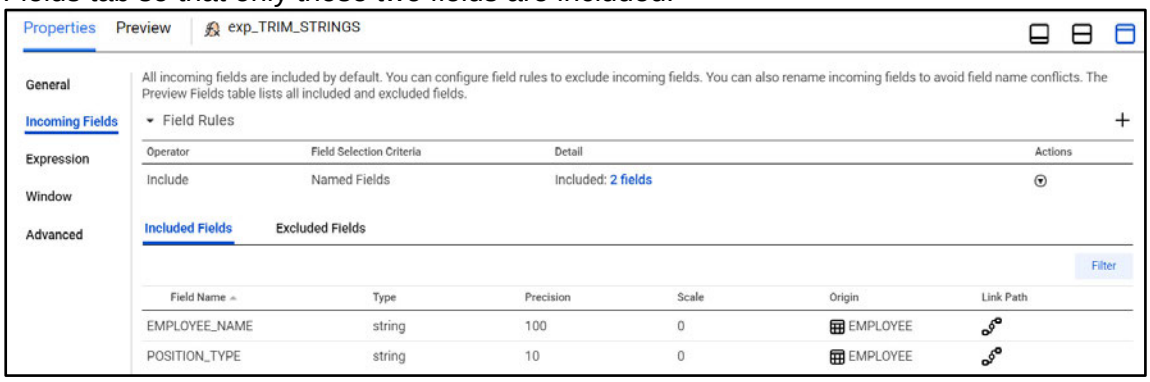
74. In the Expression field, enter the following expression:
DECODE(TRUE,
ISNULL(DEALERSHIP_REGION), 'UNKNOWN DESC',
IS_SPACES(DEALERSHIP_REGION), 'UNKNOWN DESC',
LENGTH(DEALERSHIP_REGION)= 0, 'UNKNOWN DESC',
DEALERSHIP_REGION)
75. Validate the expression and click **OK**.



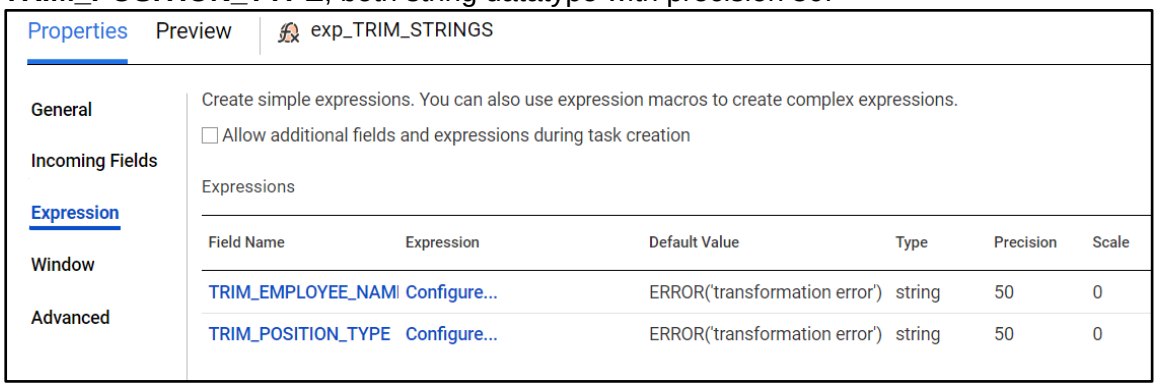
76. Add another Expression transformation to the mapping and name it **exp_TRIM_STRINGS**.
77. Link the **lkp_SRC_EMPLOYEE** Lookup to the new Expression transformation **exp_TRIM_STRINGS**.



78. The new **exp_TRIM_STRINGS** Expression transformation needs to include only two incoming fields: **EMPLOYEE_NAME** and **POSITION_TYPE**. Configure the Incoming Fields tab so that only those two fields are included.



79. In the Expression tab, create two output fields **TRIM_EMPLOYEE_NAME** and **TRIM_POSITION_TYPE**, both string datatype with precision **50**.



80. Enter these expressions for the TRIM_EMPLOYEE_NAME and TRIM_POSITION_TYPE fields, respectively:

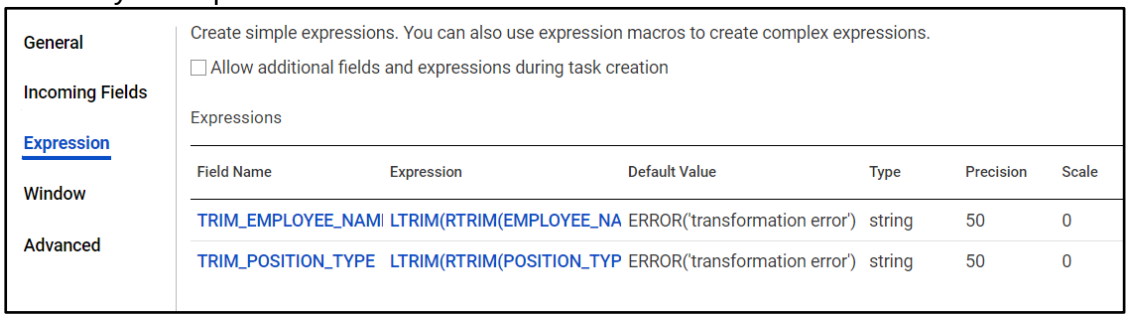
In TRIM_EMPLOYEE_NAME:

LTRIM(RTRIM(EMPLOYEE_NAME))

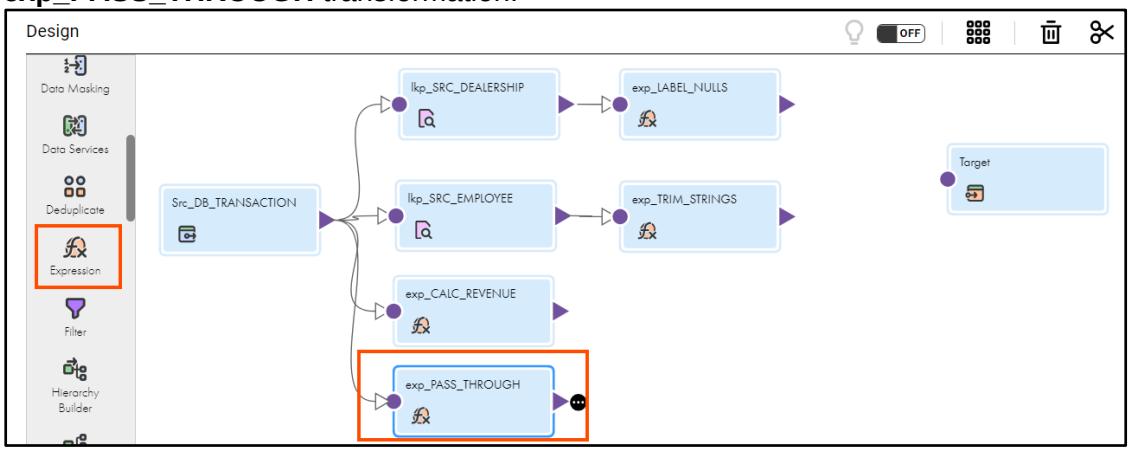
In TRIM_POSITION_TYPE:

LTRIM(RTRIM(POSITION_TYPE))

81. Confirm your Expression tab matches the screenshot below:

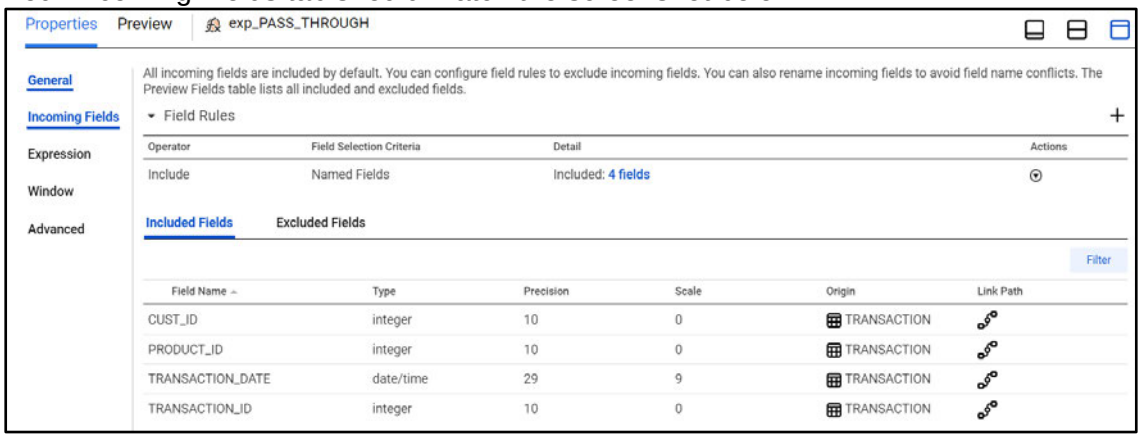


82. Create a new Expression transformation and name it **exp_PASS_THROUGH**.
83. Link the output of the **Src_DB_TRANSACTIONS** source object to the new **exp_PASS_THROUGH** transformation.



Note: The purpose of this transformation is to send four fields from the source directly to the target without any expression logic applied to those fields. The use of a single transformation for this purpose avoids name conflicts (and having to bulk rename fields) in the other transformations.

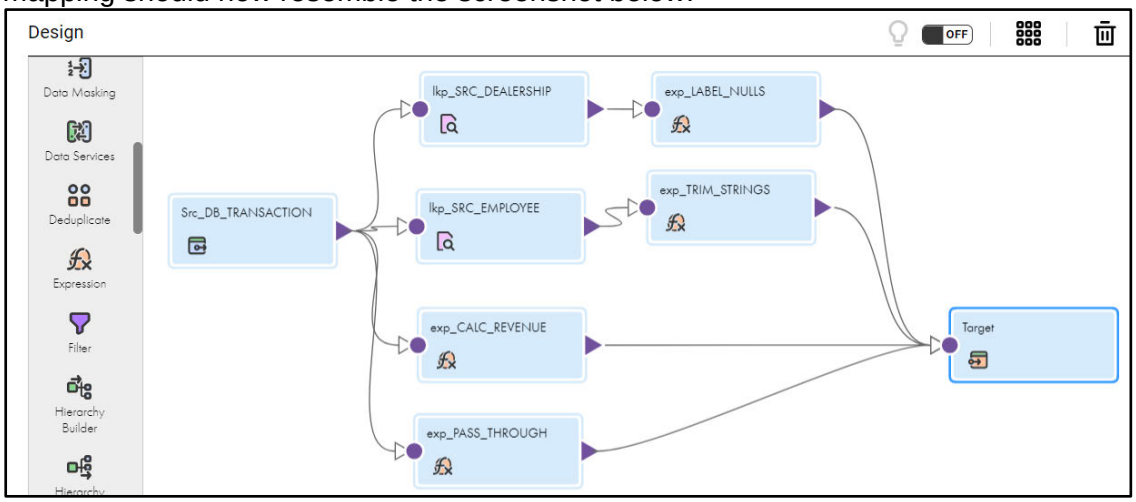
84. In the **Incoming Fields** tab, include the following fields: CUST_ID, PRODUCT_ID, TRANSACTION_DATE, and TRANSACTION_ID.
85. Your Incoming Fields tab should match the screenshot below:



86. Save the mapping.

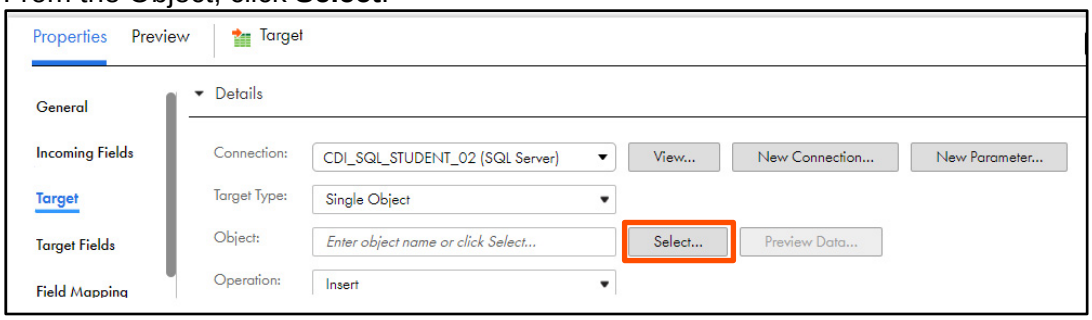
Configure Target

87. Link the outputs of **all four** of the Expression transformations to the target object. Your mapping should now resemble the screenshot below:



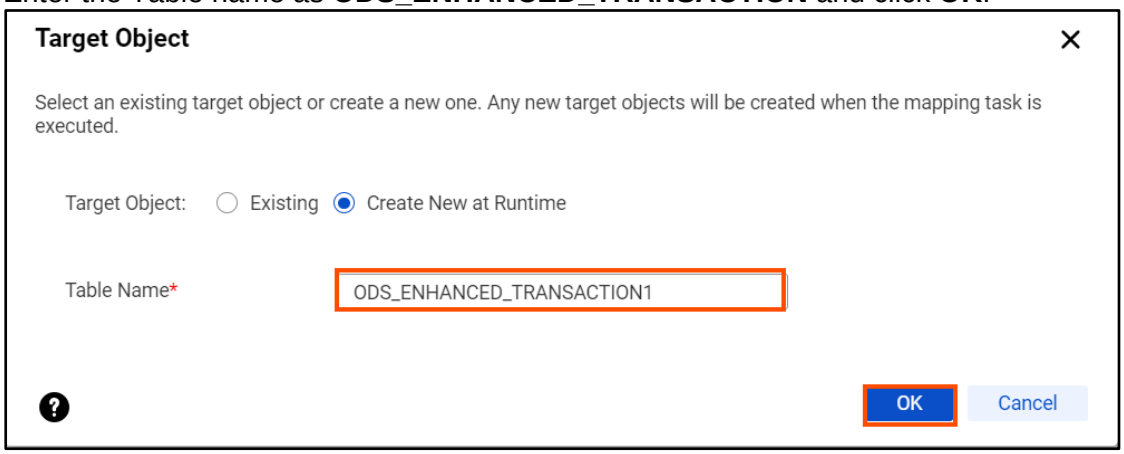
88. Select **Target** on the mapping canvas.
89. From the properties pane, select **Target**.

90. From the Connection drop down, select your **SQL Server target** connection.
91. From the Object, click **Select**.

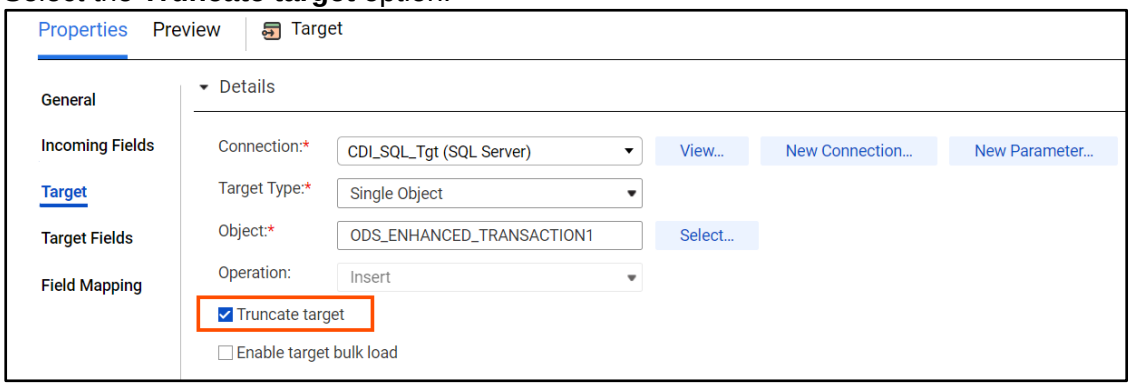


92. In the Target Object, click on **Create New at Runtime**.

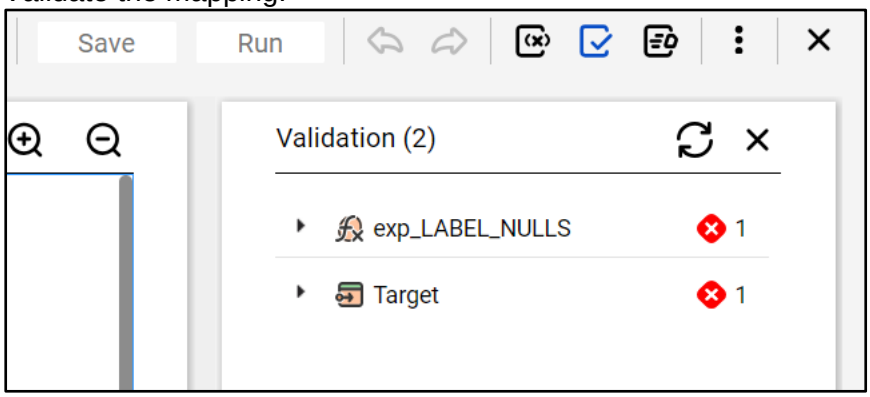
93. Enter the Table name as **ODS_ENHANCED_TRANSACTION** and click **OK**.



94. Select the **Truncate target** option.

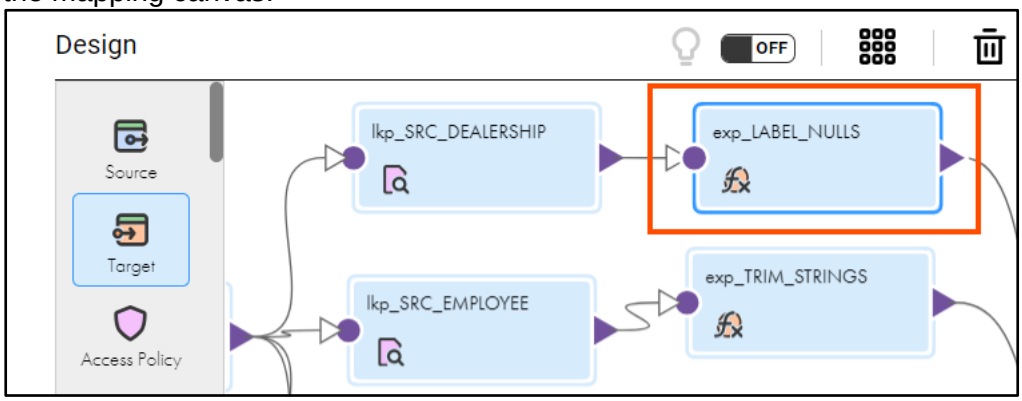


95. Validate the mapping.

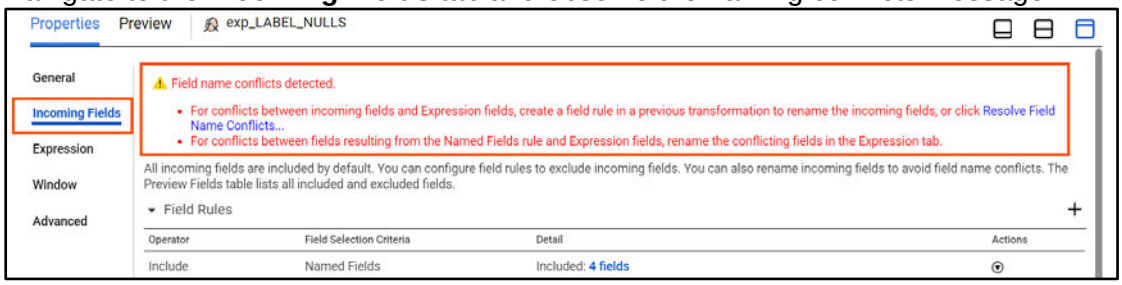


Note: If the mapping is invalid due to “field name conflicts” in any of the transformations or the target, examine the Incoming Fields tab of the invalid object and ensure they confirm to the previous instructions. Ensure only the specified fields for that object have been included. Also, ensure the mapping resembles the screenshot on the previous page with respect to the order of the transformations.

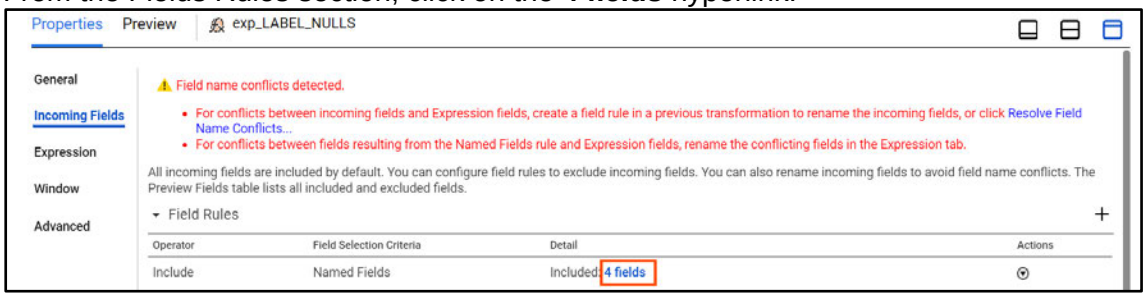
96. Click on the first object in the Validation panel. It will highlight the corresponding object in the mapping canvas.



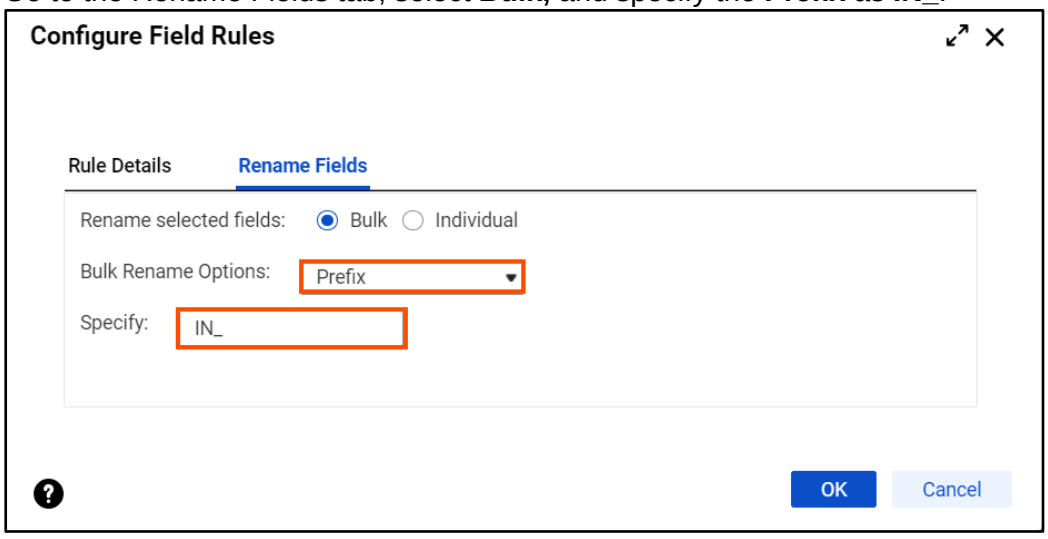
97. Navigate to the **Incoming Fields** tab and observe the naming conflicts message.



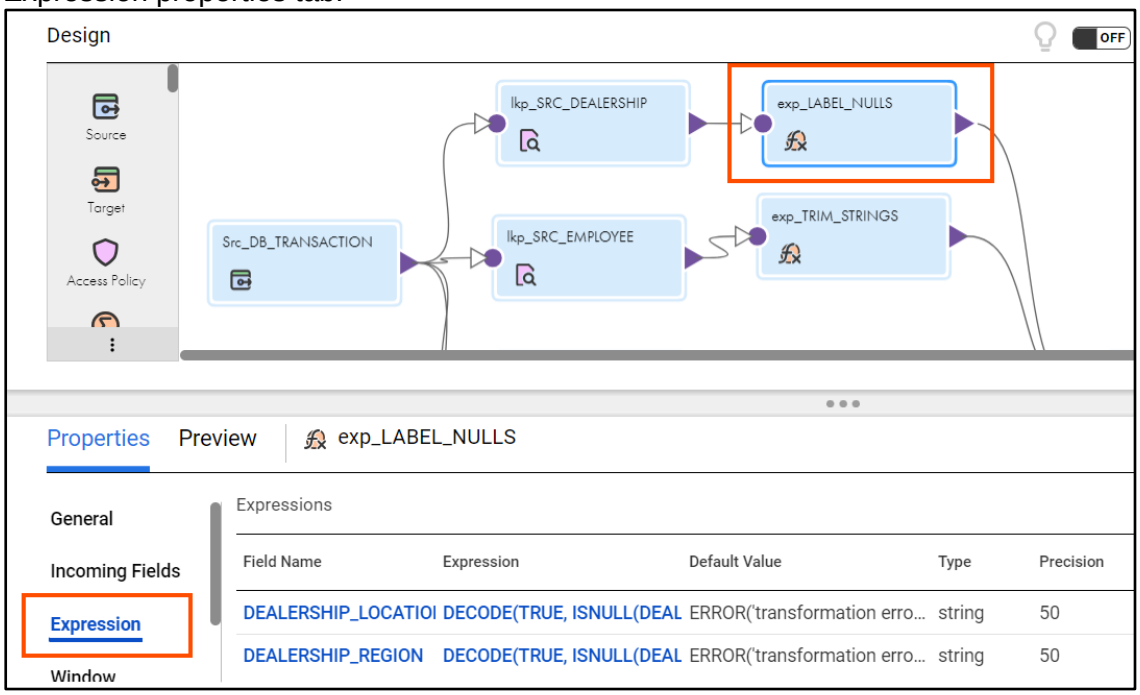
98. From the Fields Rules section, click on the **4 fields** hyperlink.



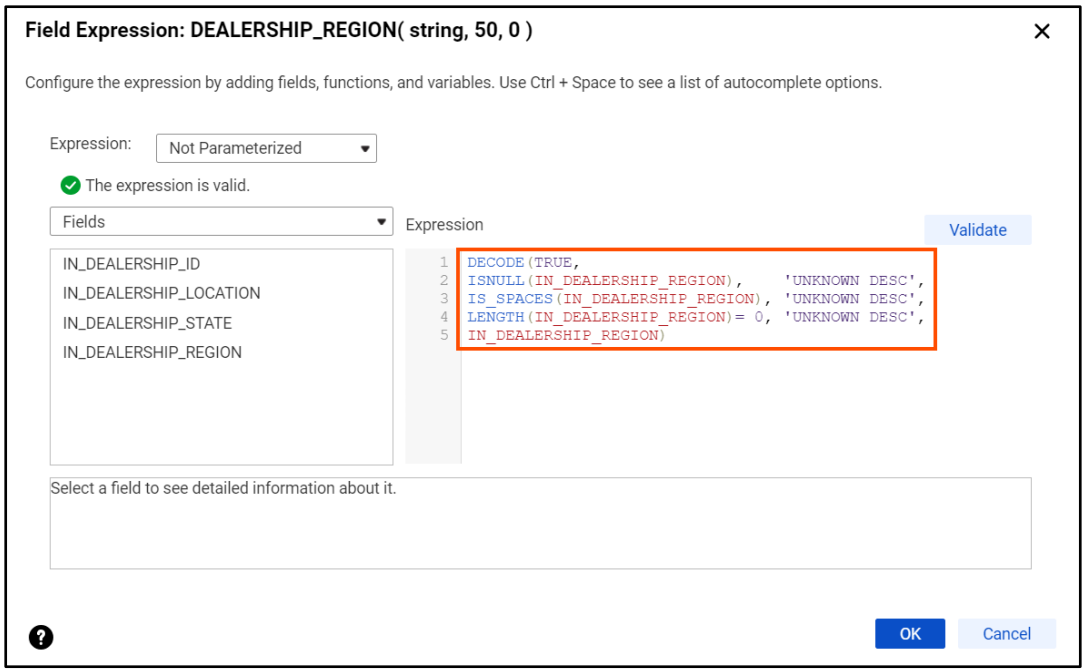
99. Go to the Rename Fields tab, select **Bulk,** and specify the **Prefix** as **IN_**.



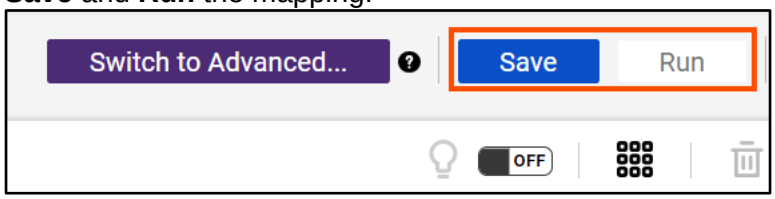
100. Click **OK**.
101. Make sure exp_LABEL_NULLS is selected from the mapping canvas and go to the Expression properties tab.



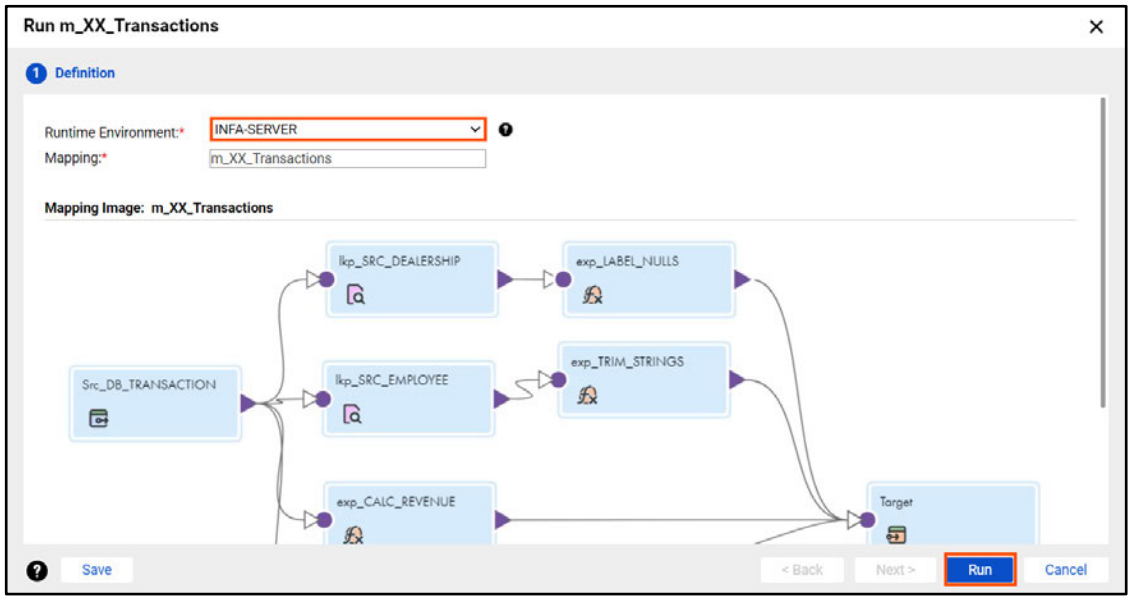
102. Click on each of the Expression fields and update the expression so that **IN**_DEALERSHIP_LOCATION is used instead of DEALERSHIP_LOCATION and **IN**_DEALERSHIP_REGION is used instead of DEALERSHIP_REGION.



103. **Save** and **Run** the mapping.

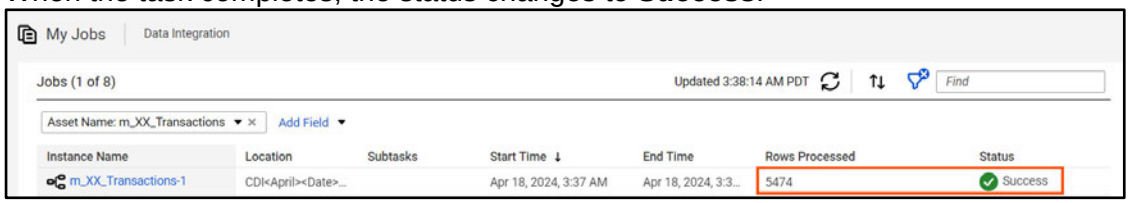


104. Select the runtime environment as INFA-SERVER and click **Run**.



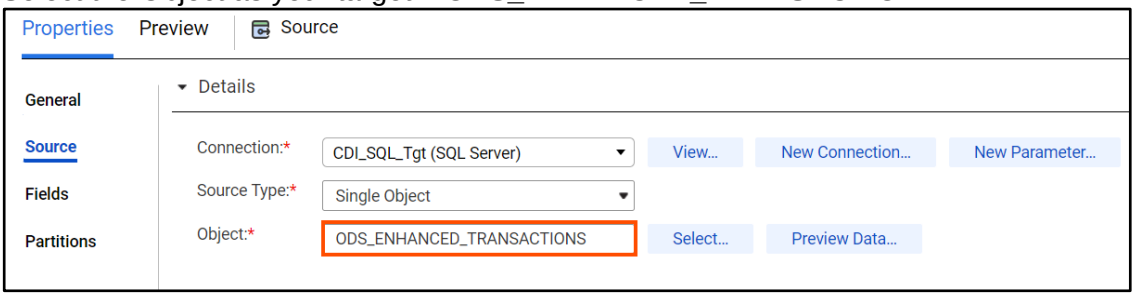
Monitor Status

105. Monitor the task status from the **My Jobs** page.
106. When the task completes, the status changes to **Success**.

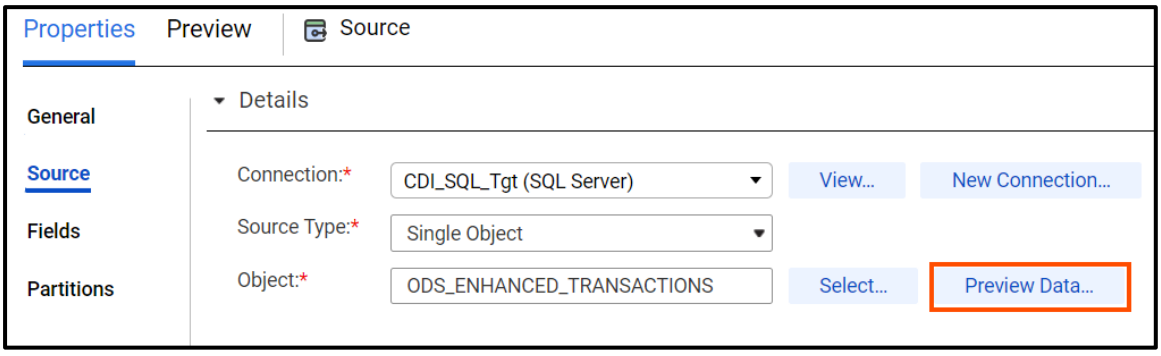


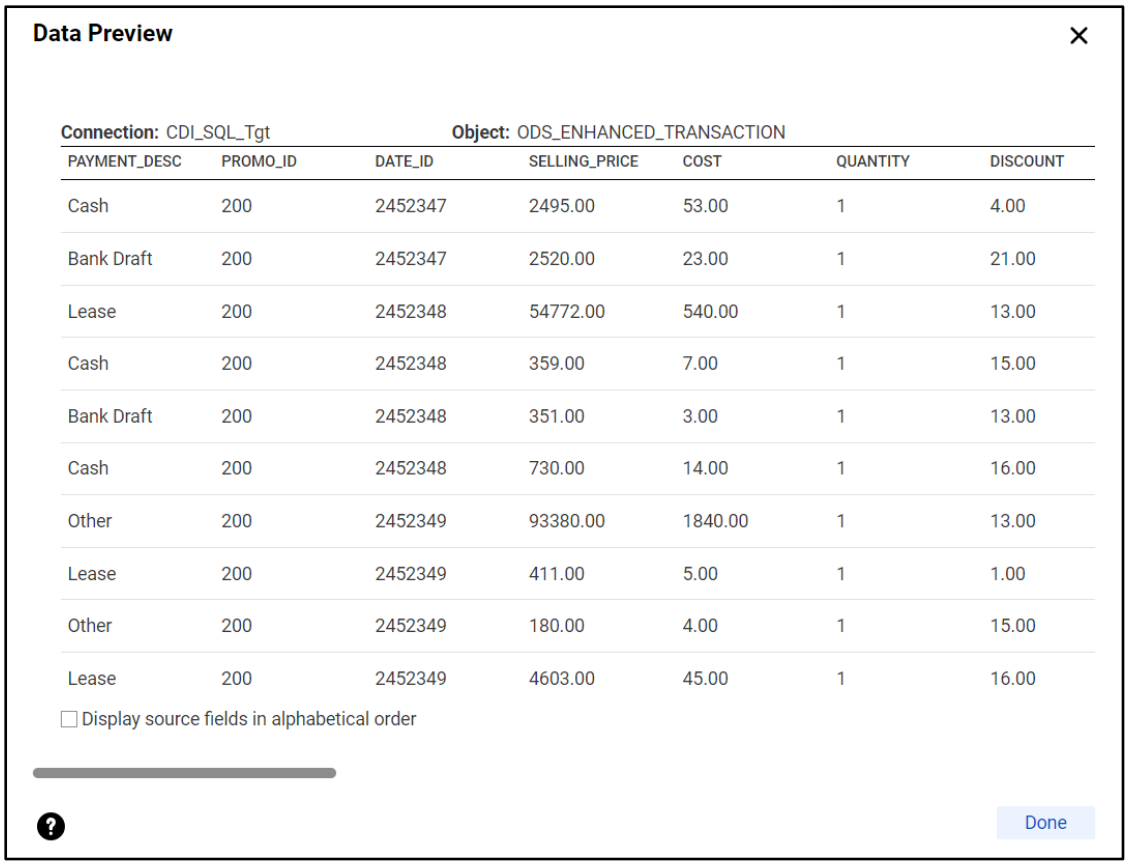
Optional - Open the Dummy Mapping and View the Data

107. Navigate to your working folder and open the dummy mapping **m_XX_DUMMY_MAPPING**.
108. From the mapping canvas, select **Source** and go to the **Source** properties tab.
109. Change the Connection to your target SQL Server DB connection. Click **Yes** in the Change Connection window.
110. Select the Object as your target – **ODS_ENHANCED_TRANSACTION**.



111. Click **Preview Data** and view the results.





112. Click **Done**.
113. Save and close all the assets from the navigation pane.

This concludes the lab.

Module 7: Joiner and Lookup Transformations

Lab 7-2: Merging Current and Previous Data Loads Using Full Outer Join

Overview:
A Joiner transformation joins two related heterogeneous sources. The Joiner transformation joins data based on the join conditions and the join type. The result set passed to the rest of the mapping is based on the join type. Full outer join includes rows with matching join conditions and all incoming data from the master pipeline and the detail pipeline.

Objective:

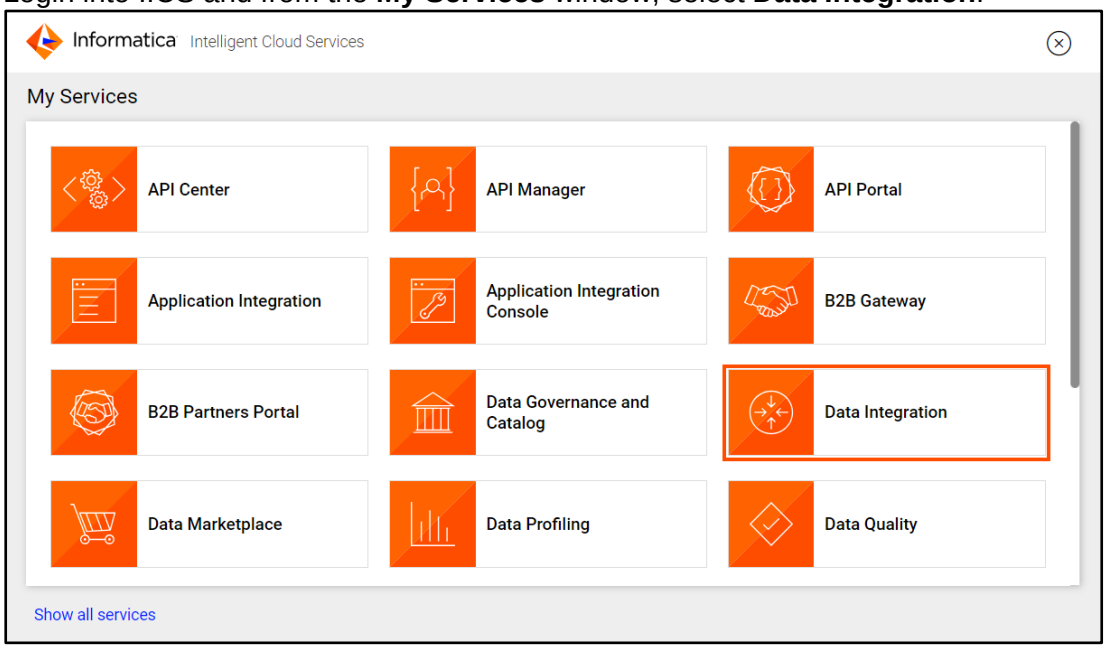
- Configure a mapping in Informatica Cloud
- Use full outer Join and complete the logic

Duration:
35 minutes

Tasks

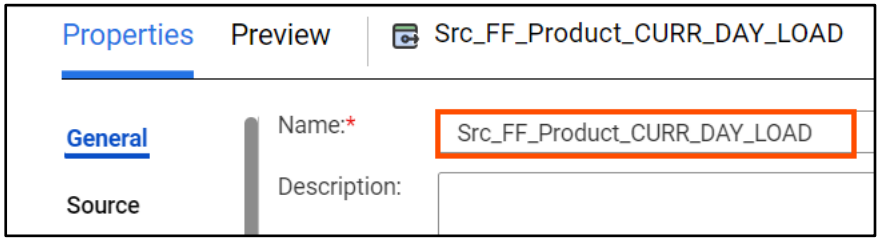
Create a Mapping

1. Login into IICS and from the **My Services** window, select **Data Integration**.

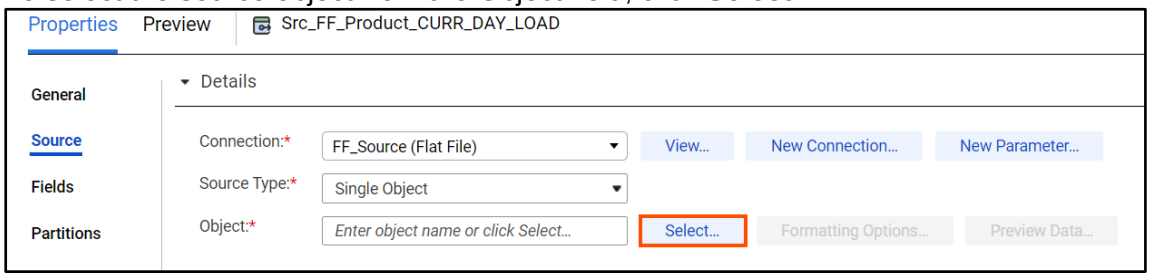


2. Create a new **Mapping**.
3. In the Name field, enter **m_XX_Full_outer_join**.

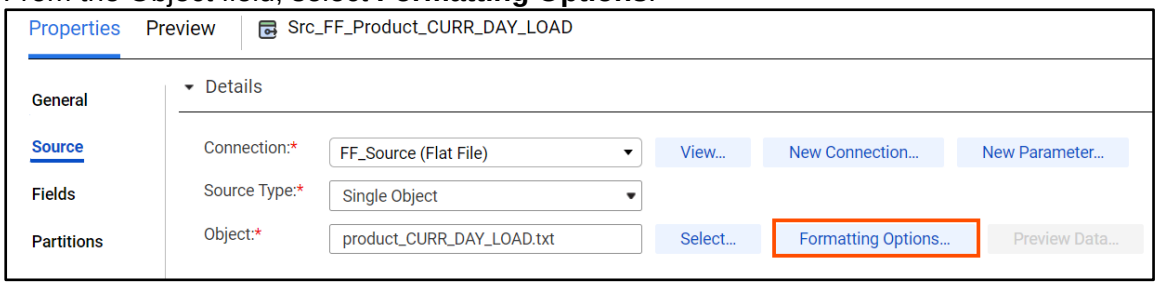
4. Verify that the asset Location is pointing to your working folder.
5. To configure the source, from the mapping canvas, click the **Source** transformation.
6. In the **General** section of the Source properties, enter the Name as **Src_FF_Product_CURR_DAY_LOAD**.



7. From the properties pane, click **Source**.
8. From the Connection drop-down, select **FF_Source**.
9. To select the source object from the Object field, click **Select**.

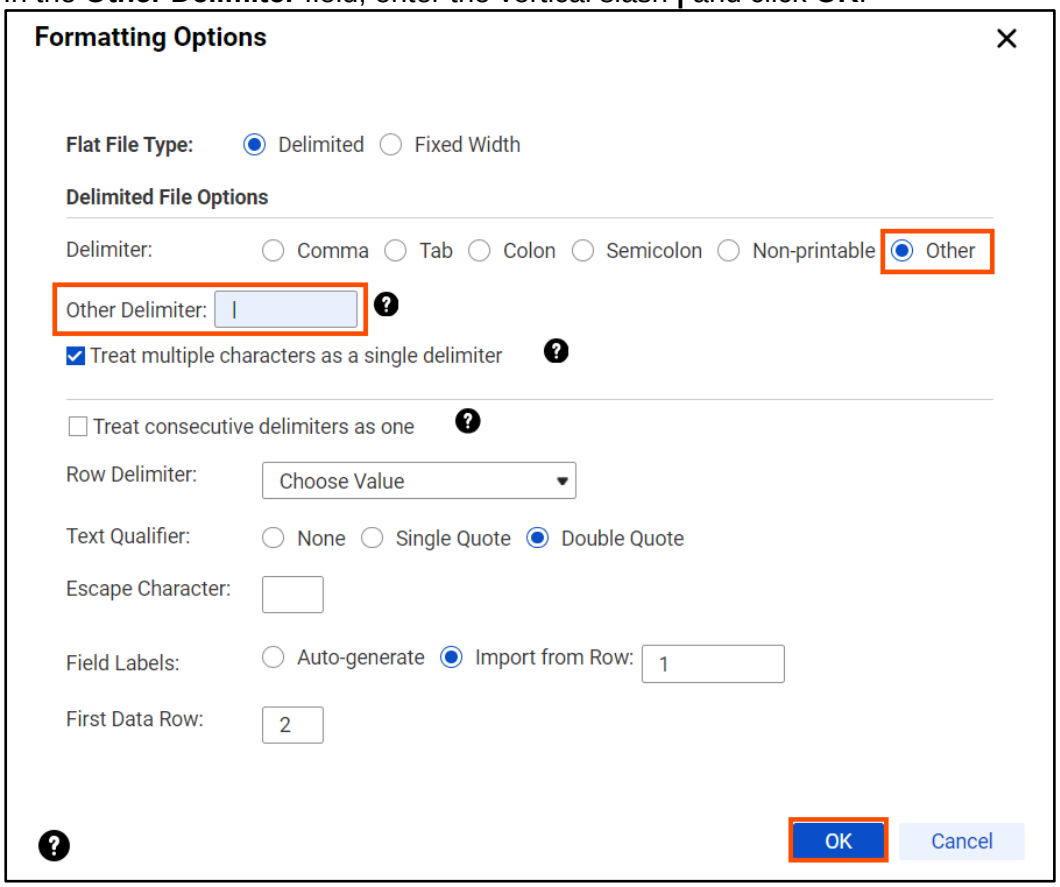


10. From the list, select **product_CURR_DAY_LOAD.txt** and click **OK**.
11. From the Object field, select **Formatting Options**.

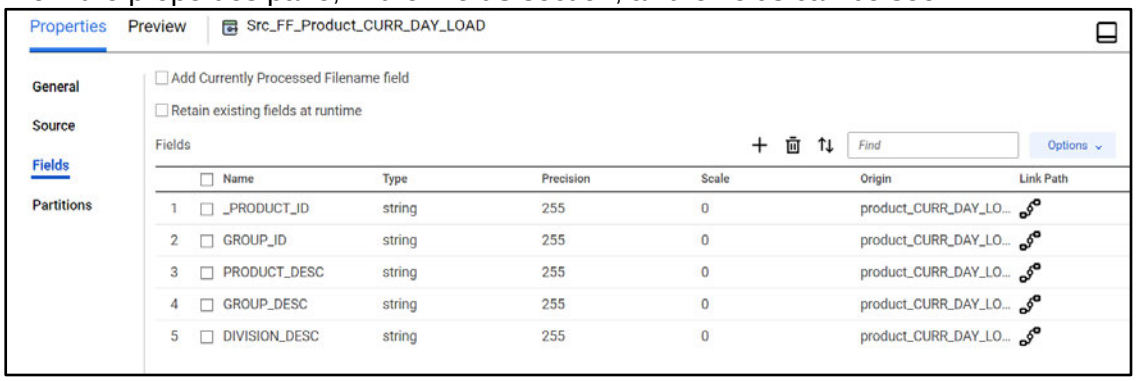


12. From the Delimiter field, select the **Other** option.

13. In the **Other Delimiter** field, enter the vertical slash **|** and click **OK**.

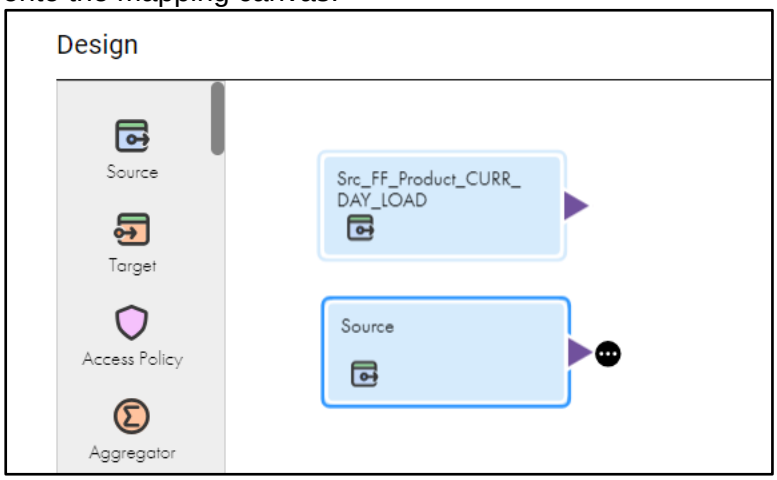


14. From the properties pane, in the **Fields** section, all the fields can be seen.

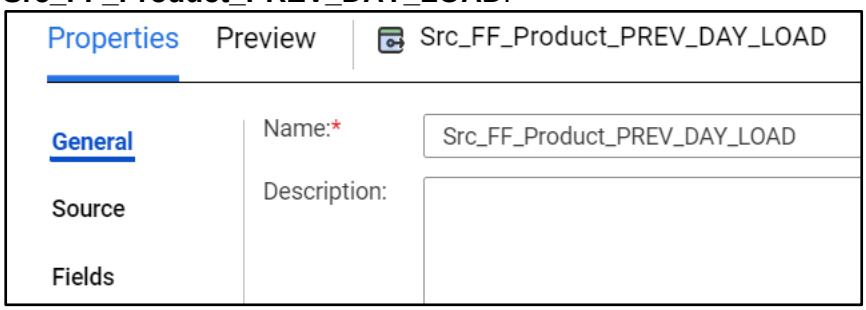


15. Delete the link between the Source and the Target.

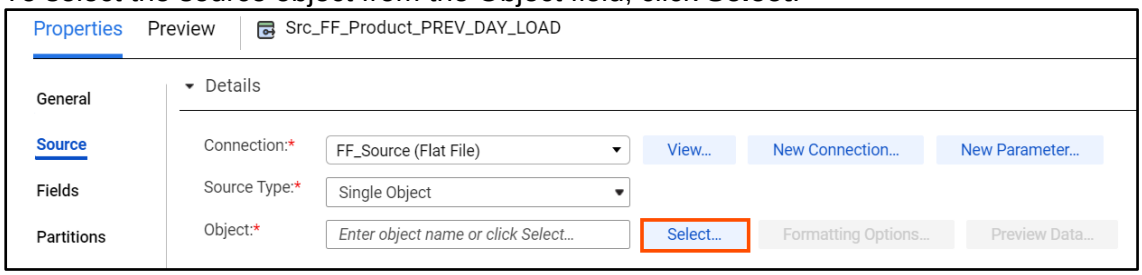
16. From the list of available transformations, drag and drop another source transformation onto the mapping canvas.



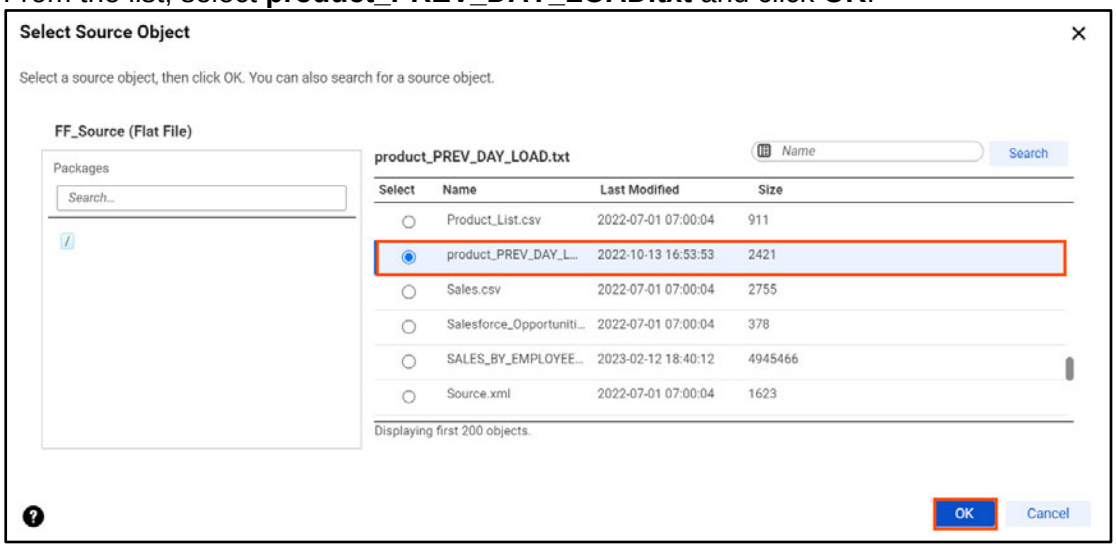
17. To configure the source, from the mapping canvas, click the **Source** transformation.
18. In the General section of the Source properties, enter the Name as **Src_FF_Product_PREV_DAY_LOAD**.



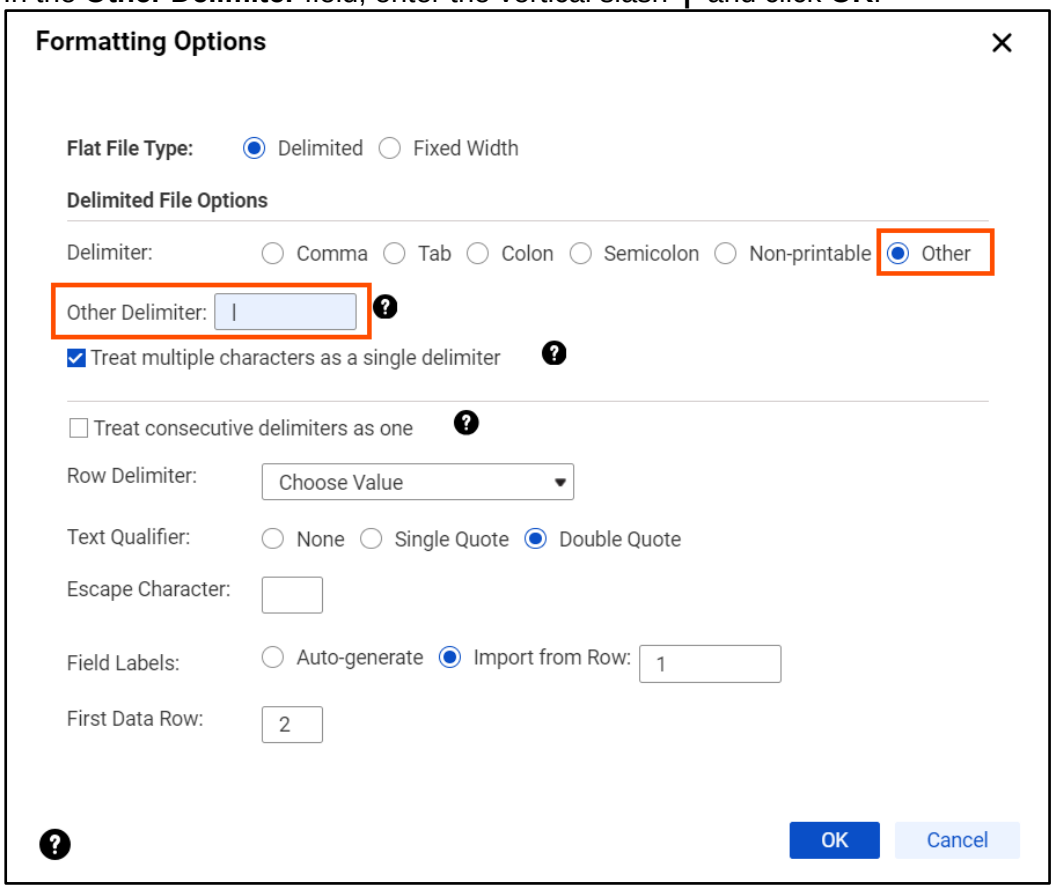
19. From the properties pane, click **Source**.
20. From the Connection drop-down, select **FF_Source**.
21. To select the source object from the Object field, click **Select**.



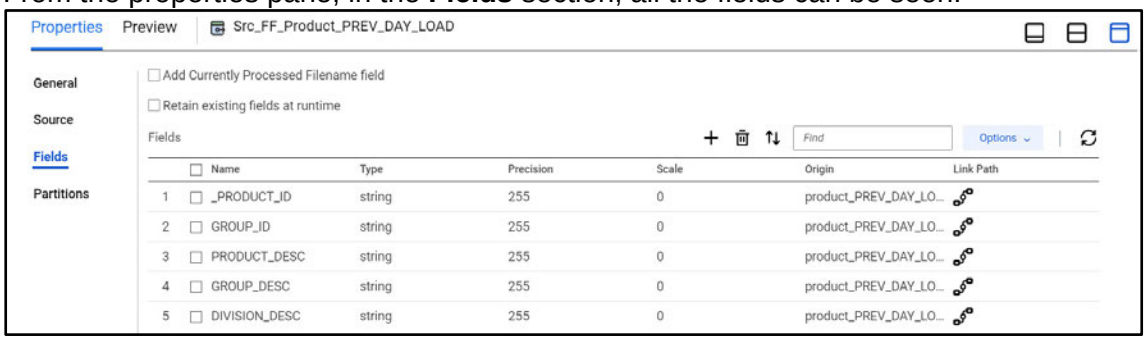
22. From the list, select **product_PREV_DAY_LOAD.txt** and click **OK**.



23. From the Object field, select **Formatting Options**.
24. From the Delimiter field, select the **Other** option.
25. In the **Other Delimiter** field, enter the vertical slash '**|**' and click **OK**.

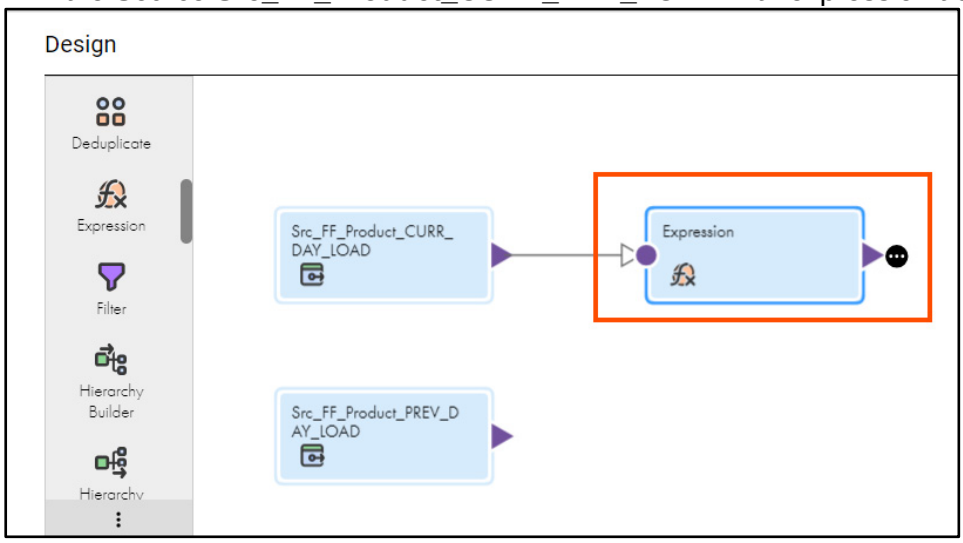


26. From the properties pane, in the **Fields** section, all the fields can be seen.



Add Expression Transformation

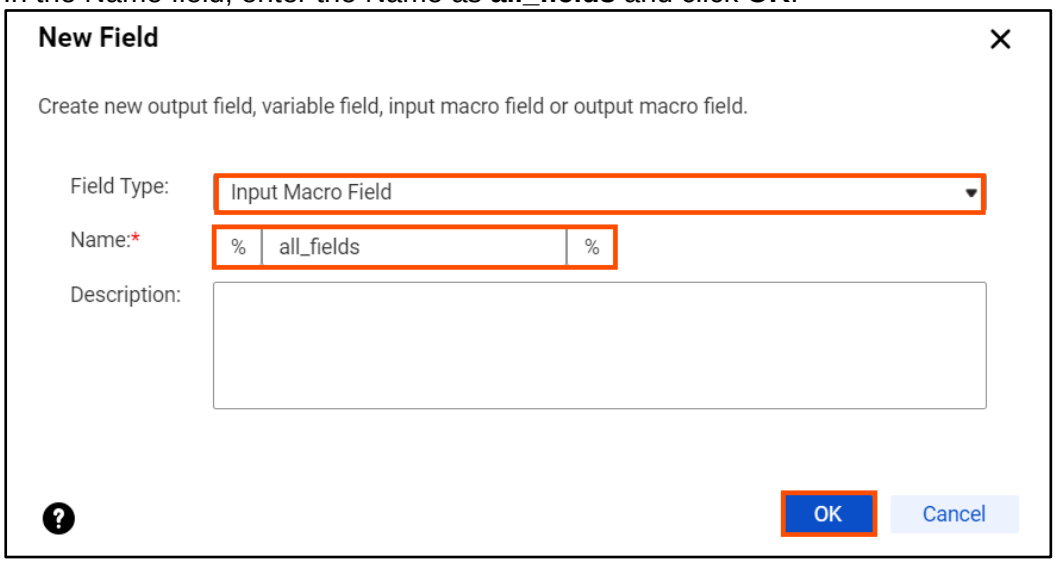
27. From the list of available transformations, drag and drop the expression transformation onto the mapping canvas.
28. Link the Source **Src_FF_Product_CURR_DAY_LOAD** with expression transformation.



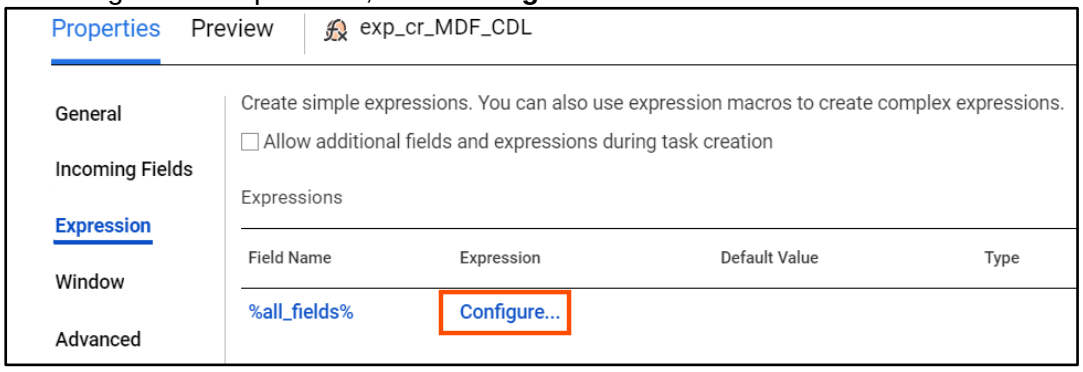
29. Select the expression transformation on the mapping canvas.
30. From the properties pane, in the General field, enter the Name as **exp_cr_MDF_CDL**.

31. From the properties pane, go to the **Expression** Field.
32. To add a new expression, click .
33. From the Field Type drop down, select **Input Macro Field**.

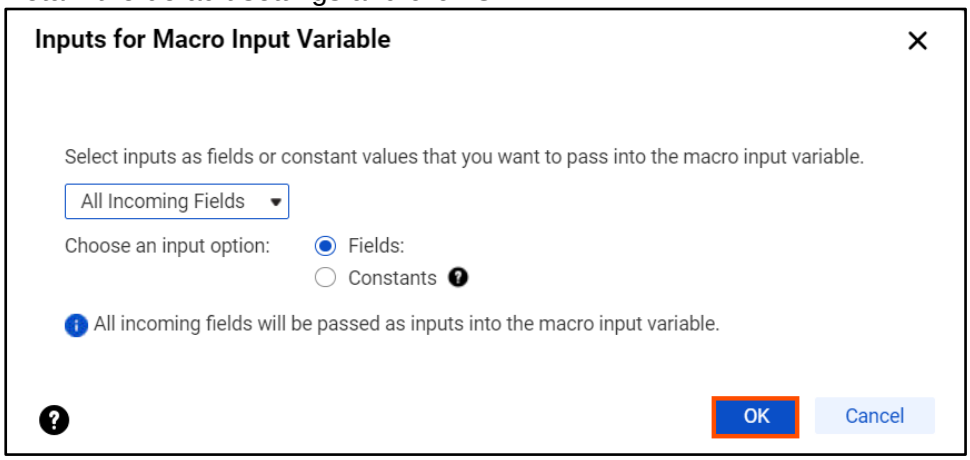
34. In the Name field, enter the Name as **all_fields** and click **OK**.



35. To configure the expression, click **Configure**.



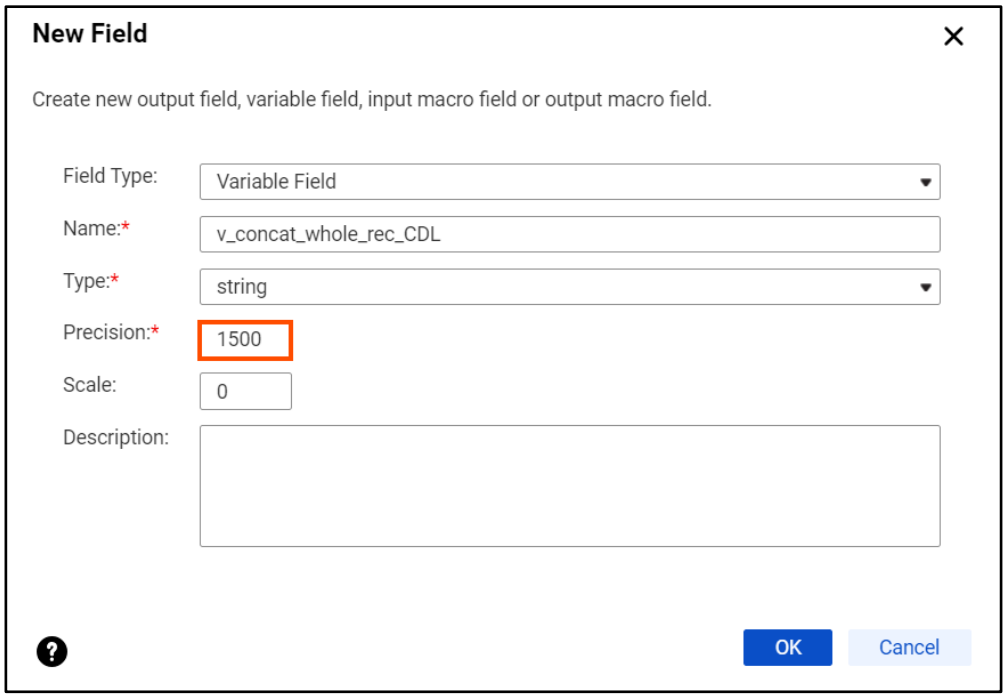
36. Retain the default settings and click **OK**.



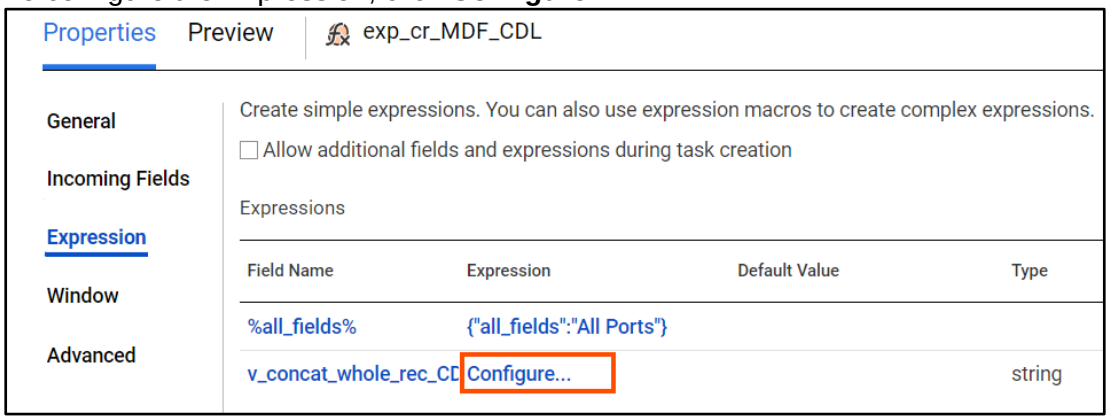
37. To add another Expression, click ⊞.

38. Enter the condition as shown in the table below and click **OK**.

Field Type	Name	Type	Precision	Scale
Variable Field	v_concat_whole_rec_CDL	string	1500	0



39. To configure the Expression, click **Configure**.

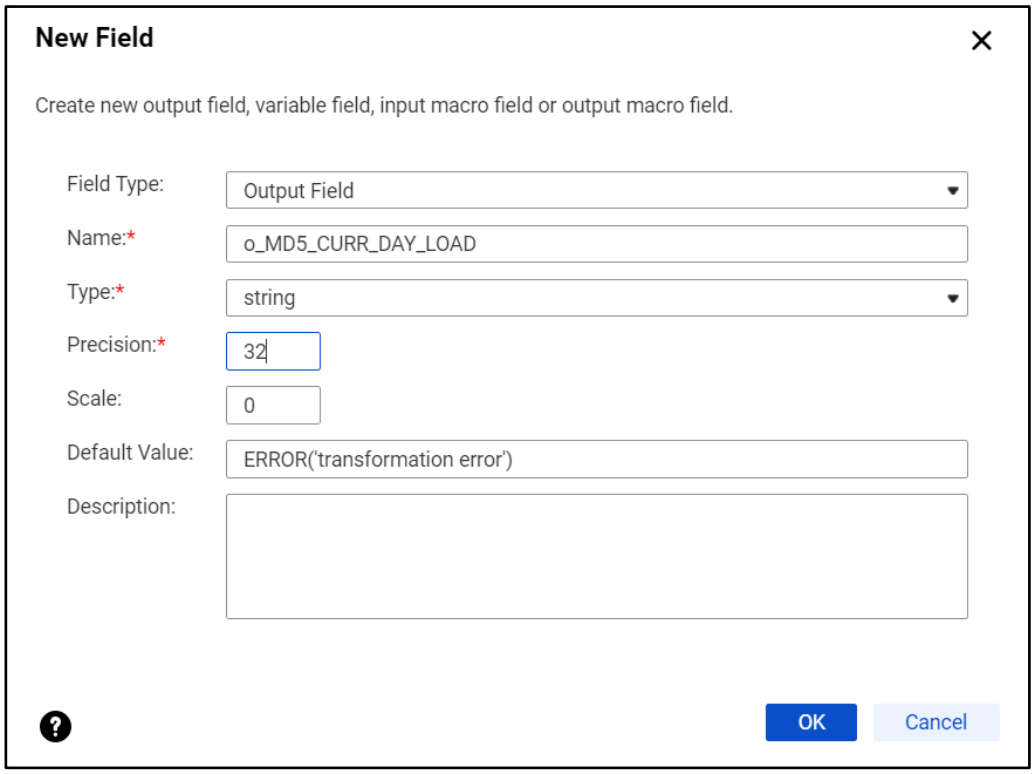


40. Enter the expression as mentioned below:
%OPR_CONCAT[%all_fields%]%

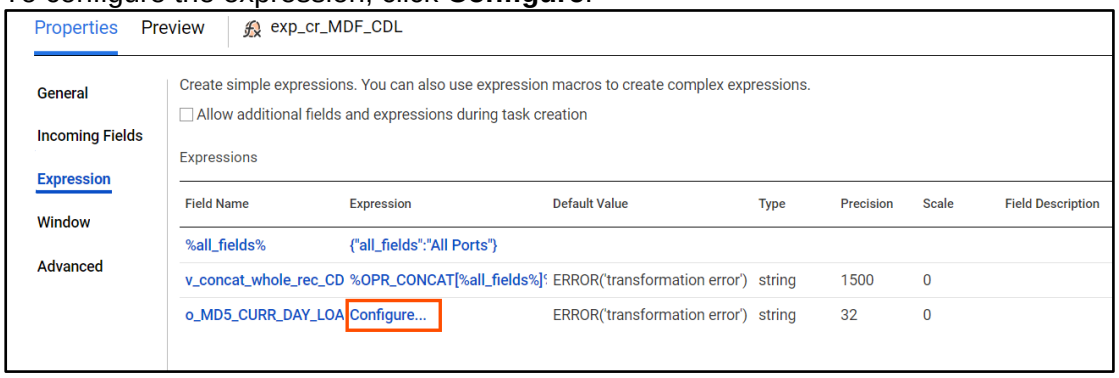
41. Click **OK**.

42. To add a new expression, click [+].
43. Enter the details as shown in the table below and click **OK**.

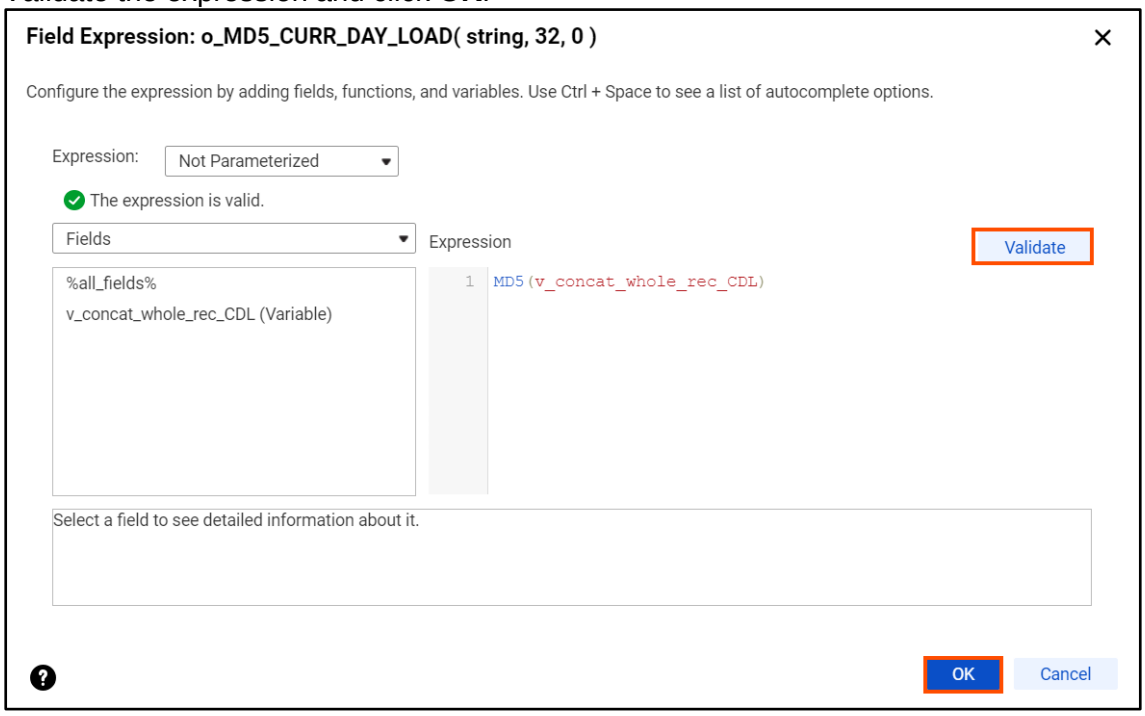
Field Type	Name	Type	Precision	Scale
Output Field	o_MD5_CURR_DAY_LOAD	string	32	0



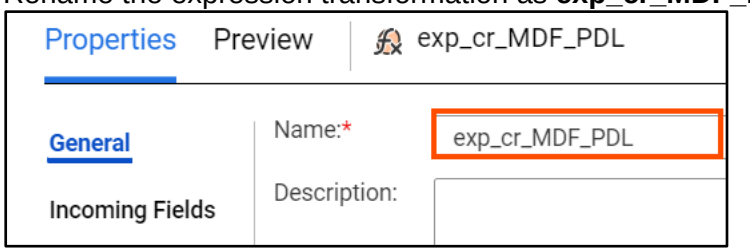
44. To configure the expression, click **Configure**.



45. In the Expression field, enter the following expression:
MD5(v_concat_whole_rec_CDL)
46. Validate the expression and click **OK**.

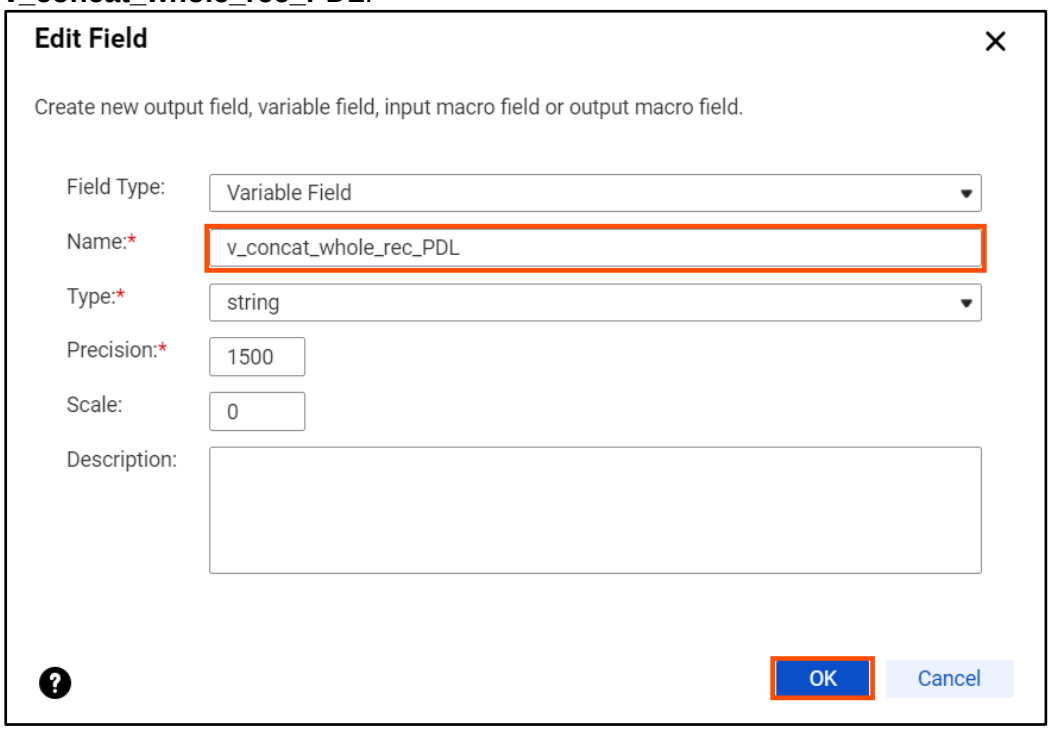


47. **Copy** the Expression transformation **exp_cr_MDF_CDL** and **Paste** it on the mapping canvas (Ctrl-C & Ctrl-V). Link the second source with the new expression transformation (in the Previous Day's load path).
48. Rename the expression transformation as **exp_cr_MDF_PDL**.

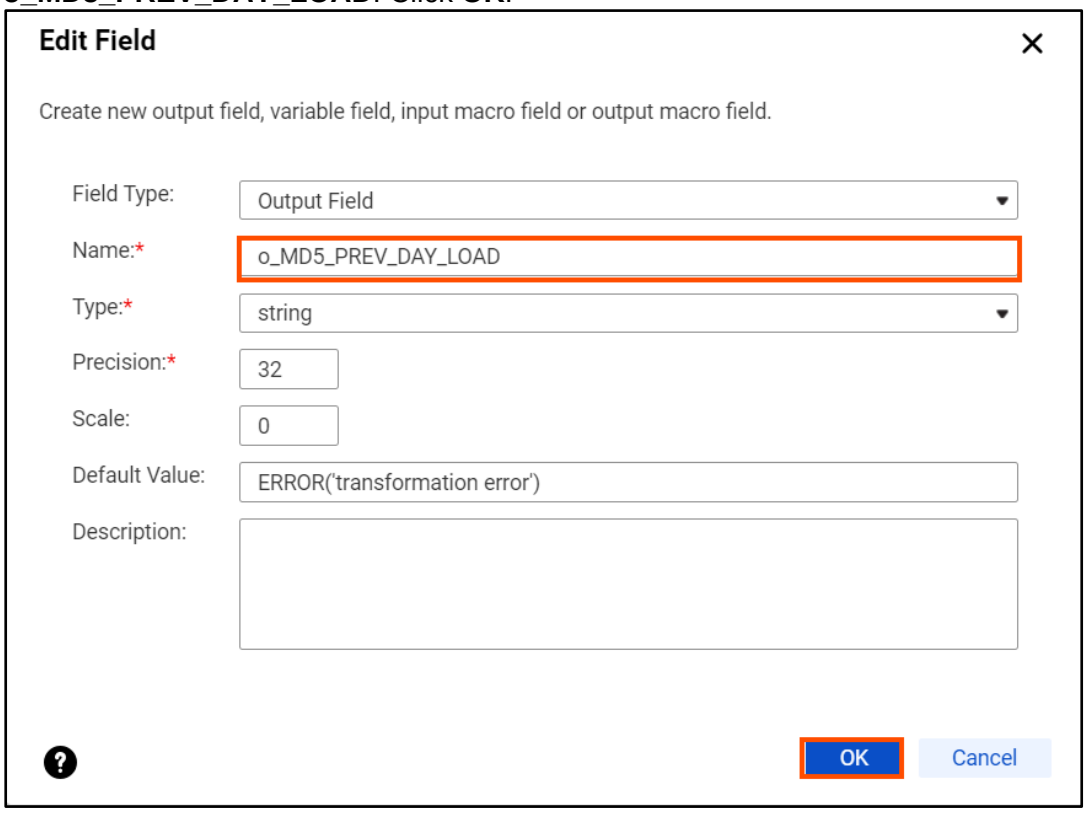


49. From the properties pane, click **Expression**.

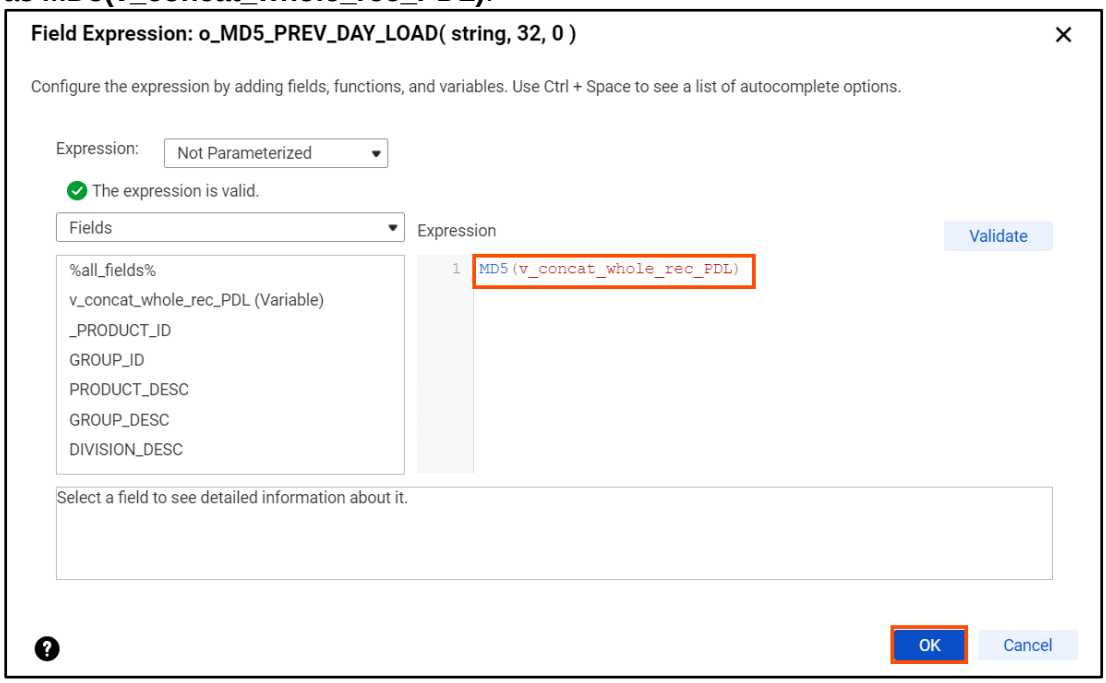
50. Select the Field name **v_concat_whole_rec_CDL** and change it to **v_concat_whole_rec_PDL**.



51. Select the field name **o_MD5_CURR_DAY_LOAD** and Rename the Name field as **o_MD5_PREV_DAY_LOAD**. Click **OK**.

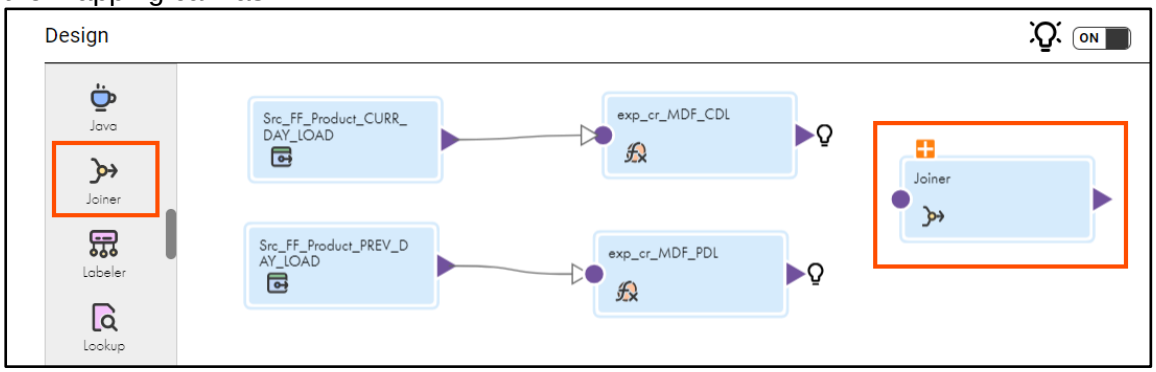


52. Click on the Expression **MD5(v_concat_whole_rec_CDL)** and modify the Expression as **MD5(v_concat_whole_rec_PDL)**.

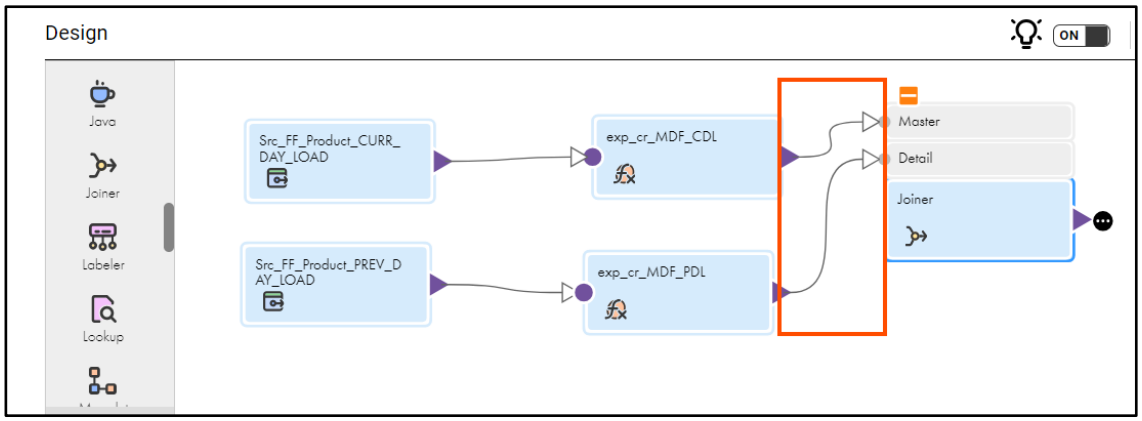


Add Joiner Transformation

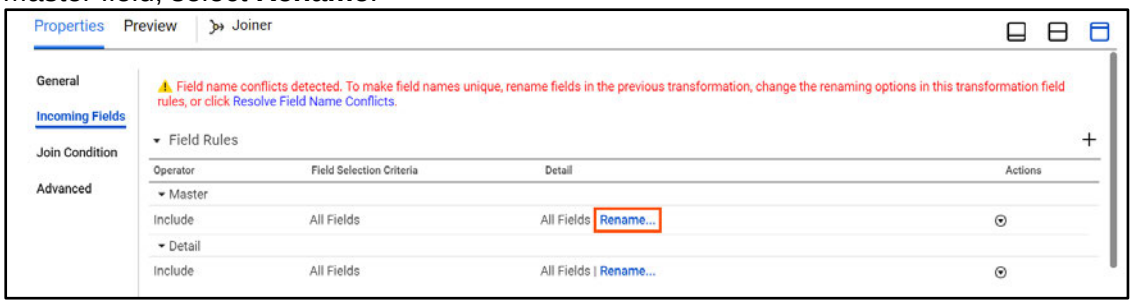
53. From the list of available transformations, drag and drop the **Joiner** transformation onto the mapping canvas.



54. Click the **+** on the Joiner transformation.
55. Link the **exp_cr_MDF_CDL** with the Master and the **exp_cr_MDF_PDL** with the detail.

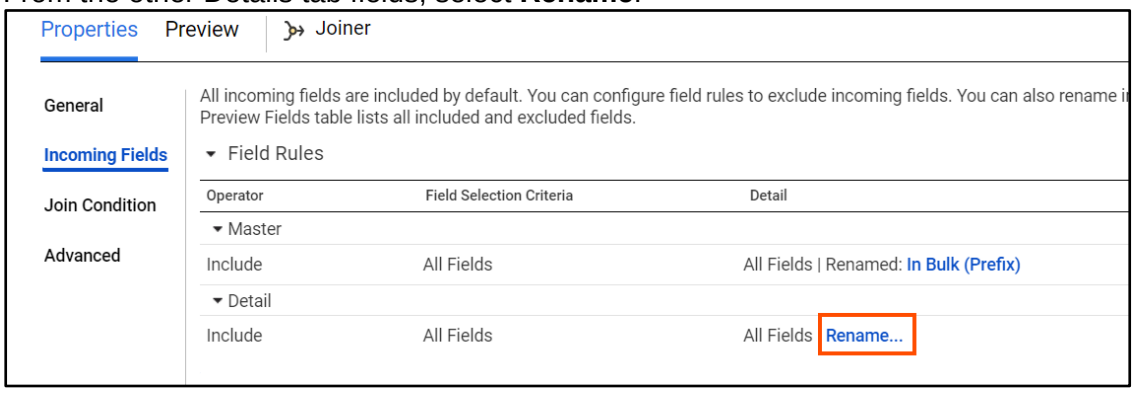


56. Select the **Joiner** transformation from the mapping canvas.
57. From the properties pane, click **Incoming Fields**.
58. If you see the **Field name conflicts detected** warning, from the Details tab in the Master field, select **Rename**.



59. In the Configure Field Rules, from the Bulk Rename Options drop down select **Prefix**.
60. In the Specify field, enter **CURR_DAY_LOAD**. Click **OK**.

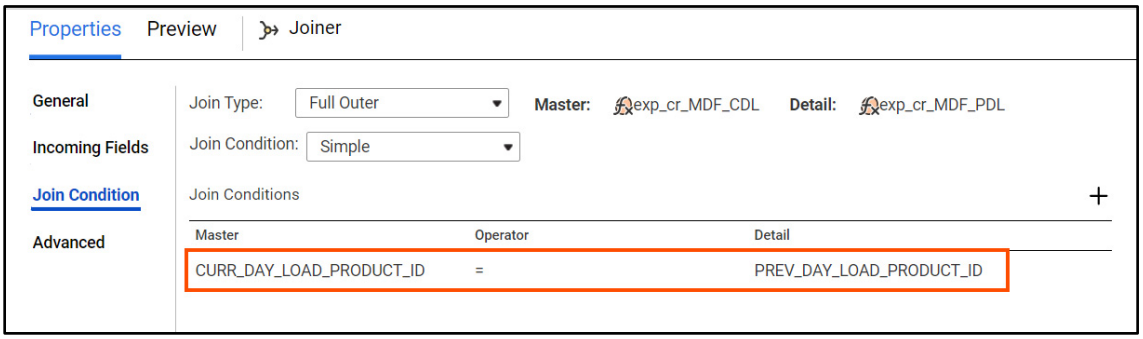
61. From the other Details tab fields, select **Rename**.



62. In the Configure Field Rules, from the Bulk Rename Options drop down select Prefix.
63. In the Specify field, enter **PREV_DAY_LOAD**. Click **OK**.

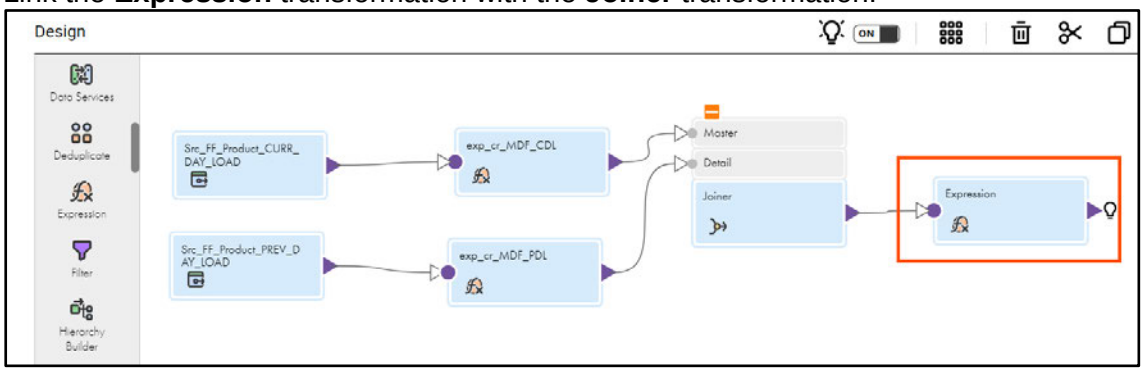
64. From the properties pane, click **Join Condition**.
65. From the Join Type drop down, select **Full Outer**.
66. To add the Join condition, click ⊞.
67. Add the join condition as shown in the table below:

Master	Operator	Detail
CURR_DAY_LOAD_ PRODUCT_ID	=	PREV_DAY_LOAD_PRODUCT _ID



68. From the list of available transformations, drag and drop another **Expression** transformation on the mapping canvas.

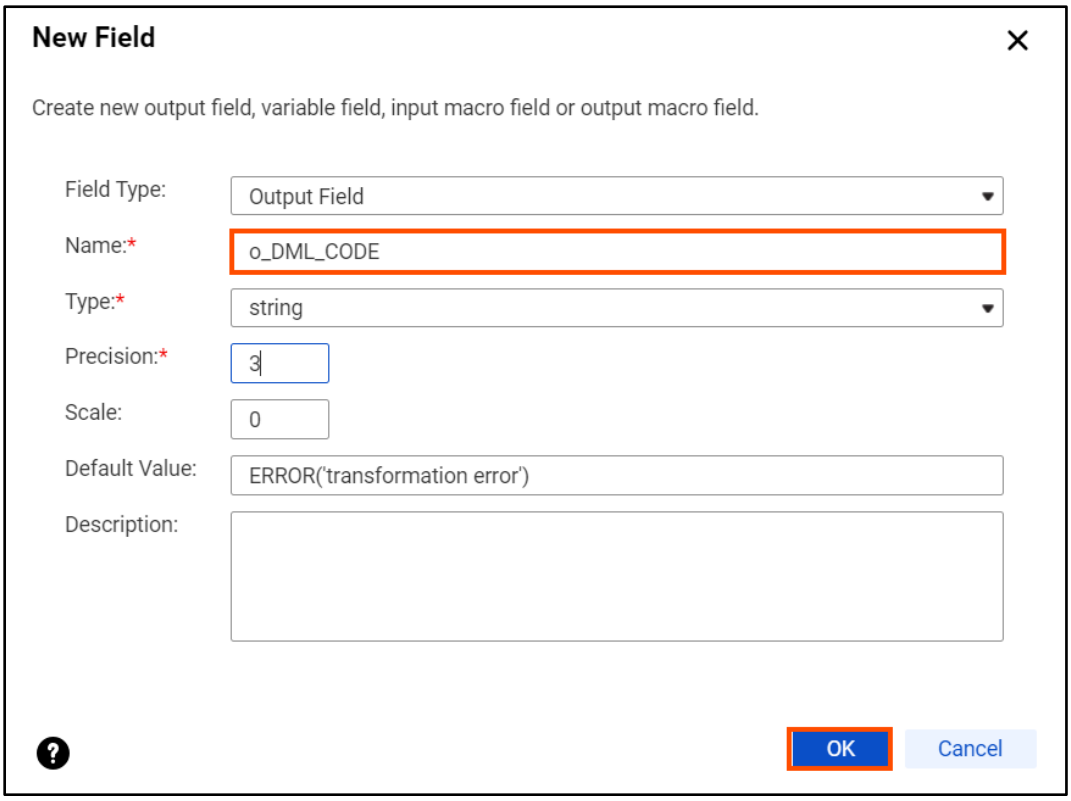
69. Link the **Expression** transformation with the **Joiner** transformation.



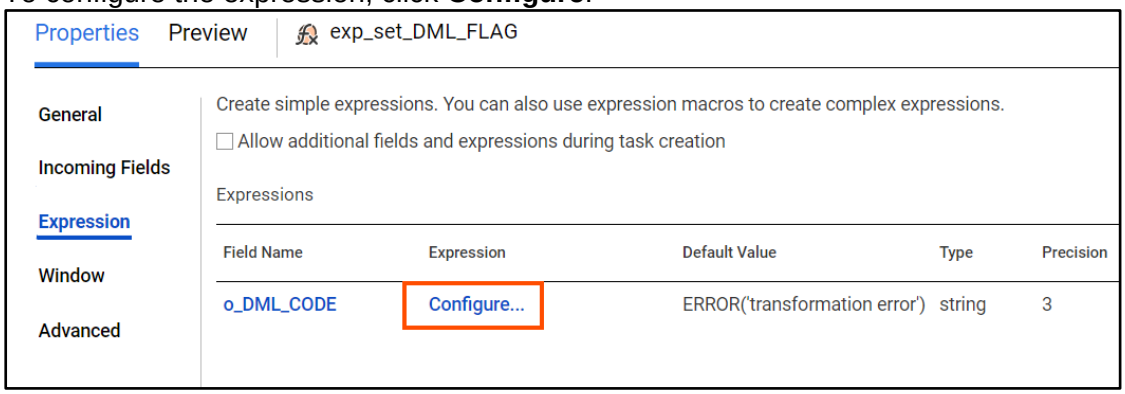
70. Select the **Expression** transformation on the mapping canvas.
71. From the properties pane, in the General section, enter the name as **exp_set_DML_FLAG**.

72. From the properties pane, click **Expression**.
73. To add the new expression, click ⊞.
74. Enter the details as shown in the table below and click **OK**.

Field Type	Name	Type	Precision	Scale
Output Field	o_DML_CODE	string	3	0



75. To configure the expression, click **Configure**.

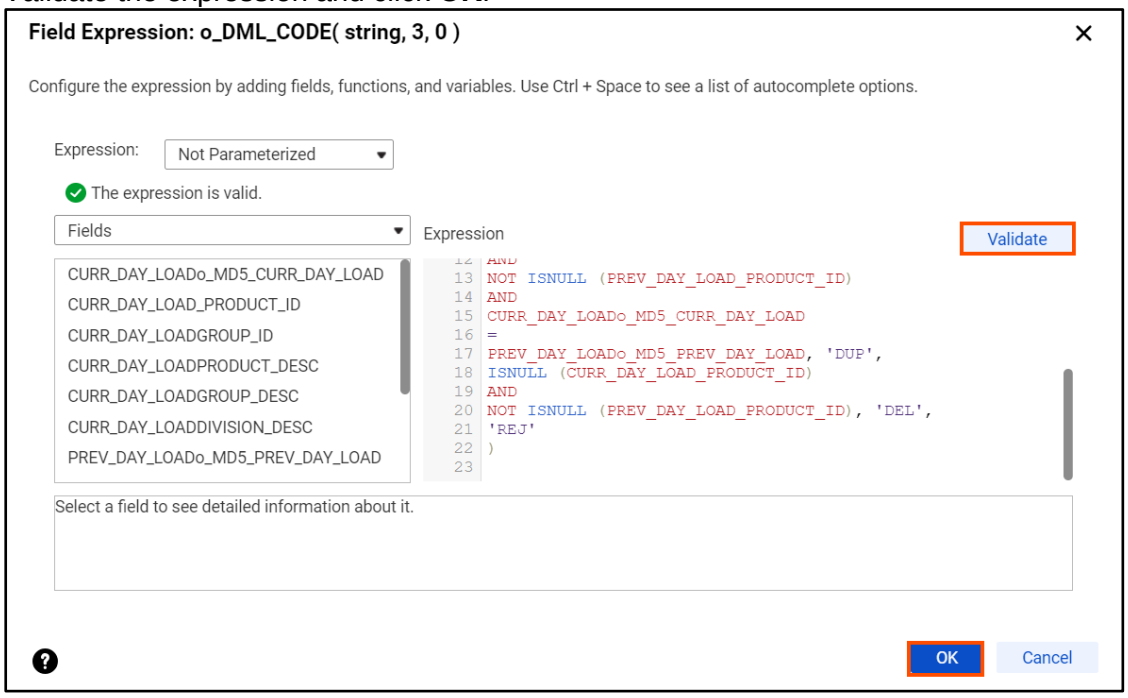


76. In the Expression field, enter the following expression:

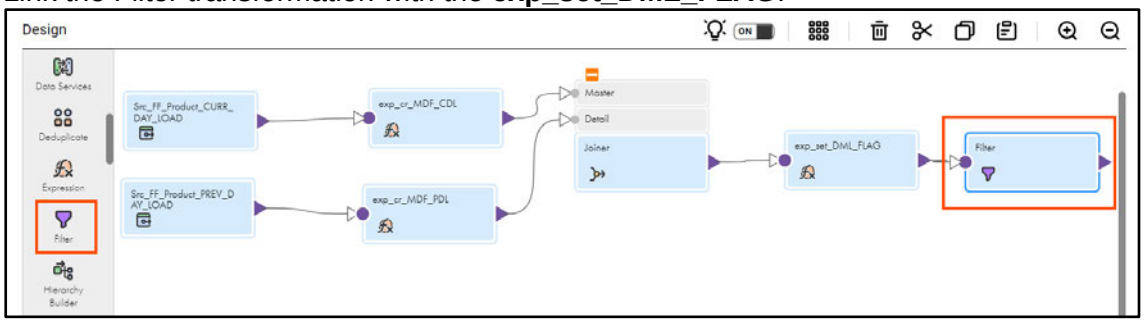
```
DECODE (TRUE,
NOT ISNULL (CURR_DAY_LOAD_PRODUCT_ID)
AND ISNULL (PREV_DAY_LOAD_PRODUCT_ID),    'INS',
NOT ISNULL (CURR_DAY_LOAD_PRODUCT_ID)
AND
NOT ISNULL (PREV_DAY_LOAD_PRODUCT_ID)
AND
CURR_DAY_LOADo_MD5_CURR_DAY_LOAD
!=
PREV_DAY_LOADo_MD5_PREV_DAY_LOAD,   'UPD',
NOT ISNULL (CURR_DAY_LOAD_PRODUCT_ID)
```

```
AND
NOT ISNULL (PREV_DAY_LOAD_PRODUCT_ID)
AND
CURR_DAY_LOADo_MD5_CURR_DAY_LOAD
=
PREV_DAY_LOADo_MD5_PREV_DAY_LOAD, 'DUP',
ISNULL (CURR_DAY_LOAD_PRODUCT_ID)
AND
NOT ISNULL (PREV_DAY_LOAD_PRODUCT_ID), 'DEL',
'REJ'
)
```

77. Validate the expression and click **OK**.



78. From the list of available transformations, drag and drop **Filter** transformation on the mapping canvas.
79. Link the Filter transformation with the **exp_set_DML_FLAG**.

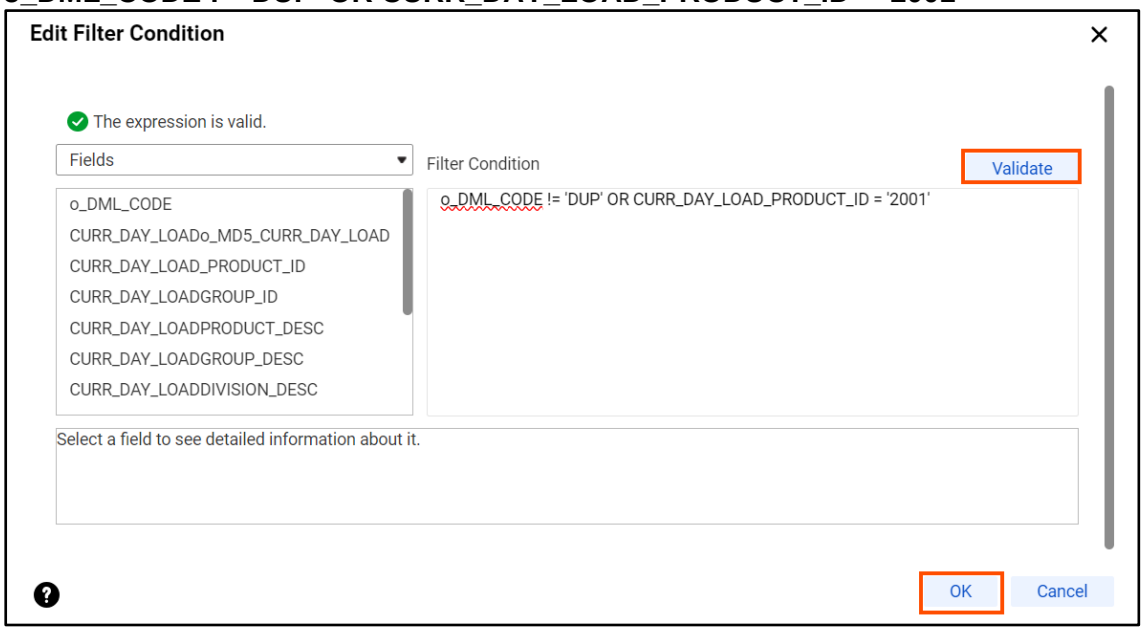


80. From the properties pane, in the General section, retain the name as Filter.
81. From the properties pane, click **Filter**.

82. From the Filter Condition drop down, select **Advanced**.

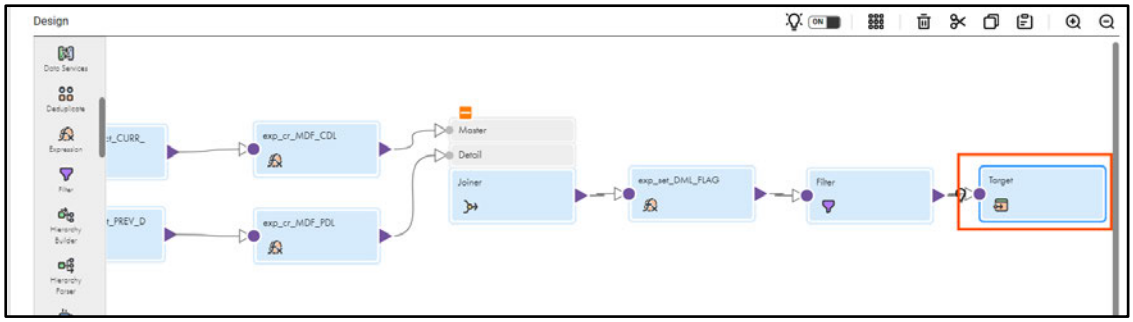
83. Click on **Define Filter Condition**.

84. In the Filter Condition window, enter the following filter condition and click **OK**.
o_DML_CODE != 'DUP' OR CURR_DAY_LOAD_PRODUCT_ID = '2001'

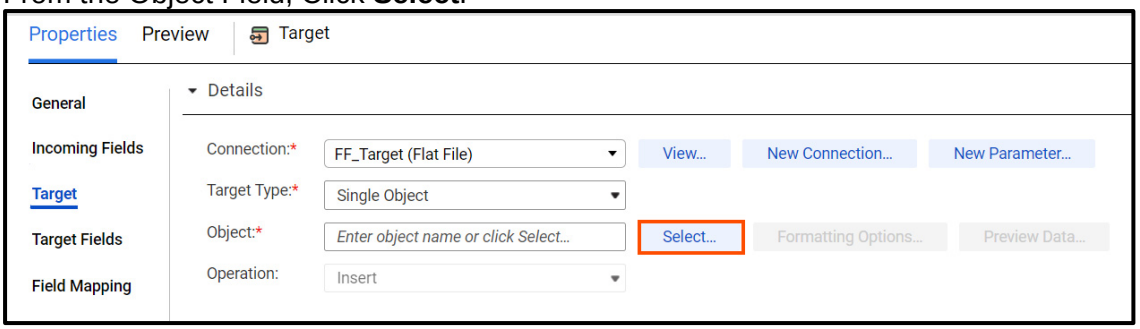


Configure Target Transformation

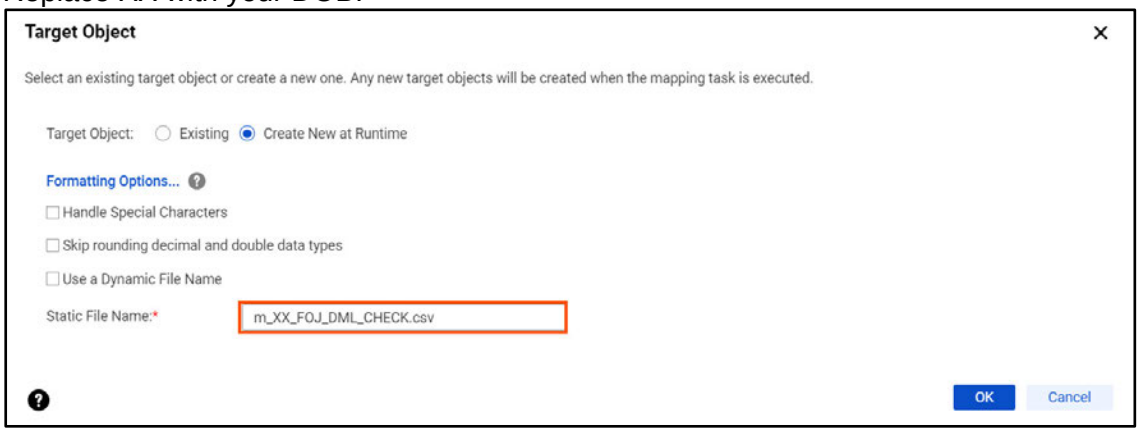
85. Link the **Filter** with the Target transformation.
86. Select the **Target** transformation from the mapping canvas.



87. From the properties pane, in the General section retain the name as **Target**.
88. From the properties pane, click **Target**.
89. From the Connection drop-down, select **your target flat file** connection.
90. From the Object Field, Click **Select**.



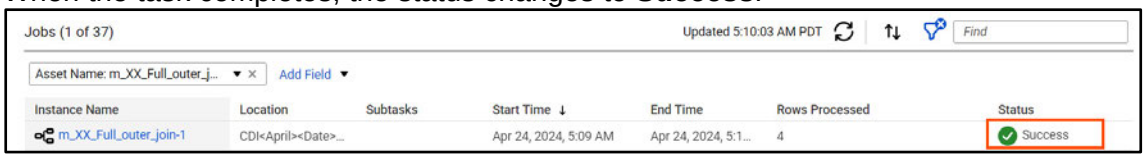
91. From the Target Object, select **Create New at Runtime**.
92. In the Static File Name, enter the name as **m_XX_FOJ_DML_CHECK.csv**. Click **OK**. Replace XX with your DOB.



93. Save and run the mapping.
94. From the Runtime Environment drop-down, select **INFA-SERVER** and click **Run**.

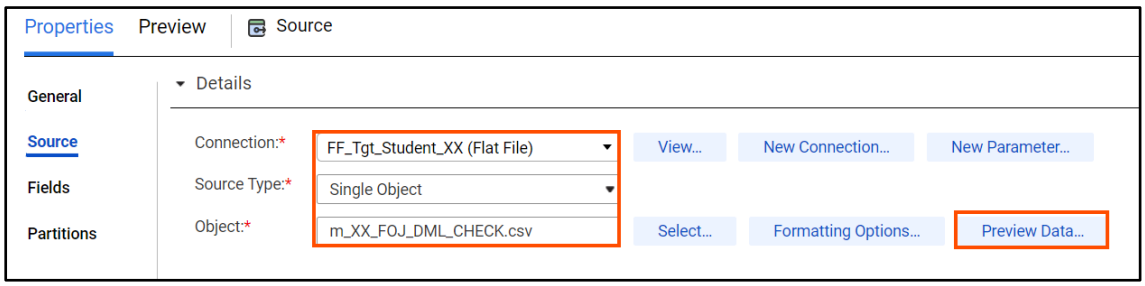
Monitor Status

95. Monitor the task status from the **My Jobs** page.
96. When the task completes, the status changes to **Success**.

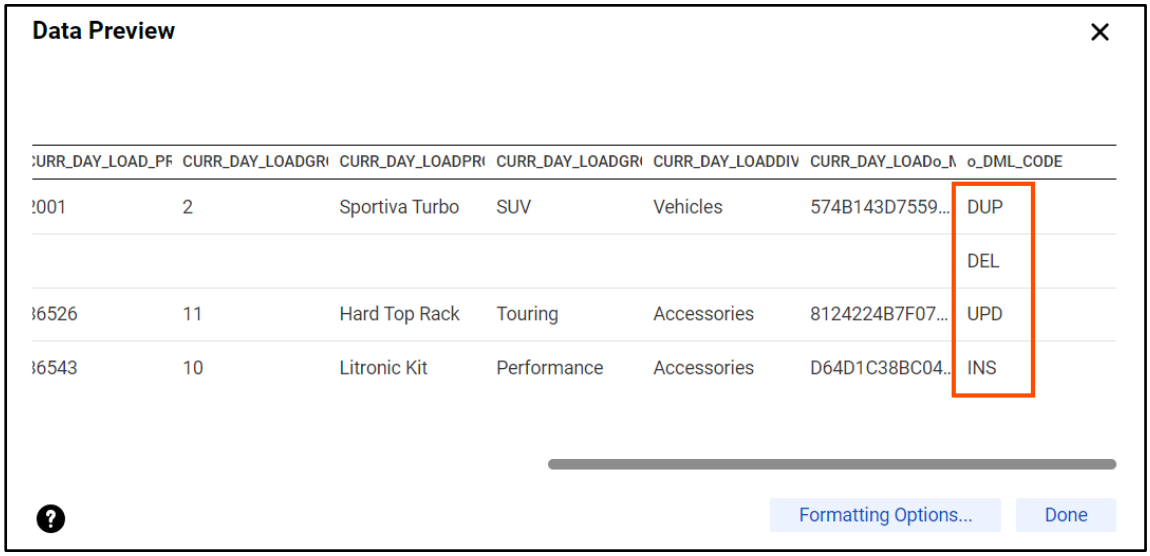


Optional - Open the Dummy Mapping and View the Data

97. Change the Source connection and object details in your dummy maping and preview the data.



There are Results for Duplicate, Delete, Update, and Insert Records.



98. Save and close all the assets in the navigation pane.

This concludes the lab.

Module 8: Mapplets

Lab 8-1: Using Mapplet Transformation in a Mapping

Overview:

In IICS, Mapplet transformation inserts a mapplet created in Data Integration, imported from PowerCenter, or generated from an SAP asset into a mapping. A mapplet can encapsulate two or more transformations.

In this lab, you will create a Mapplet and use the created Mapplet in a mapping.

Objective:

- Create a Mapplet
- Use Mapplet Transformation in a mapping

Scenario:

Ruby asks John if he can help her in automating the process of maintaining a list of product names, IDs, and costs for every product that every outlet of NH Retail offers. John informs Ruby that he can use the Mapplets features of IICS to create a set of transformations that he can use in multiple mappings to calculate the product-wise cost.

In this lab, John will create a mapplet to read product details as input. He will use the expression transformation in the mapplet to calculate the cost and pass it as the output to the mapplet. He will also use this mapplet in mapping.

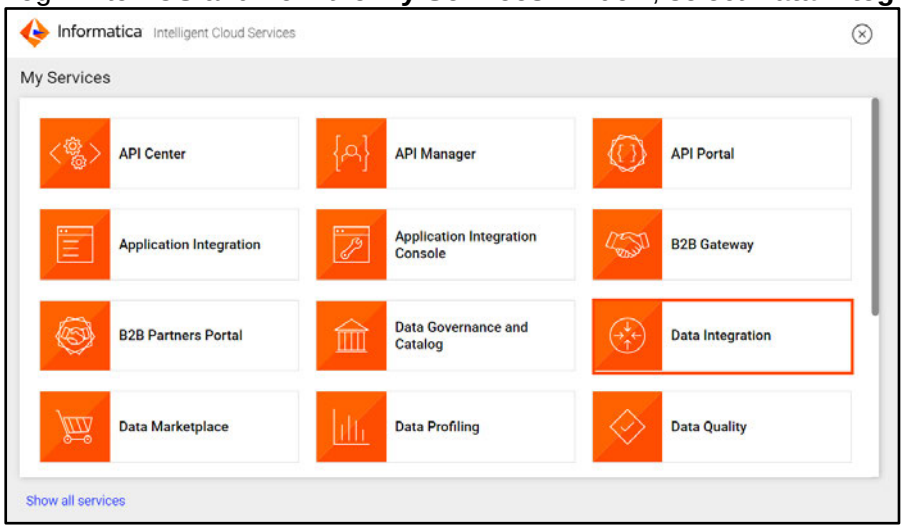
Duration:

20 minutes

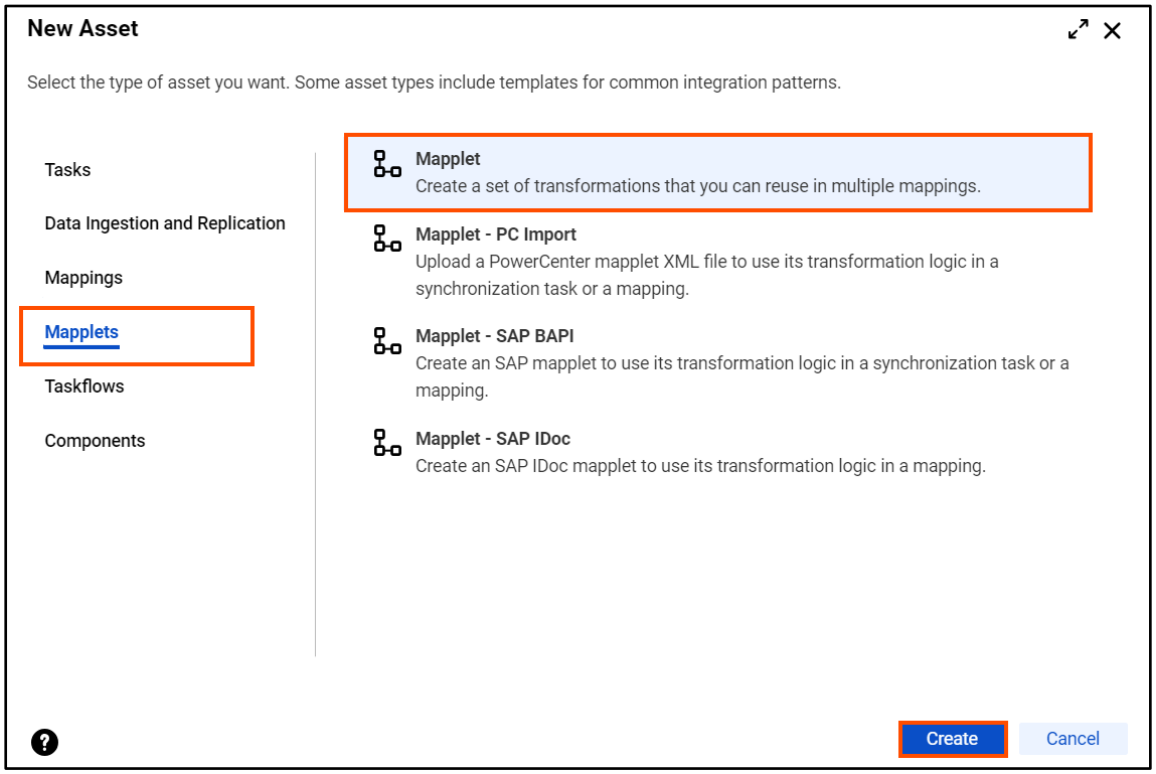
Tasks

Create Mapplet

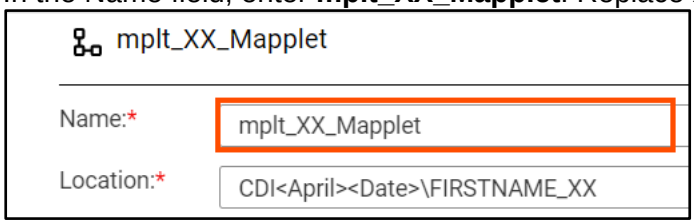
1. Login into IICS and from the **My Services** window, select **Data Integration**.



2. From the navigation pane, select **New**.
3. From the New Asset window, click the **Mapplets** tab, and select **Mapplet**.
4. Click **Create**.

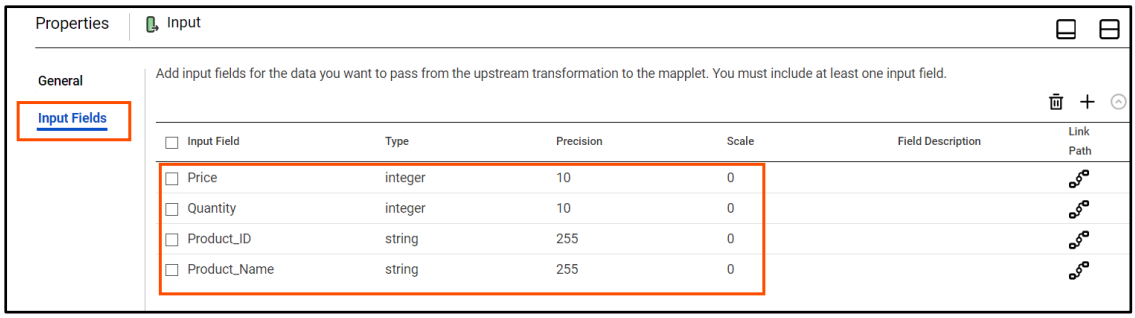


5. In the Name field, enter **mplt_XX_Mapplet**. Replace XX with your DOB.

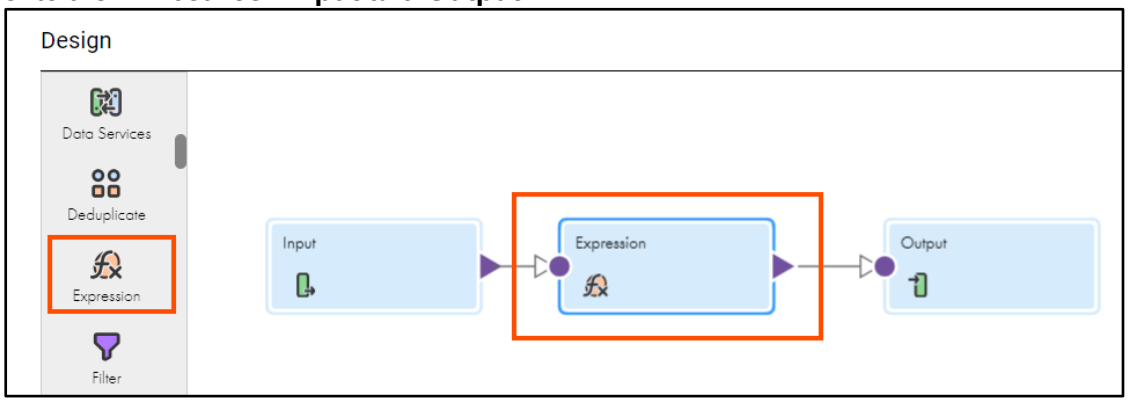


6. To configure the Input, from the mapplet canvas, click the **Input** transformation.
7. From the properties pane, click **Input Fields**.
8. To add input fields, click ⊞.
9. Define the input fields, as shown in the table below:

Name	Type	Precision	Scale
Price	integer	10	0
Quantity	integer	10	0
Product_ID	string	255	0
Product_Name	string	255	0

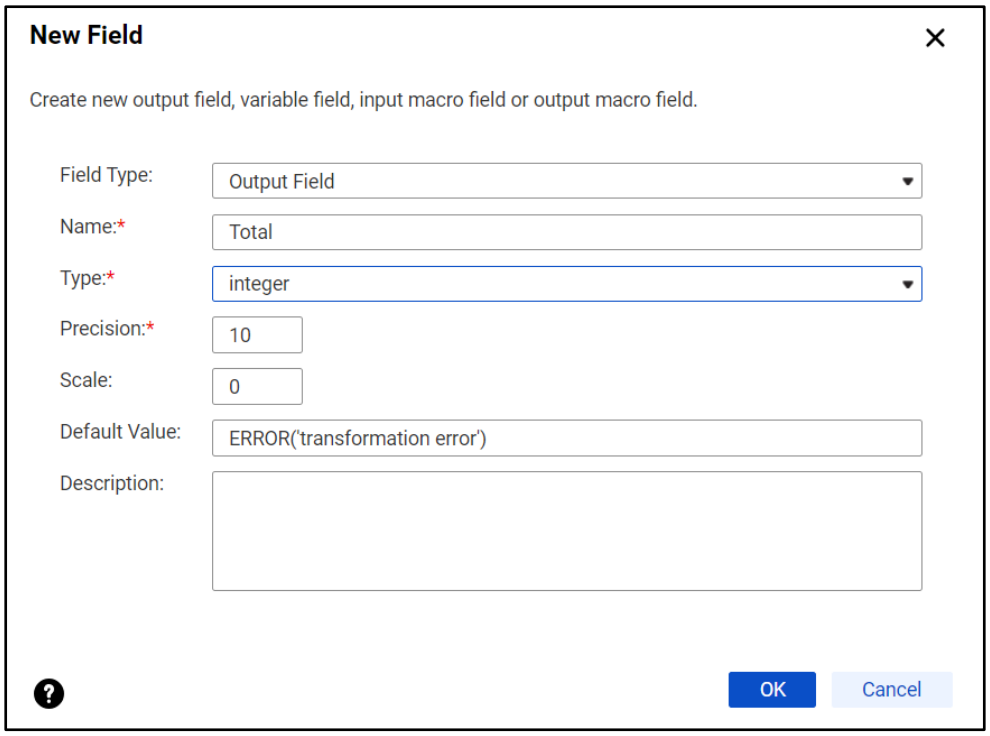


10. From the list of available transformations, drag and drop an **Expression** transformation onto the link between **Input** and **Output**.

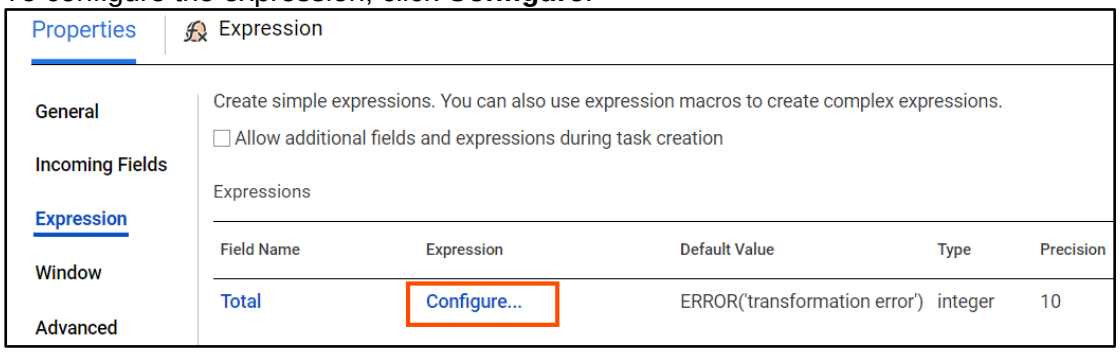


11. Select the **Expression** transformation from the mapplet canvas.
12. From the properties pane, click **Expression**.
13. To add a new expression, click ⊞.
14. Enter the details as shown in the table below and click **OK**.

Field Type	Name	Type	Precision	Scale
Output Field	Total	integer	10	0

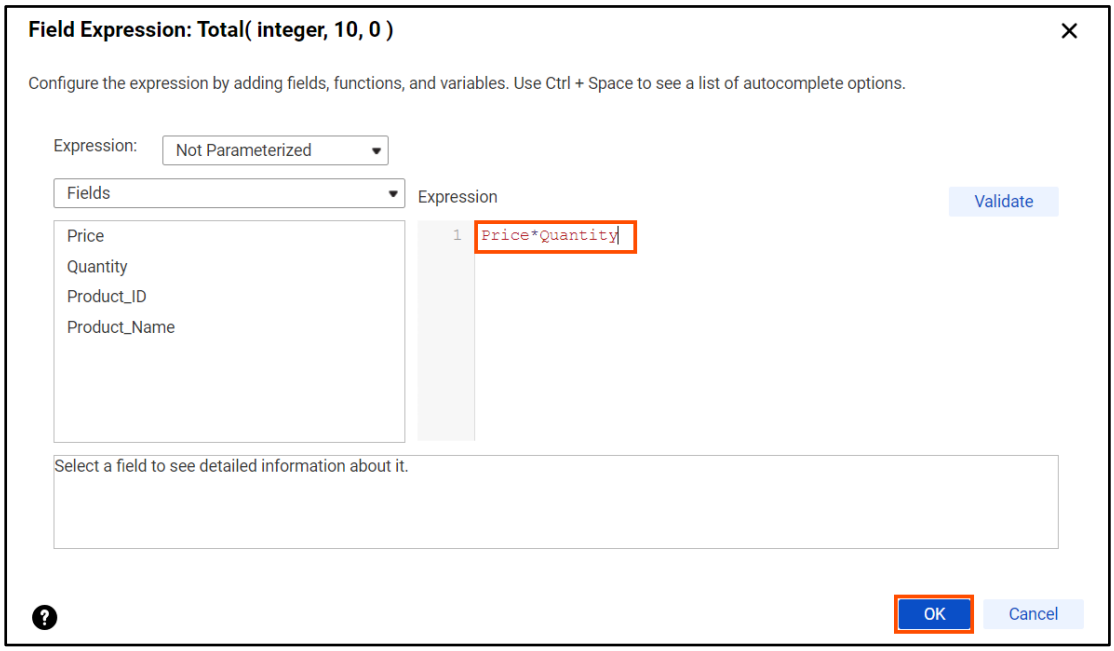


15. To configure the expression, click **Configure**.

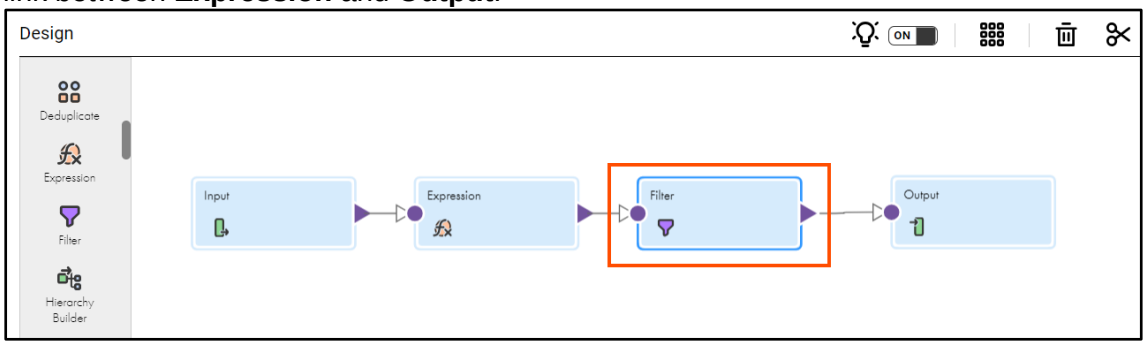


16. In the Expression field, copy and paste the following expression:
Price*Quantity

17. Click **OK**.



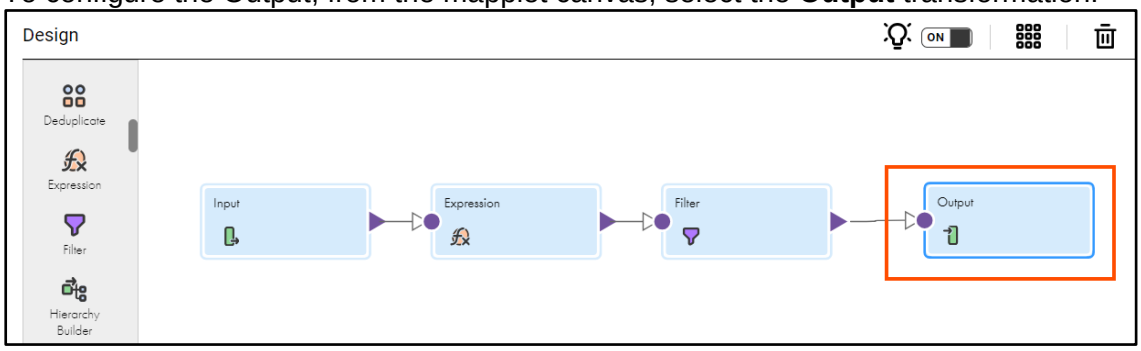
18. From the list of available transformations, drag and drop a **Filter** transformation on the link between **Expression** and **Output**.



19. Select the **Filter** transformation from the mapplet canvas.
20. From the properties pane, select **Filter**.
21. To add filter condition, click ⊞.
22. Define the filter as shown in the table below:

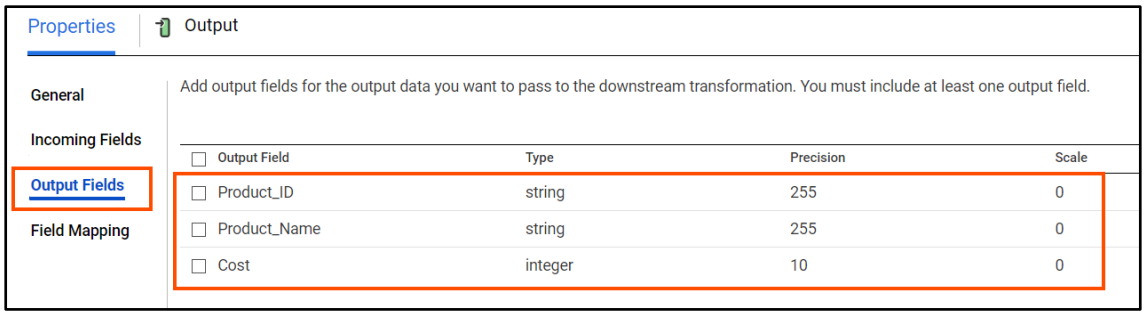
Field Name	Operator	Value
Product_ID	< (Less Than)	IN7040

23. To configure the Output, from the mapplet canvas, select the **Output** transformation.



24. From the properties pane, select **Output Fields**.
25. To add output fields, click ⊞.
26. Define the output fields as shown in the table below:

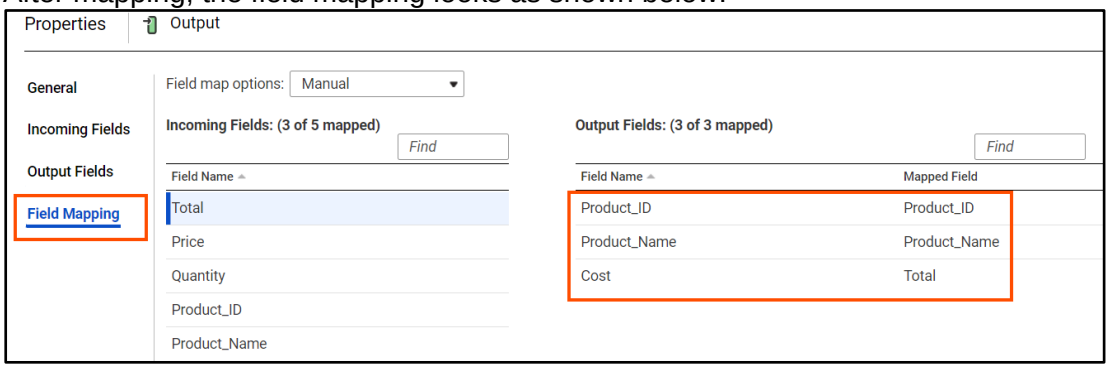
Name	Type	Precision	Scale
Product_ID	string	255	0
Product_Name	string	255	0
Cost	Integer	10	0



27. From the properties pane, select **Field Mapping**.
28. Match the fields as shown in the table below:

Incoming Field	Target Field
Product_ID	Product_ID
Product_Name	Product_Name
Total	Cost

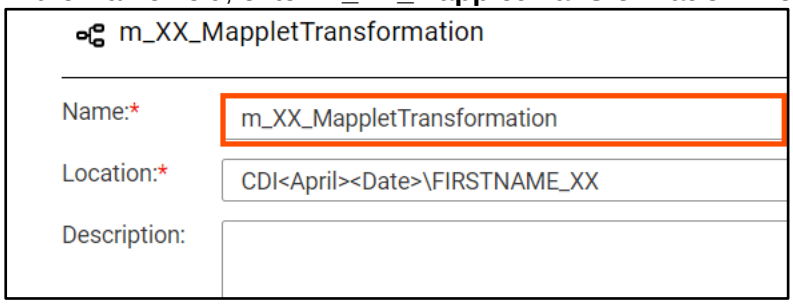
After mapping, the field mapping looks as shown below:



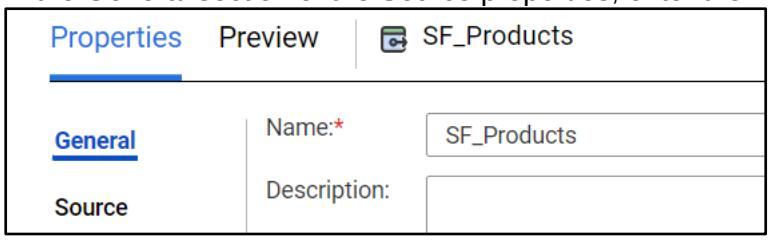
29. **Save** the mapplet.

Create Mapping

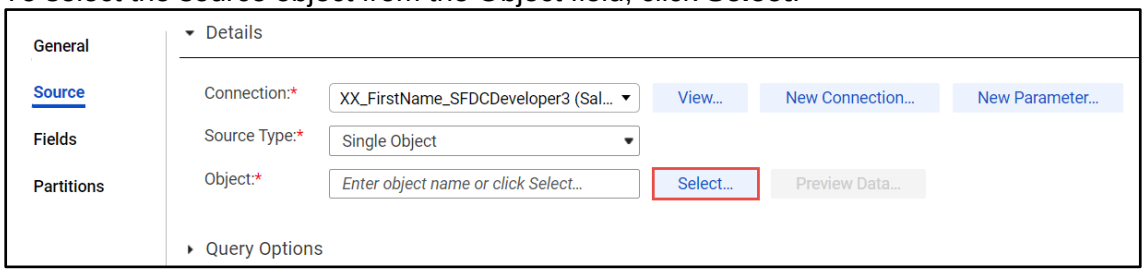
30. From the Data Integration service, create a new **Mapping**.
31. In the Name field, enter **m_XX_MappletTransformation**. Replace XX with your DOB.



32. To configure the source, from the mapping canvas, click the **Source** transformation.
33. In the General section of the Source properties, enter the Name as **SF_Products**.

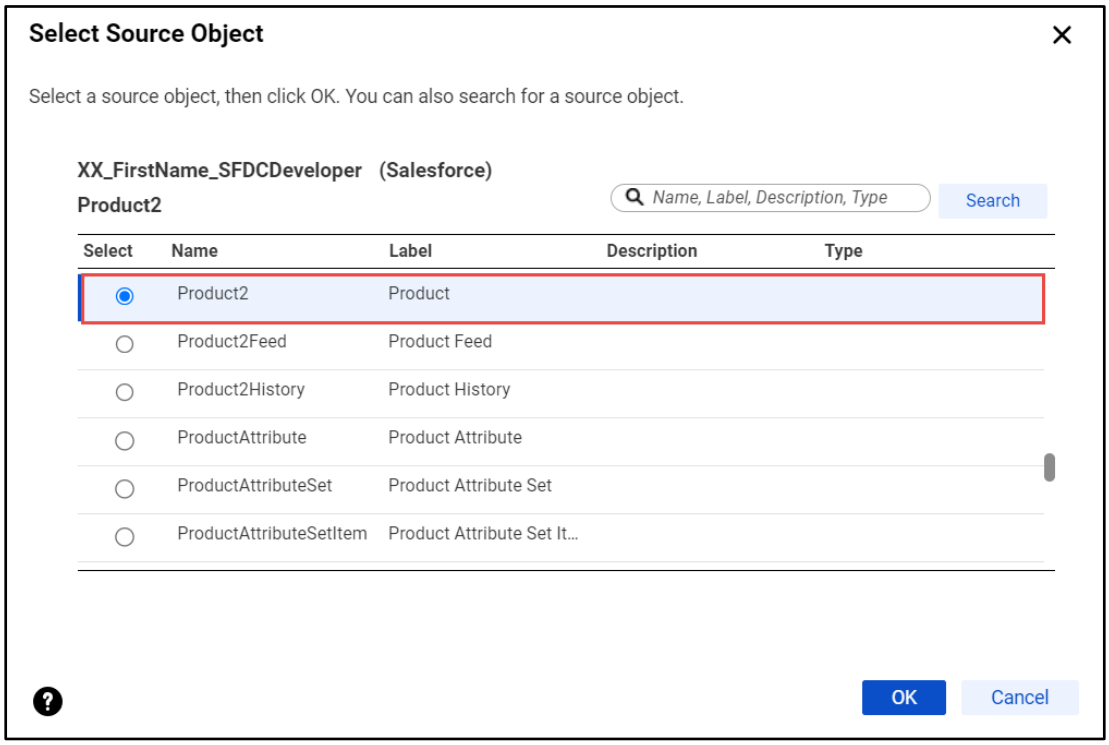


34. From the properties pane, click **Source**.
35. From the Connection drop-down, select your **Salesforce** connection (present in the format **XX_FirstName_SFDCDeveloper**).
36. To select the source object from the Object field, click **Select**.

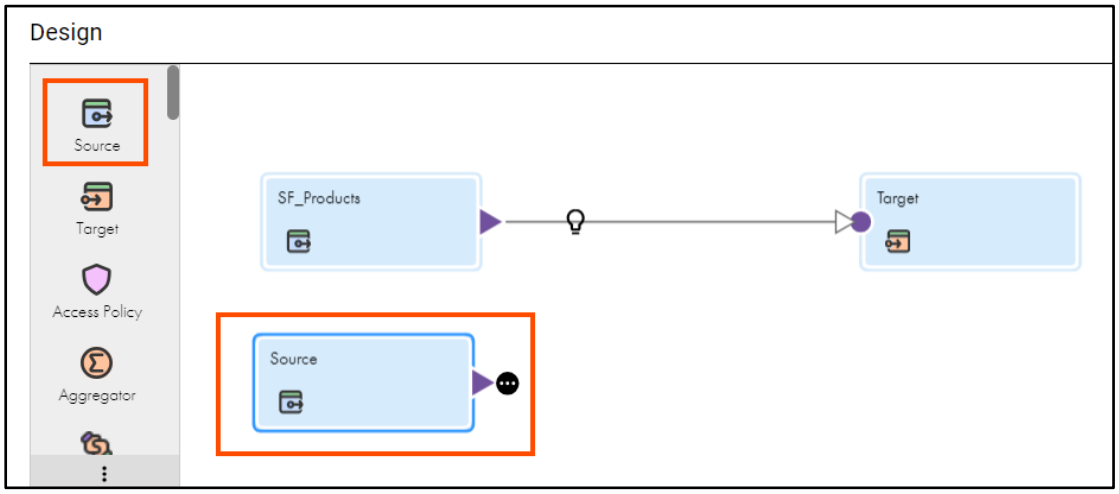


37. From the list, select **Product2**.
    Note: You can use the search option to locate the object.

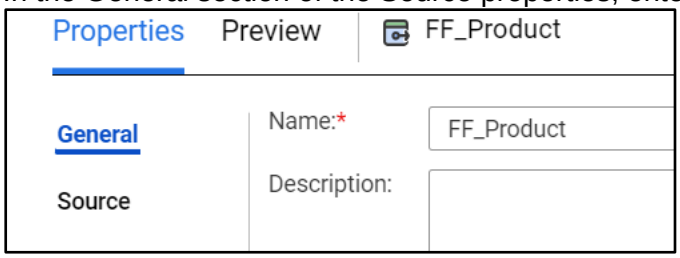
38. Click **OK**.



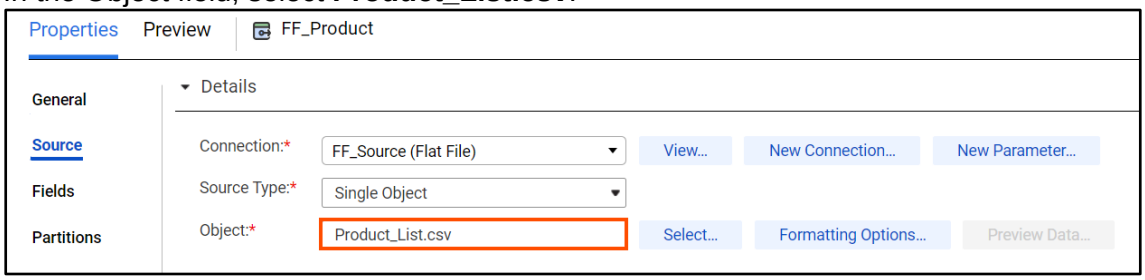
39. From the list of available transformations, drag and drop another **Source** transformation on the mapping canvas.



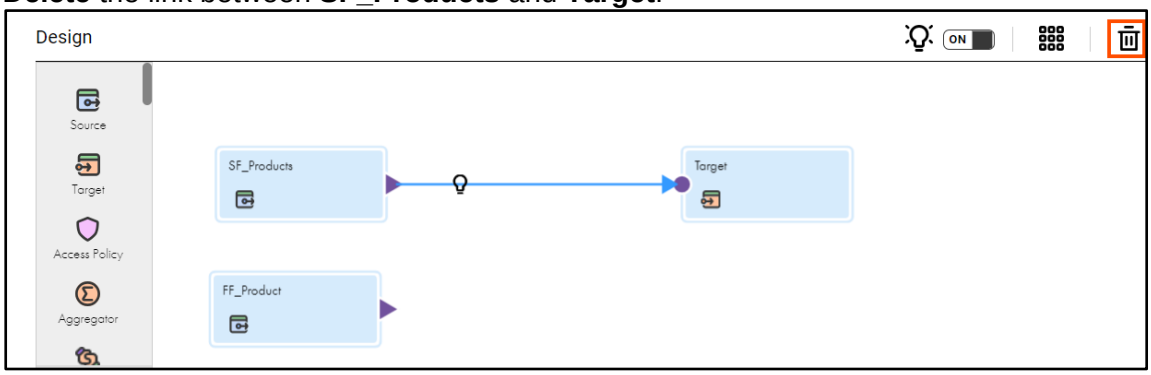
40. Select the newly added transformation from the mapping canvas.
41. In the General section of the Source properties, enter the Name as **FF_Product**.



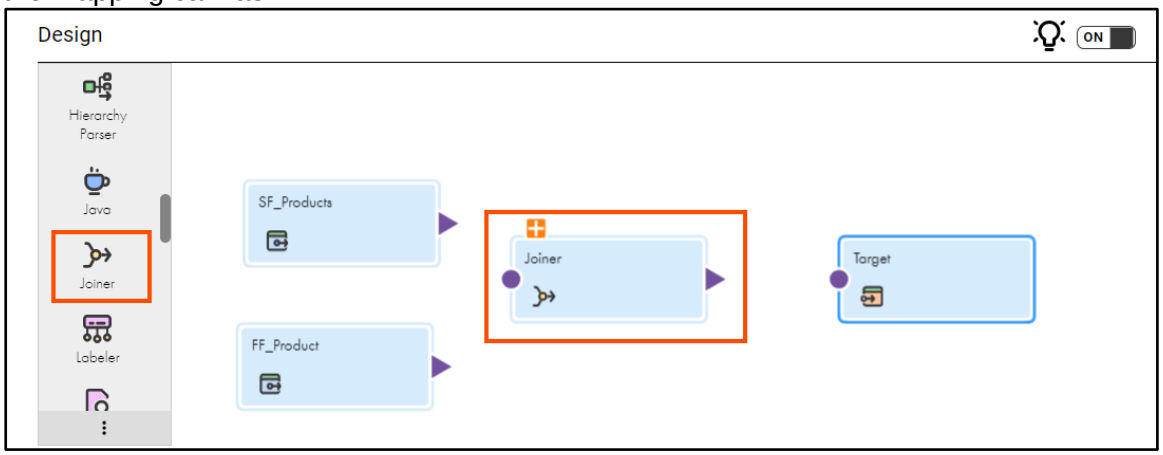
42. From the properties pane, click **Source**.
43. From the Connection drop-down, select **FF_Source**.
44. In the Object field, select **Product_List.csv**.



45. **Delete** the link between **SF_Products** and **Target**.

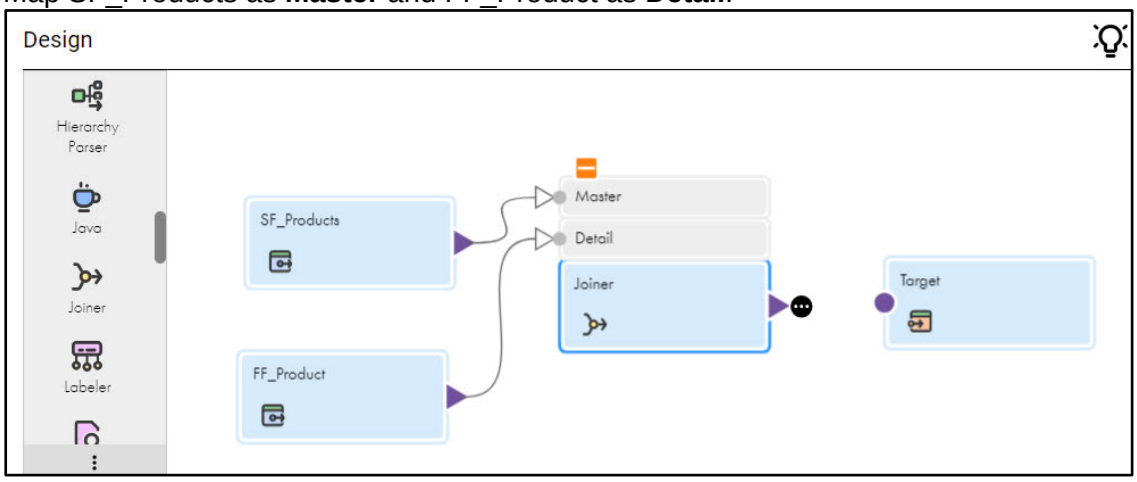


46. From the list of available transformations, drag and drop a **Joiner** transformation onto the mapping canvas.



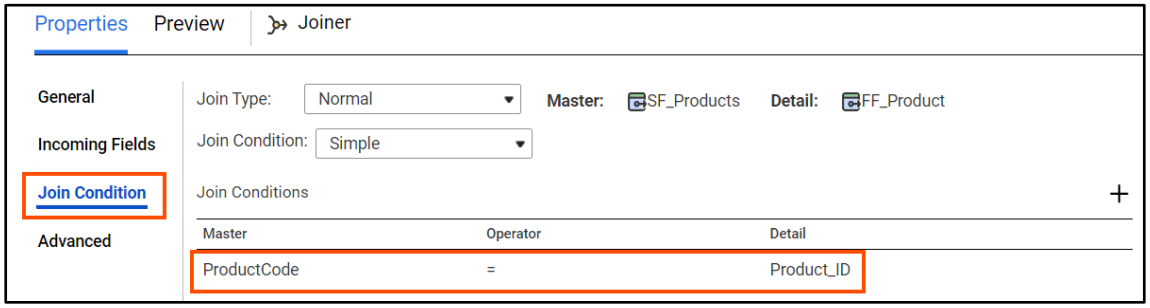
47. Select the Joiner transformation and click .

48. Map SF_Products as **Master** and FF_Product as **Detail**.

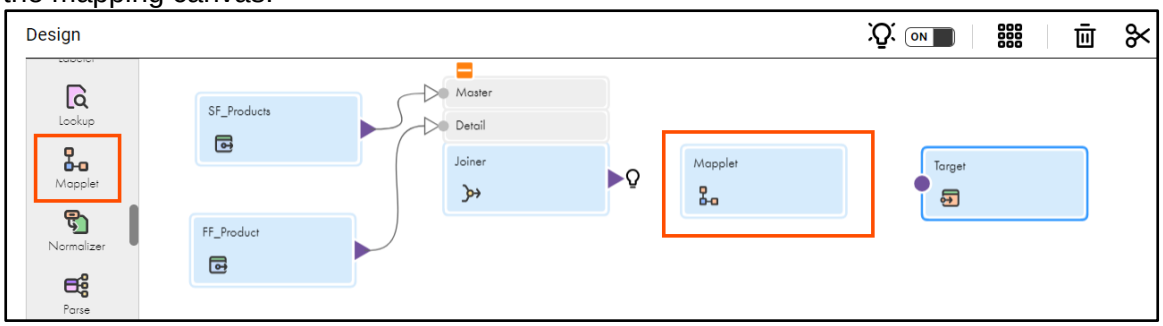


49. From the properties pane, click **Join Condition**.
50. Add the condition as shown in the table below:

Master	Operator	Detail
ProductCode	= (Equals)	Product_ID

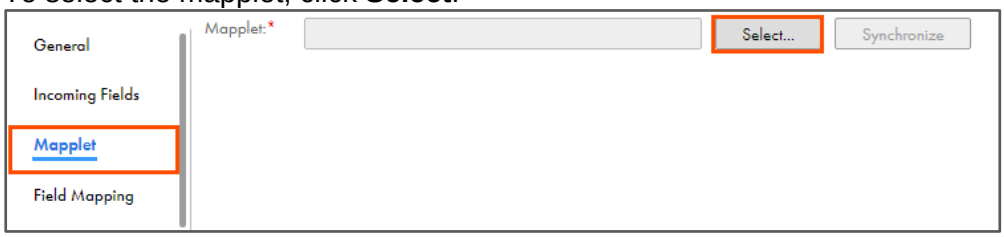


51. From the list of available transformations, drag and drop a **Mapplet** transformation onto the mapping canvas.



52. From the properties pane, select **Mapplet**.

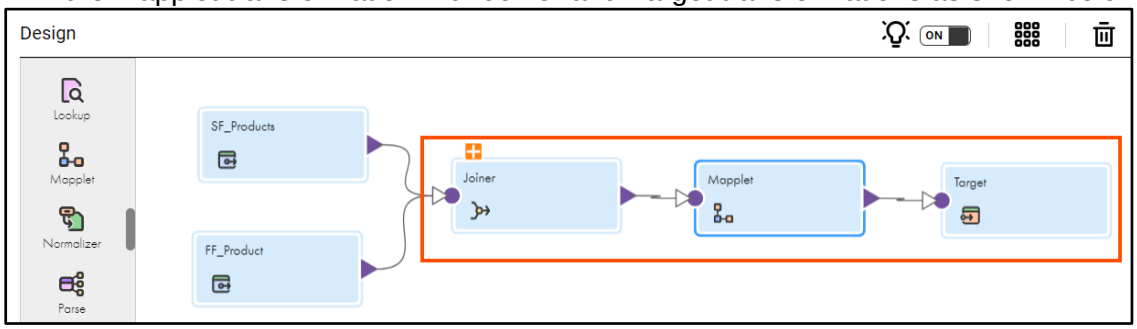
53. To select the mapplet, click **Select**.



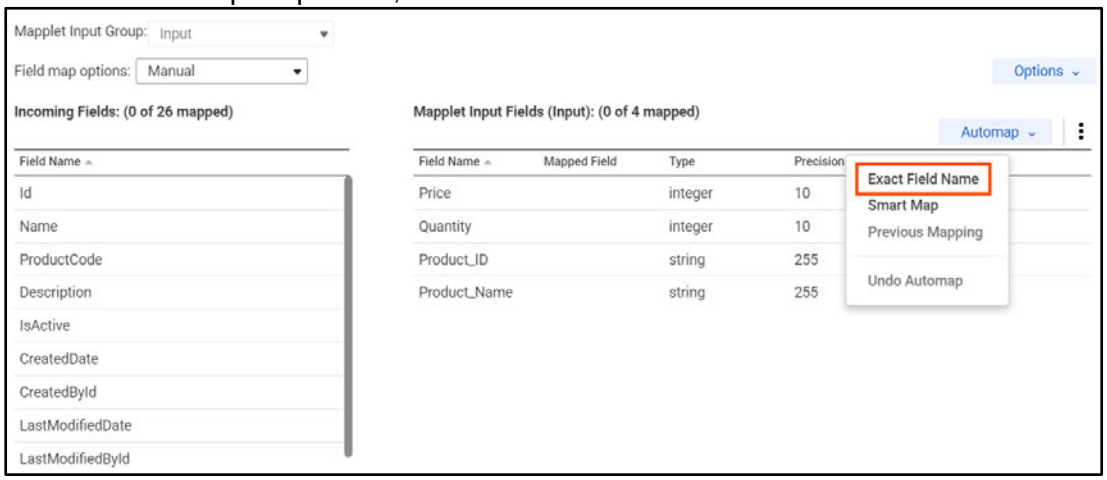
54. Navigate to your working directory and select **mplt_XX_Mapplet**.

    Note: Notice that the mapping type is **Active**. This is because, you have used active transformations like Filter in the mapplet. If you make any changes in the Mapplet, use the **Synchronize** option to update the mapplet changes in the mapping.

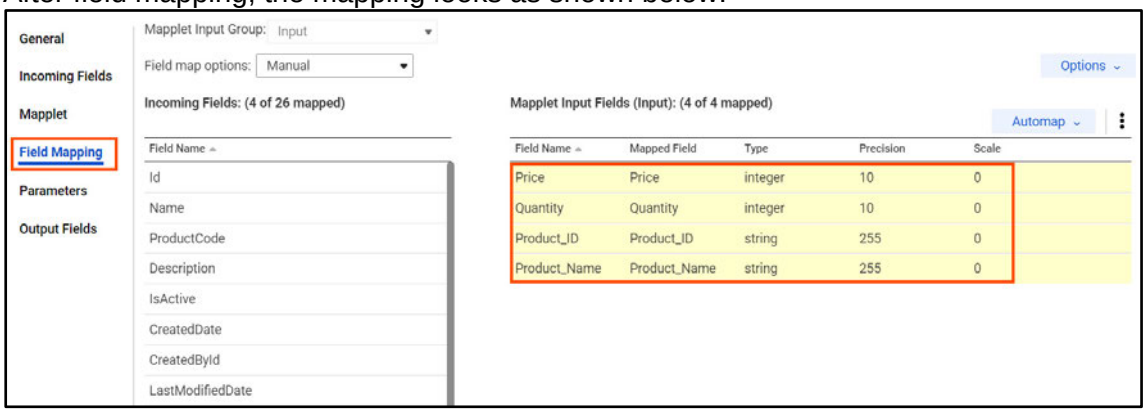
55. Link the Mapplet transformation with Joiner and Target transformations as shown below.



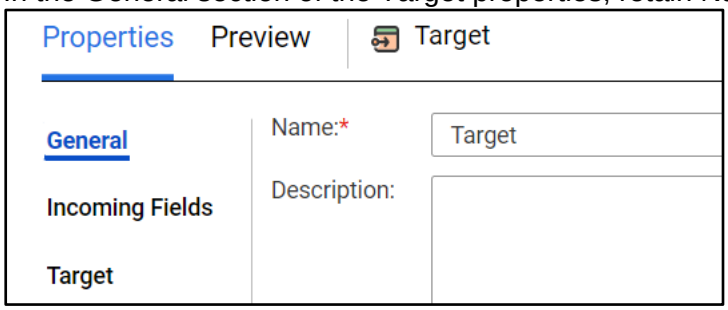
56. Select the Mapplet transformation from the mapping canvas.
57. From the properties pane, click **Field Mapping**.
58. From the Automap drop-down, select **Exact Field Name**.



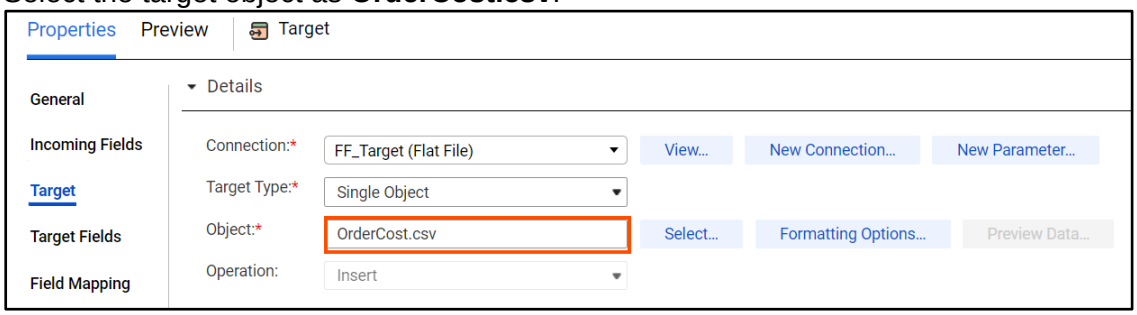
59. After field mapping, the mapping looks as shown below:



60. To configure the target, from the mapping canvas, click the **Target** transformation.
61. In the General section of the Target properties, retain Name as **Target**.

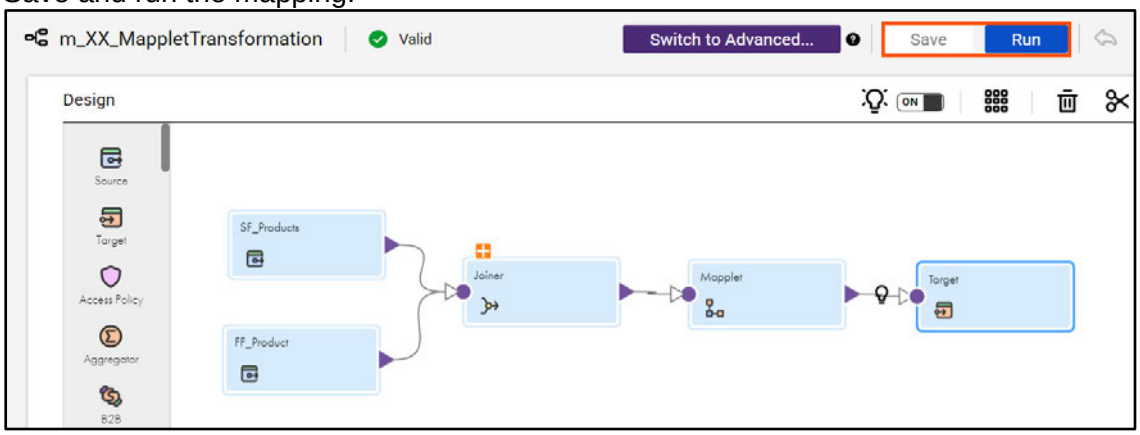


62. From the properties pane, click **Target**.
63. From the Connection drop-down, select your **target flat file** connection.
64. Select the target object as **OrderCost.csv**.

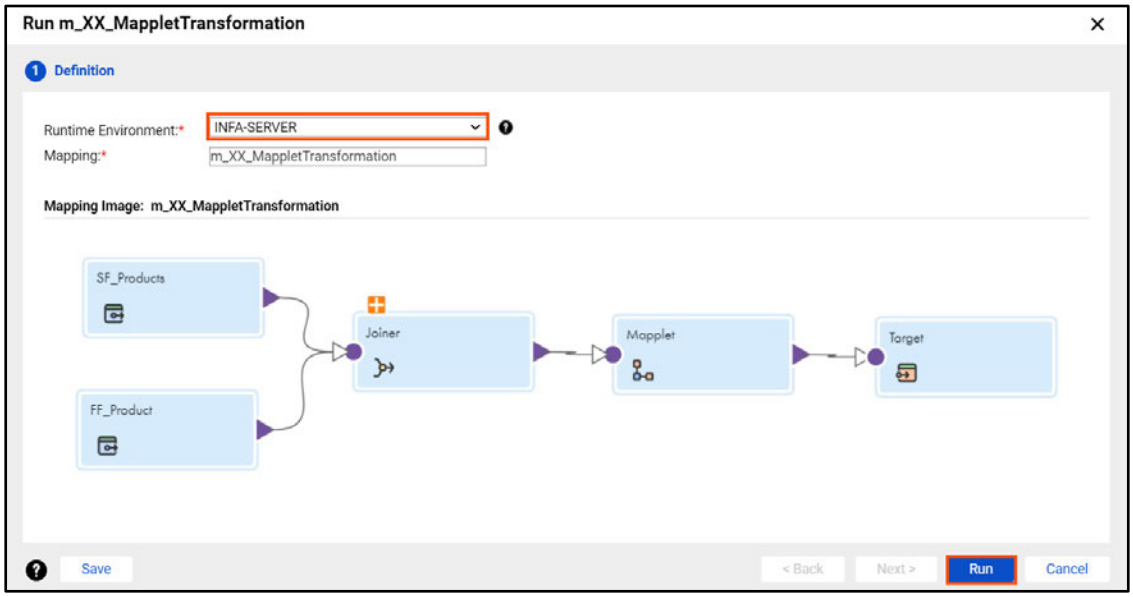


65. From the properties pane, click **Field Mapping**.
66. To match the fields automatically, from the Automap drop-down, select **Exact Field Name**.

67. Save and run the mapping.

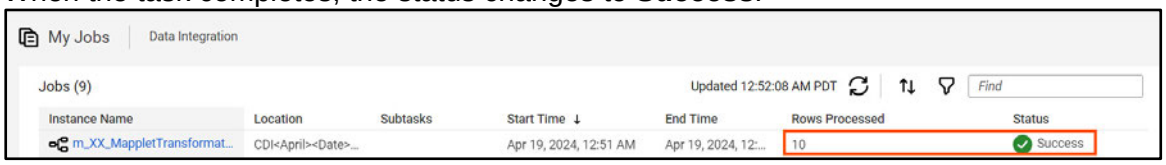


68. From the Runtime Environment drop-down, select INFA-SERVER.
69. Click **Run**.



Monitor Status

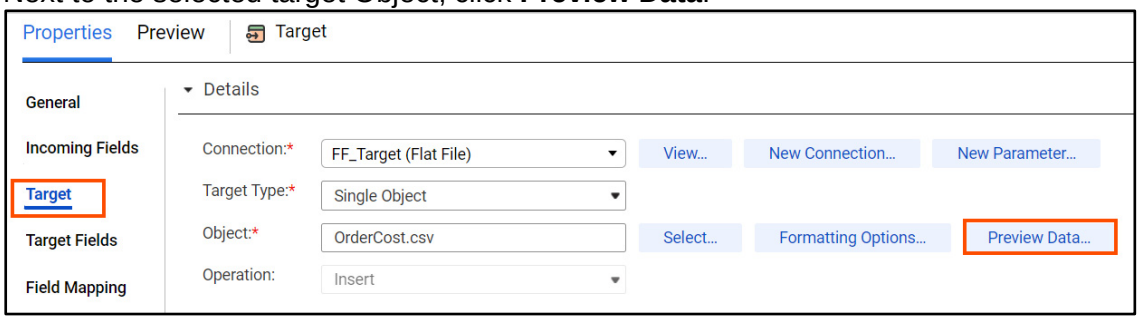
70. To monitor the mapping status, navigate to the **My Jobs** page.
71. When the task completes, the status changes to **Success**.



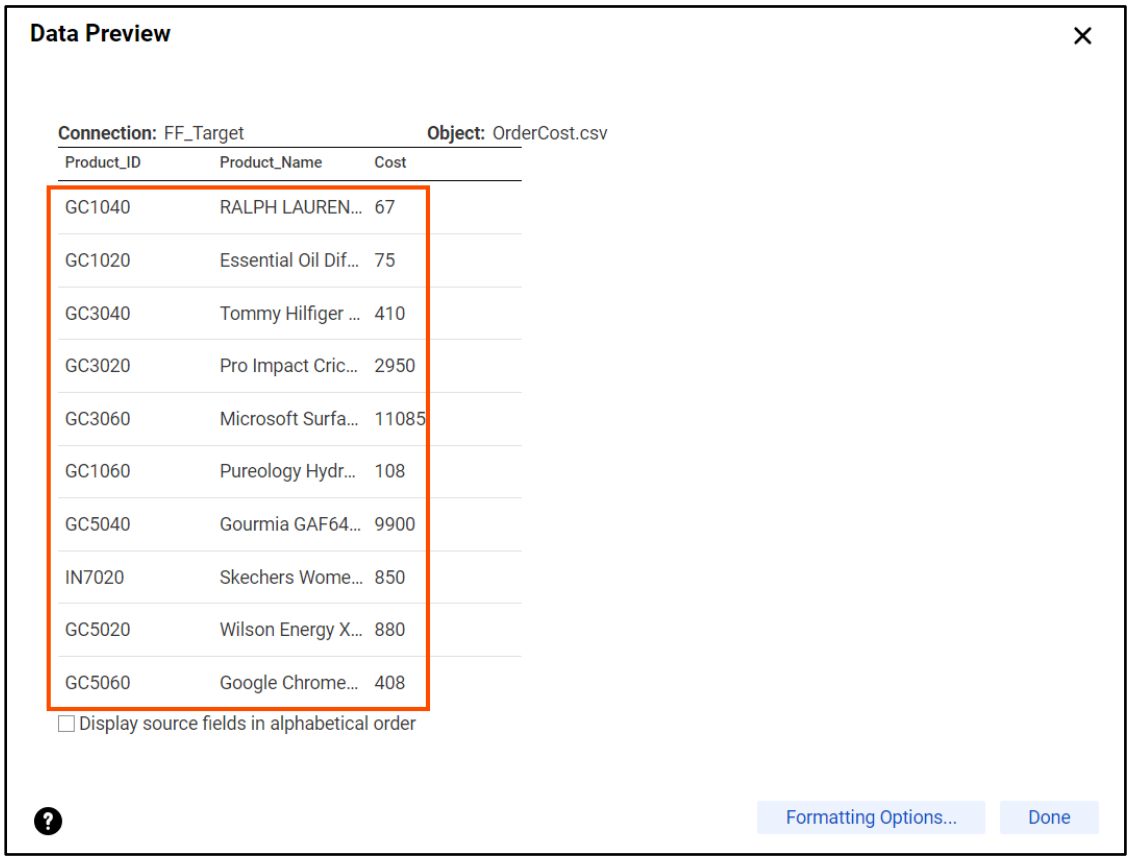
72. Close the mapplet and mapping assets from the navigation pane.

Optional - View Target Data

73. Navigate back to your mapping **m_XX_MappletTransformation**.
74. From the mapping canvas, select the target object and navigate to the **Target** properties tab.
75. Next to the selected target Object, click **Preview Data**.



76. Observe the results.



77. Click **Done**.
78. Save and close all the opened assets.

This concludes the lab.

Module 8: Mapplets

Lab 8-2: Create Mapplet and Use It in the Existing Mapping

Overview:

A mapplet is reusable transformation logic that you can use to transform source data before it is loaded into the target.

In this lab, you will create a Mapplet and use the created Mapplet in a mapping.

Objective:

- Create a Mapplet
- Replace the existing mapping objects with the mapplet
- Replace Filter transformation with Router in the mapping

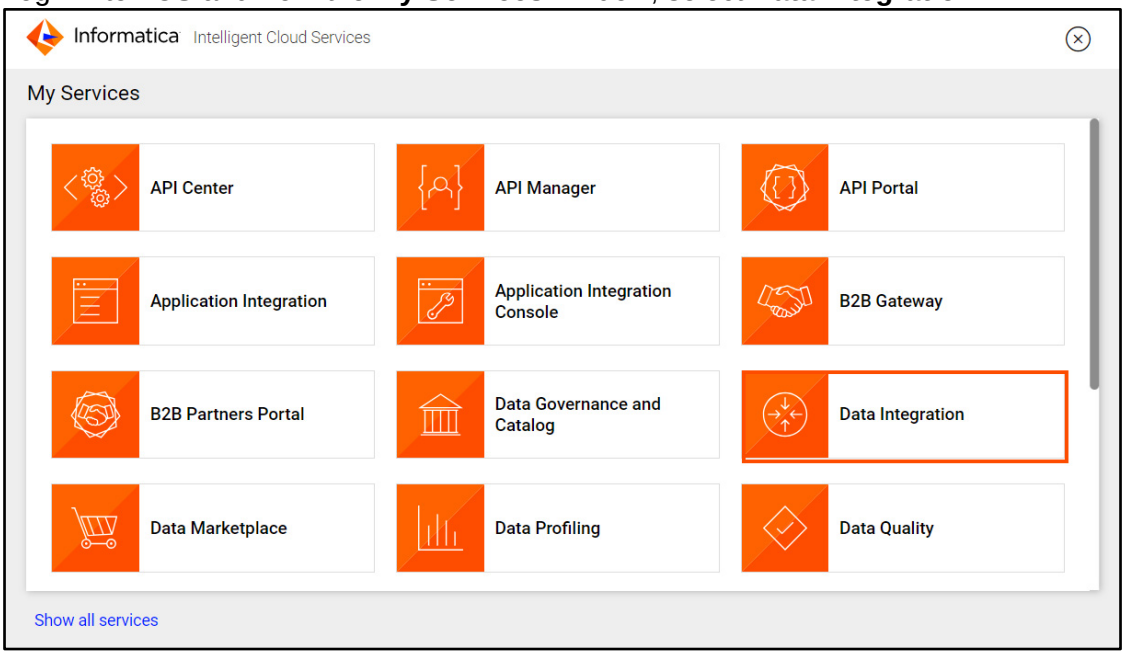
Duration:

30 minutes

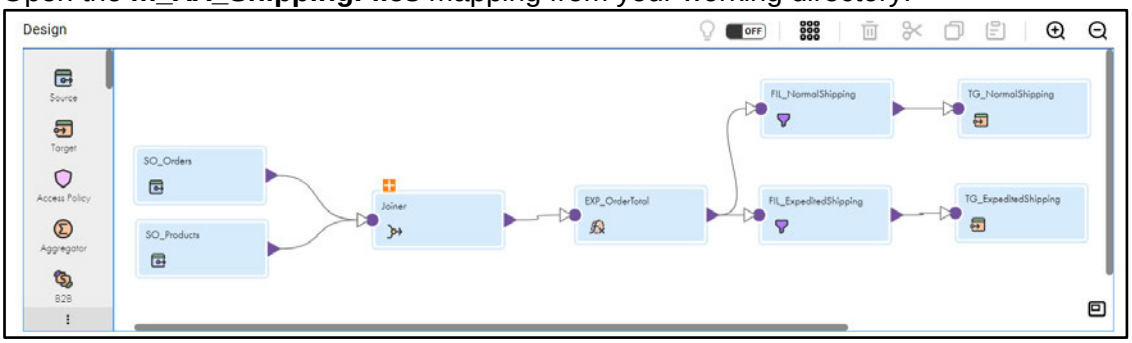
Tasks

Modify the Existing Mapping

1. Login into IICS and from the **My Services** window, select **Data Integration**.

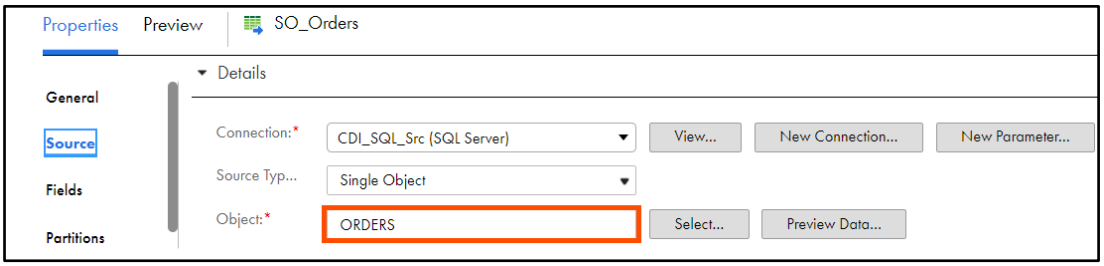


2. From the navigation pane, select **Explore**.
3. Open the **m_XX_ShippingFiles** mapping from your working directory.

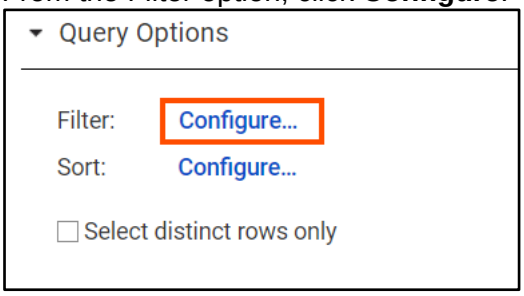


You need to modify the Mapping and make changes to it.

4. Select the **SO_Orders** source from the Mapping canvas.
5. Navigate to the **Source** properties tab and verify that the object selected is **ORDERS**.

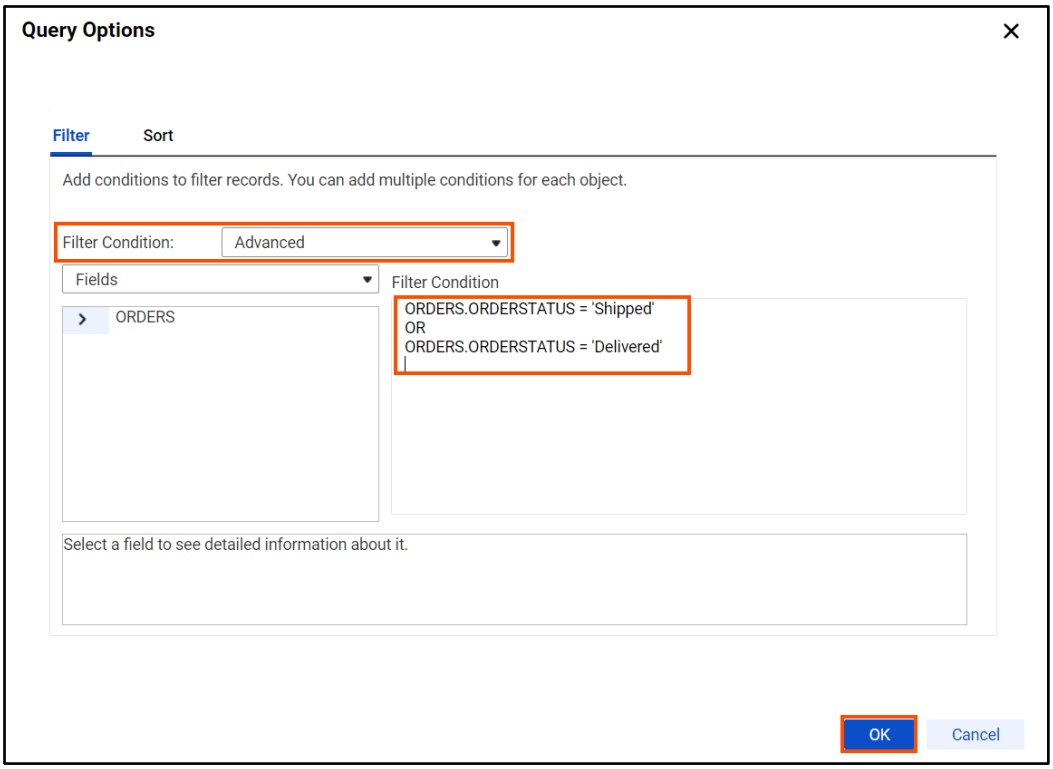


6. Scroll down and expand the **Query Options** section.
7. From the Filter option, click **Configure**.

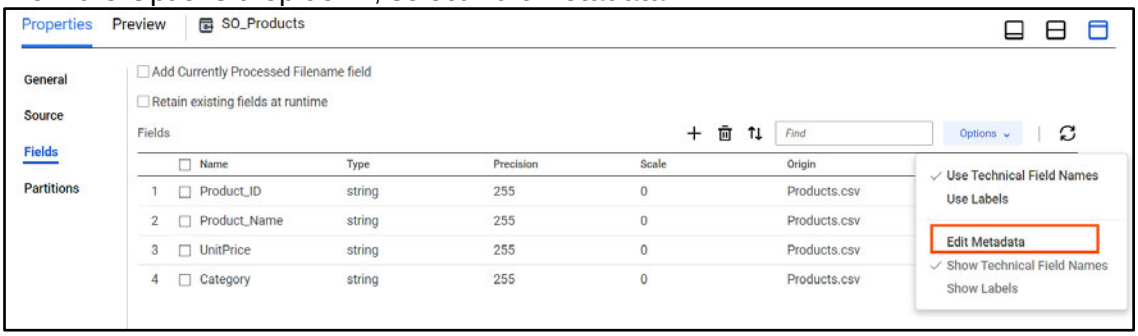


8. From the Filter Condition drop down, select the **Advanced** option.
9. In the Filter Condition expression window, enter the following expression:
   ORDERS.ORDERSTATUS = 'Shipped'
   OR
   ORDERS.ORDERSTATUS = 'Delivered'

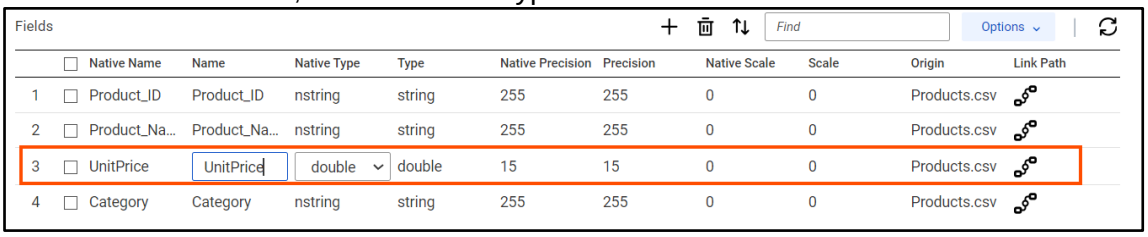
10. Click **OK**



11. Select the **SO_Products** source from the mapping canvas.
12. From the properties pane, click **Fields**.
13. From the Options drop down, select **Edit Metadata**.

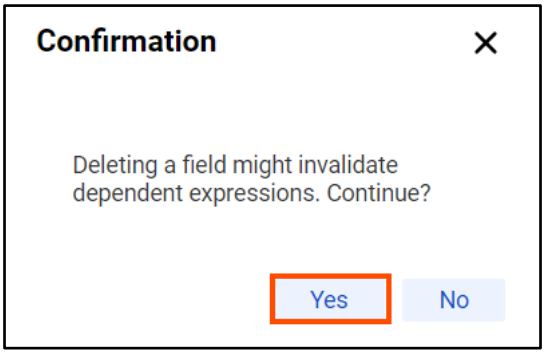


14. For the **UnitPrice** field, set the Native Type as **double** and the Native Precision as **15**.

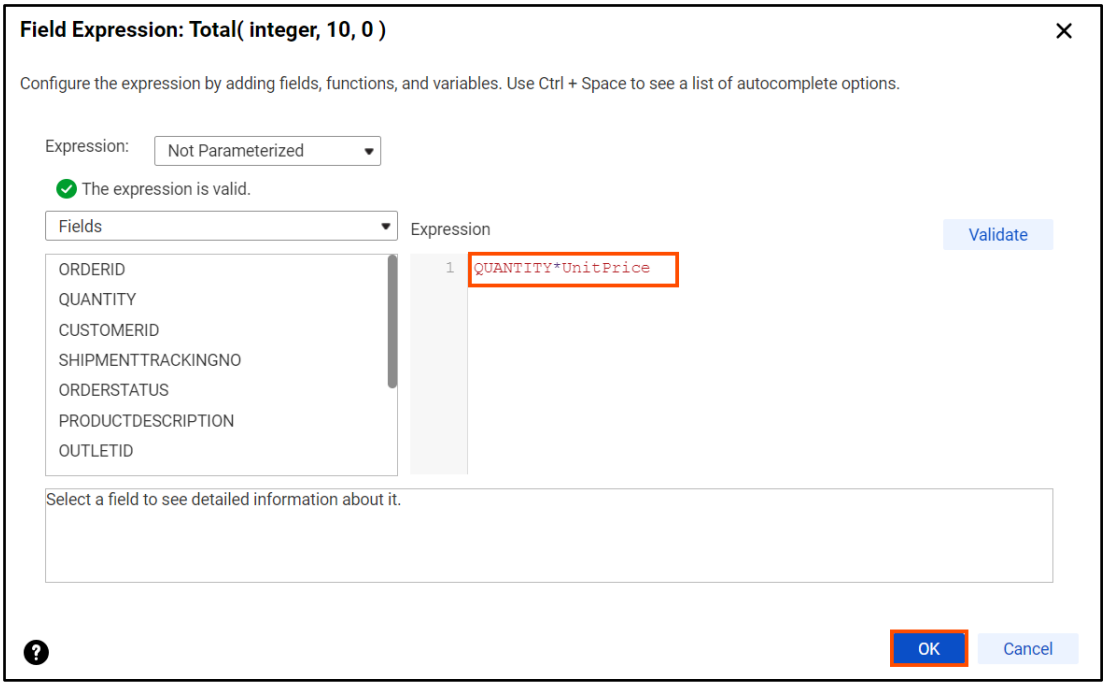


15. From the Options drop-down, uncheck **Edit Metadata**.
16. Select the **EXP_OrderTotal** from the mapping canvas.
17. From the properties pane, click **Expression**.

18. Delete the field **PPU**. Give **Yes** on the Confirmation window.



19. For the field Total, click on the Expression **PPU*QUANTITY**.
20. Delete the Expression PPU*QTY from the Expression window.
21. In the Expression window, enter the following expression:
    QUANTITY*UnitPrice
22. Click **OK**.

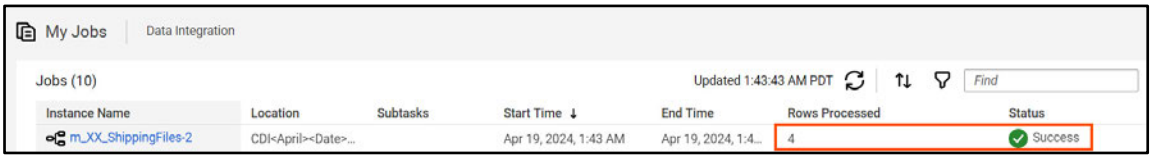


23. To save the mapping, click **Save**.
24. To run the mapping, click **Run**.

25. From the Runtime Environment drop-down, select INFA-SERVER.
26. Click **Run**.

Monitor Status

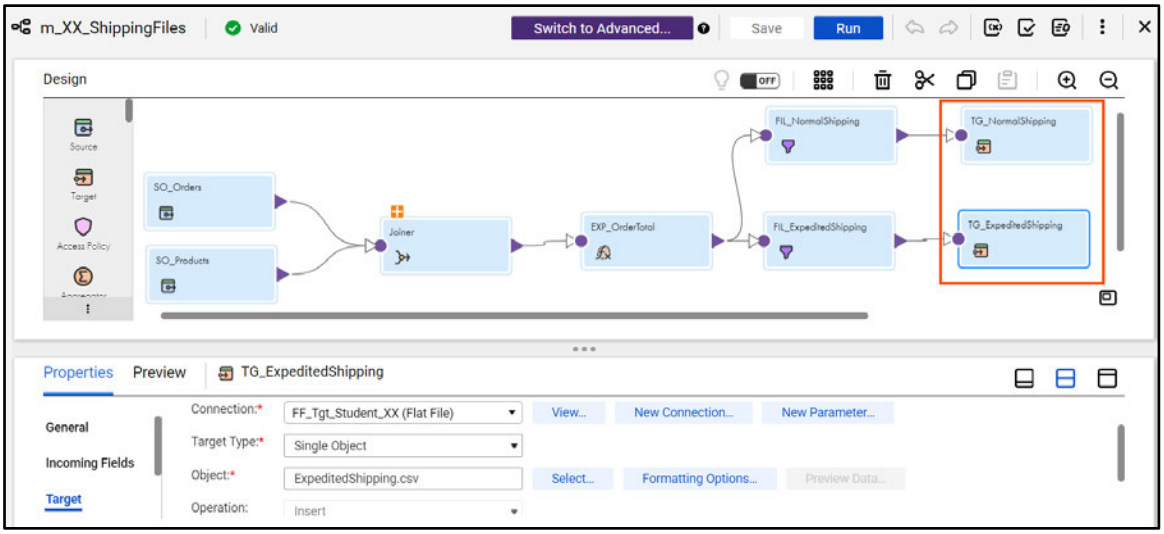
27. Monitor the task status from the **My Jobs** page.
28. When the task completes, the status changes to **Success**.



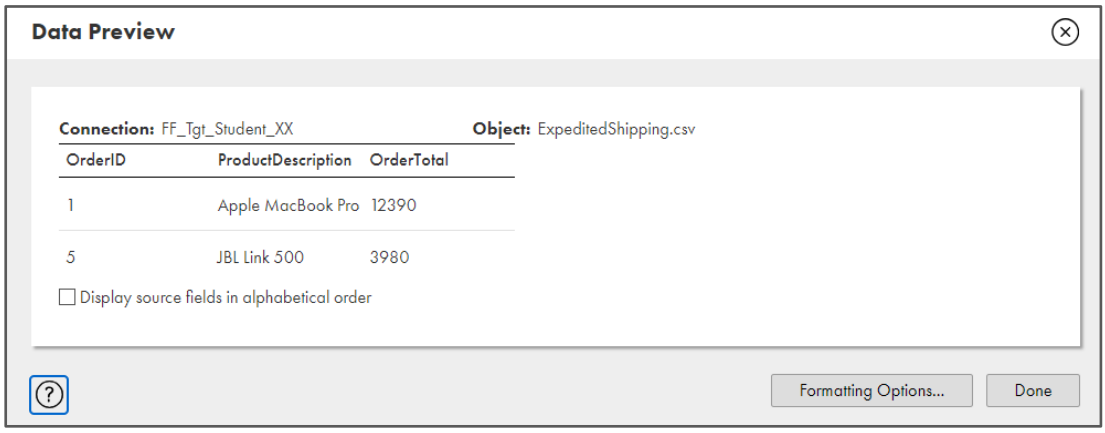
Note: You can refresh the job status if it does not change automatically.

Optional

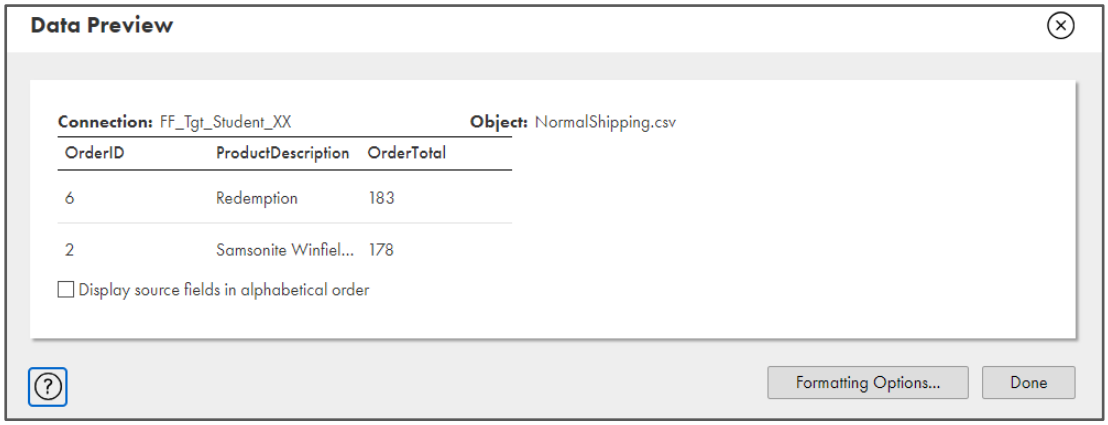
29. You can preview the target data from the targets present on the mapping canvas.



ExpeditedShipping.csv:

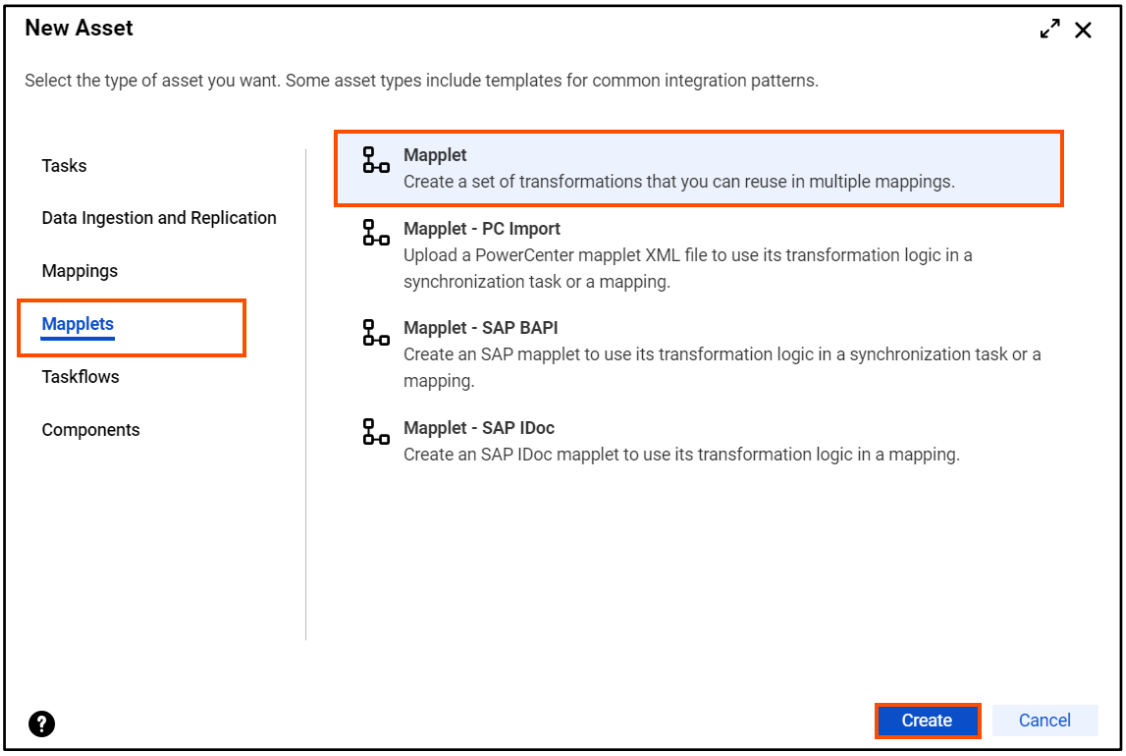


NormalShipping.csv:



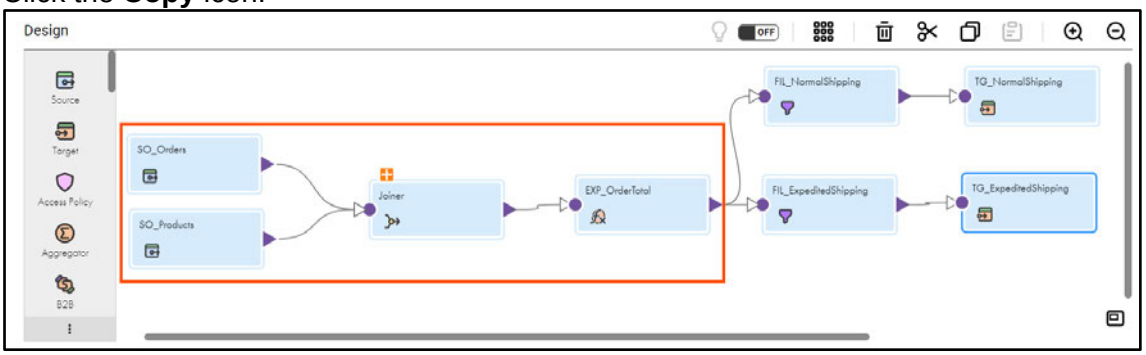
Create Mapplet

30. From the navigation pane, select **New**.
31. From the New Asset window, click the **Mapplets** tab, and select **Mapplet**.
32. Click **Create**.

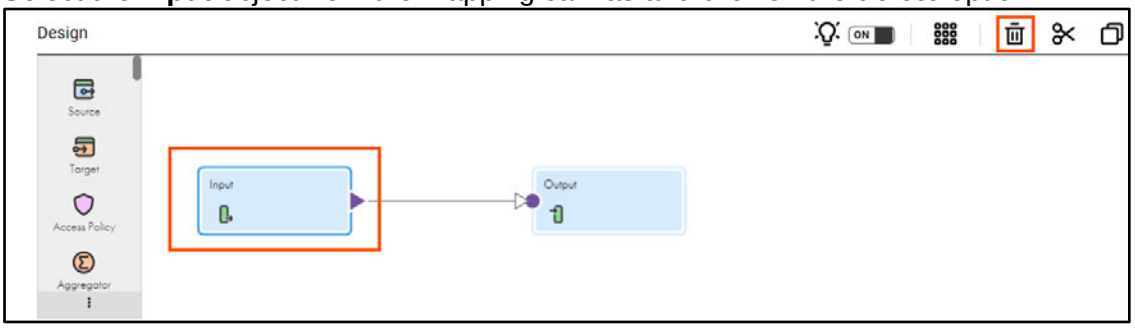


33. In the Name field, enter **mplt_XX_MappletOrders**. Replace XX with your DOB.
34. Go to your working directory and open the asset **m_XX_ShippingFiles** (if its closed).
35. From the mapping canvas, select the sources, **SO_Orders, SO_Products, Joiner,** and **EXP_OrderTotal**.

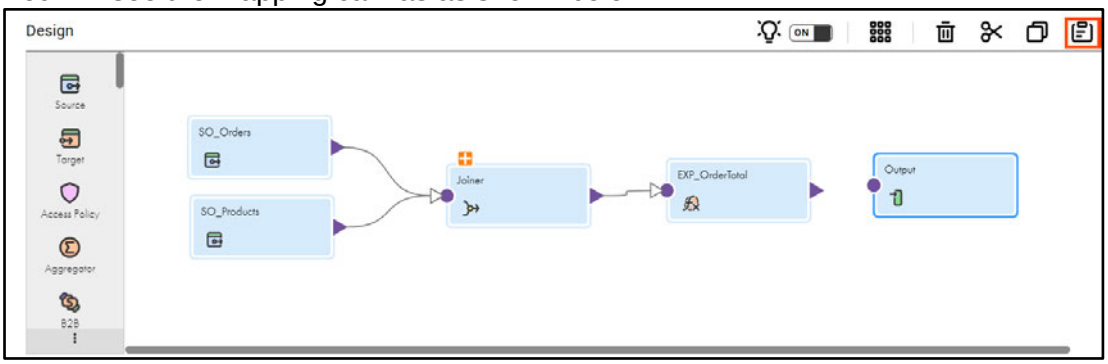
36. Click the **Copy** icon.



37. Go back to your Mapplet canvas.
38. Select the **Input** object from the mapping canvas and click on the **delete** option.



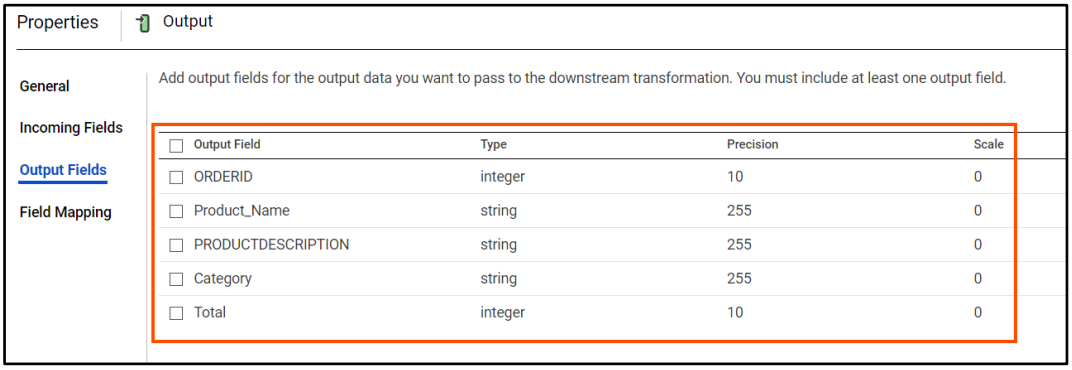
39. Click the **Paste** icon, on the mapping canvas.
40. You will see the mapping canvas as shown below:



41. Link the **EXP_OrderTotal** and the **Output**.
42. Select **Output** from your mapping canvas.
43. From the properties pane, click **Output Fields**.
44. To add a new Output Field, click ⊞.

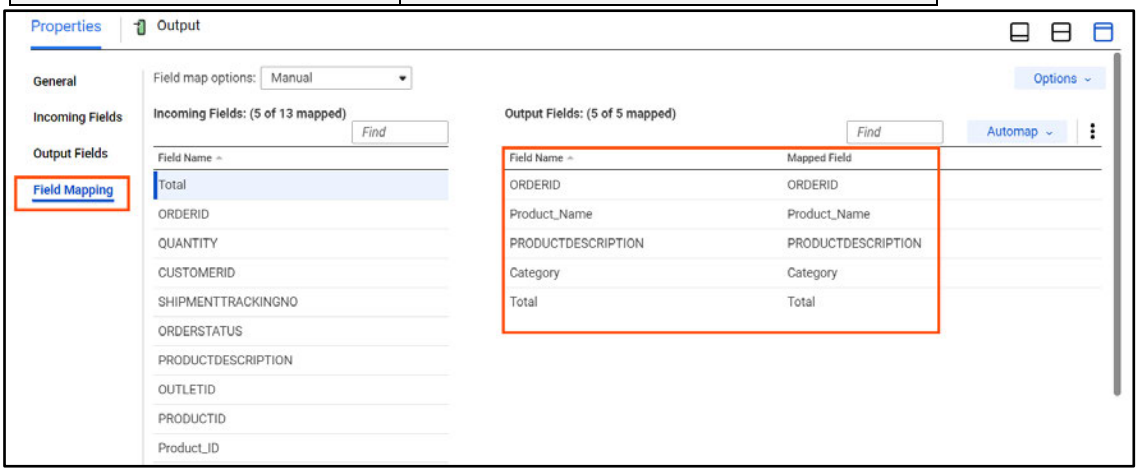
45. Define the output fields, as shown in the table below:

Name	Type	Precision	Scale
ORDERID	integer	10	0
Product_Name	string	255	0
PRODUCTDESCRIPTION	string	255	0
Category	string	255	0
Total	integer	10	0

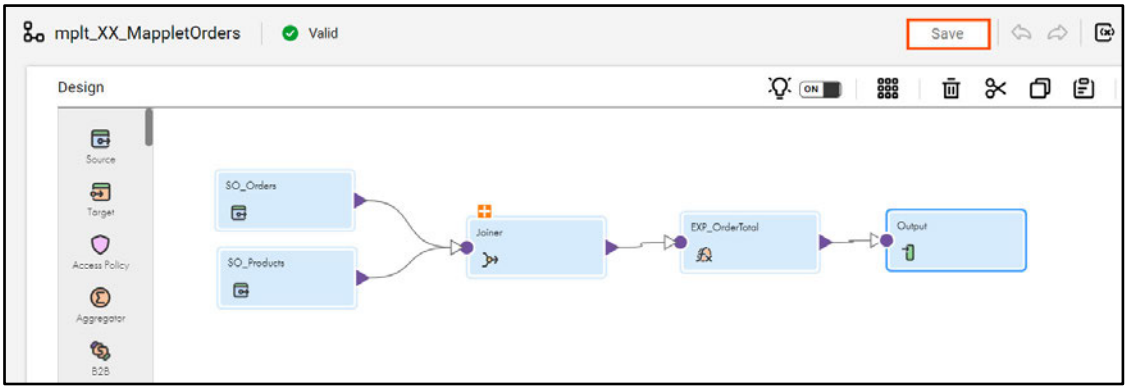


46. From the properties pane, click **Field Mapping**.
47. Match the fields as shown in the following table:
Note: Click and drag the fields to map them.

Incoming Field	Output Fields
ORDERID	ORDERID
Product_Name	Product_Name
Category	Category
Total	Total
PRODUCTDESCRIPTION	PRODUCTDESCRIPTION

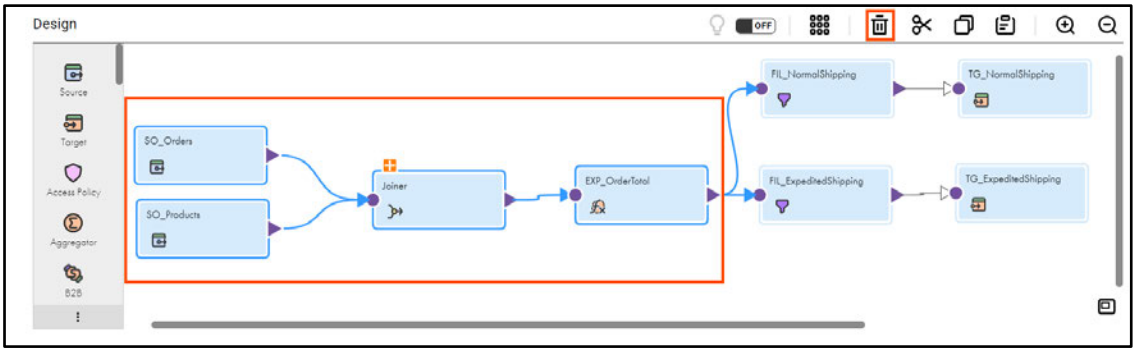


48. Click **Save** to save the Mapplet.

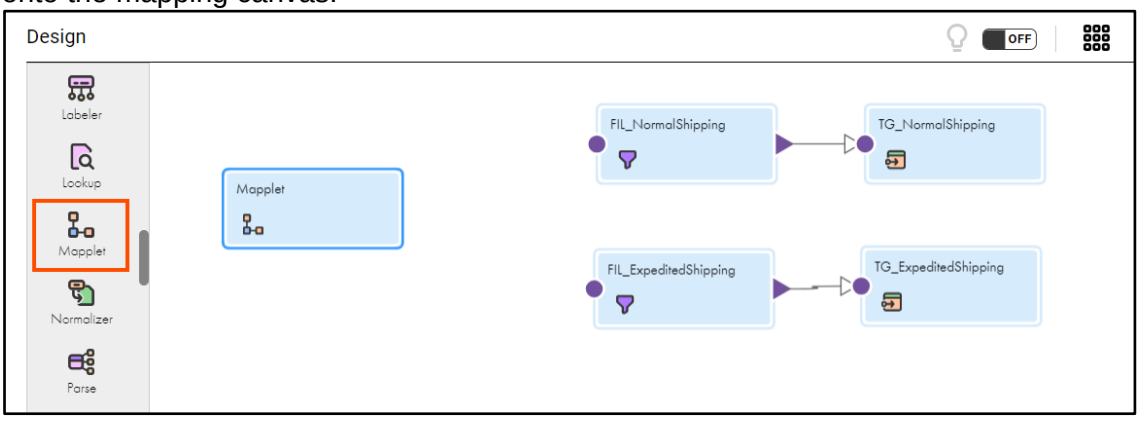


Mapping with Mapplet

49. Navigate to the **m_XX_ShippingFiles** mapping tab.
50. Now, select the **SO_Orders**, **SO_Products**, **Joiner,** and **EXP_OrderTotal** and click **Delete**.



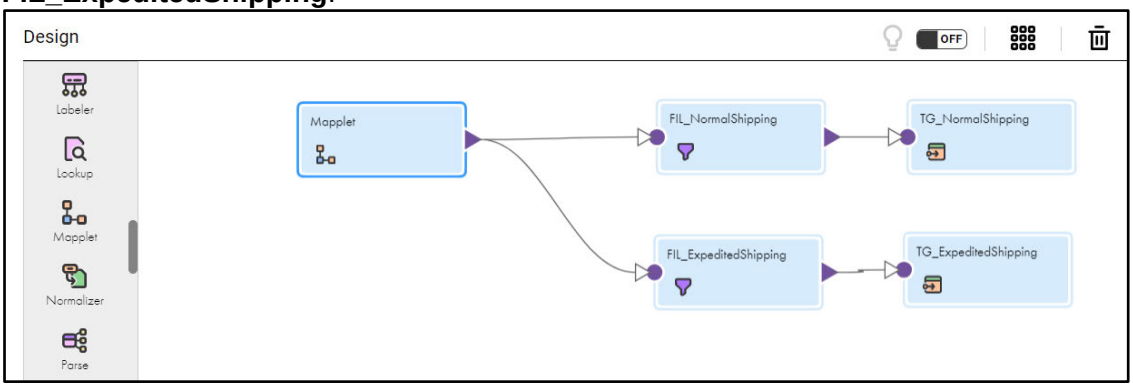
51. From the list of available transformations, drag and drop the **Mapplet** transformation onto the mapping canvas.



52. Select the **Mapplet** from the mapping canvas.
53. From the properties pane, click **Mapplet**.

54. To select the Mapplet, click **Select**.

55. From your working directory select the mapplet, **mplt_XX_MappletOrders**.
56. From the mapping canvas, link the Mapplet with the **FIL_NormalShipping** and **FIL_ExpeditedShipping**.

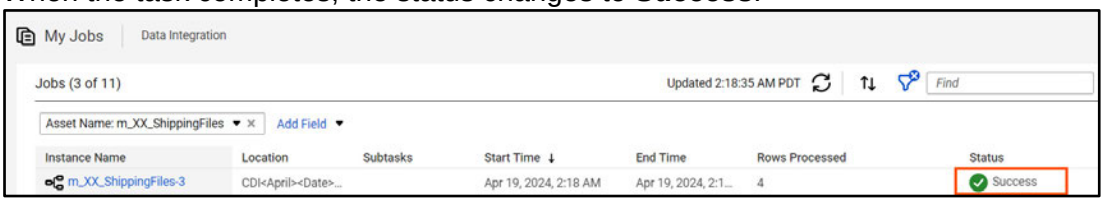


57. **Save** and **Run** the Mapping.

58. From the Runtime Environment drop down, select INFA-SERVER.
59. Click **Run**.

Monitor Status

60. Monitor the task status from the **My Jobs** page.
61. When the task completes, the status changes to **Success**.

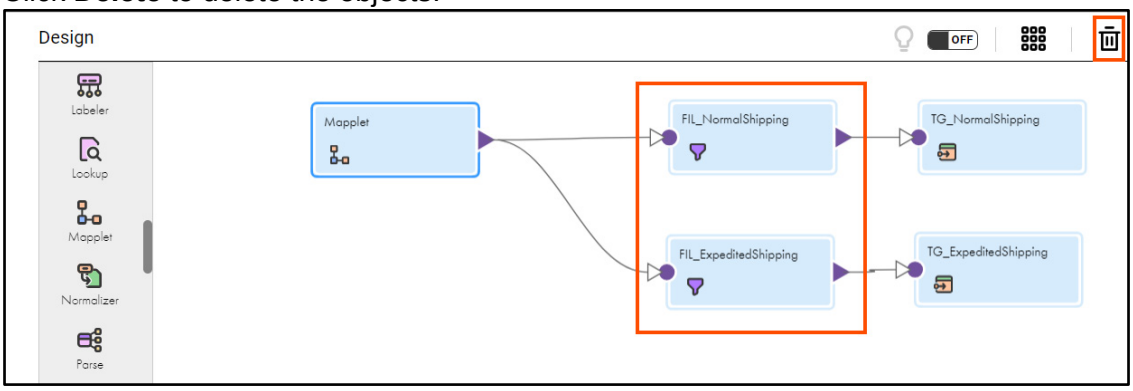


Note: You can refresh the job status if it does not change automatically.

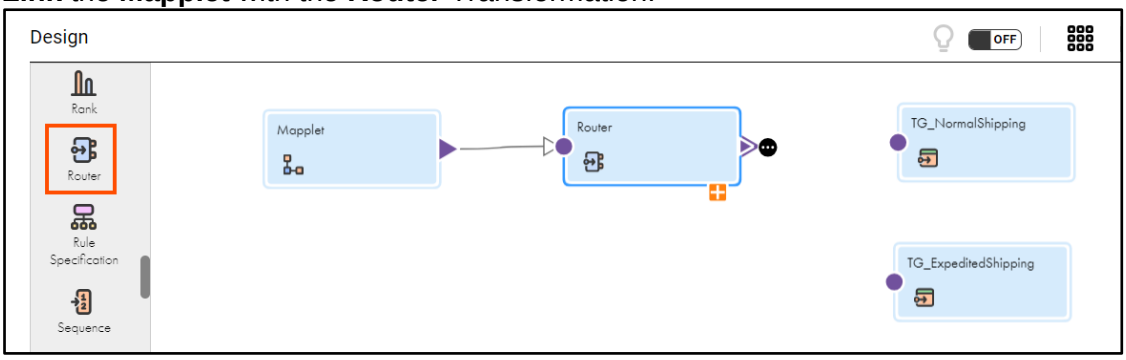
62. Optionally, you can view the preview data from the Target properties tab.

Mapping with Router Transformation

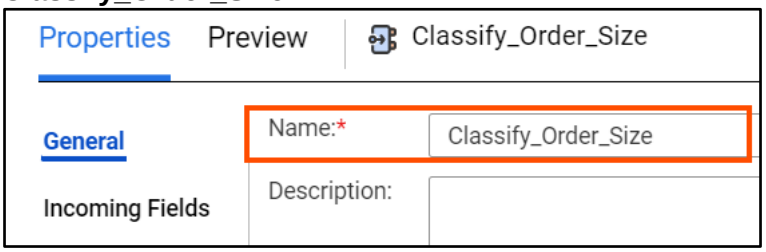
63. Navigate to the **m_XX_ShippingFiles** mapping tab.
64. From the mapping canvas, select **FIL_NormalShipping** and **FIL_ExpeditedShipping**.
65. Click **Delete** to delete the objects.



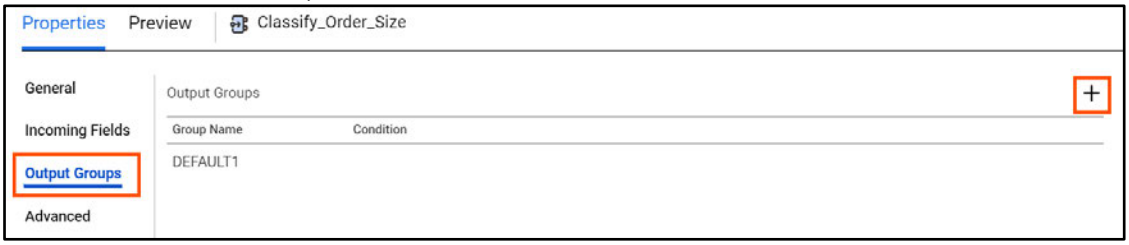
66. From the list of available transformations, drag and drop a **Router** transformation onto the mapping canvas.
67. **Link** the **Mapplet** with the **Router** Transformation.



68. In the General section of the Router properties, enter the Name as **Classify_Order_Size**.



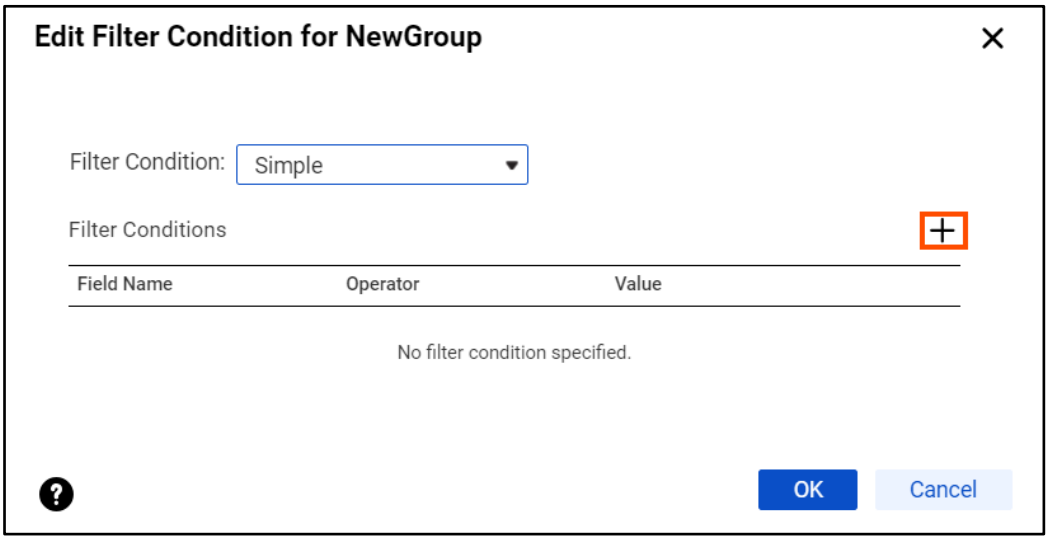
69. From the properties pane, select **Output Groups**.
70. To add a new condition, click ⊞.



71. To enter the condition, click **Configure**.

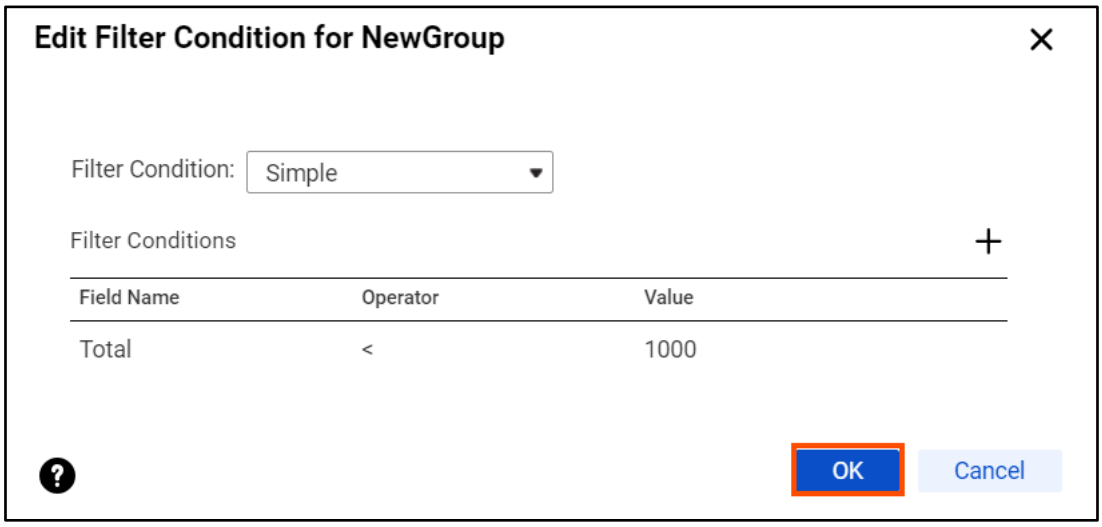
The Edit Filter Condition for NewGroup window appears.

72. Click the ⊞ icon.



73. Enter the filter condition as shown in the table below and click **OK**.

Field Name	Operator	Value
Total	< (Less than)	1000

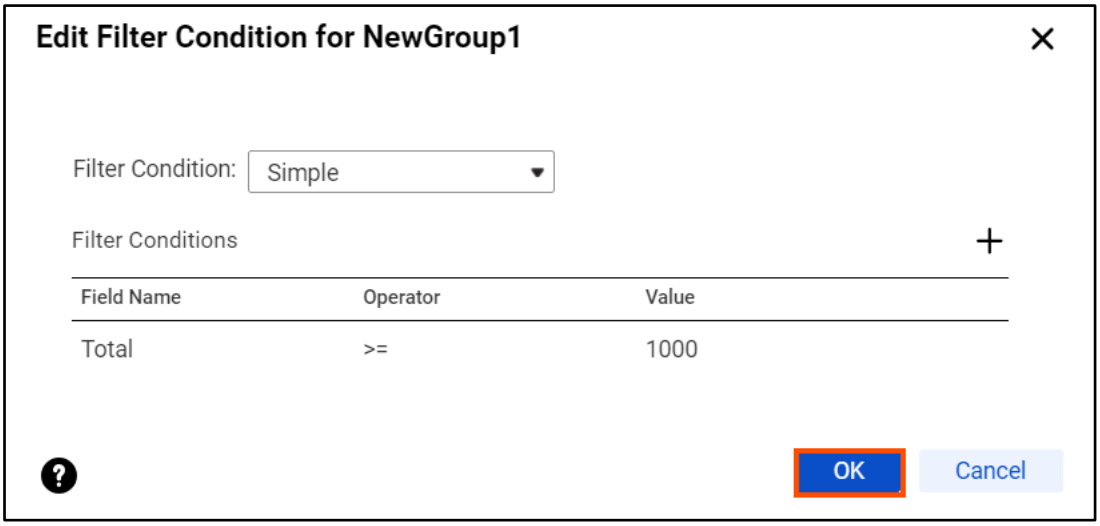


74. Change the Group Name to **Normal_Shipping**.
75. To add another filter condition, click .
76. Click **Configure**.

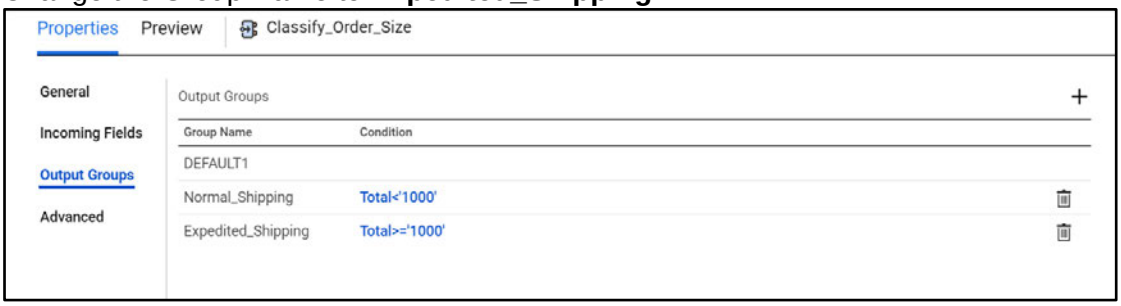
77. Enter filter condition as shown in the table below:

Field Name	Operator	Value
Total	>= (Greater than equal to)	1000

78. Click **OK**.

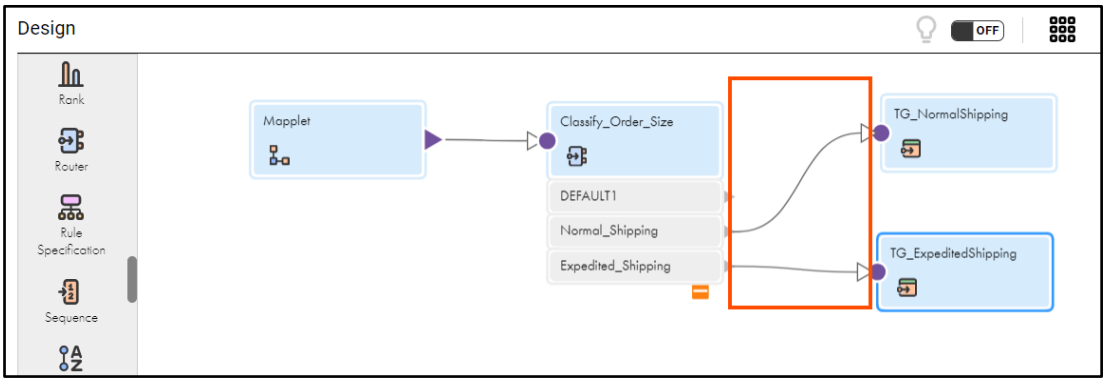


79. Change the Group Name to **Expedited_Shipping**.



80. On the mapping canvas, click the plus icon on the Router transformation to expand the output groups.

81. Link the **Classify_Order_size** to the **TG_NormalShipping** and **TG_ExpeditedShipping** respectively.

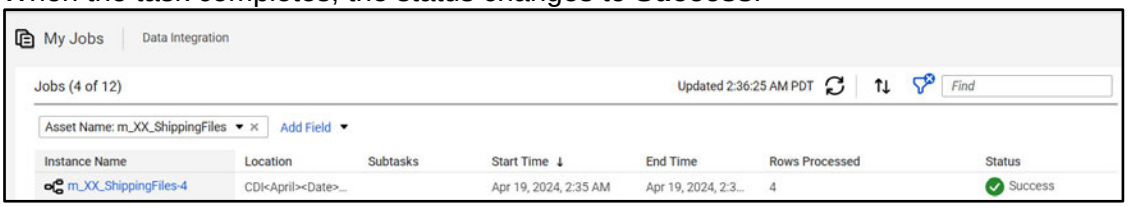


82. **Save** and **Run** the Mapping.

83. From the Runtime Environment drop down, select INFA-SERVER.
84. Click **Run**.

Monitor Status

85. To monitor the mapping status, navigate to the **My Jobs** page.
86. When the task completes, the status changes to **Success**.



87. Optionally, you can view the preview data from the Target properties tab.
88. Close the mapplet and mapping assets from the navigation pane.

This concludes the lab.

Module 8: Mapplets

Lab 8-3: Creating a Mapping Using Existing Mapplet and Aggregator Transformations

Overview:
A mapplet is reusable transformation logic that you can use to transform source data before it is loaded into the target. After you create a mapplet, you can add it to a Mapplet transformation to use its transformation logic. You can use a mapplet in another mapplet.

You can select a mapplet that you created or imported into the DI service.

In this lab, you will use the created Mapplet in a mapping and finish the login using the aggregator transformation.

Objective:

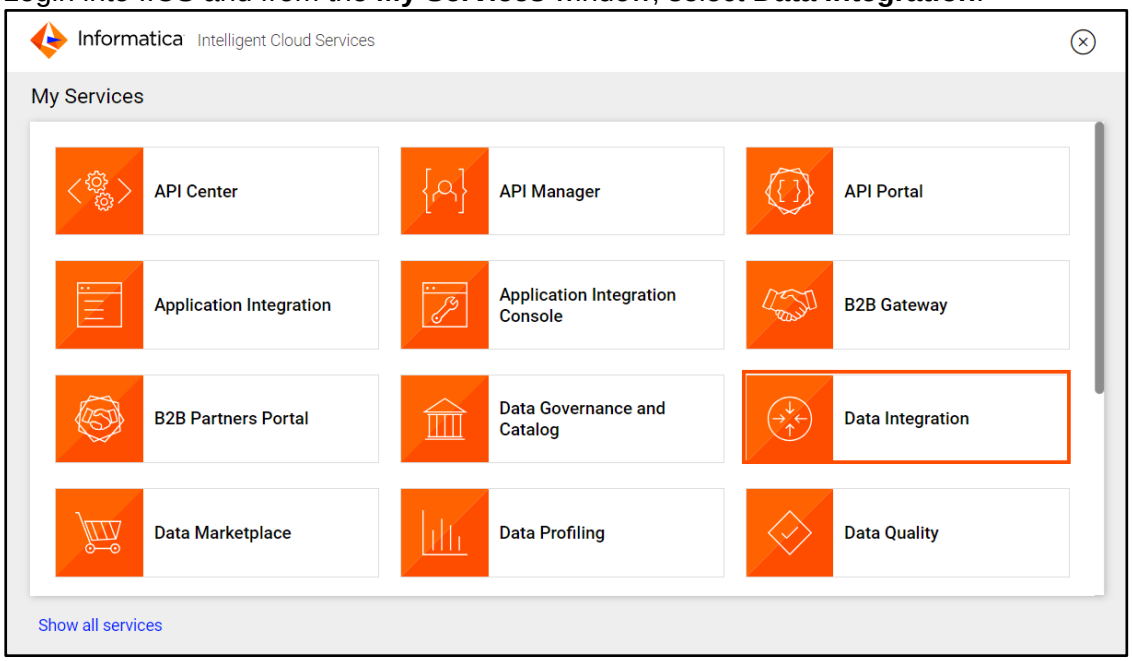
- Use the existing Mapplet
- Use the Aggregator transformation

Duration:
20 minutes

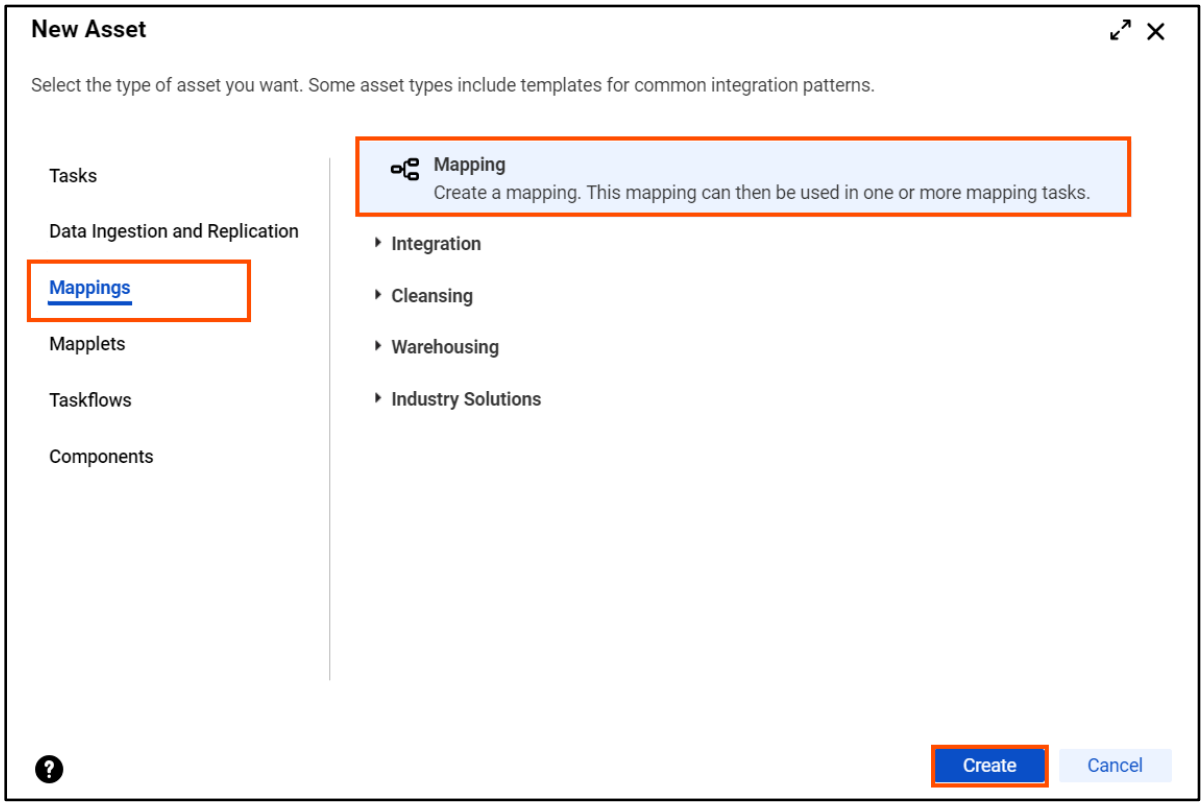
Tasks

Create Mapping

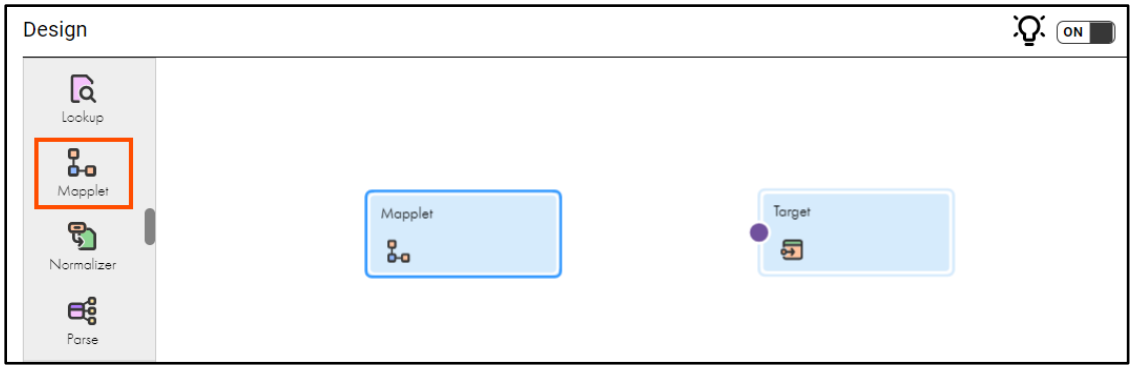
1. Login into IICS and from the **My Services** window, select **Data Integration**.



2. From the navigation pane, select **New**.
3. From the New Asset window, click the **Mappings** tab, and select **Mapping**.
4. Click **Create**.

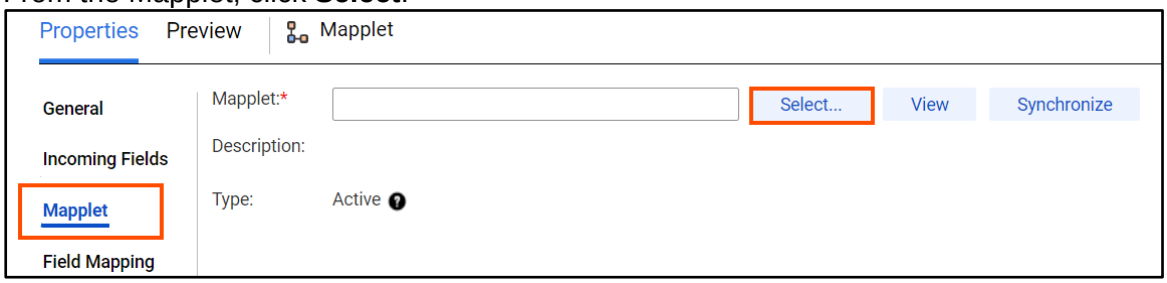


5. In the Name field, enter **m_XX_SummarizedSales**.
6. Specify the location of your working directory.
7. From the mapping canvas, select the **Source** and click the **Delete** option.
8. From the list of available transformations, drag and drop a **Mapplet** on the mapping canvas.

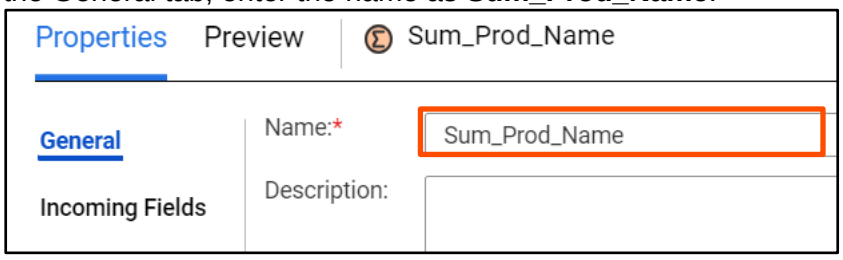


9. Select the **Mapplet** from the mapping canvas.
10. From the properties pane, click **Mapplet**.

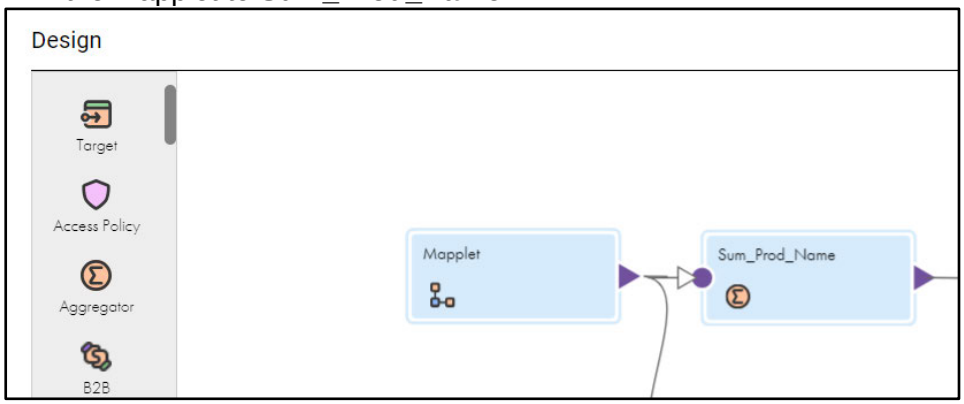
11. From the Mapplet, click **Select**.



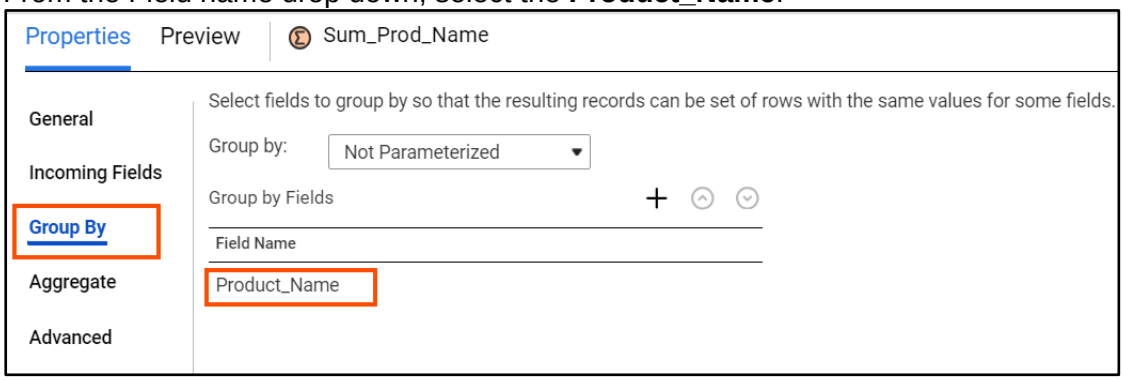
12. From your working directory, select the mapplet **mplt_XX_MappletOrders**.
13. From the list of available transformations, drag and drop the **Aggregator** transformation between Mapplet and Target on the mapping canvas.
14. With the Aggregator transformation being selected, navigate to the properties pane, in the General tab, enter the name as **Sum_Prod_Name**.



15. Link the Mapplet to **Sum_Prod_Name**.



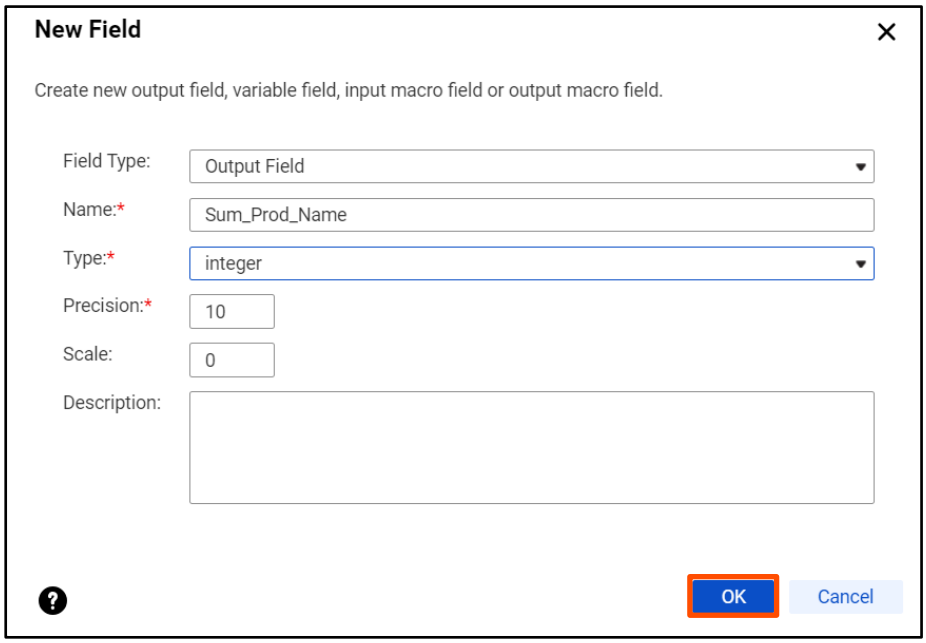
16. From the properties pane, click **Group By**.
17. To add a new expression, click ⊞.
18. From the Field name drop down, select the **Product_Name**.



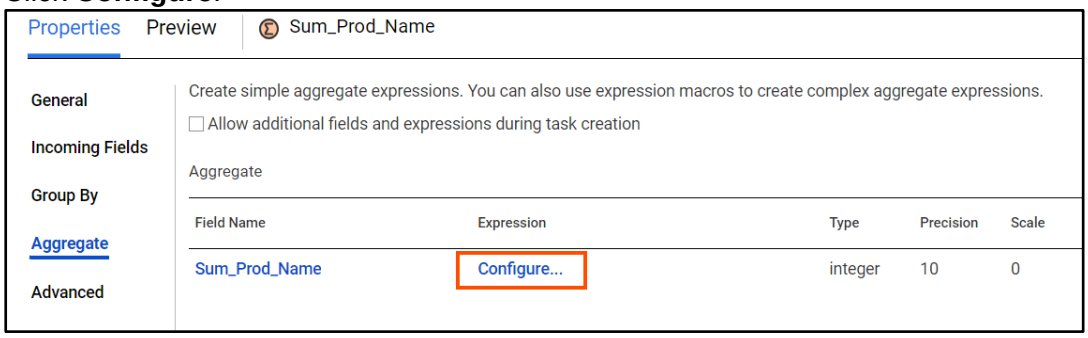
19. From the properties pane, click **Aggregate**.
20. To add the new Aggregate field, click ⊞.
21. Enter the details as shown below:

Field Type	Name	Type	Precision	Scale
Output Field	Sum_Prod_Name	integer	10	0

22. Click **OK**.



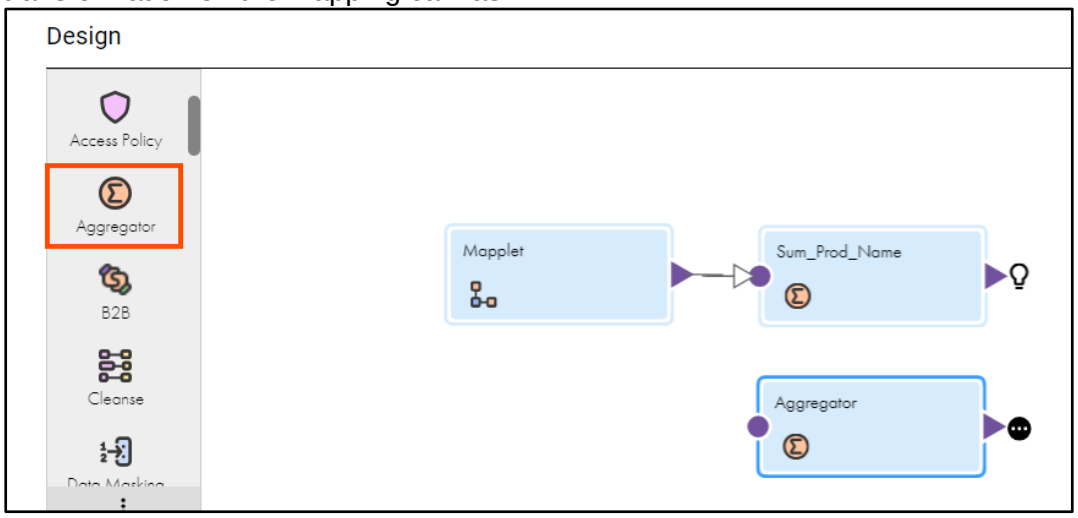
23. Click **Configure**.



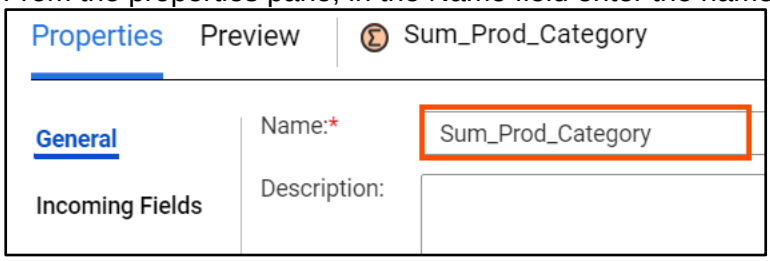
24. In the Expression window, enter the following expression:
    SUM(Total)

25. Click **OK**.

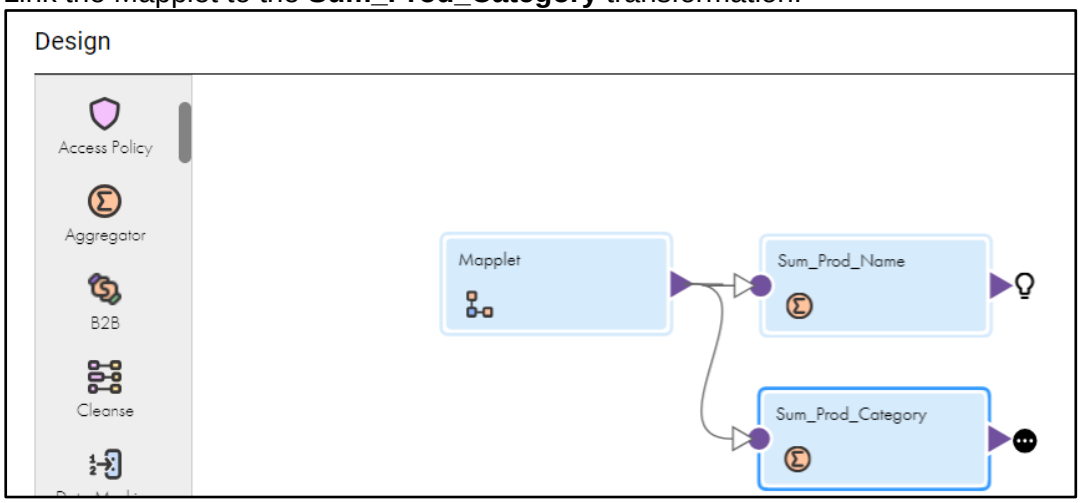
26. From the list of available transformations, drag and drop another aggregate transformation on the mapping canvas.



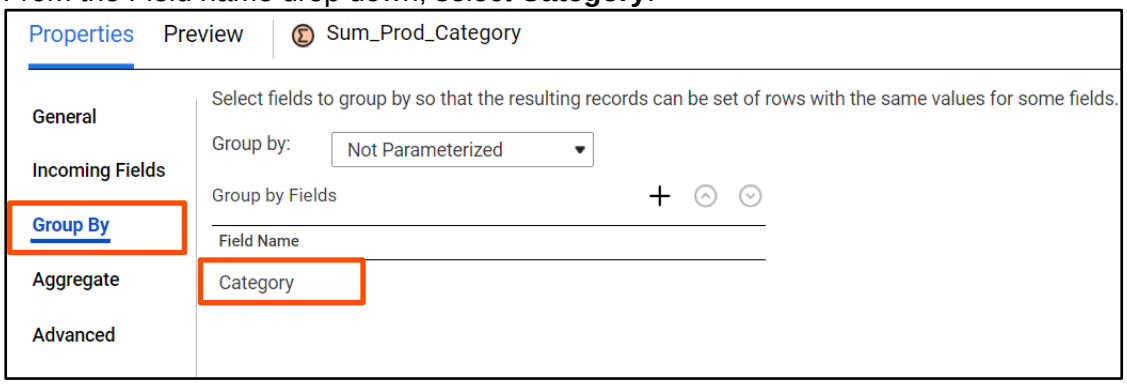
27. From the properties pane, in the Name field enter the name as **Sum_Prod_Category**.



28. Link the Mapplet to the **Sum_Prod_Category** transformation.



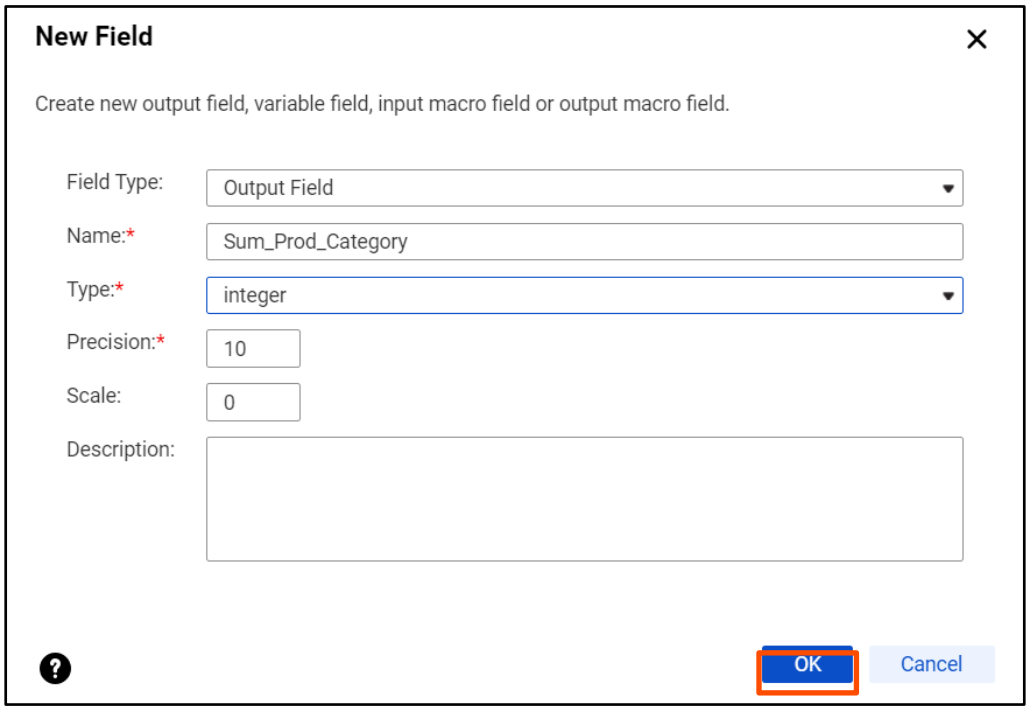
29. From the properties pane, click **Group By**.
30. To add a new expression, click ⊞.
31. From the Field name drop down, select **Category**.



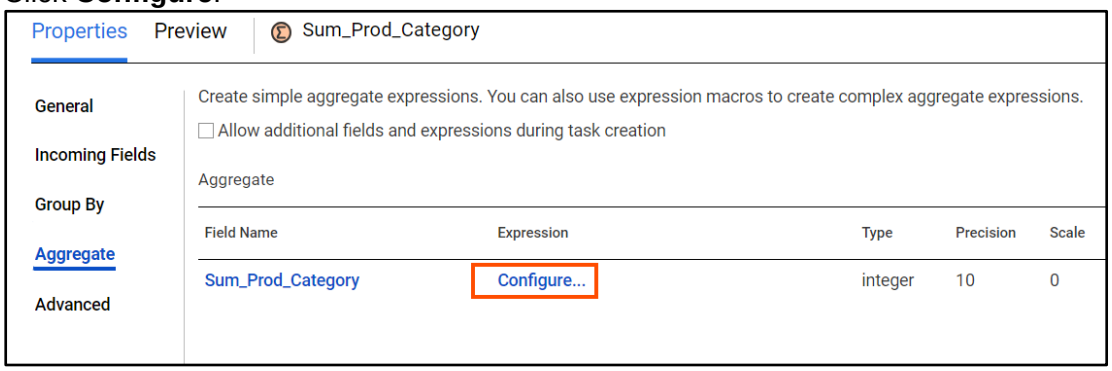
32. From the properties pane, click on **Aggregate**.
33. To add the new Aggregate Field, click ⊞
34. Enter the details as shown below:

Field Type	Name	Type	Precision	Scale
Output Field	Sum_Prod_Category	integer	10	0

35. Click **OK**.

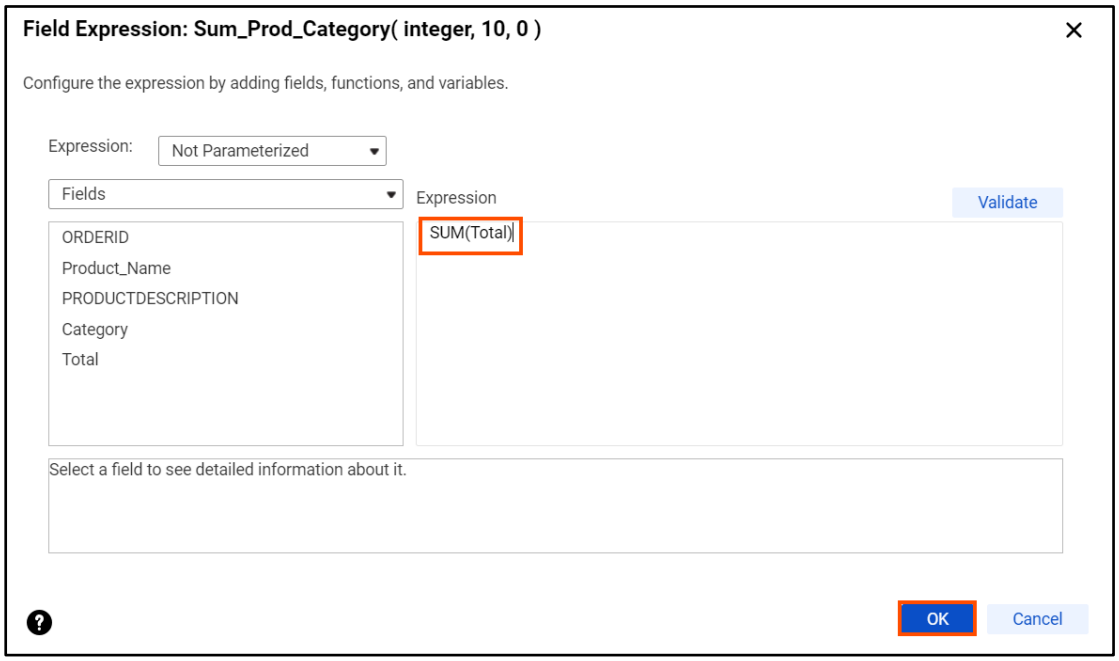


36. Click **Configure**.

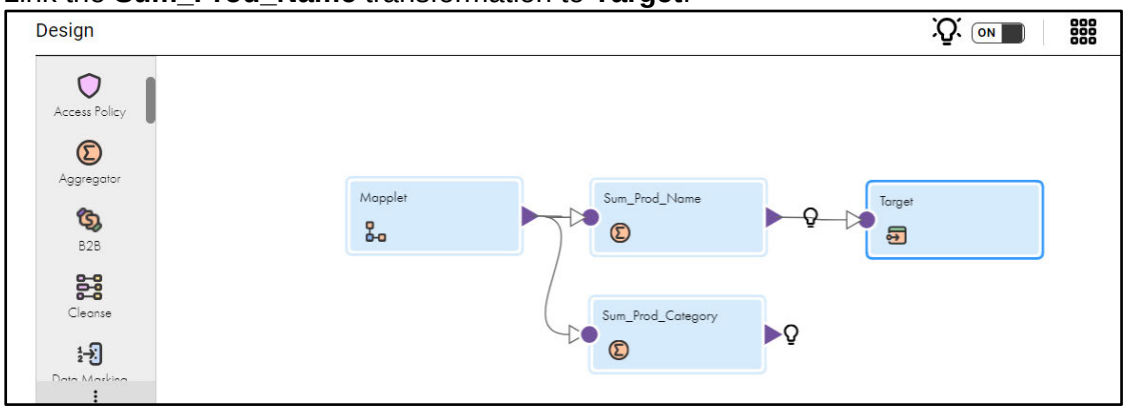


37. In the Expression window, enter the following expression:
SUM(Total)

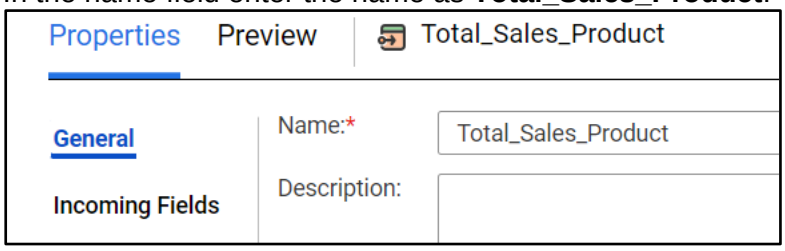
38. Click **OK**.



39. Select the Target from the mapping canvas.
40. Link the **Sum_Prod_Name** transformation to **Target**.

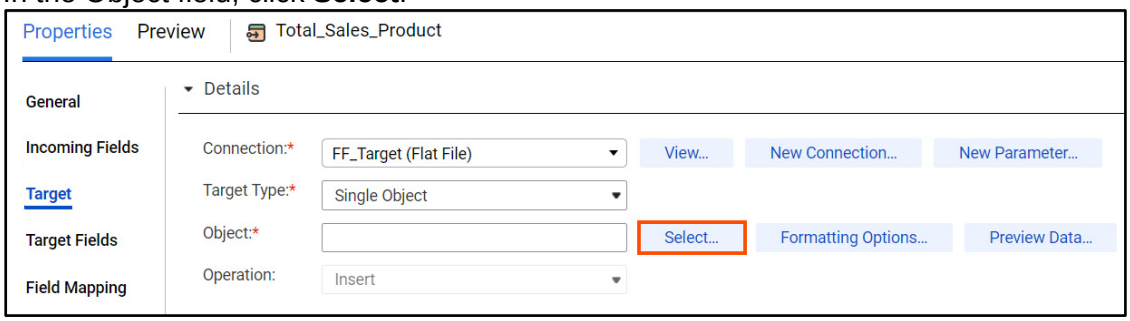


41. In the name field enter the name as **Total_Sales_Product**.



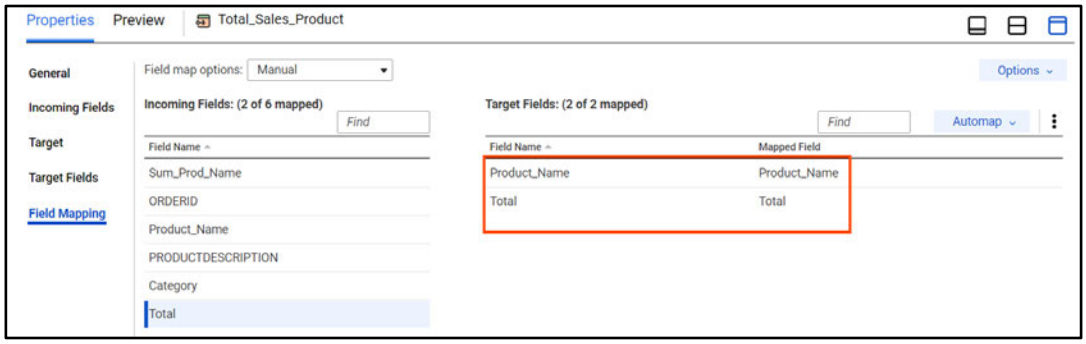
42. From the properties pane, select **Target**.
43. From the Connection drop down, select your **target Flat File** connection.
44. Retain the Target Type as **Single Object**.

45. In the Object field, click **Select**.

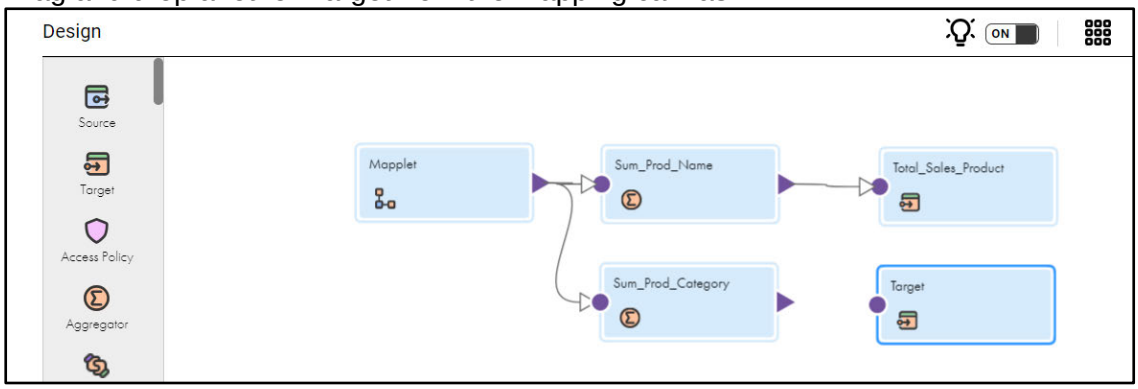


46. From the list, select the **Total_Sales_Product.csv**. Click **OK**.
47. From the properties pane, select **Field Mapping**.
48. Map the fields as shown in the table below (If they are mapped already, verify, and proceed with the next step):

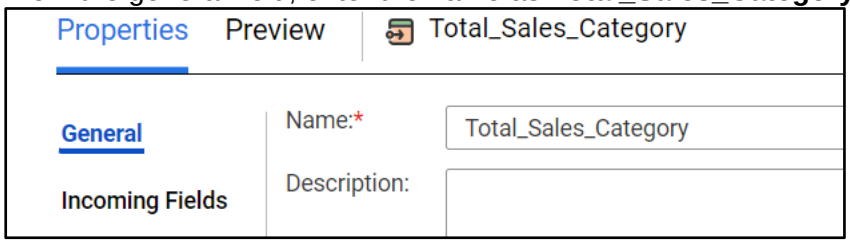
Incoming Field	Target Field
Product_Name	Product_Name
Total	Total



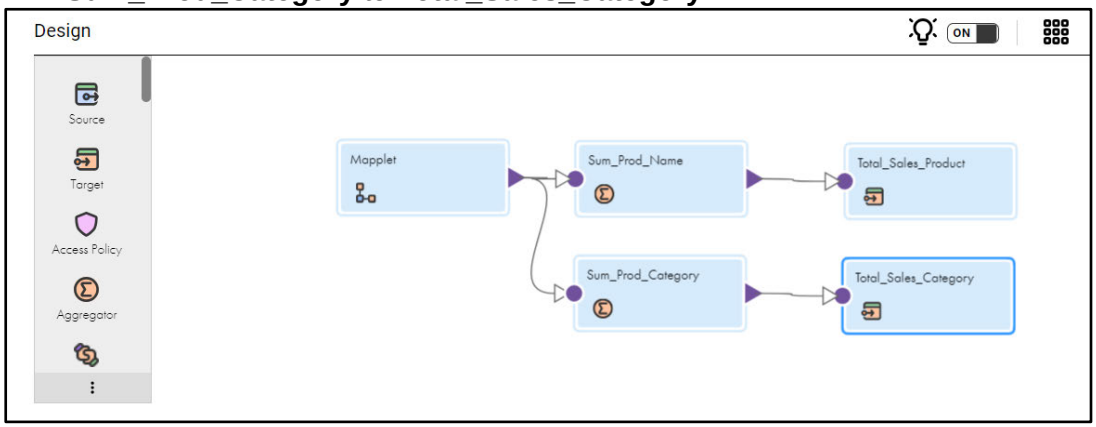
49. Drag and drop another Target from the mapping canvas.



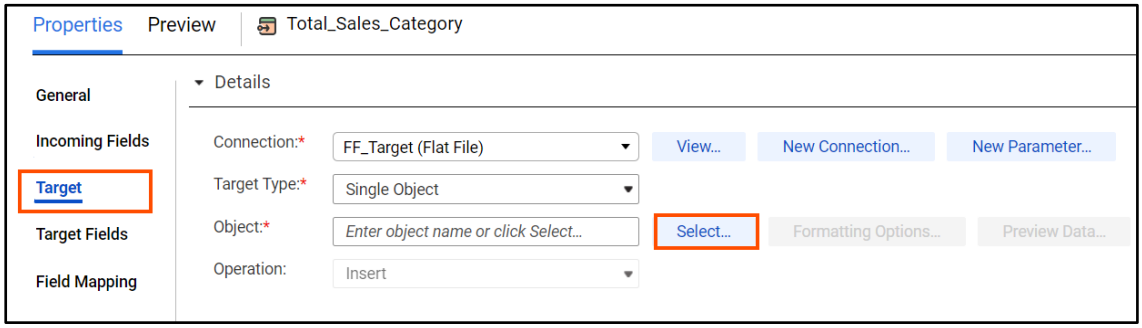
50. From the general field, enter the name as **Total_Sales_Category**.



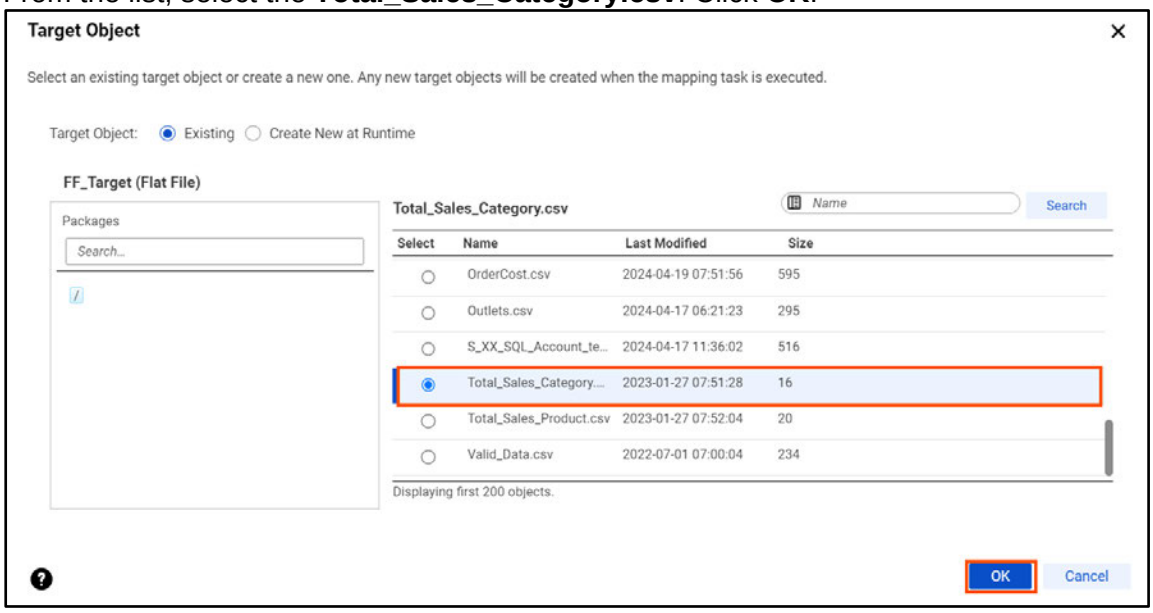
51. Link **Sum_Prod_Category** to **Total_Sales_Category**.



52. Select the **Total_Sales_Category** from the mapping canvas and from the properties pane, select **Target**.
53. From the Connection drop-down, select your **target Flat File** connection.
54. To select the source object from the Object field, click **Select**.



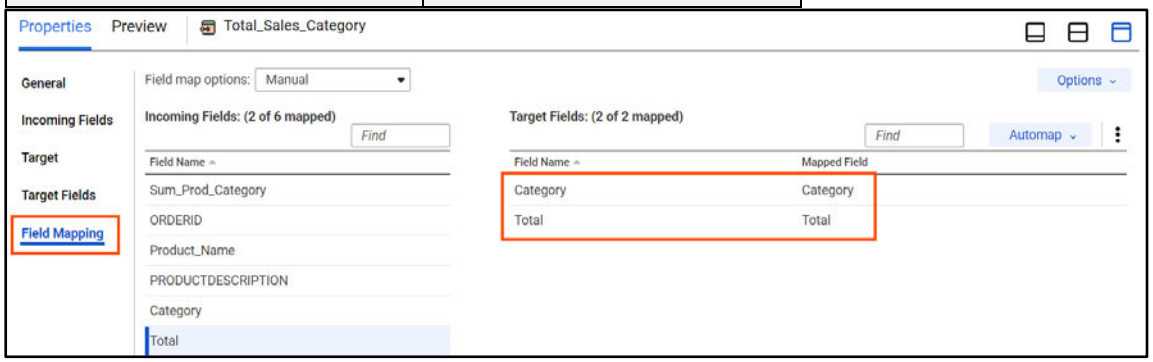
55. From the list, select the **Total_Sales_Category.csv**. Click **OK**.



56. From the properties pane, click **Field Mapping**.

57. Map the fields, as shown in the following table:

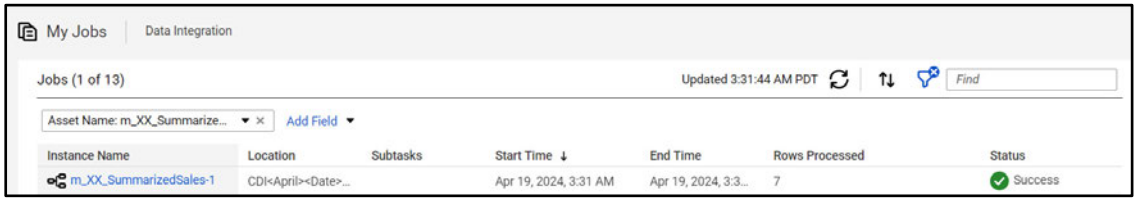
Incoming Fields	Target Fields
Category	Category
Total	Total



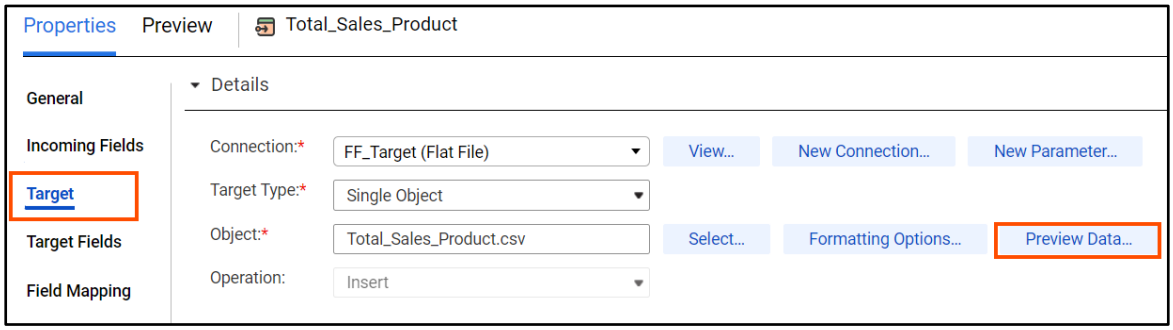
58. **Save** and **Run** the mapping.
59. From the Runtime Environment drop down, select **INFA-SERVER**.
60. Click **Run**.

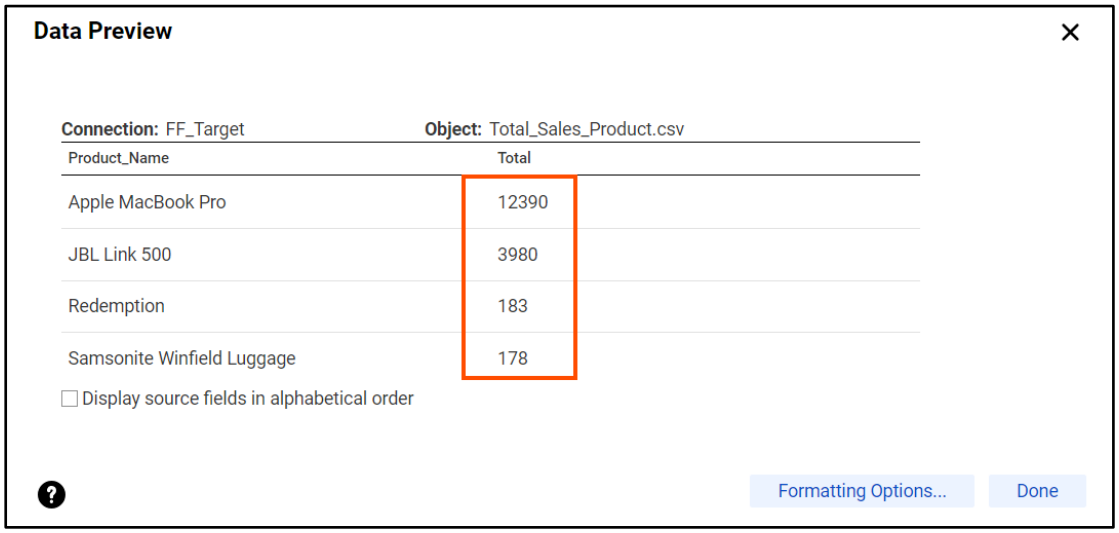
Monitor Status

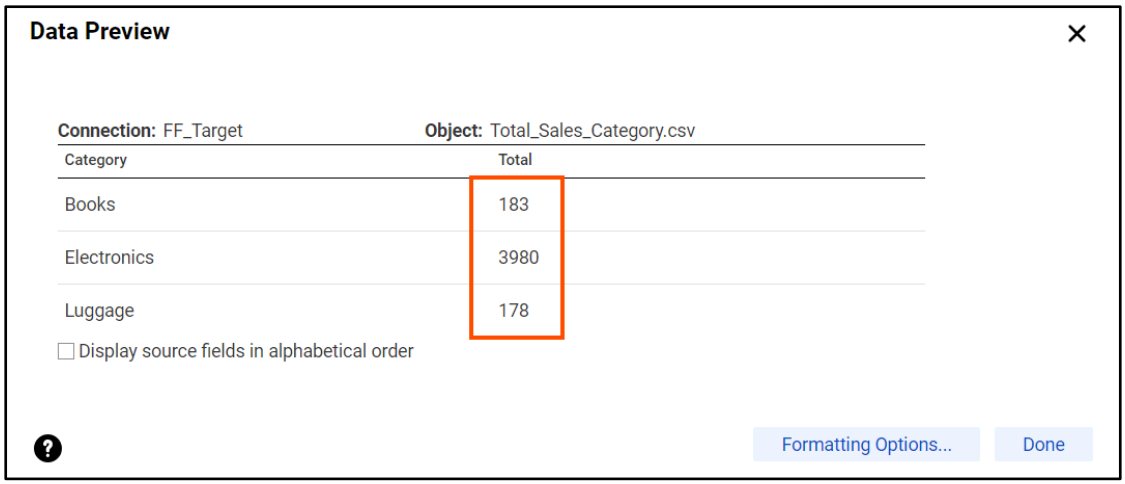
61. Monitor the task status from the **My Jobs page**.
62. When the task completes, the status changes to **Success**.



63. Optionally, preview the data on the targets.







64. Close the asset from the navigation pane.

This concludes the lab.

Module 9: Mapping Parameterization

Lab 9-1: Performing Complete Parameterization

Overview:
Parameters are placeholders that represent values in a mapping. In IICS, you can make a mapping reusable with the help of parameters.

Objective:

- Create a completely parameterized mapping

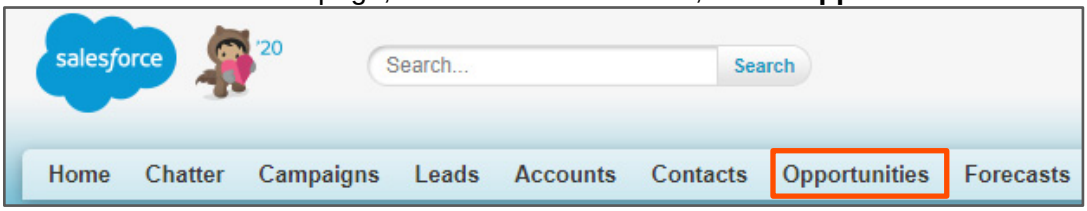
Scenario:
Ruby wants to filter the Salesforce opportunities based on probability. However, creating a mapping every time an opportunity is added is time consuming. When she explained this concern to John, he recommends creating a completely parameterized mapping.

Duration:
30 minutes

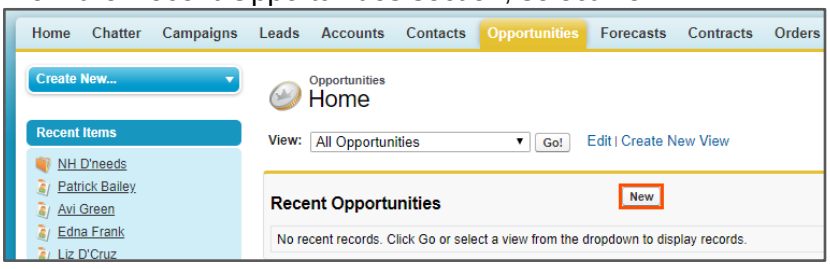
Tasks

Create Opportunities in Salesforce

1. Log in to your Salesforce Developer account using your credentials.
   Note: You can use the below mentioned link to log in:
   https://login.salesforce.com/
2. On the Salesforce homepage, from the available tabs, select **Opportunities**.



3. From the Recent Opportunities section, select **New**.



4. Enter the details, as shown in the table below:

Field Name	Values
Opportunity Name	Team Training
Account Name	NH Digital
Amount	700
Close Date	Choose a date greater than the current date
Stage	Prospecting
Probability	10

5. Click **Save & New**.

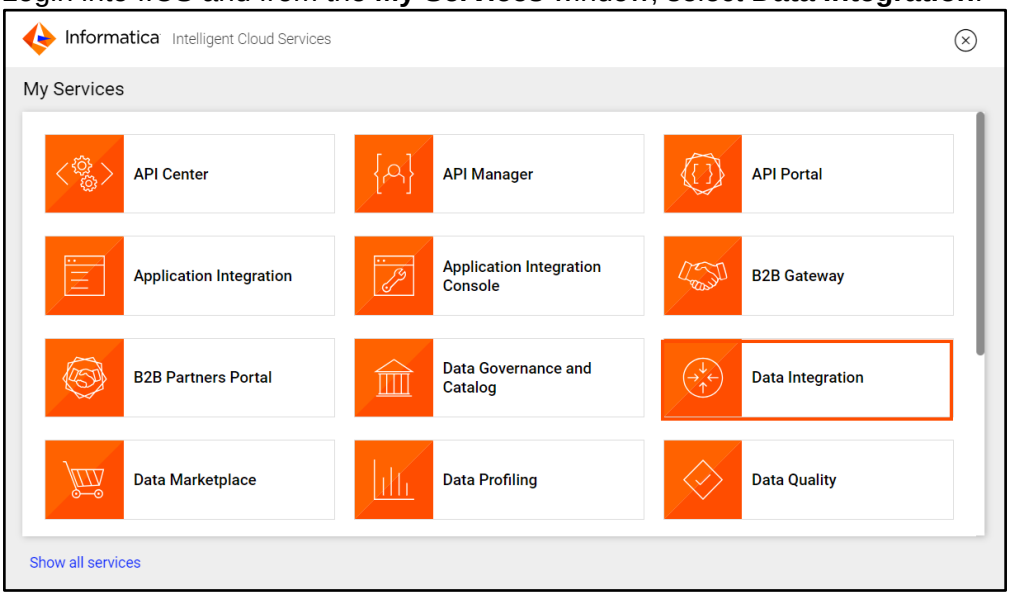
6. Similarly, create three or more opportunities in Salesforce. You can refer to the image below to create opportunities:

	A	B	C	D	E	F
1	Opportunity Name	Account Name	Amount	Close Date	Probability	Stage
2	Stock Supply	NH Trends	1000000	Choose a date greater than the current date	75	Proposal/Price Quote
3	Service Contract Renewal	NH Everyday	95000	Choose a date greater than the current date	60	Id. Decision Makers
4	Annual Maintenance	NH Mart	50000	Choose a date greater than the current date	90	Negotiation/Review

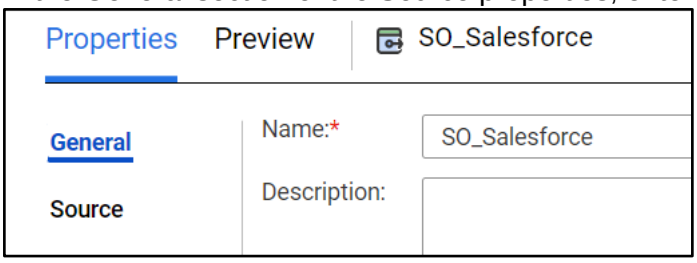
7. After the last opportunity is added, click **Save**.

Create Mapping

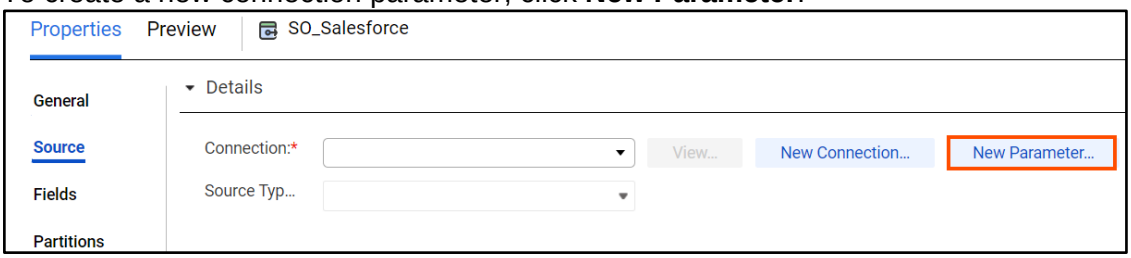
8. Login into IICS and from the **My Services** window, select **Data Integration**.



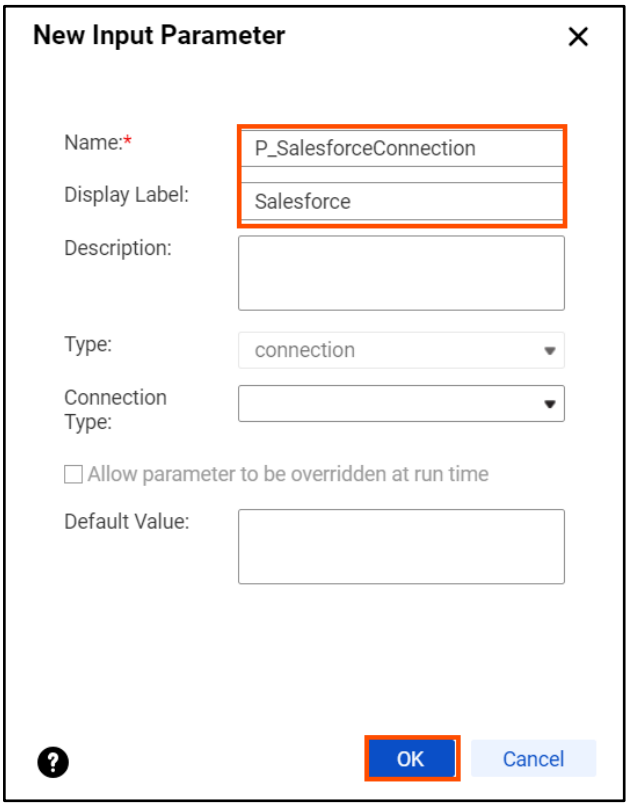
9. Create a new Mapping and name it **m_XX_MappingParameters**. Replace XX with your student number.
10. Verify the asset location.
11. To configure the source, from the mapping canvas, click the **Source** transformation.
12. In the General section of the Source properties, enter Name as **SO_Salesforce**.



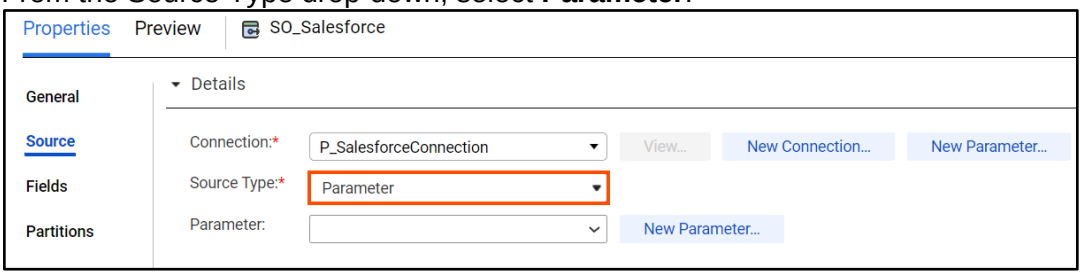
13. From the properties pane, click **Source**.
14. To create a new connection parameter, click **New Parameter**.



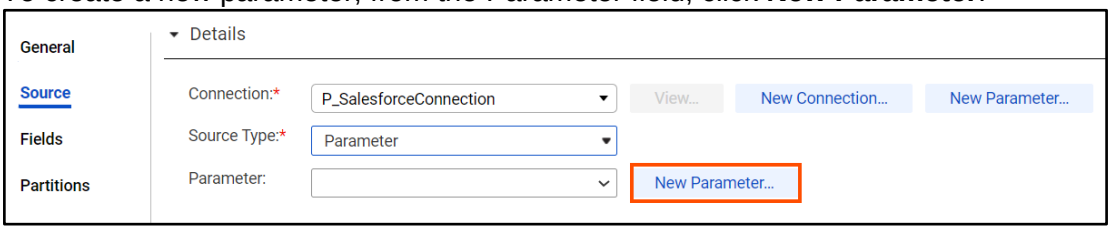
15. Enter Name as **P_SalesforceConnection**, and Display Label as **Salesforce**.
16. Click **OK**.



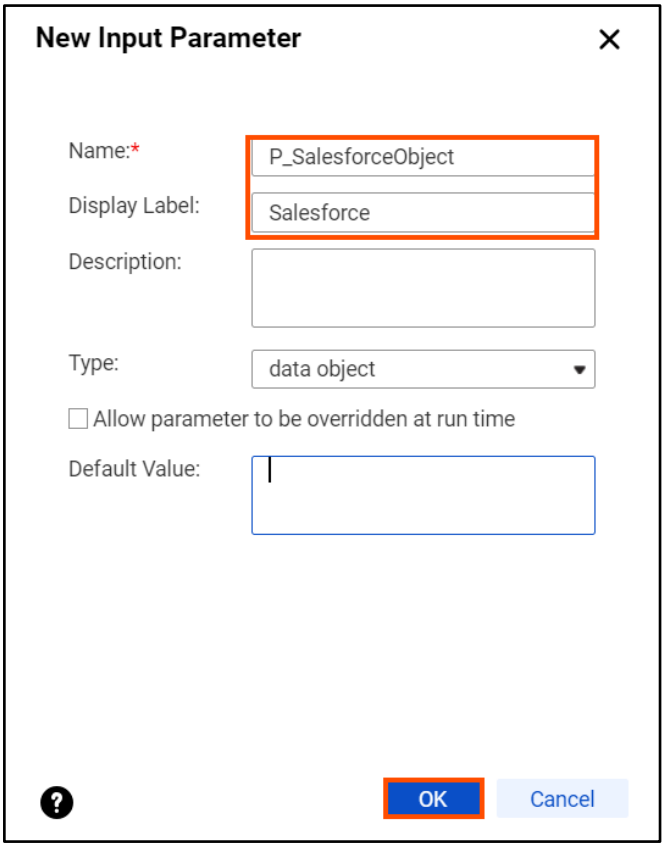
17. From the Source Type drop-down, select **Parameter**.



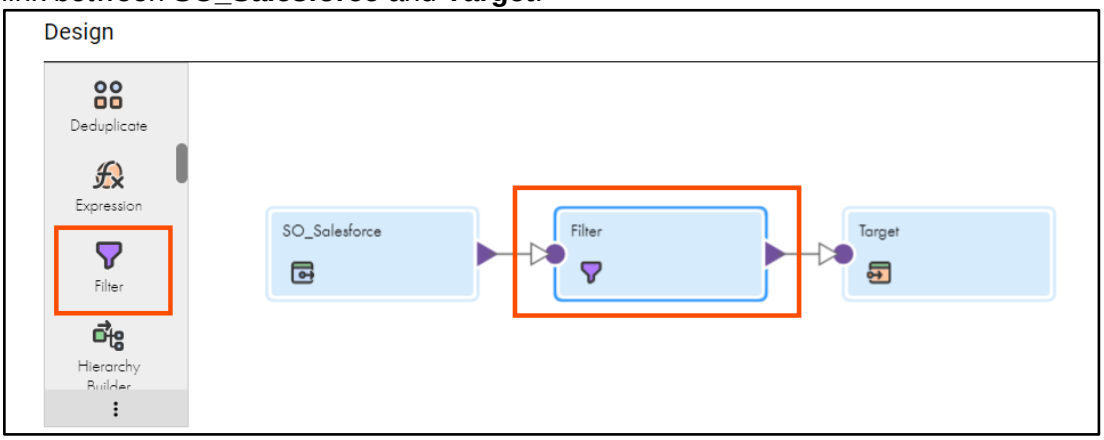
18. To create a new parameter, from the Parameter field, click **New Parameter**.



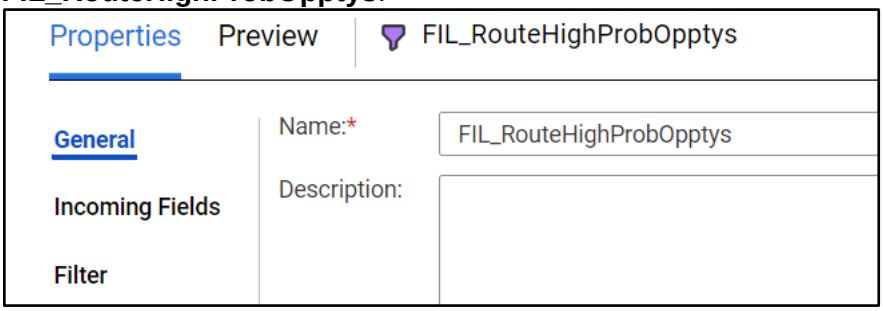
19. Enter Name as **P_SalesforceObject**, and the Display Label as **Salesforce**.
20. Click **OK**.



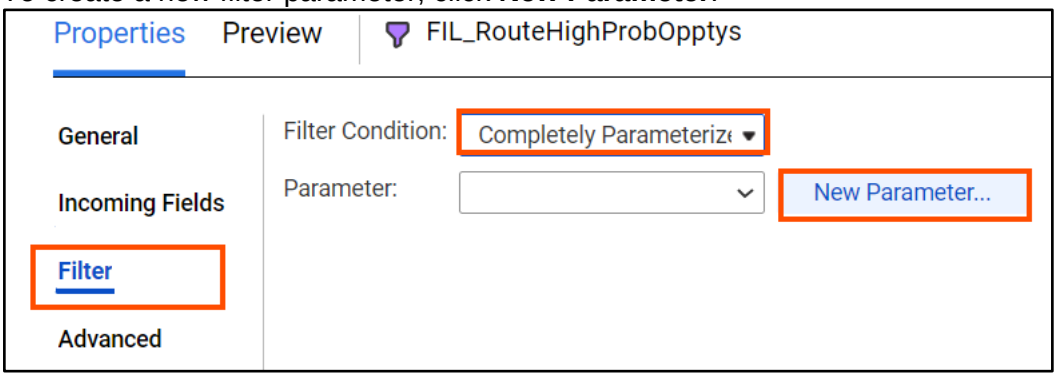
21. From the list of available transformations, drag and drop a **Filter** transformation onto the link between **SO_Salesforce** and **Target**.



22. In the General section of the Filter properties, enter the Name as **FIL_RouteHighProbOpptys**.

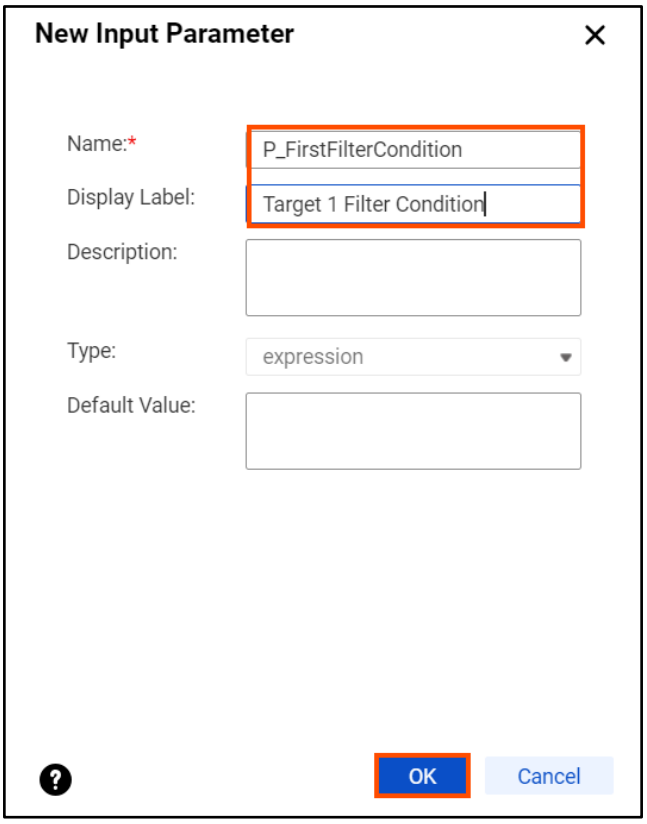


23. From the properties pane, click **Filter**.
24. From the Filter Condition drop-down, select **Completely Parameterized**.
25. To create a new filter parameter, click **New Parameter**.

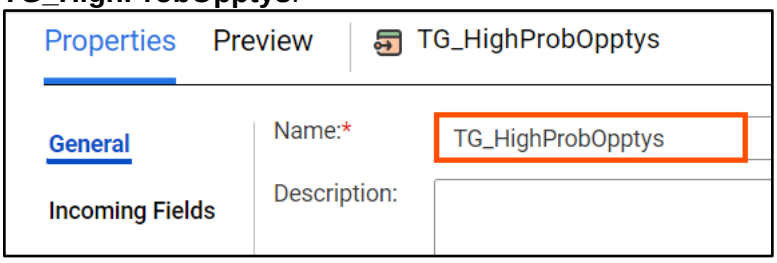


26. Enter Name as **P_FirstFilterCondition**, and Display Label as **Target 1 Filter Condition**.

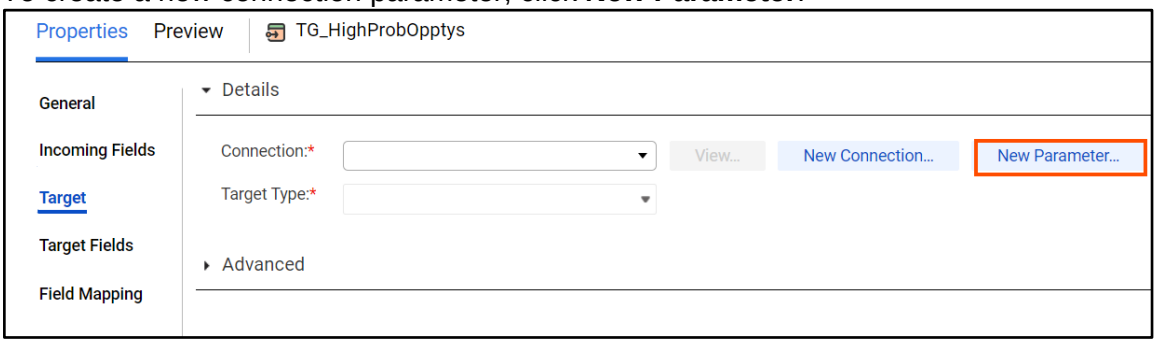
27. Click **OK**.



28. To configure the target, from the mapping canvas, click the **Target** transformation.
29. In the General section of the Target properties, enter the Name as **TG_HighProbOpptys**.

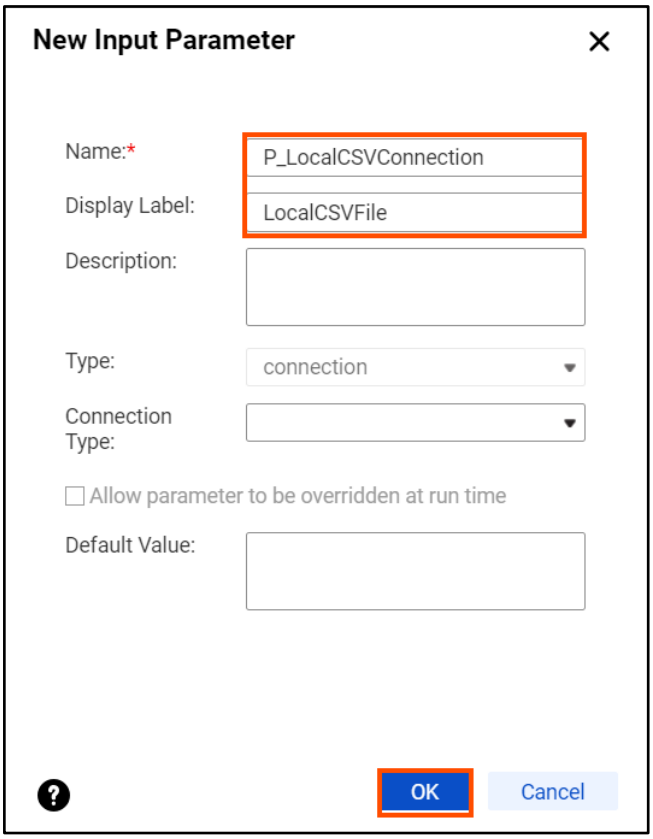


30. From the properties pane, click **Target**.
31. To create a new connection parameter, click **New Parameter**.

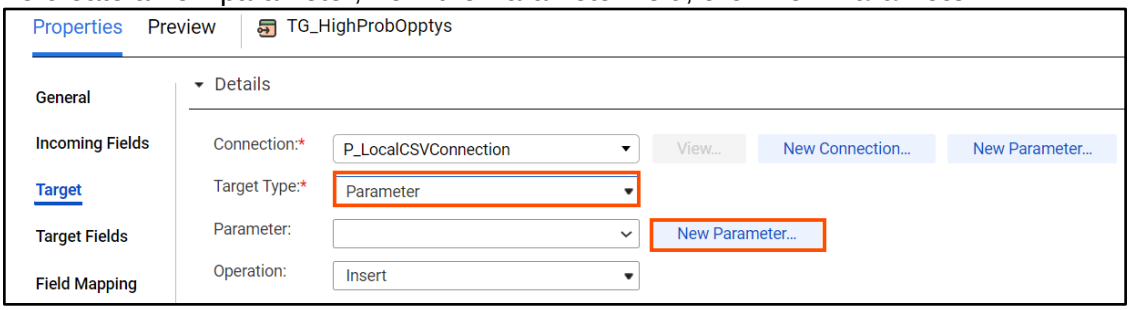


32. Enter Name as **P_LocalCSVConnection**, and Display Label as **LocalCSVFile**.

33. Click **OK**.

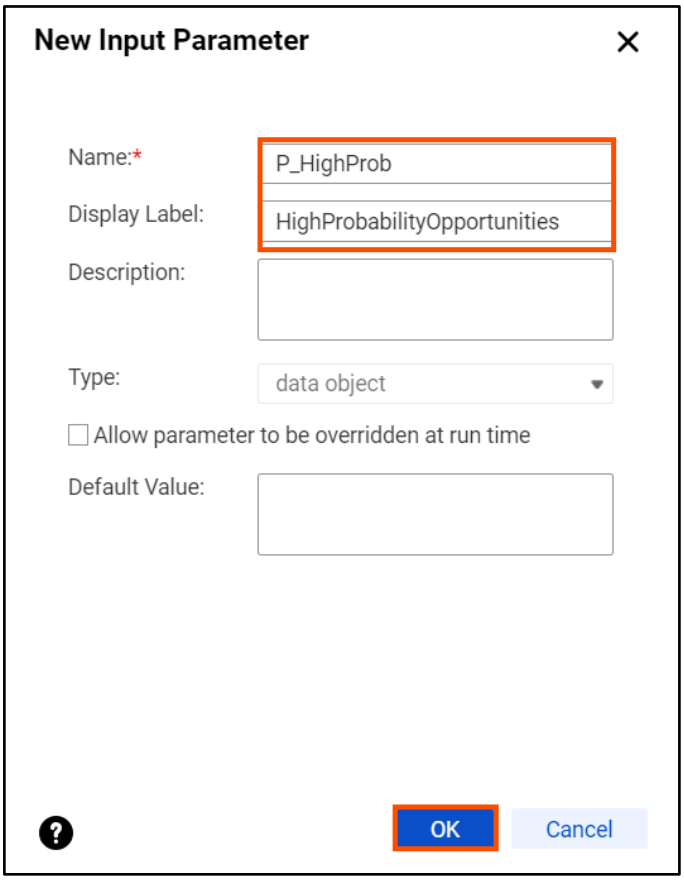


34. From the Target Type drop-down, select **Parameter**.
35. To create a new parameter, from the Parameter field, click **New Parameter**.

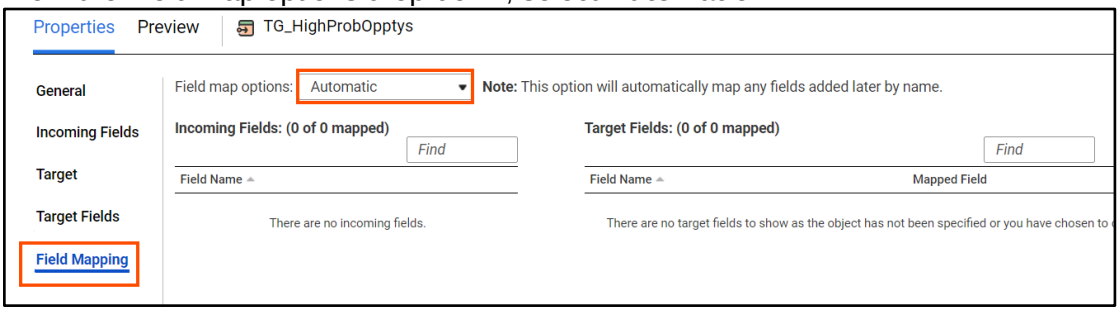


36. Enter Name as **P_HighProb**, and the Display Label as **HighProbabilityOpportunities**.

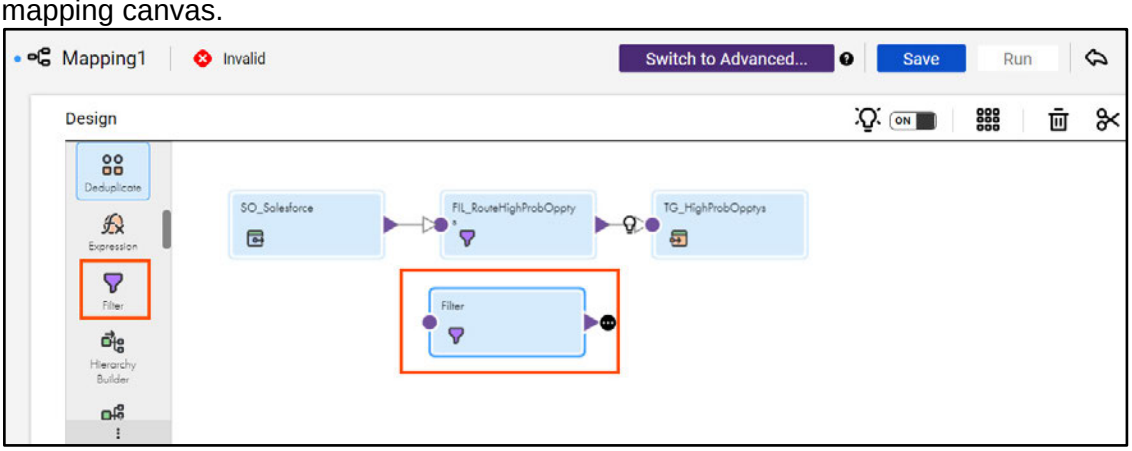
37. Click **OK**.



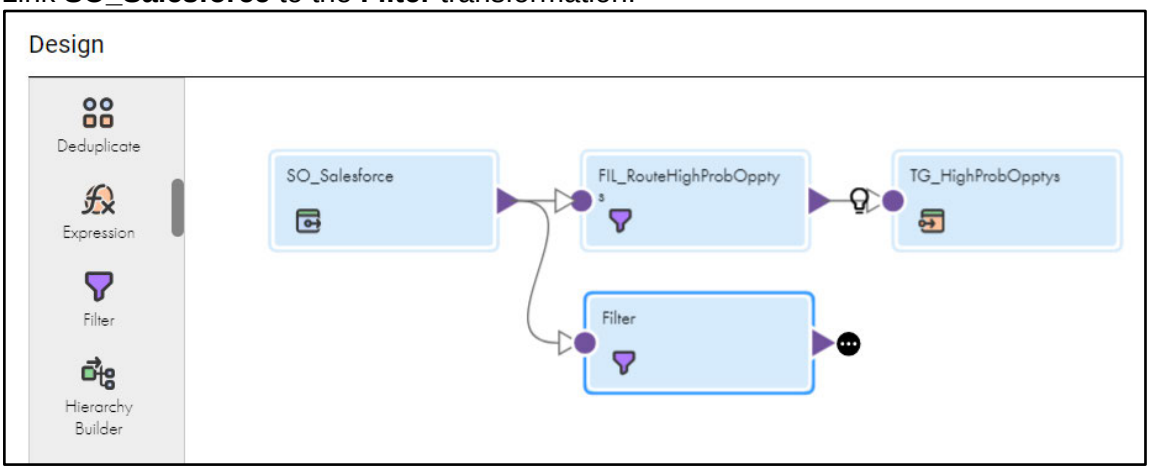
38. From the properties pane, click **Field Mapping**.
39. From the Field map options drop-down, select **Automatic**.



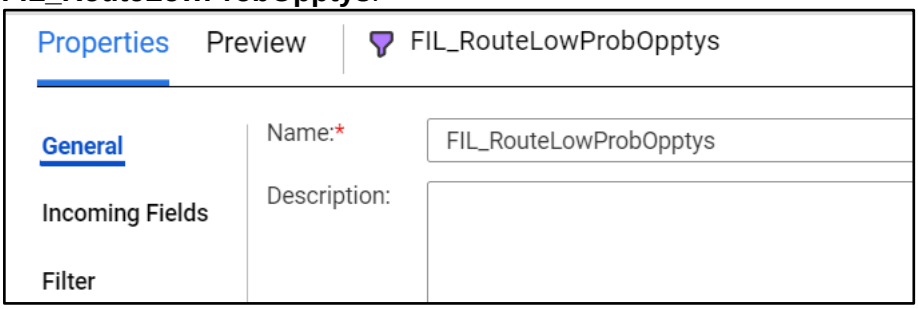
40. From the list of available transformations, drag and drop a **Filter** transformation onto the mapping canvas.



41. Link **SO_Salesforce** to the **Filter** transformation.

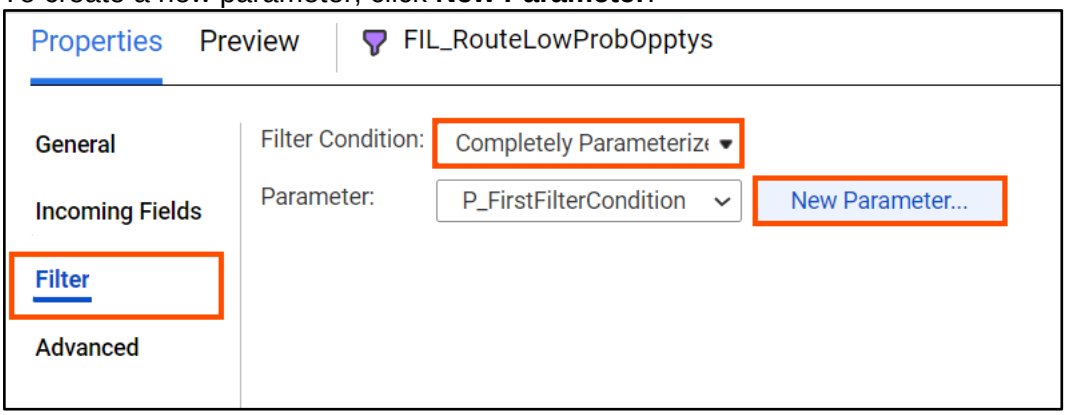


42. Select the **Filter** transformation from the mapping canvas.
43. In the General section of the Filter properties, enter the Name as **FIL_RouteLowProbOpptys**.

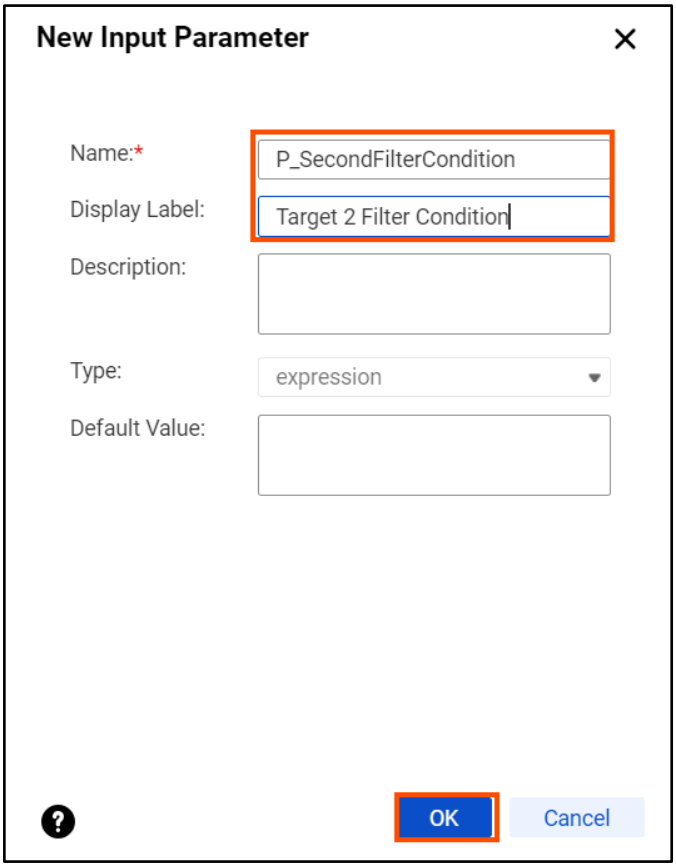


44. From the properties pane, click **Filter**.
45. From the Filter Condition drop-down, select **Completely Parameterized**.

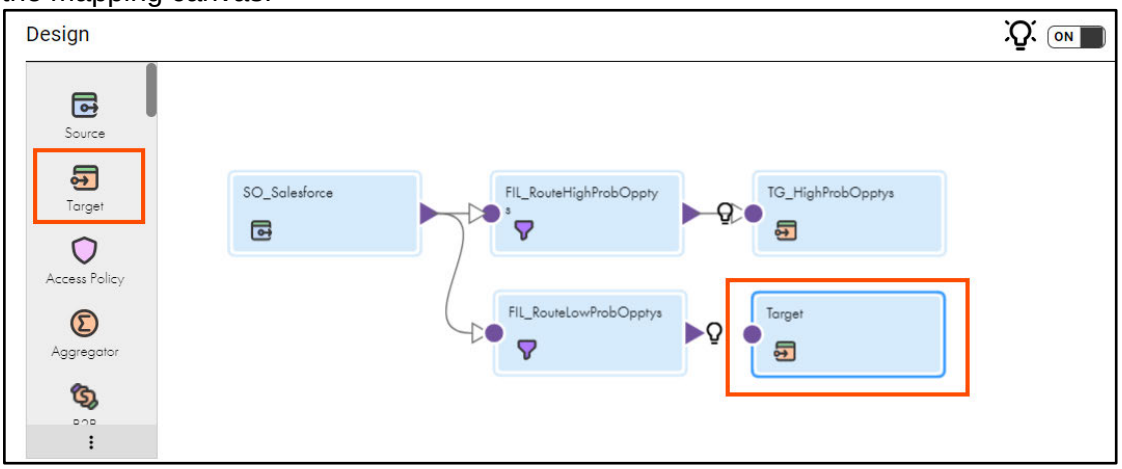
46. To create a new parameter, click **New Parameter**.



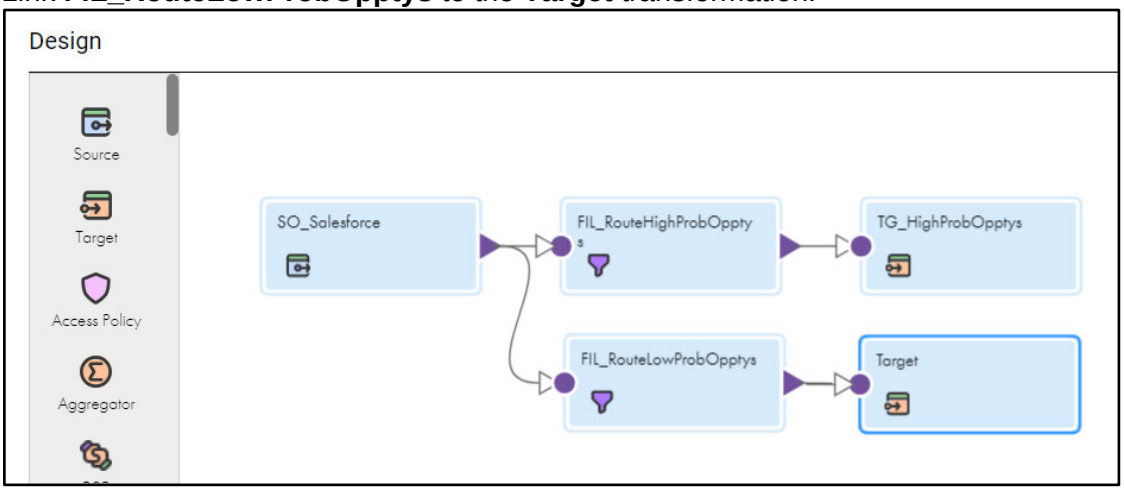
47. Enter Name as **P_SecondFilterCondition**, and Display Label as **Target 2 Filter Condition**.
48. Click **OK**.



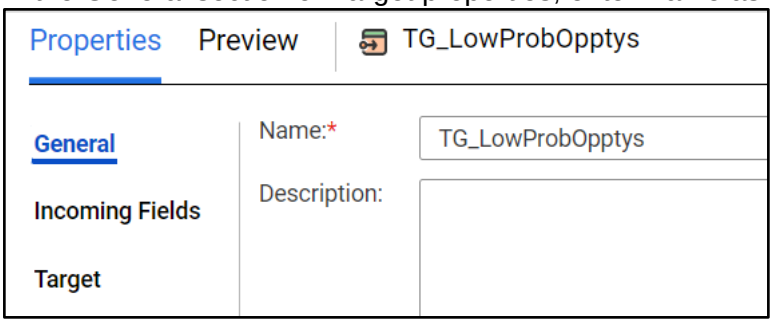
49. From the list of available transformations, drag and drop a **Target** transformation onto the mapping canvas.



50. Link **FIL_RouteLowProbOpptys** to the **Target** transformation.

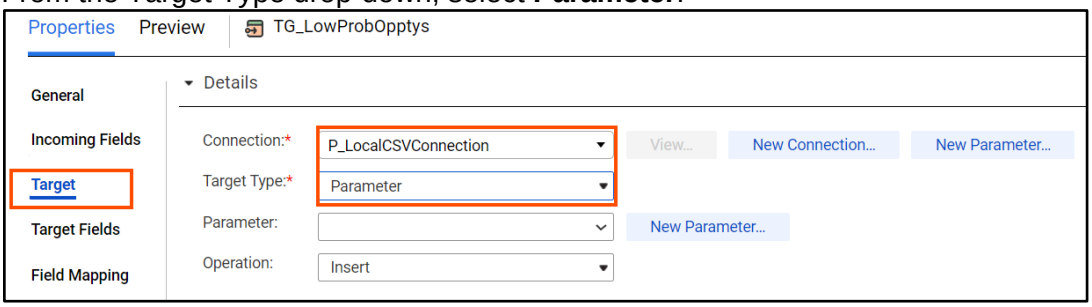


51. Select the **Target** transformation from the mapping canvas.
52. In the General section of Target properties, enter Name as **TG_LowProbOpptys**.

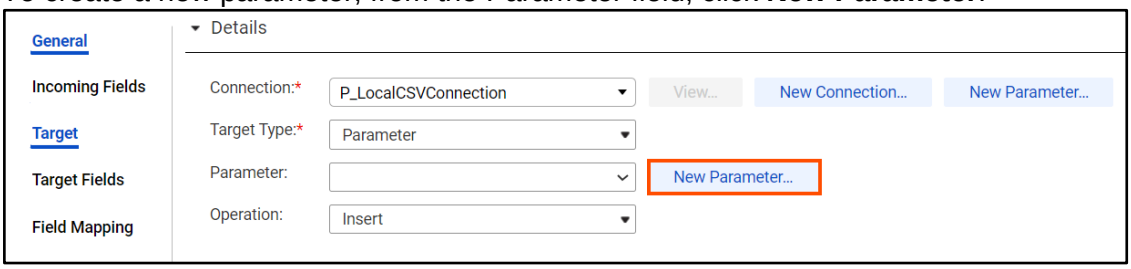


53. From the properties pane, click **Target**.
54. From the Connection drop-down, select **P_LocalCSVConnection**.

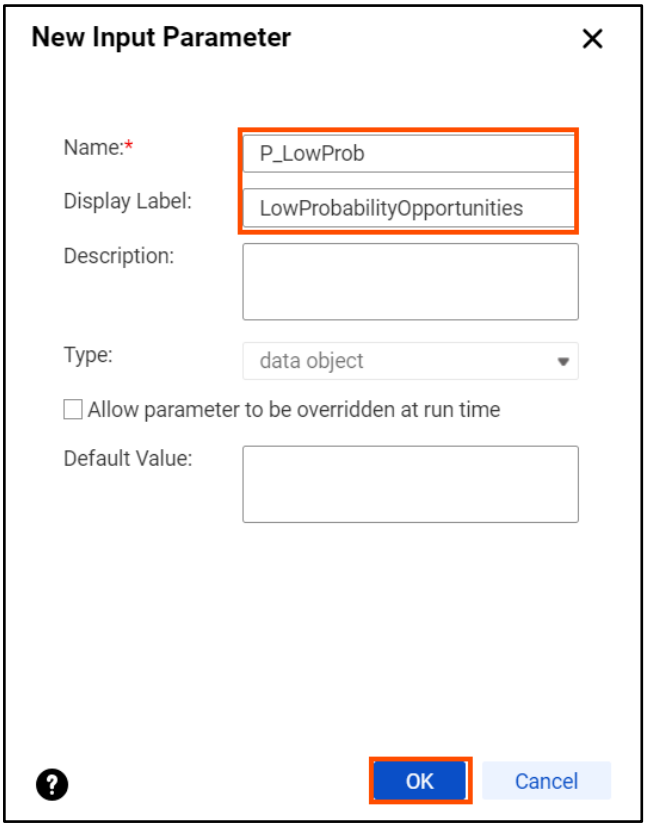
55. From the Target Type drop-down, select **Parameter**.



56. To create a new parameter, from the Parameter field, click **New Parameter**.

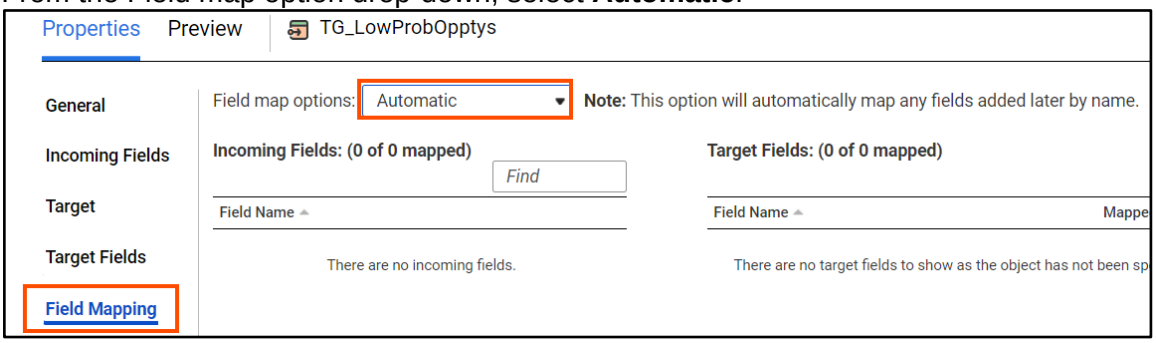


57. Enter Name as **P_LowProb**, and the Display Label as **LowProbabilityOpportunities**.
58. Click **OK**.

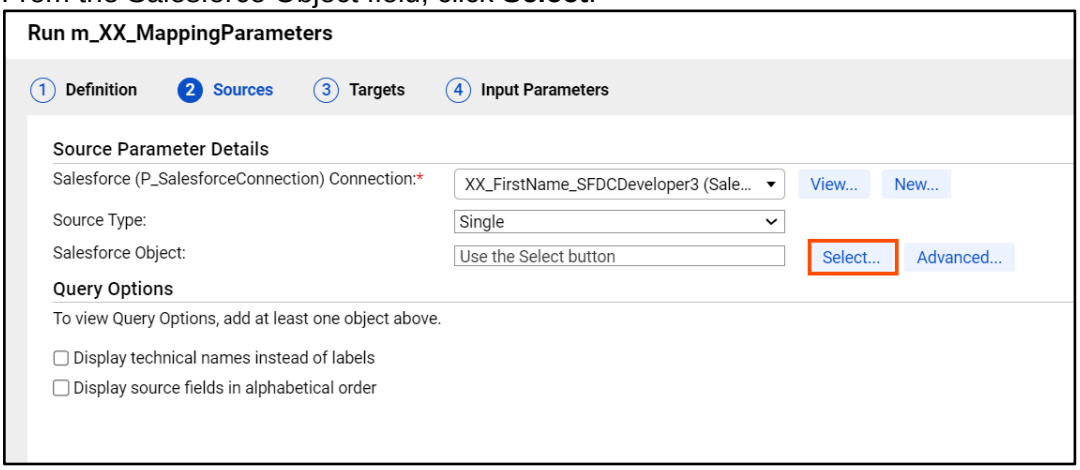


59. From the properties pane, click **Field Mapping**.

60. From the Field map option drop-down, select **Automatic**.

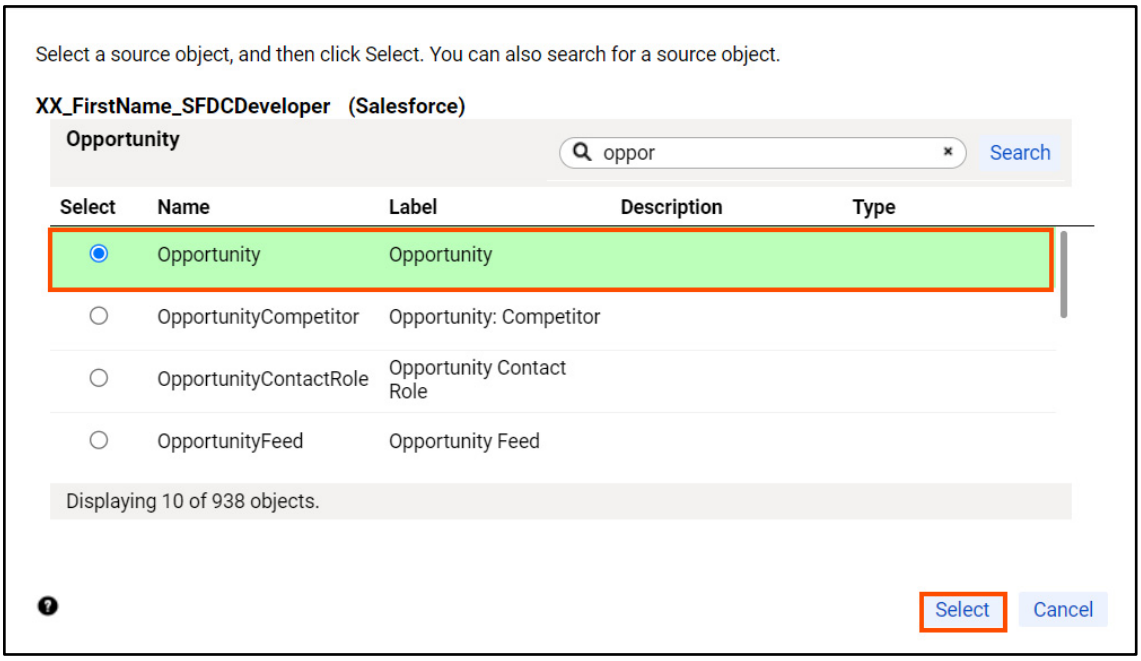


61. Save and run the mapping.
62. From the Runtime Environment drop-down, select INFA-SERVER.
63. Click **Next**.
64. From the Salesforce (P_SalesforceConnection) Connection drop-down, select your **Salesforce** connection.
65. From the Salesforce Object field, click **Select**.

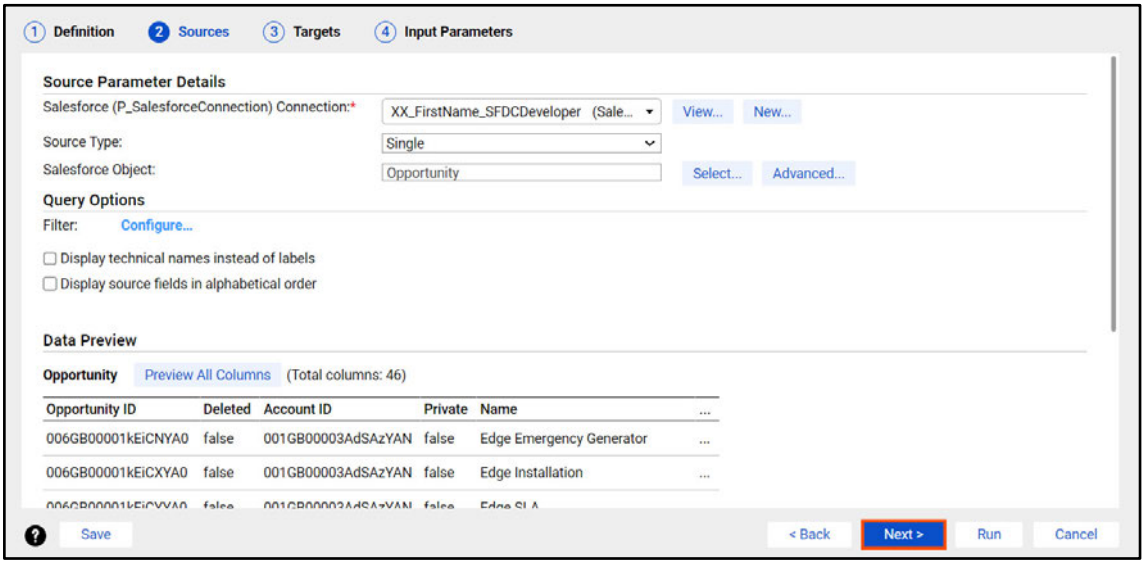


66. From the list, select **Opportunity**.
Note: You can also use the search feature.

67. Click **Select**.

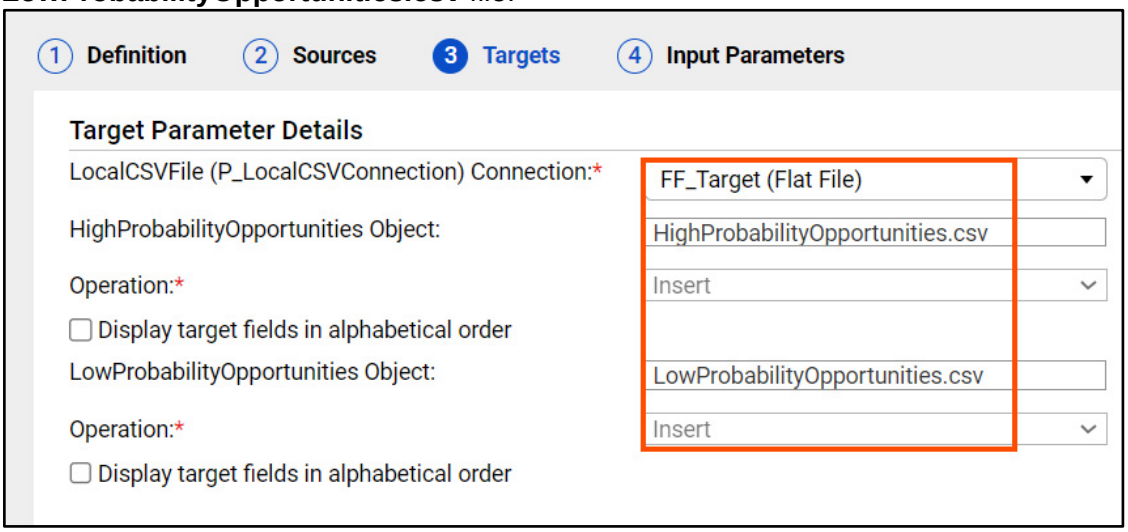


68. Click **Next**.

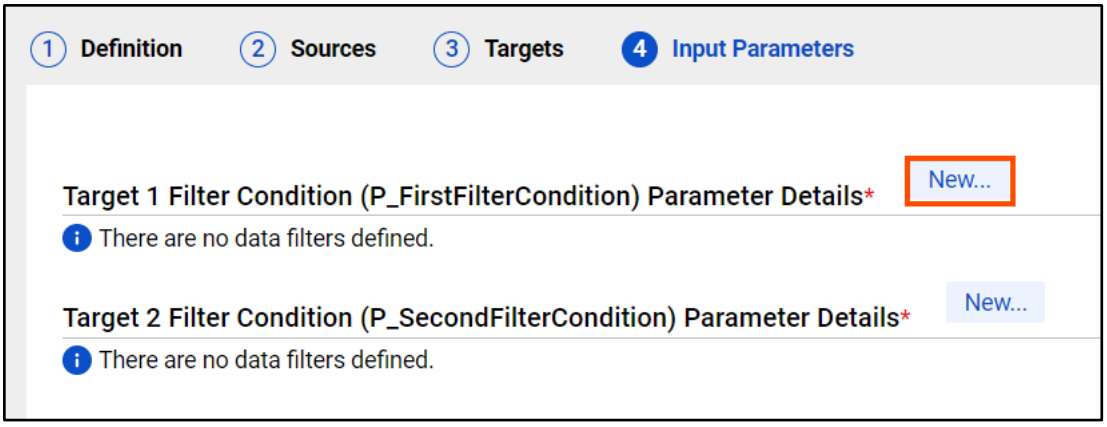


69. From the LocalCSVFile (P_LocalCSVConnection) Connection drop-down, select your **target Flat File** connection.
70. To select the HighProbabilityOpportunities Object, click **Select** and choose the **HighProbabilityOpportunities.csv** file.

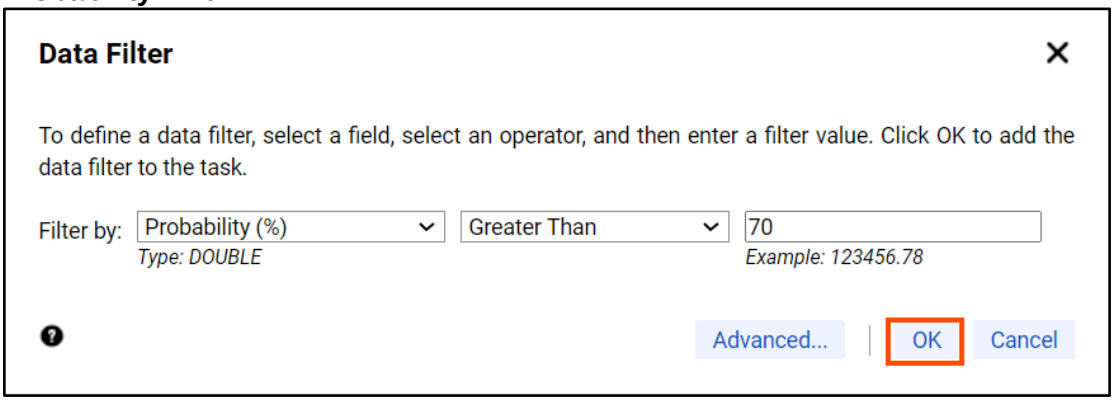
71. From the LowProbabilityOpportunities Object field, click **Select** and choose the **LowProbabilityOpportunities.csv** file.



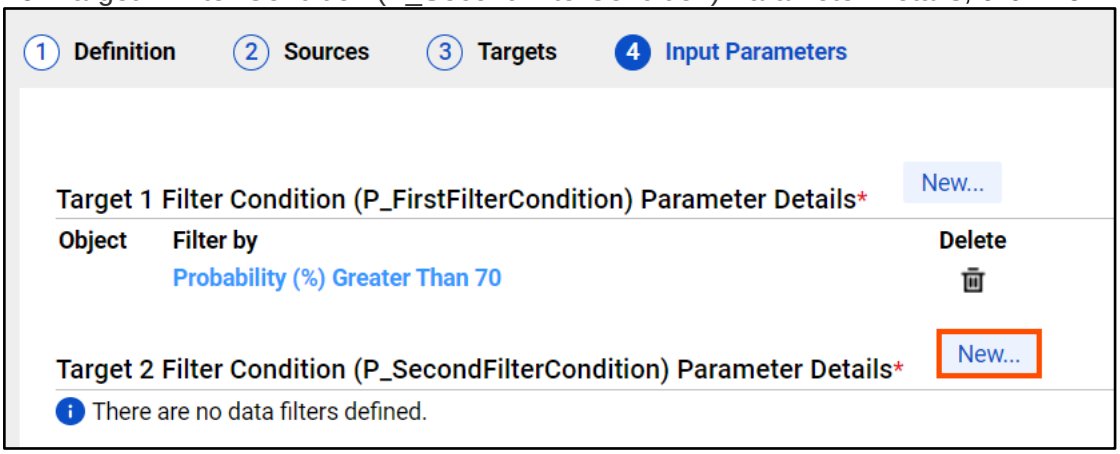
72. Click **Next**.
73. To define Target 1 Filter Condition (P_FirstFilterCondition) Parameter Details, click **New**.



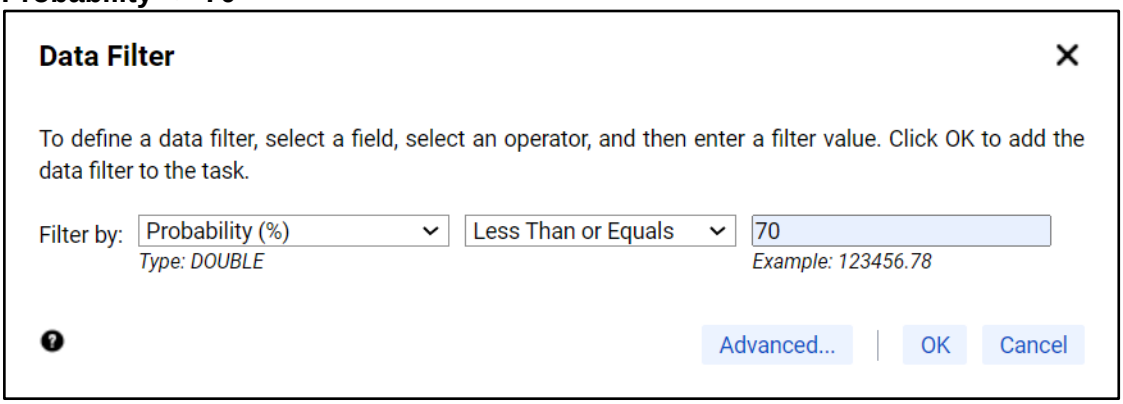
74. Enter the filter condition as shown below and click **OK**.
Probability > 70



75. For Target 2 Filter Condition (P_SecondFilterCondition) Parameter Details, click **New**.



76. Enter the filter condition as shown below and click **OK**.
Probability <= 70



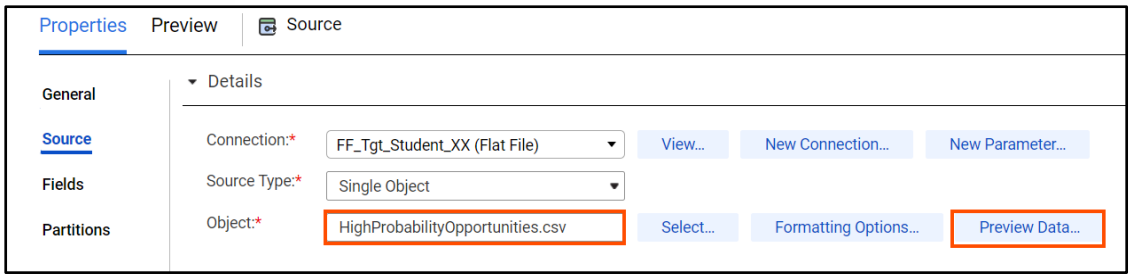
77. Click **Run**.

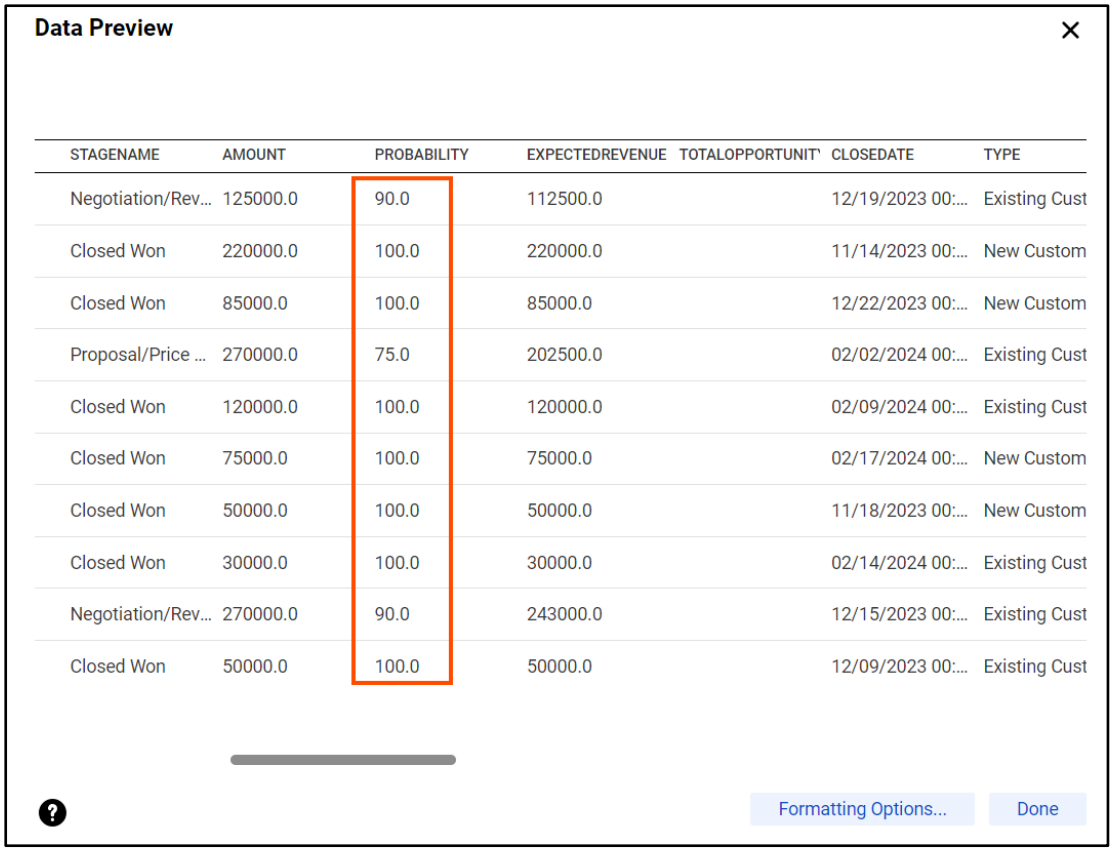
Monitor Status

78. To monitor the task status, navigate to the **My Jobs** page.
79. The task status changes to **Success**.
Note: You can refresh the job status if it does not change automatically.
80. Close the asset from the navigation pane.

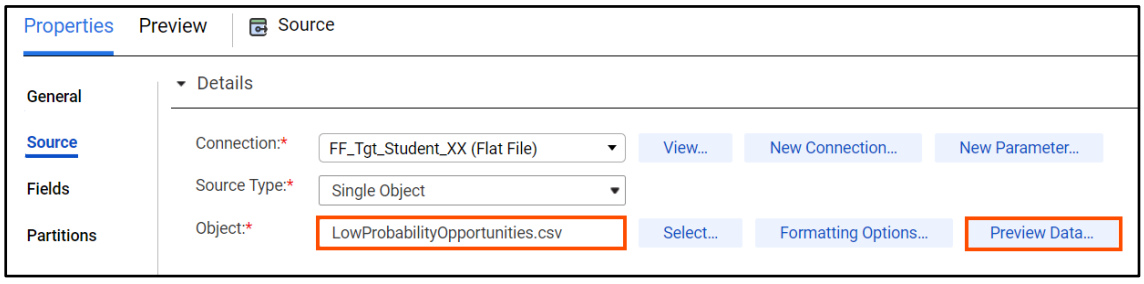
Optional - Open the Dummy Mapping and View the Data

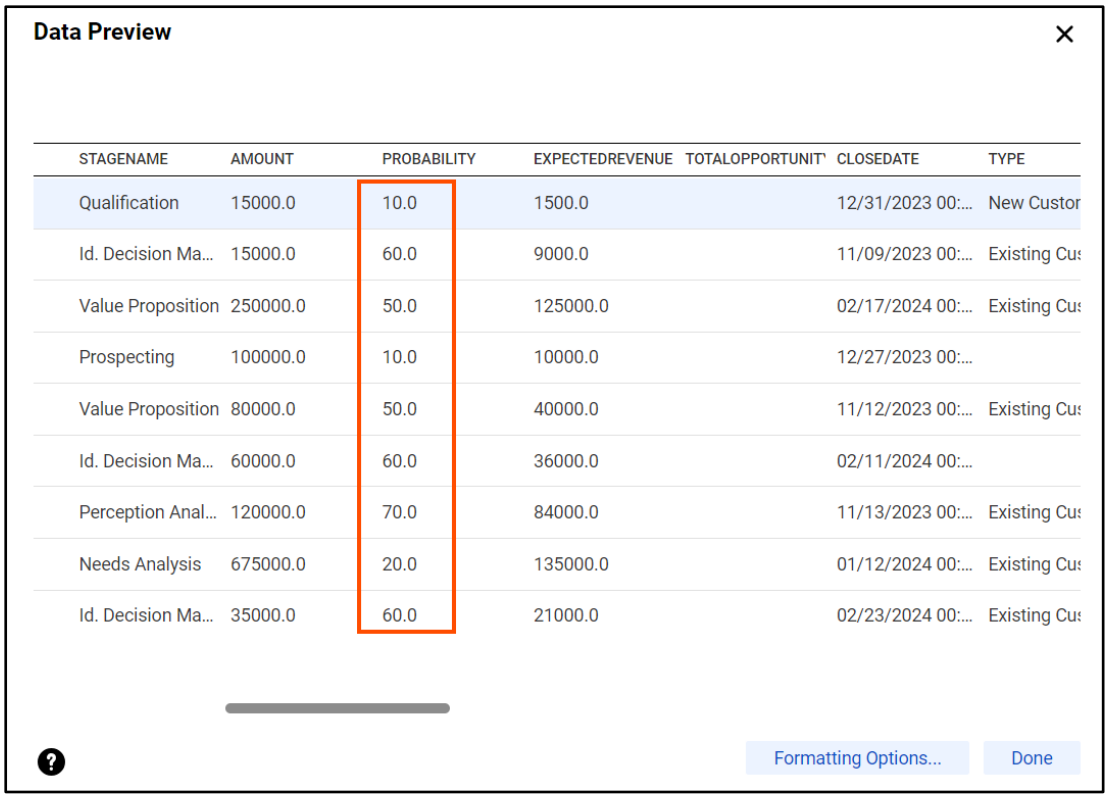
81. Navigate to your working folder and open the dummy mapping **m_XX_DUMMY_MAPPING**.
82. From the mapping canvas, select **Source** and go to the **Source** properties tab.
83. Change the Connection to your target Flat File connection. Click **Yes** in the Change Connection window.
84. Select the Object as your target – **HighProbabilityOpportunities.csv**, and preview the data.





85. Similarly, observe the results of the target - **LowProbabilityOpportunities.csv**.





86. Save and close all the assets from the navigation pane.

This concludes the lab.

Module 9: Mapping Parameterization

Lab 9-2: Using In-Out Parameters for Incremental Data Loading

Overview:

In IICS, you can use an in-out parameter as a persistent task variable to manage an incremental data load.

In this lab, you will create a mapping with in-out parameter.

Objective:

- Use Input-Output parameters in a mapping

Scenario:

The Alaska outlet of NH suppliers is facing issues in maintaining sales data and wants to archive the older data. They also want to maintain the data based on the order Ids. So, they contact Ruby for help. Ruby discusses the issue with John, who suggests using the In-Out parameters in IICS for incremental data load.

In this lab, John uses the in-out parameters to define an orderid parameter in the mapping.

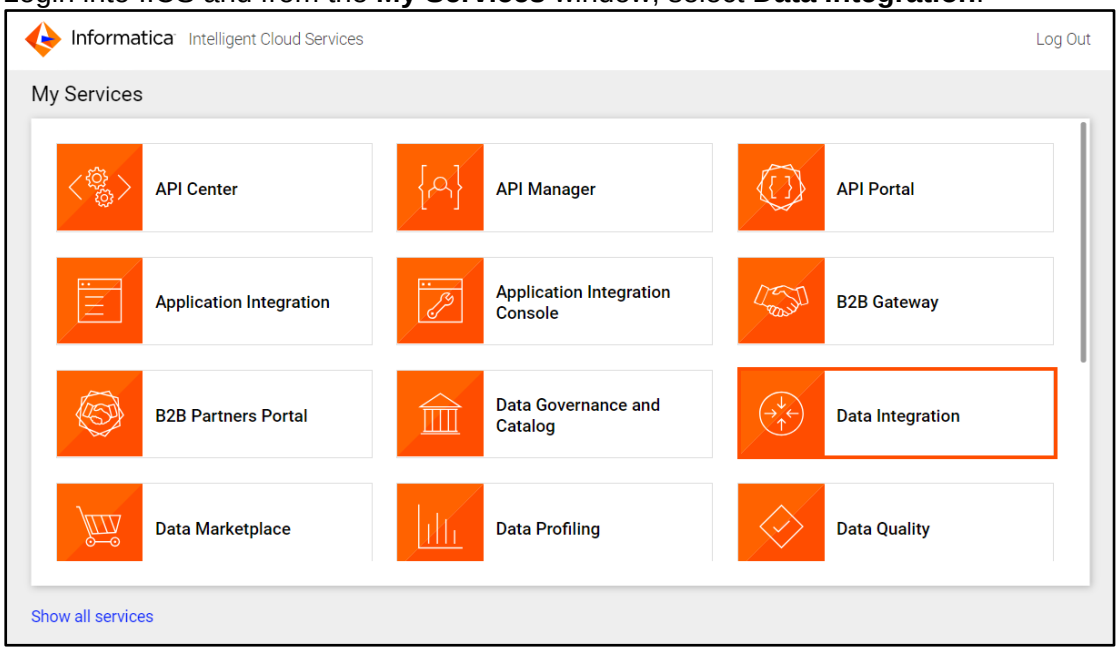
Duration:

15 minutes

Tasks

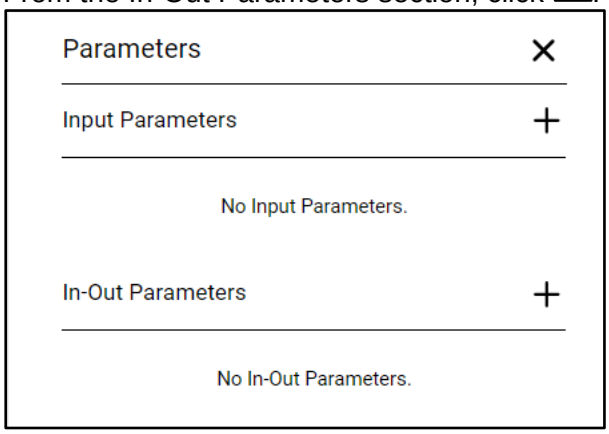
Create a Mapping

1. Login into IICS and from the **My Services** window, select **Data Integration**.



2. Create a new Mapping and name it **m_XX_InOutParameter**. Replace XX with your DOB.
3. To create an input-output parameter, click .

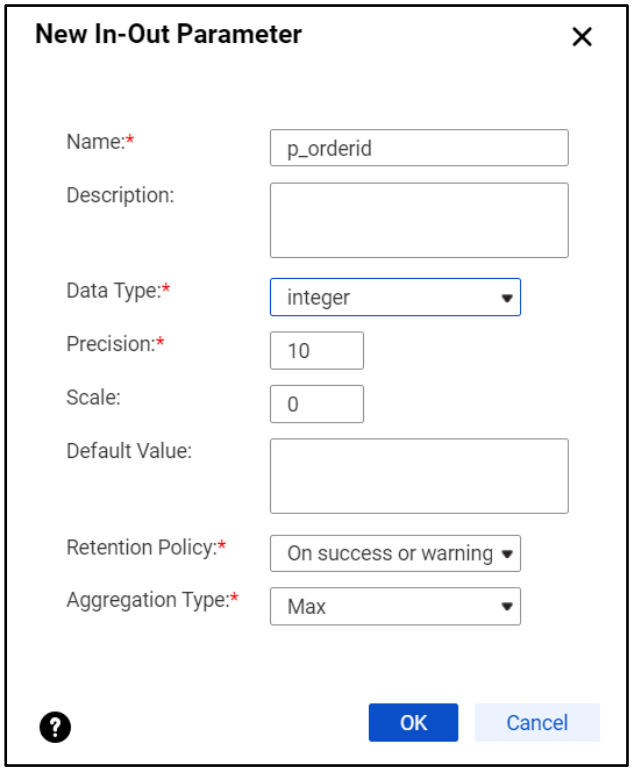
4. From the In-Out Parameters section, click .



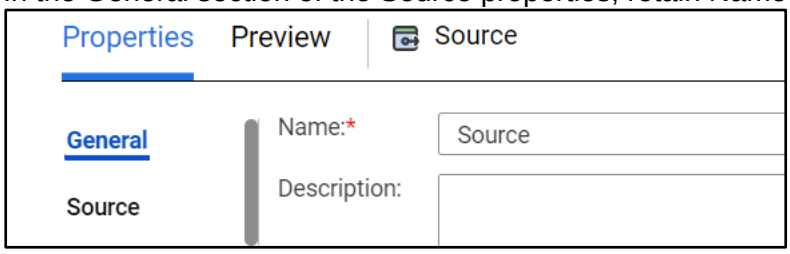
5. Create the field as shown in the table below:

Name	Data Type	Precision	Default Value
p_orderid	integer	10	0

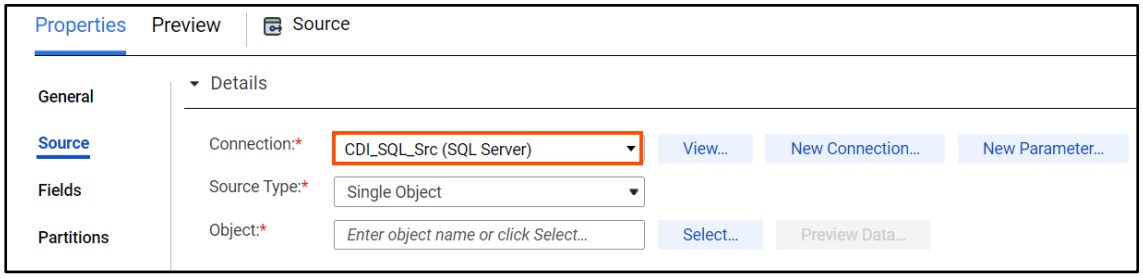
6. Click **OK**.



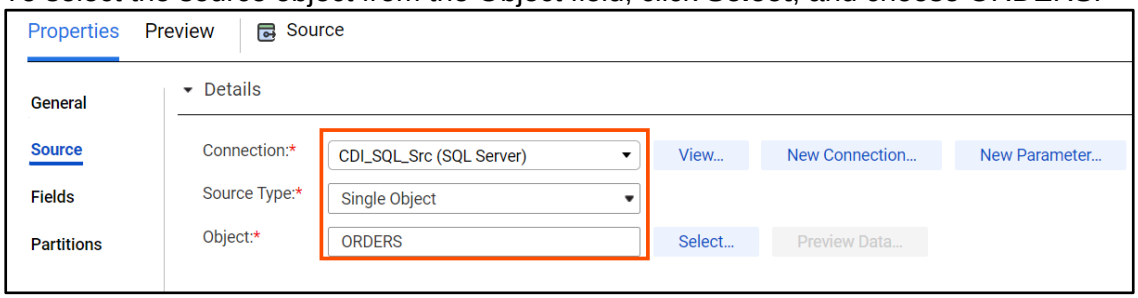
7. Close the Parameters window.
8. To configure the source, from the mapping canvas, click the **Source** transformation.
9. In the General section of the Source properties, retain Name as **Source**.



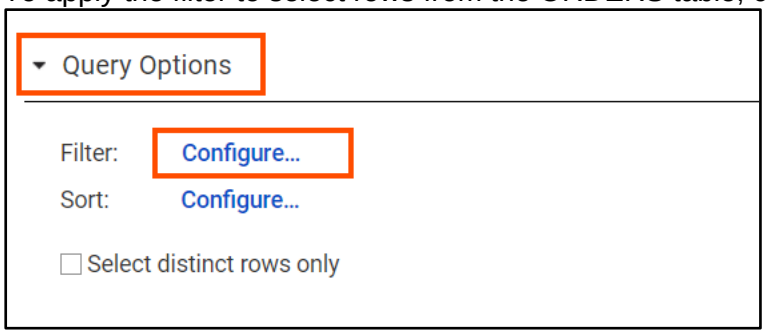
10. From the properties pane, click **Source**.
11. From the Connection drop-down, select **CDI_SQL_Src**.



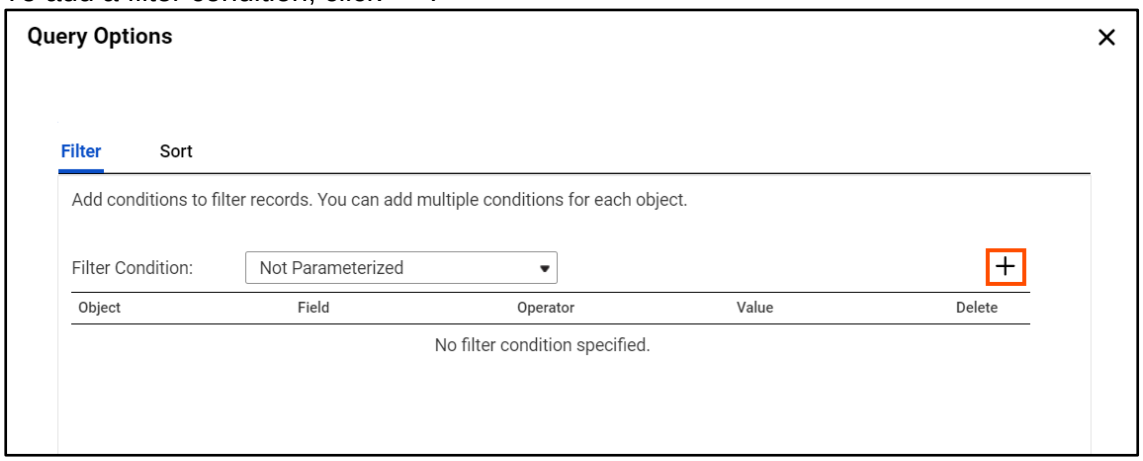
12. To select the source object from the Object field, click **Select**, and choose ORDERS.



13. Scroll down and expand **Query Options**.
14. To apply the filter to select rows from the ORDERS table, click **Configure**.

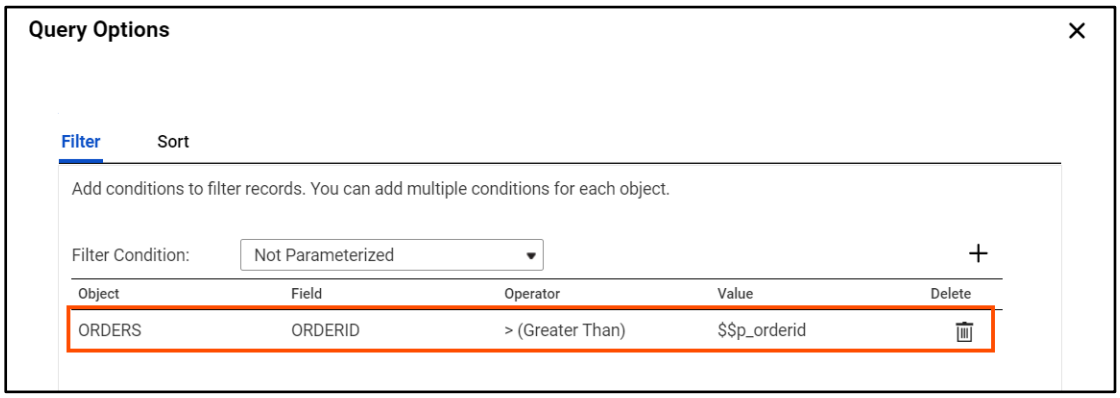


15. To add a filter condition, click ⊞.



16. Enter the details as shown in the table below:

Object	Field	Operator	Value
ORDERS	ORDERID	Greater Than (>)	$$p_orderid

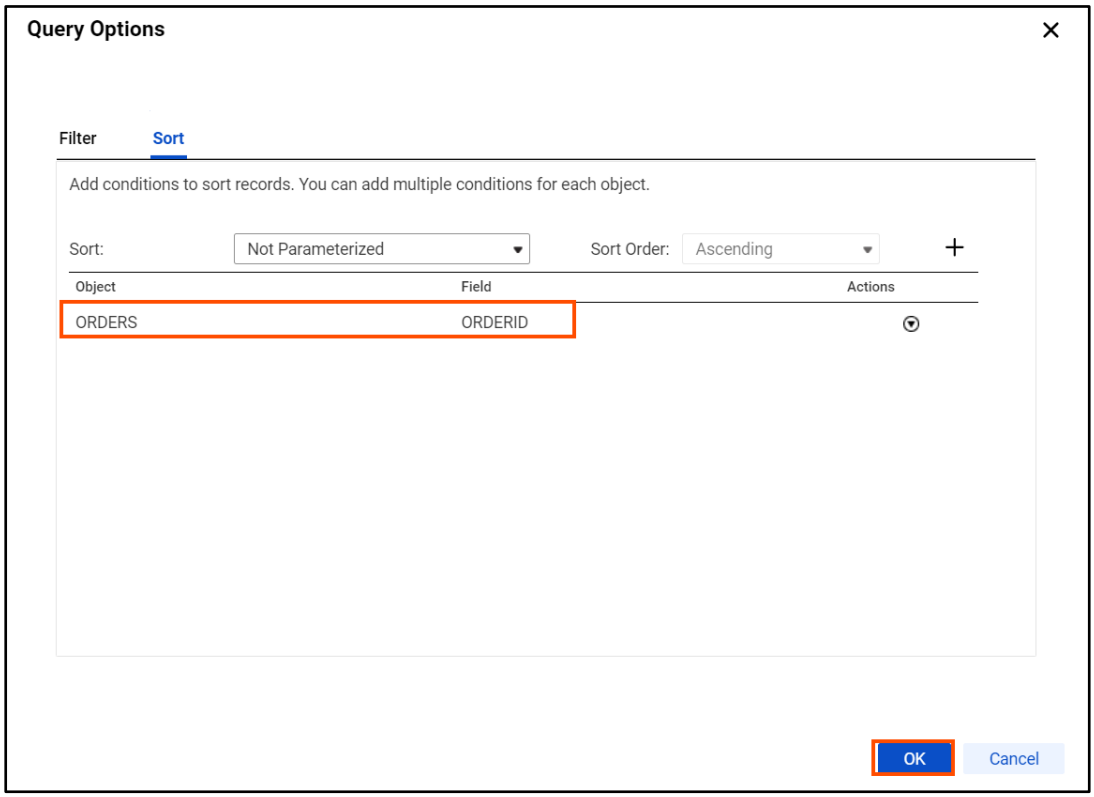


17. Click **OK**.
18. To configure the sort condition, click **Configure**.

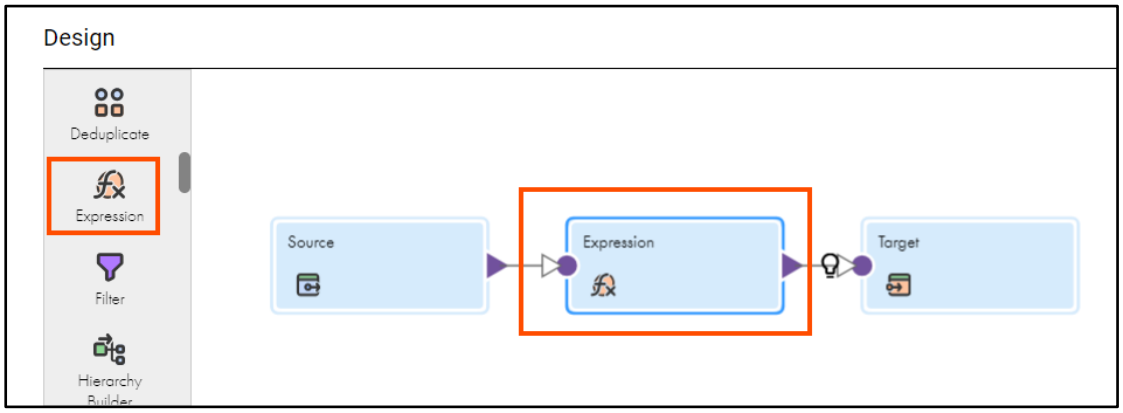
19. To add a sort condition, click ⊞.
20. Enter the details as shown in the table below:

Object	Field
ORDERS	ORDERID

21. Click **OK**.

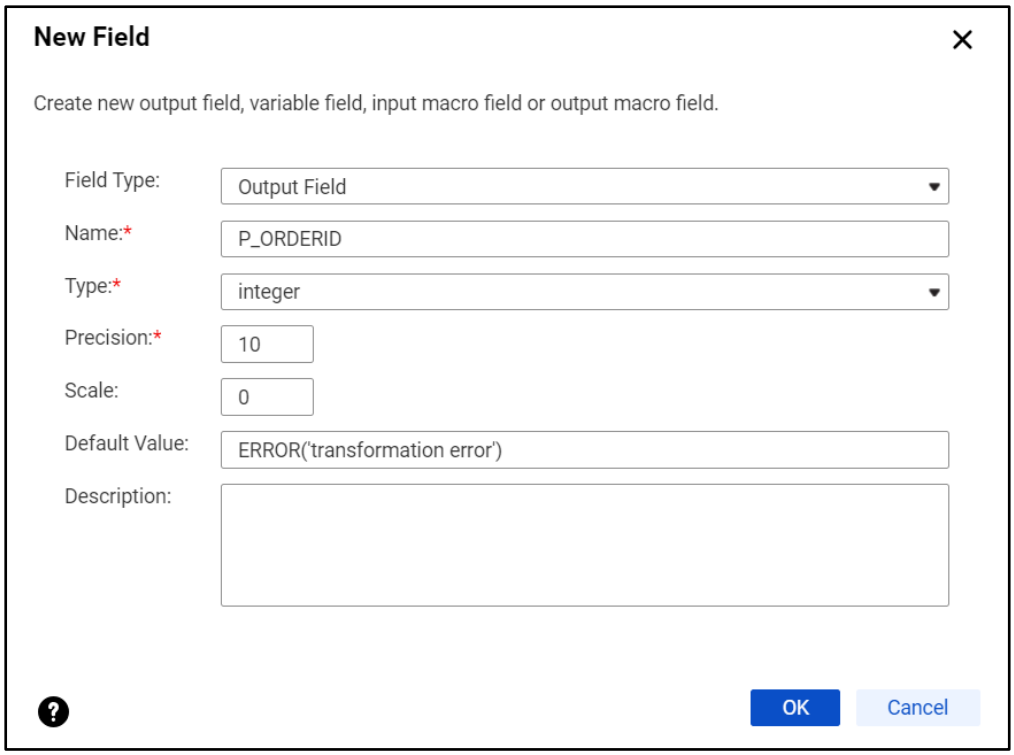


22. From the list of available transformations, drag and drop an **Expression** transformation on the link between **Source** and **Target**.

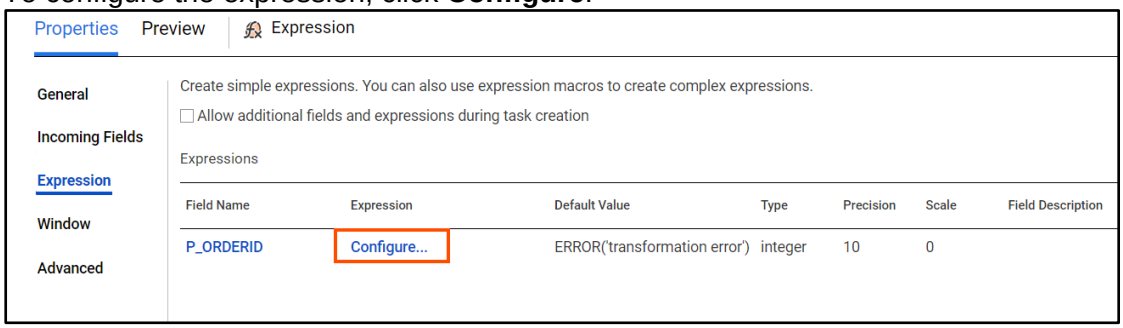


23. Select the **Expression** transformation from the mapping canvas.
24. From the properties pane, click **Expression**.
25. To add a new expression, click ⊞.
26. Enter the details as shown in the table below and click **OK**.

Field Type	Name	Type	Precision	Scale
Output Field	P_ORDERID	integer	10	0



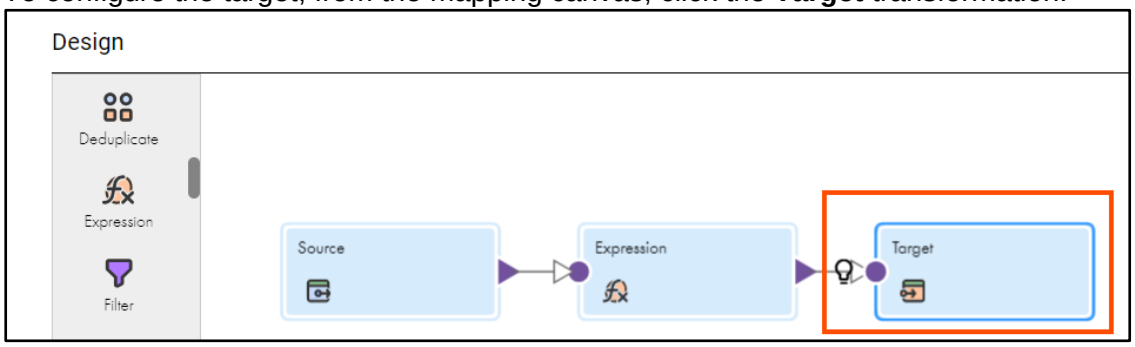
27. To configure the expression, click **Configure**.



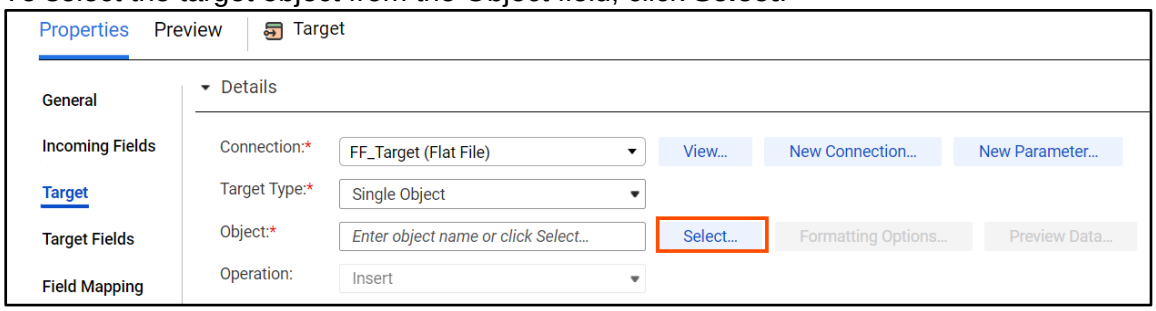
28. In the Expression field, copy and paste the following expression:
SetMaxVariable($$p_orderid,ORDERID)

29. Validate the expression and click **OK**.

30. To configure the target, from the mapping canvas, click the **Target** transformation.

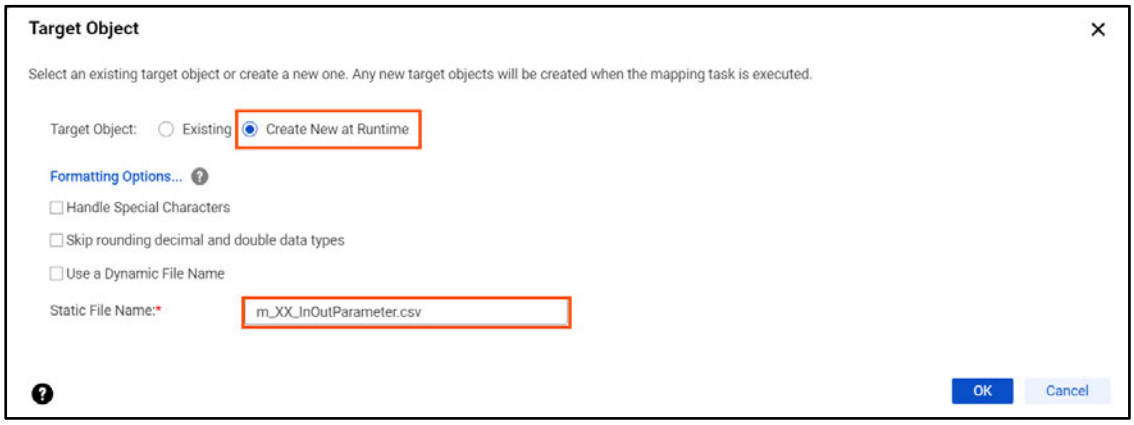


31. From the properties pane, click **Target**.
32. From the Connection drop-down, select your **target Flat File** connection.
33. To select the target object from the Object field, click **Select**.

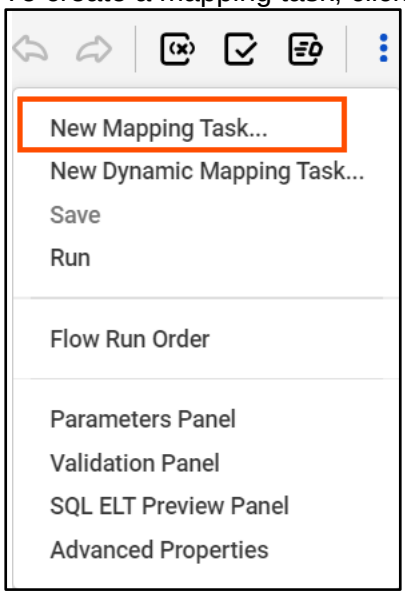


34. In the Target Object window, select **Create New at Runtime**.

35. In the Table Name field, enter **m_XX_InOutParameter.csv** (replace XX with your DOB) and click **OK**.

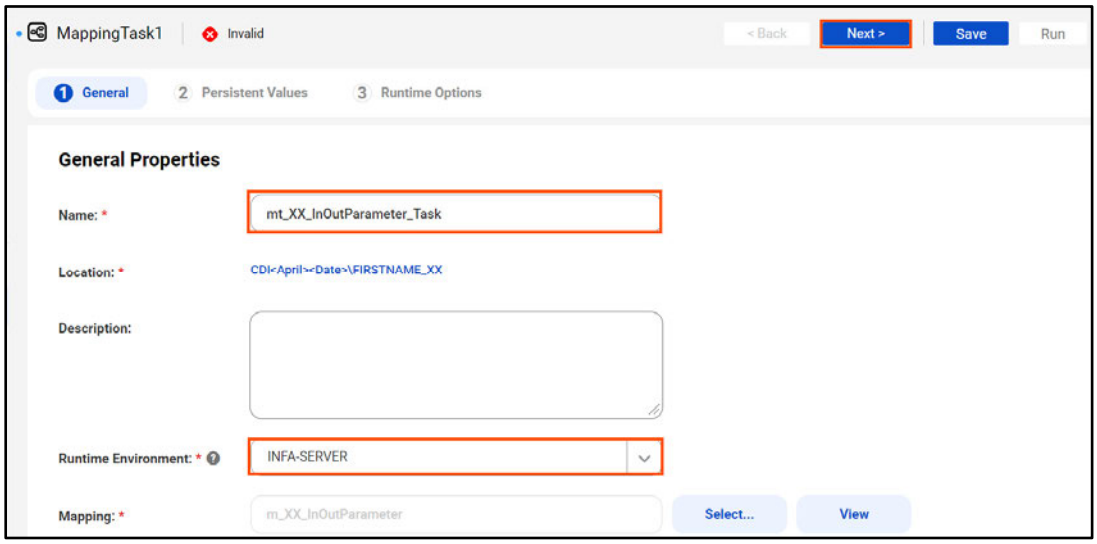


36. **Save** the mapping.
37. To create a mapping task, click on the ellipsis icon and select **New Mapping Task**.

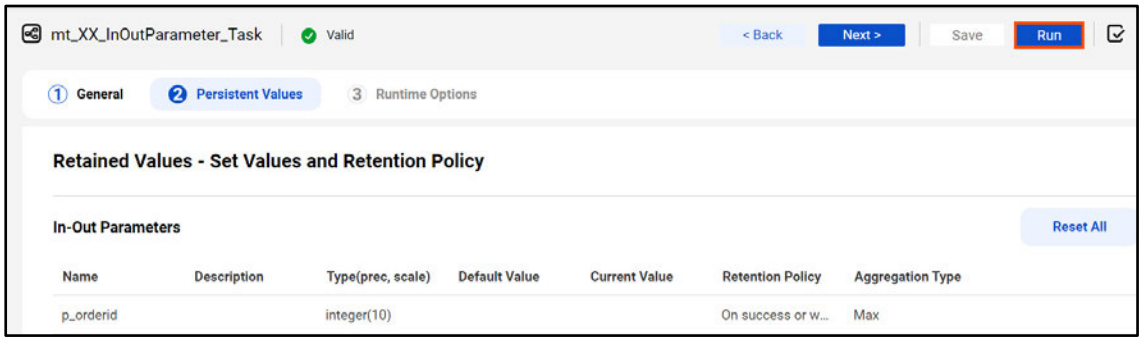


38. In the Task Name field, enter **mt_XX_InOutParameter_Task**. Replace XX with your DOB.
39. From the Runtime Environment drop-down, select **INFA-SERVER**.

40. Click **Next**.

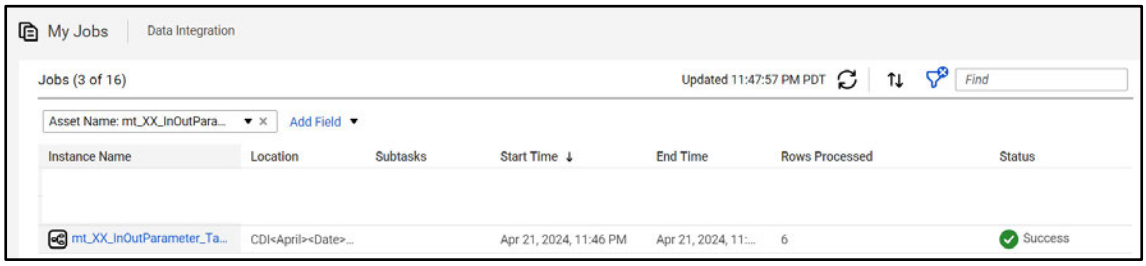


41. Retain the In-Out Parameters configuration and click **Save**.
42. Click **Run**.

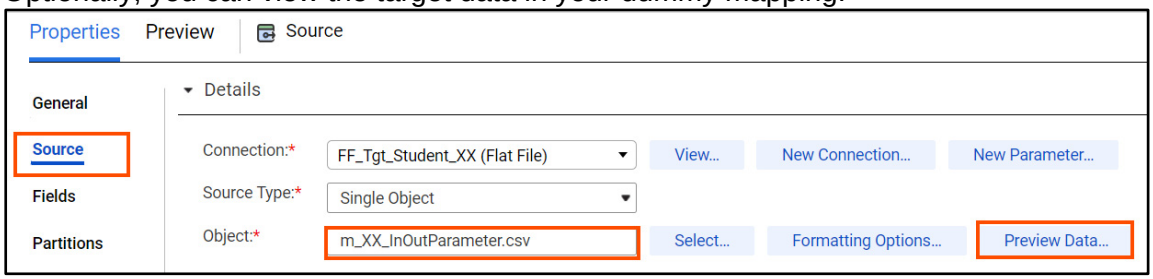


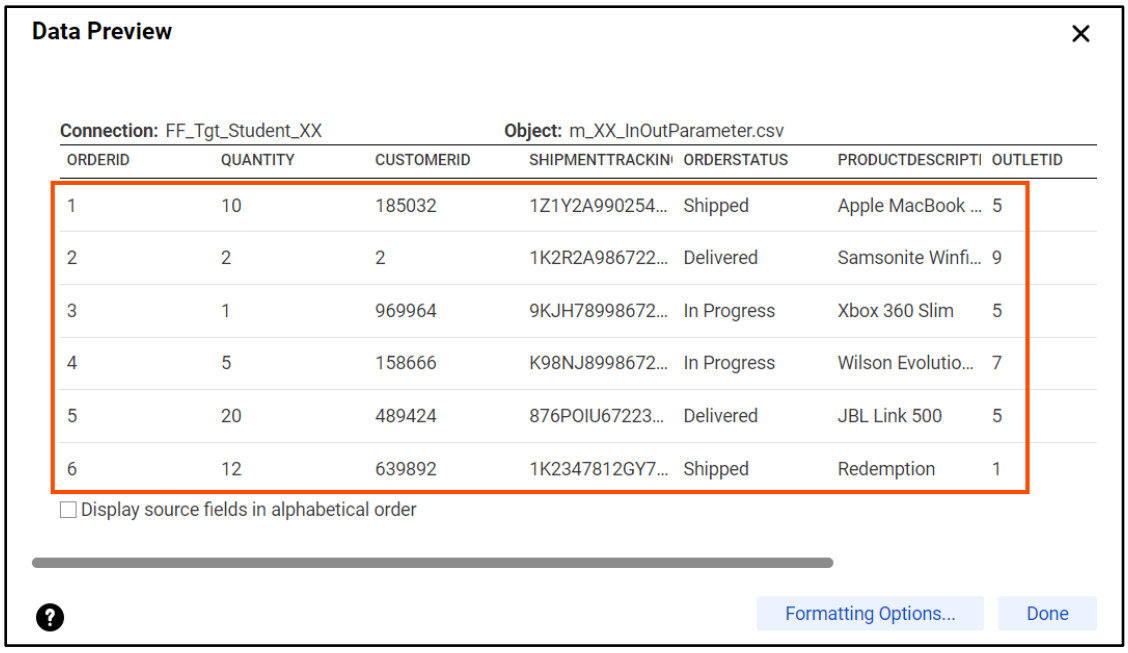
Monitor Status

43. Monitor the job status and verify that 6 rows are processed by the mapping.



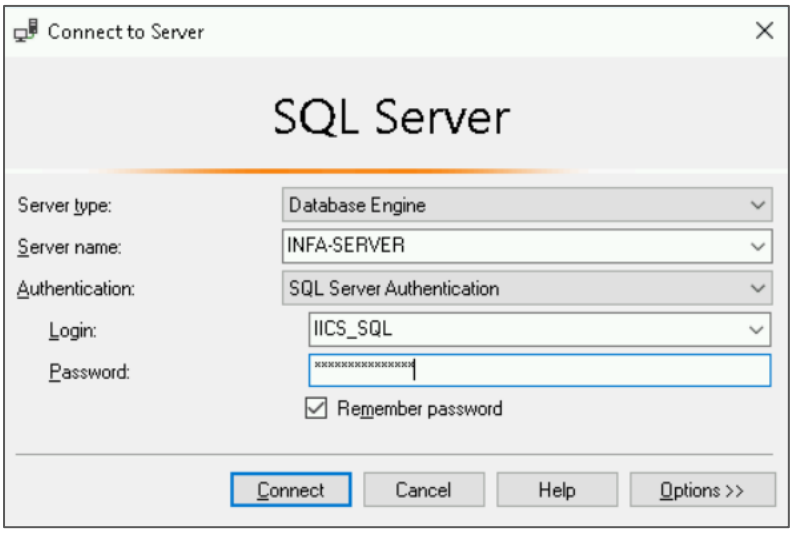
44. Optionally, you can view the target data in your dummy mapping.



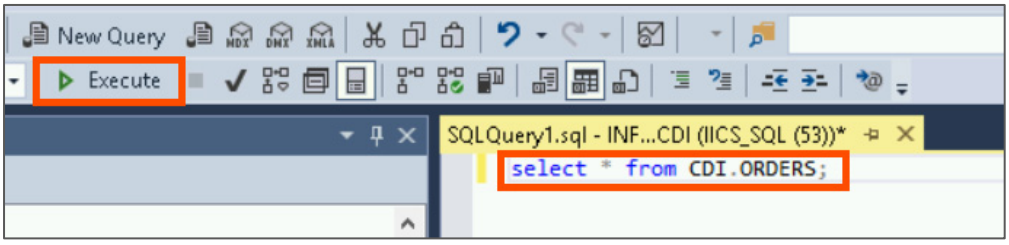


45. Save and close all the assets from the navigation pane.

46. To verify the incremental load of orders data, from the Windows Start menu, open **SQL Server Management Studio**.
47. From the **Connections** section, enter the Login and Password as **IICS_SQL**.
48. Click **Connect**.



49. Expand the **Databases** > **CDI** > **Tables** and click on **CDI.ORDERS** table.
50. Click **New Query**.
51. To check the existing data in the **CDI.ORDERS** table, execute the below query.
    select * from CDI.ORDERS;



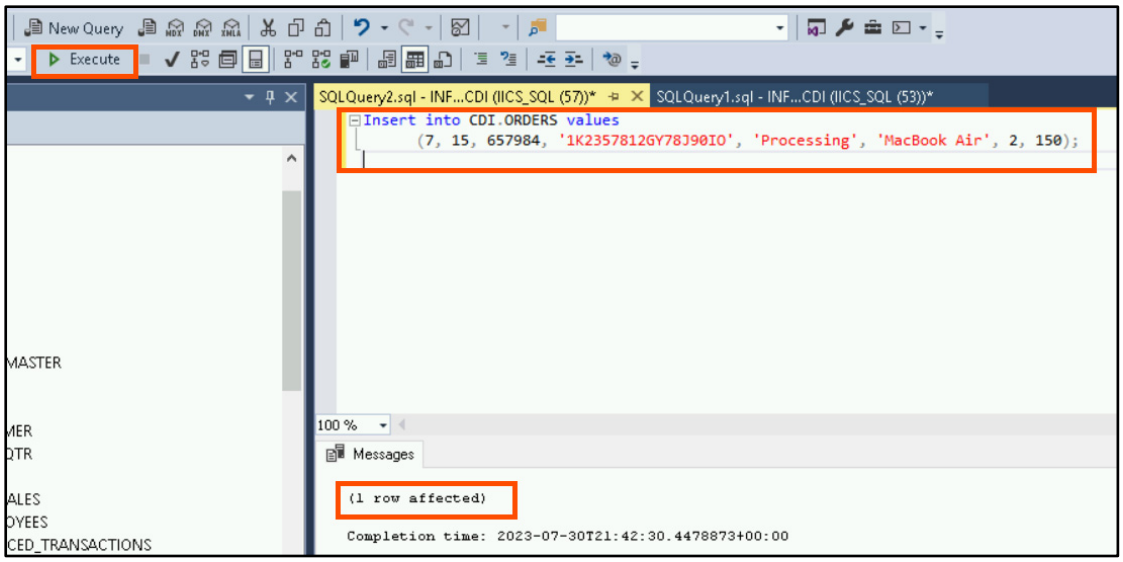
52. Add a new row and enter the below data.

ORDER ID	QUANTITY	CUSTOMER ID	SHIPMENT TRACKING NO	ORDER STATUS	PRODUCT DESCRIPTION	OUTLET ID	PRODUCT ID
7	15	657984	1K2357812GY78J90IO	Processing	MacBook Air	2	150

For Adding the new row, use the query shown below in a new query window:

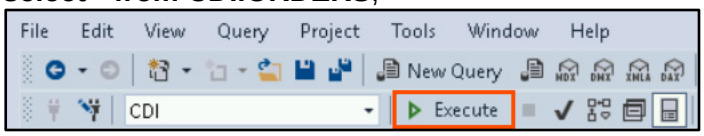
Insert into CDI.ORDERS values

(7, 15, 657984, '1K2357812GY78J90IO', 'Processing', 'MacBook Air', 2, 150);

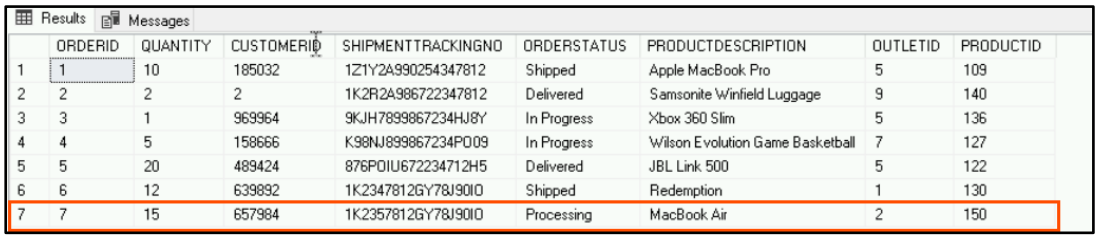


53. Now, to check the newly inserted row, navigate back to the previous query window and re-run the command or enter the below query and click **Execute**.

    select * from CDI.ORDERS;



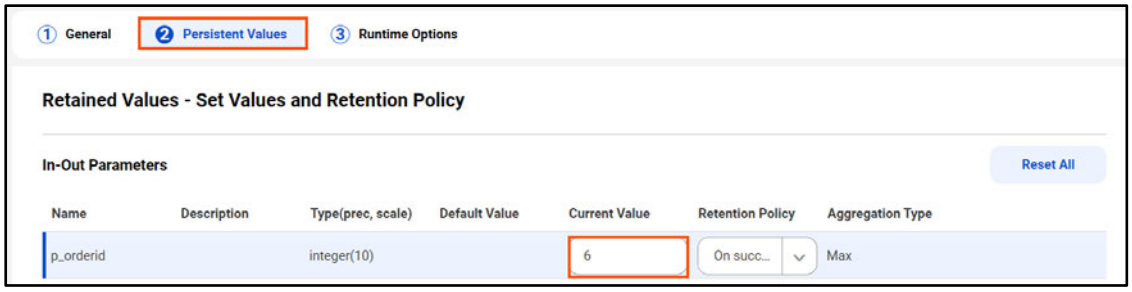
54. A new row is inserted in the table.



Results | Messages

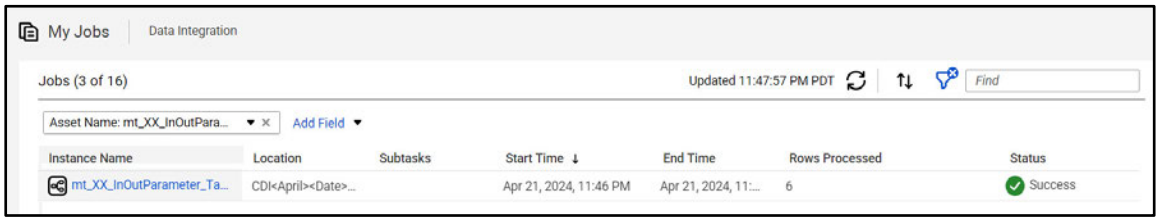
	ORDERID	QUANTITY	CUSTOMERID	SHIPMENTTRACKINGNO	ORDERSTATUS	PRODUCTDESCRIPTION	OUTLETID	PRODUCTID
1	1	10	185032	1Z1Y2A990254347812	Shipped	Apple MacBook Pro	5	109
2	2	2	2	1K2R2A986722347812	Delivered	Samsonite Winfield Luggage	9	140
3	3	1	969964	9KJH7899867234HJ8Y	In Progress	Xbox 360 Slim	5	136
4	4	5	158666	K98NJ899867234PO09	In Progress	Wilson Evolution Game Basketball	7	127
5	5	20	489424	876POIU672234712H5	Delivered	JBL Link 500	5	122
6	6	12	639892	1K2347812GY78J90IO	Shipped	Redemption	1	130
7	7	15	657984	1K2357812GY78J90IO	Processing	MacBook Air	2	150

55. Navigate back to IICS, from your working directory, and edit the **mt_XX_InOutParameter_Task** asset.
56. Go to **Persistent Values** step and verify that the **Current Value** is set to **6**. This is because the task read the 6 rows in the first run.



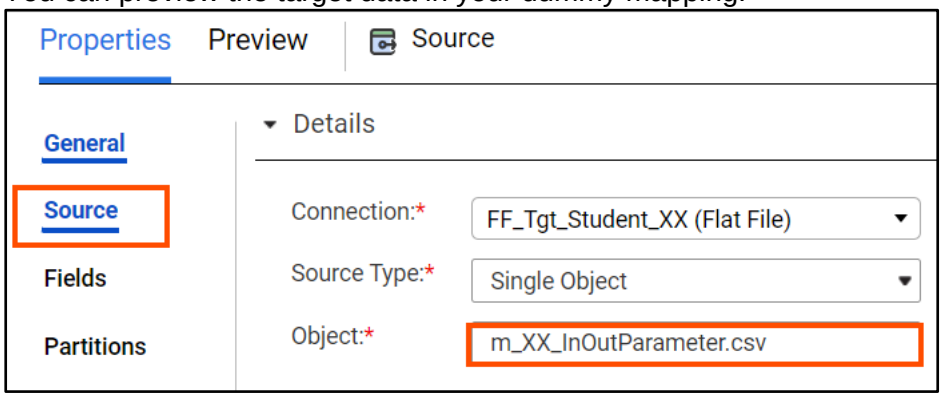
57. Click **Save** and **Run** the task again.

58. Go to the **My Jobs** page to verify the task status. Notice that only 1 row is processed in the second run of the task.

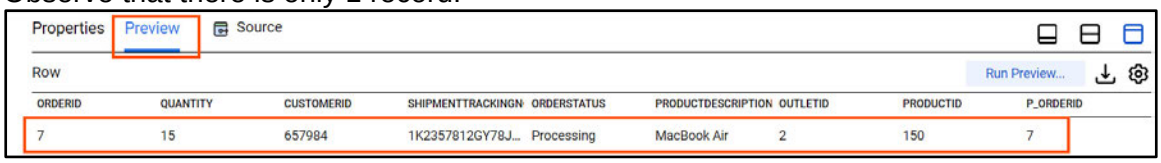


Optional

59. You can preview the target data in your dummy mapping.



60. To view the updated target records, navigate to the Preview tab and click **Run Preview**. Observe that there is only 1 record.



61. Save and close all the assets.

This concludes the lab.

Module 10: Mapping Task

Lab 10-1: Creating a Mapping Task to Perform Basic History Tracking And Configure Advanced Settings

Overview:

The Mapping task allows you to process data based on the data flow logic defined in a Mapping. When you create a mapping task, you select the mapping for the task to use. On the Schedule page of the mapping task, you can define to run a mapping task manually or schedule it to run at a specific time or interval.

Objective:

- Create a Mapping to perform basic history tracking
- Create a Mapping Task
- Configure Email Notifications

Scenario:

Ruby wants to track the incoming data records and update the changes accordingly.

The process of updating or inserting records efficiently involves using the target table as a lookup to determine if a record represents a new or existing customer. This requires two instances of the target table in the mapping. New customer records are routed to a target object for INSERT operations, with a Sequence Generator providing surrogate keys. Existing customer records are routed to a target object for UPDATE operations, with any data changes captured. An Expression transformation outputs a timestamp for record currency. The Sequence Generator saves the last surrogate key value in the cloud repository for the next run, requiring a Mapping Task for this functionality.

Later, Ruby wants to monitor the job statuses of a few important jobs timely for 48 hours. Instead of verifying whether the job status is success or fail, Ruby will configure the email notifications property to get notified about the job statuses.

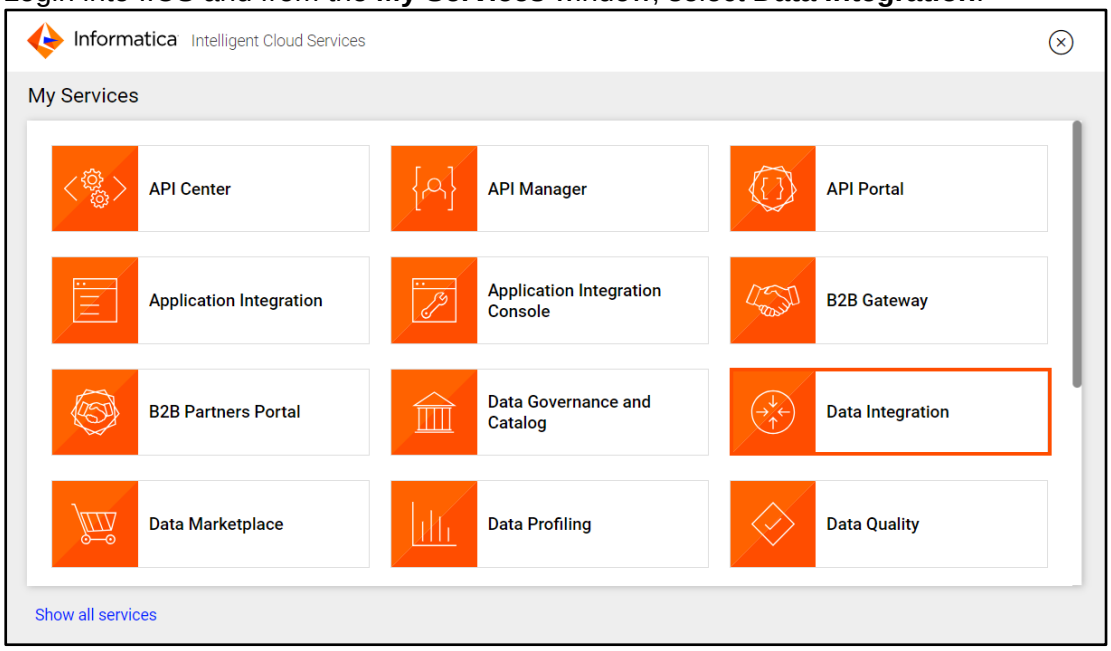
Duration:

40 minutes

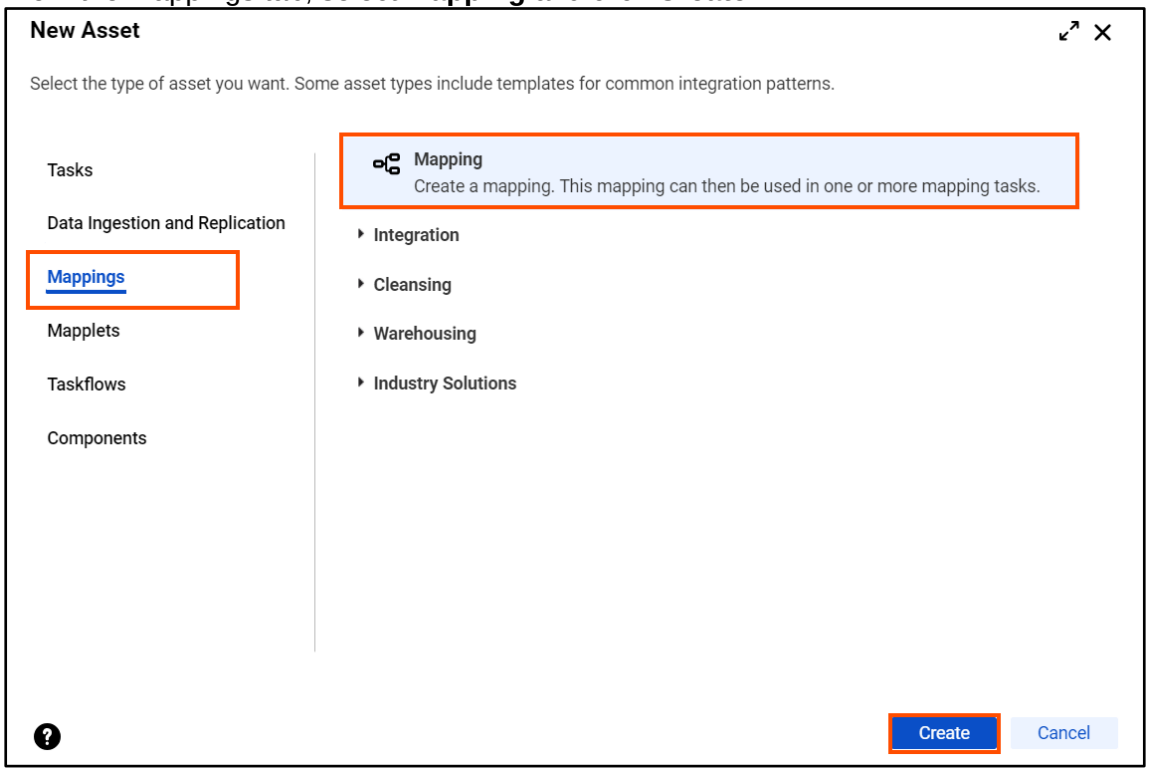
Tasks

Create a Mapping

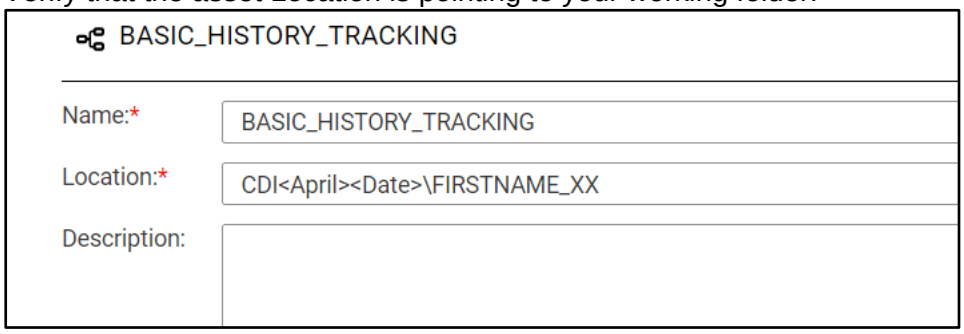
1. Login into IICS and from the **My Services** window, select **Data Integration**.



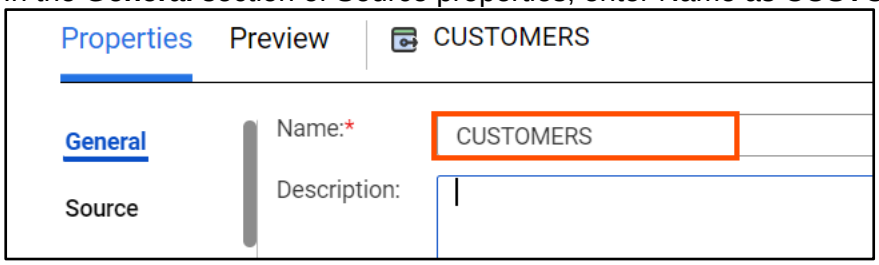
2. To create a new asset, from the navigation pane, select **New**.
3. From the Mappings tab, select **Mapping** and click **Create**.



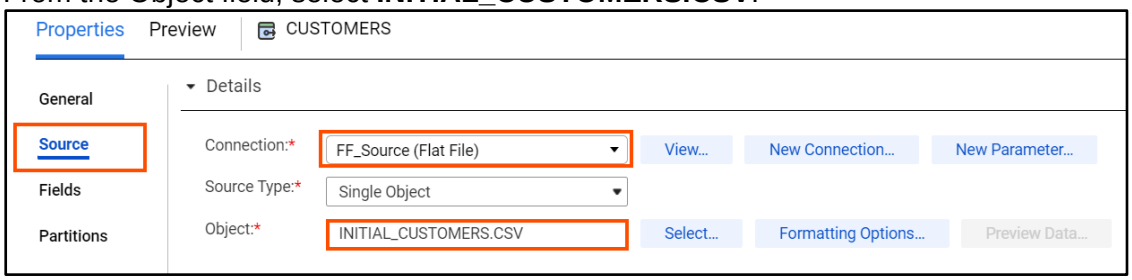
4. In the Name field of the mapping properties, enter **BASIC_HISTORY_TRACKING**.
5. Verify that the asset Location is pointing to your working folder.



6. To configure the source, from the mapping canvas, click the **Source** transformation.
7. In the **General** section of Source properties, enter Name as **CUSTOMERS**.

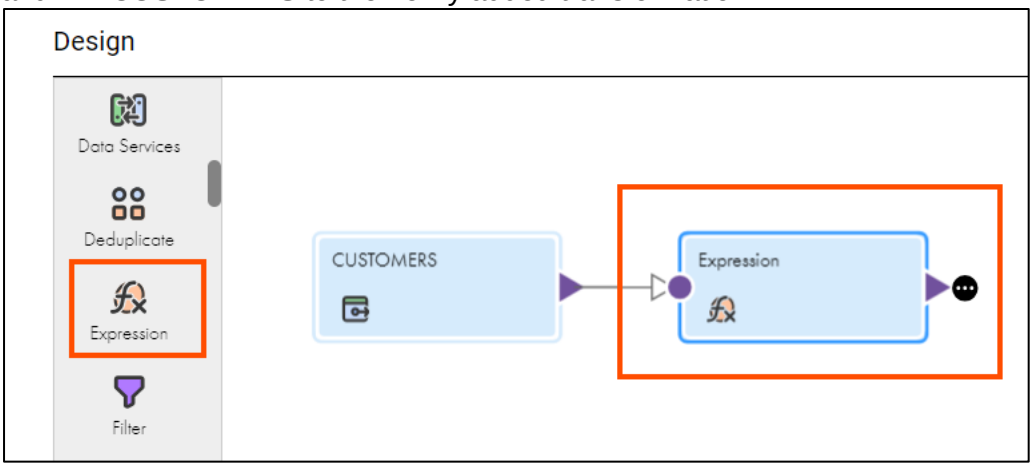


8. **Delete** the link between Source and Target transformations in the mapping canvas.
9. From the properties pane, click **Source**.
10. In the Connection drop-down, select **FF_Source**.
11. From the Object field, select **INITIAL_CUSTOMERS.CSV**.

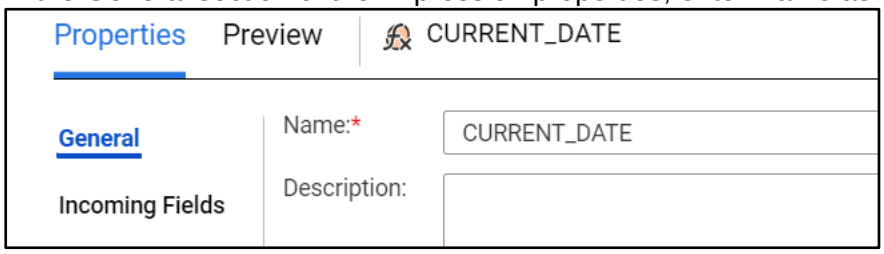


Add Expression Transformation

12. From the list of available transformations, drag and drop the **Expression** transformation and link **CUSTOMERS** to the newly added transformation.

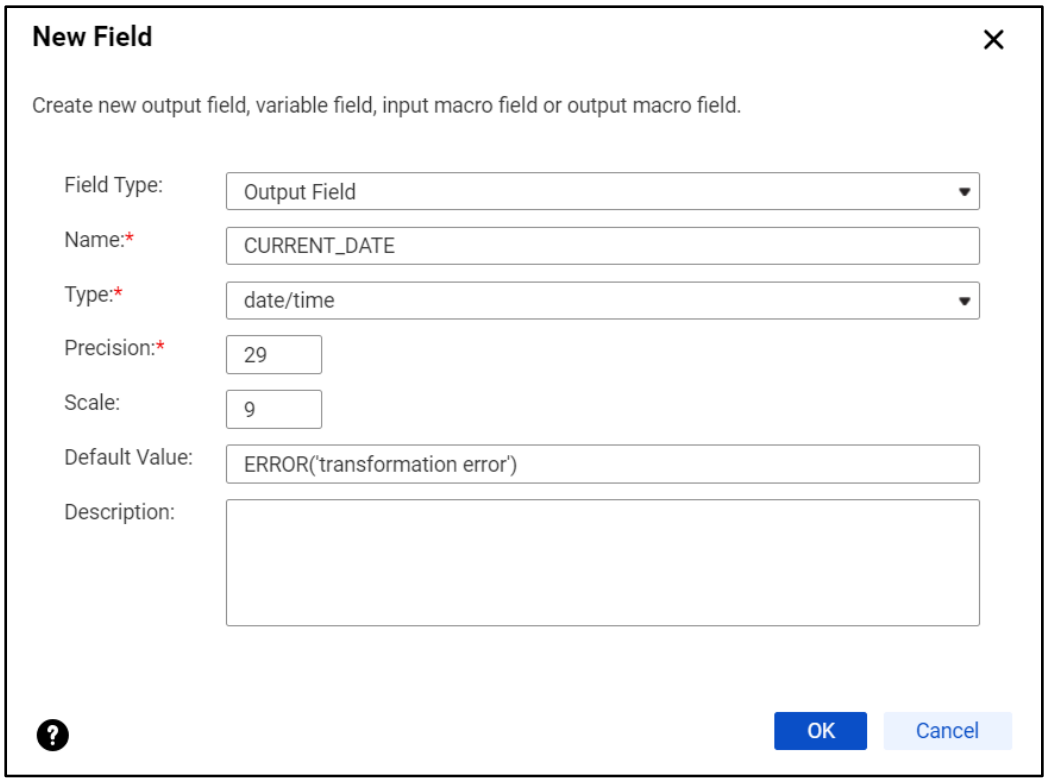


13. Select **Expression** transformation from the mapping canvas.
14. In the General section of the Expression properties, enter Name as **CURRENT_DATE**.

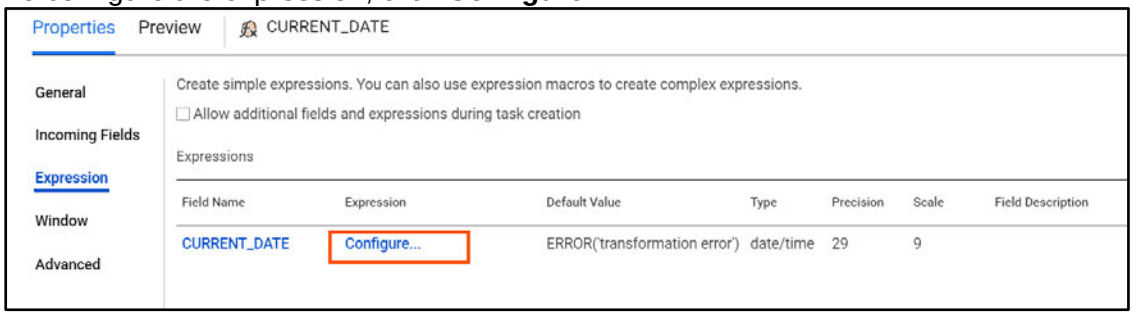


15. From the properties pane, click **Expression**.
16. To add a new expression, click ⊞ .
17. Enter the details as shown in the table below and click **OK**.

Field Type	Name	Type	Precision	Scale
Output Field	CURRENT_DATE	date/time	29	9

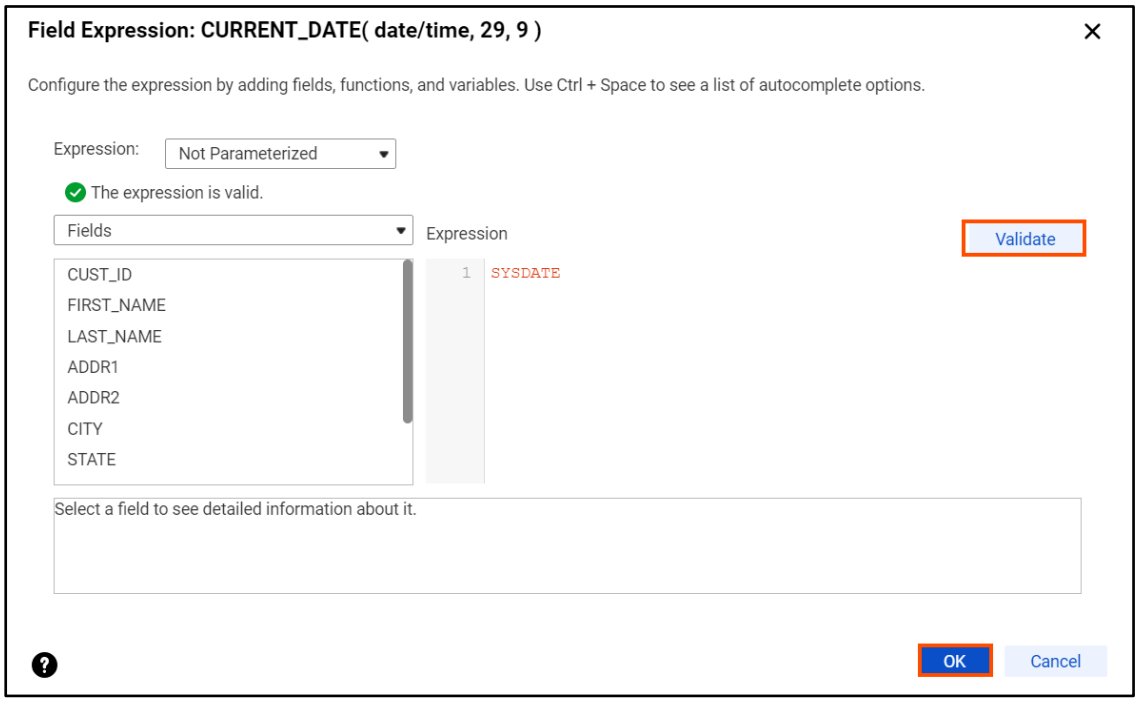


18. To configure the expression, click **Configure**.



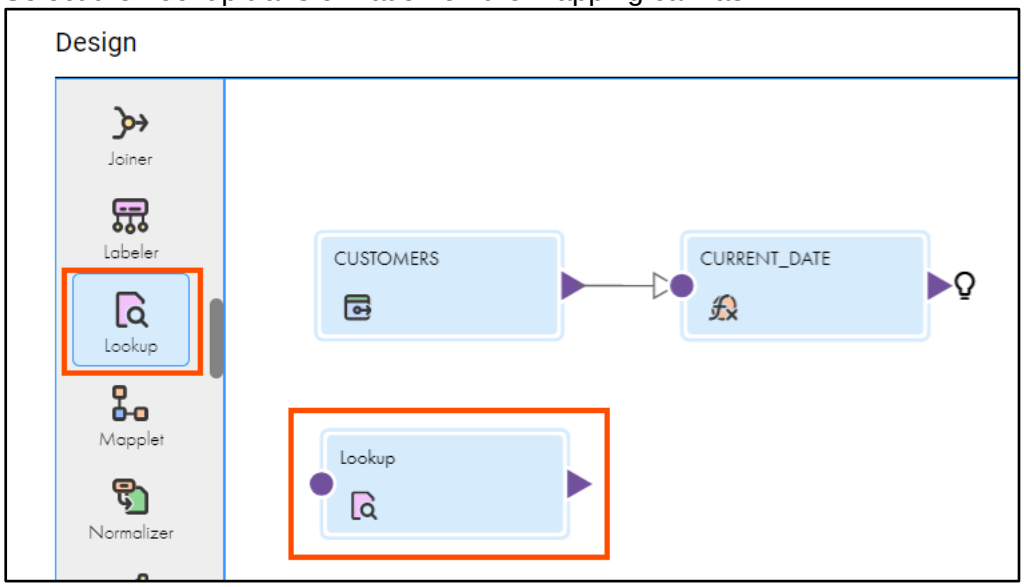
19. In the Expression field, copy and paste the following expression:
SYSDATE

20. Click **Validate** and then click **OK**.



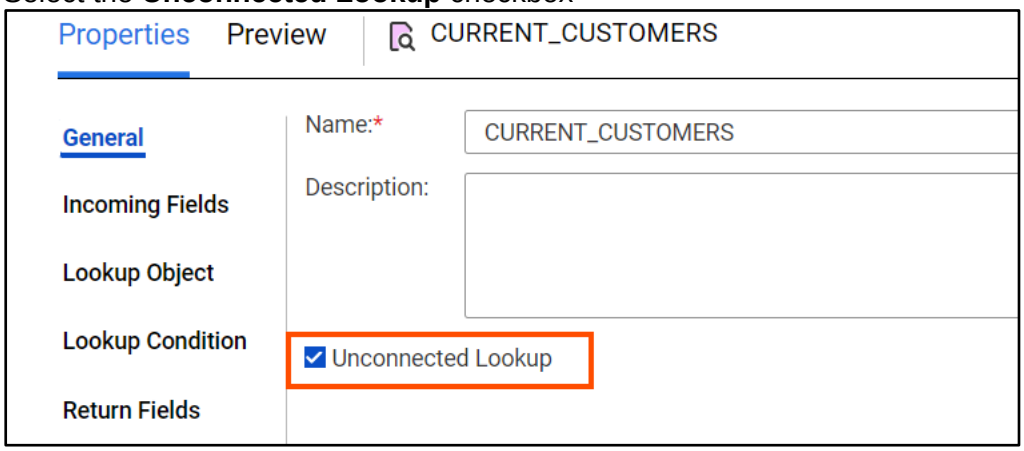
Add Lookup Transformation

21. From the list of available transformations, drag and drop a **Lookup** transformation on the mapping canvas.
22. Select the Lookup transformation on the mapping canvas.

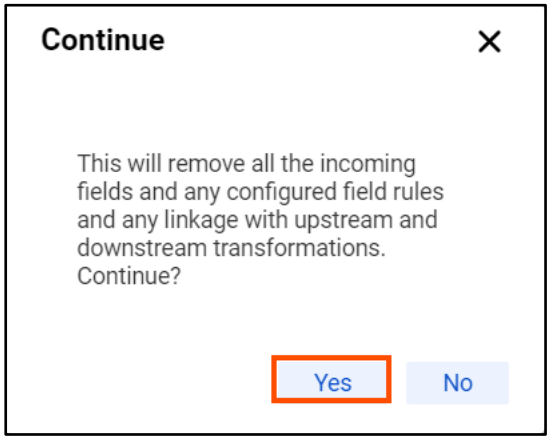


23. In the **General** section of Lookup properties, enter Name as **CURRENT_CUSTOMERS**.

24. Select the **Unconnected Lookup** checkbox

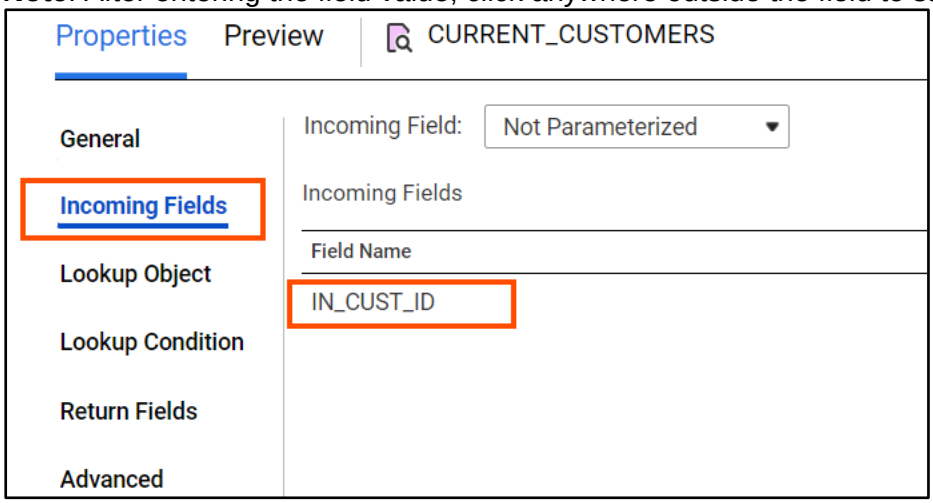


25. In the **Continue** window, select **Yes**.



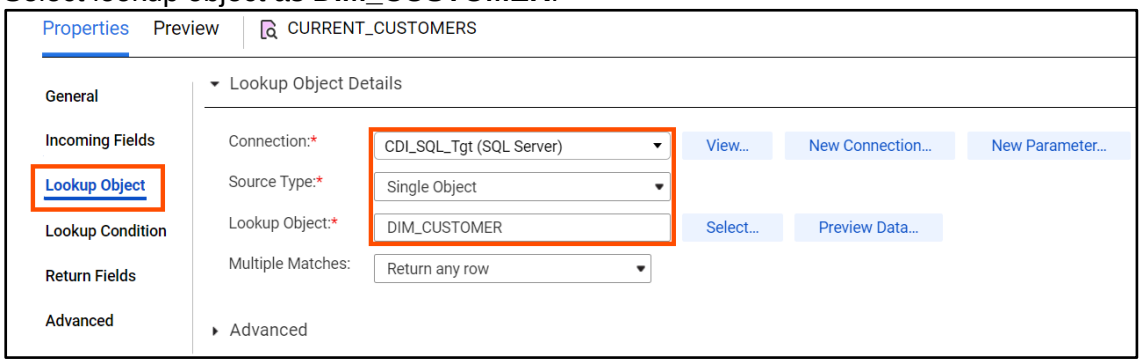
26. From the properties pane, click **Incoming Fields**.
27. To add a new incoming field, click ⊞.
28. In Field Name, enter **IN_CUST_ID**.

    Note: After entering the field value, click anywhere outside the field to save the details.



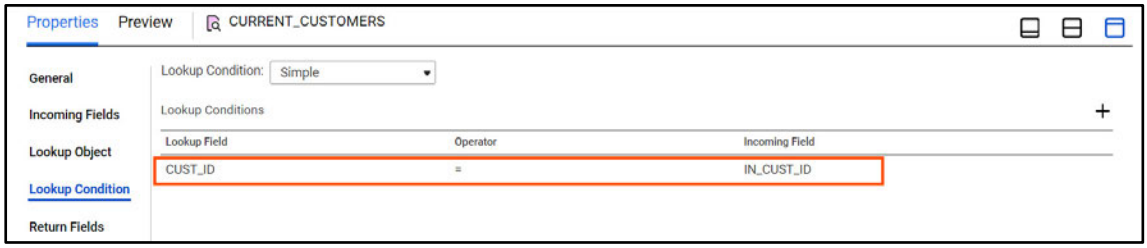
29. From the properties pane, click **Lookup Object**.

30. From the Connection drop-down, select your target SQL connection **CDI_SQL_Tgt**.
31. Select lookup object as **DIM_CUSTOMER**.

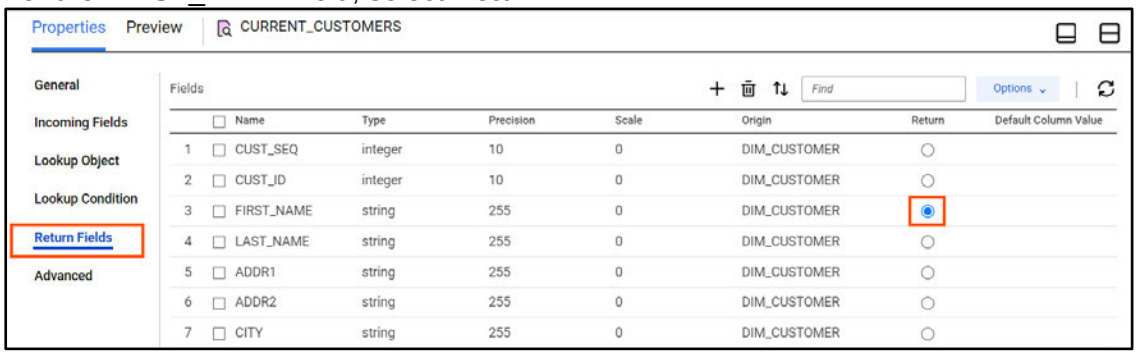


32. From the properties pane, click **Lookup Condition**.
33. To add a new lookup condition, click ⊞.
34. Set the lookup condition as shown in the table below:

Lookup Field	Operator	Incoming Field
CUST_ID	=	IN_CUST_ID

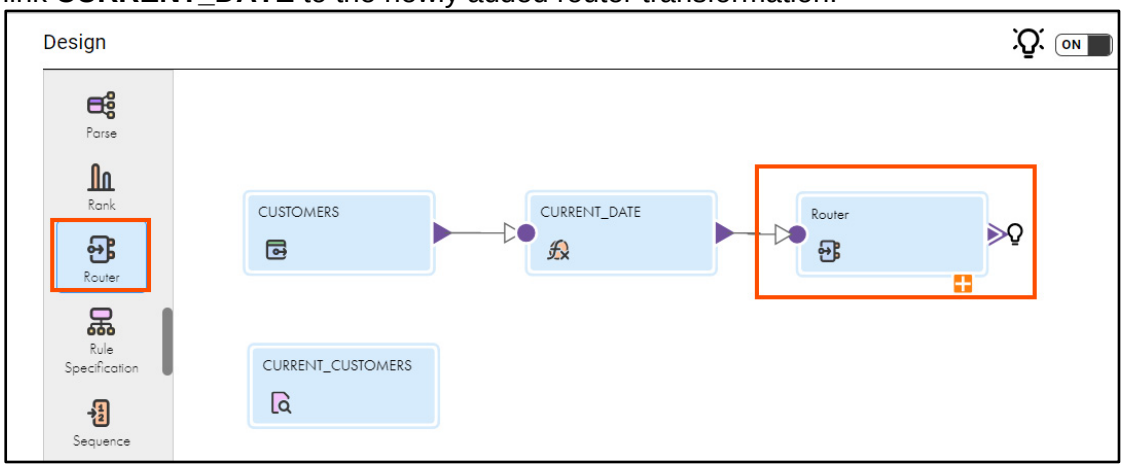


35. From the properties pane, click **Return Fields**.
36. For the FIRST_NAME field, select **Return**.

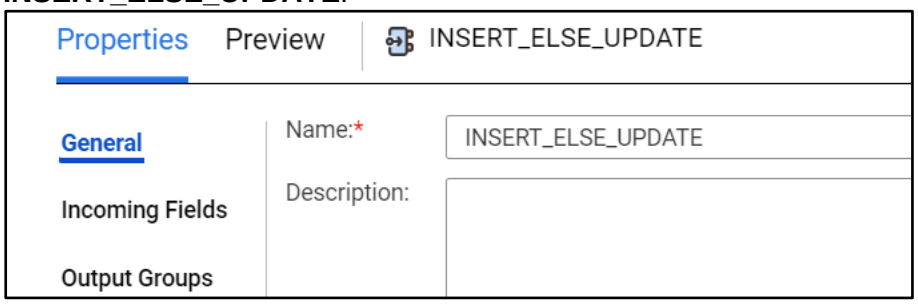


Add Router Transformation

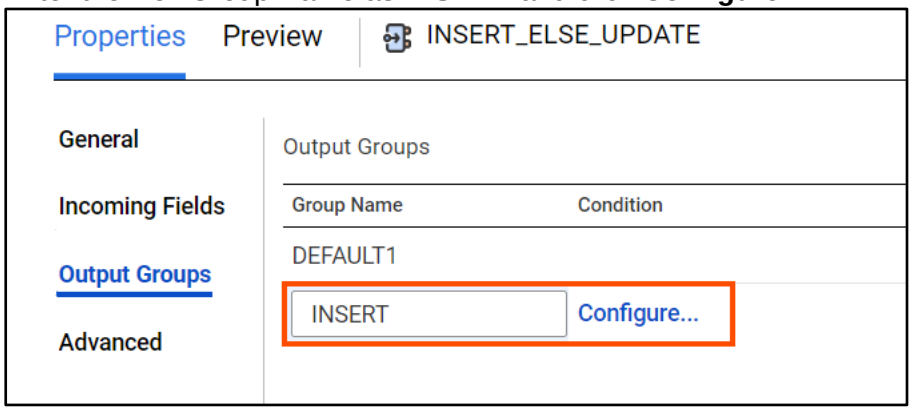
37. From the list of available transformations, drag and drop the **Router** transformation and link **CURRENT_DATE** to the newly added router transformation.



38. Select **Router** transformation from the mapping canvas.
39. In the **General** section of the Expression properties, enter Name as **INSERT_ELSE_UPDATE**.

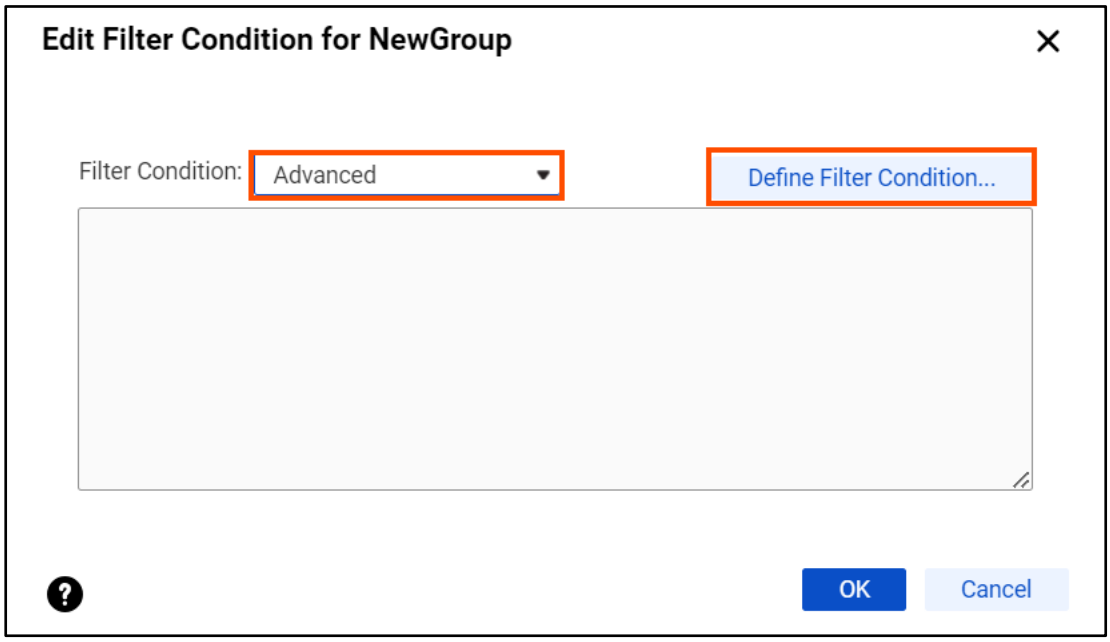


40. From the properties pane, click **Output Groups**.
41. To add a new Output Group, click ⊞.
42. Enter the NewGroup Name as **INSERT** and click **Configure**.



43. From the Filter Condition drop down, select **Advanced**.

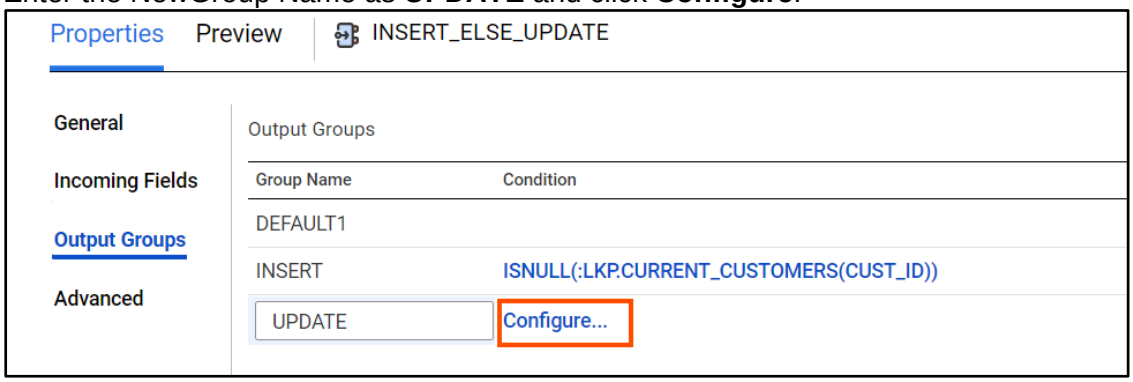
44. Click on the **Define Filter Condition** button.



45. In the Filter Condition field, enter the following expression:
    ISNULL(:LKP.CURRENT_CUSTOMERS(CUST_ID))
46. Click **Validate** and then click **OK.**

47. Click **OK**.
48. To add another Output Group, click ⊞ .

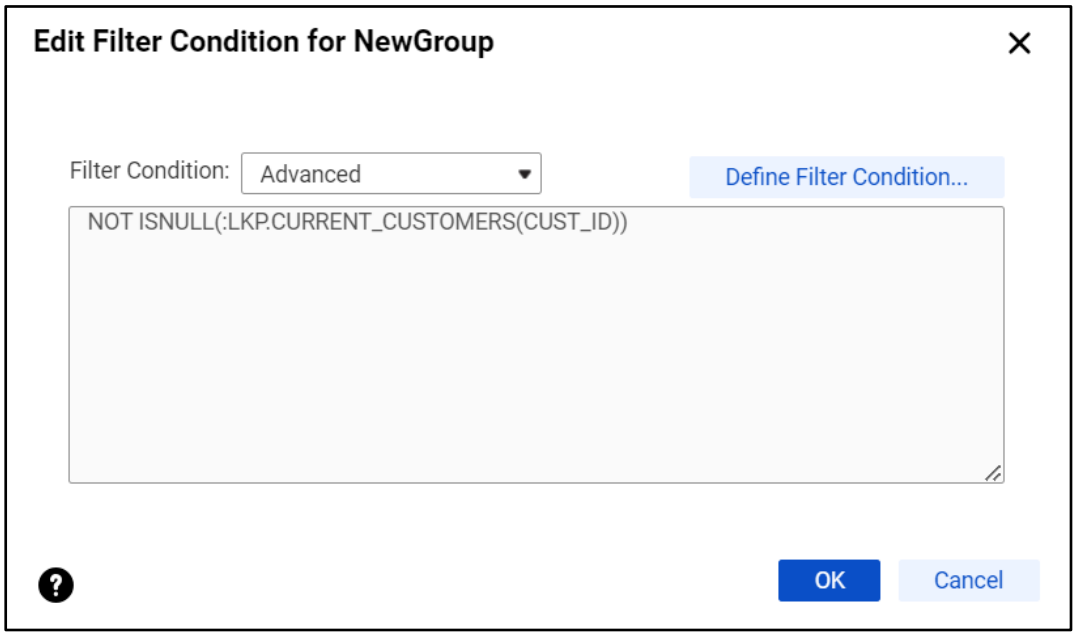
49. Enter the NewGroup Name as **UPDATE** and click **Configure**.



50. From the Filter Condition drop down, select **Advanced**.
51. Click on **Define Filter Condition**.
52. In the Filter Condition field, enter the following expression:
    NOT ISNULL(:LKP.CURRENT_CUSTOMERS(CUST_ID))

    Tip: You can copy the previous field's filter condition and paste it into this Update field. Edit the filter condition to include **NOT** at the beginning.
53. Click **Validate** and then click **OK**.

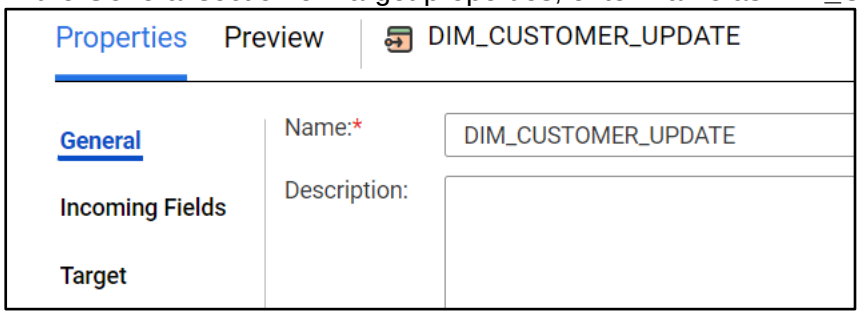
54. Click **OK**.



Configure Target

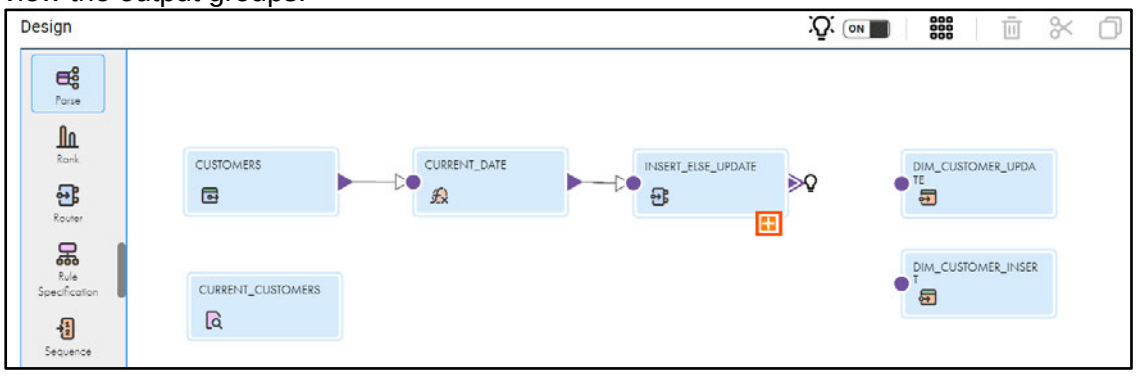
53 Select **Target** transformation on the mapping canvas.

55. In the General section of Target properties, enter Name as **DIM_CUSTOMER_UPDATE**.

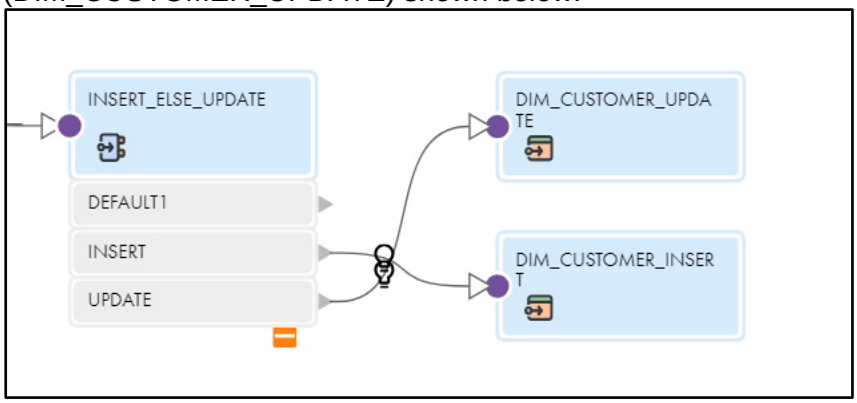


56. Copy-paste the Target Object and name the copy **DIM_CUSTOMER_INSERT**.

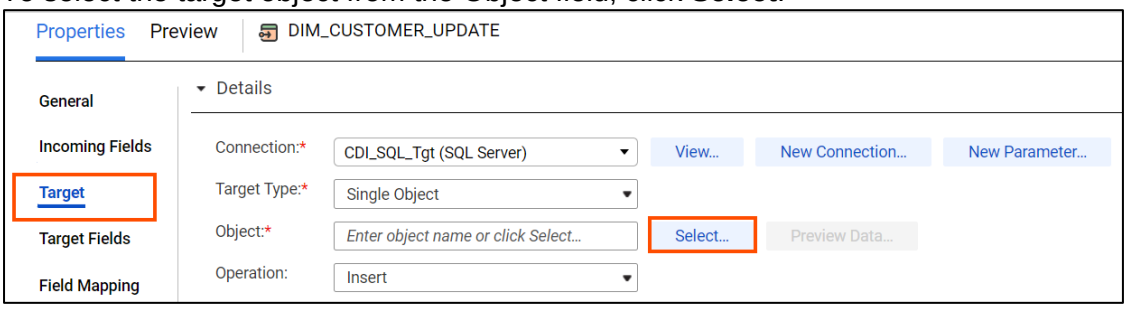
57. Click the icon present at the bottom of the Router transformation to expand it and view the output groups.



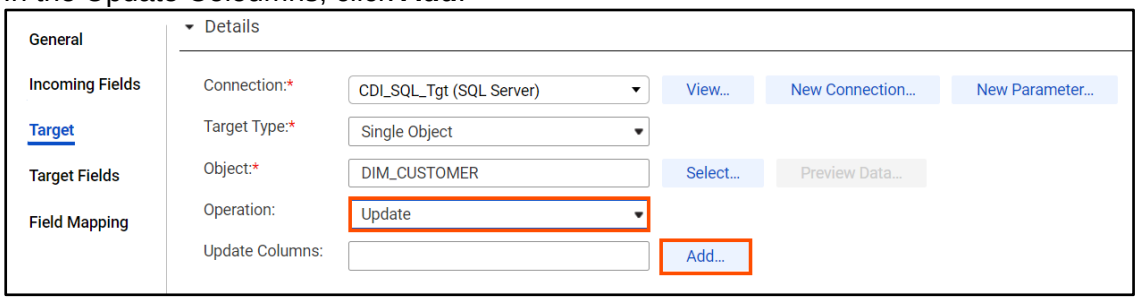
58. Link the **INSERT** output group of Router transformation to the INSERT target (DIM_CUSTOMER_INSERT).
59. Similarly, link the **UPDATE** output group of Router transformation to the UPDATE target (DIM_CUSTOMER_UPDATE) shown below:



60. From the properties pane of the DIM_CUSTOMER_UPDATE, click **Target**.
61. From the Connection drop-down, select **CDI_SQL_Tgt**.
62. To select the target object from the Object field, click **Select**.

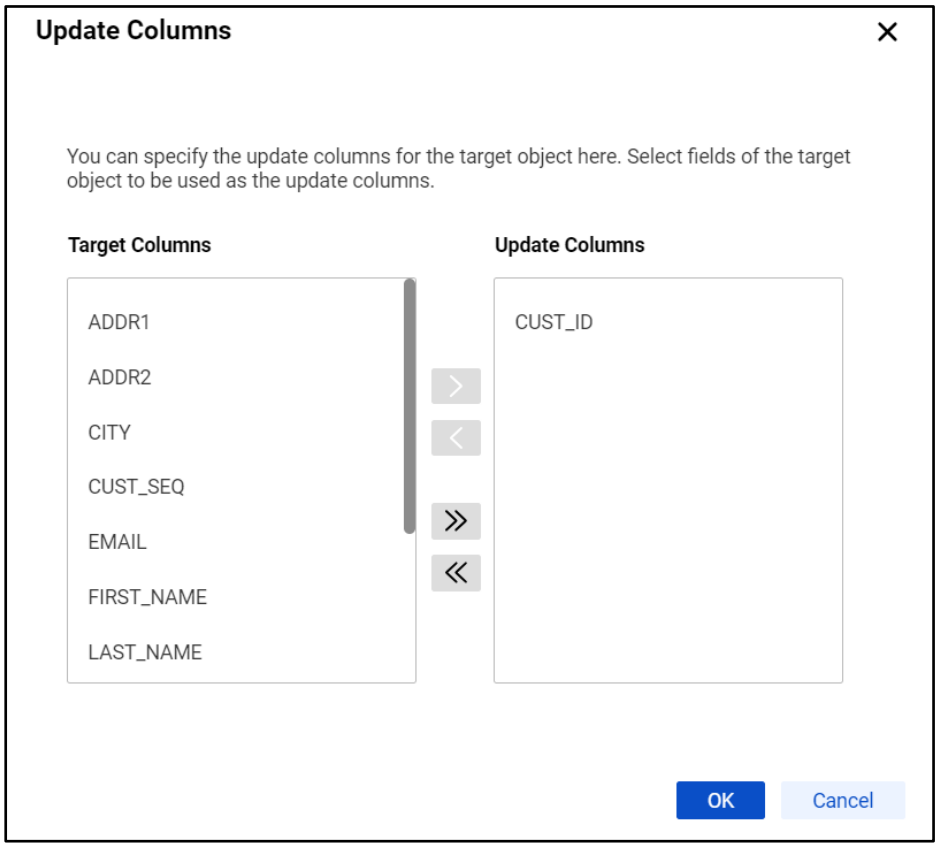


63. From the list, select **DIM_CUSTOMER** and click **OK**.
64. From the Operation drop down, select **Update**.
65. In the Update Coloumns, click **Add**.



66. From the Target Columns, select **CUST_ID** and click >.

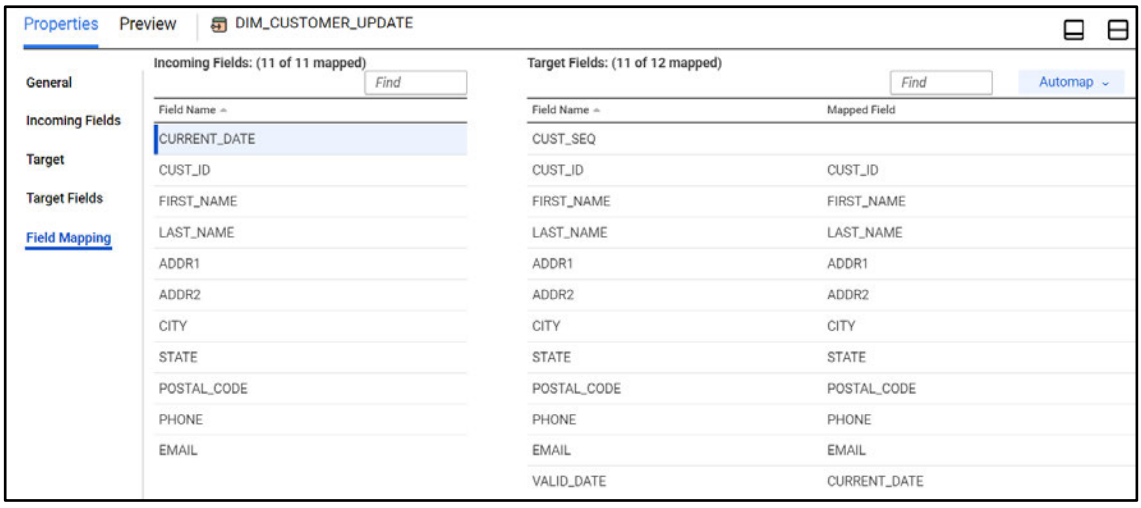
67. Click **OK**.



68. Select **Field Mapping** from the properties pane.
69. Map the fields as shown below:

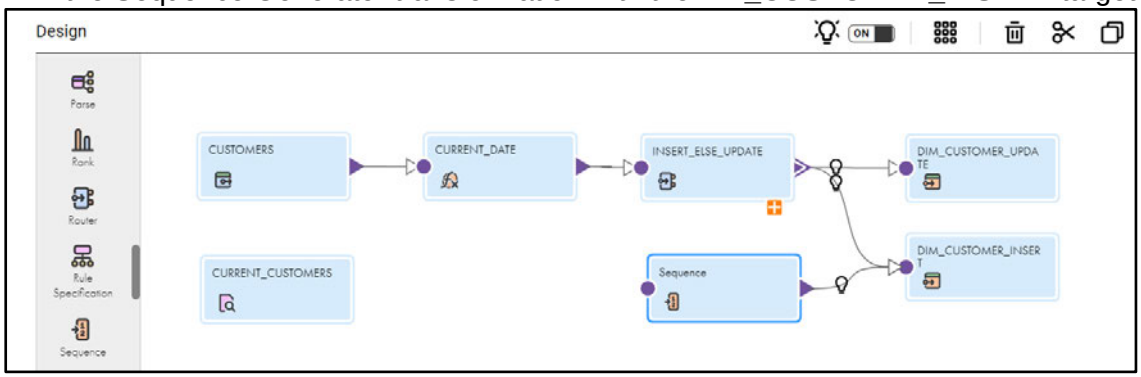
    Tip: You can use the down arrow at Automap and choose **Exact Field Name** to map the fields, and manually map the **CURRENT_DATE** field to the **VALID_DATE** field.

Incoming Field	Target Fields
CUST_ID	CUST_ID
FIRST_NAME	FIRST-NAME
LAST_NAME	LAST_NAME
ADDR1	ADDR1
ADDR2	ADDR2
CITY	CITY
STATE	STATE
POSTAL_CODE	POSTAL_CODE
PHONE	PHONE
EMAIL	EMAIL
CURRENT_DATE	VALID_DATE

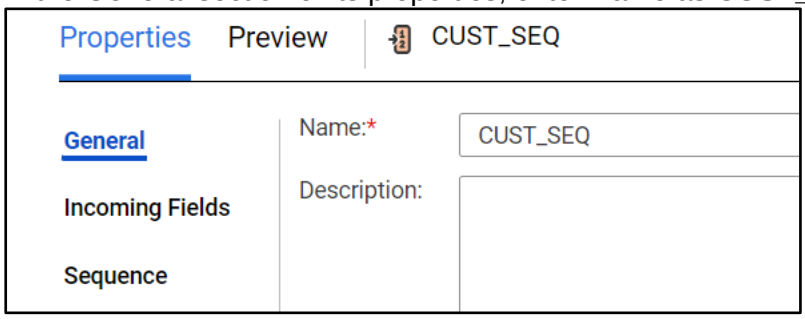


Add a Sequence Generator transformation

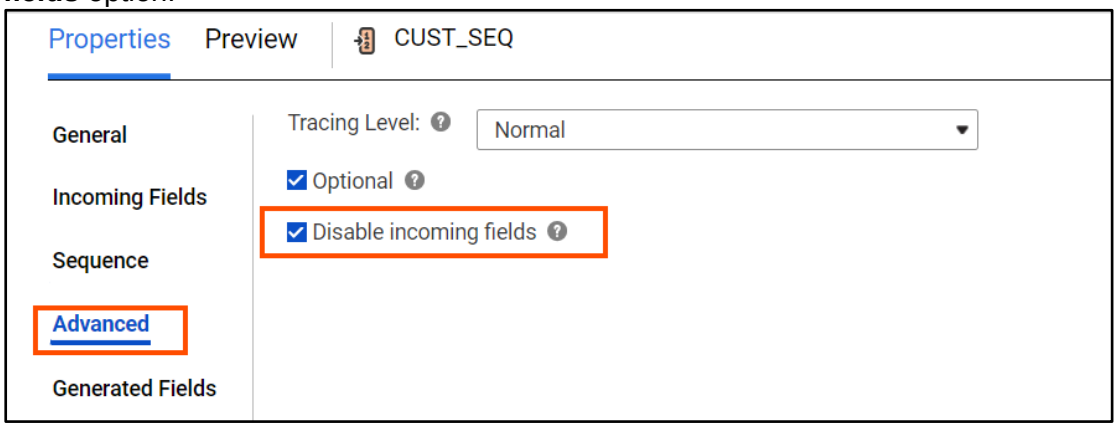
70. From the list of available transformations, drag and drop the Sequence Generator transformation.
71. Link the Sequence Generator transformation with the **DM_CUSTOMER_INSERT** target.



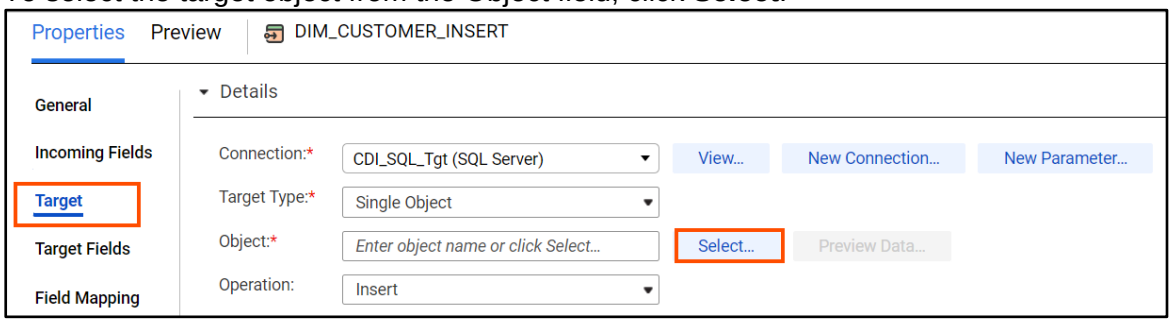
72. Select the Sequence Generator transformation from the mapping canvas.
73. In the General section of its properties, enter Name as **CUST_SEQ**.



74. From the properties pane, select **Advanced** field and check the **Disable incoming fields** option.



75. Now, select the target **DIM_CUSTOMER_INSERT**.
76. From the properties pane, click **Target**.
77. From the Connection drop-down, select **CDI_SQL_Tgt**.
78. To select the target object from the Object field, click **Select**.

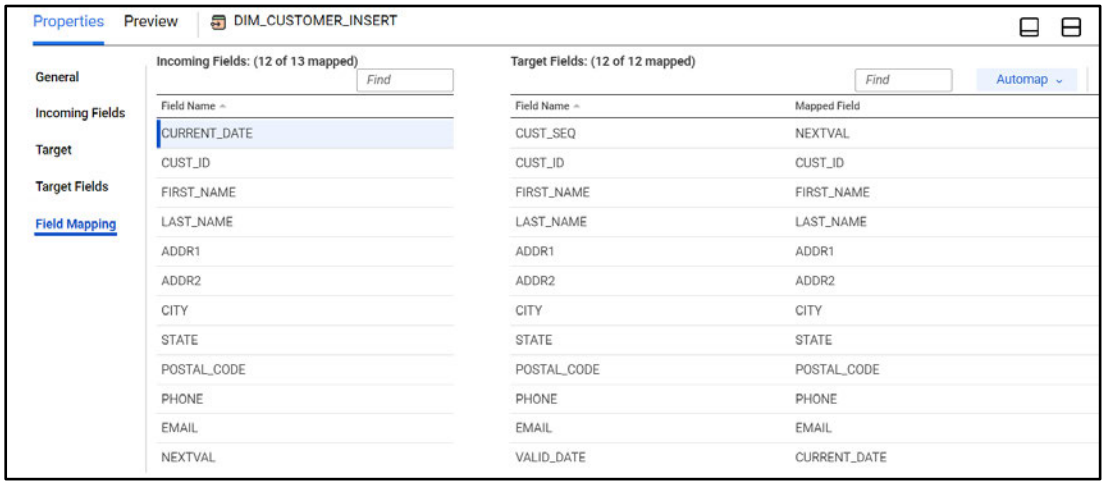


79. From the list, select **DIM_CUSTOMER** and click **OK**.
80. In the Operation field, retain the field as **Insert.**

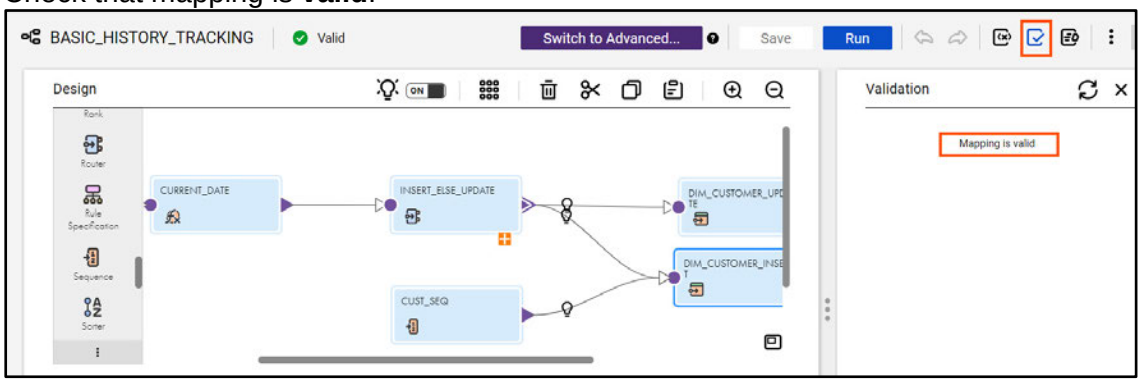
81. From the properties pane, click **Field Mapping**.
82. Map the following fields as shown below:

Incoming Field	Target Fields
NEXTVAL	CUST_SEQ
CUST_ID	CUST_ID
FIRST_NAME	FIRST_NAME

LAST_NAME	LAST_NAME
ADDR1	ADDR1
ADDR2	ADDR2
CITY	CITY
STATE	STATE
POSTAL_CODE	POSTAL_CODE
PHONE	PHONE
EMAIL	EMAIL
CURRENT_DATE	VALID_DATE

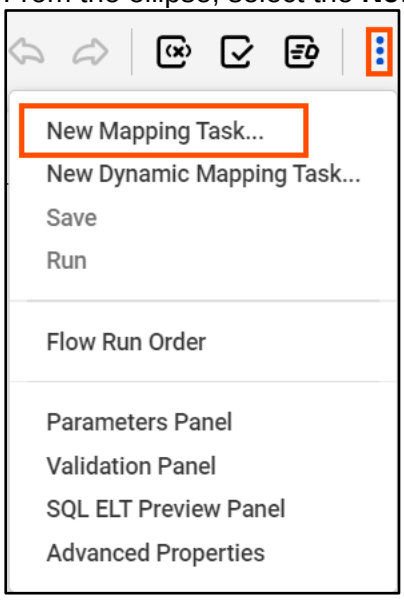


83. **Save** the mapping.
84. Check that mapping is **valid**.

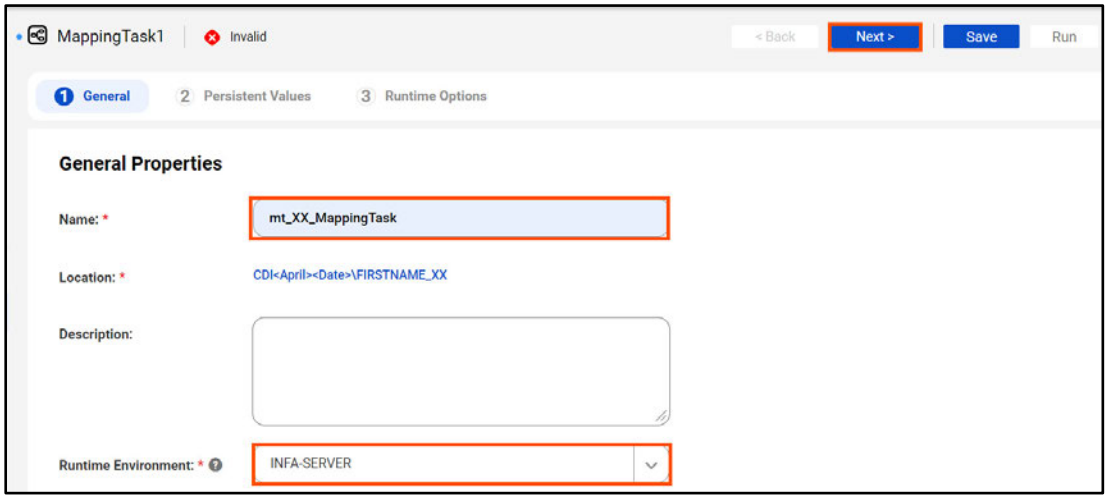


Create a new Mapping Task

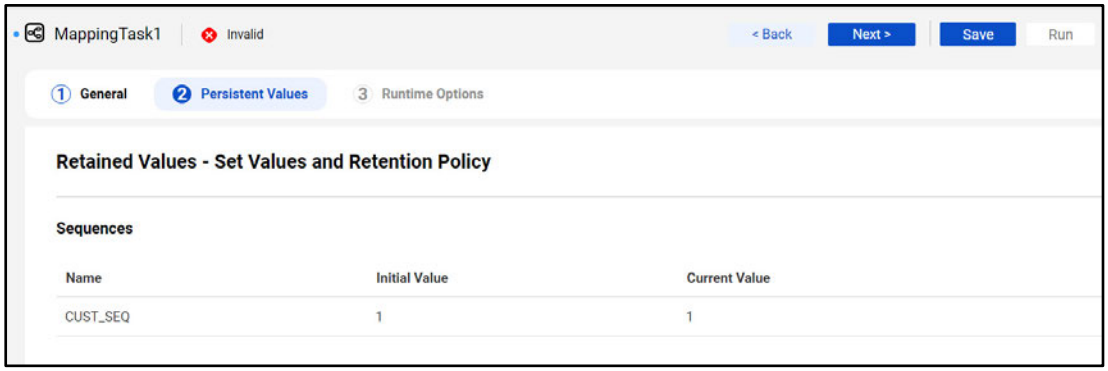
85. From the ellipse, select the **New Mapping Task**.



86. In the Task Name field, enter the Task Name as **mt_XX_MappingTask**.
87. From the Runtime Environment drop down, select **INFA-SERVER**.
88. Click **Next**.



89. In the Persistent Values, retain the values and click **Next**.

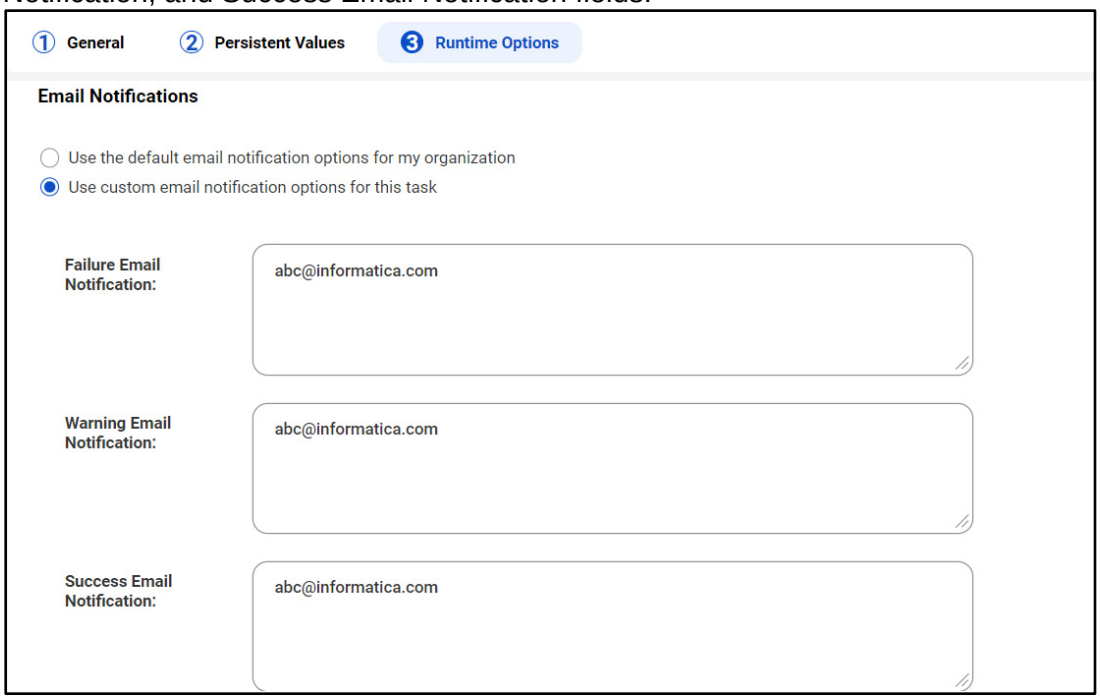


Configure the Email Notifications

90. Scroll down to **Email Notification Options**, and select **Use custom email notification options for this task.**

    Note: You can set it to receive emails if the mapping or Mapping task is successful, failed, or finished with some warnings.

91. Specify a valid and active email ID for the Failure Email Notification, Warning Email Notification, and Success Email Notification fields.

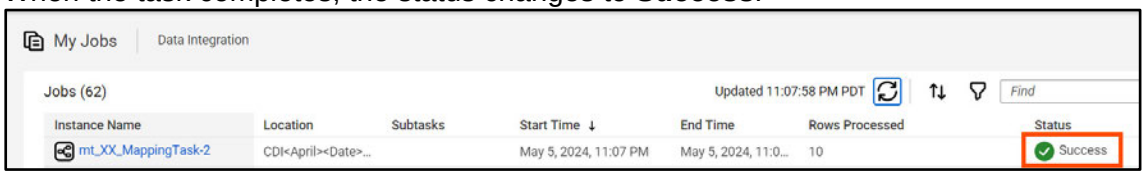


92. Click **Save**. The mapping task will be saved.

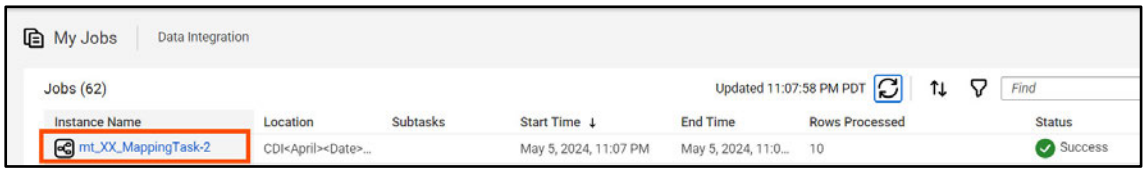
93. **Run** the Mapping task.

Monitor Status

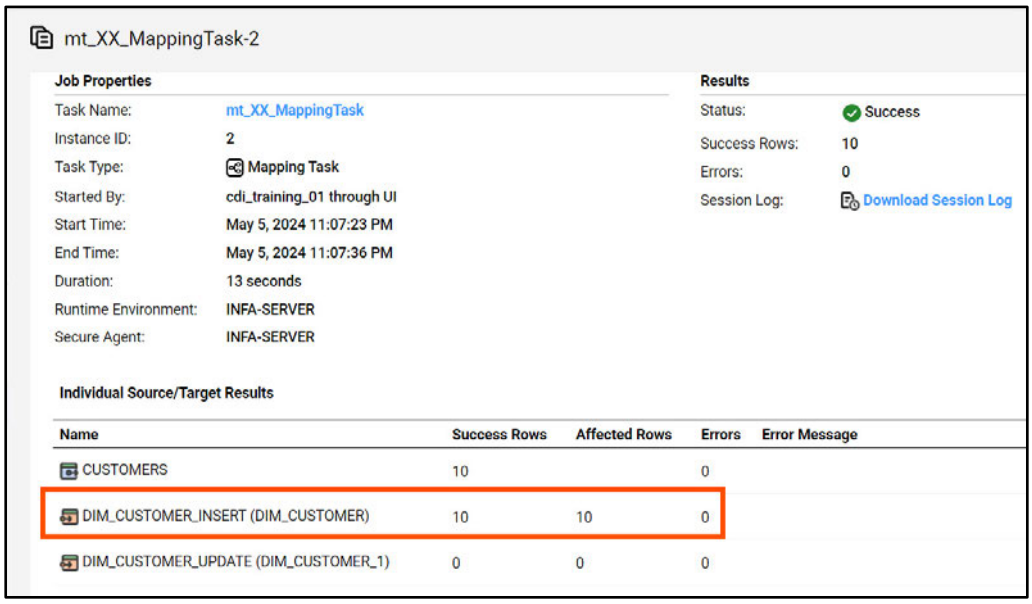
94. Monitor the task status from the **My Jobs** page.
95. When the task completes, the status changes to **Success**.



96. Click on the **Instance Name** of the job.

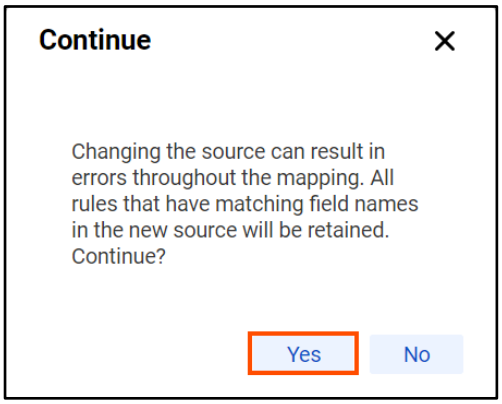


97. Scroll down to the Individual Source/Target Results section and analyze the results. Confirm that all the records have been inserted and none of them are updated (Since we began with an empty target).



98. Navigate to the BASIC_HISTORY_TRACKING mapping and select the source object **CUSTOMERS**.
99. From the properties section, go to the Source tab and change the source object to **UPDATE_CUSTOMERS.csv**.

100. Click **Yes** in the Continue window.



101. **Save** the mapping.

Note: Ensure that neither of the Target objects should have the **Truncate target** checkbox checked. Make sure that they are **UNCHECKED**.

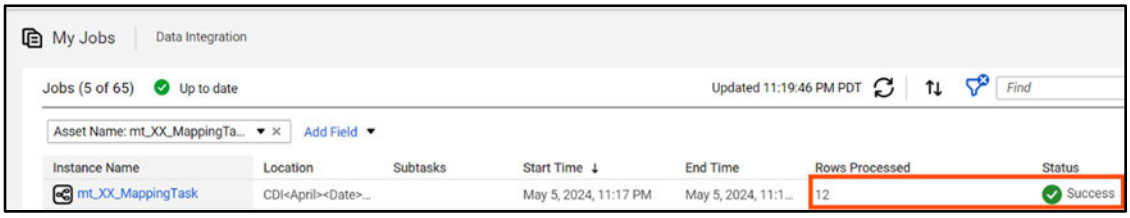
102. Navigate to the mapping task (mt_XX_MappingTask) window.

103. Run the mapping task.

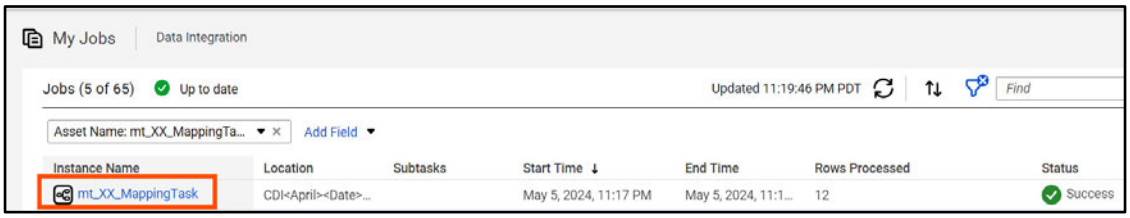
IMPORTANT: You must run the Mapping Task and not the mapping! If you run only the mapping, the Sequence Generator will start at "1" and the target database will reject all the records due to uniqueness constraint.

104. Navigate to the My Jobs page and wait till the job runs successfully.

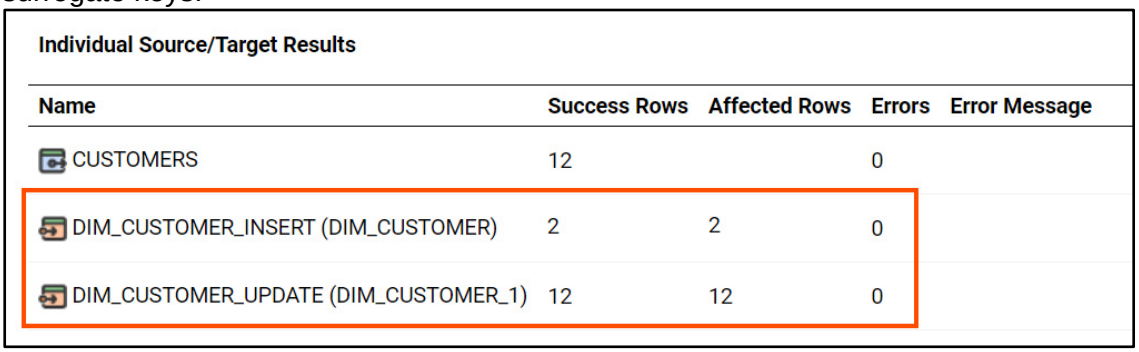
105. Observe the no.of Rows Processed.



106. Click on the Instance Name to view more information about results.



107. Under the Individual Source/Target Results section, observe the result that most of the records have been updated, while two new customers have been inserted with the new surrogate keys.

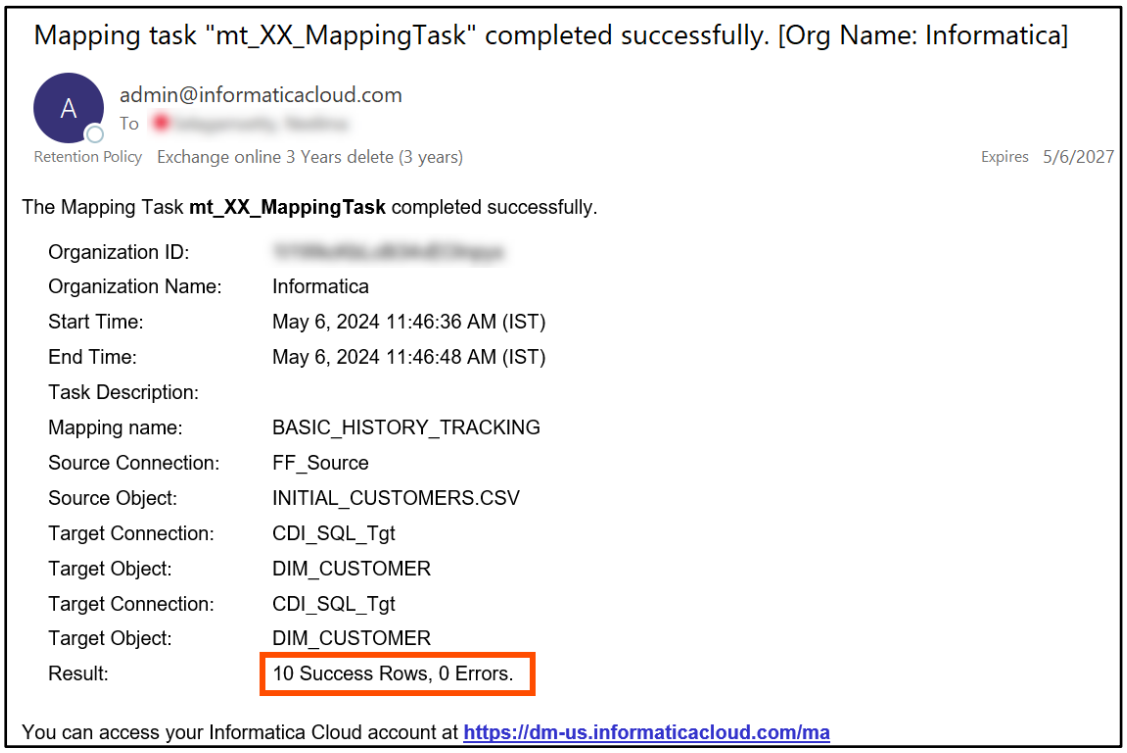


Individual Source/Target Results

Name	Success Rows	Affected Rows	Errors	Error Message
CUSTOMERS	12		0	
DIM_CUSTOMER_INSERT (DIM_CUSTOMER)	2	2	0	
DIM_CUSTOMER_UPDATE (DIM_CUSTOMER_1)	12	12	0	

108. Close all the assets from the navigation pane.

109. Also, as configured, you will get emails on the registered email address with the task details. Verify your email inbox.



Mapping task "mt_XX_MappingTask" completed successfully. [Org Name: Informatica]

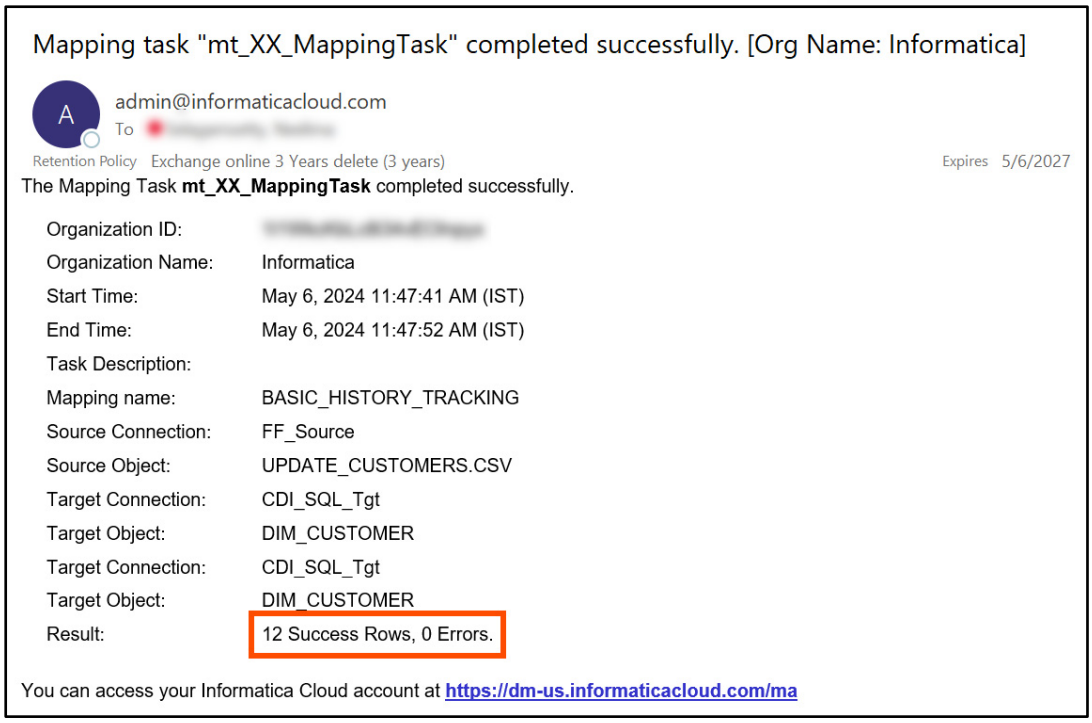
A admin@informaticacloud.com
To [redacted]

Retention Policy Exchange online 3 Years delete (3 years) Expires 5/6/2027

The Mapping Task **mt_XX_MappingTask** completed successfully.

Organization ID:	[redacted]
Organization Name:	Informatica
Start Time:	May 6, 2024 11:46:36 AM (IST)
End Time:	May 6, 2024 11:46:48 AM (IST)
Task Description:	
Mapping name:	BASIC_HISTORY_TRACKING
Source Connection:	FF_Source
Source Object:	INITIAL_CUSTOMERS.CSV
Target Connection:	CDI_SQL_Tgt
Target Object:	DIM_CUSTOMER
Target Connection:	CDI_SQL_Tgt
Target Object:	DIM_CUSTOMER
Result:	10 Success Rows, 0 Errors.

You can access your Informatica Cloud account at https://dm-us.informaticacloud.com/ma



Thus, you have built a logic to perform the basic history tracking of the data records. This technique can be useful even if there is no traditional data warehouse involved.

This concludes the lab.

Module 11: Replication Task

Lab 11-1: Replicating Data to a Flat File

Overview:
A replication task replicates data from a source to a target.

Objective:

- Create a replication task to replicate data to a CSV file

Scenario:

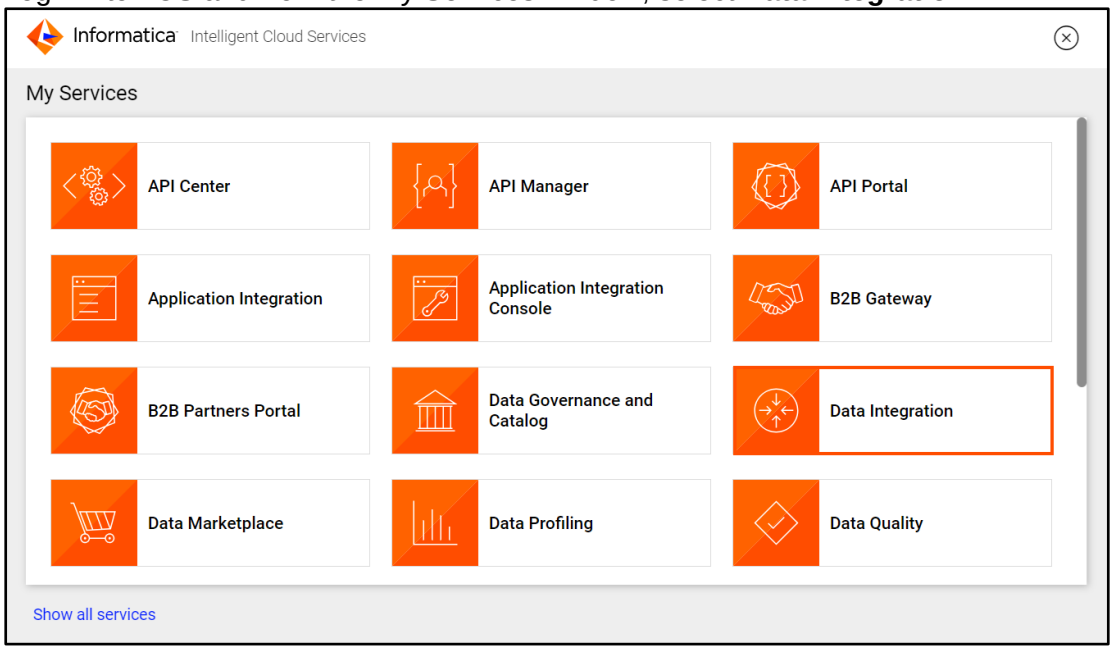
Ruby is concerned that the updated data on the Salesforce Account object is not saved anywhere for backup purposes. So, John suggests creating a replication task to load Account objects in Salesforce to a flat file.

Duration:
10 minutes

Tasks

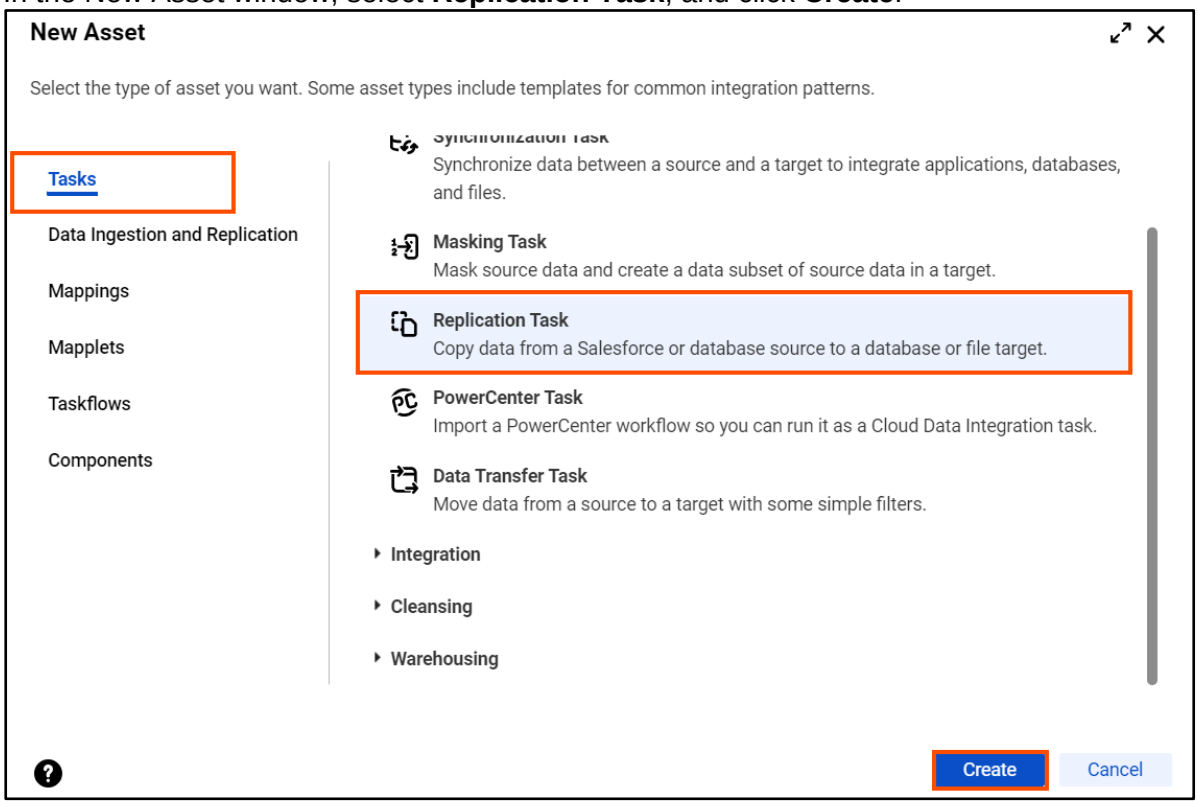
Create Replication Task

1. Login into IICS and from the My Services window, select **Data Integration**.

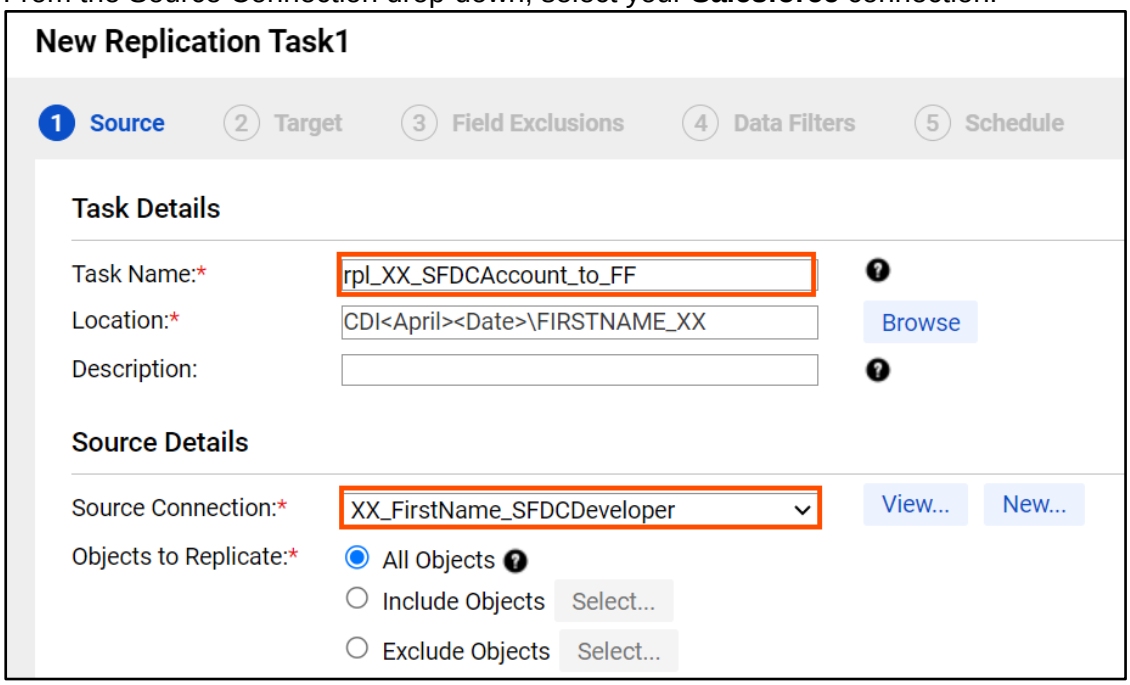


2. To create a new asset, from the navigation pane, select **New**.

3. In the New Asset window, select **Replication Task**, and click **Create**.

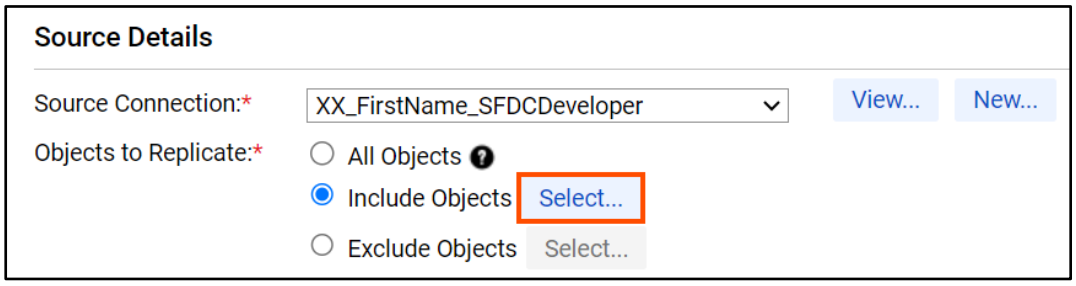


4. In the Task Name field, enter **rpl_XX_SFDCAccount_to_FF**. Replace XX with your DOB.
5. Verify that the asset Location is pointing to your working folder.
6. From the Source Connection drop-down, select your **Salesforce** connection.

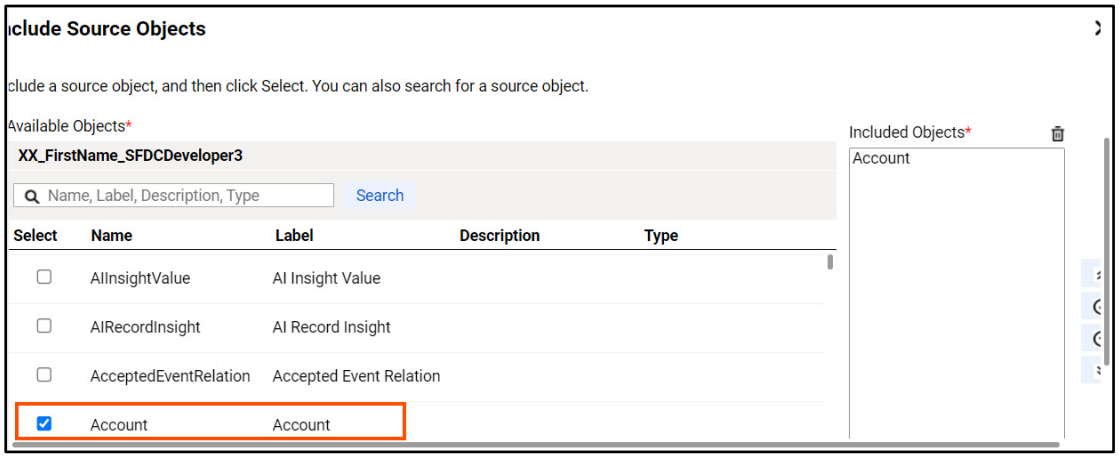


7. From the Objects to Replicate field, select **Include Objects**.

8. Click **Select**.

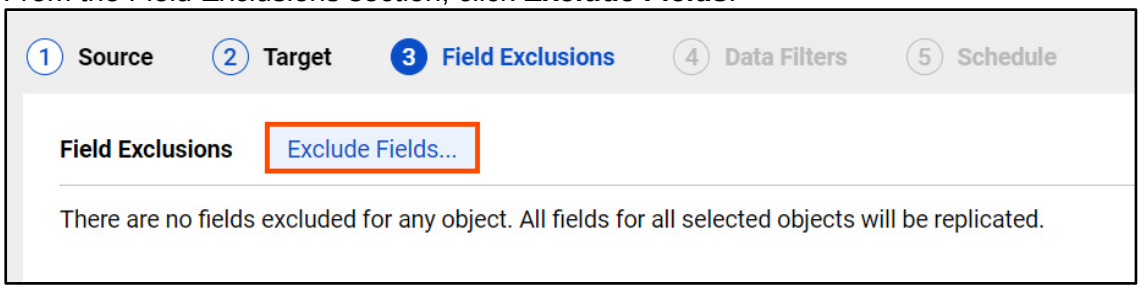


9. From the list, select **Account**.
10. Click **Select**.



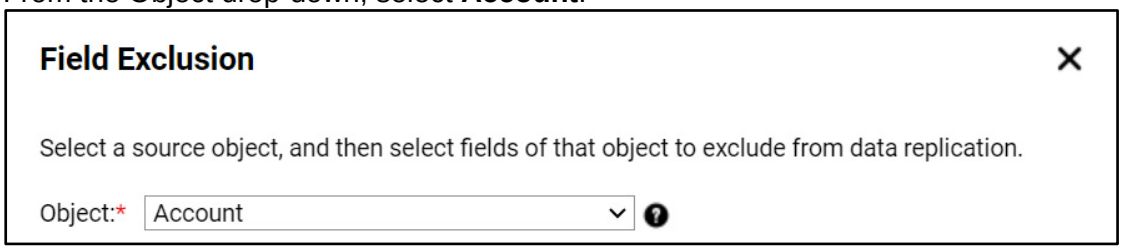
11. Click **Next**.
12. From the Connection drop-down, select your **target Flat File** connection.
13. Enter **SXX_** in the Target Prefix field.

14. Click **Next**.
15. From the Field Exclusions section, click **Exclude Fields**.



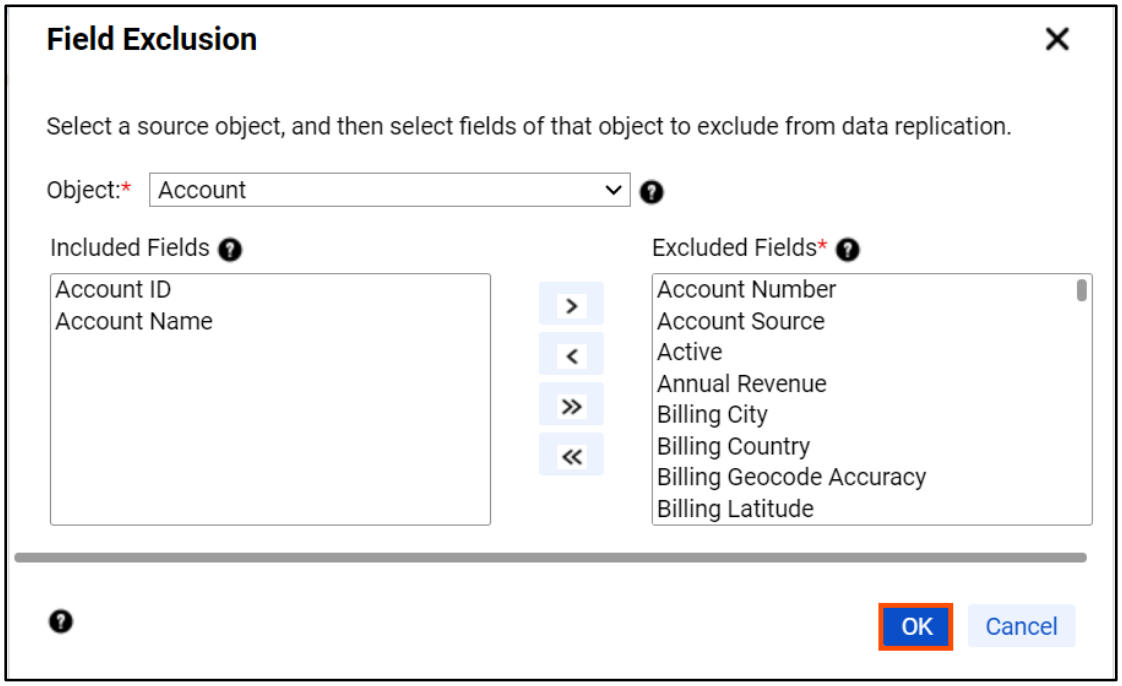
The Field Exclusion window appears.

16. From the Object drop-down, select **Account**.

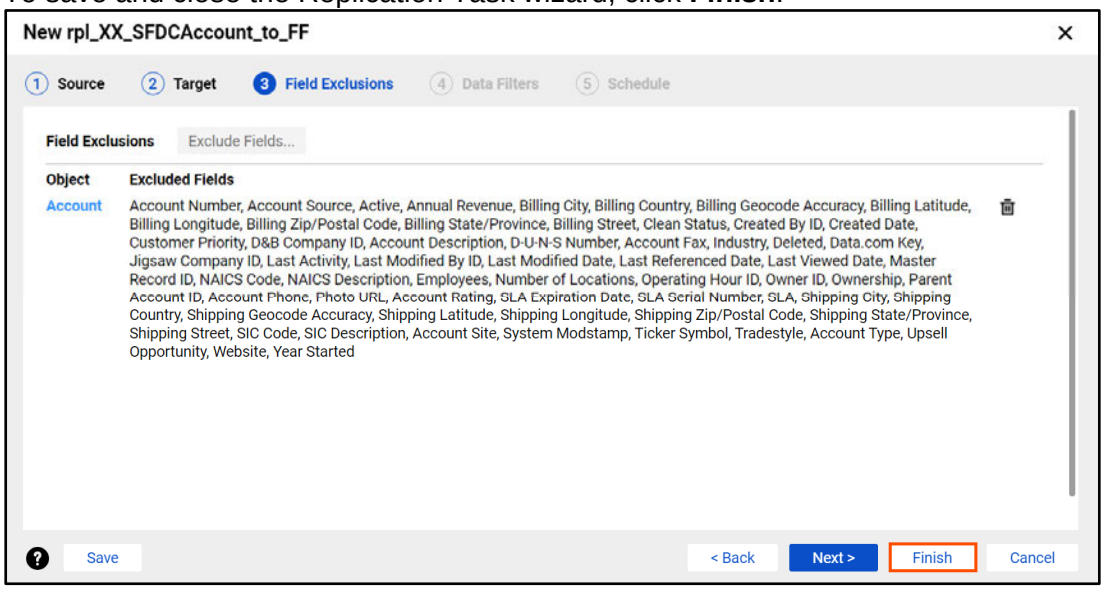


17. In the Included Fields section, retain only the **Account Name** and **Account ID**.
Note: To retain the Account Name and Account ID, select the rest of the fields, and click .

18. Click **OK**.



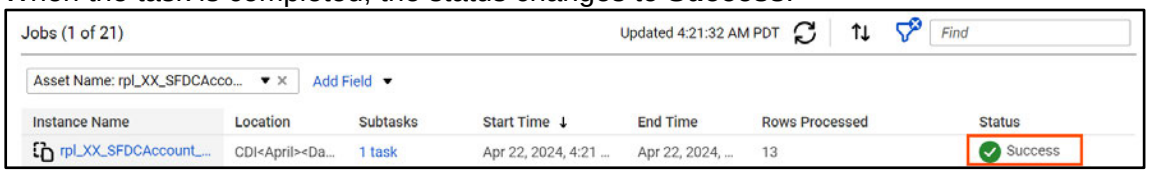
19. To save and close the Replication Task wizard, click **Finish**.



20. To run the Replication task, click **Run**.

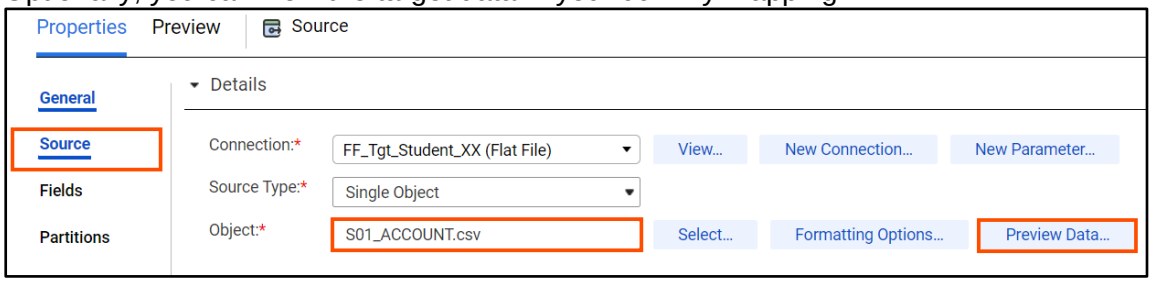
Monitor Task

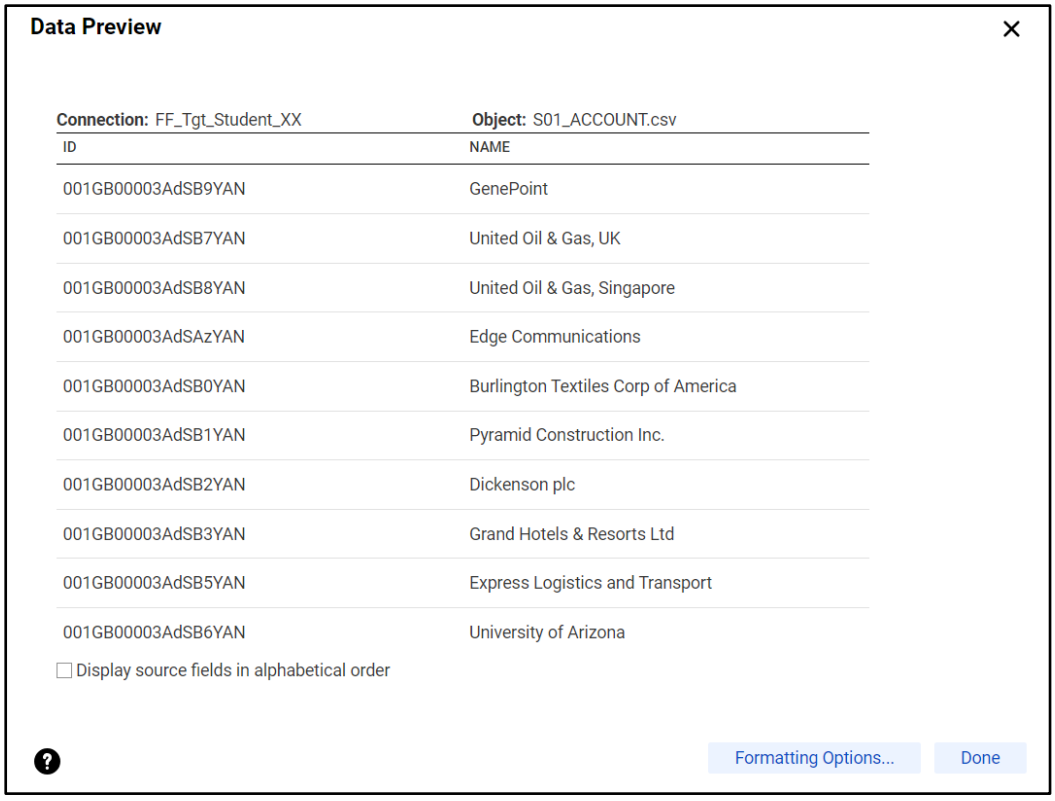
21. To monitor the task, from the navigation pane, click **My Jobs**.
22. When the task is completed, the status changes to **Success**.



Note: The number of processed rows can change depending on the data in the Salesforce Account object.

23. Close the asset from the navigation pane.
24. Optionally, you can view the target data in your dummy mapping.





This concludes the lab.

Module 12: Masking Task

Lab 12-1: Creating a Masking Task

Overview:
A masking task masks source data and creates a data subset of the source data in the target.

In this lab, you will mask the account phone number using a masking task.

Objective:

- Create a masking task to mask the phone number

Scenario:

Ruby wants to use the customer data for development and testing purposes. However, as customer information is sensitive data, John suggests using the Masking task to mask customer phone number details.

Duration:
10 minutes

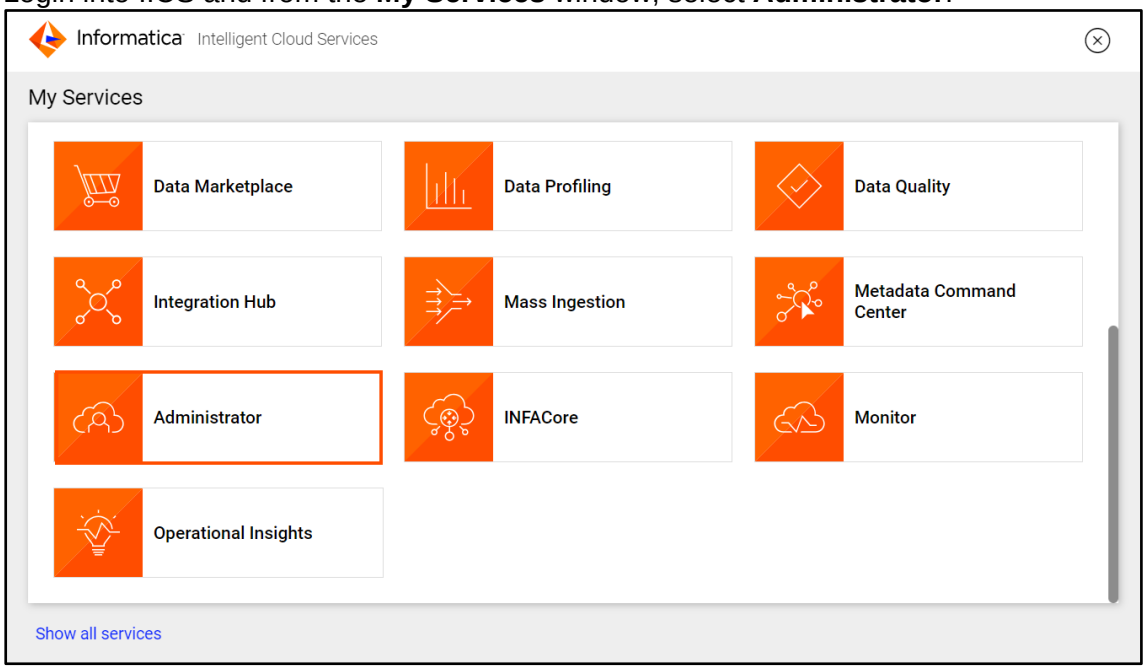
Tasks

Create Masking Task

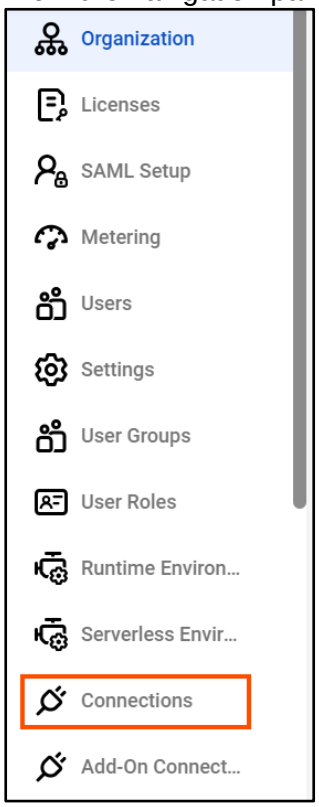
1. The following screenshot shows the data of the **Outlets.csv** file used in creating the Synchronization Task lab. Note the phone numbers for some of the accounts.

A	B	C	D	E	F	G	H	I
Outlet_ID	Outlet_Name	Phone_Number	Street	City	State	ZipCode	Country	Employees
1	NH D'needs	754-3010	11 West Park Road	Pasadena	CA	91001	USA	4
2	NH Mart	281-6000	6620 W. Broad St.	Rampart	AK	99767	USA	4
3	NH Groceries	522-8175	7227 W Harry St	Winslow	AZ	86047	USA	4
4	NH Lifestyle	864-8000	2145 Hamilton Ave	Salida	CO	81201	USA	5
5	NH Digital	314-3600	6601 Hawkinsville Rd	Milford	DE	19963	USA	5
6	NH Trends	782-9000	95 N Moorland Rd	Miami	FL	33101	USA	4
7	NH Everyday	454-3071	114 Division Ave N	Bloomigton	IN	47401	USA	4
8	NH Digiworld	547-2518	651 Holiday Dr	Pineville	LA	71359	USA	5
9	NH Supplies	612-6700	32 Old Slip	Grand Rapids	MI	49501	USA	5

2. Login into IICS and from the **My Services** window, select **Administrator**.

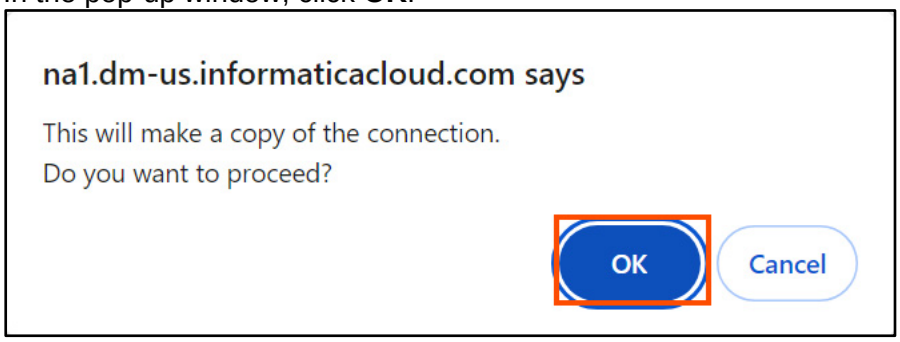


3. From the navigation pane, select **Connections**.

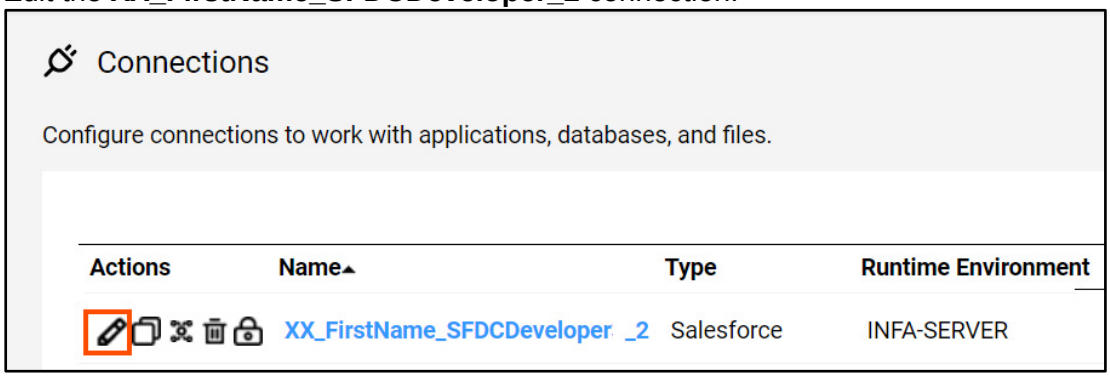


4. Locate your **Salesforce** connection and click on to make a copy of the connection.

5. In the pop-up window, click **OK**.



6. Edit the **XX_FirstName_SFDCDeveloper_2** connection.

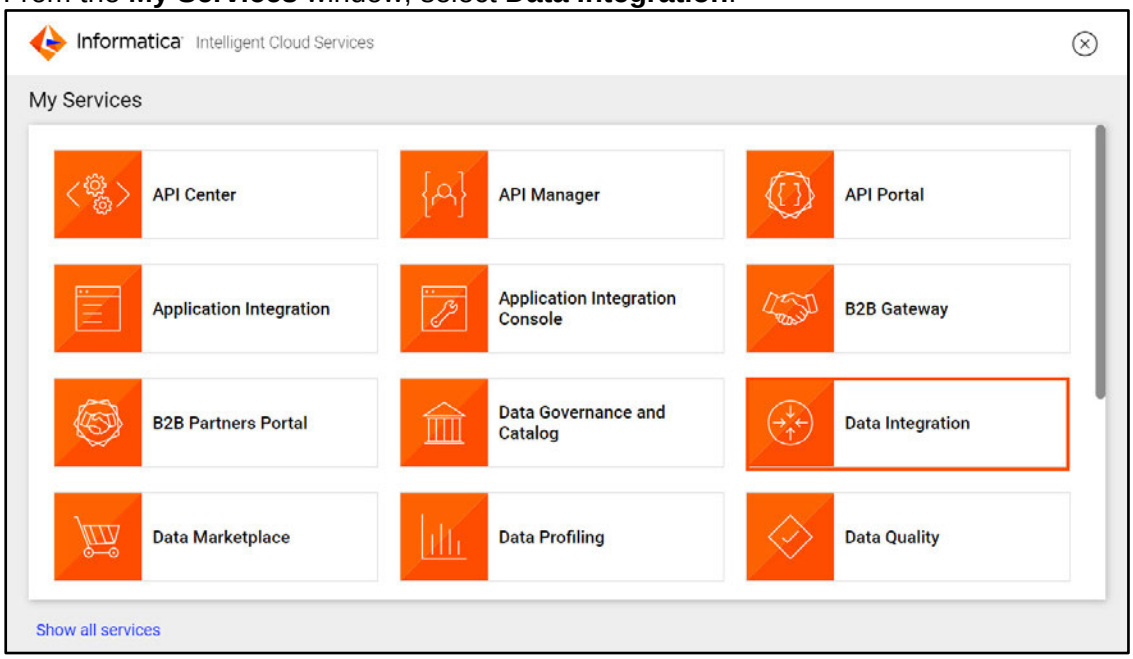


7. Update the connection name to **SXX_SFDCDeveloper**. Make sure to replace XX with your DOB.

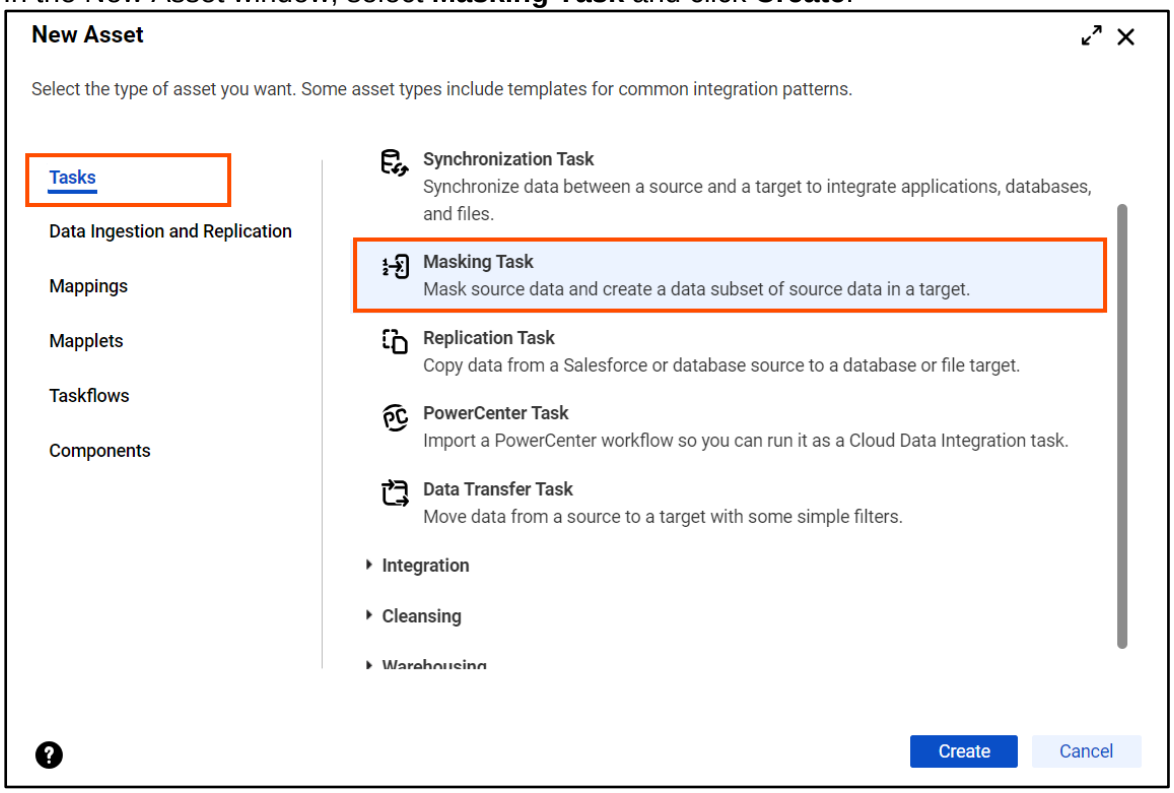
Note: When you create a Masking Task, the connection name cannot start with a number. So, you must update the connection name to use it in the Masking task.

8. **Test** the connection and **save** the changes. Make sure that the connection is valid. If you encounter any issues while testing the connection, for example: "**..username, password, or security token is not valid..**", reset your security token and specify the valid details.
9. To switch between the available services, from the toolbar, select the drop-down next to **Administrator**.

10. From the **My Services** window, select **Data Integration**.

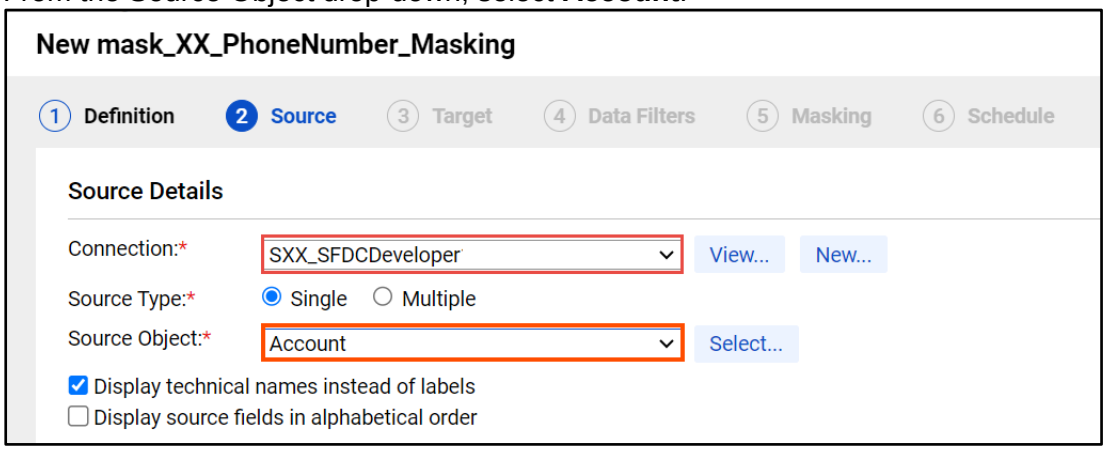


11. To create a new asset, from the navigation pane, select **New**.
12. In the New Asset window, select **Masking Task** and click **Create**.

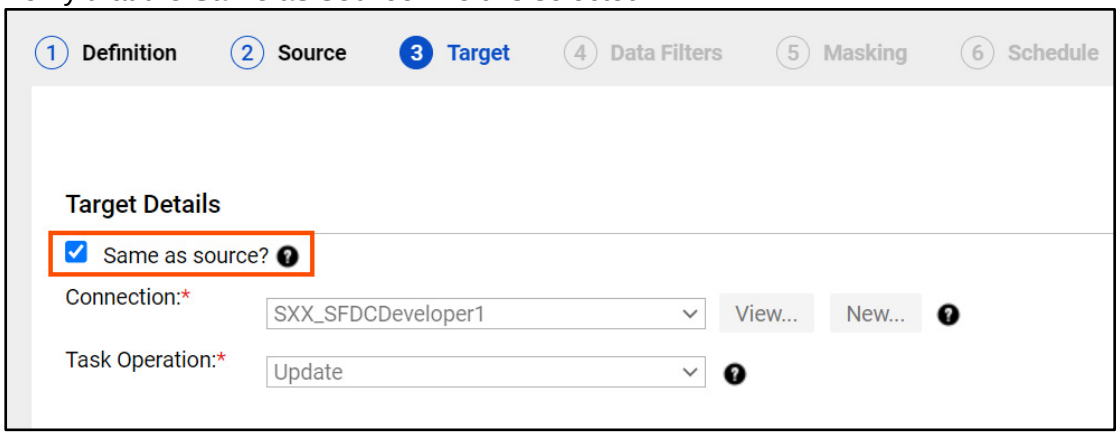


13. In the Task Name field, enter **mask_XX_PhoneNumber_Masking**. Replace XX with your DOB.
14. Verify that the asset Location is pointing to your working folder.
15. Click **Next**.

16. From the Source Connection drop-down, select your duplicate **Salesforce** connection **SXX_SFDCDeveloper**.
17. From the Source Object drop-down, select **Account**.

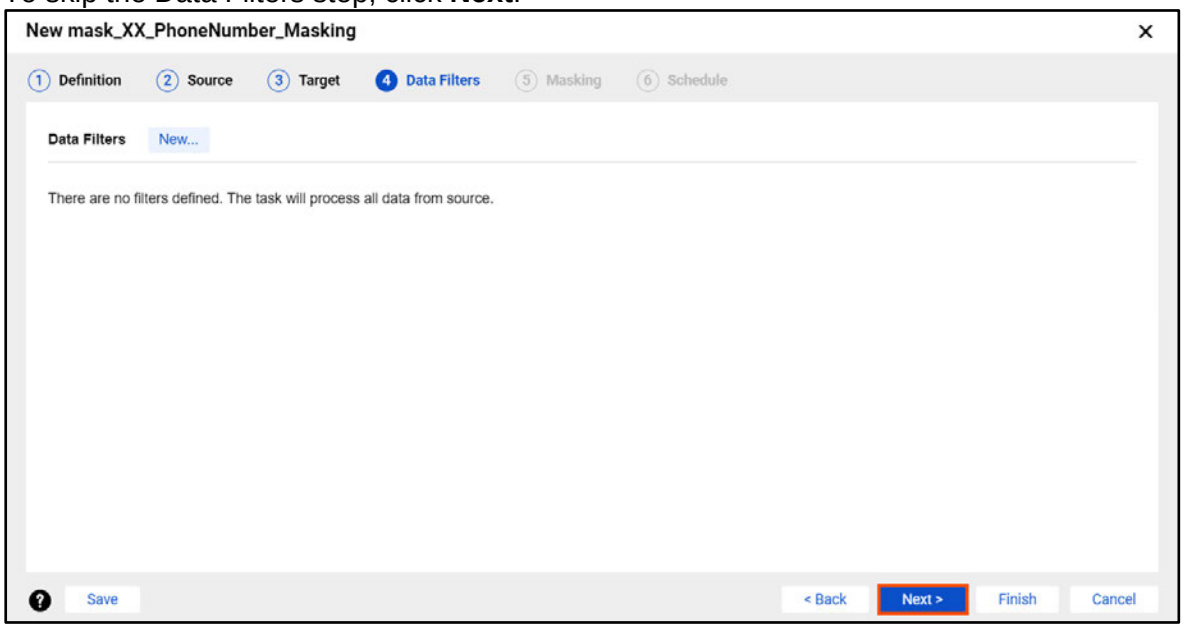


18. Click **Next**.
19. Verify that the **Same as source?** field is selected.

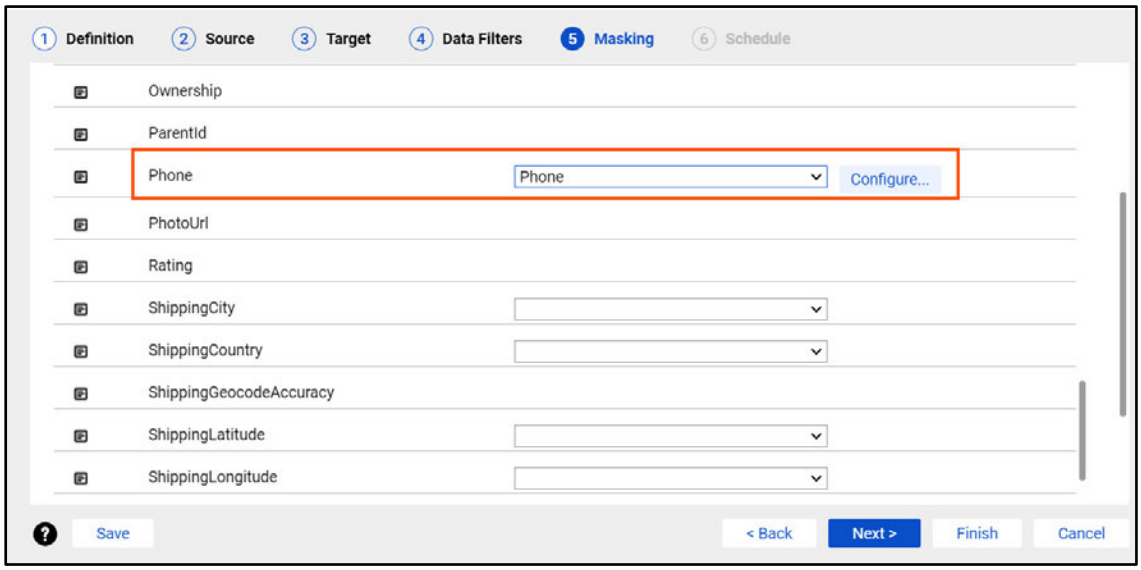


20. Click **Next**.

21. To skip the Data Filters step, click **Next**.

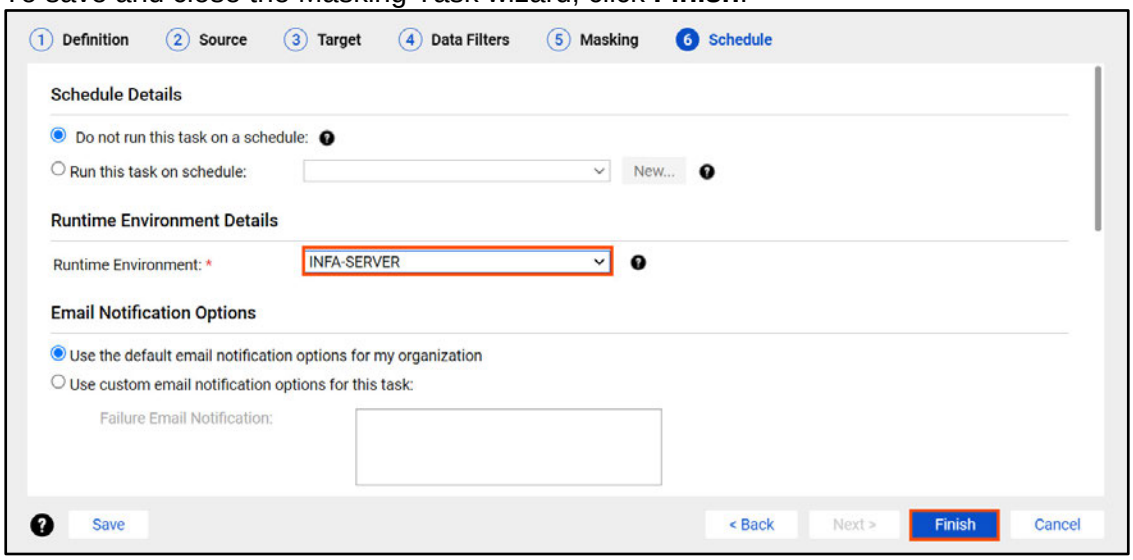


22. In the Define Masking Rules section, from the Phone field drop-down, select **Phone**.
Note: To mask all the fields of the Account object, you can use the Default Assignment option. For this lab, we will just mask the Phone field.



23. Click **Next**.
24. From the Runtime Environment drop-down, select **INFA-SERVER**.

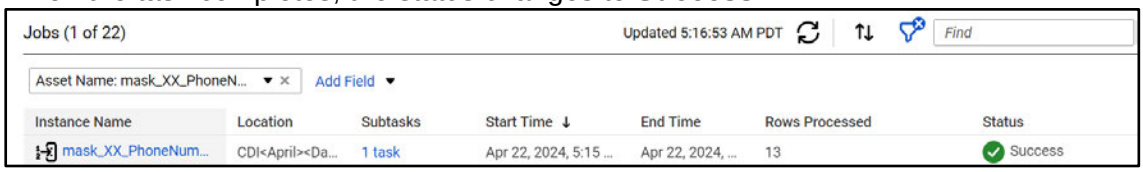
25. To save and close the Masking Task wizard, click **Finish**.



26. To run the Masking task, click **Run**.

Monitor the Status

27. To monitor the task, from the navigation pane, click **My Jobs**.
28. When the task completes, the status changes to **Success**.



Note: The number of processed rows can change depending on the data in the Salesforce Account object.

29. Close the asset from the navigation pane.

Examine Results

30. Log in to your Salesforce Developer account using your credentials.
Note: You can use the below mentioned link to login into Salesforce:
https://login.salesforce.com/
31. On the Salesforce homepage, from the available tabs, select **Accounts**.

32. Observe the phone number for accounts noted earlier.
33. Verify that the phone numbers have been masked.

This concludes the lab.

Module 12: Masking Task

Lab 12-2: Using Data Masking Transformation in a Mapping

Overview:
Data Masking transformation changes the sensitive production data into realistic test data for non-production environments. The Data Masking transformation modifies source data based on masking rules that you configure for each column.

Objective:

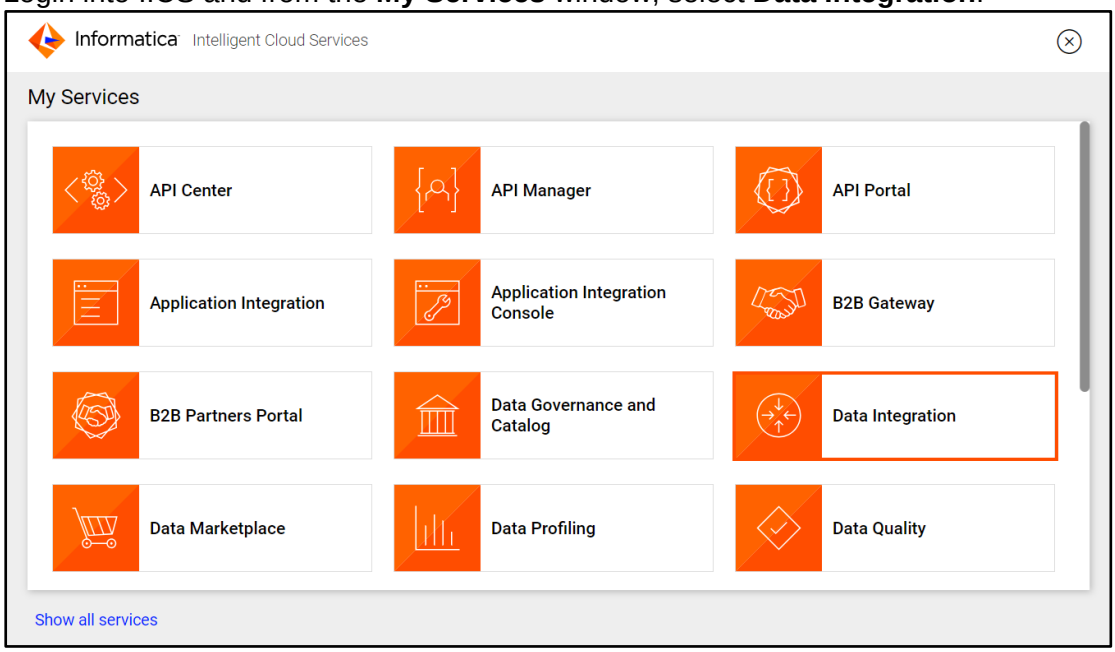
- Configure a mapping using Data Masking transformation
- Mask the Surname, DOB, and State field values in the source data

Duration:
10 minutes

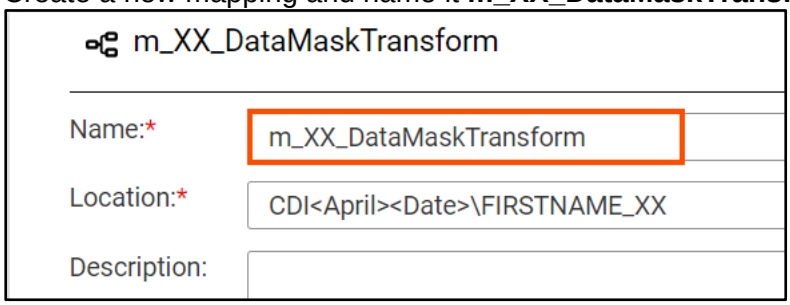
Tasks

Create Mapping

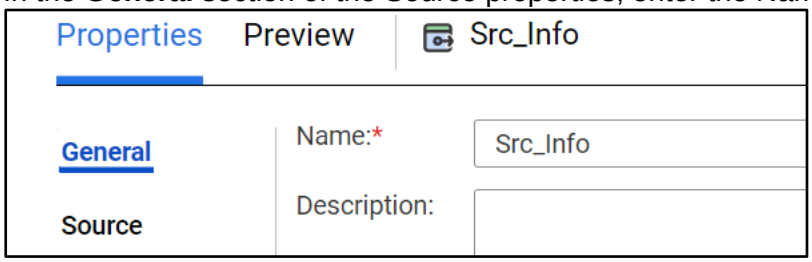
1. Login into IICS and from the **My Services** window, select **Data Integration**.



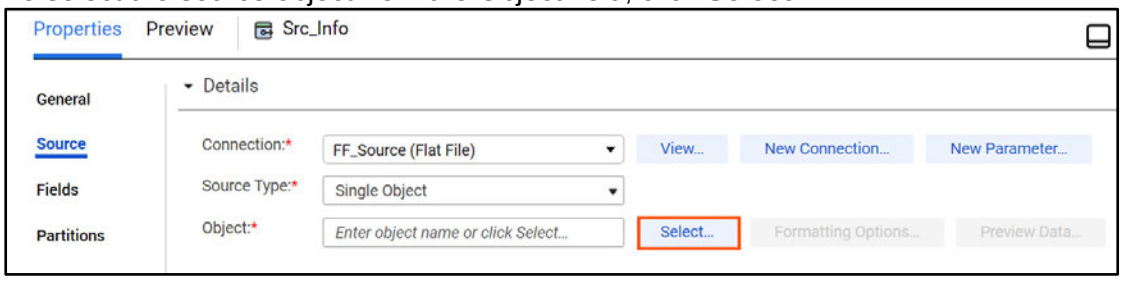
2. Create a new mapping and name it **m_XX_DataMaskTransform**.



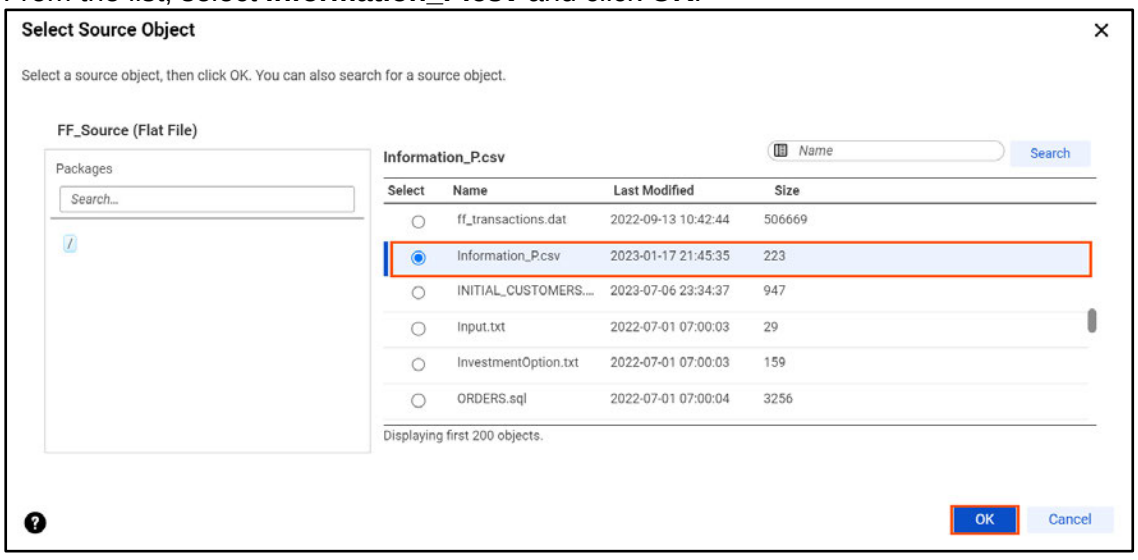
3. To configure the source, from the mapping canvas, click the **Source** transformation.
4. In the **General** section of the Source properties, enter the Name as **Src_Info**.



5. From the properties pane, click **Source**.
6. From the Connection drop-down, select **FF_Source**.
7. To select the source object from the Object field, click **Select**.

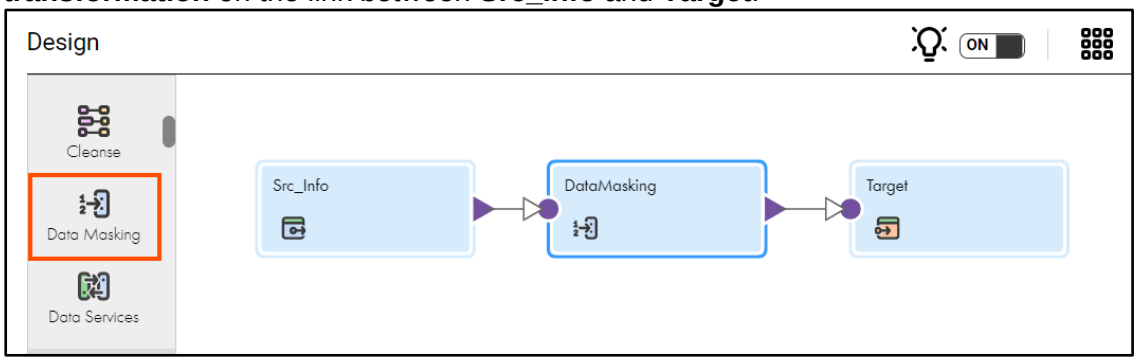


8. From the list, select **Information_P.csv** and click **OK**.

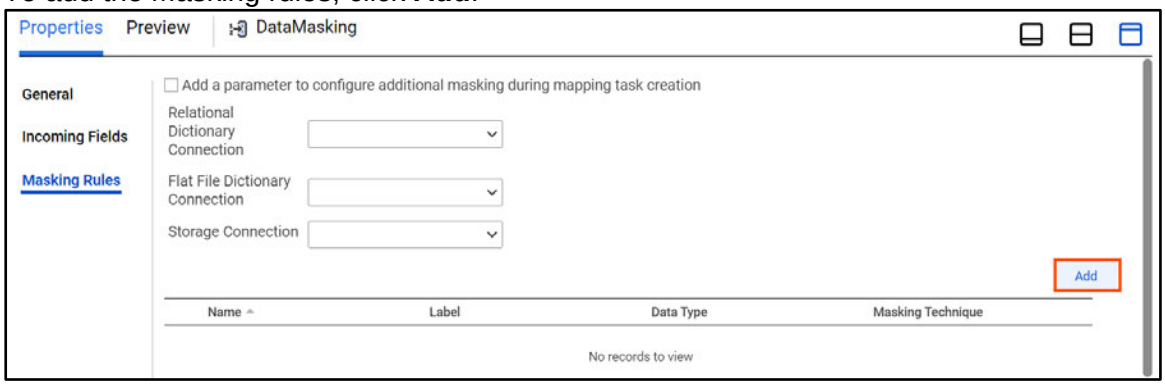


Add Data Masking Transformation

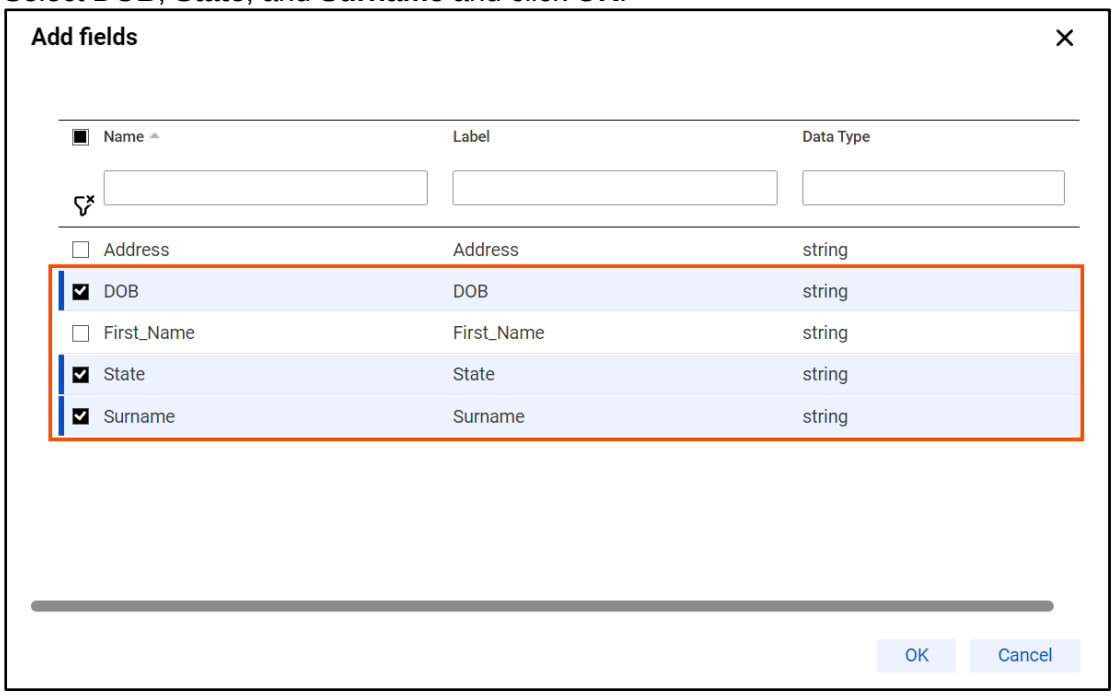
9. From the list of available transformations, drag and drop the **Data Masking transformation** on the link between **Src_Info** and **Target**.

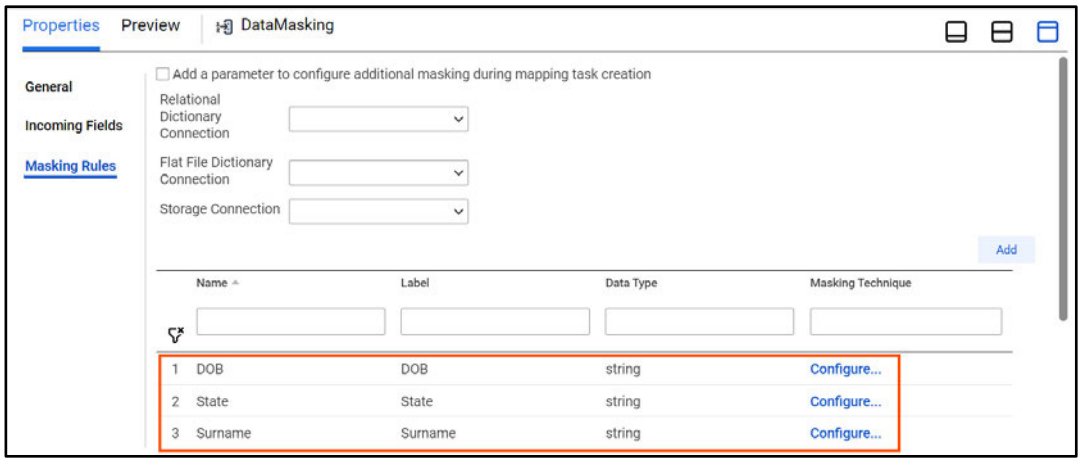


10. Select the **DataMasking** transformation from the mapping canvas.
11. From the properties pane, click **Masking Rules**.
12. To add the Masking rules, click **Add**.

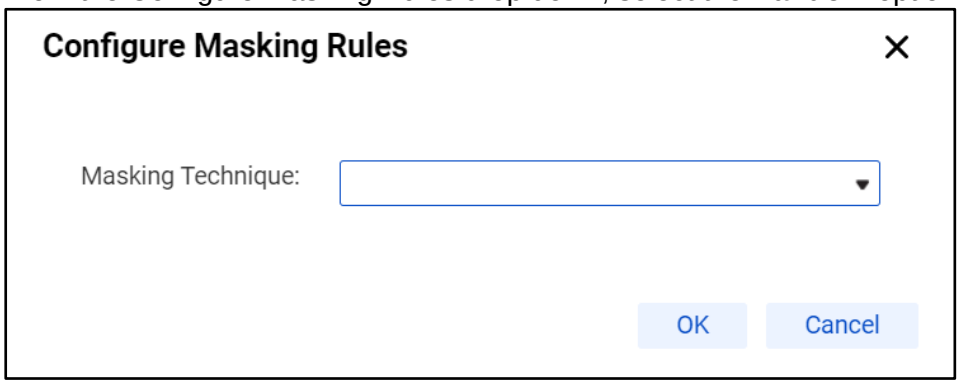


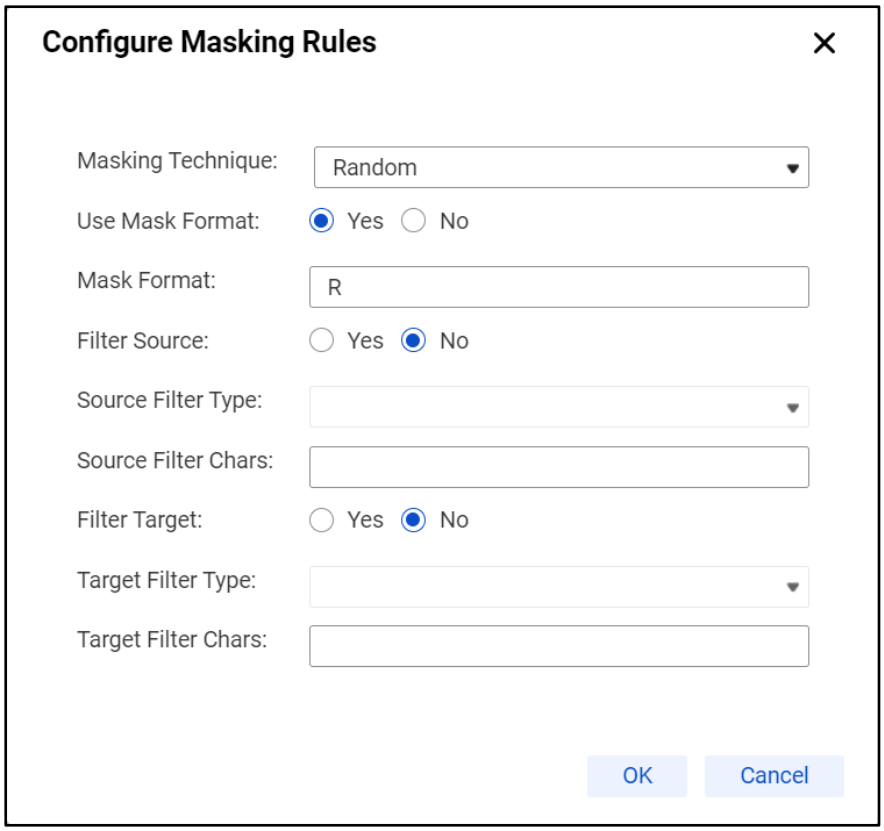
13. Select **DOB**, **State**, and **Surname** and click **OK**.





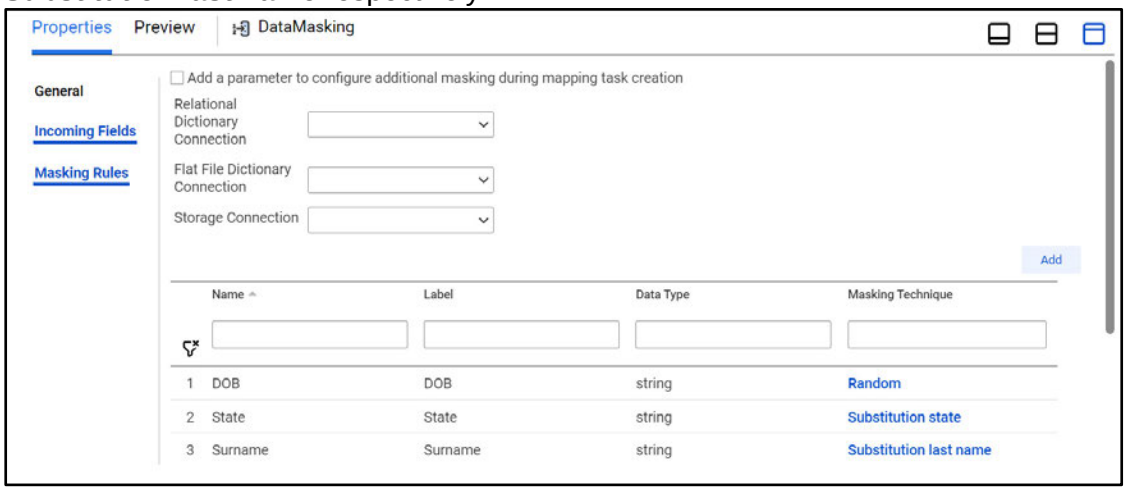
14. To configure the DOB, click **Configure**.
15. From the Configure Masking Rules drop down, select the **Random** option





Note: The masking technique is not available in the trial version. To enable this feature, you will need to purchase a license.

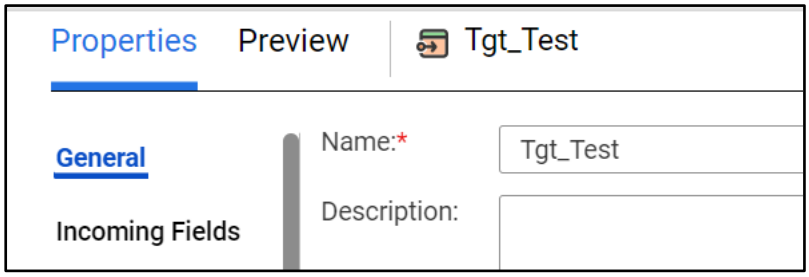
16. Review all the other default settings and click **OK**.
17. Similarly, configure the **State** and **Surname** fields, by selecting **Substitution state** and **Substitution last name** respectively.



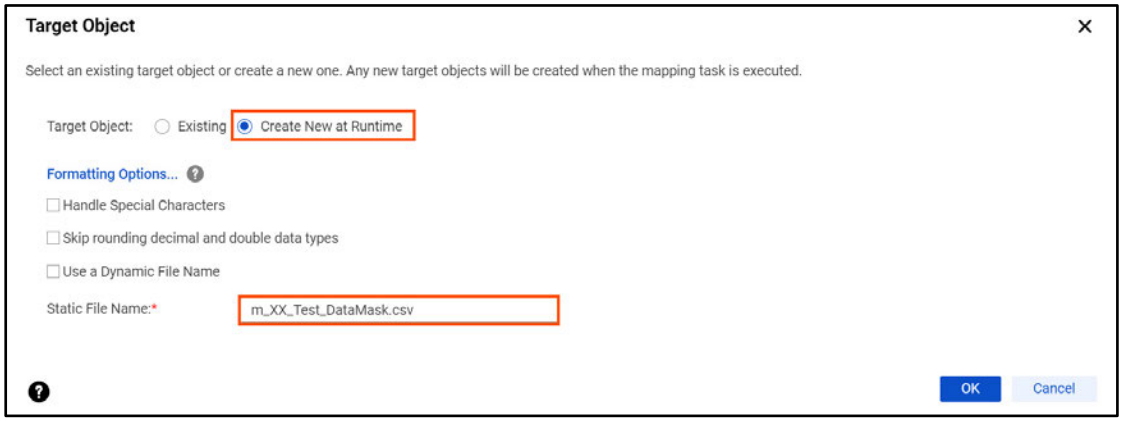
Configure Target Transformation

18. Link the **Data Masking** transformation with the **Target** transformation if it is not linked.
19. Select the **Target** transformation from the mapping canvas.

20. In the General section of Target properties, enter Name as **Tgt_Test**.



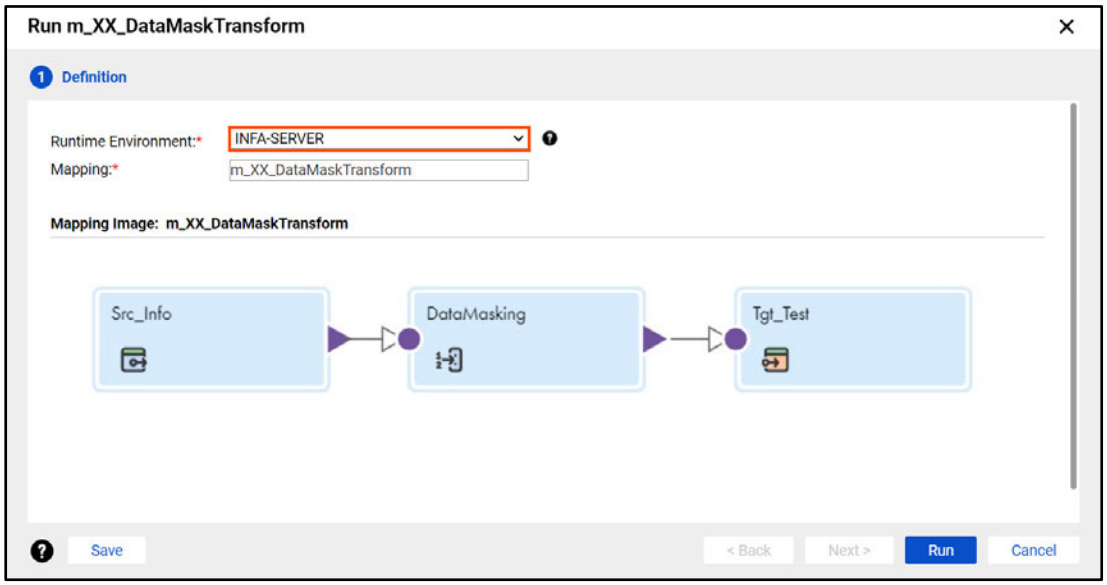
21. From the properties pane, select **Target**.
22. From the Connection drop-down, select your **target Flat File** connection.
23. To choose an object, click **Select**.
24. In the Objects fields, select **Create New at Runtime**.
25. In the File Name, enter the name as **m_XX_Test_DataMask.csv** and click **OK**.



26. Save and run the mapping.

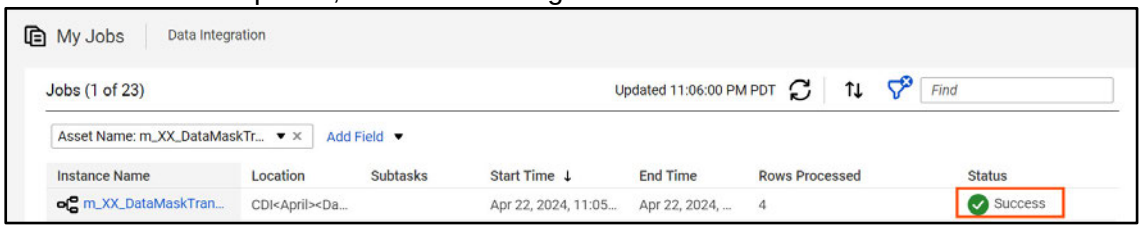
27. From the Runtime Environment drop-down, select **INFA-SERVER**.

28. Click **Run**.

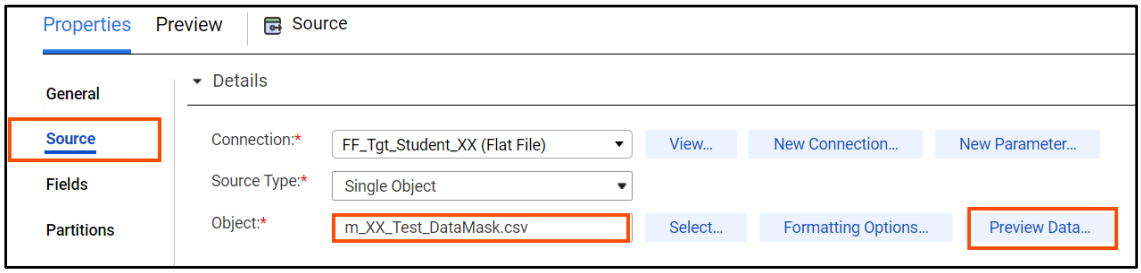


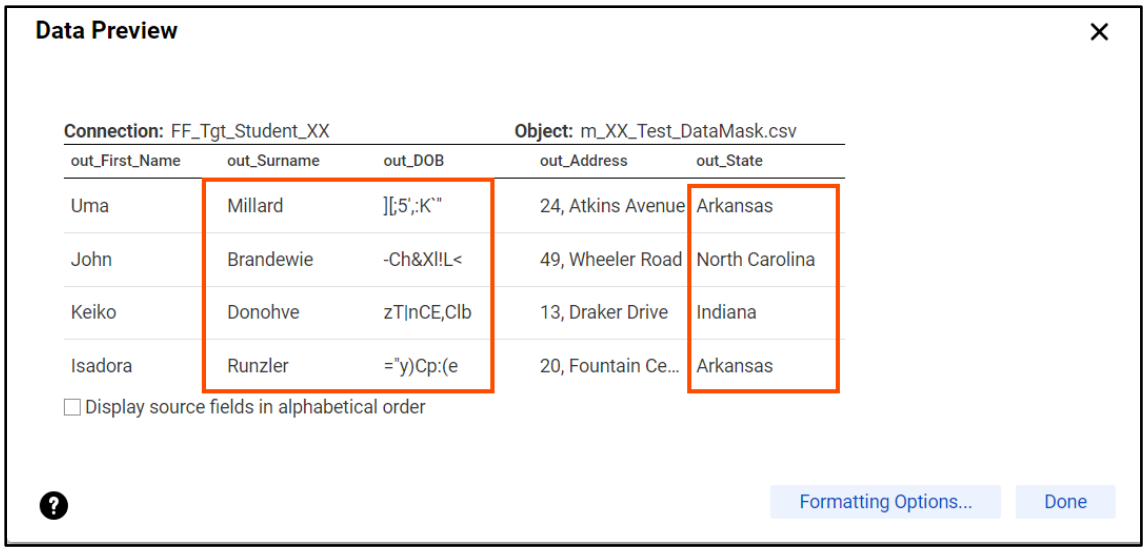
Monitor Status

29. Monitor the task status from the **My Jobs** page.
30. When the task completes, the status changes to **Success**.



31. Optionally, you can view results using your dummy mapping.





32. Save and close all the assets from the navigation pane.

This concludes the lab.

Module 13: Taskflows

Lab 13-1: Creating a Linear Taskflow

Overview:
A Taskflow enables you to add multiple data integration tasks and run them in a specific sequence or in parallel.

In this lab, you will create Linear Taskflow to group multiple data integration tasks and run them in the order that is specified.

Objective:

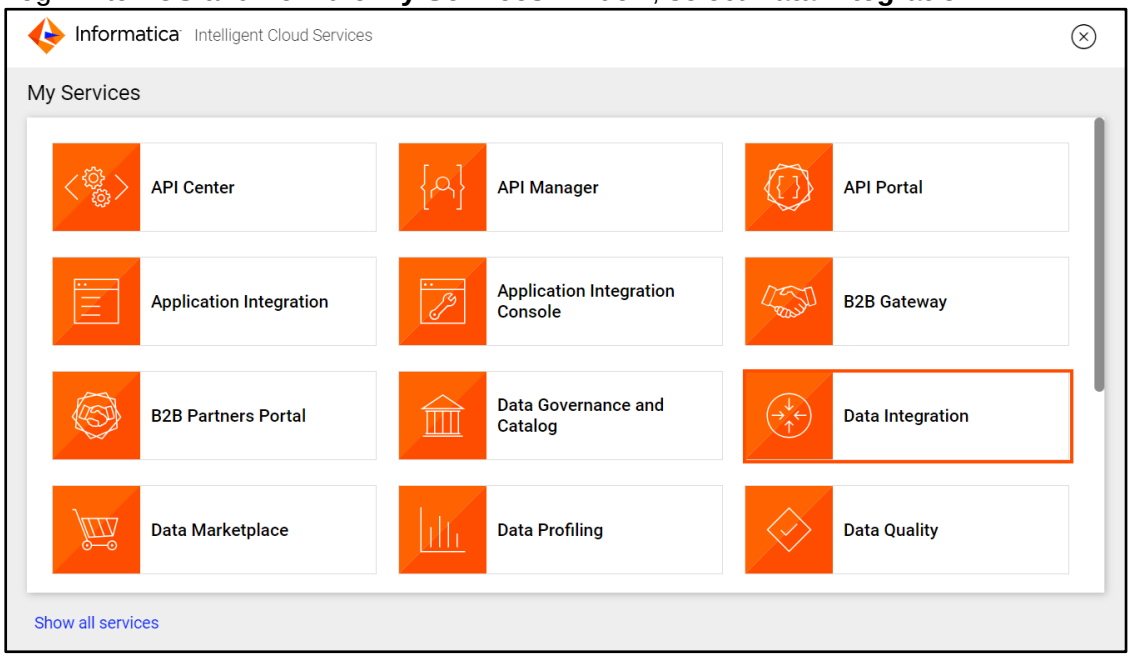
- Create a Linear taskflow
- Configure email notifications for Linear taskflow

Duration:
20 minutes

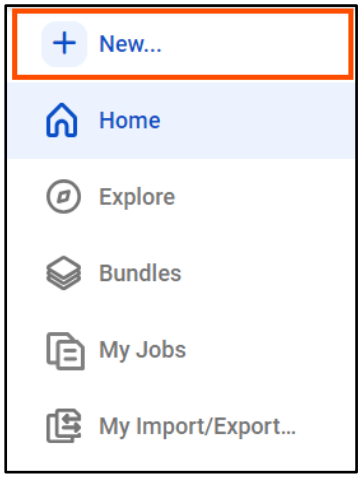
Tasks

Create Taskflow

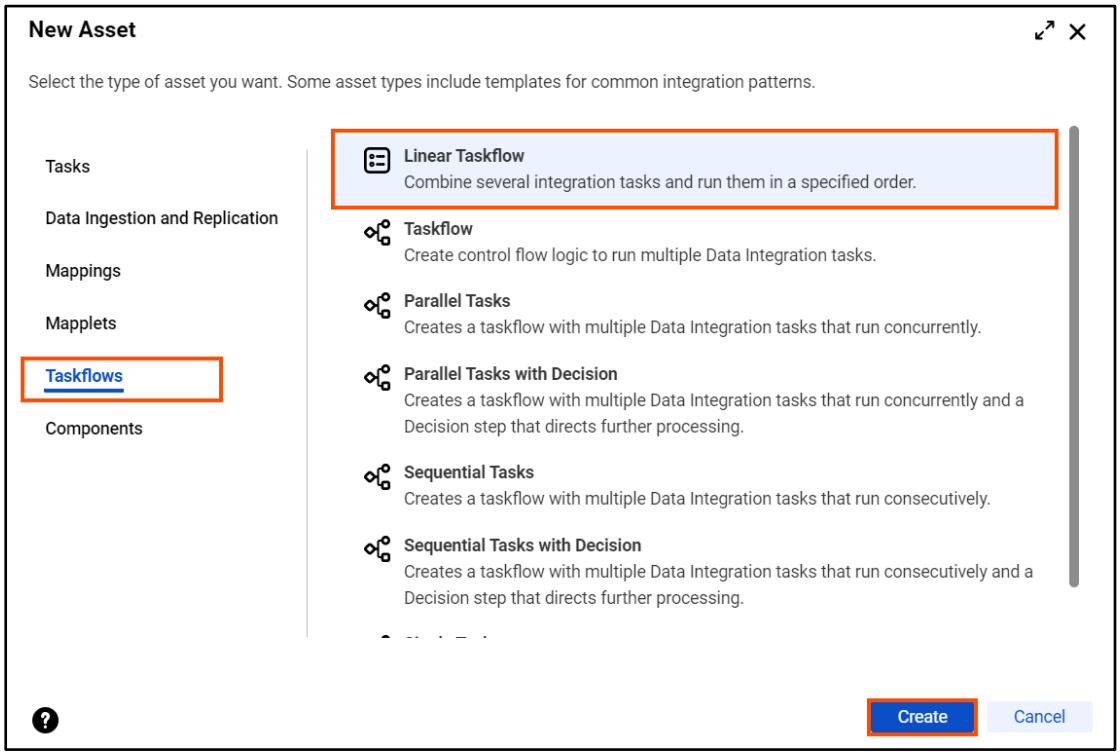
1. Login into IICS and from the **My Services** window, select **Data Integration**.



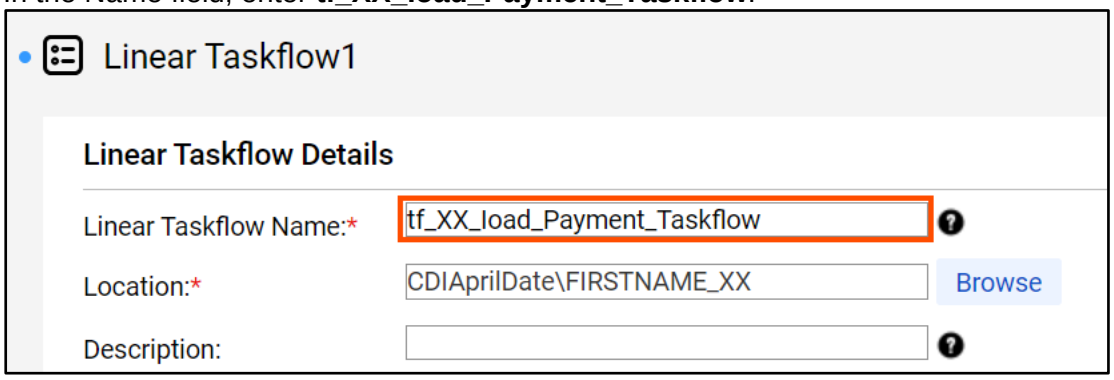
2. From the navigation pane, select **New**.



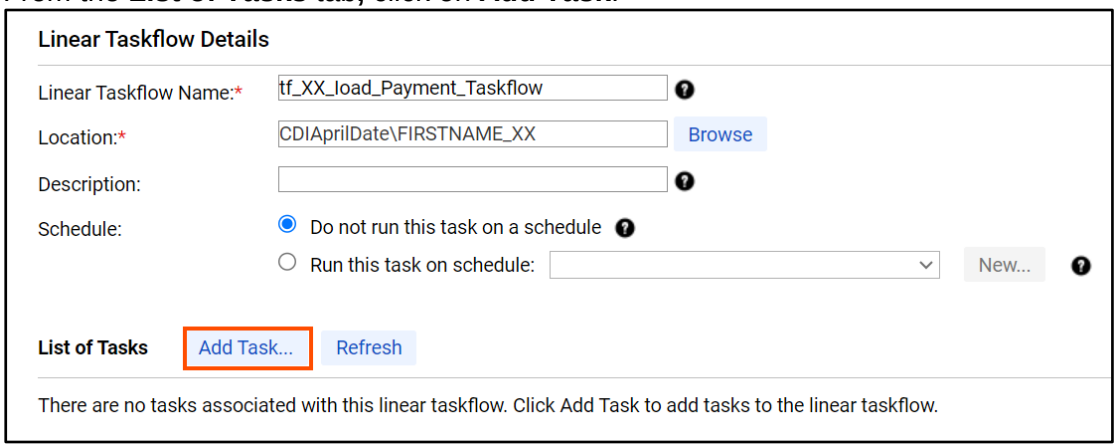
3. From the New Asset window, click the **Taskflows** tab.
4. Select **Linear Taskflow** and click **Create**.



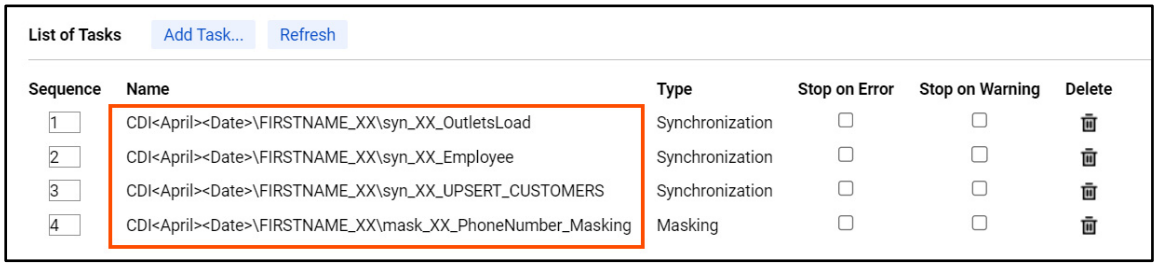
5. In the Name field, enter **tf_XX_load_Payment_Taskflow**.



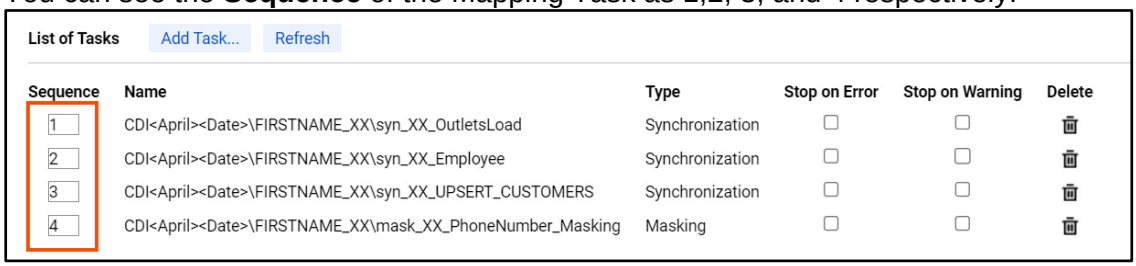
6. Verify that the asset Location is pointing to your working folder.
7. From the **List of Tasks** tab**,** click on **Add Task**.



8. Go to your work folder. You will add the following pre-created mapping tasks from this folder:
   - syn_XX_OutletsLoad
   - syn_XX_Employee
   - syn_XX_UPSERT_CUSTOMERS
   - mask_XX_PhoneNumber_Masking
9. Select the task **syn_XX_OutletsLoad** and click **Select**.
10. Click on **Add Task** again and go to your working folder.
11. Select the task **syn_XX_Employee** and click **Select**.
12. Similarly, add other tasks – **syn_XX_UPSERT_CUSTOMERS** and **mask_XX_PhoneNumber_Masking**.

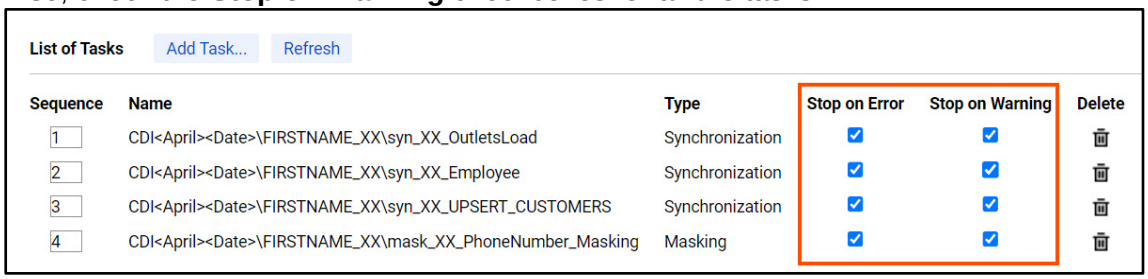


13. You can see the **Sequence** of the Mapping Task as 1,2, 3, and 4 respectively.

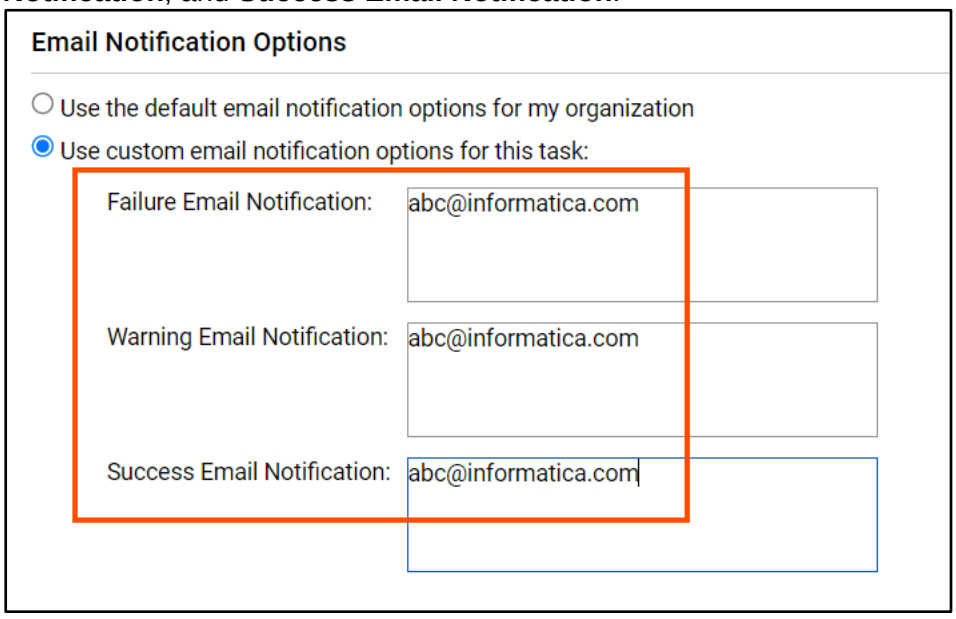


Note: You can change the Sequence number of the Mapping Tasks to any other order of your preference.

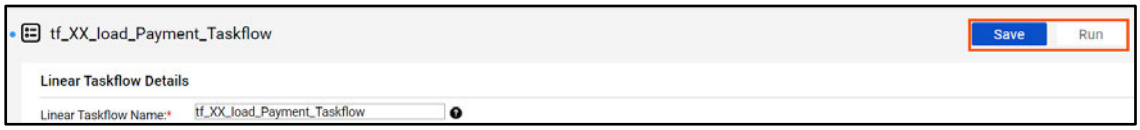
14. Check the **Stop on Error** checkboxes for all the 4 tasks.
15. Also, check the **Stop on Warning** checkboxes for all the tasks.



16. From the Email Notification Options, select the **Use custom email notification options for this task** option.
17. Enter your valid email address in the **Failure Email Notification**, **Warning Email Notification**, and **Success Email Notification**.

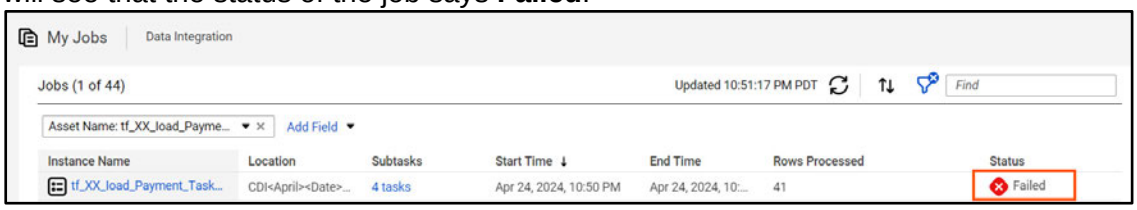


18. Save and run the Linear Taskflow.



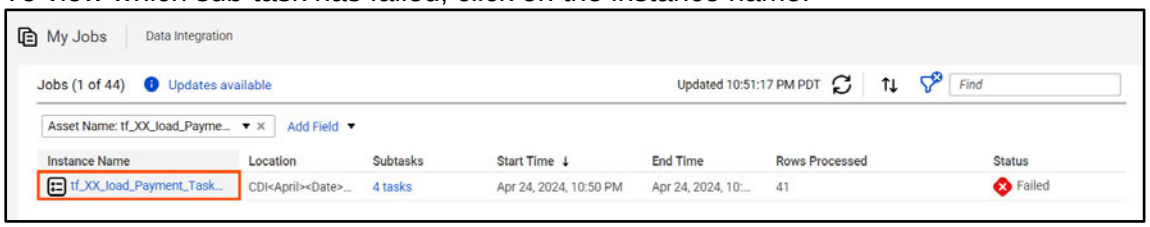
Monitor the Status

19. To monitor the task, from the navigation pane, click **My Jobs**.
20. It may take 1 to 2 minutes for the task to complete. When the task run is complete, you will see that the status of the job says **Failed**.

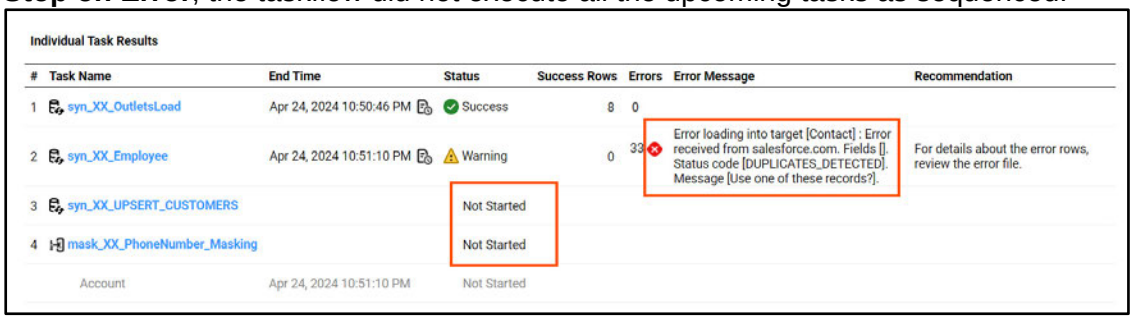


Note: You can refresh the page if the status does not change automatically.

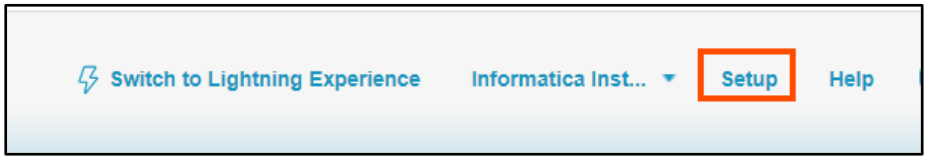
21. To view which sub-task has failed, click on the instance name.



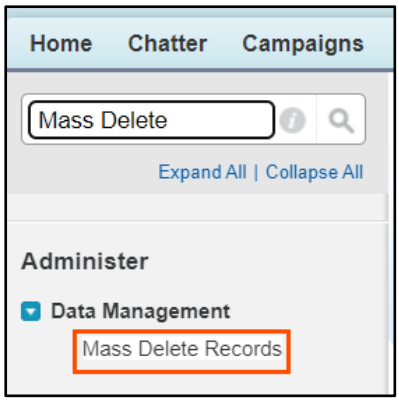
22. Scroll down and observe that the error has occurred because of loading duplicate records into the salesforce object **Contact** (as you have previously run the task and loaded data into the Contact object). Also, note that, as you selected the checkbox – **Stop on Error**, the taskflow did not execute all the upcoming tasks as sequenced.



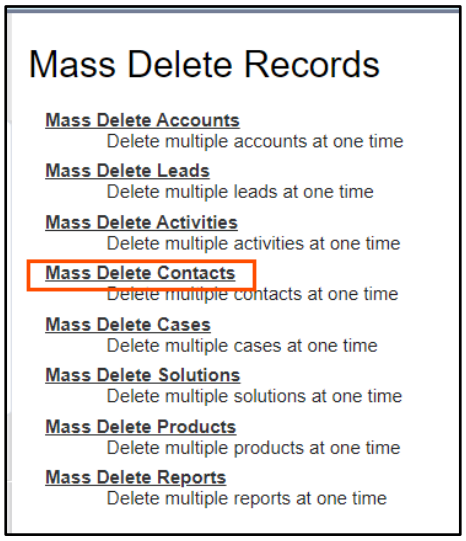
23. To troubleshoot the issue, you need to clear all the contacts records from Salesforce. Login into your Salesforce account and go to the **Setup** page.



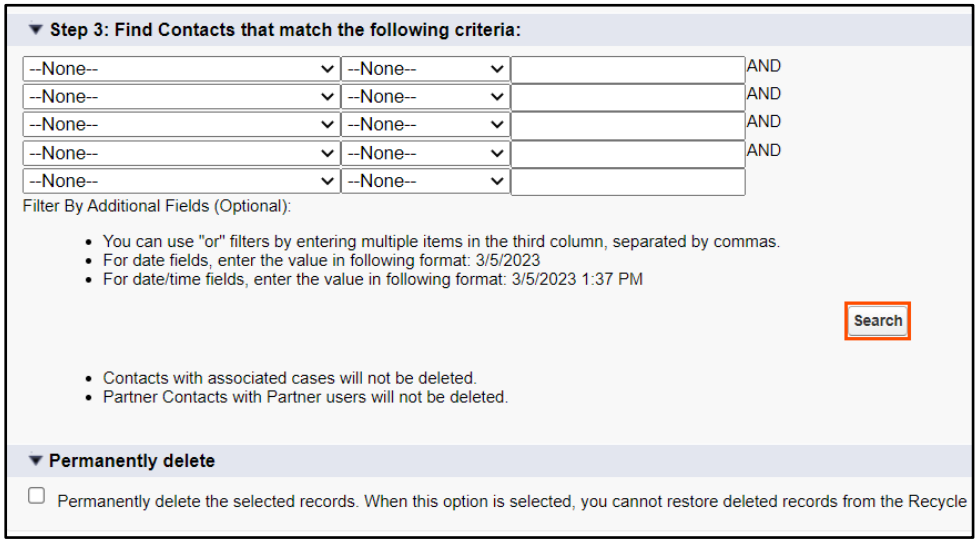
24. From the Search box on your left, enter Mass Delete and click the **Mass Delete Records** section.



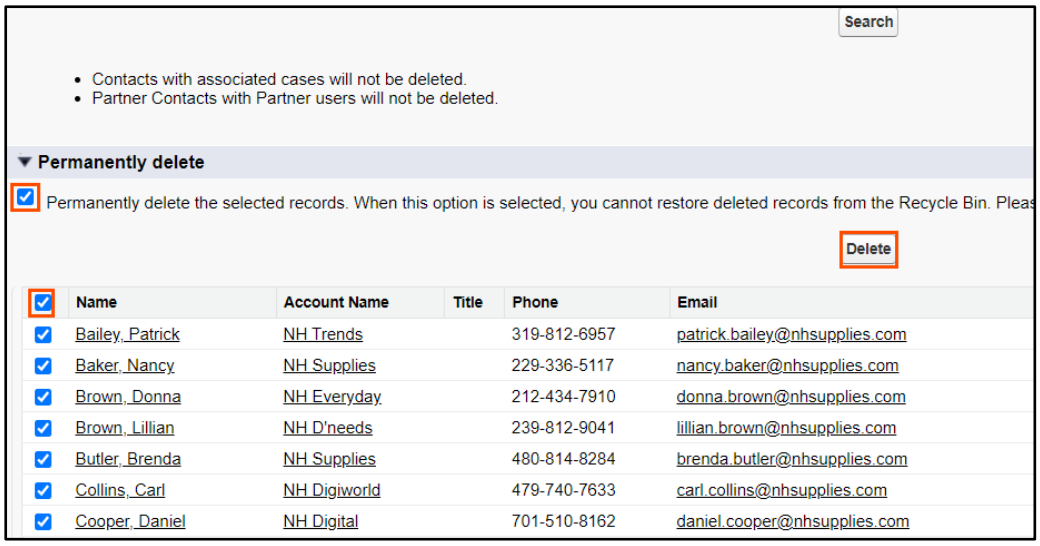
25. Click on the **Mass Delete Contacts** link.



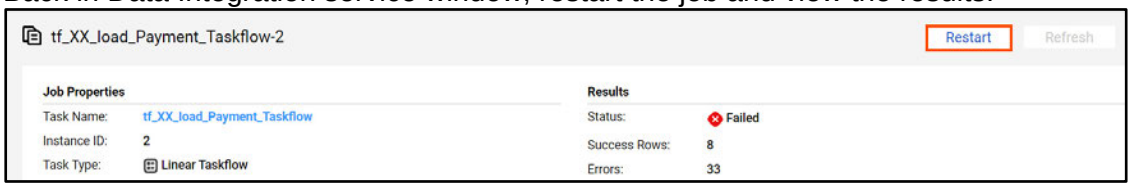
26. Scroll down and click **Search**.

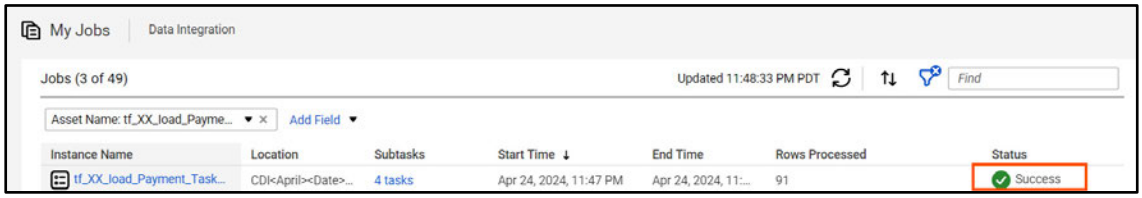


27. Now you will see all the records that belong to Contacts. Select all the records including the permanent delete checkbox and click **Delete**.

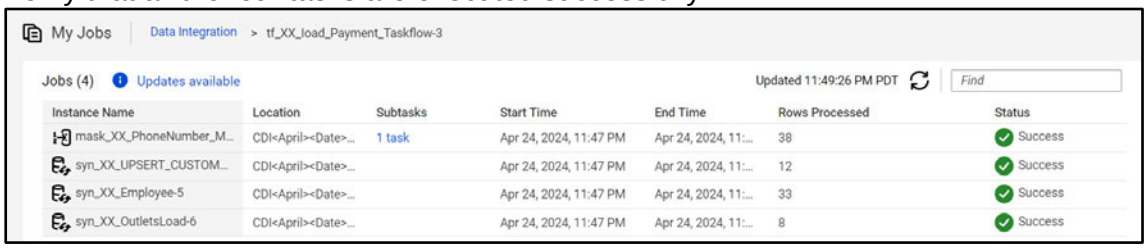


28. Back in Data Integration service window, restart the job and view the results.





29. To view the status of the subtask, click on **4 tasks**.
30. Verify that all the four tasks are executed successfully.



31. Open your email inbox and check the task notification emails. It may take 2 to 3 minutes for the email notifications to initiate.
32. Close the assets from the navigation pane.

This concludes the lab.

Module 13: Taskflows

Lab 13-2: Creating a Taskflow with Conditional Logic

Overview:
A Taskflow enables you to add multiple data integration tasks and run them in a specific sequence or in parallel.

Use a taskflow to control the execution sequence of a data transfer task, dynamic mapping task, mapping task, PowerCenter task, or synchronization task based on the output of the previous task.

Objective:

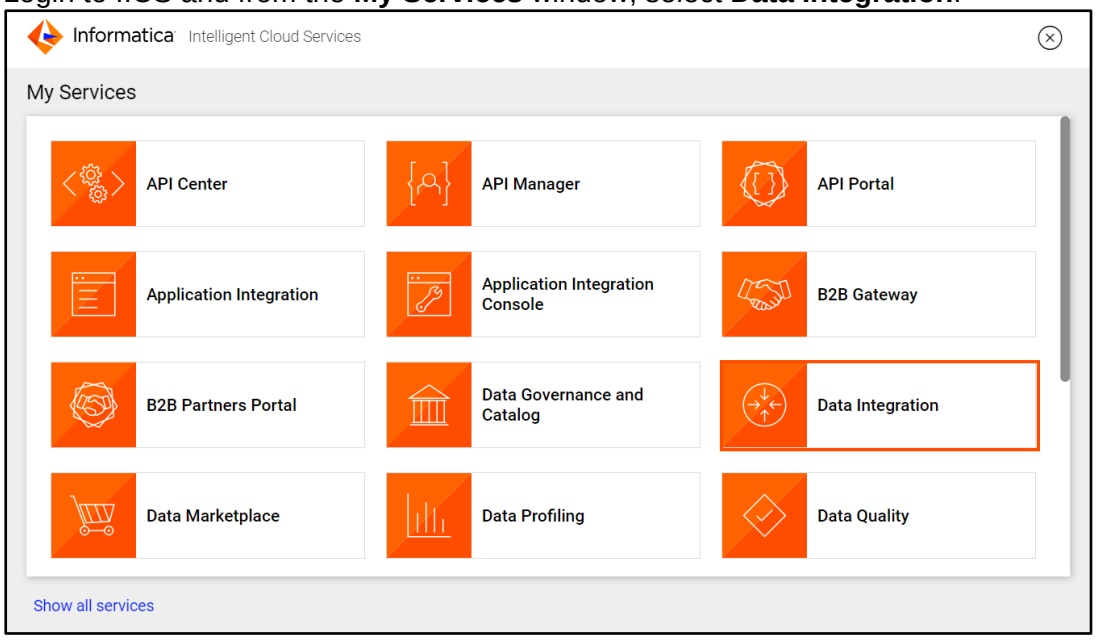
- Create a taskflow to run the Task.
- Get an email notification.
- Use conditional logic and take decision based on output of the task.

Duration:
40 minutes

Tasks

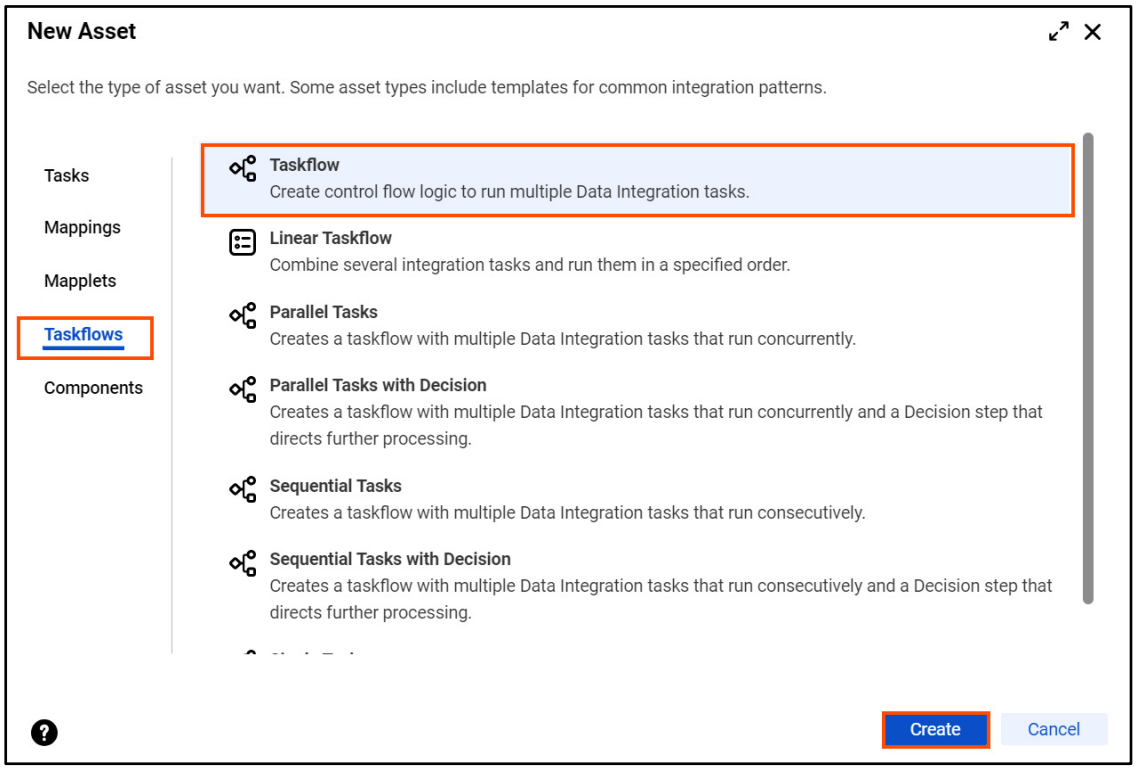
Create Taskflow

1. Login to IICS and from the **My Services** window, select **Data Integration**.

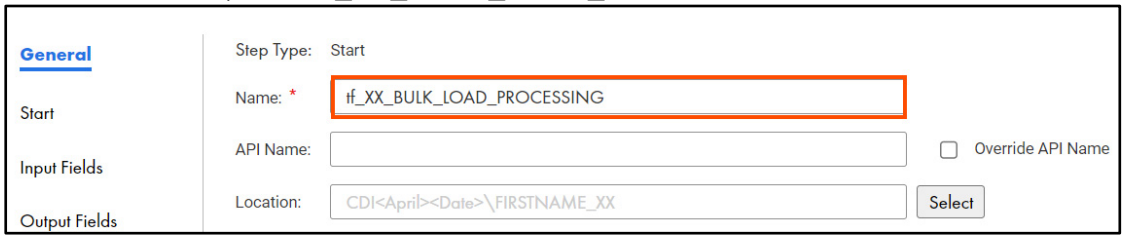


2. From the navigation pane, select **New**.
3. From the New Asset window, click the **Taskflows** tab.

4. Select **Taskflow** and click **Create**.



5. From the Taskflow properties pane, select **General**.
6. In the Name field, enter **tf_XX_BULK_LOAD_PROCESSING**.



7. From the properties pane, navigate to **Temp Fields**.
8. To add new Temp Field, click .

9. Enter the details as shown in table below.

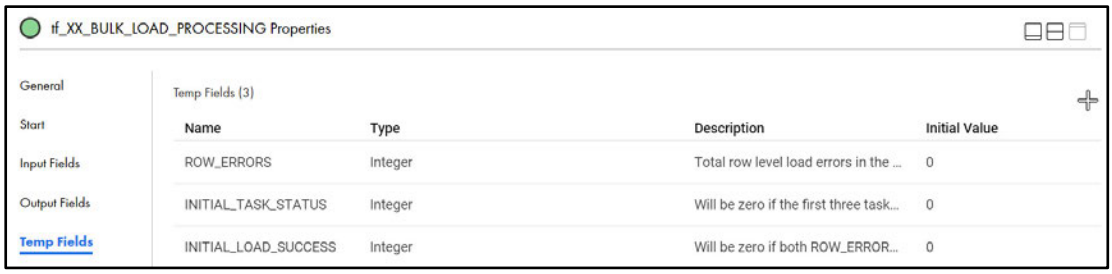
Name	ROW_ERRORS
Type	Integer
Description	Total row level load errors in the first three tasks
Initial Value	0 (click the edit icon and enter the value)

10. To add another Temp Field, click .
11. Enter the details as shown in table below.

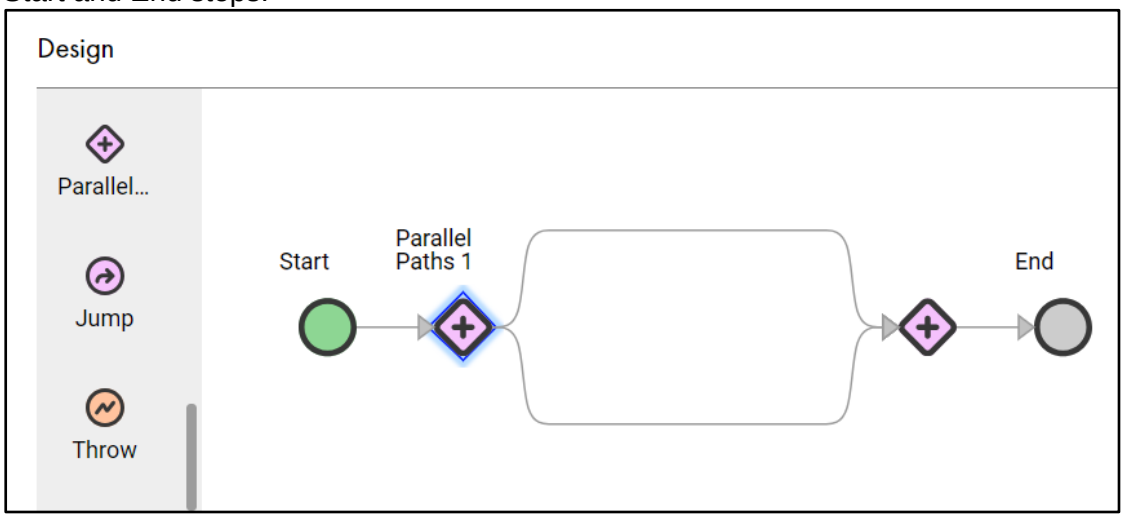
Name	INITIAL_TASK_STATUS
Type	Integer
Description	Will be zero if the first three tasks succeed
Initial Value	0

12. Similarly add another Temp field with the following configurations:

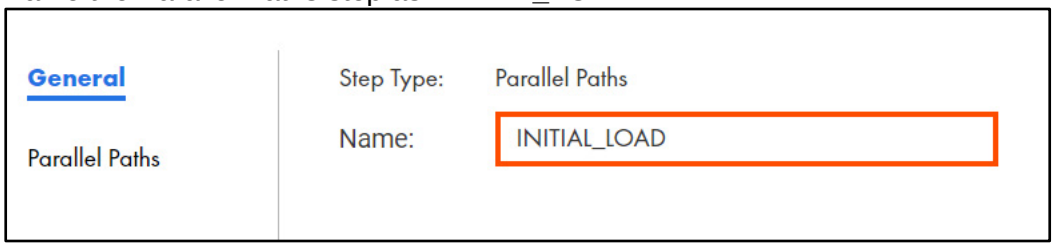
Name	INITIAL_LOAD_SUCCESS
Type	Integer
Description	Will be zero if both ROW_ERRORS and INITIAL_TASK_STATUS are zero
Initial Value	0



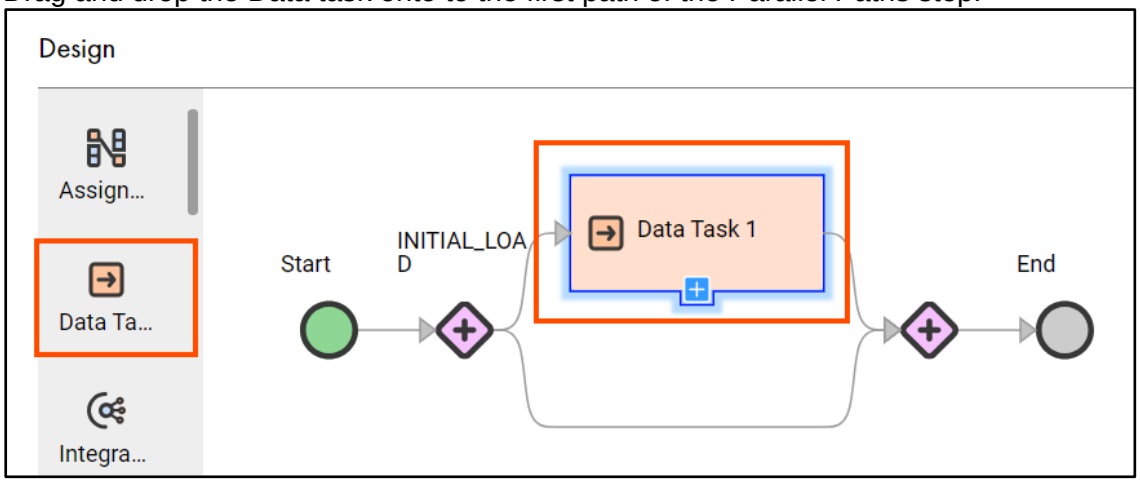
13. From the Design pane, drag and drop the **Parallel Paths** step onto the link between Start and End steps.



14. Name the Parallel Paths step as **INITIAL_LOAD**.

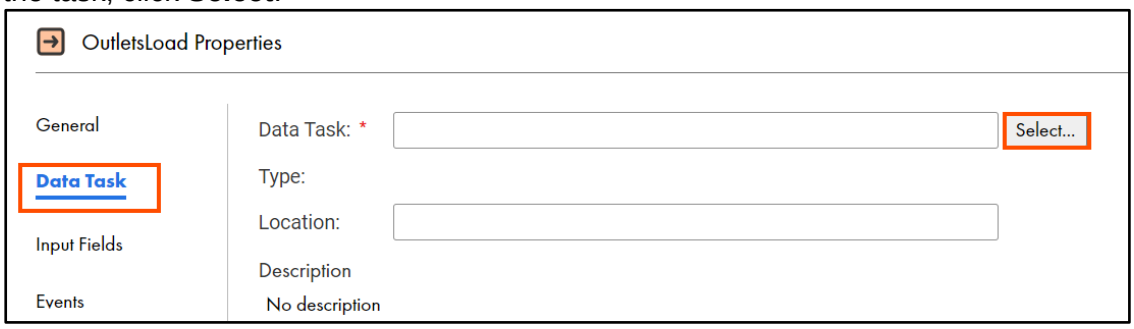


15. Drag and drop the Data task onto to the first path of the Parallel Paths step.

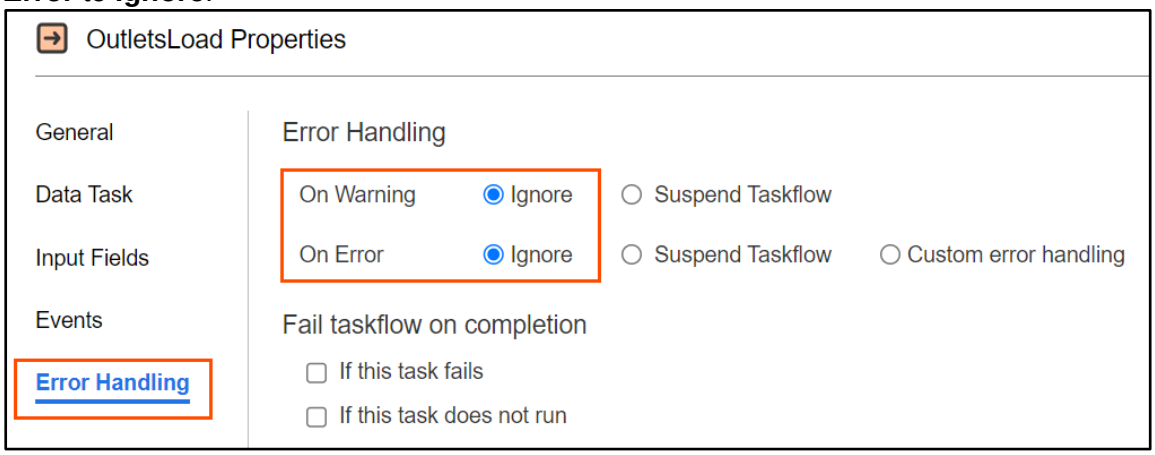


16. In the General properties section, enter **OutletsLoad** as the name of the Data task.

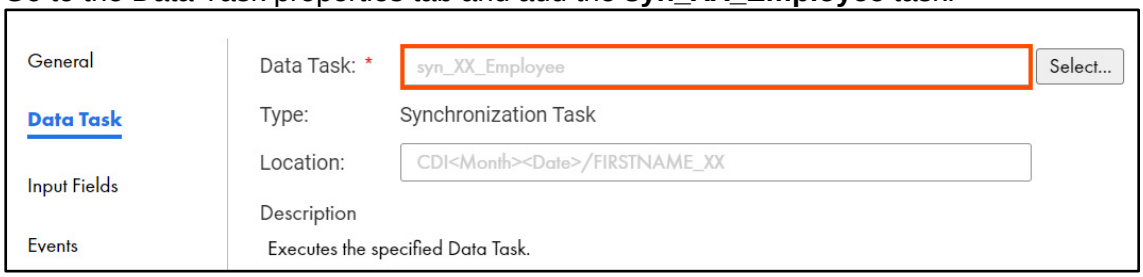
17. With the Data task being selected, navigate to the **Data Task** properties tab. To choose the task, click **Select**.



18. Navigate to your working folder and select the **syn_XX_OutletsLoad** task.
19. From the properties pane, select **Error Handling** and set both **On Warning** and **On Error** to **Ignore**.



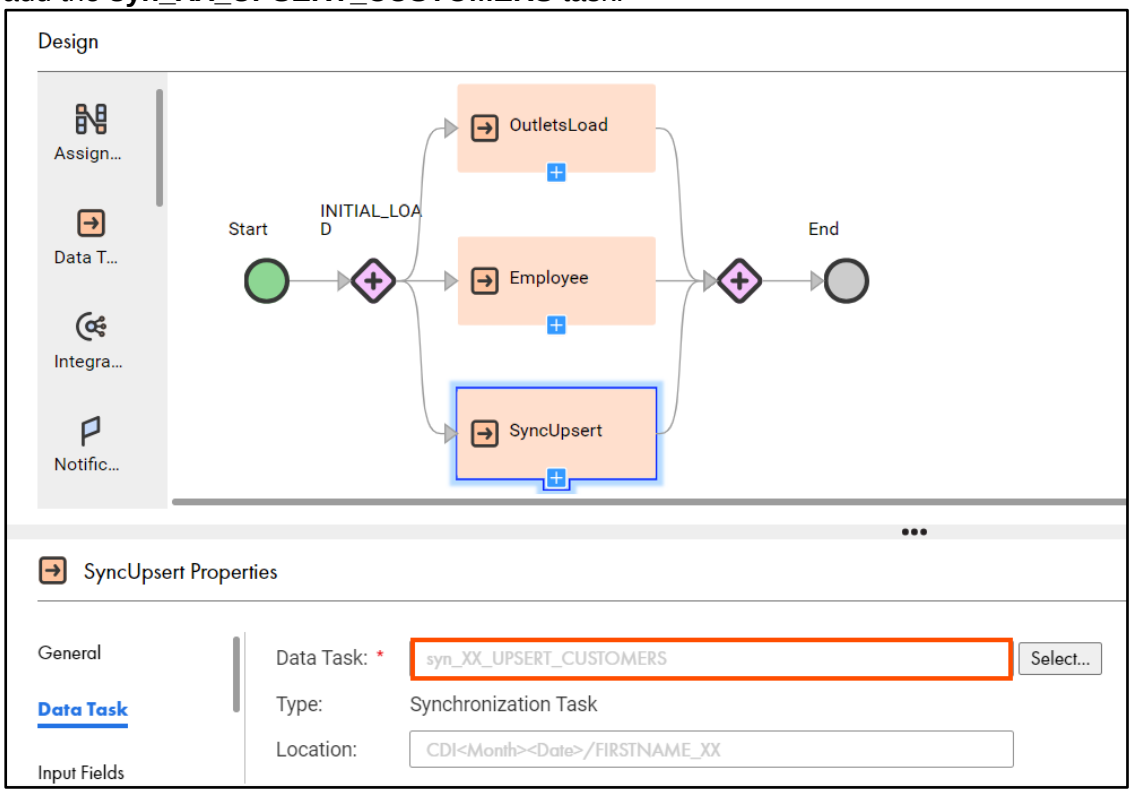
20. Drag and drop another data task onto the 2nd Parallel path and name it as **Employee**.
21. Go to the Data Task properties tab and add the **syn_XX_Employee** task.



22. From the properties pane, select **Error Handling** and set both **On Warning** and **On Error** to **Ignore**.

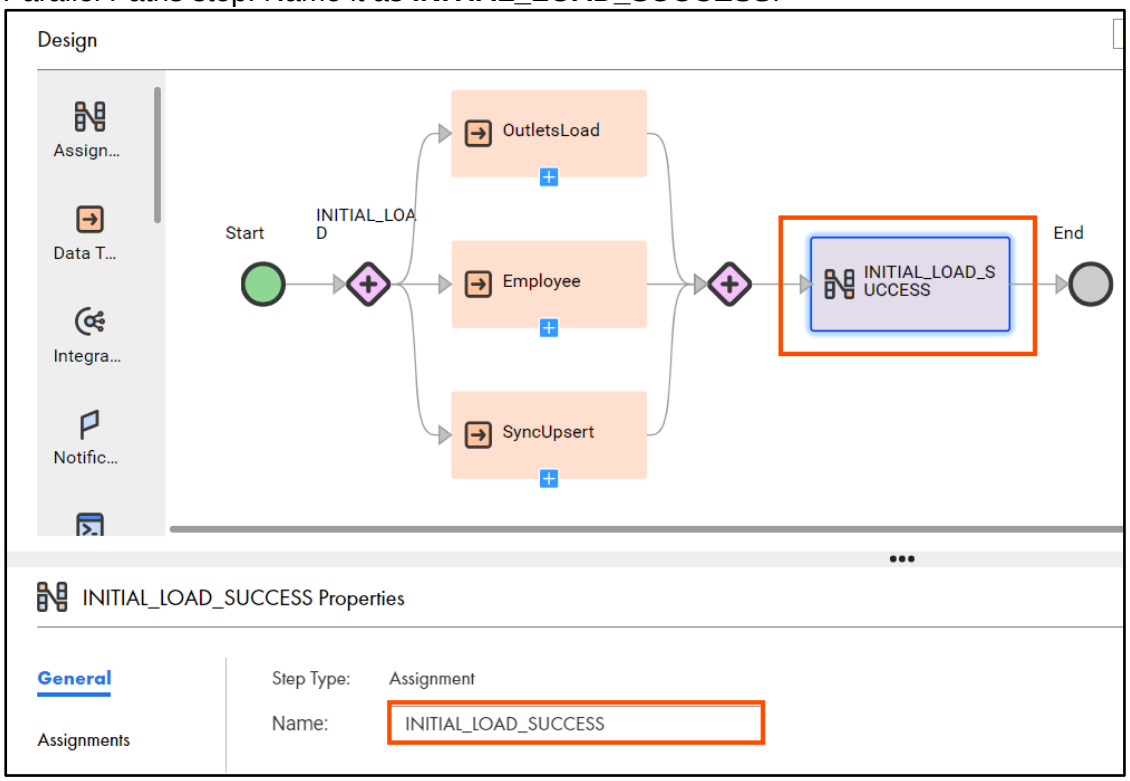
23. Select the Parallel Paths step from the canvas and navigate to the Parallel Paths tab. You can view 2 default paths. Click the plus icon to add the 3rd path to it.

24. Drag and drop another Data task step onto the 3rd parallel path and name it as **SyncUpsert**.
25. Navigate to the Data Task properties tab and from your working directory, select and add the **syn_XX_UPSERT_CUSTOMERS** task.

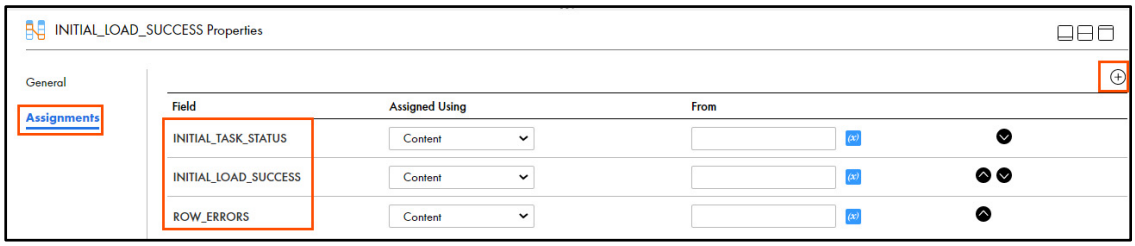


26. From the properties pane, select **Error Handling** and set both **On Warning** and **On Error** to **Ignore**.
27. Save your work.

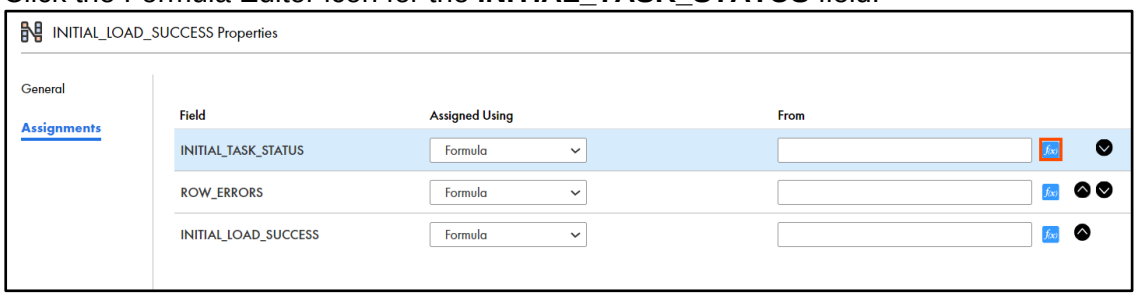
28. Drag and drop an Assignment step from the Design panel and place it right after the Parallel Paths step. Name it as **INITIAL_LOAD_SUCCESS**.



29. Go to the Assignments properties tab and using the plus icon, add **INITIAL_TASK_STATUS**, **ROW_ERRORS**, and **INITIAL_LOAD_SUCCESS** fields.



30. Change the **Assigned Using** column property for all the three fields to **Formula**.
31. Click the Formula Editor icon for the **INITIAL_TASK_STATUS** field.



32. Set the Expression as shown below:

    $temp.OutletsLoad[1]/output[1]/Task_Status

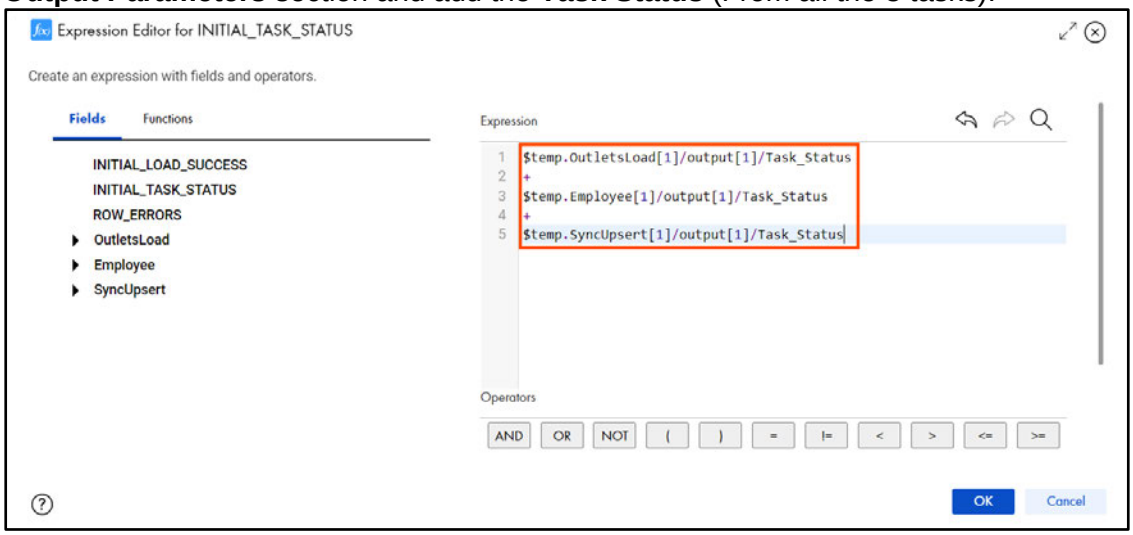
    +

    $temp.Employee[1]/output[1]/Task_Status

    +

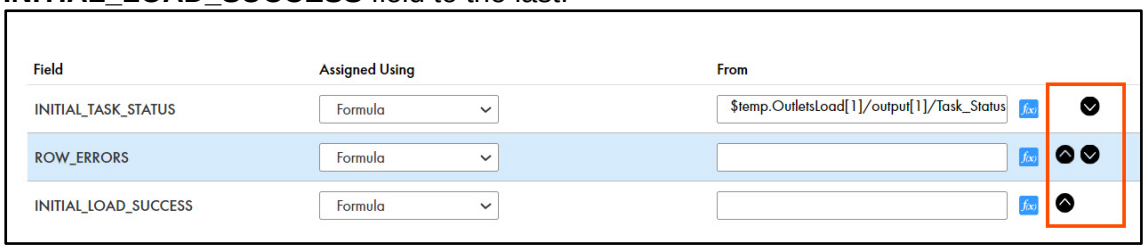
    $temp.SyncUpsert[1]/output[1]/Task_Status

    Tip: To configure the expression, from the Fields section, expand each of the Object's **Output Parameters** section and add the **Task Status** (From all the 3 tasks).



33. Click **OK**.

    Note: Using the up and down arrows next to the fields, keep the **INITIAL_LOAD_SUCCESS** field to the last.



34. Similarly, for the ROW_ERRORS field, go to the expression editor and set the following expression using the Output Parameters:

    $temp.OutletsLoad[1]/output[1]/Failed_Target_Rows

    +

    $temp.OutletsLoad[1]/output[1]/TotalTransErrors

    +

    $temp.Employee[1]/output[1]/Failed_Target_Rows

    +

    $temp.Employee[1]/output[1]/TotalTransErrors

    +

    $temp.SyncUpsert[1]/output[1]/Failed_Target_Rows

    +

    $temp.SyncUpsert[1]/output[1]/TotalTransErrors

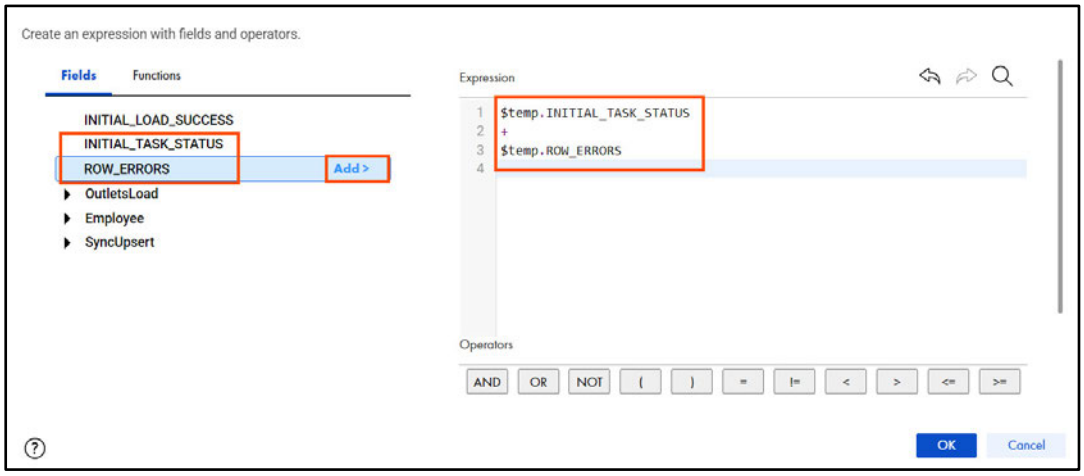
35. Click **OK**.

36. Similarly, for **INITIAL_LOAD_SUCCESS**, set the following expression:

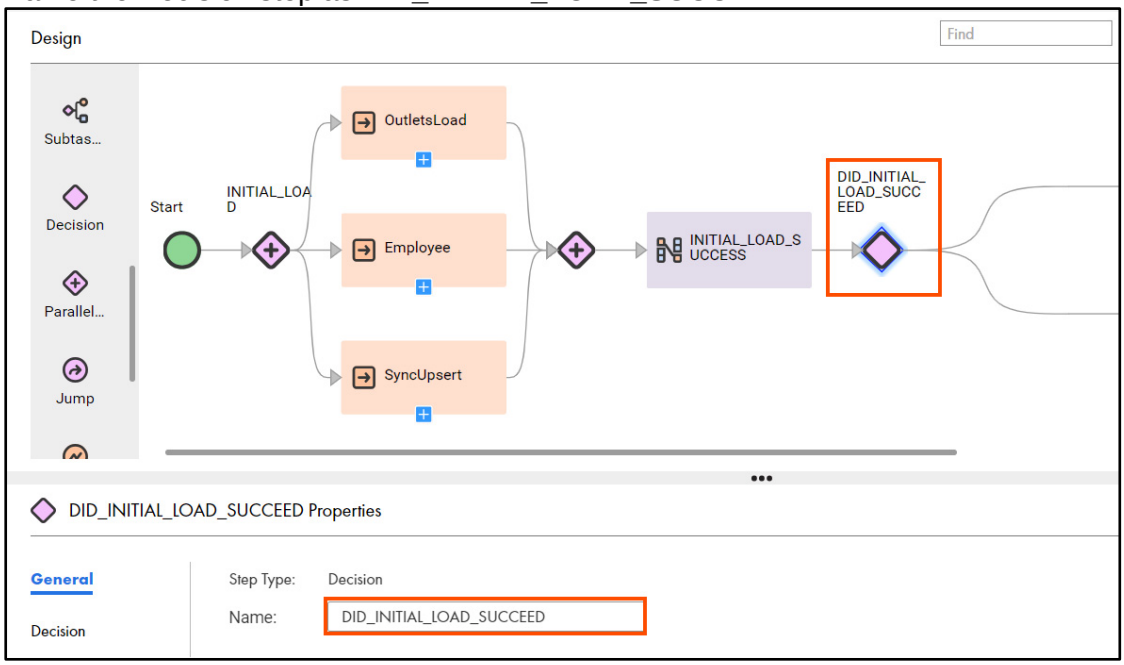
    $temp.INITIAL_TASK_STATUS

    +

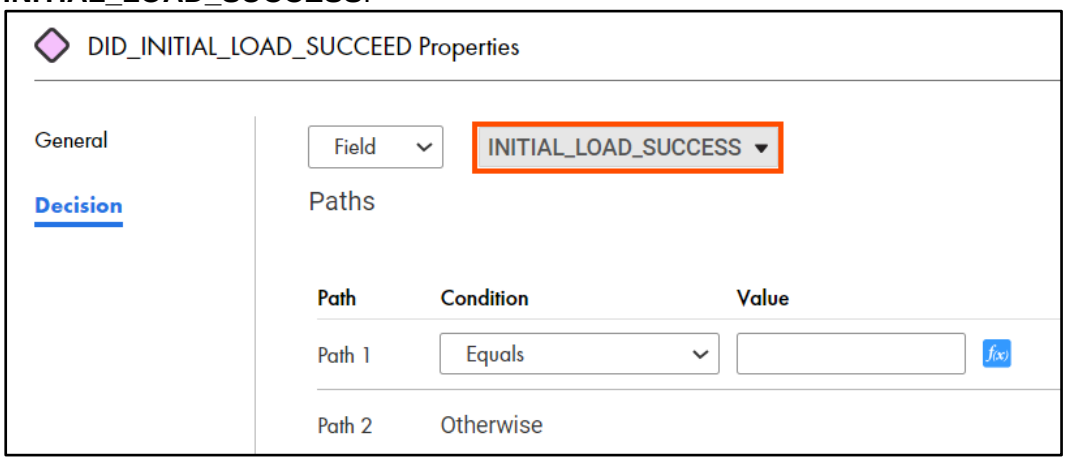
    $temp.ROW_ERRORS



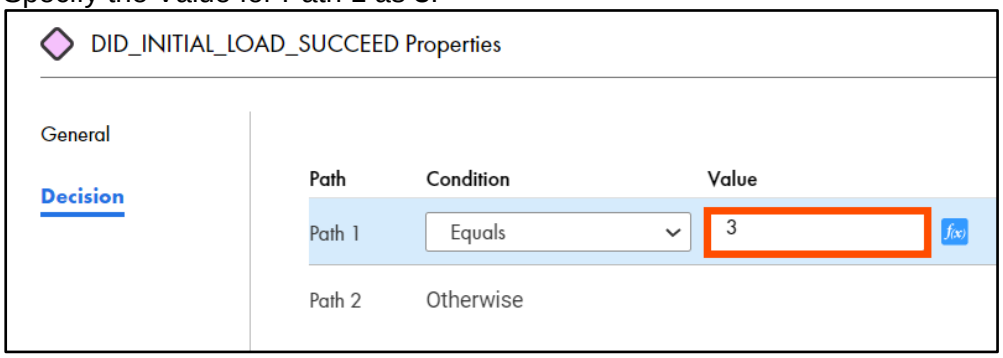
37. Click **OK**.
38. Save your work.
39. Drag and drop a Decision step onto the canvas and place it next to the Assignment step.
40. Name the Decision step as **DID_INITIAL_LOAD_SUCCEED**.



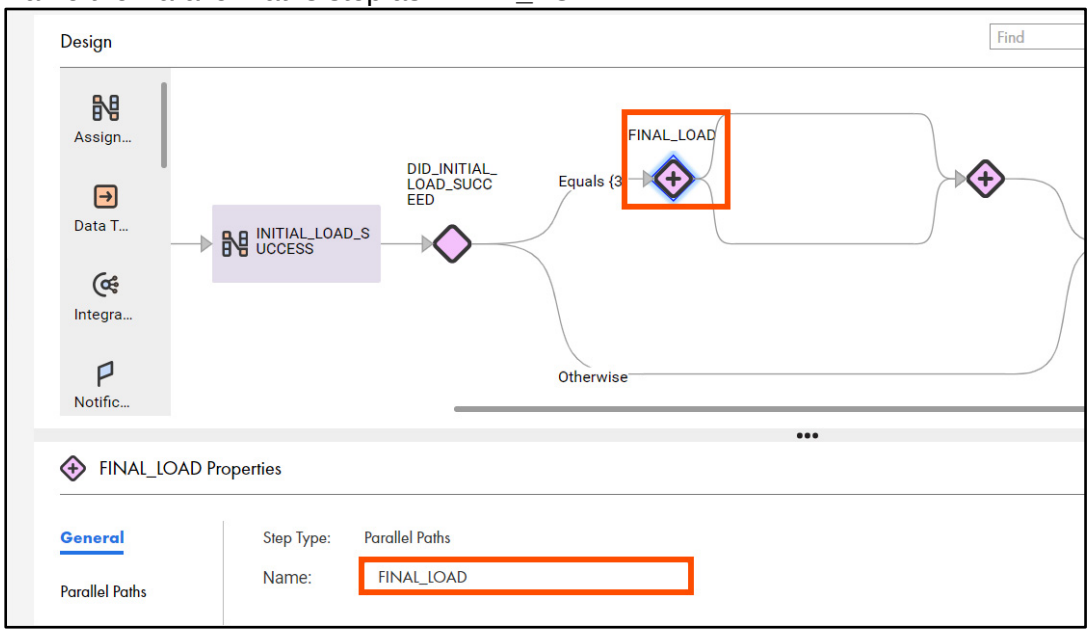
41. Navigate to the Decision properties tab and select the decision field as **INITIAL_LOAD_SUCCESS**.



42. Specify the Value for Path 1 as **3**.

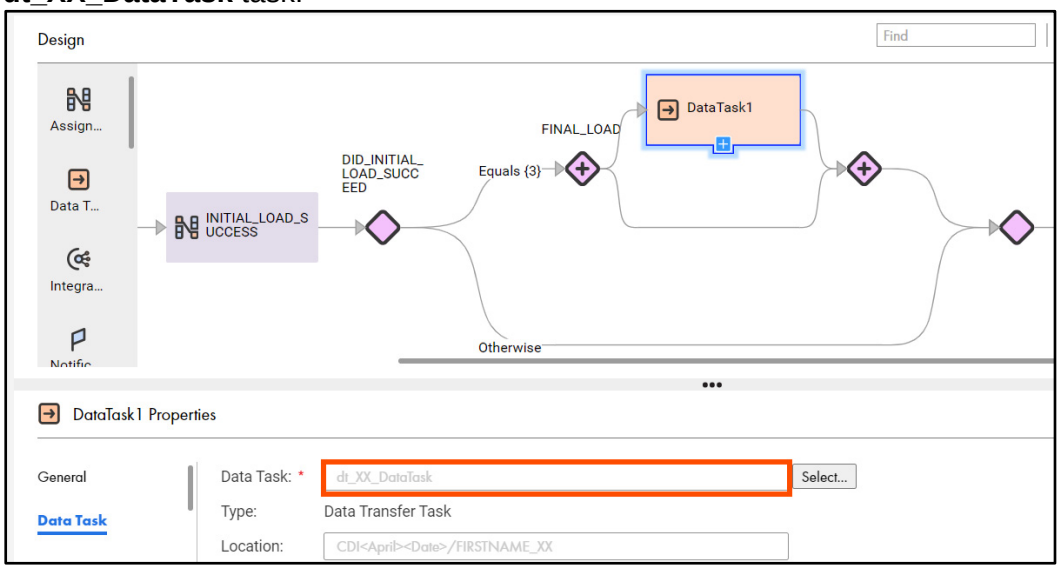


43. In path 1 of decision step (value = 3 path), drag and drop a Parallel Paths step.
44. Name the Parallel Paths step as **FINAL_LOAD**.

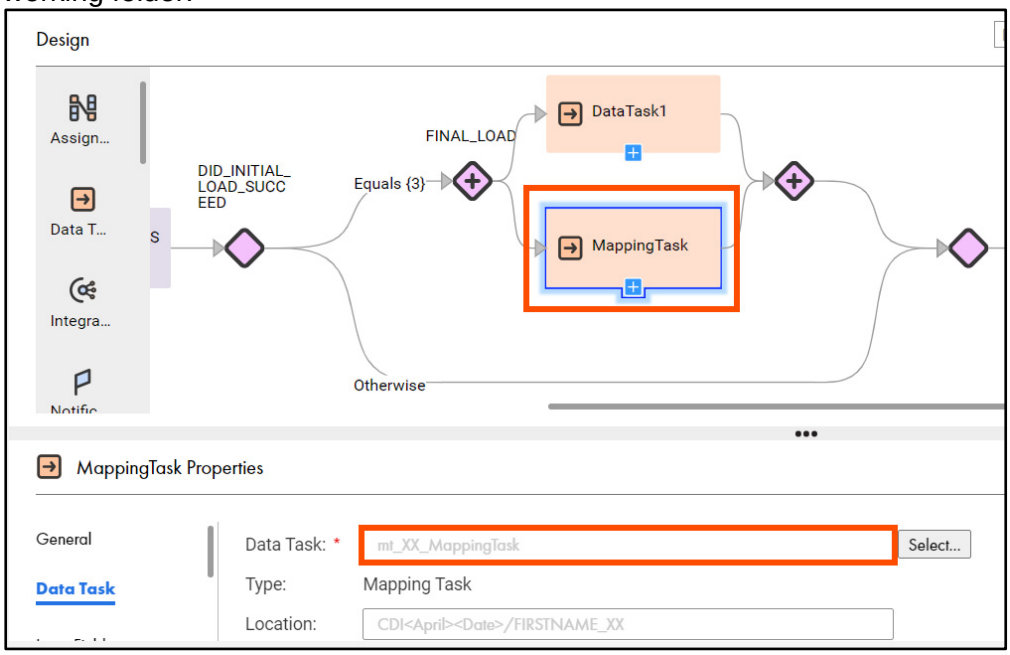


45. Save your work. To the first parallel path, add a Data task step and retain the default name.

46. From the Data Task properties tab, navigate to your working folder and select the **dt_XX_DataTask** task.

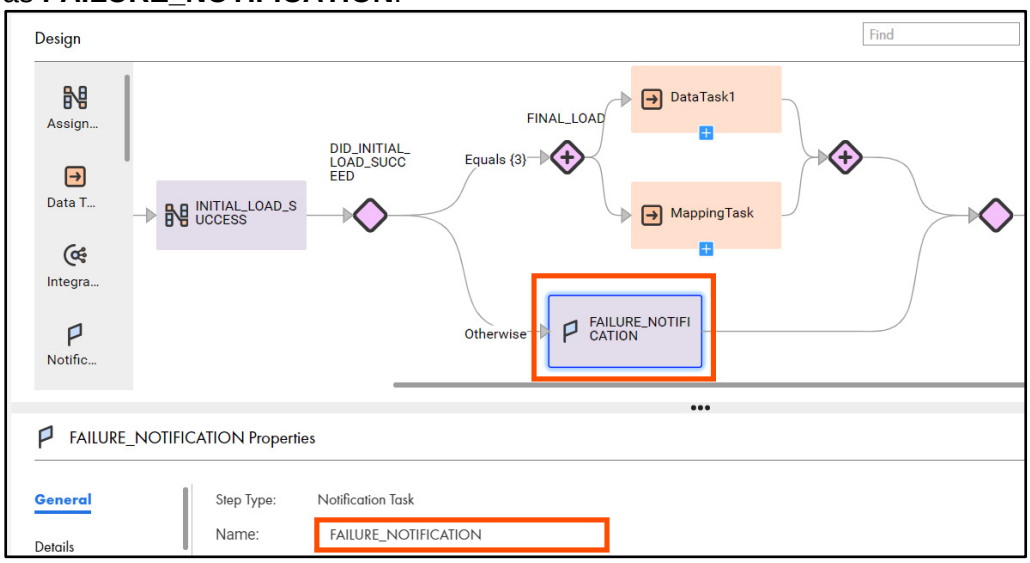


47. From the properties pane, select **Error Handling** and set both **On Warning** and **On Error** to **Ignore**.
48. Onto the second parallel path, add another data task step and name it as **MappingTask**.
49. From its Data Task properties tab, select the **mt_XX_MappingTask** task from your working folder.

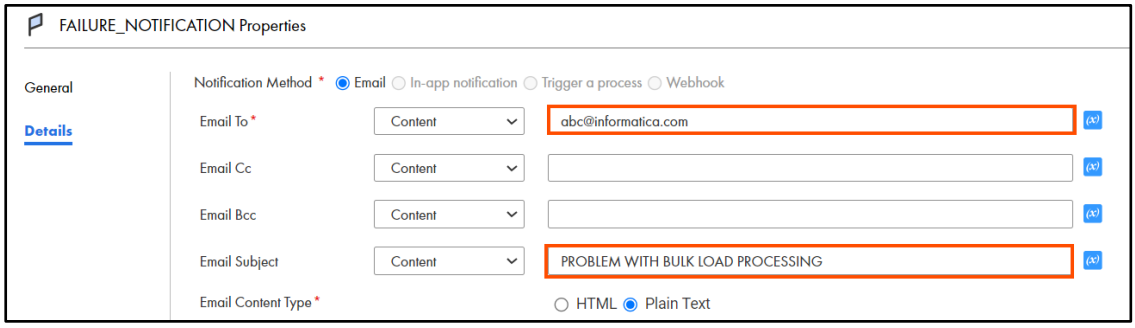


50. From the properties pane, select **Error Handling** and set both **On Warning** and **On Error** to **Ignore**.

51. Onto the Otherwise path of Decision step, drag and drop a Notification step and name it as **FAILURE_NOTIFICATION**.

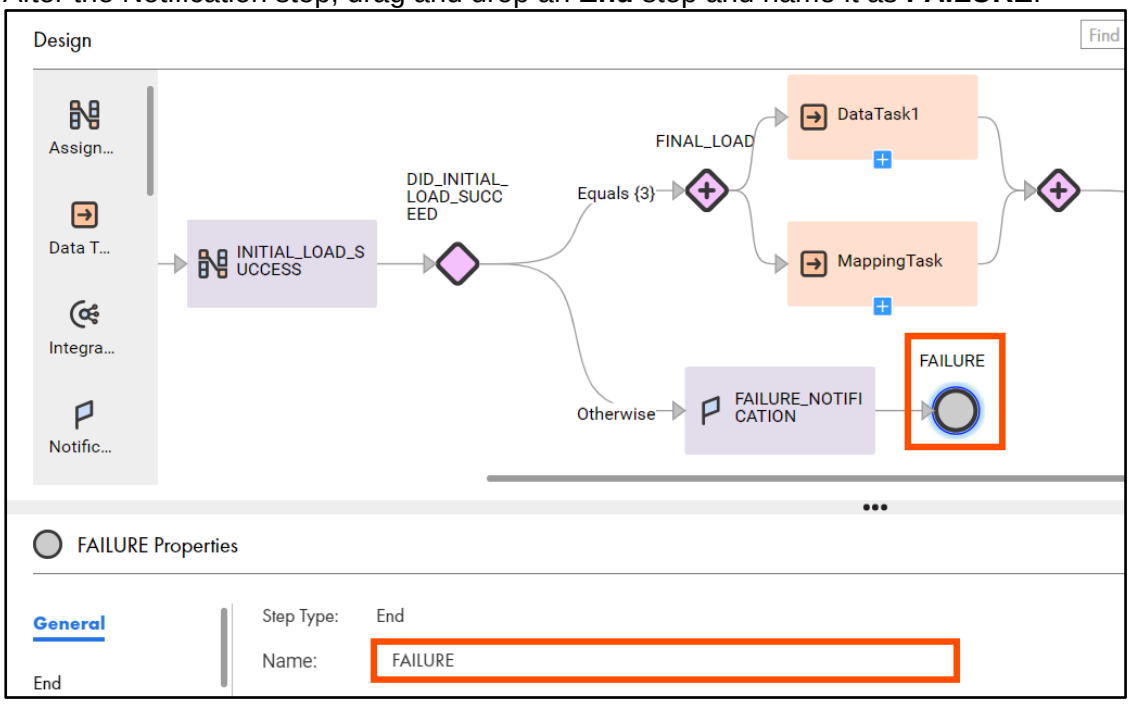


52. Go to the Details properties tab of the Notification step.
53. Provide a valid email Id for the Email To property field and Email Subject as **PROBLEM WITH BULK LOAD PROCESSING**.

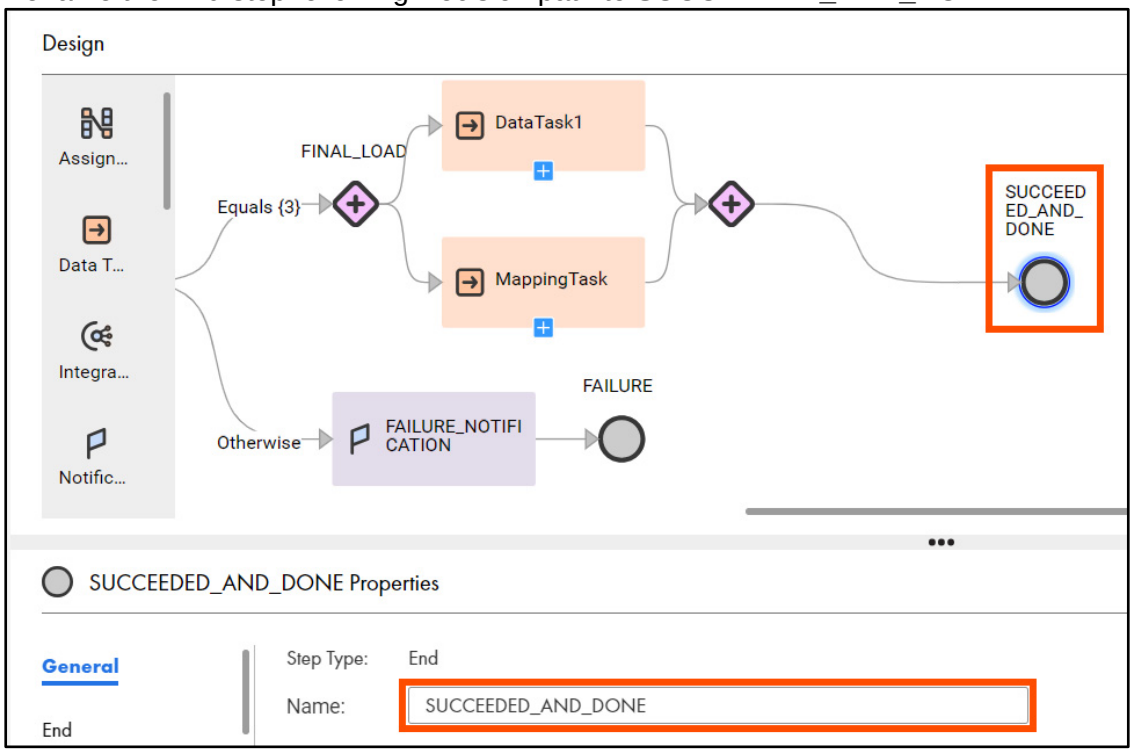


54. Save your work at regular intervals.

55. After the Notification step, drag and drop an **End** step and name it as **FAILURE**.



56. Rename the End step following Decision path to **SUCCEEDED_AND_DONE**.



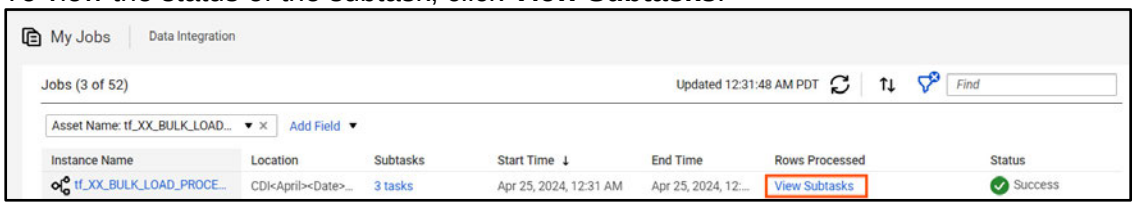
57. **Save** the Taskflow.
58. Publish the Taskflow by clicking on the **Publish** option next to Save.

59. Run the Taskflow.

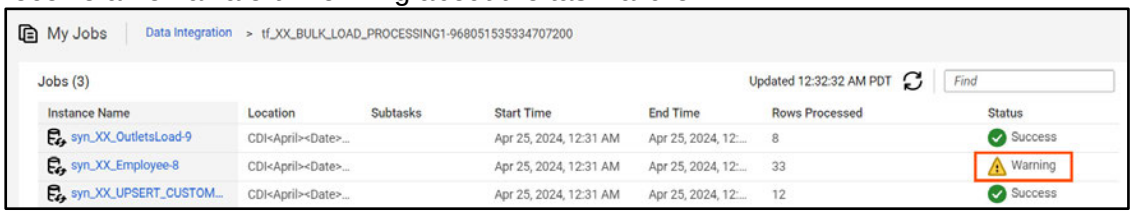
Monitor Status

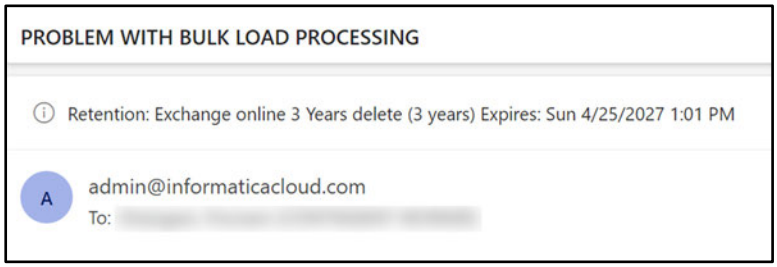
60. To monitor the task, from the navigation pane, click **My Jobs**.
61. When the task is completed, the status changes to **Success**.

    Note: You can refresh the page if the status does not change automatically.
62. To view the status of the subtask, click **View Subtasks**.

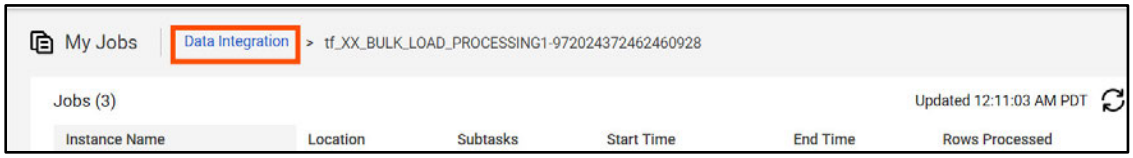


63. Observe that you see a warning status for 1 task, and it is expected. This is the same duplication error that was observed in the previous lab exercise. In this case, you will receive an email alert informing about the task failure.

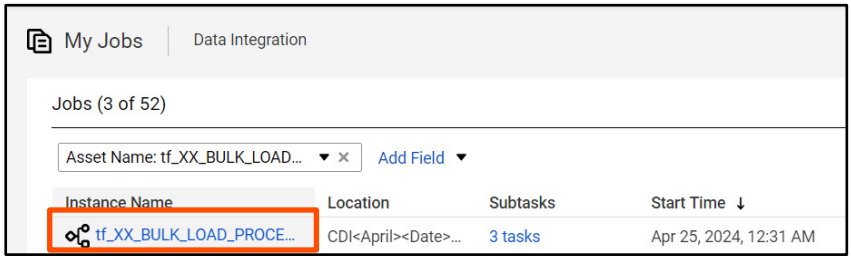




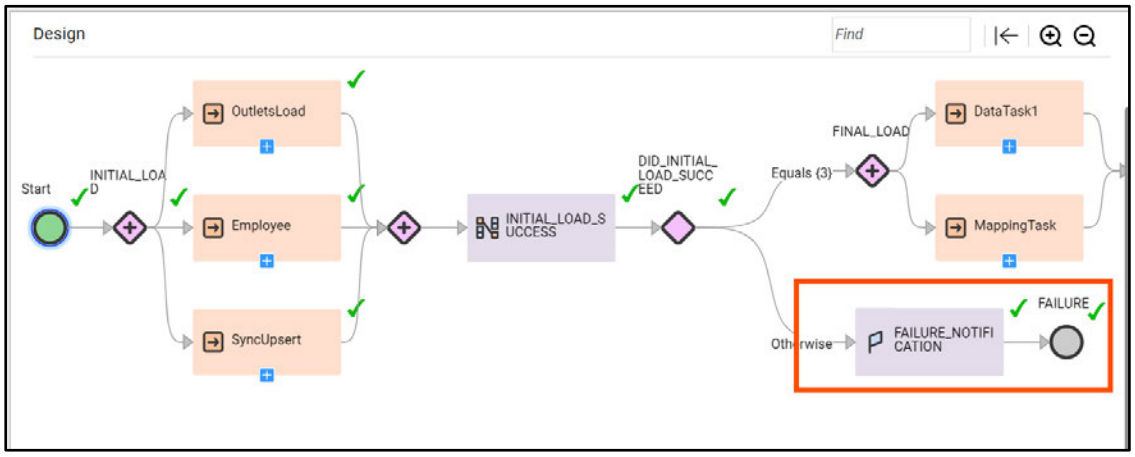
64. To go back to the previous window, click **Data Integration** next to My Jobs.



65. Click on the instance name.



66. Observe the executed path. It has taken the failure path since one of the associated subtasks has failed due to duplicate records (as seen in step 58).



Note: If you want to run your taskflow in the success path1, delete the existing records (as mentioned in the previous lab exercise) in the target of the failed subtask and re-run the taskflow.

This concludes the lab.

Module 14: IICS REST APIs

Lab 14-1: Creating a Mapping Using a REST V2 Connector

Overview:
REST is a web standards-based architecture that uses HTTP Protocol for data communication. These commonly used APIs allow you to create web-based applications.

In this lab, you will use the REST V2 Connector to connect to OpenWeather API and to get the current weather information.

Objective:

- Create a REST connection using the REST V2 connector
- Get the JSON message and write it to a Flat File

Scenario:

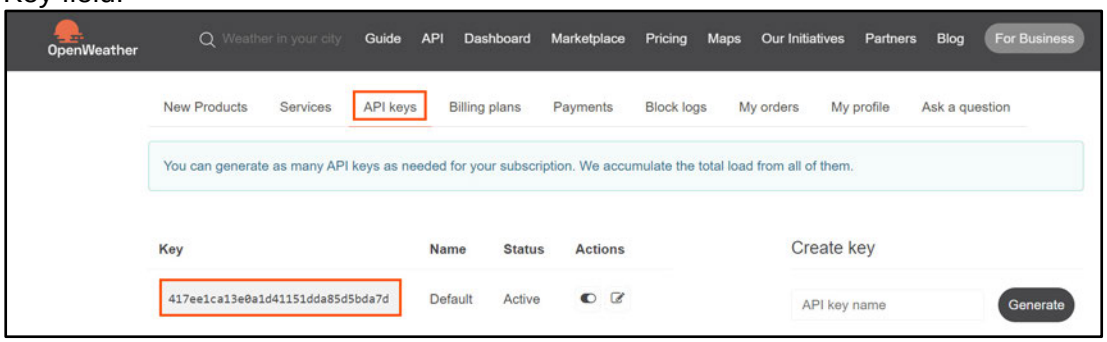
John receives a request to implement weather information for requested location. He uses a REST V2 Connector to connect to the OpenWeather API. He uses the unique API key to fetch the current weather information for the city specified in the request message. After the information is received from the API in the JSON format, he creates a target file at runtime to write the output.

Duration:
30 minutes

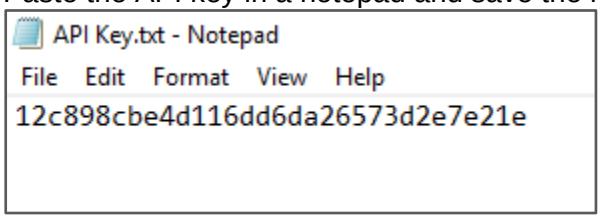
Tasks

Sign up to Access OpenWeatherAPI

1. Open a web browser and enter the following URL in a new tab:
   https://home.openweathermap.org/users/sign_up
   Note: You can bookmark this link for future use.
2. Create a new account in OpenWeather using the link provided.
   Note: After you create an account, you can access your unique API key.
3. To access your unique API Key, select the **API Keys** tab, and copy the API key from the Key field.

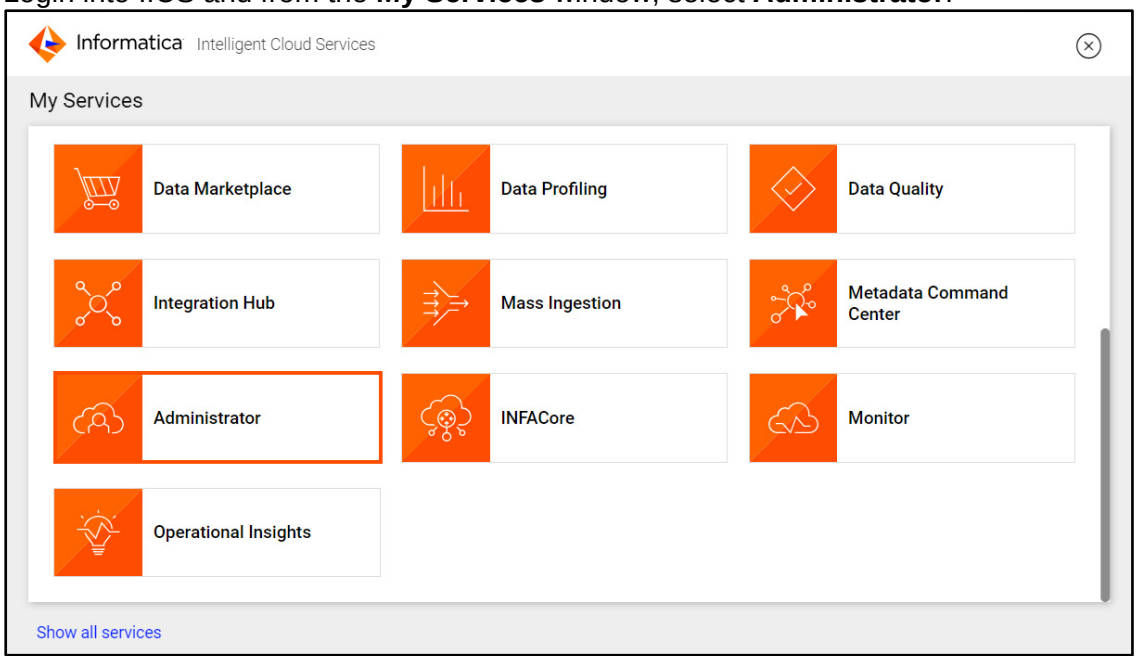


4. Paste the API key in a notepad and save the notepad file on your system.

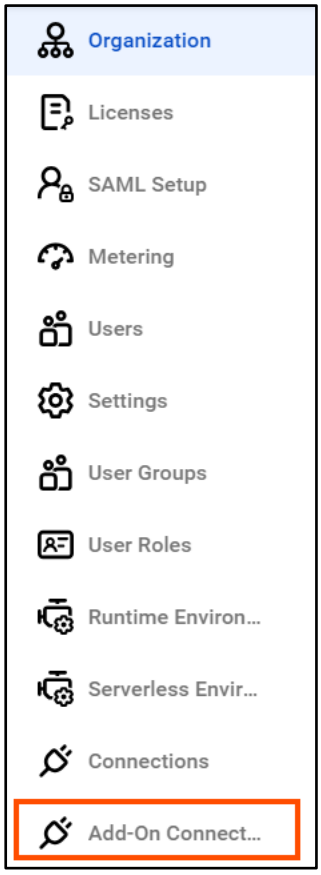


Enable REST V2 Connector

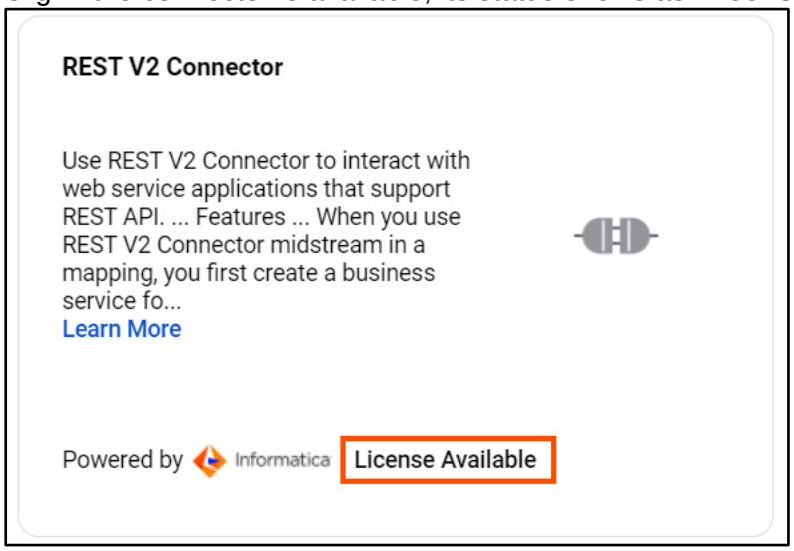
5. Login into IICS and from the **My Services** window, select **Administrator**.



6. From the navigation pane, select **Add-On Connectors**.



7. Locate the **REST V2 Connector** and check if the REST V2 connector is available for the Org. If the connector is available, its status shows as '**License Available**'.

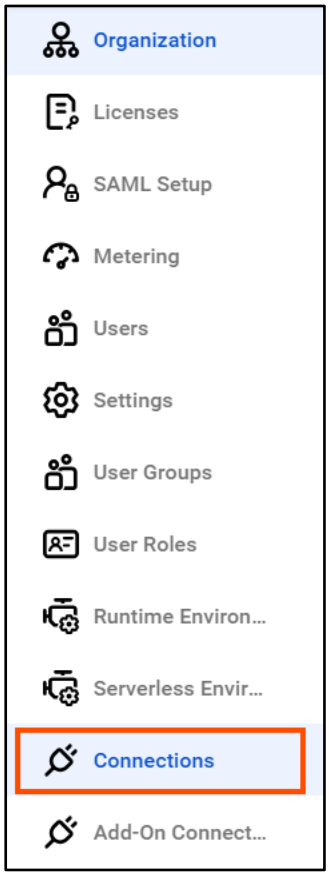


Note: If the connector is already available for the org, you can skip the steps to enable the connector.

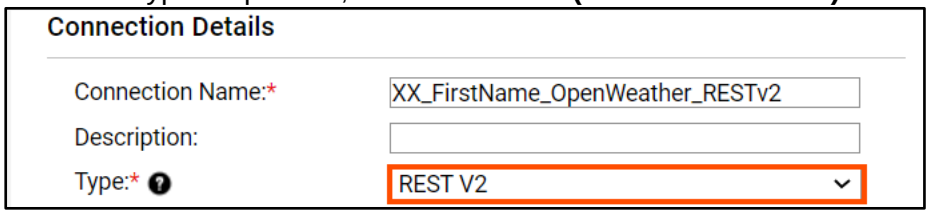
8. If the REST V2 Connector is not available, click **Start Free Trial** to enable the connector.

Create a REST Connection

9. From the navigation pane, select **Connections**.

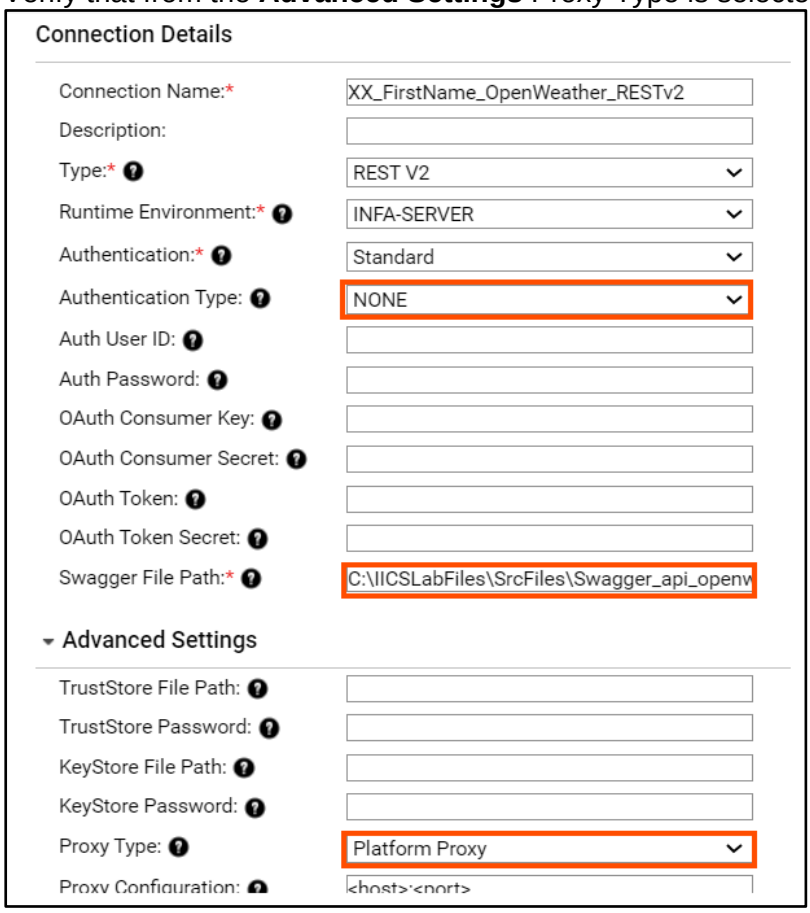


10. Create a new connection.
11. In the Name field, enter **XX_FirstName_OpenWeather_RESTv2**.
12. From the Type drop-down, select **REST V2 (Informatica Cloud)**.

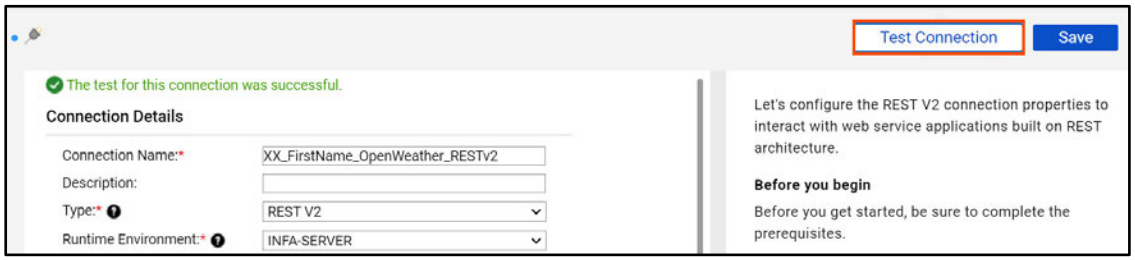


13. From the **Runtime Environment** drop-down, select **INFA-SERVER**.
14. From the Authentication drop-down, select **Standard**.
15. Verify that the Authentication Type is set to **NONE**.
16. In the Swagger File Path field, enter the following path of the json file: **C:\IICSLabFiles\SrcFiles\Swagger_api_openweathermap_org_072248.json**

17. Verify that from the **Advanced Settings** Proxy Type is selected as **Platform Proxy**.



18. Test the connection.

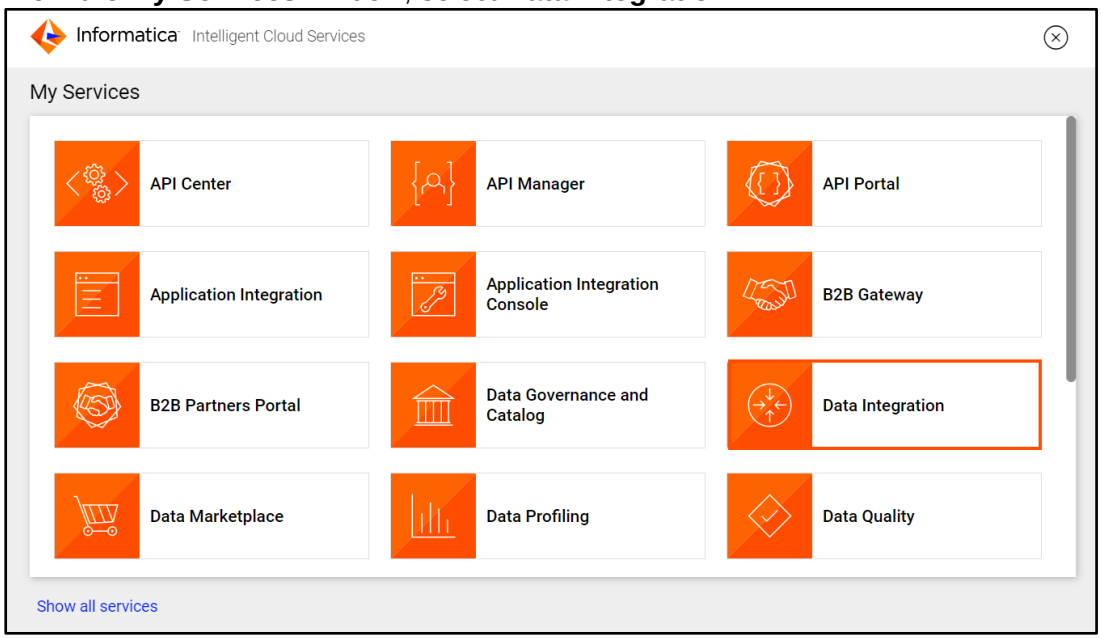


19. Save the connection.

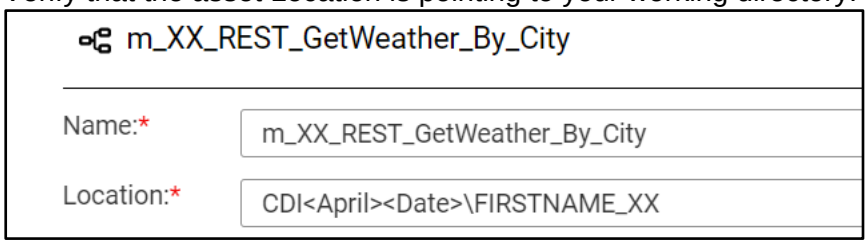
Create a Mapping

20. To switch between the available services, from the toolbar, select the current service **Administrator**.

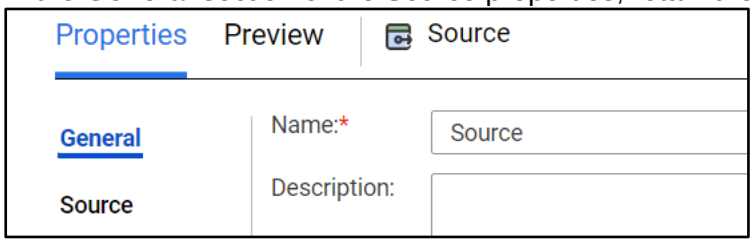
21. From the **My Services** window, select **Data Integration**.



22. Create a new Mapping.
23. In the Name field, enter **m_XX_REST_GetWeather_By_City**.
24. Verify that the asset Location is pointing to your working directory.

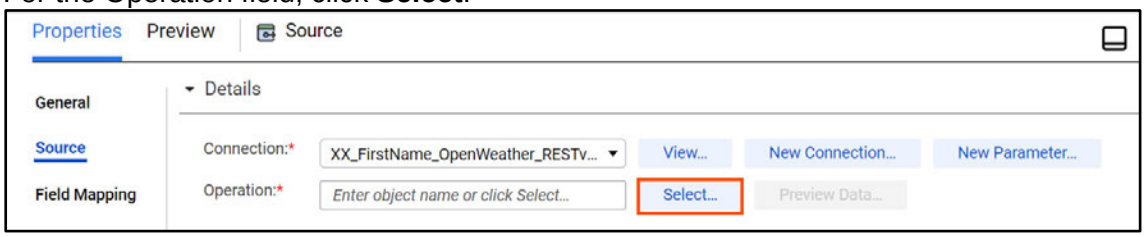


25. To configure the source, from the mapping canvas, click the **Source** transformation.
26. In the **General** section of the Source properties, retain the Name as **Source**.

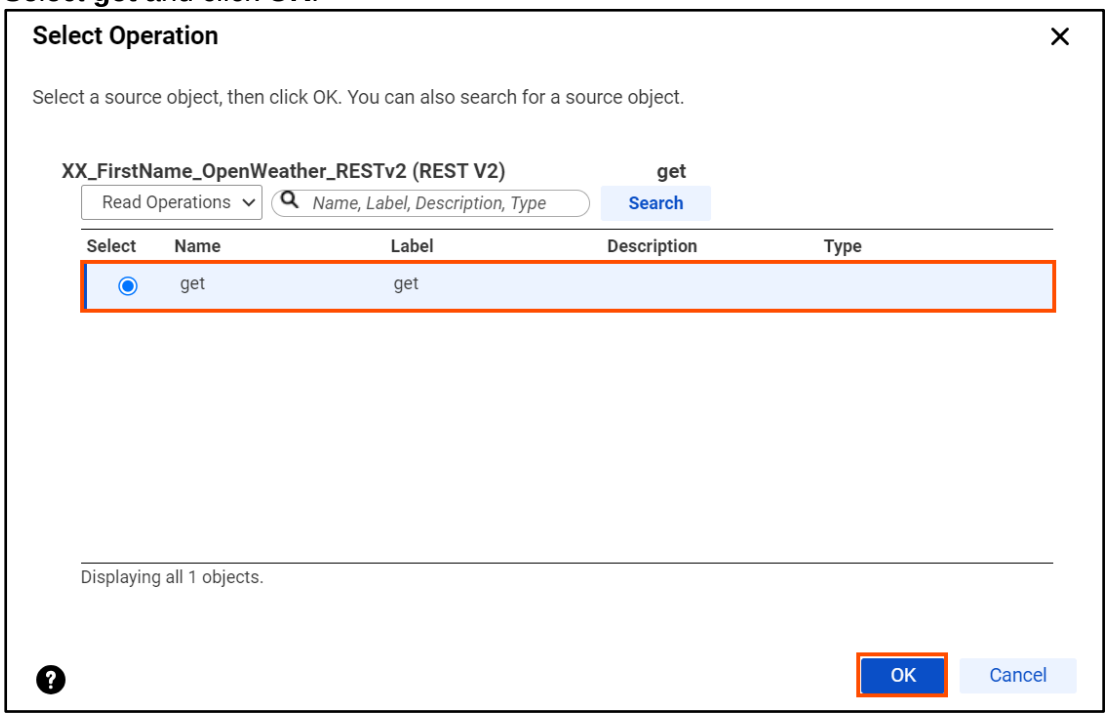


27. From the properties pane, click **Source**.
28. From the Connection drop-down, select **XX_FirstName_OpenWeather_RESTv2**.

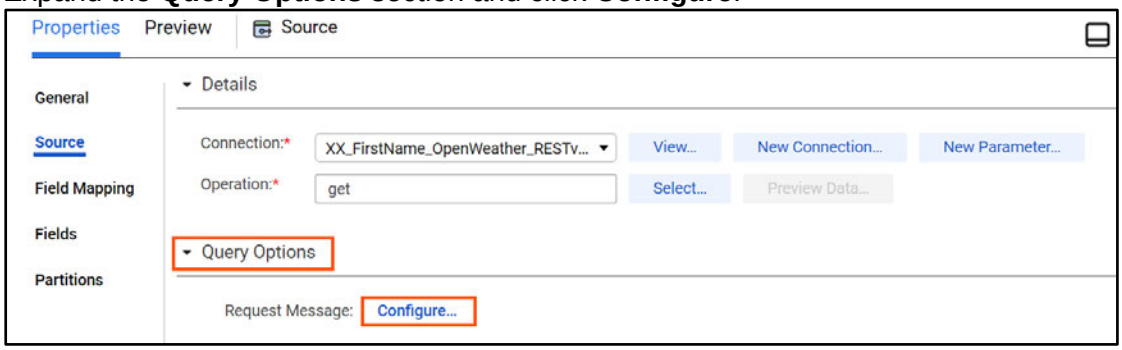
29. For the Operation field, click **Select**.



30. Select **get** and click **OK**.

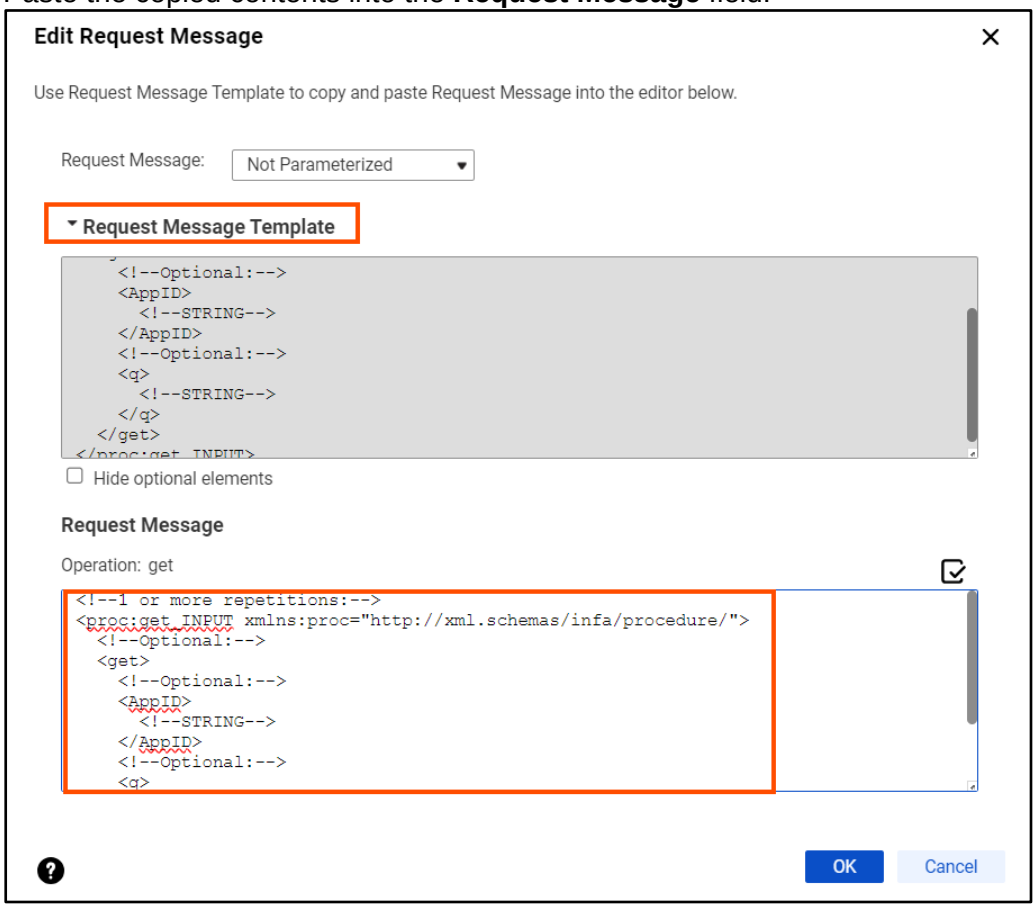


31. Expand the **Query Options** section and click **Configure**.



32. Expand the **Request Message Template** section and copy all the contents from the field.

33. Paste the copied contents into the **Request Message** field.

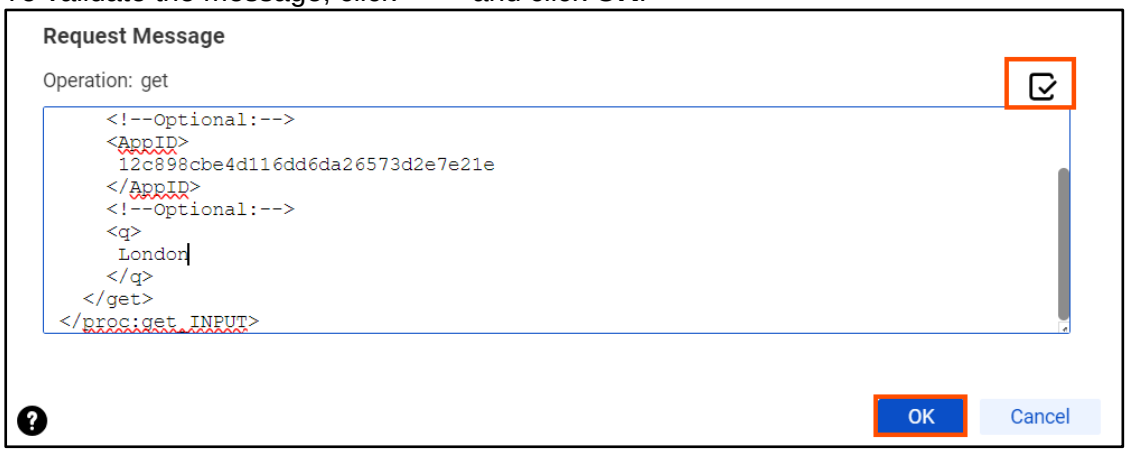


34. In the Request Message field, in between **<AppID> </AppID>**, enter the unique API Key saved earlier.
35. In the Request Message field, in between **<q> </q>**, enter **London**.

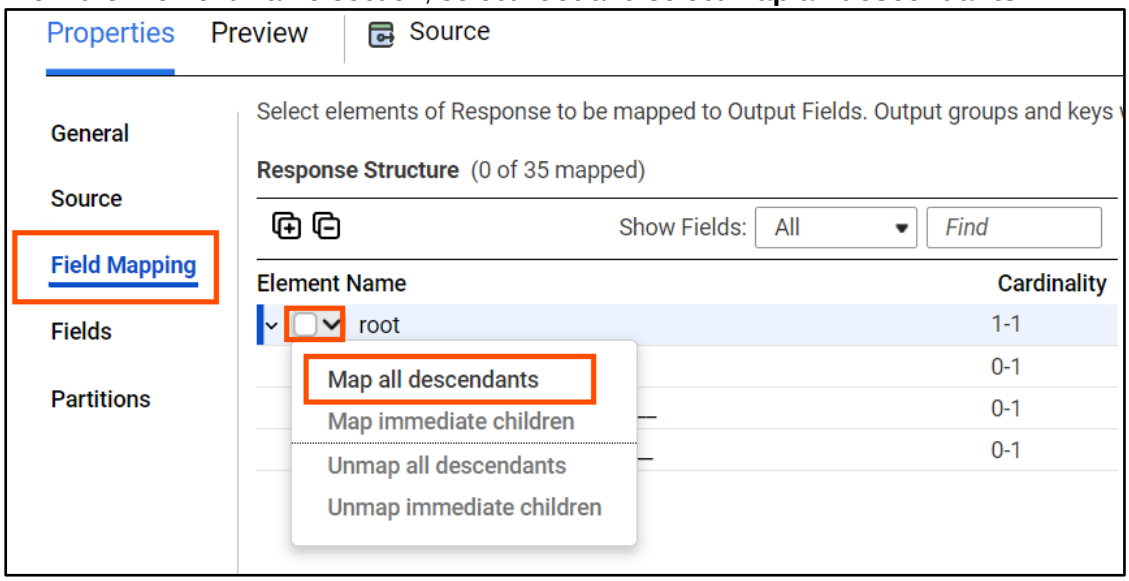
Example:

<AppID>
12c898cbe4d116dd6da26573d2e7e21e
</AppID>
<q>
London
</q>

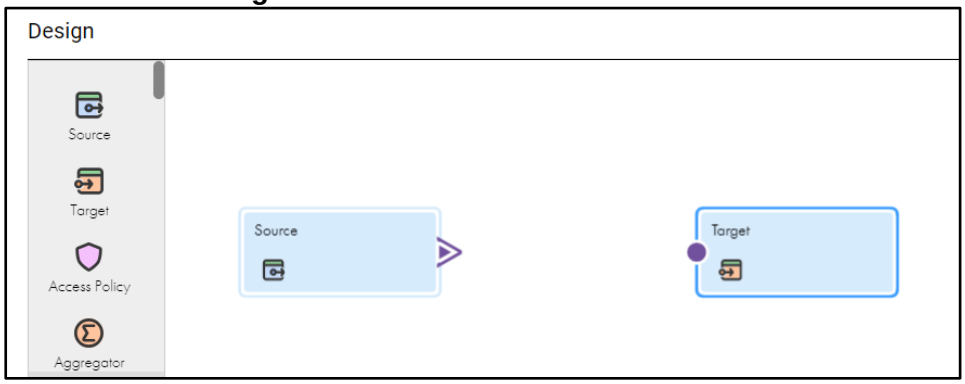
36. To validate the message, click and click **OK**.



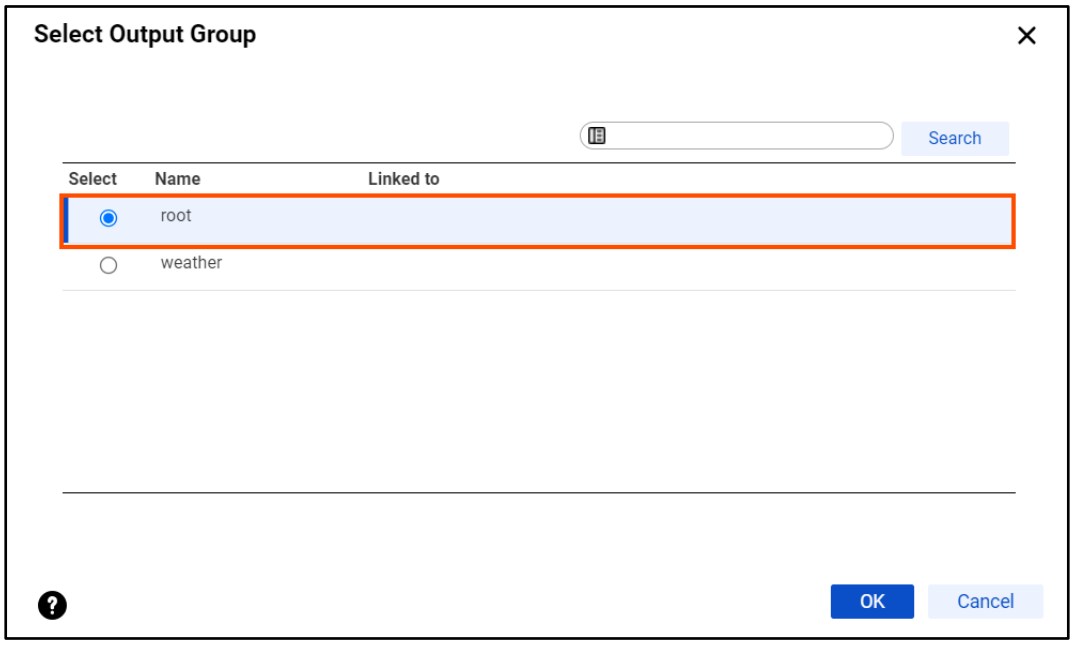
37. From the properties pane, click **Field Mapping**.
38. From the Element Name section, select **root** and select **Map all descendants**.



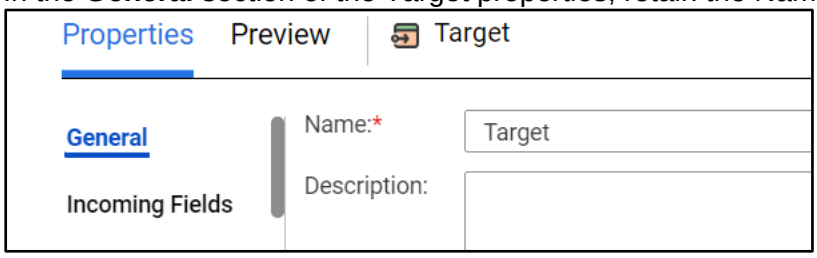
39. Link **Source** to **Target**.



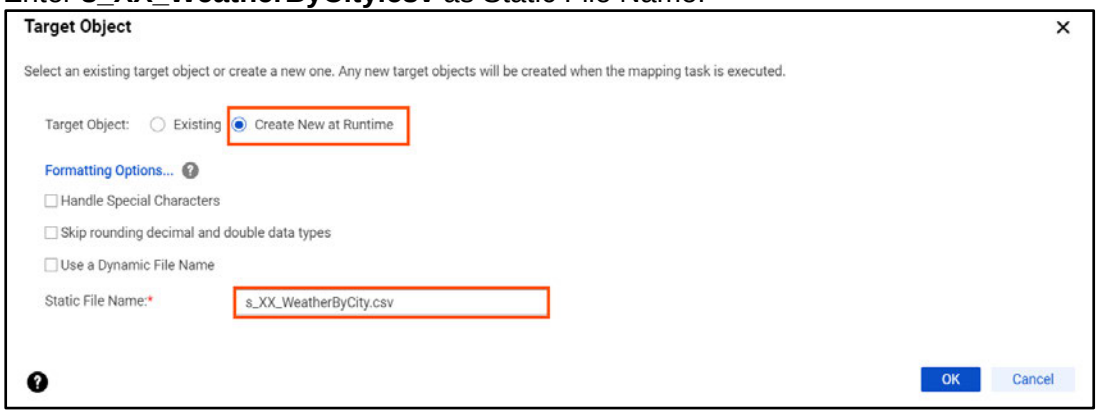
40. Select **root** and click **OK**.



41. To configure the target, from the mapping canvas, click the **Target** transformation.
42. In the **General** section of the Target properties, retain the Name as **Target**.



43. From the properties pane, click **Target**.
44. From the Connection drop-down, select your **target Flat File** connection.
45. To select the target object, from the Object field, click **Select**.
46. In the Target Object window, select **Create New at Runtime**.
47. Enter **s_XX_WeatherByCity.csv** as Static File Name.



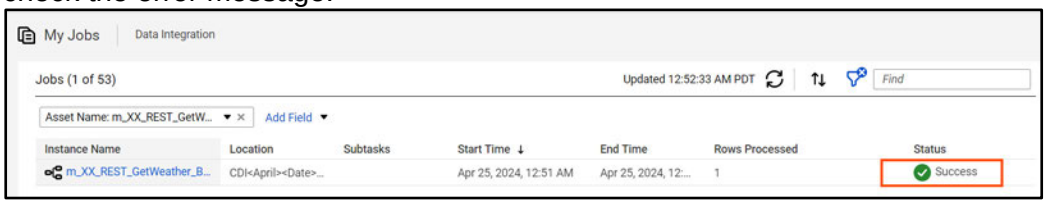
48. Click **OK**.

49. Save and run the mapping.

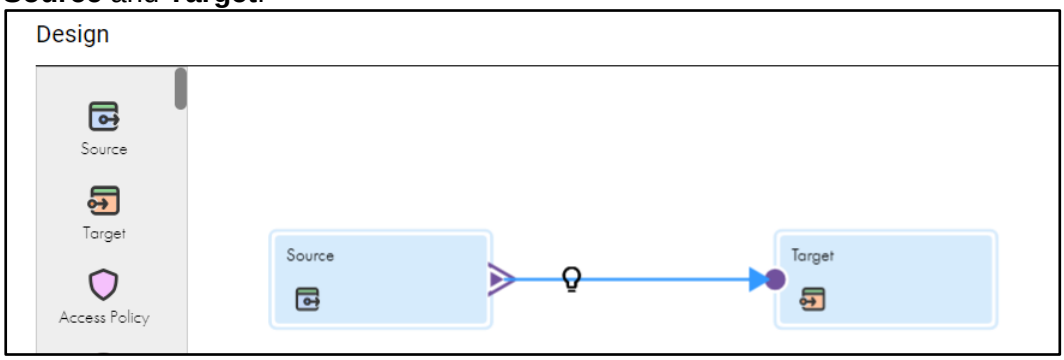
50. From the Runtime Environment drop-down, select INFA-SERVER.
51. Click **Run**.

Monitor Status

52. To monitor the mapping status, from the navigation pane, click **My Jobs**.
53. When the task completes, the status changes to **Success**.
    Note:
    i. If the task completes with 0 processed rows, click on the Instance Name, and check the error message.

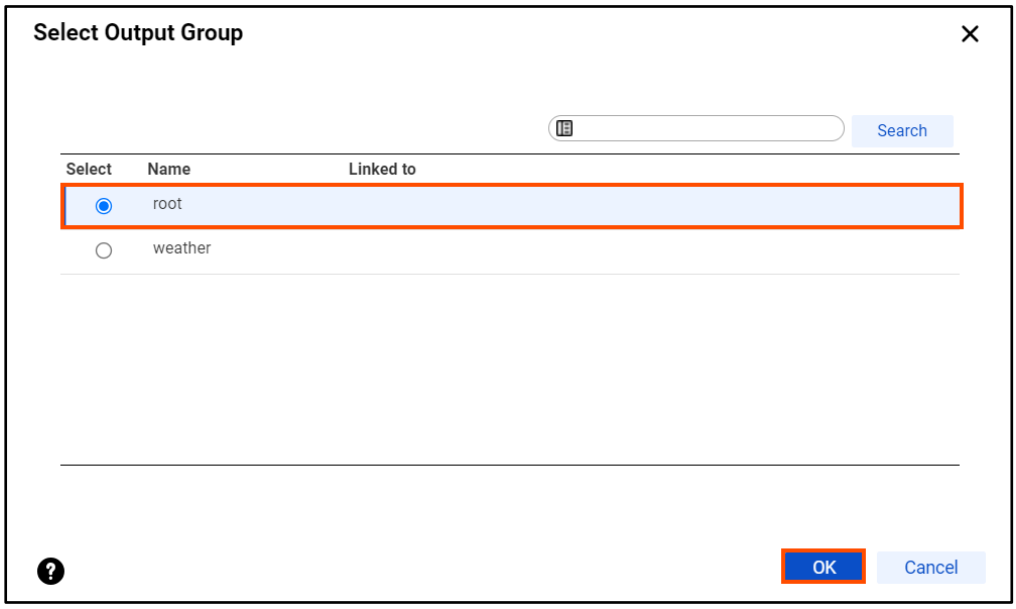


    ii. If you see the following error message, perform the below mentioned steps:
    [ERROR] log4j:ERROR A "org.apache.log4j.ConsoleAppender" object is not assignable to a "org.apache.log4j.Appender" variable,
    then, open the REST_GetWeather_By_City mapping and delete the link between **Source** and **Target**.



    iii. Again, link Source to Target.
    iv. In the **Select Output Group** window, select **root**.

v. Click **OK**.



vi. Save and run the mapping again.

54. Close the asset from the navigation pane.
55. Optionally, you can preview the target data records using your dummy mapping.

This concludes the lab.

Module 14: IICS REST APIs

Lab 14-2: Using Web Services Transformation in a Mapping

Overview:

A web service is a system software that enables machine-to-machine interaction over a network. It makes applications platform-and-technology independent.

In this lab, you will create a business service. You will also create a mapping that calls the business service in a mapping using web service transformation.

Objective:

- Create a Business Service
- Use Web Services transformation in a mapping

Scenario:

Ruby wants to change the format of billing for NH Retail outlets. She wants to convert the price information from numbers to words. So, John suggests using a business service to convert numbers to words.

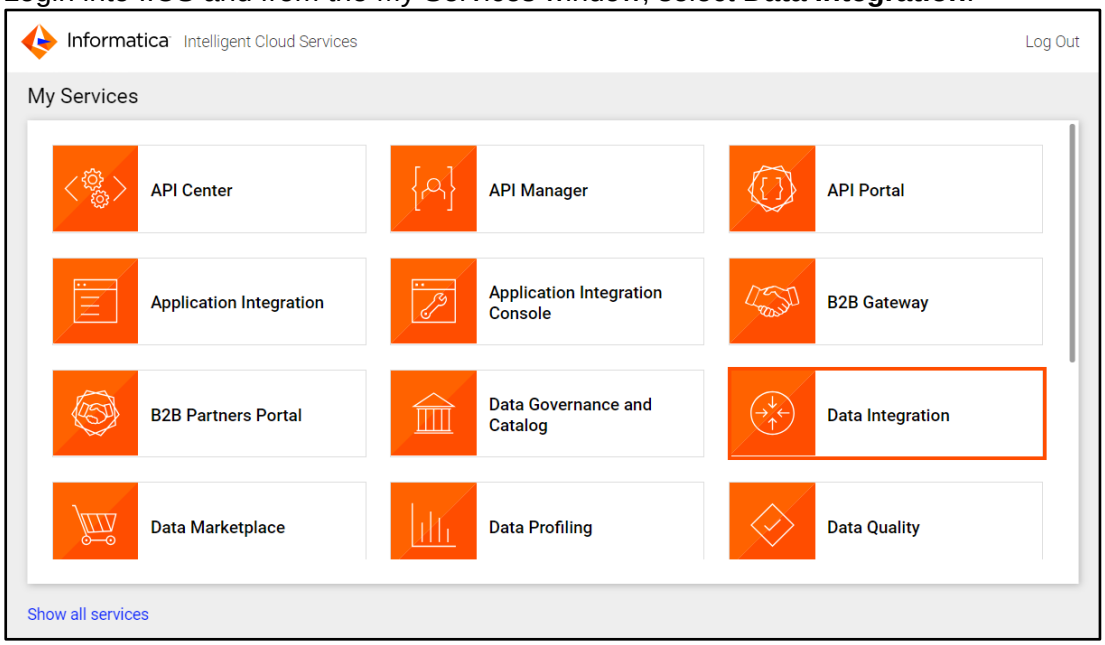
In this lab, John creates a business service that connects to the NumberToDollars web service. He also creates a mapping in IICS and calls the business service in the mapping using the web service transformation. The transformation converts a list of numbers received as input into words to generate a flat file output.

Duration:

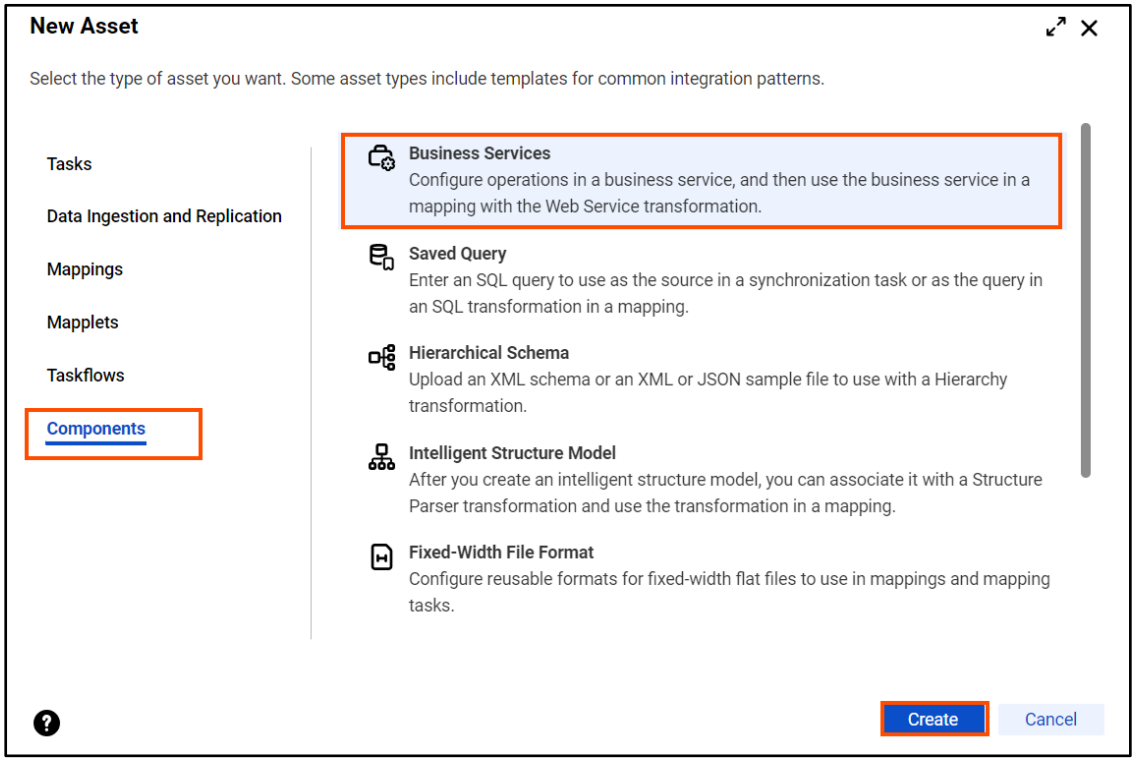
25 minutes

Tasks
Create Business Service

1. Login into IICS and from the My Services window, select **Data Integration**.

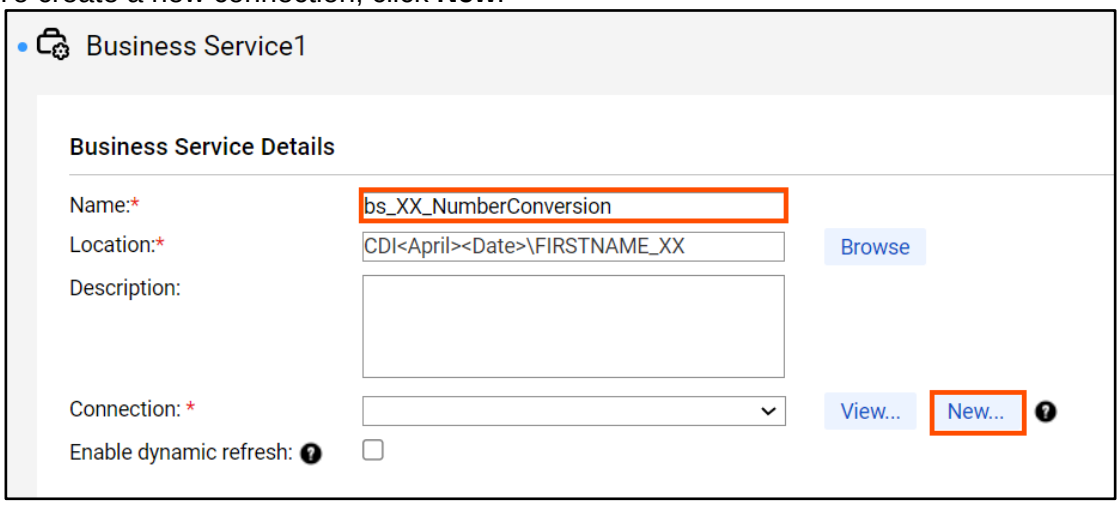


2. From the navigation pane, select **New**.
3. From the New Asset window, click the **Components** tab.
4. Select **Business Services** and click **Create**.

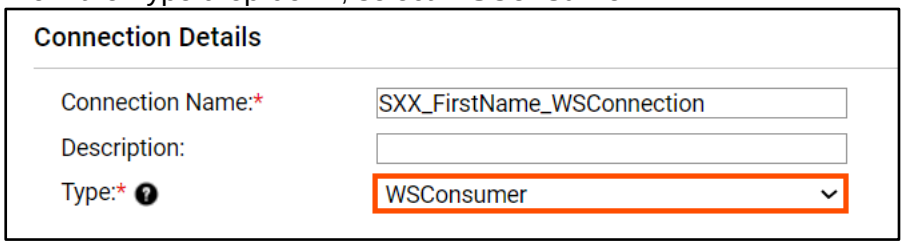


5. In the Name field, enter **bs_XX_NumberConversion**.

6. To create a new connection, click **New**.



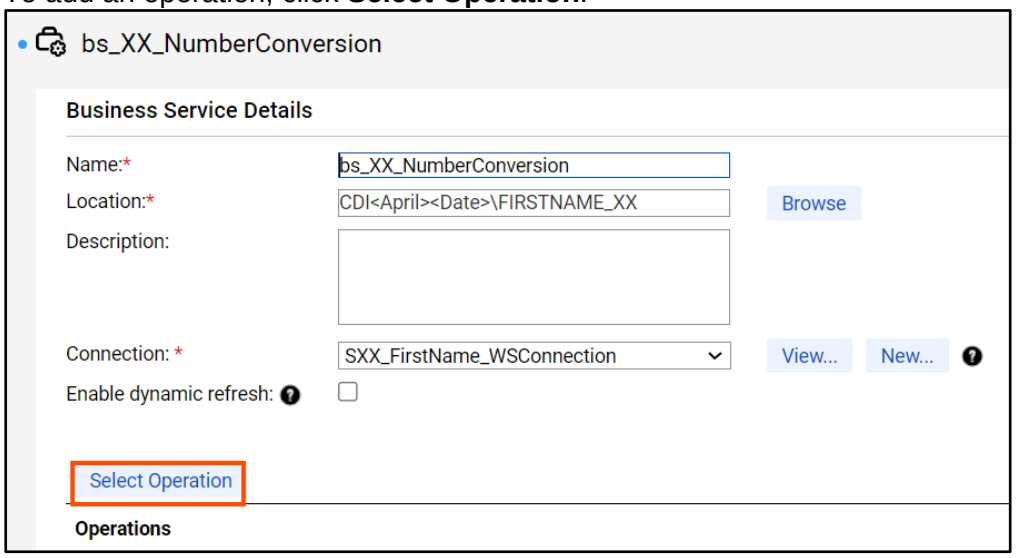
7. In the Name field, enter **SXX_FirstName_WSConnection**.
8. From the Type drop-down, select **WSConsumer**.



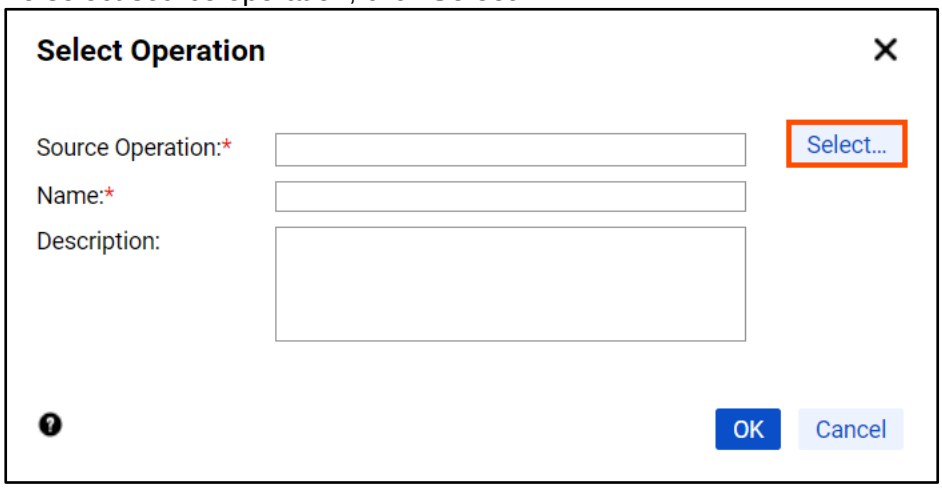
9. From the Runtime Environment drop-down, select INFA-SERVER.
10. From the Authentication drop-down, select **Other Authentication**.
11. In the WSDL URL field, enter the following URL:
    https://www.dataaccess.com/webservicesserver/numberconversion.wso?WSDL
12. In the Endpoint URL field, enter the following URL:
    https://www.dataaccess.com/webservicesserver/numberconversion.wso
13. Click **OK**.

14. From the Connection drop-down, select **SXX_FirstName_WSConnection**.
    Note: Skip this step if the created connection is already selected.

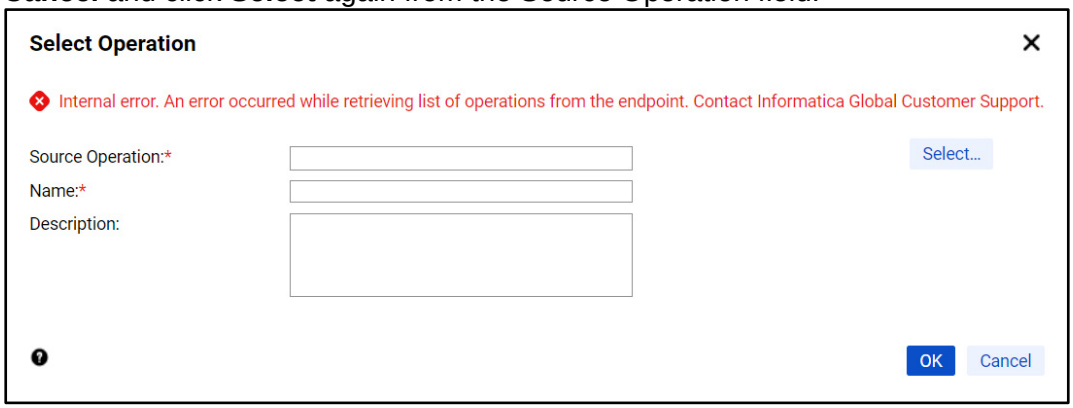
15. To add an operation, click **Select Operation**.



16. To select source operation, click **Select**.

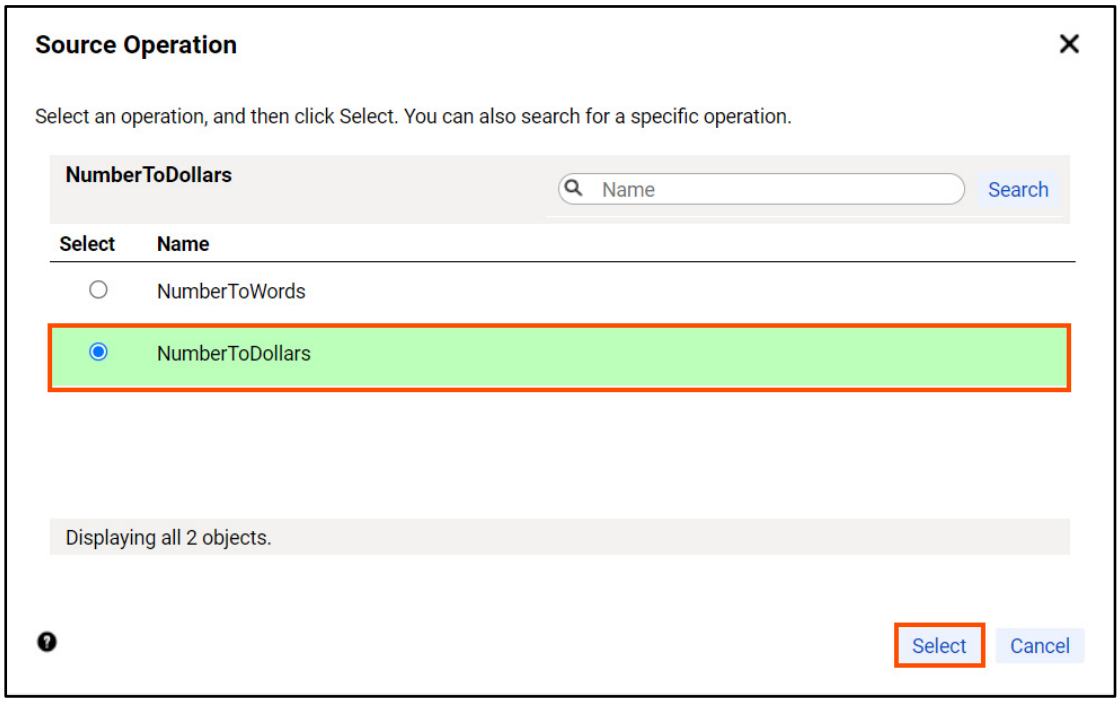


Note: If you get the Internal error message as shown in the screenshot below, click **Cancel** and click **Select** again from the Source Operation field.



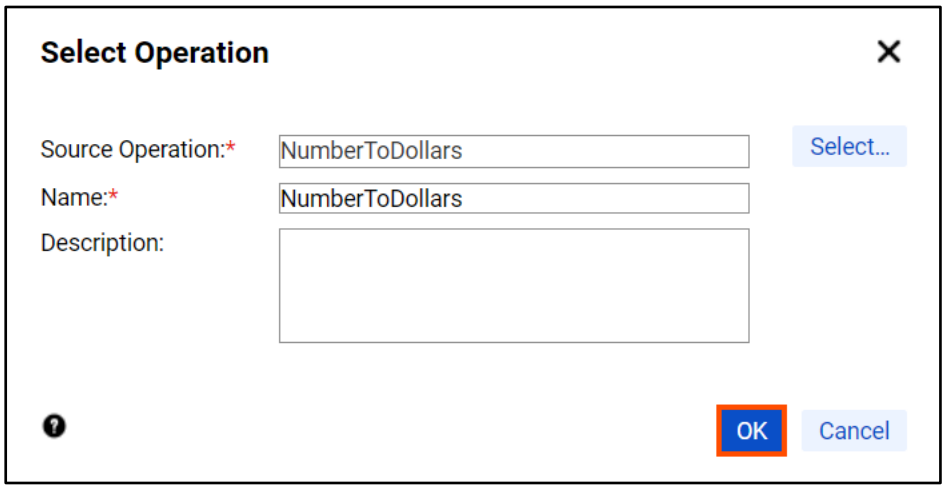
17. From the list, select **NumberToDollars**.

18. Click **Select**.

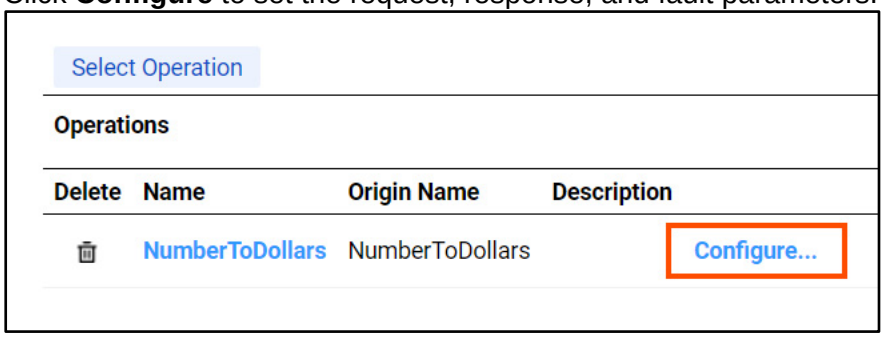


Note: This action redirects you to the Select Operation window.

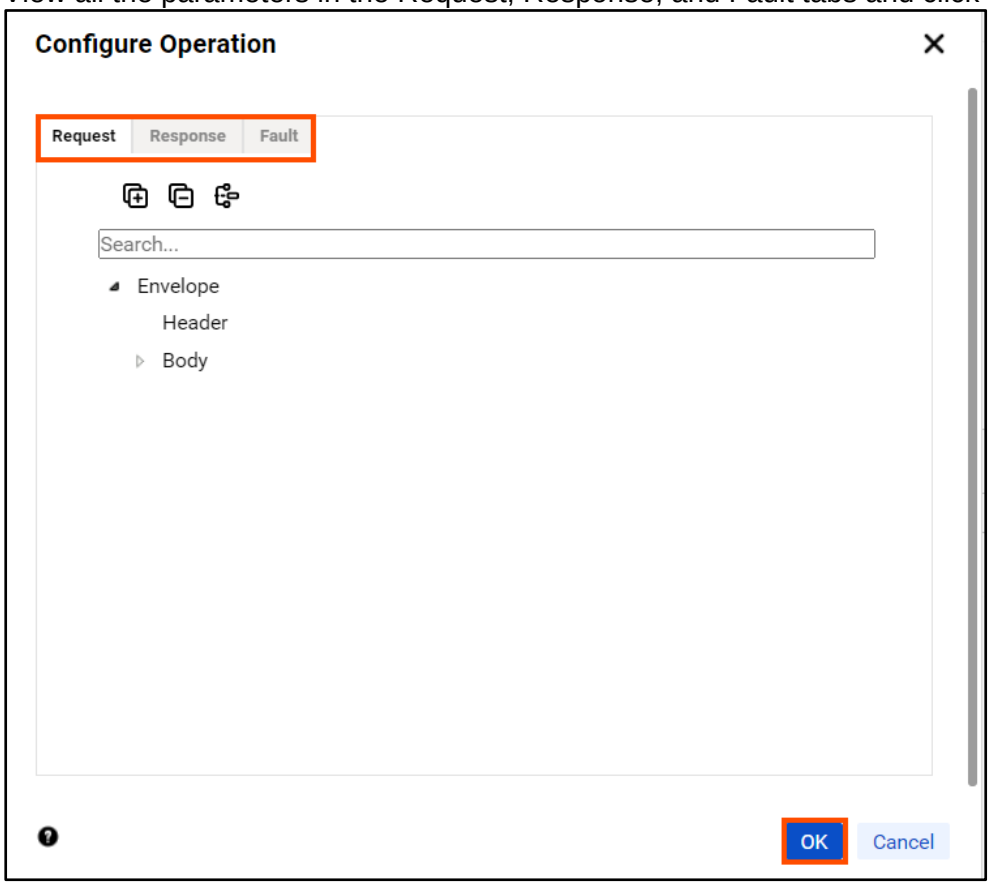
19. Click **OK**.



20. Click **Configure** to set the request, response, and fault parameters.



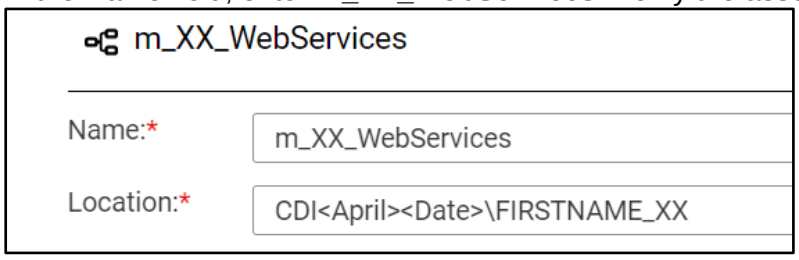
21. View all the parameters in the Request, Response, and Fault tabs and click **OK**.



22. Save the business service.

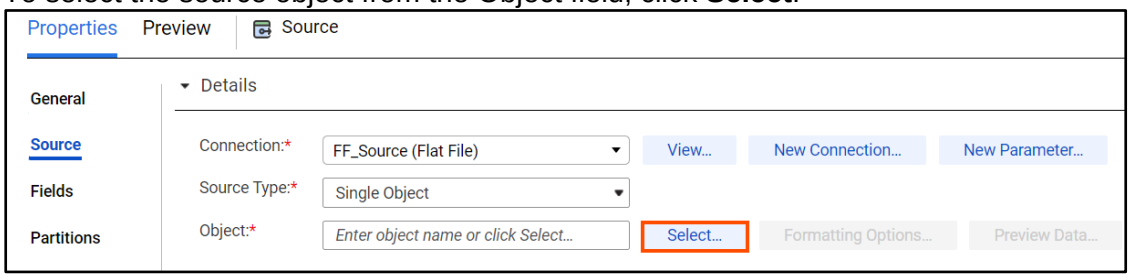
Create a Mapping

23. Create a new Mapping.
24. In the Name field, enter **m_XX_WebServices**. Verify the asset location.

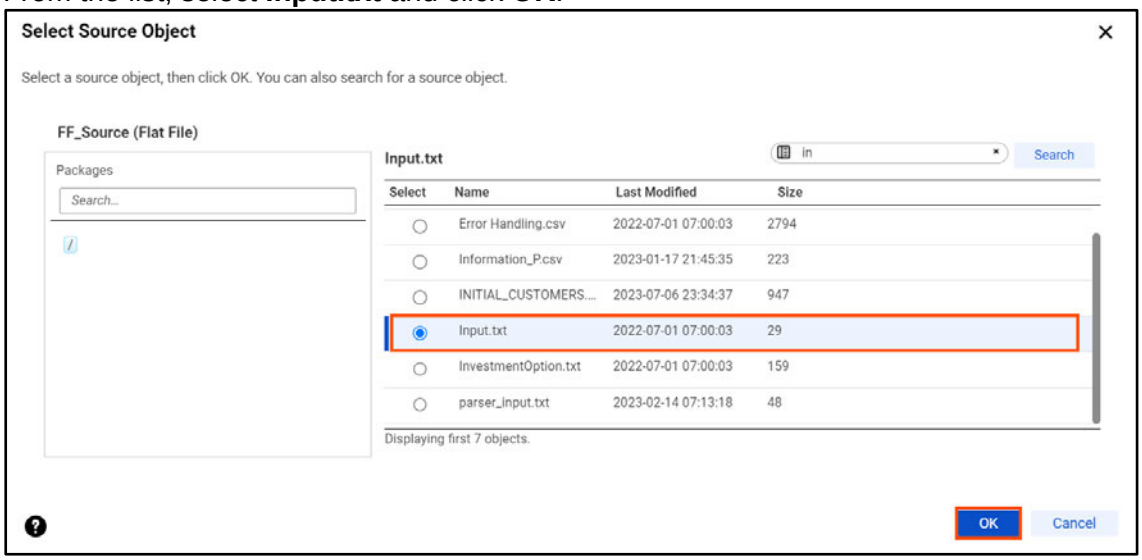


25. To configure the source, from the mapping canvas, click the **Source** transformation.
26. From the properties pane, click **Source**.
27. From the Connection drop-down, select **FF_Source**.

28. To select the source object from the Object field, click **Select**.

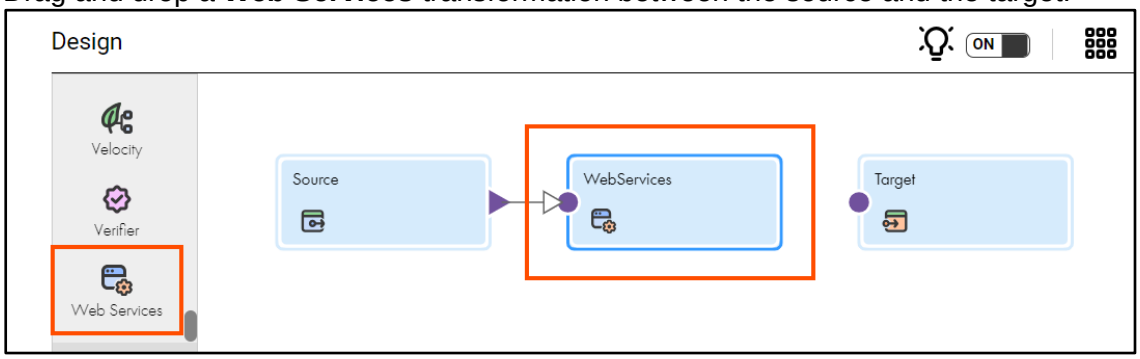


29. From the list, select **Input.txt** and click **OK**.



Add Web Services Transformation

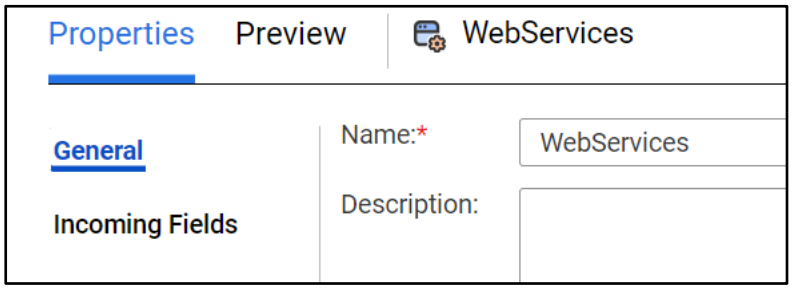
30. Drag and drop a **Web Services** transformation between the source and the target.



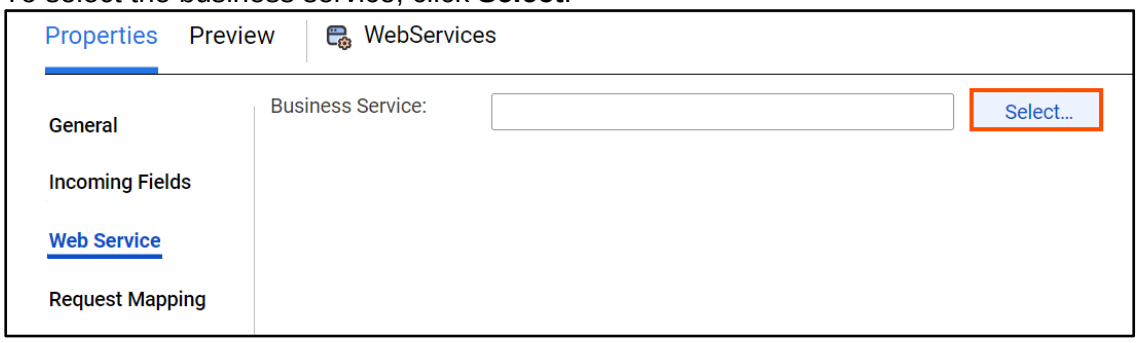
Note: The Source transformation gets linked to the Web Service transformation.

31. Select the **WebServices** transformation on the mapping canvas.

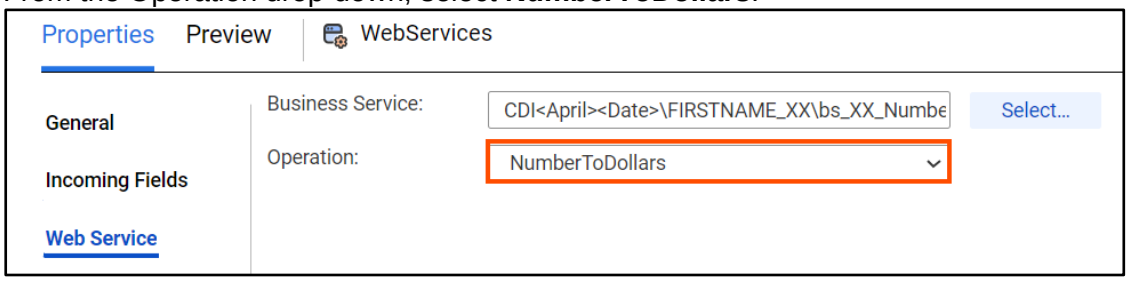
32. In the **General** section of the WebServices properties, retain the Name as **WebServices**.



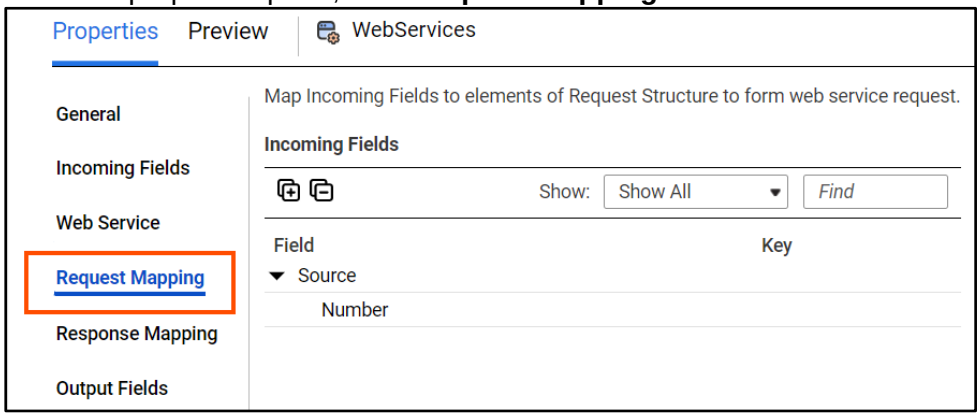
33. From the properties pane, click **Web Service**.
34. To select the business service, click **Select**.



35. Navigate to your working directory and select the **bs_XX_NumberConversion** business service.
36. From the Operation drop-down, select **NumberToDollars**.

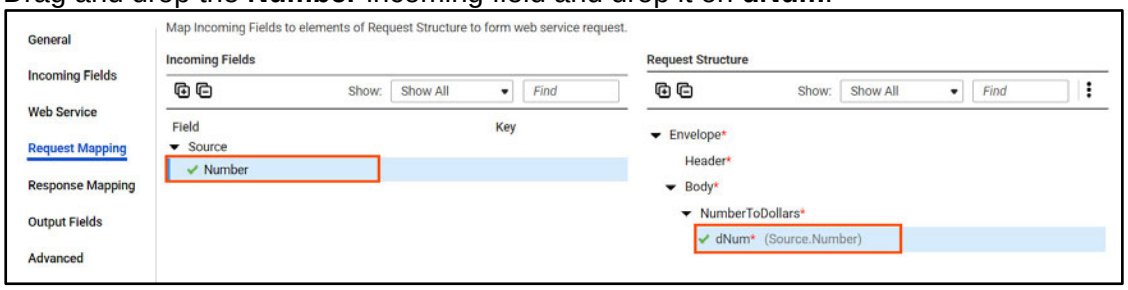


37. From the properties pane, click **Request Mapping**.

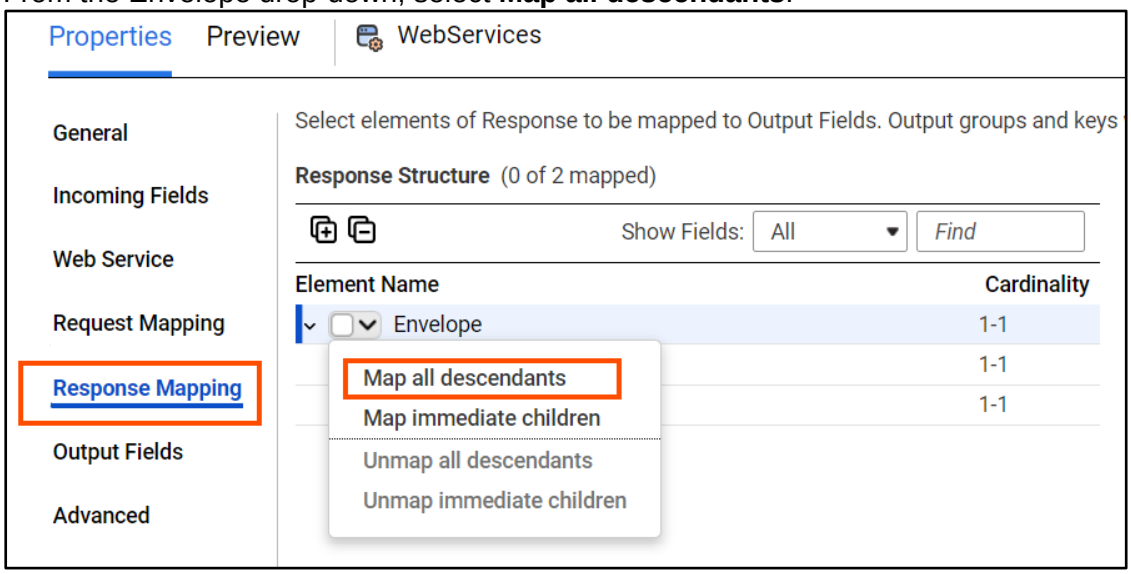


38. From the Request Structure section, drill-down to **Body** > **NumberToDollars** > **dNum**.

39. Drag and drop the **Number** incoming field and drop it on **dNum**.

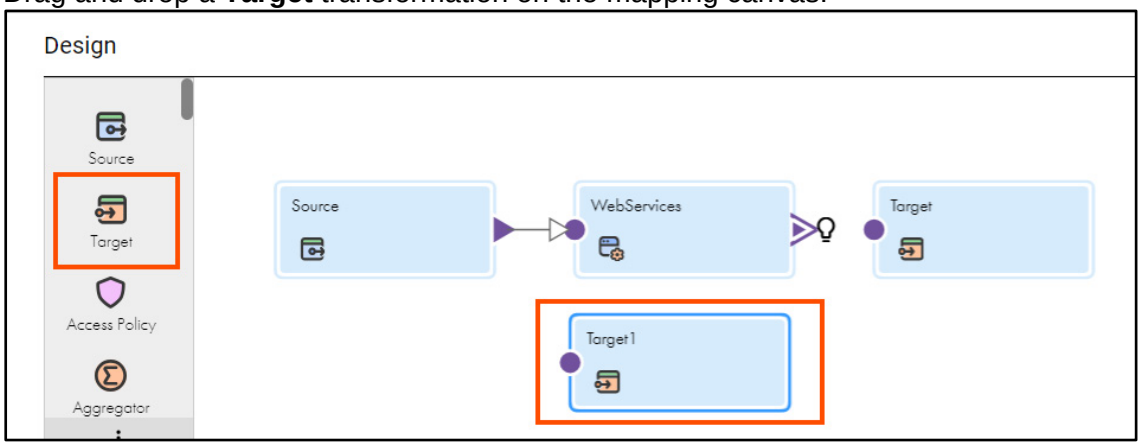


40. From the properties pane, click **Response Mapping**.
41. From the Envelope drop-down, select **Map all descendants**.



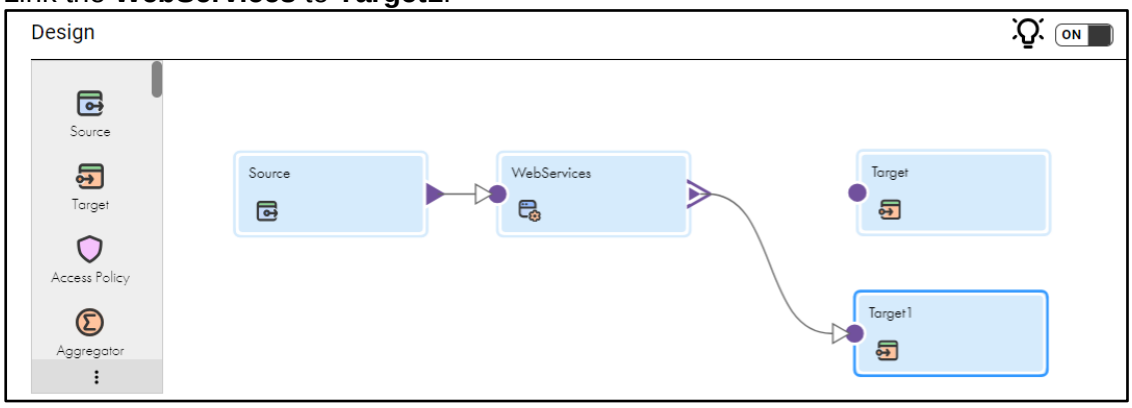
Add a Target Transformation

42. Drag and drop a **Target** transformation on the mapping canvas.

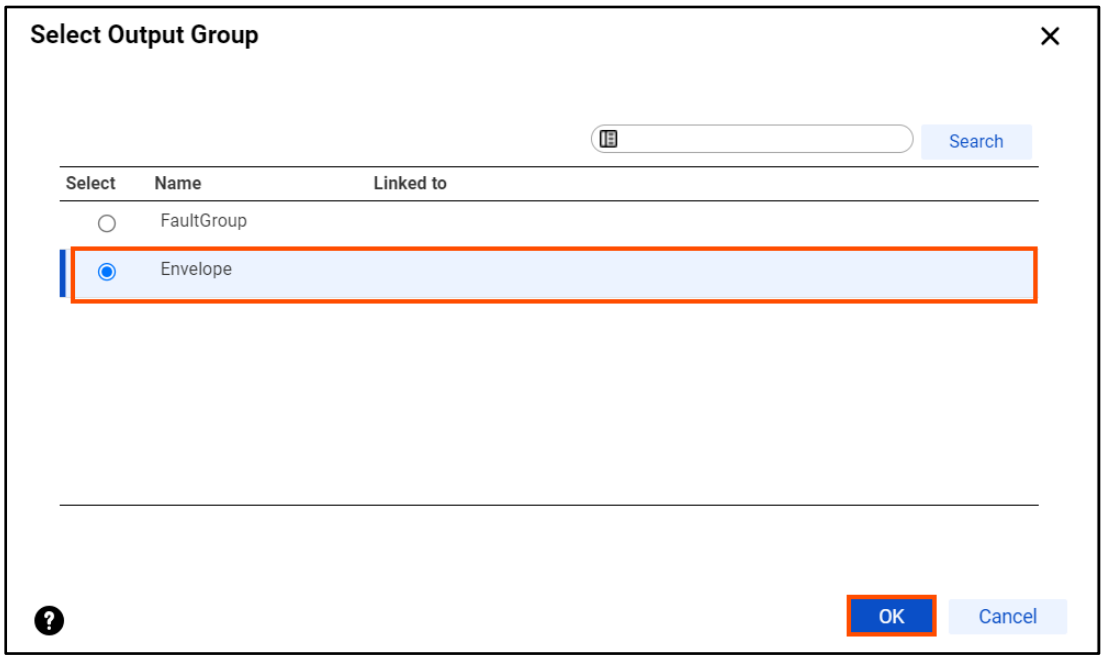


43. Select the added Target transformation (**Target 1**) on the mapping canvas.

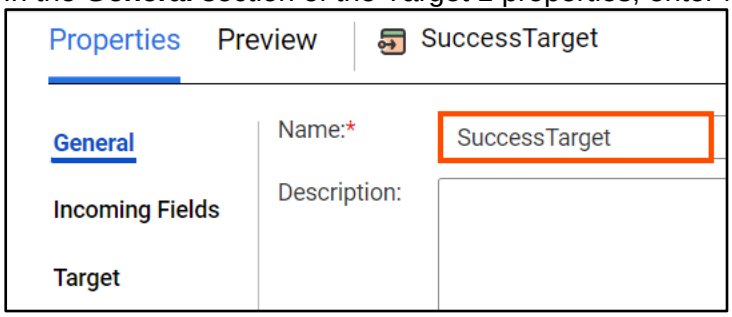
44. Link the **WebServices** to **Target1**.



45. From the list, select **Envelope**.
46. Click **OK**.

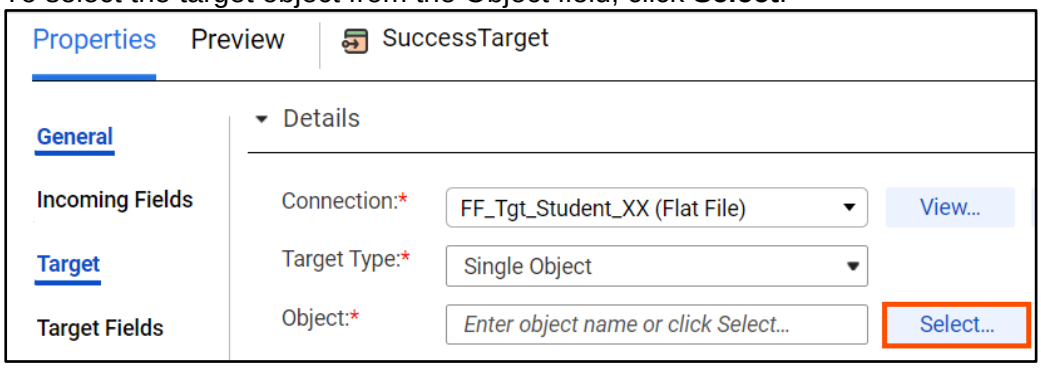


47. In the **General** section of the Target 1 properties, enter Name as **SuccessTarget**.

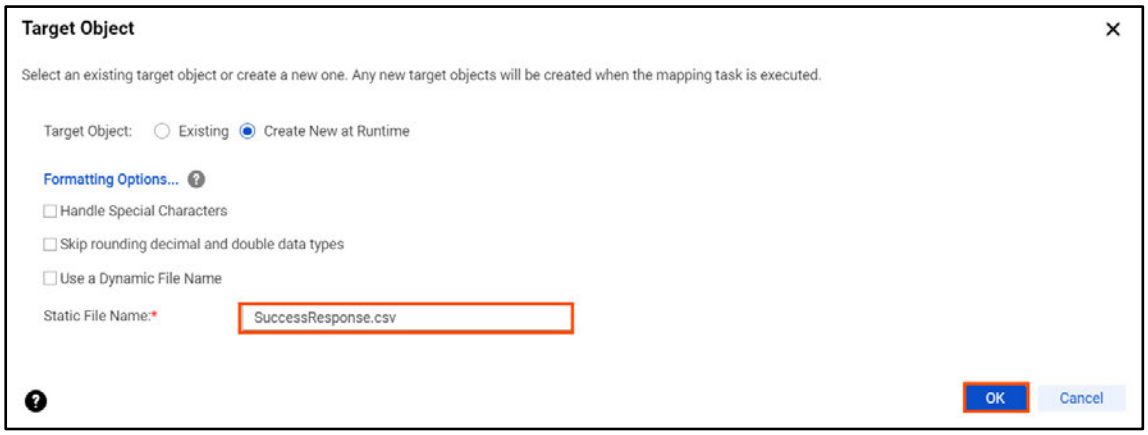


48. From the properties pane, click **Target**.
49. From the Connection drop-down, select your **target Flat File** connection.

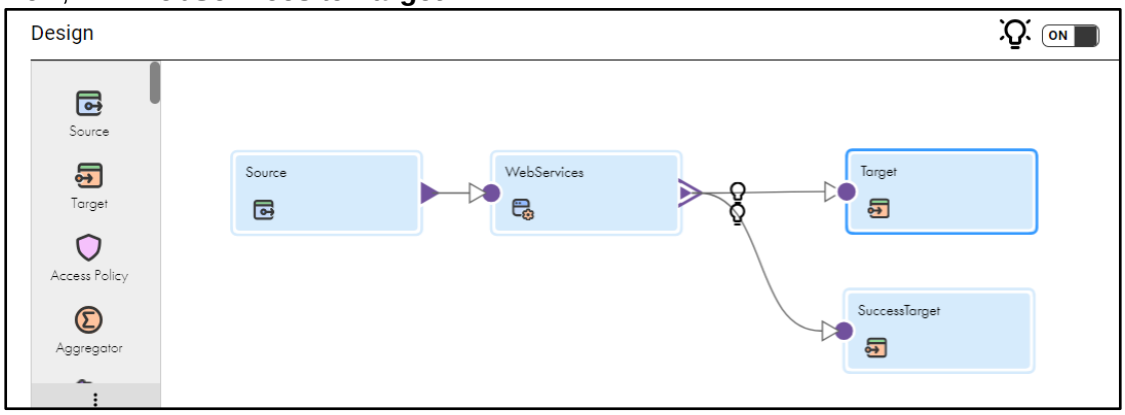
50. To select the target object from the Object field, click **Select**.



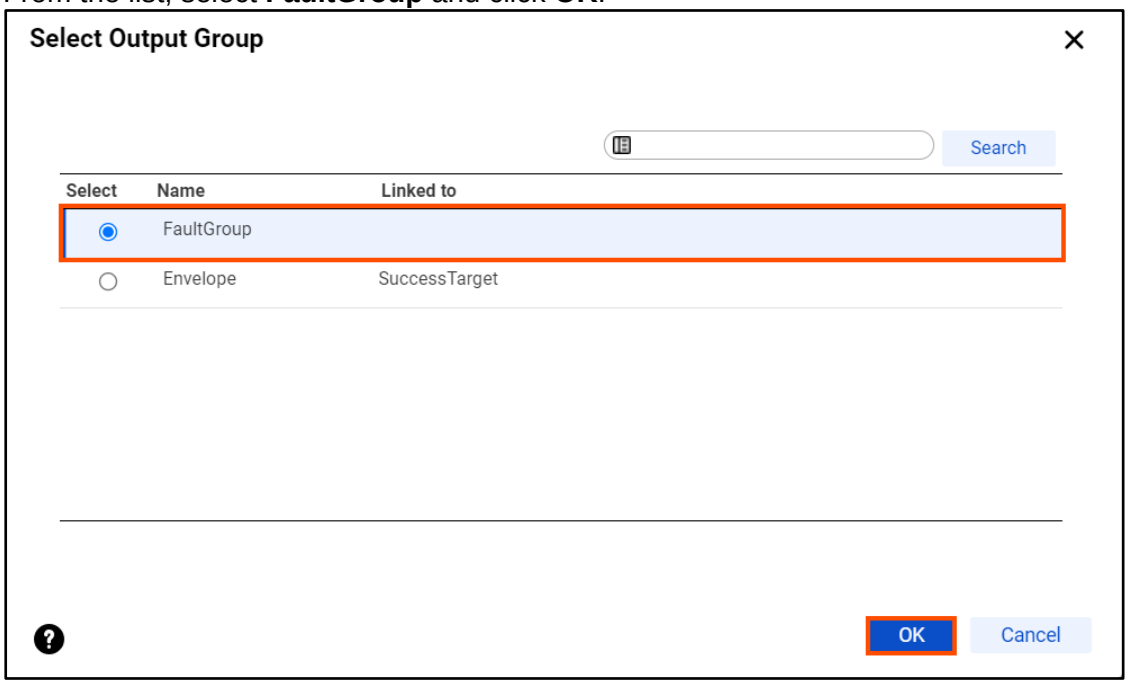
51. In the Target Object window, select **Create New at Runtime**.
52. Enter **SuccessResponse.csv** as Static File Name.
53. Click **OK**.



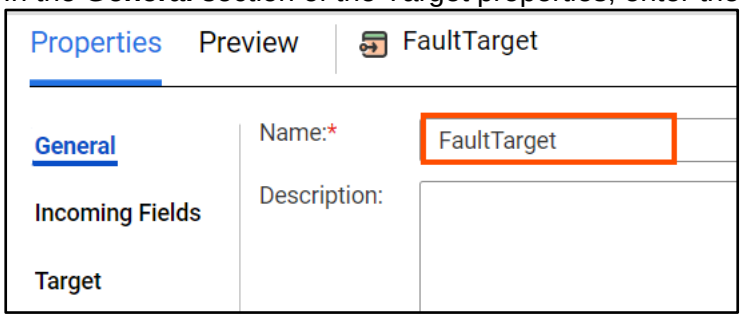
54. Now, link **WebServices** to **Target**.



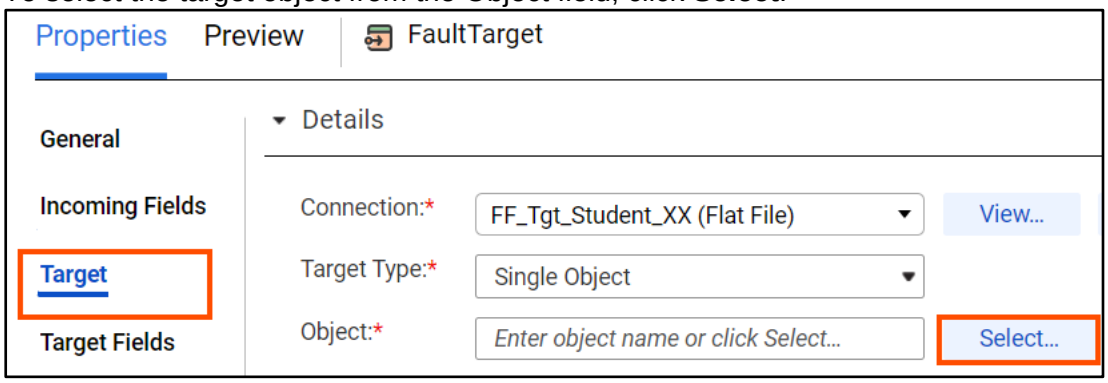
55. From the list, select **FaultGroup** and click **OK**.



56. Select the Target transformation on the mapping canvas.
57. In the **General** section of the Target properties, enter the Name as **FaultTarget**.

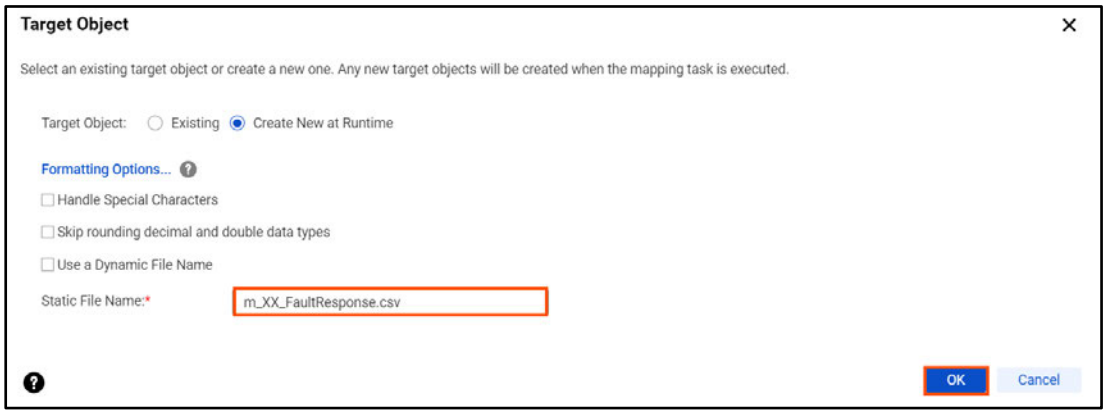


58. From the properties pane, click **Target**.
59. From the Connection drop-down, select your **target Flat File** connection.
60. To select the target object from the Object field, click **Select**.

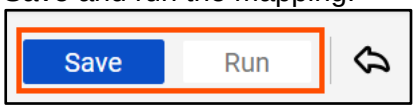


61. In the Target Object window, select **Create New at Runtime**.
62. Enter **m_XX_FaultResponse.csv** as Static File Name.

63. Click **OK**.



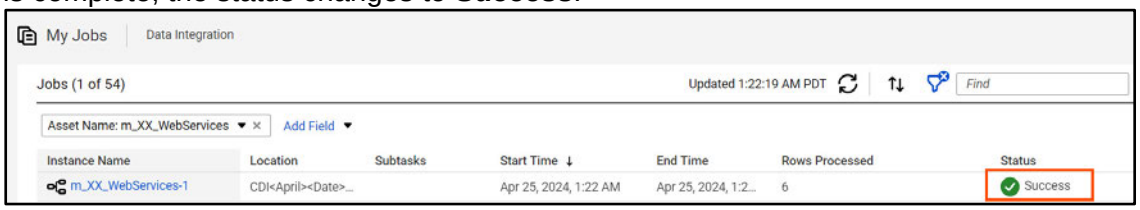
64. Save and run the mapping.



65. From the Runtime Environment drop-down, select INFA-SERVER.
66. Click **Run**.

Monitor Status

67. To monitor the mapping status, from the navigation pane, click **My Jobs**. When the task is complete, the status changes to **Success**.



68. Close the asset from the navigation pane.
69. Optionally, you can preview data in the dummy mapping.

This concludes the lab.

Module 14: IICS REST APIs

Lab 14-3: Running a Mapping Task Using REST API

Overview:
An API Connector uses the REST API to access programs and activities in Informatica Cloud and perform various tasks like logging into IICS, running a task, and much more.

In this lab, you will use the REST API to log in to your Org and run tasks.

Objective:

- Use a REST client application to call the login resource and obtain a session ID
- Start a Mapping task
- Log out of the Informatica Cloud API session

Scenario:

John demonstrates to Ruby how she can use REST APIs to log in and run tasks in IICS. He uses the Postman client to log in to the Org and obtain a session ID. He will use the session ID to run a mapping task using the REST API and then log out from the Org.

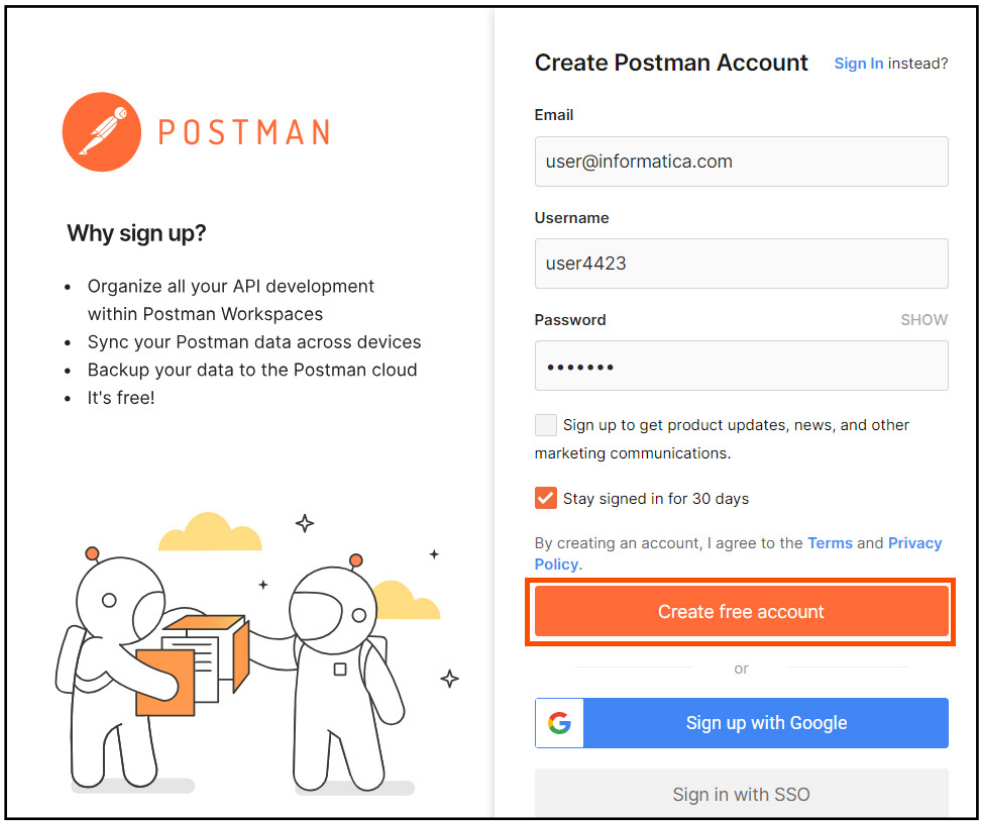
Duration:
25 minutes

Tasks

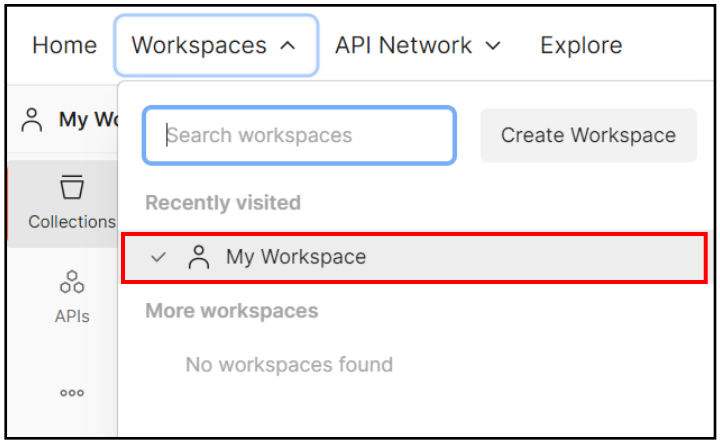
Create a Postman Account

1. Open a web browser and enter the following link to open Postman: https://identity.getpostman.com/
2. Click **Create Account**.

3. Provide a valid email Id and fill in all the required details. Click **Create free account**.



4. You will get an account confirmation email to the registered email id. Confirm your postman account.
5. Now you will be logged into Postman. If not, login into Postman using your credentials.
6. From the Workspace drop-down, select **My Workspace**.

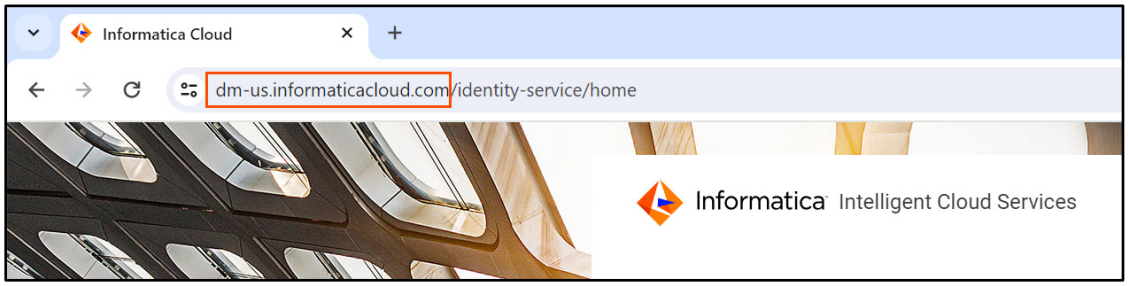


Send a Login Request

7. Click on a new tab in the workspace section and from the drop-down, select **POST**.
8. Enter the URL in the request URL field in the following format:
   https://<Informatica URL>/ma/api/v2/user/login

   Note: The Informatica URL can change based on the POD (Point of Deployment) region of your IICS Org. Your POD region is based on the location of your Informatica Intelligent Cloud Services data center. Use one of the following POD regions
   - For North America, use **us**
   - For Europe, use **em**
   - For Asia, use **ap**

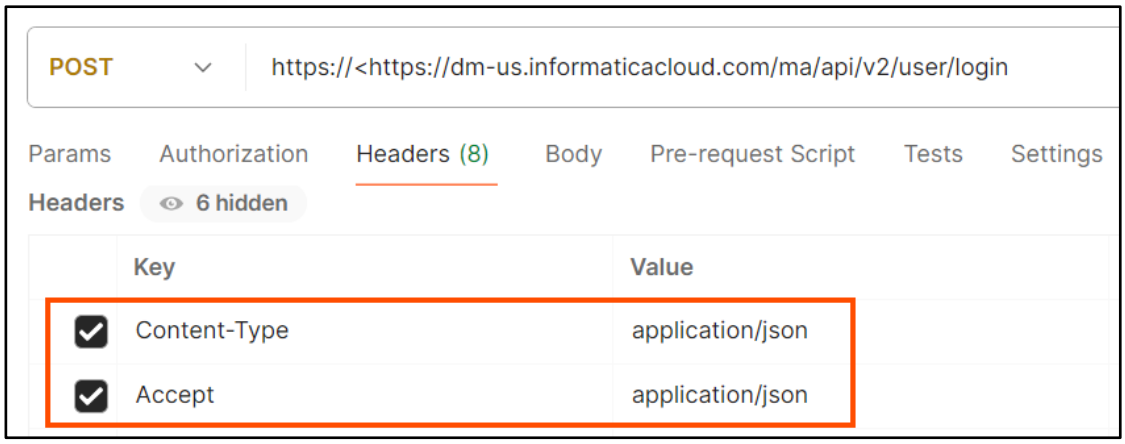
   You can identify the Informatica URL from the IICS login page, as shown in the image below:



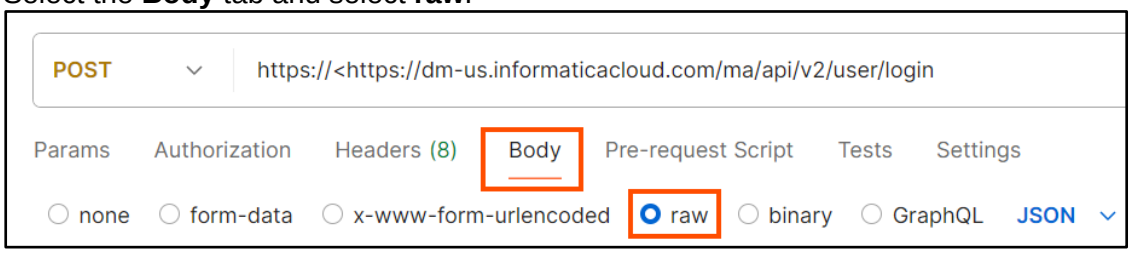
9. For the screenshot above, following is the URL (you must modify the URL based on your IICS Account link):
   https://dm-us.informaticacloud.com/ma/api/v2/user/login

   Note: Make sure to remove extra space from the URL after pasting the URL into the request field.
10. Select the **Headers** tab.
11. Enter Key and Value, as shown in the table below:

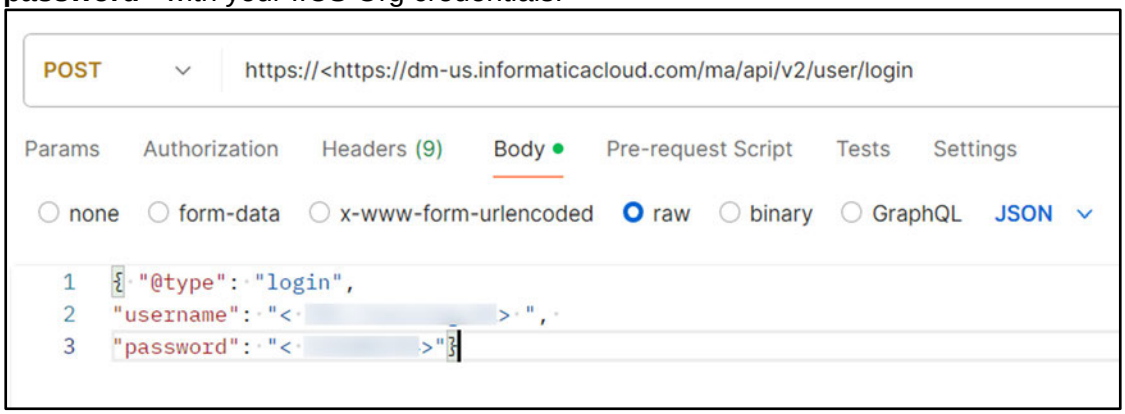
Key	Value
Content-Type	application/json
Accept	application/json



12. Select the **Body** tab and select **raw**.

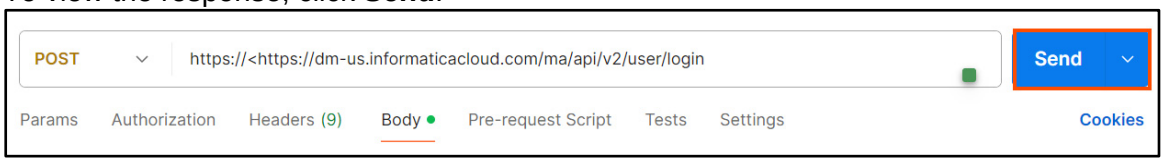


13. Enter the following syntax:
    { "@type": "login",
    "username": "< Informatica Cloud username> ",
    "password": "< Informatica Cloud password>"}
14. In the syntax, replace **< Informatica Cloud username>** and **< Informatica Cloud password>** with your IICS Org credentials.



Note: For this lab, we have masked the username and password in the screenshot.

15. To view the response, click **Send**.

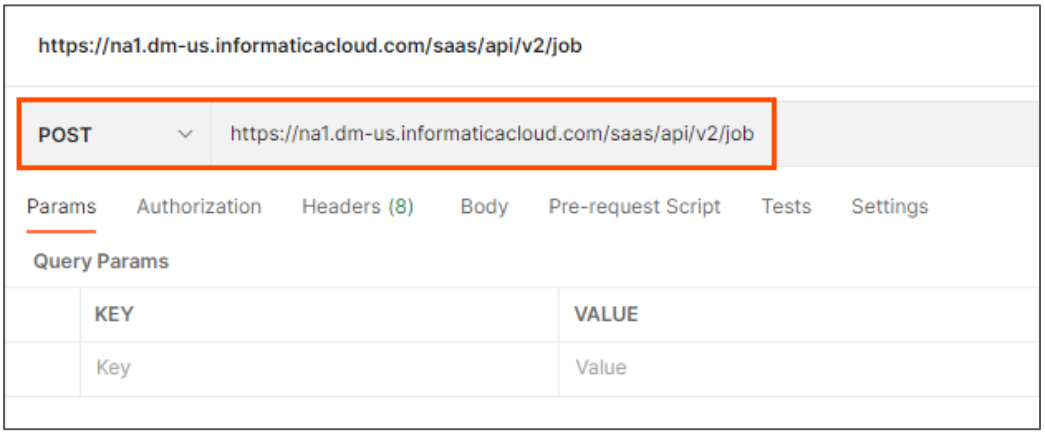


16. The status of the response is **200 OK**.

17. From the response, note down the values for **ServerUrl** and **icSessionId** in a notepad.

Send Request to Run a Job

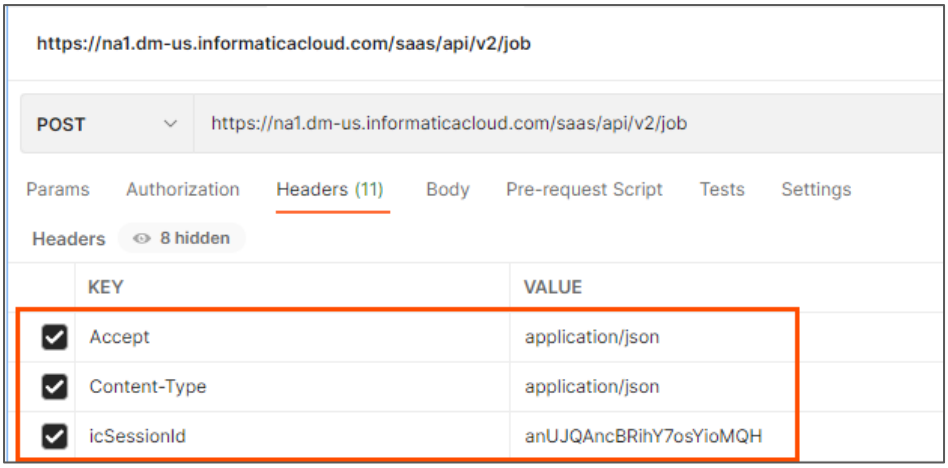
18. To add a new tab, click .
19. From the drop-down, select **POST**.
20. Enter the URL in the request URL field in the following format:
    <Copied ServerUrl>/api/v2/job



Note: Make sure to remove extra space from the URL after pasting the URL in the request field.

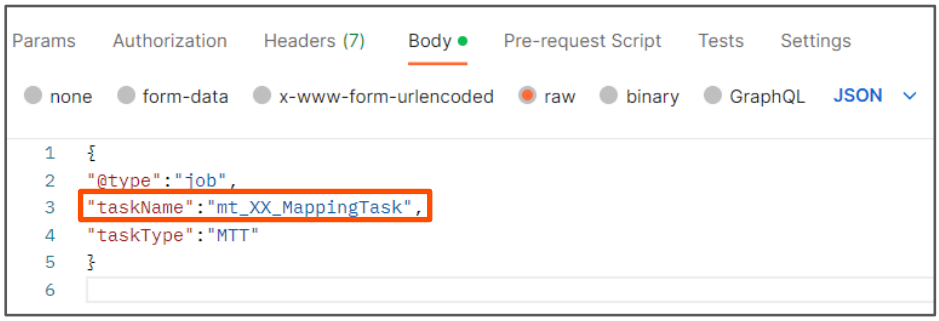
21. Select the **Headers** tab.
22. Enter Key and Value, as shown in the table below:

Key	Value
Accept	application/json
Content-Type	application/json
icSessionId	Enter the icSessionId noted earlier



23. Select the **Body** tab and select **raw**.
24. Enter the following syntax:
    Note: In this syntax, use the exact asset name you have used while creating this request. Replace XX with your DOB.

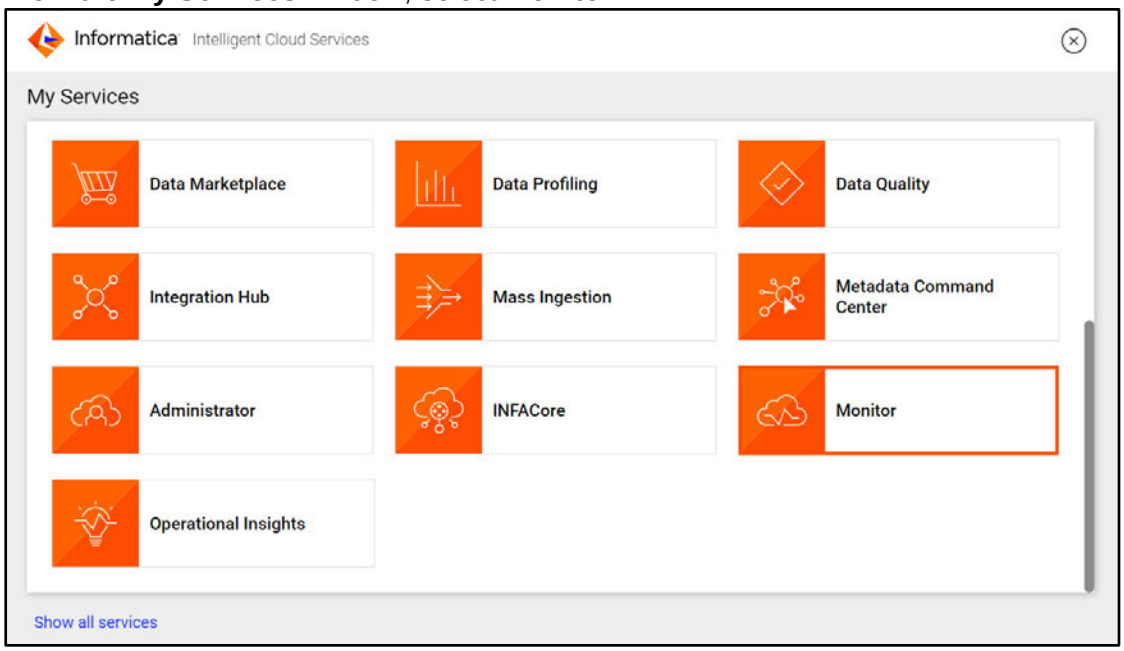
    {
    "@type":"job",
    "taskName":"mt_XX_MappingTask",
    "taskType":"MTT"
    }
25. Ensure that the **taskName** is same as the task name in IICS.



    Note: The task name shown in the screenshot is for demonstration purposes and you must specify your exact task name as a value to the taskName parameter.
26. To view the response, click **Send**.

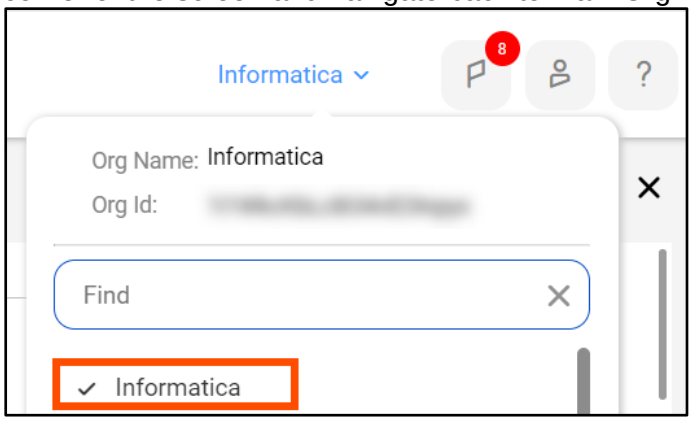
27. The status of the response is **200 OK**.

28. Open the IICS Login page from the Bookmarks bar.
29. Log in to your IICS Org.
    Note: If you are already logged in, log out and log in again.
30. From the **My Services** window, select **Monitor**.



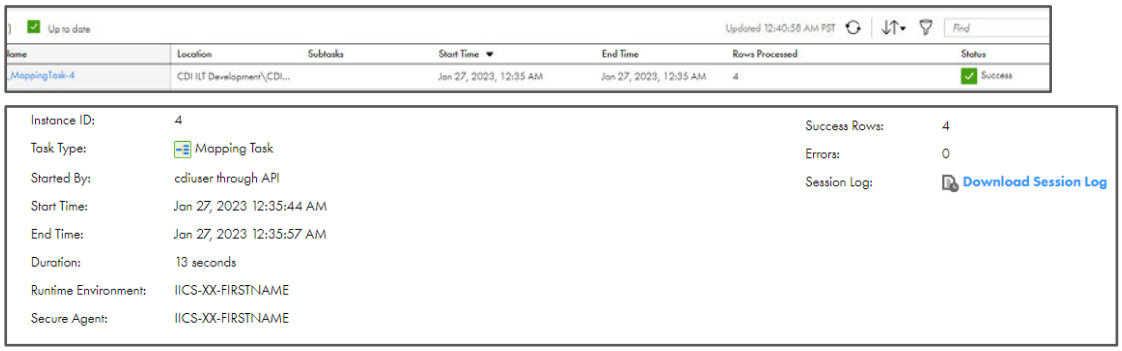
Note: Ensure that you are accessing the service from the main org.

Important: If you are in the sub-org, click on the Org name available on the top right corner of the screen and navigate back to main Org.



31. From the navigation pane, select **My Jobs**.

32. Open the **mt_XX_MappingTask** and verify that the task is started by API.

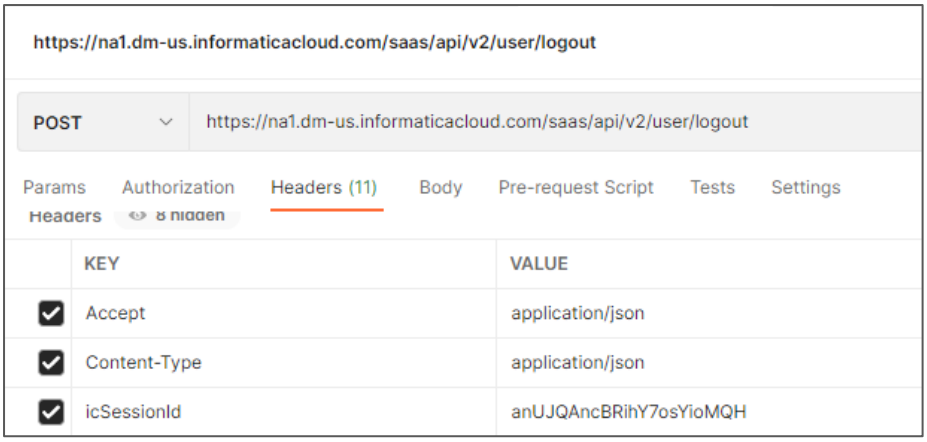


Send a Request to Logout

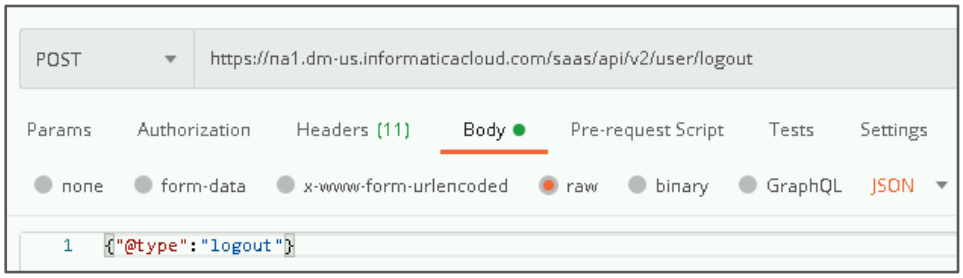
33. In Postman, to add a new tab, click +.
34. From the drop-down, select **POST**.
35. Enter the following URL in the request URL field:
    <Copied ServreUrl>/api/v2/user/logout

    Note: Make sure to remove extra space from the URL after pasting the URL in the request field.
36. Select the **Headers** tab.
37. Enter Key and Value, as shown in the table below:

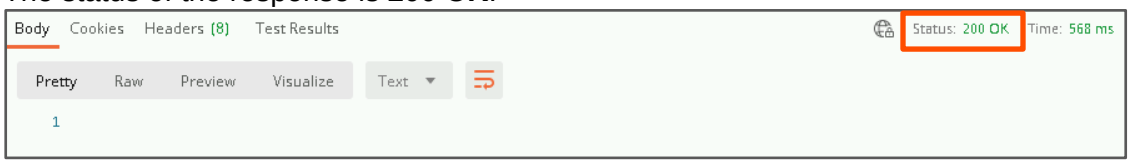
Key	Value
Accept	application/json
Content-Type	application/json
icSessionId	Enter the icSessionId noted earlier



38. Select the **Body** tab and select **raw**.
39. Enter the following syntax:
    {"@type":"logout"}



40. To view the response, click **Send**.
41. The status of the response is **200 OK**.



Note: There is no response in body for log out request.

This concludes the lab.

Module 14: IICS REST APIs

Lab 14-4: Monitoring Tasks Using REST API

Overview:
Informatica Intelligent Cloud Services supports the platform REST API version 2 and version 3 resources, and service-specific resources.
In this lab, you will use the REST API to request log information for running jobs from the Monitor service.

Objective:

- Use a REST client application to call the login resource and obtain a session ID
- Get the log information about the running jobs

Duration:
15 minutes

Tasks

Open Postman

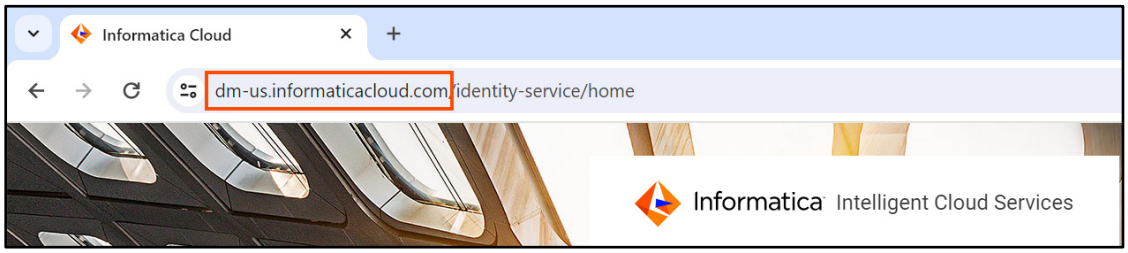
1. Navigate to the web Postman application.
2. Click ⊞ to open a new tab.

Send a Login Request

3. From the drop-down, select **POST**.
4. Enter the URL in the request URL field in the following format:
   https://<Informatica URL>/ma/api/v2/user/login

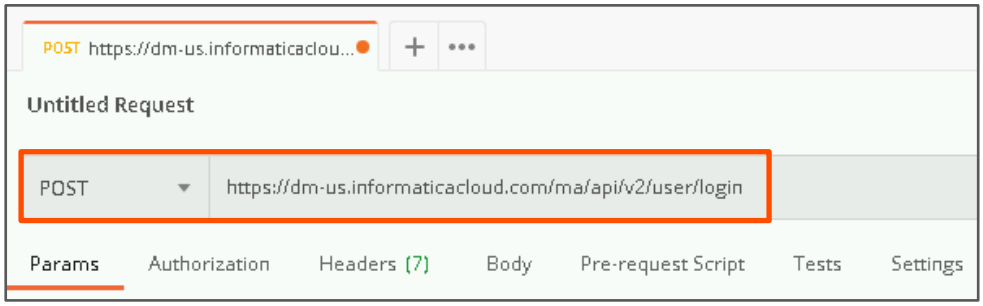
   Note: The Informatica URL can change based on the POD (Point of Deployment) region of your IICS Org. Your POD region is based on the location of your Informatica Intelligent Cloud Services data center. Use one of the following POD regions
   - For North America, use **us**
   - For Europe, use **em**
   - For Asia, use **ap**

You can identify the Informatica URL from the IICS login page, as shown in the image below:



5. For the screenshot provided above, the following is the URL (you must enter your IICS Account URL details):

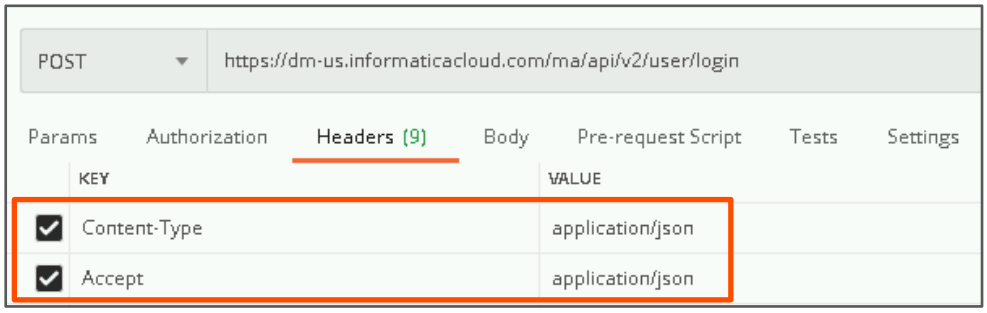
   https://dm-us.informaticacloud.com/ma/api/v2/user/login



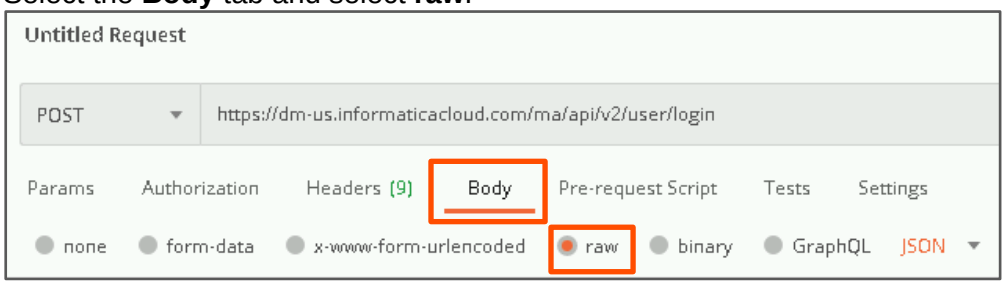
   Note: Make sure to remove extra space from the URL after pasting the URL in the request field.

6. Select the **Headers** tab.
7. Enter Key and Value, as shown in the table below:

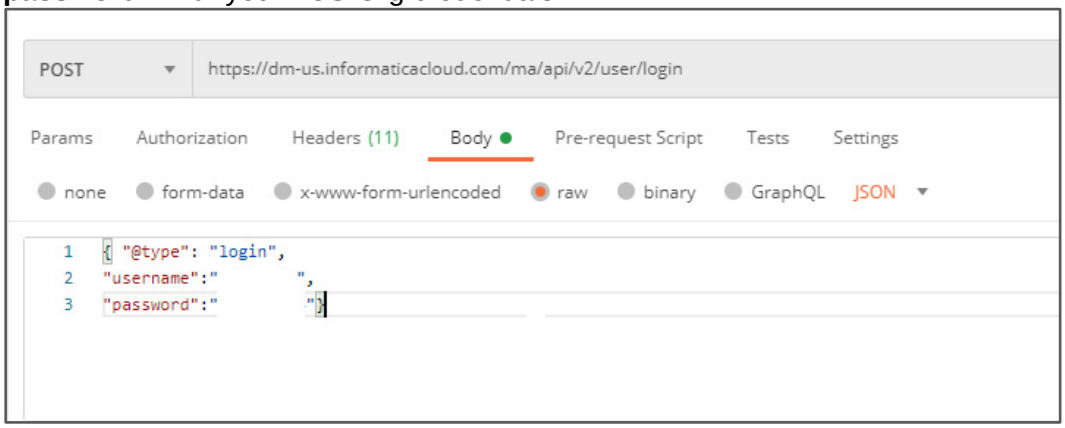
Key	Value
Content-Type	application/json
Accept	application/json



8. Select the **Body** tab and select **raw**.

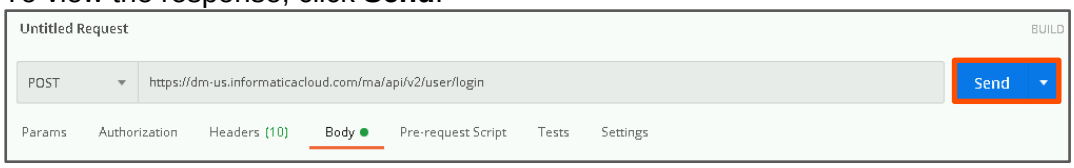


9. Enter the following syntax:
   { "@type": "login",
   "username": "< Informatica Cloud username> ",
   "password": "< Informatica Cloud password>"}
10. In the syntax, replace **< Informatica Cloud username>** and **< Informatica Cloud password>** with your IICS Org credentials.



Note: For this lab, we have masked the username and password in the screenshot.

11. To view the response, click **Send**.

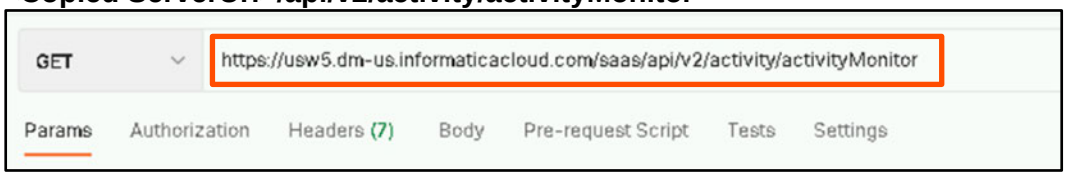


12. The status of the response is **200 OK**.
13. From the response, note down the values for **ServerUrl** and **icSessionId** in a notepad.

Request the information from the activity monitor

14. To add a new tab, click .
15. From the drop-down, select **GET**.
16. Enter the URL in the request URL field in the following format:

<Copied ServerUrl>/api/v2/activity/activityMonitor

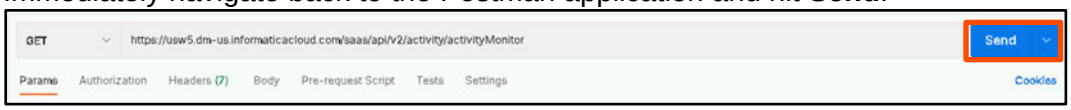


17. Select the **Headers** tab.
18. Enter Key and Value, as shown in the table below:

Key	Value
Accept	application/json
Content-Type	application/json
icSessionId	<enter the copied session ID>

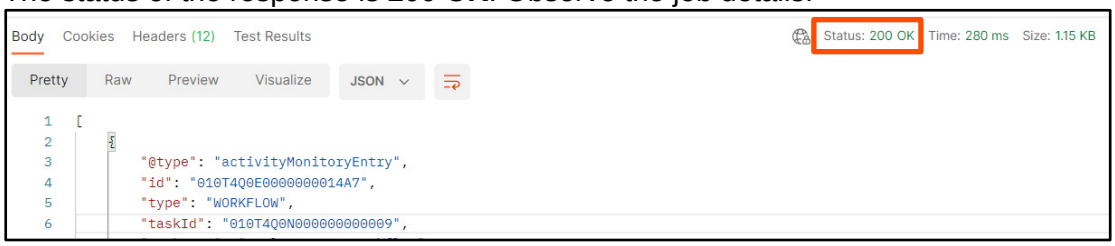
Before you send the request, you must ensure that at least 1 job is running in the DI service. This API will then fetch the details of the running jobs. Let us start a job in DI and view the results in Postman.

19. Navigate back to IICS > Data Integration service and open your working folder.
20. Locate the task **tf_XX_BULK_LOAD_PROCESSING** and run it (you can right-click on the task to view the Run option).
21. Immediately navigate back to the Postman application and hit **Send**.



Note: You will view the results in the Body response tab only if you send the request as soon as you hit the run button in DI. Because, the ActivityMonitor API will only fetch the details of running jobs.

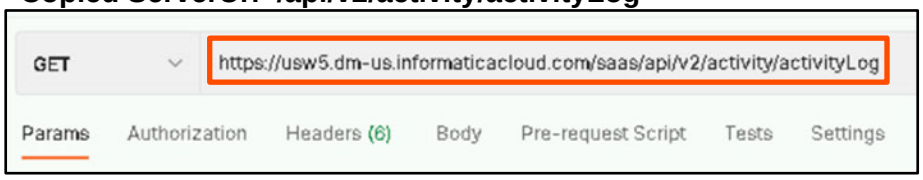
22. The status of the response is **200 OK**. Observe the job details.



Retrieve the entries from the activityLog REST API resource.

23. To add a new tab, click .
24. From the drop-down, select **GET**.
25. Enter the URL in the request URL field in the following format:

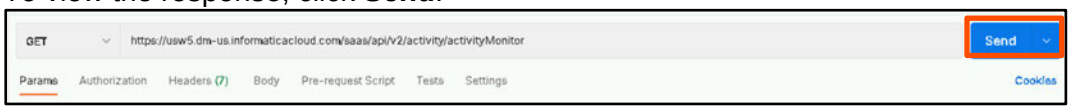
<Copied ServerUrl>/api/v2/activity/activityLog



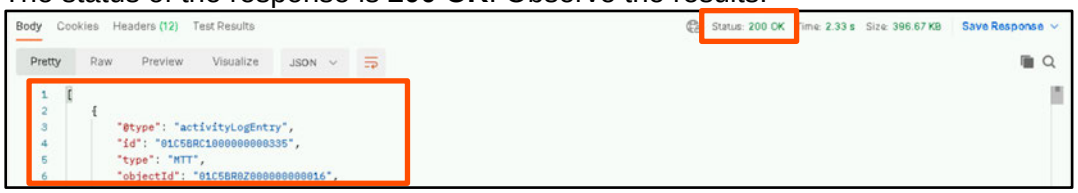
26. Select the **Headers** tab.
27. Enter Key and Value, as shown in the table below:

Key	Value
Accept	Application/json
Content-Type	application/json
icSessionId	<enter the copied session ID>

28. To view the response, click **Send**.



29. The status of the response is **200 OK**. Observe the results.



This concludes the lab.